HEM / HEMK / PCSM / PCSK GEN3

UTILITY SCALE CENTRAL STRING INVERTER

Operation Manual

for Power Electronics' staff
SW versions up to
HEM_K_PCSM_KG3SW5.21.0

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ABOUT THIS DOCUMENT

PURPOSE

This manual contains important instructions for the use and operation of the Power Electronics solar and battery GEN3 inverters (HEM, HEMK, PCSM and PCSK). From now on, the term “product” will be used to refer to HEM, HEMK, PCSM and PCSK GEN3.

Throughout this document, in those tasks where access via Power Electronics remote management tool is indicated, the configuration and monitoring tool developed by Power Electronics must be used. The manuals for these tools can be found on the digital platform of Power Electronics.

Power Electronics reserves the right to modify product features.

TARGET AUDIENCE

This manual is intended for personnel who have the current certification issued by Power Electronics who will use and operate Power Electronics’ HEM, HEMK, PCSM and PCSK GEN3 products.

Only authorized technical personnel validated by Power Electronics may commission, configure and operate inverters. The instructions assume that the installer is familiar with electrical installation rules and regulations.

NOTICE

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REVISIONS CONTROL

DATE (DD / MM / YYYY)	REVISION	DESCRIPTION
22 / 12 / 2025	A	First edition.
06 / 03 / 2026	B	Update to SW version 5.1.0: “Status Messages, Events, Faults and Warnings”, “Visualization and Status Parameters”, “Programming Parameters”, and “Service Group Parameters”. Other changes in “Safety Warnings” and “Interface Description (Modbus Communication)”. Other improvements.
11 / 06 / 2026	C	Update to SW version 5.4.0: “Status Messages, Events, Faults and Warnings”, “Visualization and Status Parameters”, “Programming Parameters”, “Service Group Parameters”, “Parameter Initialization”, “Important Safety Instructions” and “Safety Warnings”. Other improvements.
25 / 06 / 2026	D	Update of faults list: reset type and reset role added. Minor changes.
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Note: changes in this manual, with respect to the previous version, are highlighted in blue.

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ACRONYMS

The terms commonly used in the documentation of Power Electronics' products of the *Solar & Storage* division are listed in the table below.

ACRONYM	MEANING
AASS	Auxiliary Services.
AC	Alternating Current.
AI	Analogue Input.
AO	Analogue Output.
BESS	Battery Energy Storage System.
BMS	Battery Manager Solution.
CP	Control Pilot.
CPU	Central Processing Unit.
dBA	Weighted decibel level.
DC	Direct Current.
DCL	Discharge Current Limit.
DI	Digital Input.
DSP	Digital Signal Processor.
DO	Digital Output.
EMS	Energy Management System.
FPGA	Programmable device (Field-Programmable Gate Array).
FRU	Field Replaceable Unit.
GFDI	Ground Fault Detector Interrupter.
GPRS	General Packet Radio Services, a data transmission system.
HVAC	Heating, Ventilation, and Air Conditioning.
HW	Hardware
IGBT	Insulated Gate Bipolar Transistor.
IMI	Insulation Monitoring Device.
IK	Degree of protection against mechanical impact.
IP	Degree of protection against solid/fluid objects.
IT	Grid system where the power supply is kept isolated and the electrical equipment system is grounded.
LOTOTO	Locking, tagging and verification procedure (Lock Out – Tag Out – Try Out).
LVR	Low Voltage Regulation.
MCB	Miniature Circuit Breaker.
MCCB	Moulded Case Circuit Breaker.
MPCS	Multi Power Conversion System.
MID	Measuring Instrument Directive.
MV	Medium Voltage. This term is used to refer to high voltage in general.
OF	Optical Fiber.
PCB	Printed Circuit Board.

ACRONYM	MEANING
PE	Ground connection (Protective Earth).
PI	Proportional and Integral.
PLC	Programmable Logic Controller.
POI	Point Of Interconnection.
PowerComms	Configuration and monitoring interface developed by Power Electronics.
PPE	Personal Protection Equipment.
PV	Photovoltaic energy.
RAL	Color code (Reichsausschuß für Lieferbedingungen und Gütesicherung).
RCD	Residual Current Device.
RCM	Residual Current Monitor.
RFID	Radio Frequency Identification.
SOC	State Of Charge – referred to battery.
SOH	State Of Health – referred to battery. It compares the actual state of the battery to its initial conditions. It is measured in percentage.
STO	Safe Torque Off.
SW	Software.
THD	Total Harmonic Distortion.
TN	Grid system where the power supply is grounded, and the electrical equipment system is brought to the same ground through the neutral connector.
TT	Grid system where both the power supply and the electrical devices are connected to the ground via separate connections.
UPS	Uninterruptible Power Supply.
Vac	Alternating current voltage.
Vdc	Direct current voltage.

IMPORTANT SAFETY INSTRUCTIONS

The user must read and understand all product-related instructions described below before performing any operation task.

SAVE THESE INSTRUCTIONS

This manual contains important instructions for models **HEM/K** and **PCSM/K GEN3** that shall be followed during installation, operation, maintenance and troubleshooting of the inverter.

Read carefully all documentation before manipulating the product and pay special attention to safety recommendations to maximize the performance of this product and to ensure its safe use and installation.

It is the responsibility of the user to follow the instructions provided in this manual and all product documentation, follow good electrical practices, and identify all warnings and recommendations before starting up, operating, or servicing the product.

Follow the health and safety regulations of each country / area and the good practice / recommendations of each industry. The most restrictive health and safety standards shall be followed in case of discrepancy. When adverse weather conditions exist, work restrictions shall be applied.

In this manual, safety messages are classified as follows:

 **WARNING**	Identifies potentially hazardous situations where dangerous voltage may be present, which if not avoided, could result in minor personal injury, serious injury or death. Be extremely careful and follow the instructions to avoid the risk of electrical shocks.
 **CAUTION**	Identifies potentially hazardous situations, which if not avoided, could result in product damage or minor or moderate personal injury. Read the message and follow the instructions carefully.
 **NOTICE**	Identifies important measures to take in order to prevent damages to the product.

Other symbols used in this manual for safety messages are:

	Hot surface. Be careful and follow the instructions to avoid burns and personal injuries.
	Caution, risk of electric shock. Energy storage timed discharge. Wait for the indicated time to avoid electrical hazards.

WARNING

If the product is used in a manner other than that specified by the manufacturer, the protection provided by the product may not be effective.

When the photovoltaic array is exposed to light, it supplies a DC voltage to the product.

Keep all product documentation accessible throughout the life of the product for future reference.
Read all instructions and caution signs before the installation, operation, maintenance and troubleshooting of the product. Failure to comply with these warnings may result in serious electrical shock or death. Always pay attention to avoid possible accidents.

National/local restrictions for the product must be considered. The product must be used only for its intended application.
Power Electronics is not liable for damages resulting from improper use of the product or failure to comply with national or local regulations.

Do not apply voltage to a product that is damaged or with missing parts, even if the installation has been completed.

Ensure that the product is in the corresponding safety status before performing any action on it. Refer to the Safety Instructions for Operating, Troubleshooting and Maintenance document.

Follow the documentation of the product regarding the safety instructions to bring the product to the indicated safety status for the procedures that are going to be carried out. Hazardous energy lockout/tagout by applying the 5 golden rules or the equivalent procedure according to the applicable region or country.
Failure to comply with these warnings may result in serious electrical shock or death. Always pay attention to avoid possible accidents.
If the product is equipped with cutting devices, perform the maneuvers of these externally before accessing the product and/or performing any action.

Always use personal protective equipment (PPE) when opening a live door. In the case of **HEMK or PCSK** products, it is prohibited to open the doors of the MV transformer if there is voltage inside it.

Always handle switches with dry hands.

Wear personal protective equipment (PPE). The required PPE must be in accordance with, but not limited to, what is indicated in this document. The safety instructions (risk assessment and method statements) of the user must specify what specific PPE is required and when and how it is to be used in accordance with the specific safety plan for the work to be performed.

All work on the product must be carried out by at least two operators. The product must never be operated by only one person. There must be a plant supervisor on the work team, both for technical and health and safety reasons.

Do not use cables with damaged insulation. Do not subject cables to abrasion, excessive stress, overloading or clamping.

If the product stops due to a loss of power, do not perform any work on it.
The automatic restart function may be active.



5 min

The product contains capacitors on the AC and DC sides. The discharge time of the capacitors is 5 minutes. Wait until the capacitors have been discharged before carrying out any action.

EN

WARNING

In the case that any intervention or maneuver on the product is allowed, in the presence of voltage and with the appropriate PPE, consider that there is a potential risk of dropping a tool if work with metal tools inside the product is carried out. This can cause sparks or short circuits leading to an explosion.

After completing any work, thoroughly inspect the product and the work area to ensure that there are no loose metal parts (screws, washers, or other items). The presence of these parts can cause short circuits when the product is commissioned or put into service.

The enclosure must be properly sealed; otherwise, it may not adequately protect people or property in the event of any abnormal situation inside the product.

The output airflow can reach high temperatures during product operation, especially when the outside temperature and power supply are high.

PRODUCTS WHICH INCLUDE GFDI PROTECTION (HEM/K with negative pole to ground configuration):

- For more information on the operation and configuration of the GFDI protection, please refer to this manual.
- When a ground fault is indicated, normally grounded conductors may be ungrounded and hazardous, and energized or normally ungrounded conductors may be grounded.
- In case of GFDI activation, the fault LED will light up and a fault will be activated, which can be read via communications. Refer to this manual for more information.

PRODUCTS WITH UNGROUNDED POLES CONFIGURATION:

- When a ground fault is indicated, normally ungrounded conductors may be grounded.

In the cabinets where high voltage is present, the installer must add a warning signal “DANGER – HIGH VOLTAGE”. [1]

After the disconnection of the input voltage, the product may remain warm for several minutes. Be aware of the hot surfaces to avoid possible injury or burns.

GROUND CONNECTION IN HEM OR PCSM PRODUCTS
Electric shock prevention:

- The product chassis must be properly grounded to prevent a possible electric shock if a leakage current flows through the enclosure. Disconnect all the voltage supplies before proceeding with the corresponding actions within the product. Make sure the product is discharged before performing any operation.
- Connect the earth grounding to the earth plate of the product. Do not use the case or the chassis screw for grounding.
- The earth protection cable must be connected first and disconnected last.
- The product must be connected to ground by using the ground busbar located in the medium voltage switchgear.

In the case of **HEMK or PCSK** products:

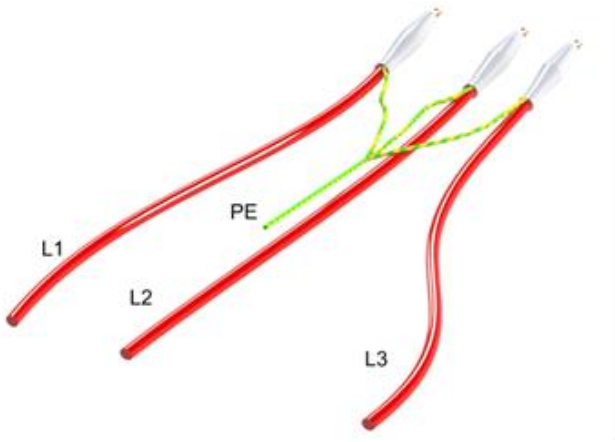
- The AC output of the product must be connected to an IT system and requires the use of an external isolation transformer between the inverter and the grid. This transformer is NOT integrated in the product.

[1] Medium or high voltage depending on the regulations of the country of installation.

WARNING

In the case of **HEM or PCSM** products:

- The isolation transformer between the product and the grid is integrated in the product and provides basic isolation between the HV and LV winding.
- AC output cables are medium voltage cables. As shown in the figure below, they are coaxial cables, each one including earth. To carry out a correct connection, it is necessary to bond the earth of each phase and connect it to the ground busbar of the product (located in the MV voltage).



In the case of **PCSM or PCSK** products**:**

- Adjustments to the charging algorithm controls must be made according to the recommendations provided by the battery manufacturer for the particular battery type.

The transformer may contain hazardous voltages even when the main system has been disconnected.

- Before performing any intervention on the transformer, the safe status of all associated circuits must be verified, both on the medium-voltage and high-voltage side and on the low-voltage side.
- The transformer may reach high temperatures during normal operation. Wait the required time to allow the components to cool before performing any intervention.

Medium-voltage and high-voltage switchgear contain conductors, busbars, switches, disconnectors, and associated equipment capable of maintaining lethal energy levels.

- Access to the medium-voltage/high-voltage compartment is restricted exclusively to personnel specifically qualified to work on medium-voltage and high-voltage installations.
- Access to medium-voltage/high-voltage compartments shall not be permitted until it has been verified, both documentally and physically, that the installation is de-energized, isolated, locked out, and grounded where applicable.
- Mechanical, electrical, or electronic interlocks of the switchgear shall never be bypassed, modified, or disabled.
- Do not open doors or covers of medium-voltage and high-voltage compartments while there is any possibility of the presence of voltage.
- Before accessing a switchgear cell, the correct position of circuit breakers, disconnectors, and grounding devices shall be verified in accordance with the approved operating procedure.
- The presence of voltage cannot always be determined visually. The absence of indicator lights or alarms does not guarantee the absence of voltage.
- Maintain the safety distances established by the applicable regulations at all times when working in the vicinity of medium-voltage and high-voltage installations.

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HOT SURFACE

After the disconnection of the input voltage, the product may remain warm for several minutes.
Be aware of the hot surfaces to avoid possible injury or burns.

The product is to be installed so that it is not expected to be contacted by persons.



NOTICE

CYBERSECURITY DISCLAIMER

This product is designed to be connected to and to communicate information and data via a network interface. Access to the system is restricted to those employees who legitimately need it for reasons of maintenance and/or updating of the system.
It is sole responsibility of the user for providing and continuously ensuring a secure connection between the product and user network or any other network (as the case may be). User must establish and maintain any appropriate measures (such as but not limited to the installation of firewalls, application of authentication measures, encryption of data, installation of antivirus programs, etc.) to protect the product, the network, its system and the interface against any kind of security breaches, unauthorized access, interference, intrusion, leakage and/or theft of data or information.
Power Electronics and its affiliates are not liable for damages and/or losses related to such security breaches, any unauthorized access, interference, intrusion, leakage and/or theft of data or information.

Safety statuses definition

To carry out operation, fault diagnosis, repair, and/or maintenance tasks, certain safety statuses have been defined.

The following table shows how to interpret the status coding used throughout the product documentation, along with a brief description of each status.

EXAMPLE: INVMG3_ST3B						
PRODUCT		**GENERATION**		**STATUS**		**DESCRIPTION**
INV	All inverters	G3	GEN3	ST1A	Status 1A	Inverter powered and feeding into the grid.
INVK	HEMK / PCSK inverters			ST1B	Status 1B	Inverter powered and not feeding into the grid.
INVM	HEM / PCSM inverters			ST2	Status 2	Fully isolated inverter. External and internal AC and DC cut-off. No AASS voltage (internal cut-off).
				ST3A	Status 3A	Inverter with AC and DC terminal voltage and auxiliary services (AASS).
				ST3B	Status 3B	Inverter with AC and DC terminal voltage, but without AASS (internal cut-off).
				ST3C[1]	Status 3C	Inverter with AC and DC terminal voltage, but without AASS (AC busbar protection tool).
				ST4	Status 4	Inverter with no voltage at AC terminals (external cut-off) but existing voltage at DC terminals. No AASS internal voltage (internal cut-off)
				ST5A	Status 5A	Inverter without voltage at DC terminals (external cut-off) in this status, the DC cabinet is grounded and short-circuited. There is existing voltage at AC terminals. There are no AASS voltages (internal cut-off).
				ST5B	Status 5B	Inverter without voltage at DC terminals (external cut-off), in this status the DC cabinet is not grounded nor short-circuited. Voltage present at AC terminals. No AASS voltage (internal cut-off).
				ST6[1]	Status 6	Inverter with no voltage at AC terminals but existing voltage at DC terminals and AASS.

[1] Applies to HEMK / PCSK.

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To identify the areas of the product that are allowed to be accessed in each safety status and the required conditions (always using the appropriate PPE and the necessary tools for each intervention), as well as the procedures to reach any safety status, please refer to the *Safety Instructions for Operating, Troubleshooting and Maintenance* document.

Recommended PPE and auxiliary elements

The operation, repair and maintenance of products involve a risk that may be mechanical, electrical, chemical, or related to work at heights. For this reason, **it is mandatory to wear the appropriate personal protective equipment (PPE) for each task.**

The PPE listed are a reference, and it will never be the definitive full list. The definition of all PPE is responsibility of the service of the customer, according to the regulations in force.

The list of PPE required for each type of risk is shown below:

Mechanical risk

				
Safety footwear	Mechanical gloves	Hard hat	Safety glasses Safety sunglasses	High visibility vest ARC rated

Electrical risk

				
Electrical shock proof boots	Electrical safety gloves	Safety helmet	Arc flash face shield	Clothing with arc flash protection 40 cal/cm2

Chemical risk

	
Safety glasses	Chemical resistant nitrile glove or akin

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Work at heights risk

				
Safety footwear	Mechanical and anti-cut gloves	Safety helmet	High visibility vest ARC rated	Safety harness with double rope

Auxiliary elements

			
First aid kit	Fire extinguisher	Insulating platform	Insulating mat
			
Vinyl fabric and dielectric clamps	Blocking devices	Blocking devices	LOTO kit
			
Ground and earthing kit	Danger tags	Security tape	Safety signaling element

SAFETY WARNINGS

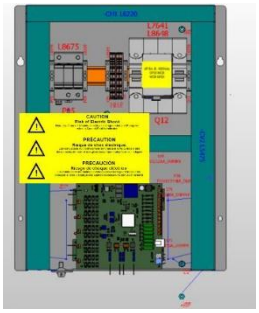
The product is labeled with safety warnings indicating electrical hazards. It is necessary to locate all the warnings before handling the product.

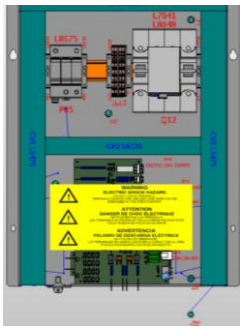
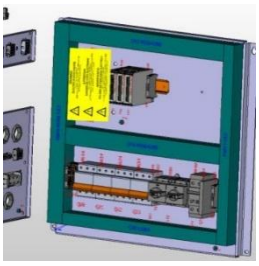
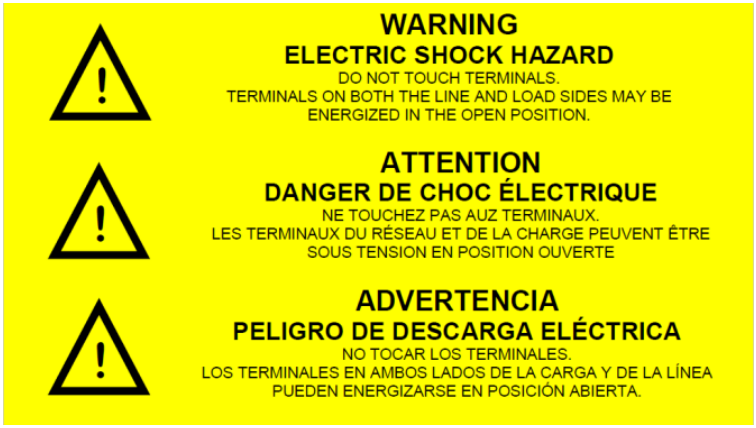
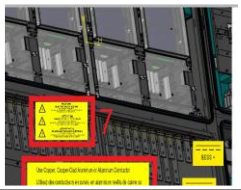
Safety warnings can be found either on labels or marked on the product itself and may be in different languages.

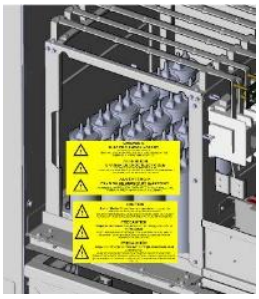
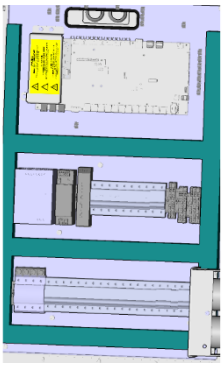
The following table describes and locates the safety labeling.

Note: the panels that show the location of the labels may vary slightly depending on the configuration of the project. For the accessories “Japan Kit” or “VDE test interface”, refer to the *Accessories Manual* for the corresponding translations of the safety warnings in Japanese and German, respectively.

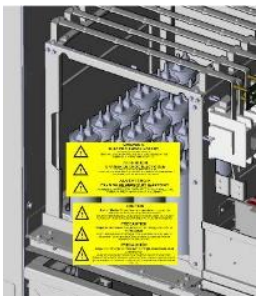
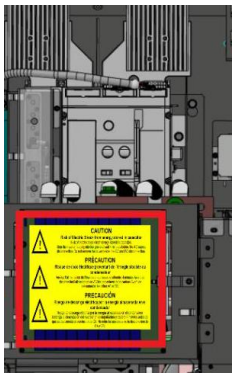
EN

SAFETY WARNINGS	LOCATION
LABELS	
CAUTION ELECTRICAL HAZARD DUE TO GROUND FAULT In the event of a ground fault, there is a risk of electrical hazard or shock due to a possible fault in the wiring. It is possible that the grounding cables are not correctly connected to the grounding circuit and voltage may be present in them. **CAUTION** **Risk of Electric Shock** Normally Grounded Conductors May be Ungrounded and Energized when a Ground-Fault is indicated **PRÉCAUTION** **Risque de choc électrique.** Les conducteus qui normalement sont mis à la terre purraient être déconnectés de terre et énergisés lorsqu´une fauteà terre est indiquée **PRECAUCIÓN** **Riesgo de choque eléctrico** Los conductores normalmente conectados a tierra pueden estar sin conexión a tierra y energizados cuando se indica una derivación a tierra	Near the GFDI of the DC cabinet (HEM / HEMK) (1 unit) 

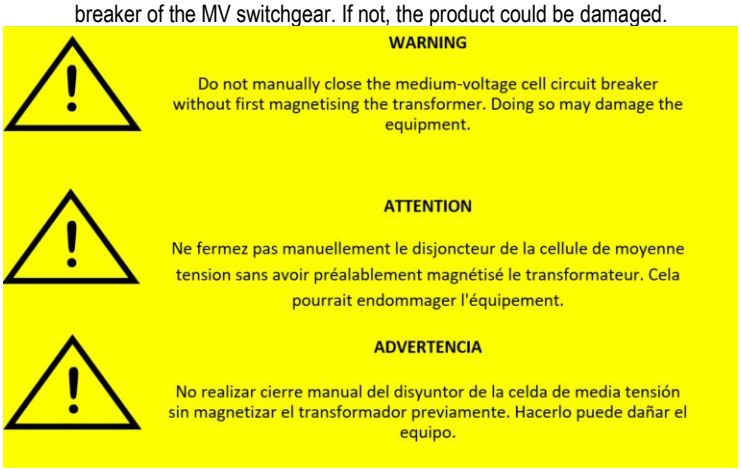
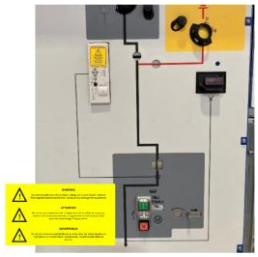
SAFETY WARNINGS	LOCATION
LABELS	
	DC protection panel of the DC cabinet (1 unit)
	
	MV switchgear panel of the protections and measurements cabinet (HEM / PCSM) (1 unit)
CAUTION DANGER OF ELECTRIC SHOCK ON TERMINALS Indicates electrical risk due to the presence of voltage on the terminals of the inverter.	
	Power supply panel of the main control cabinet (1 unit)
	
	On the pre-assembly of the DC fuses of the DC cabinet (number 7) (2 units)
	

SAFETY WARNINGS	LOCATION
LABELS	
CAUTION DANGER OF ELECTRIC SHOCK ON TERMINALS Indicates electrical risk due to the presence of voltage on the terminals of the inverter. **WARNING** **ELECTRIC SHOCK HAZARD** DO NOT TOUCH TERMINALS. TERMINALS ON BOTH THE LINE AND LOAD SIDES MAY BE ENERGIZED IN THE OPEN POSITION. **ATTENTION** **DANGER DE CHOC ÉLECTRIQUE** NE TOUCHEZ PAS AUZ TERMINAUX. LES TERMINAUX DU RÉSEAU ET DE LA CHARGE PEUVENT ÊTRE SOUS TENSION EN POSITION OUVERTE **ADVERTENCIA** **PELIGRO DE DESCARGA ELÉCTRICA** NO TOCAR LOS TERMINALES. LOS TERMINALES EN AMBOS LADOS DE LA CARGA Y DE LA LÍNEA PUEDEN ENERGIZARSE EN POSICIÓN ABIERTA.	On the protection cover of the AASS transformer of the AASS transformer cabinet (1 unit) Air circuit breaker (ACB) (HEMK / PCSK) (1 unit) C filter panel of the LC filter cabinet (1 unit)  Control panel of the main control cabinet (1 unit) 

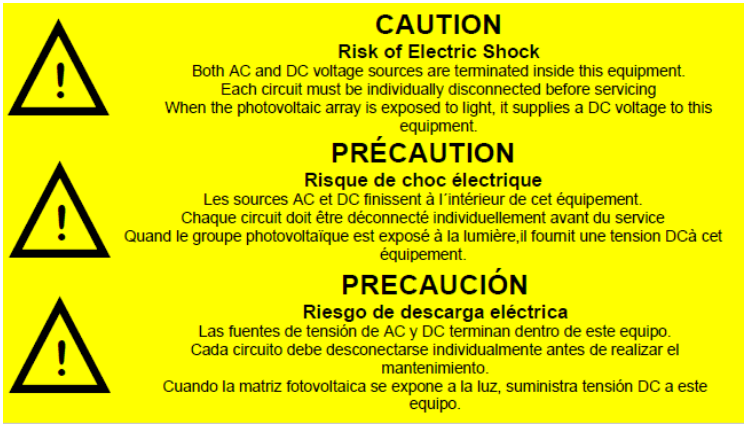
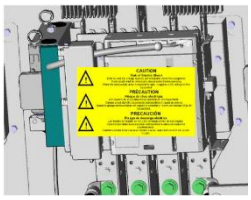
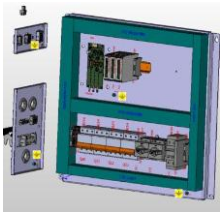
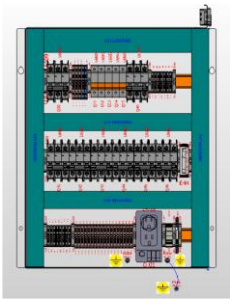
EN

SAFETY WARNINGS	LOCATION
LABELS	
	C filter panel of the LC filter cabinet (1 unit) 
ELECTRICAL HAZARD DUE TO CAPACITOR DISCHARGE The inverter contains energized capacitors which may be powering parts of the inverter until they are completely discharged. It is necessary to wait for a certain time (indicated on the label) after the inverter has stopped in order to be able to access the different components without risk of electrical hazard. 	Control panel capacitors of the main control cabinet (1 unit) 
	At the height of the PCB boards of the DC cabinet (4 units) 
	On each power stage (2, 3 or 4 units [1])

[1] The number of labels depends on the frame of the product.

SAFETY WARNINGS	LOCATION
LABELS	
CAUTION FUSE INSTALLATION Indicates that the replacement fuse must be of the same type and rating as the fuse being replaced. **Note**: there are labels inside the inverter indicating the characteristics of the fuses.  **WARNING** For continued protection against risk of fire, replace only with the same type and ratings of fuse **ATTENTION** Pour la protection continue fase au risque d´incendie. Remplacez les fusibles uniquement avec ceux de même type et gamme **ADVERTENCIA** Para una protección continua contra el riesgo de incendio, reemplace solo con el mismo tipo y clasificación de fusible.	Near the fuses of the DC cabinet (number 3) (4 units)  PSA panel of the main control cabinet (1 unit)  MPP panel of the AC cabinet (HEMK / PCSK) or LC filter cabinet (HEM / PCSM) (1 unit) 
CAUTION MV SWITCHGEAR HANDLING Indicates that the transformer must be magnetized before manually closing the circuit breaker of the MV switchgear. If not, the product could be damaged.  **WARNING** Do not manually close the medium-voltage cell circuit breaker without first magnetising the transformer. Doing so may damage the equipment. **ATTENTION** Ne fermez pas manuellement le disjoncteur de la cellule de moyenne tension sans avoir préalablement magnétisé le transformateur. Cela pourrait endommager l'équipement. **ADVERTENCIA** No realizar cierre manual del disyuntor de la celda de media tensión sin magnetizar el transformador previamente. Hacerlo puede dañar el equipo.	MV switchgear of the MV switchgear cabinet (HEM / PCSM) (1 unit) 

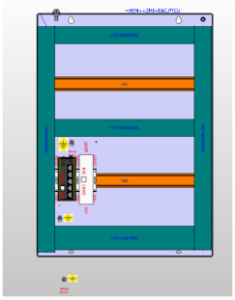
EN

SAFETY WARNINGS	LOCATION
LABELS	
CAUTION AC/DC ELECTRIC SHOCK RISK Indicates that before performing any operation or maintenance task, the corresponding LOTOTO procedure must be carried out ensuring the absence of voltage on both AC and DC. Note that even if the AC and DC circuits are open, external voltage may be present at the inverter terminals.  **CAUTION** **Risk of Electric Shock** Both AC and DC voltage sources are terminated inside this equipment. Each circuit must be individually disconnected before servicing When the photovoltaic array is exposed to light, it supplies a DC voltage to this equipment. **PRÉCAUTION** **Risque de choc électrique** Les sources AC et DC finissent à l'intérieur de cet équipement. Chaque circuit doit être déconnecté individuellement avant du service Quand le groupe photovoltaïque est exposé à la lumière,il fournit une tension DCà cet équipement. **PRECAUCIÓN** **Riesgo de descarga eléctrica** Las fuentes de tensión de AC y DC terminan dentro de este equipo. Cada circuito debe desconectarse individualmente antes de realizar el mantenimiento. Cuando la matriz fotovoltaica se expone a la luz, suministra tensión DC a este equipo.	On the front of the DC switch disconnectors of the DC cabinet (2, 3 or 4 units [2])  On the front of the AC circuit breaker of the AC cabinet (HEMK / PCSK) (1 unit) 
PROTECTIVE CONDUCTOR TERMINAL Indicates the DC user grounding points and the different grounding points of all the panels of the inverter. Note: the layouts on the DC user grounding points will depend on the configuration of the product. 	MV switchgear panel of the protections and measurements cabinet (HEM / PCSM) (4 units)  User panel [3] of the user cabinet (HEM / HEMK) (3 units) 

[2] The number of labels depends on the frame of the product.
[3] The user panel may vary depending on the configuration of the product.

SAFETY WARNINGS	LOCATION
LABELS	

User control panel of the user cabinet (3 units)



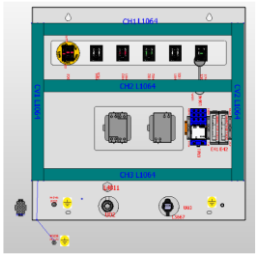
Pushbuttons panel of the main control cabinet (3 units)

PROTECTIVE CONDUCTOR TERMINAL

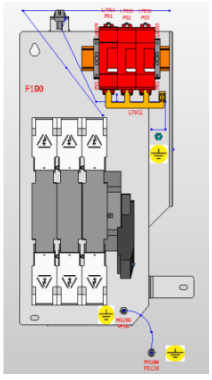
Indicates the DC user grounding points and the different grounding points of all the panels of the inverter.

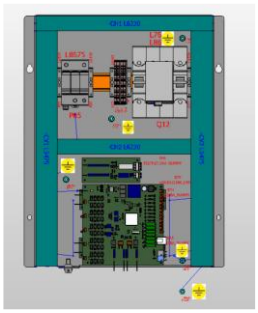
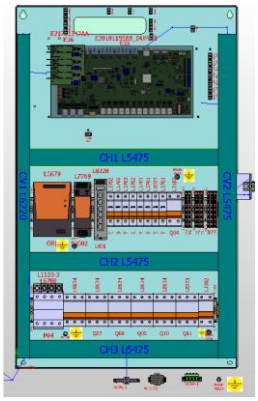
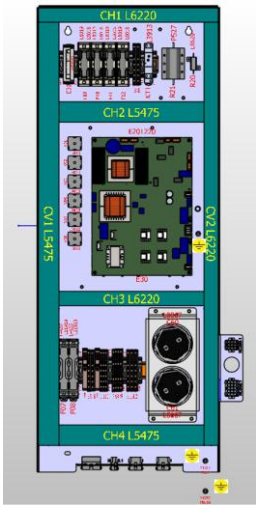
Note: the layouts on the DC user grounding points will depend on the configuration of the product.

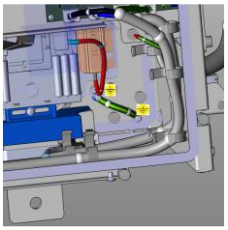
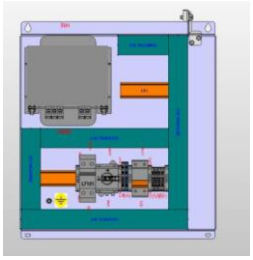




AC and SPD protection panel of the AC cabinet (HEMK / PCSK) or LC filter cabinet (HEM / PCSM) (3 units)

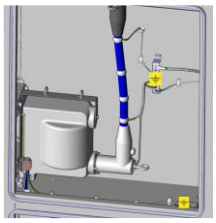
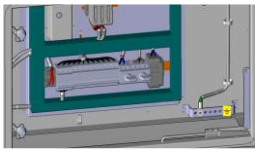
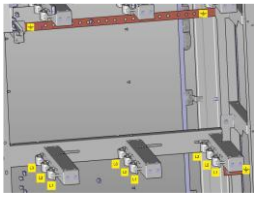


SAFETY WARNINGS	LOCATION
LABELS	
	DC protection panel of the DC cabinet (5 units)
	
	Control panel of the main control cabinet (5 units)
	
PROTECTIVE CONDUCTOR TERMINAL Indicates the DC user grounding points and the different grounding points of all the panels of the inverter. **Note**: the layouts on the DC user grounding points will depend on the configuration of the product. 	Control panel capacitors of the main control cabinet (3 units)
	

SAFETY WARNINGS	LOCATION
LABELS	
PROTECTIVE CONDUCTOR TERMINAL Indicates the DC user grounding points and the different grounding points of all the panels of the inverter. **Note**: the layouts on the DC user grounding points will depend on the configuration of the product. 	Inside each power stage of the modules cabinet (4, 6 or 8 units [4])  POCU panel of the user cabinet (1 unit) 

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[4] The number of labels depends on the frame of the product.

SAFETY WARNINGS	LOCATION
LABELS	
GROUND TERMINAL Indicates the chassis grounding points of the inverter. 	On all the doors next to the grounding mesh (4, 14 or 20 units [5])  Next to the Econ Mode optional transformer of the MV switchgear cabinet (HEM / PCSM) (2 units)  MV switchgear panel of the protections and measurements cabinet (HEM / PCSM) (1 unit)  Ground terminal of the MV switchgear of the MV switchgear cabinet (HEM / PCSM) (3 units) 

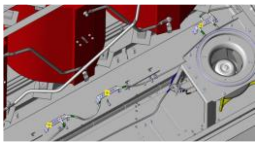
[5] The number of labels depends on the configuration of the product.

SAFETY WARNINGS	LOCATION
LABELS	

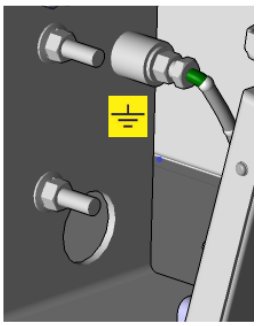
GROUND TERMINAL
Indicates the chassis grounding points of the inverter.



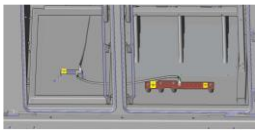
On the plates of the MV transformer of the MV cabinet (HEM / PCSM) (3 units)



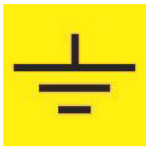
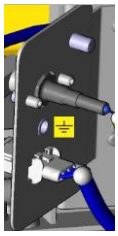
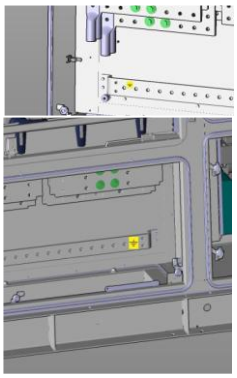
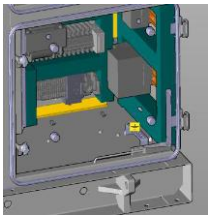
On the two opposite corners of the MV transformer of the MV cabinet (HEM / PCSM) (2 units)



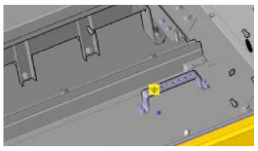
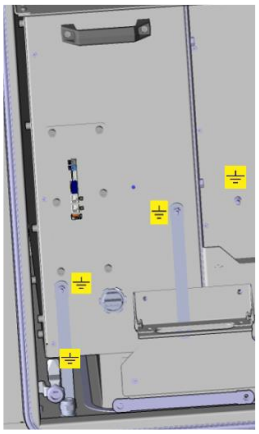
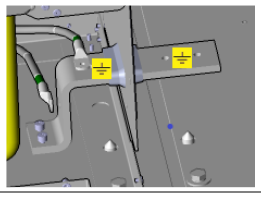
On the lower switch plate and lower ground plate of the MV switchgear of the MV switchgear cabinet (HEM UL) (3 units)



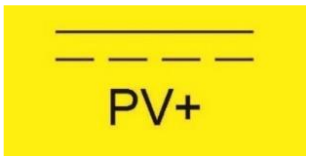
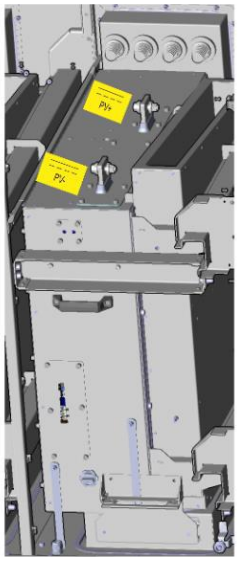
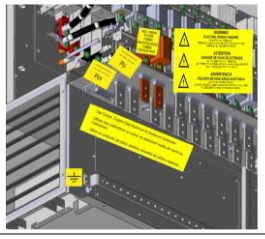
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SAFETY WARNINGS	LOCATION
LABELS	
GROUND TERMINAL Indicates the chassis grounding points of the inverter. 	On the ground of each fan of the DC cabinet (2 units), modules cabinet (1, 2, 3 or 4 units [6]), MV cabinet (HEM / PCSM) (4 units) and LC filter cabinet (1 unit)  At each end of the DC ground busbar of the access cover for DC cables connection (2 units)  On the ground plate of the user cabinet (1 unit) 

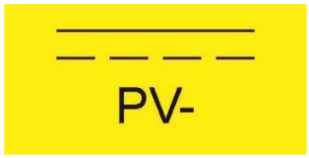
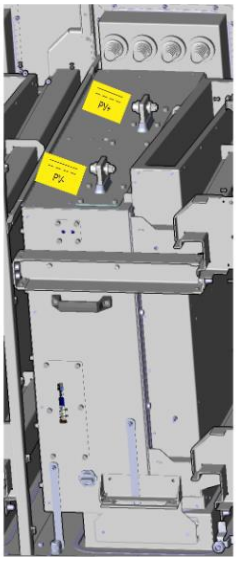
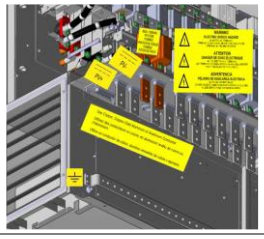
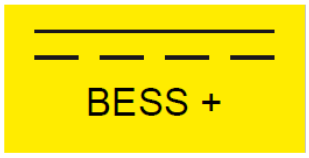
[6] The number of labels depends on the frame of the product.

SAFETY WARNINGS	LOCATION
LABELS	
	On the ground plate of both the AASS transformer and inductance of the inductance cabinet (1 unit) 
	On each power stage of the modules cabinet (8, 12 or 16 units [7]) 
GROUND TERMINAL Indicates the chassis grounding points of the inverter. 	On the interconnection ground plate between the MV transformer and the AC busbar of the MV and LC filter cabinets (2 units) 

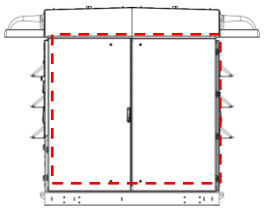
[7] The number of labels depends on the frame of the product.

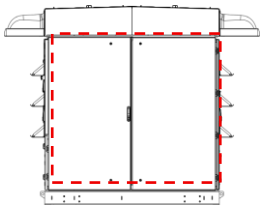
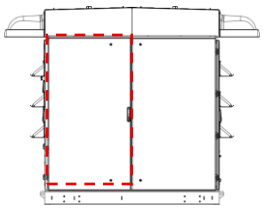
SAFETY WARNINGS	LOCATION
LABELS	
POSITIVE PHOTOVOLATAIC VOLTAGE Indicates that the DC voltage that comes from the photovoltaic field is positive.  	On each power stage of the modules cabinet (2, 3 or 4 units [8])  On the polycarbonates of the DC busbar of the DC cabinet (4 units per row of plates) 

[8] The number of labels depends on the frame of the product.

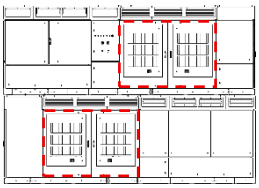
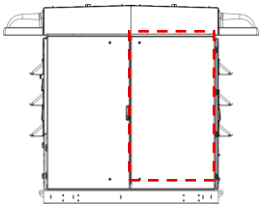
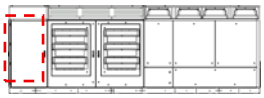
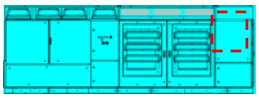
SAFETY WARNINGS	LOCATION
LABELS	
NEGATIVE PHOTOVOLATAIC VOLTAGE Indicates that the DC voltage that comes from the photovoltaic field is negative. 	On each power stage of the modules cabinet (2, 3 or 4 units [9])  On the polycarbonates of the DC busbar of the DC cabinet (4 units per row of plates) 
POSITIVE BESS VOLTAGE Indicates that the BESS DC voltage is positive. 	Close to each end of the DC busbar of the access cover for DC cables connection (PCSM / PCSK) (2 units) 

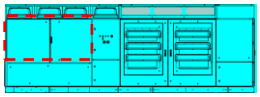
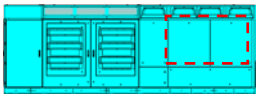
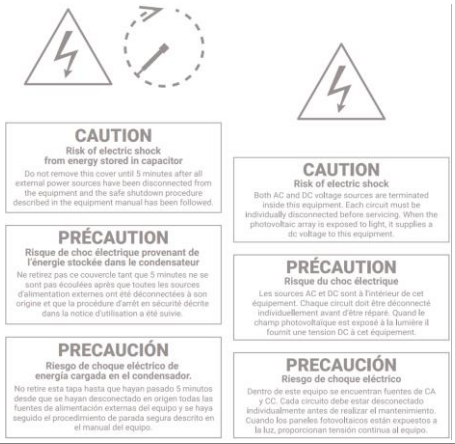
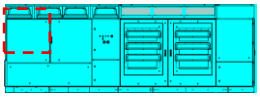
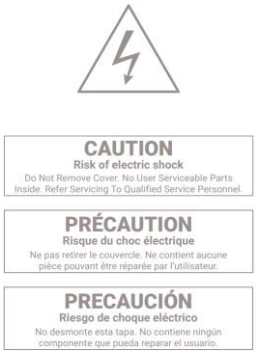
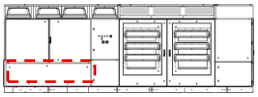
[9] The number of labels depends on the frame of the product.

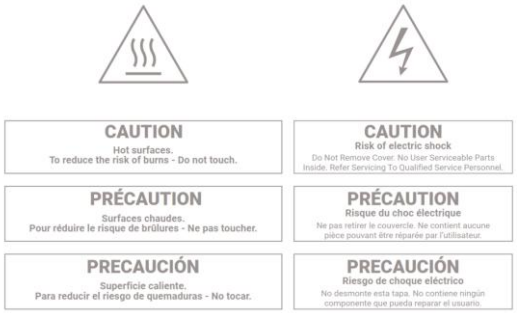
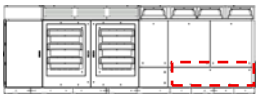
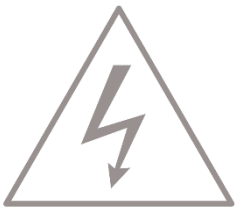
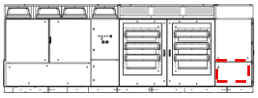
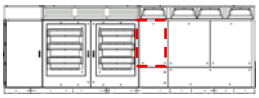
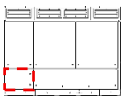
SAFETY WARNINGS	LOCATION
LABELS	
NEGATIVE BESS VOLTAGE Indicates that the BESS DC voltage is negative. 	Close to each end of the DC plates of the access cover for DC cables connection (PCSM / PCSK) (2 units) 
FIRST AID Indicates the steps to follow for first aid in the event of an accident. 	Inside of the MV switchgear cabinet (HEM / PCSM) (1 unit) 

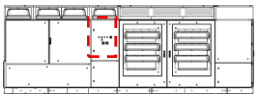
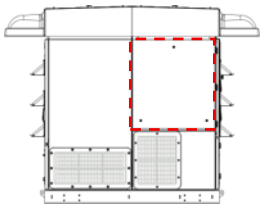
SAFETY WARNINGS	LOCATION
LABELS	
FIVE GOLDEN RULES INDICATIONS Indicates the five golden rules that must be followed when working under voltage. 	Inside of the MV switchgear cabinet (HEM / PCSM) (1 unit) 
SERIGRAPHY	
There is a risk of electric shock. Both AC and DC circuits must be individually disconnected before servicing. 	Left door of the MV switchgear cabinet (HEM / PCSM) 

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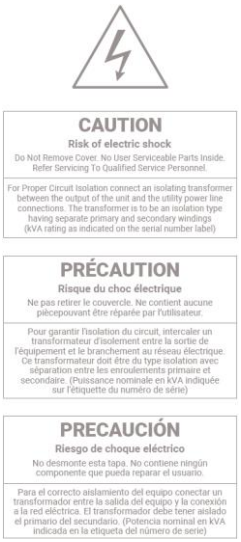
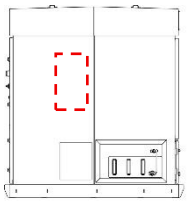
SAFETY WARNINGS	LOCATION
SERIGRAPHY	
High voltage may be present. Failure to take the necessary precautions may result in death, minor or serious personal injury.  	MV cabinet (HEM / PCSM)  Right door of the MV switchgear cabinet (HEM / PCSM)  Access door to MV AC cables installation (HEM / PCSM)  Econ Mode (HEM / PCSM) 

SAFETY WARNINGS	LOCATION
SERIGRAPHY	
There is a risk of electric shock. Both AC and DC circuits must be individually disconnected before servicing. It is necessary to wait a certain time (indicated on the serigraphy) from the moment the power supply is cut off and the safe shutdown has been carried out before being able to access the different components without risk of electrical hazard. 	Left door of the DC cabinet (PCSM / PCSK) and right door of the DC cabinet  Modules cabinet 
There is a risk of electric shock. When the photovoltaic array is exposed to light. It supplies a DC voltage to the product. Both AC and DC circuits must be individually disconnected before servicing. It is necessary to wait a certain time (indicated on the serigraphy) from the moment the power supply is cut off and the safe shutdown has been carried out before being able to access the different components without risk of electrical hazard. 	DC cabinet left door (HEM / HEMK) 
Do not open the cover, there is a risk of electric shock. 	Access cover for DC cables connection 

SAFETY WARNINGS	LOCATION
SERIGRAPHY	
Do not open the cover, there is a risk of electric shock. Hot surfaces. To reduce the risk of burns, do not touch. 	Main inductance access cover 
There is a potential hazardous situation where dangerous voltage may be present, which if not avoided, could result in minor personal injury, serious injury or death. 	Protections and measurements cabinet (HEM / PCSM) 
There is a risk of electric shock from energy stored in capacitor. It is necessary to wait a certain time (indicated on the serigraphy) from the moment the power supply is cut off before being able to access the different components without risk of electrical hazard. 	C filter access cover 
There is a risk of electric shock. Both AC and DC circuits must be individually disconnected before servicing. 	AC busbar (HEM / PCSM)  AC circuit breaker access door (HEMK / PCSK) 

SAFETY WARNINGS	LOCATION
SERIGRAPHY	
Read all documentation carefully before handling the product. The product has the interactive grid support utility. It also indicates the degree of protection of the product. **WARNING** To reduce the risk of injury, read all instructions. Grid support utility interactive inverter | NEMA 3R **AVERTISSEMENT** Pour réduire le risque de blessure, lisez toutes les instructions. Onduleur raccordée au réseau | NEMA 3R **ADVERTENCIA** Para reducir el riesgo de lesión, lea todas las instrucciones. Inversor con funcionalidad de soporte a red | NEMA 3R	Main control cabinet 
Do not open the cover, there is a risk of electric shock. It is necessary to wait a certain time (indicated on the label) from the moment the power supply is cut off and the safe shutdown has been carried out before being able to access the different components without risk of electrical hazard.  **CAUTION** **Risk of electric shock** Do Not Remove Cover. No User Serviceable Parts Inside. Refer Servicing To Qualified Service Personnel. **PRÉCAUTION** **Risque du choc électrique** Ne pas retirer le couvercle. Ne contient aucune pièce pouvant être réparée par l'utilisateur. **PRECAUCIÓN** **Riesgo de choque eléctrico** No desmonte esta tapa. No contiene ningún componente que pueda reparar el usuario.  **CAUTION** **Risk of electric shock from energy stored in capacitor** Do not remove this cover until 5 minutes after all external power sources have been disconnected from the equipment and the safe shutdown procedure described in the equipment manual has been followed. **PRÉCAUTION** **Risque de choc électrique provenant de l'énergie stockée dans le condensateur** Ne retirez pas ce couvercle tant que 5 minutes ne se sont pas écoulées après que toutes les sources d'alimentation externes ont été déconnectées à son origine et que la procédure d'arrêt en sécurité décrite dans la notice d'utilisation a été suivie. **PRECAUCIÓN** **Riesgo de choque eléctrico de energía cargada en el condensador** No retire esta tapa hasta que hayan pasado 5 minutos desde que se hayan desconectado en origen todas las fuentes de alimentación externas del equipo y se haya seguido el procedimiento de parada segura descrito en el manual del equipo.	DC cabinet side 

EN

SAFETY WARNINGS	LOCATION
SERIGRAPHY	
Do not open the cover/door, there is a risk of electric shock. The transformer to be connected is to be an isolation type having separate primary and secondary windings. 	AC side access cover (HW1.3 onwards) (HEMK / PCSK) 

Any of the following symbols may also be used in the labeling or marking of the solar and battery GEN3 inverters:

	Alternating current supply.
	Product phase conductor.
	ON (up) and OFF (down) for contact signaling.

NOTICE

ARC-FLASH PROTECTION MARKING

The arc flash protection marking must be defined on a project specific basis, considering the actual electrical characteristics, short circuit levels, and installation conditions of the plant. The final arc flash hazard analysis and the corresponding determination of labels, boundaries, and PPE requirements are not within the scope of Power Electronics. Each customer is responsible for performing a dedicated arc flash study for their project and for implementing the applicable safety markings.

The calculation must be done for AC and DC side of the inverter. For each case, a different calculation method has been used, both based on NFPA 70E:2021 (Standard for Electrical Safety in the Workplace), from the National Fire Protection Association.

In table 130.7(C) (16) from NFPA 70E 4 PPE type classes are presented based on incident energy:

Arc Flash Hazard PPE Category	Minimum Arc Rating	PPE Layers	Arc Rated Clothing (1)	Protective Equipment
PPE CAT 1	4 cal/cm^2	1 Layer	Arc-rated long-sleeve shirt and pants or arc-rated coverall Arc-rated face shield (2) or arc flash suit hood Arc-rated jacket, parka, rainwear, or hard hat liner (AN) (FR long sleeve shirt, FR coveralls, FR pants, FR face shields, FR arc flash hoods, FR jackets, FR coats & parkas & FR hard hat liners)	Hard hat Safety glasses or safety goggles (SR) Hearing protection (ear canal inserts) Heavy duty leather gloves (3) Leather footwear (AN)
PPE CAT 2	8 cal/cm^2	1 or 2 Layers	Arc-rated long-sleeve shirt and pants of arc-rated coverall Arc-rated flash suit hood or arc-rated face shield (see Note 2) and arc-rated balaclava Arc-rated jacket, parka, rainwear, or hard hat liner (AN)	Hard hat Safety glasses or safety goggles (SR) Hearing protection (ear canal inserts) Heavy duty leather gloves (3) Leather footwear (AN)
PPE CAT 3	25 cal/cm^2	2 or 3 Layers	Arc-rated long-sleeve shirt (AR) Arc-rated pants (AR) Arc-rated coverall (AR) Arc-rated arc flash suit jacket (AR) Arc-rated arc flash suit pants (AR) Arc-rated arc flash suit hood Arc-rated gloves (1)	Hard hat Safety glasses or safety goggles (SR) Hearing protection (ear canal inserts) Leather footwear
PPE CAT 4	40 cal/cm^2	3 + Layers	Arc-rated long-sleeve shirt (AR) Arc-rated pants (AR) Arc-rated coverall (AR) Arc-rated arc flash suit jacket (AR) Arc-rated arc flash suit pants (AR) Arc-rated arc flash suit hood Arc-rated gloves (1) Arc-rated jacket, parka, rainwear, or hard hat liner (AN)	Hard hat Safety glasses or safety goggles (SR) Hearing protection (ear canal inserts) (4) Leather footwear

Notes:
(1) Arc rating is defined in Article 100.
(2) Face shields are to have wrap-around guarding to protect not only the face but also the forehead, ears, and neck, or, alternatively, an arc-rated arc flash suit hood is required to be worn.
(3) If rubber insulating gloves with leather protectors are used, additional leather or arc-rated gloves are not required. The combination of rubber insulating gloves with leather protectors satisfies the arc flash protection requirement.
(4) Other types of hearing protection are permitted to be used in lieu of or in addition to ear canal inserts provided they are worn under an arc-rated arc flash suit hood.
(AN) As Needed (optional)
(AR) As Required
(SR) Selection Required

EN

1. INTRODUCTION AND GENERAL CONSIDERATIONS

1

CAUTION

According to IEC 62920 type A products are not designed to be used in a residential environment.
This product may cause radio frequency interference, in which case the user may be required to take additional mitigation measures against electromagnetic interference.

NOTICE

It may be necessary to obtain authorization from the relevant local authority before making changes to the settings indicated in this document.

1.1. Regulatory compliance

This product complies the 3rd edition of UL1741, which includes applicable requirements of SA (SA8, SA9, SA10, SA11, SA12, SA13, SA14 and SA15 with SRD Electric Rule No. 21, Hawaiian Rule 14H for Hawaii and PRC 024-2) and IEEE 1547.1 2005 required for full compliance for Support Utility Interactive Inverter requirements.

1.2. Preliminary steps

WARNING

Before using and operating the product, it should have been installed and commissioned in accordance with the official Power Electronics documentation.
The user must ensure that the minimum essential basic configuration of the product has been carried out during the initial commissioning procedure.

1.2.1. Firmware check

The product firmware can be checked using the Power Electronics remote management tool. For more information, refer to the *Operation Manual* for the corresponding application.

1.3. Sign criteria

The following are the sign criteria for identifying whether a setpoint is for charging or discharging, or whether it is inductive or capacitive reactive energy:

- Charging: **positive sign**.
- Discharging: **negative sign**.
- Inductive reactive power: **positive sign**.
- Capacitive reactive power: **negative sign**.

1.4. Technical characteristics

The technical characteristics of the product can be found in the datasheets included in the *Hardware and Installation Manual.*

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2.INTERFACE DESCRIPTION

2

2.1. Control elements and indicators

The user can interact directly with Power Electronics inverters through the different control elements and indicators. Please refer to the *Hardware and Installation Manual* to find out where they are located.

- **ON/OFF key:** allows to stop or start the power stages of the inverter.
- **AC DISCONNECT:** disconnector. Allows the AC disconnection of the product. The manipulation of this selector must only be used for the LOTOTO procedure of the inverter.
- **DC DISCONNECT:** disconnector. Allows the DC disconnection of the product. The manipulation of this selector must only be used for the LOTOTO procedure of the inverter.
- **EMERGENCY STOP:** pushbutton. Allows to stop and isolate the inverter in case of an emergency situation. Auxiliaries are not isolated with this pushbutton.

NOTICE

Do not use the emergency stop to perform regular stops. It must only be used when an emergency occurs. Otherwise, longevity of the main components could be shortened and result in product damage.

- **LEDs:** show the operation status of the products, according to the following table:

	COLOR	LED ON	LED OFF
AC Voltage presence	White	Presence of AC voltage	No presence of AC voltage
DC Voltage presence	White	Presence of DC voltage	No presence of DC voltage
Run	Green	Inverter in operation	Inverter stopped
Fault / Warning	Red	Flashing every second: Warning Still: Fault	Without faults / warnings

In the case of **PCS** battery inverters, the number of DCx VOLTAGE PRESENCE LEDs depends on the number of busbars. These LEDs turn on when DC voltage is present:

- PCS of two busbars: two LEDs.
- PCS of four busbars: four LEDs.

2.2. Communication interfaces

The user can access the Power Electronics inverter via a direct connection through the **RJ45 port** located on the control cabinet door, which allows an **Ethernet** cable to be connected. It is also possible to access the inverter remotely via the plant's communications network using the connection enabled in the user cabinet.

The product may incorporate the PSAMS (*Power Station Advanced Monitoring System*), which provides a security boundary between the external network (WAN) and the trusted internal network (LAN). In that case, remote access to the inverter through the plant network is routed through the PSAMS, which controls communications between both networks. For information on PSAMS features, read and configuration parameters, operational scenarios, and communication and access protection measures, refer to Annex I: PSAMS.

- **Communication protocols**: communication with the inverter is via Ethernet using Modbus TCP/IP.
- **Modbus clients**: the inverter software parameters can be accessed using the following Modbus clients:
  - **Power Electronics remote management tool:** there are different models of applications, each of which allows different functionalities depending on the role assigned to the user. For more information about the use and configurations allowed by these applications, please refer to the *Operation Manual* for the corresponding application.

APPLICATION MODELS			
MODEL / USER ROLE	**POWER ELECTRONICS STAFF**	**QUALIFIED CUSTOMER**	**CUSTOMER**
PowerCommsV	X		
Power Supervisor		X	
Power Monitoring			X

  - **External PPC / SCADA**.

    The "Modbus Communication" section outlines best practices for Modbus communication, as well as supported Modbus function codes, remote control functions, and power display parameters.

    If the communication properties need to be configured, as well as the definition of communications with the external PPC/SCADA, refer to the parameter group **[G6: Communications]**.

EN

2.2.1. Modbus communication

2.2.1.1. Best practices for Modbus communication

Modbus TCP is a widely used industrial communication protocol that enables efficient data exchange over Ethernet networks. To ensure optimal performance, stability, and reliability, it is crucial to follow best practices when implementing Modbus TCP in an application. This section highlights key considerations and recommendations for using Modbus TCP efficiently.

- **Maintain a single persistent socket connection**

One of the most important best practices when using Modbus TCP is to maintain a single, persistent socket connection for all Modbus requests instead of opening and closing sockets for each transaction. This approach has several advantages:

- Reduces network overhead and improves performance.
- Avoids excessive connection establishment delays.
- Minimizes resource consumption on both client and server.

Implementation tip:

- Open a single TCP socket at the beginning of the communication session and reuse it for all subsequent Modbus requests.
- Only close the socket when the communication session is complete or if an error occurs that necessitates a reconnect.

- **Ensure a minimum delay of 50 milliseconds between requests**

To prevent network congestion and avoid overloading the Modbus server, a minimum delay of 50 milliseconds should be maintained between consecutive Modbus requests. Rapid consecutive polling can lead to increased CPU load and potential data loss due to excessive traffic.

Implementation tip:

- Implement a request scheduler with a minimum inter-request delay of 50 milliseconds.
- Use a non-blocking approach to manage timing efficiently.

- **Optimize read operations for non-consecutive addresses**

In applications where it is necessary to read multiple non-consecutive addresses periodically, issuing separate read requests for each address can lead to increased communication overhead and inefficiencies. Instead, the use of Custom Modbus mapping is recommended. This feature involves grouping non-consecutive registers into a contiguous range in the Modbus server so they can be read in a single request.

Implementation tip:

- Configure the Modbus server to map non-contiguous registers into a contiguous range.
- Use a single read request to fetch all required data, reducing the number of transactions.

➢ **Disable Nagle's algorithm for low latency communication**

Nagle's Algorithm is designed to reduce the number of small packets sent over a network by buffering data before transmission. However, in Modbus TCP applications, this can introduce unwanted latency when multiple requests are sent in quick succession. To achieve real-time performance, it is recommended to disable Nagle's Algorithm.

Implementation tip:

- Enable the TCP_NODELAY flag on the socket to disable Nagle's Algorithm.
- This ensures that each Modbus request is transmitted immediately without delay.
- Sending requests immediately also prevents the CPU from receiving large packets containing multiple requests, reducing processing overhead and improving response times.
- Disabling Nagle's Algorithm is particularly important in time-sensitive industrial applications where low-latency communication is required.

➢ **Conclusion**

By following these best practices, Modbus TCP applications can achieve improved efficiency, reliability, and reduced network load. Maintaining a persistent socket, ensuring appropriate timing between requests, and optimizing read operations for non-consecutive addresses will result in a more robust and stable Modbus TCP communication system.

For further technical assistance, refer to the Modbus specification or contact Power Electronics for support.

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2.2.1.2. Supported Modbus function codes

Serial communications protocol provided by FREESUN inverters adheres to Modicon's industrial standard communications protocol, Modbus. The inverter uses the following reading and writing functions from the Modbus protocol:

Function	Description	Registers number
3	Registers reading	120
6	Writing of a single register	1
16	Multiple writing of registers	120
23	Reading and writing in multiple registers	120

The implementation of this function code in the inverter allows reading up to 120 registers of a parameter group in a single frame. To access consecutive memory registers, but belonging to different groups, the request should be performed using as many frames as groups are involved.

2.2.1.3. Modbus function code No. 3: Registers reading

This function code allows the Modbus controller (master) reading the content of the data registers indicated in the inverter (slave). This function code only admits unicast addressing. Broadcast or groupcast addressing are not possible with this function code.

The implementation of this function code in the inverter allows reading up to 120 registers with consecutive addresses of the inverter in a single frame.

In the next example, a frame is shown where the master tries to read the minimum DC voltage via communications. The information that should be attached in the request frame is the following one:

- Modbus address of the slave.
- Modbus function code (3 Registers reading).
- Starting address of the register (Modbus offset).
- Number of registers to read.
- CRC-16 code.

The answer of the inverter (slave) should contain the following fields:

- Modbus address of the slave.
- Modbus function code (3 Registers reading).
- Number of bytes of data to read.
- Data read (2 bytes per register).
- CRC-16 code.

Each register consists of 2 bytes (2x8bits=16 bits). This is the default registry length of the inverter.

➢ **Operation example of Modbus function code No. 3 (registers reading)**

Suppose we want to read via communications the minimum DC voltage to start the inverter. This data corresponds to parameter [G2.1.1-Vdc low]. The frame that should be transmitted is:

Modbus Address of the slave	Modbus Function Code	Starting Address of the register	Registers Number	CRC-16
0x0A	0x03	0x000C	0x0001	0xB449

Suppose that minimum DC Voltage of the inverter is 1050V (Modbus value 1050 in decimal is equivalent to 0x041A in hexadecimal). The answer of the slave would be:

Modbus Address of the slave	Modbus Function Code	Bytes Number	Data read (address 20)	CRC-16
0x0A	0x03	0x02	0x041A	0x9D84

2.2.1.4. Modbus function code No. 6: Registers single writing

This function code allows the Modbus controller (master) to write the content of one single register indicated in the inverter (slave), as long as that register is not “Read only”. Registers writing by the master does not impede the later modification of those registers by the slave.

The implementation of this function code in the inverter allows writing one register of the inverter in a single frame.

Up next, a frame is shown where the master tries to write the content of a register. The information that must be attached to the request frame is:

- Modbus address of the slave.
- Modbus function code (6, Register single writing).
- Data address of the register.
- Content to write.
- CRC code.

The answer of the slave includes:

- Modbus address of the slave.
- Modbus function code (6, Register single writing).
- Data address of the register.
- Content written.
- CRC code.

2.2.1.5. Modbus function code No. 16: Registers writing

This function code allows the Modbus controller (master) to write the content of the data registers indicated in the inverter (slave), as long as those registers are not “Read only”. Registers writing by the master does not impede the later modification of those registers by the slave.

The implementation of this function code in the inverter allows writing up to 5 registers of the inverter in a single frame.

Up next, a frame is shown where the master tries to write the content of a register. The information that should be attached in the request frame is:

- Modbus address of the slave.
- Modbus function code (16, Registers writing).
- Starting address of the register.
- Number of registers to write.
- Number of bytes of data to write.
- Content of registers to write.
- CRC-16 code.

The answer of the slave includes:

- Modbus address of the slave.
- Modbus function code (16, Registers writing).
- Starting address of the register.
- Written registers number.
- CRC-16 code.

2.2.1.6. Modbus function code No.23: Reading and writing in multiple registers

This function code allows the Modbus controller (master) to read and write the contents of several specified data registers in the inverter (slave) in a single request, instead of making two separate requests. This saves time and ensures that both operations are executed in a coordinated manner.

An example frame is shown below. The information that must be contained in the request frame is:

- Modbus address of the slave
- Modbus function code (23, Reading and writing in multiple registers).
- Starting address of the register.
- Number of registers to read.
- Number of registers to write.
- Number of bytes of data to write.
- Content of registers to write.
- CRC-16 code.

The answer of the slave only contains the read values; it does not confirm the written values:

- Modbus address of the slave.
- Modbus function code (23, Reading and writing in multiple registers).
- Number of bytes of data to read.
- Data read (2 bytes per register).
- CRC-16 code.

2.2.1.7. Remote Control Functions

START	
Screen	-
Range	0 – 1
Modbus address	**40040**
Modbus range	0 to 1
READ-ONLY / Write	Yes
Description	Manual start: Send the start command to the inverter through the communications network.

STOP	
Screen	-
Range	0 – 1
Modbus address	**40041**
Modbus range	0 to 1
Read / Write	Yes
Description	Sends the stop command to the inverter through the communications network.

RESET	
Screen	-
Range	0 – 1
Modbus address	**40042**
Modbus range	0 to 1
Read / Write	Yes
Description	Sends a reset command to the inverter through the communications network.

EMERGENCY STOP	
Screen	-
Range	0 – 1
Modbus address	**43038**
Modbus range	0 to 1
Read / Write	Yes
Description	Sends an emergency stop command to the inverter through the communications network. To perform a remote emergency stop, the following steps must be followed: 1. Enter the password at Modbus address 43038. 2. Execute the emergency stop. Write the value 1 to Modbus address 43039. 3. Once the emergency stop has been executed, the Modbus address 43038 will return the value 1571 as confirmation of the action. The device will also trigger the fault [F56: Emergency stop] corresponding to the remote emergency stop.

PPC HEARTBEAT	
Screen	-
Range	0 – 65535
Modbus address	**41437**
Modbus range	0 to 65535
Read / Write	Yes
Description	Sends a heartbeat PPC command to the inverter through the communications network. To establish communication with the inverter, previously the SCADA or PPC must write in the variable a value greater than zero (valid range from 0 to 65535) and update the value regularly. If the PPC or SCADA writes a zero in the PPC Heartbeat variable, the system is disabled.

EN

2.2.1.8. Energy visualization parameters

The user can access the active and reactive energy readings of all modules via Modbus (registers 42676 to 42755). To visualize information from several modules at the same time, refer to the table below:

<table>
<tr><th>MODULE NUMBER</th><th>DESCRIPTION</th><th>ADDRESSES</th></tr>
<tr><td rowspan="2">Module 1</td><td>Module 1 active and reactive energy parameters.</td><td>42676 to 42695</td></tr>
<tr><td colspan="2">Each module has 10 parameters, five related to active energy and five to reactive energy. See SV5.1 and SV5.2 for further details [1]:
<table>
<tr><th>PARAMETER</th><th>UNITS</th><th>ADDRESS</th><th>MODBUS CONVERSION</th></tr>
<tr><td>Daily active energy [*]</td><td rowspan="10">kWh</td><td>42676 – MSB
42677 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Yesterday active energy [*]</td><td>42678 – MSB
42679 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Current month active energy [*]</td><td>42680 – MSB
42681 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Last month active energy [*]</td><td>42682 – MSB
42683 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Total active energy [*]</td><td>42684 – MSB
42685 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Daily reactive energy [*]</td><td>42686 – MSB
42687 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Yesterday reactive energy [*]</td><td>42688 – MSB
42689 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Current month reactive energy [*]</td><td>42690 – MSB
42691 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Last month reactive energy [*]</td><td>42692 – MSB
42693 – LSB</td><td>Real value = Modbus value</td></tr>
<tr><td>Total reactive energy [*]</td><td>42694 – MSB
42695 – LSB</td><td>Real value = Modbus value</td></tr>
</table>
[*] The value of these parameters is shown as a 32-bits DWord, two 16-bits words (MSB and LSB).
The high byte (MSB) corresponds with the lowest Modbus address, and the low byte (LSB) corresponds with the highest Modbus address.</td></tr>
</table>

[1] Modules 2 to 4 follow the same structure; the user simply must enter the Modbus address that corresponds to each parameter. The address range for modules 2 to 4 is:

MODULE	PARAMETER ADDRESSES
Module 2	42696 to 42715
Module 3	42716 to 42735
Module 4	42736 to 42755

2.2.2. User access level

Software parameters are divided into groups, depending on their functionality:

- **Visualization and status parameters** (marked with SVx) cannot be modified and display useful information about the inverter, such as current and voltage per phase, amount of energy generated by the inverter, fault messages and many more. Please refer to section "Visualization parameters" for a detailed list of all parameters, their default values and their attributes.
- **Configuration parameters** (marked with Gx) allow setting and modifying the inverter operating conditions and general parameters which are explained in section "Programming parameters" which include their default values and adjustment ranges.
- Parameters from the **service group** (marked with SGx) are reserved for Power Electronics' staff. They allow setting parameters for commissioning and configuring all the functionalities of the inverter. They are explained in section, "Service group parameters" which include their default values and adjustment ranges.

The Modbus addresses of remotely accessible parameters are provided in the parameters table for the corresponding software version.

NOTICE

This document includes all login roles. Throughout this document, information highlighted in coral is that reserved to roles Power Electronics staff and Developer.

The user modes are **Customer** and **Qualified Customer**, being the latter named in the document as Administrator (Admin, from now on). To prevent unintentional modifications that could lead to malfunctioning and even damage to the inverter, access to parameters is password protected and managed by Power Electronics.

The available **user modes** are explained in parameter [G.1.3-User], for further information refer to its description at "Programming parameters" section. By default, the user mode is Customer. If assistance is required, please contact Power Electronics.

3. STATUS MESSAGES, EVENTS, FAULTS AND WARNINGS

3

The software provides, at all times, information about the inverter status, warnings and faults:

- **STATUS** messages provide information about the current state of the inverter (e.g. ready to start, operating…).
- **EVENT** messages report a non-hazardous action or event occurring in the inverter (e.g., AntiPID has been activated).
- A **FAULT** is a condition that, if satisfied, will force the inverter to stop (e.g. an islanding condition has been detected).
- **WARNING** messages provide useful information about what is going on in the inverter (e.g. derating, a module has stopped, heating is active…). They should be checked periodically, and the corresponding actions should be taken (if required).

3.1. Status messages

3.1.1. Module status messages

The following table lists the possible module statuses and their acronyms. These messages are displayed using the visualization parameters [SV12.x.1-Status], where x is the module number.

CODE	NAME	DESCRIPTION
1	Stop	The module is stopped and has not performed the soft charge.
2	Softcharge	The module is performing the soft charge.
3	Ready	The module is enabled and has the soft charge performed.
5	Pre Run	The module has received run command, and it is in starting process.
6	Run	The module is running and the IGBTs are switching.
7	Discharge	The module is actively discharging the bus.
10	Wait Over-Current	The module is stopped due to an overcurrent. Next status will be wait state (12).
11	Wait Under-Voltage	The module is stopped due to a bus undervoltage. Next status will be wait state (13).
12	Over-Current Wait DSP	Wait status to confirm fault. Rearm inverter or report fault.
13	Under-VDC Wait DSP	Wait status to confirm fault. Rearm inverter or report fault.
14	Unbalance VDC Wait DSP	The module is stopped due to a bus unbalance.
22	IDLE	The module starts up in this status after power-up.
55	Init	The module has not communicated with the main control board.

CODE	NAME	DESCRIPTION
65	Fault Overcurrent	Overcurrent detected in AC area.
66	Fault Unbalance Voltage	DC bus unbalanced fault.
67	Fault Overcurrent DC	Overcurrent detected in DC area.
69	Fault Overvoltage DC	DC bus overvoltage detected.
71	Fault Desturation out HR	A desaturation has been detected in positive outer IGBT of phase R of the power module.
72	Fault Desturation in HR	A desaturation has been detected in positive inner IGBT of phase R of the power module.
73	Fault Desturation in LR	A desaturation has been detected in negative outer IGBT of phase R of the power module.
74	Fault Desturation out LR	A desaturation has been detected in negative inner IGBT of phase R of the power module.
75	Fault Desturation out HS	A desaturation has been detected in positive outer IGBT of phase S of the power module.
76	Fault Desturation in HS	A desaturation has been detected in positive inner IGBT of phase S of the power module.
77	Fault Desturation in LS	A desaturation has been detected in negative outer IGBT of phase S of the power module.
78	Fault Desturation out LS	A desaturation has been detected in negative inner IGBT of phase S of the power module.
79	Fault Desturation out HT	A desaturation has been detected in positive outer IGBT of phase T of the power module.
80	Fault Desturation in HT	A desaturation has been detected in positive inner IGBT of phase T of the power module.
81	Fault Desturation in LT	A desaturation has been detected in negative outer IGBT of phase T of the power module.
82	Fault Desturation out LT	A desaturation has been detected in negative inner IGBT of phase T of the power module.
83	Fault Multiple	Desaturation detected in at least two IGBTs.
84	Fault Comms Timeout	Communications have been lost between the main control board and module board.
85	Fault SoftCharge Timeout	A module is in soft charge mode and the voltage has not reached the set value within the defined time.
86	Fault SoftCharge FB OpenClose	The DC switch disconnector is not closing/opening when required.
90	Fault Over Temperature	The temperature measurement of an IGBT has exceeded the maximum threshold defined by hardware.
95	Fault power supply	The output voltage of the internal power supplies is not correct.
98	Fault DC Leakge	DC current has been detected when the module is stopped.
99	Fault AC Leakge	AC current has been detected when the module is stopped.
102	Unbalance current AC	Output current is unbalanced between phases (handled by the DSP).
103	IGBT overtemperature	Overtemperature detected in IGBT (handled by the DSP).
104	Over temperature PCB	Overtemperature detected on the module control board (handled by the DSP).
106	Fault Unbalance V DC Emer	DC bus unbalance fault.
107	Fault unable reach reference	The output current has not exceeded at least 50% of the reference current.
108	Fault Over Voltage DC Emer	At least one of the DC buses has reached the emergency fault voltage.
109	Fault Under Voltage DC	The DC bus voltage has reached the configured minimum value when the module is running.
112	Fault Overcurrent Emer	The module has detected an overcurrent that has been sustained for more than 500us.

EN

CODE	NAME	DESCRIPTION
113	Fault comms crc	Four frames in a row with invalid CRC have been detected.
114	Fault sense R	The current sensor of phase R has an invalid offset.
115	Fault sense S	The current sensor of phase S has an invalid offset.
116	Fault sense T	The current sensor of phase T has an invalid offset.
118	Module disabled by DSP	The module has been disabled by the DSP (handled by the DSP).
119	Fault Desturation mid HR	A desaturation has been detected in positive clamp IGBT of phase R of the power module.
120	Fault Desturation mid LR	A desaturation has been detected in negative clamp IGBT of phase R of the power module.
121	Fault Desturation mid HS	A desaturation has been detected in positive clamp IGBT of phase S of the power module.
122	Fault Desturation mid LS	A desaturation has been detected in negative clamp IGBT of phase S of the power module.
123	Fault Desturation mid HT	A desaturation has been detected in positive clamp IGBT of phase T of the power module.
124	Fault Desturation mid LT	A desaturation has been detected in negative clamp IGBT of phase T of the power module.
125	Fault homopolar AC	AC zero sequence current has been detected.
128	Charge feedback fault	DC circuit breaker load feedback fault.
129	Module Isolation	A fault has been detected during self-diagnostics and the module is disabled (handled by the DSP).
130	Softcharge feedback fault	Fault of soft charge contactor feedback (MPCS only).
131	Faults limit exceeded	The module has exceeded faults threshold, and the DSP has disabled it (handled by the DSP).
132	Overvoltage DC sum	DC overvoltage fault on full bus.
134	Fault hum/temp sensor comms	Fault of the humidity and temperature sensor.
139	Mod. low Vdc recover	Total bus voltage drop fault.
200	RU IDLE	RU (Remote Upgrade) in idle status.
201	RU MUTE	A different device has been selected.
202	RU ERASING	RU is erasing the FLASH memory.
203	RU ERASE COMPLETE	RU has erased the full FLASH memory.
204	RU ERASE ERROR	Timeout error during FLASH memory erase.
205	RU WRITE	RU is writing data in FLASH memory.
206	RU WRITE ERROR	Writing error in the FLASH memory.
207	RU VERIFYING	RU is checking FLASH memory.
208	RU VERIFY OK	FLASH memory correct.
209	RU VERIFY ERROR	FLASH memory wrong.
210	RU EXIT	RU disconnection.
214	RU WRITE COMPLETE	RU writing process completed.
215	RU WRONG PART CHOOSE	Incorrect RU FPGA selected.

3.1.2. Inverter status messages

The following table lists the possible inverter statuses and their acronyms. These messages are displayed using the visualization parameter [SV1.1.4-Current status].

CODE	NAME	DESCRIPTION
0	Power Up	The inverter starts up at this state after power-up.
1	Init	The inverter is neither connected to the grid nor to the DC power supply source.
2	Off	The inverter is stopped waiting for a command to move forward. Refer to "Group G2: Running conditions" for more information on start-up and stop configurations of the inverter.
3	Standby DC	The inverter is checking DC conditions to move forward.
4	Precharge DC	DC precharge to connect to AC grid without connecting to DC power source. In case of inverters with MV cell management, grid connection could be done without pre-magnetizing if allowed.
5	Softcharge DC	Operation depending on the state of connection to the AC grid. • Disconnected from AC grid: DC self-diagnosis and connection to DC power source with soft charge. • Connected from AC grid: connection to DC power source with soft charge.
6	DC Ready	The inverter is not connected to the grid but to the DC power supply source.
7	Standby AC	The inverter is checking the AC and DC conditions for progress.
8	Black Start	Connection to the AC grid.
9	Ready	The inverter is connected to the AC grid and to the DC power source. The inverter is waiting for a command to move forward. Refer to "Group G2: Running conditions" for more information on start-up and stop configurations of the inverter.
10	Wait	The inverter is checking the AC conditions according to the configured grid connection conditions for the set time.
11	Pre ON	The inverter has completed the start-up conditions and is in the process of starting up.
12	ON	The inverter is running.
13	Stop	The inverter has completed the stop conditions and it is in the process of being stopped.
14	Ready AC	The inverter is connected to the AC power grid. The inverter is stopped waiting for a command to move forward. Refer to "Group G2: Running conditions" for more information on start-up and stop configurations of the inverter.
15	AC Diag	The inverter is disconnected from the DC power source but is connected to the AC power grid. AC self-diagnosis is executed.
16	Discharge	Active discharge of the inverter to de-energize internal elements. The inverter is disconnected from the AC mains and DC power supply.
17	Fault	The inverter is stopped because there is an active fault in the inverter.

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3.2. Events

Below is a summary of possible inverter events and their descriptions. The status of each event, as well as the history, is displayed using the [SV7.3: Events] visualization group.

CODE	EVENT	DESCRIPTION
1	LVRT	Algorithm Low Voltage Ride Through is running. Low voltage grid support (voltage dips).
2	OVRT	Algorithm Over Voltage Ride Through is running. High voltage grid support.
3	P(f) curve	Grid support by means of active power curve as a function of frequency.
4	Q(V) curve	Grid support by means of reactive power curve as a function of voltage.
5	P(V) curve	Grid support by means of active power curve as a function of voltage.
6	Premagnetization	The inverter is premagnetizing the MV transformer.
7	Econ Mode	The inverter has activated the Econ mode functionality.
8	Night mode	The inverter is in night mode.
9	Curve Q(P)	Grid support by means of reactive power curve as a function of active power.
10	Cosphi (P)	Grid support by means of power factor (cos φ) as a function of active power.
11	Iac CTs	The auto-detection algorithm has detected that the current transformers measure correctly.
12	Iac intern	The current transformers auto-detection measurement algorithm has detected a malfunctioning of them.
13	Blackstart conditions	The conditions for the unit to start in blackstart are fulfilled.
14	MT close blackstart	The conditions for the unit to close the medium voltage switchgear for performing the blackstart are met.
15	MT open blackstart	The conditions for the unit to open the medium voltage switchgear during the blackstart process are fulfilled.
16	New HW filter	The new HW filter has been configured. The unit will limit the current when working in double-signal to 828 A before giving the improper modulation fault. This fault will remain in case the power stage exceeds 828 A current.
17	Standby mode	The inverter has entered Standby mode.
18	Precharge conditions	The inverter has detected precharge conditions.
19	Precharge	The inverter is in precharge.
20	Soft-charge AC	Soft charging from AC is in progress.
21	Inductive self-limiting	The inductive reactive power limitation due to undervoltage has been activated.
22	Capacitive self-limiting	The capacitive reactive power limitation due to overvoltage has been activated.
23	P(ROCOF)	The P(ROCOF) algorithm has been activated.
33	IMI GFDI MV	Opening / closing maneuver of the GFDI in progress.
34	AntiPID ON	The AntiPID functionality is activated.
35	Autoverification	The Autoverification mode has been activated during Hot / Cold Commissioning.
36	Troubleshooting	Troubleshooting mode is activated.
47	Watchdog	A watchdog error has occurred: the process has not finished on time or there has been an error in the microprocessor.

CODE	EVENT	DESCRIPTION
63	Set MCU ACT FAULT	Indicates special fault activation of the digital output wired to the uP. This digital output is located on the RJ21 connector of the central expander board.
64	Clear MCU ACT FAULT	Indicates the special fault clearance of the digital output wired to the uP. This digital output is located on the RJ21 connector of the central expander board.

3.3. Faults

The table below summarizes all possible inverter faults and their descriptions. For more information on root causes and diagnostics, refer to the guidelines indicated in the corresponding documentation for the resolution of faults, warnings and troubleshooting. The status, history, and information for each fault is displayed using the [SV7.1: Faults] visualization group.

Each fault has an associated reset type and reset role, indicated in the **RESET TYPE** and **ROL** columns respectively:

- Faults with "**Manual" reset type:** they must be reset by a user with the role indicated in the **ROL** column.
- Faults with "**Auto-reset" reset type:** they are reset automatically by the inverter for the configured number of retries and time between retries; if the number of retries is exceeded, the inverter is blocked and must be cleared manually by a user with the role indicated in the **ROL** column.
- Faults with "**Auto-reset by measurement" reset type:** they are reset automatically when the value of the variable that originated the fault returns within the fault activation threshold.

CODE	FAULT	DESCRIPTION	RESET TYPE	ROL
F1	Watchdog	Error in software update or response out of time.	Manual	Power electronics staff
F2	HW Vbus	The voltage measured on the DC channels has exceeded the threshold (1600 Vdc).	Auto-reset by measurement	N/A
F3	SoftCharge	The inverter has exceeded the configured timeout during soft charge.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F4	Discharge	The inverter has taken more than 60 seconds to go below 10 Vdc since the module buses were discharged.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F5	High Vac	One of the programmable voltage thresholds has been exceeded in its associated time.	Auto-reset by measurement	N/A
F6	Low Vac		Auto-reset by measurement	N/A
F7	High frequency		Auto-reset by measurement	N/A
F8	Low frequency		Auto-reset by measurement	N/A
F10	Comms SCI FPGA-DSP	The FPGA has not been configured correctly.	Auto-reset by measurement	N/A
F11	Active Anti-island	A possible island condition has been detected in the AC part of the network.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F13	No modules	All modules stopped for some reason while the inverter was running.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F14	Drive-Select DSP	Wrong drive select configuration.	Manual	Power Electronics staff

CODE	FAULT	DESCRIPTION	RESET TYPE	ROL
F15	Synchronization	Error in the working conditions of the inverter.	Manual	Power Electronics staff
F23	Unbalanced Vac	There is an unbalance between phases greater than the protection of the inverter.	Auto-reset by measurement	N/A
F25	Low Vdc	The inverter has read a DC voltage lower than the rectified mains voltage.	Manual	Power Electronics staff
F27	Modules start fault	The inverter has not been able to start any module during start-up.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F28	Passive Anti-island	The algorithm has detected a possible island condition in the AC part of the grid.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F31	Selfdiagnosis fault	The self-test has not been successfully completed before the timeout.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F32	Error Mod selfdiagnosis	One of the modules (or more) has triggered a fault while the self-diagnosis was running.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F33	Unable reconnect	After a network event the modules have tried to boot 'N' times without success.	Auto-reset (5 retries; 10 sec between retries)	Power Electronics staff
F35	MT Premag	A problem occurred during pre-magnetization of the MV transformer.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F40	Internal overtemperature	The main control board has measured a temperature higher than the set limit.	Auto-reset by measurement	N/A
F41	GFDI	The GFDI circuit breaker is in TRIP status.	Auto-reset by measurement	N/A
F43	Emergency stop	It indicates that the emergency stop button has been pressed.	Manual	Qualified customer
F44	Drive-Select MCU	Incorrect drive select group configuration. The main board does not have the Drive select programmed properly. Drive select parameters are out of range.	Auto-reset by measurement	N/A
F45	General Isolation	The inverter has measured an earth resistance lower than the configured threshold.	Auto-reset by measurement	N/A
F46	Data Fault	The system has detected out-of-range parameters.	Manual	Power Electronics staff
F47	Watchdog uP	The process has not finished on time or there has been an error in the microprocessor. **Note:** from version 5.0.0, this fault has been replaced by Event 47.	Auto-reset (5 retries; 20 sec between retries)	Power Electronics staff
F48	Internal comms	Internal communication problem between MCU and DSP.	Manual	Power Electronics staff
F49	IMD Selfdiagnosis Error	The IMD's periodic hardware self-diagnosis has reported an error.	Manual	Power Electronics staff
F51	PPC Comms	Inverter has stopped receiving signals from the PPC or SCADA.	Manual	Qualified customer
F54	DU Overcurrent	Overcurrent in one of the channels of the DC cabinet.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F55	External fault/OVGR	The inverter has detected an emergency stop ordered by the customer.	Manual	Qualified customer
F56	Remote emergency stop	The inverter has received an emergency stop remotely from the customer.	Manual	Qualified customer
F58	Control SW mismatch	Incompatibility between MCU and DSP versions.	Manual	Power Electronics staff
F59	Module SW Mismatch	The power modules do not have the same software version.	Manual	Power Electronics staff

CODE	FAULT	DESCRIPTION	RESET TYPE	ROL
F62	DU Comms	There is a communications problem between the main control board and the DU currents measurement board.	Auto-reset by measurement	Power Electronics staff
F63	IMD Earth Connection	The ground resistance measured is greater than the earth fault resistance.	Auto-reset (5 retries; 5 sec between retries)	N/A
F64	Invalid MAC	The MAC address entered is invalid.	Manual	Power Electronics staff
F65	Overcurrent AC	One of the AC currents of a module has exceeded the protection threshold.	Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff
F66	Module VDC Unbalance	There is a voltage unbalance between the DC bus half-buses.	Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff
F69	Overvoltage DC+	Overvoltage on the positive side of the DC bus (F69) or on the negative side (F70).	Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff
F70	Overvoltage DC-		Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff
F71	Desaturation R1(H)	A desaturation has been triggered in one of the IGBT (inners or outers) of the power module.	Manual	Power Electronics staff
F72	Desaturation R2(H)		Manual	Power Electronics staff
F73	Desaturation R2(L)		Manual	Power Electronics staff
F74	Desaturation R3(L)		Manual	Power Electronics staff
F75	Desaturation S1(H)		Manual	Power Electronics staff
F76	Desaturation S2(H)		Manual	Power Electronics staff
F77	Desaturation S2(L)		Manual	Power Electronics staff
F78	Desaturation S3(L)		Manual	Power Electronics staff
F79	Desaturation T1(H)		Manual	Power Electronics staff
F80	Desaturation T2(H)		Manual	Power Electronics staff
F81	Desaturation T2(L)		Manual	Power Electronics staff
F82	Desaturation T3(L)		Manual	Power Electronics staff
F83	Multiple desaturation	Several of the IGBT have reported desaturation.	Manual	Power Electronics staff
F84	Comunications	There is a communication frame lost between the FPGA of the main control board and module board.	Manual	Power Electronics staff
F85	Timeout soft charge	A module is in soft charge mode, and the voltage has not reached the set value within the defined time.	Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff
F86	Feedback breaker	The DC switch disconnector is not closing/opening when required.	Manual	Power Electronics staff
F90	Tª IGBTS HF	The temperature measurement of an IGBT has exceeded the maximum threshold defined by hardware.	Manual	Power Electronics staff
F95	Source PCB	It has been detected that the voltage of the 24 Vdc and 15 Vdc sources has dropped.	Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff

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CODE	FAULT	DESCRIPTION	RESET TYPE	ROL
F98	Idc derivation	The module has measured an average current greater than that set in the DC current sensor.	Manual	Power Electronics staff
F99	Mod. Iac derivation	The module has measured an RMS current greater than that configured in one of the AC sensors.	Manual	Power Electronics staff
F102	Unbalanced current	There is a current unbalance between phases higher than the protection of the equipment configured.	Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff
F103	IGBTs temperature	The temperature of one or more IGBT has exceeded the maximum permissible value according to the software configuration.	Manual	Power Electronics staff
F104	PCB temperature	A temperature has been reached on the PCB of one of the modules higher than the fault threshold configured.	Manual	Not required
F106	Mod. Vdc unbalanced	Critical bus unbalance.	Manual	Power Electronics staff
F107	Iac not reached	The output current has not exceeded at least 50% of the reference current.	Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff
F108	Mod. High Vdc	Critical overvoltage of a semibus.	Manual	Power Electronics staff
F109	Mod. Low Vdc	The module has detected a very low DC voltage or has remained at a low threshold for 500us.	Manual	Power Electronics staff
F112	OCAC emergency	The module has detected an overcurrent that has been sustained for more than 500us.	Manual	Power Electronics staff
F113	CRC	Four frames in a row with invalid CRC have been detected.	Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff
F114	Measure Ir	The module FPGA has read an unexpected current on one of its hall sensors.	Manual	Power Electronics staff
F115	Measure Is		Manual	Power Electronics staff
F116	Measure It		Manual	Power Electronics staff
F117	Stage HW2.0	A GEN3 HW2.0 module control board has been detected in a GEN3 inverter with a previous hardware version (HW1.4 or lower).	Manual	Power Electronics staff
F118	Module Disabled	The module is disabled due to an anomalous situation.	Manual	Power Electronics staff
F119	Desaturation R1(L)	A desaturation has been triggered in one of the IGBT (clamps).	Manual	Power Electronics staff
F120	Desaturation R3(H)		Manual	Power Electronics staff
F121	Desaturation S1(L)		Manual	Power Electronics staff
F122	Desaturation S3(H)		Manual	Power Electronics staff
F123	Desaturation T1(L)		Manual	Power Electronics staff
F124	Desaturation T3(H)		Manual	Power Electronics staff
F125	Diff Iac mod run	The zero sequence current calculation is higher than the set value for more than 500 µs.	Manual	Power Electronics staff
F128	Spring loaded	The spring of the DC switch disconnector is not charged when it should be.	Manual	Power Electronics staff
F129	Module Isolation	An insulation fault has been detected in one of the power modules.	Manual	Power Electronics staff

CODE	FAULT	DESCRIPTION	RESET TYPE	ROL
F130	Softcharge Feedback	Soft charge feedback not received.	Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff
F131	Faults limit exceeded	The module has given more than 'N' faults in the configured time and the DSP has blocked it.	Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff
F132	VDC HW overvoltage	The module has detected an overvoltage on the entire module bus. The fault threshold is set to 1550 V total bus voltage.	Manual	Power Electronics staff
F134	Module sensor comms	The module FPGA cannot communicate with the temperature sensor.	Auto-reset (1 retry; 1 sec between retries)	Power Electronics staff
F139	Mod. low Vdc recover	While the module is running, a drop of its total bus voltage below the average threshold set in [G5.10.15-Low module Vdc] has been detected and, within 500 µs, the voltage has recovered above the upper threshold ([G5.10.15-Module Low Vdc] + [G5.10.15a-Low module Vdc High]).	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F150	MV cell wrong state	Inconsistency between the status of the MV switchgear and the inverter command.	Manual	Power Electronics staff
F152	DGPT2 Temp failure	The oil temperature of the MV transformer has exceeded 100 ºC.	Manual	Power Electronics staff
F153	DGPT2 Pressure failure	The MV Skid protection switchgear has reported a high oil pressure fault.	Manual	Power Electronics staff
F155	DGPT2 oil or gas level	The MV Skid protection switchgear has reported a low oil or gas level fault.	Manual	Power Electronics staff
F157	Feedback AC breaker	Inconsistency between the AC circuit breaker switching order and the feedback.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F158	Low pressure SF6	The pressure switch indicating the SF6 gas pressure in the MV switchgear has been activated.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F159	Overtemperature DU	The temperature read by the DU board has exceeded the threshold configured internally.	Auto-reset by measurement	N/A
F161	LOTO AC	AC LOTOTO is activated, and the inverter must be disconnected from AC and DC.	Manual	Qualified customer
F162	LOTO DC	DC LOTOTO is activated, and the inverter must be disconnected from AC and DC.	Manual	Qualified customer
F163	WD MV Cell	The DSP has detected a WD fault of the MV switchgear.	Manual	Power Electronics staff
F164	Excessive Maneuvers	The DSP has detected 6 soft charge maneuvers in 50 minutes or 15 filter C contactor maneuvers in 30 minutes.	Manual	Power Electronics staff
F165	Improper Modulation	The inverter has been working longer than allowed with improper modulation.	Manual	Power Electronics staff
F166	E-STOP	The customer has triggered an emergency stop.	Manual	Power Electronics staff
F167	Open DU	A DC switch disconnector was opened while the inverter was running during the day.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F169	LCL Feedback	The LCL capacitor contactor has been activated and is not responding.	Manual	Power Electronics staff

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CODE	FAULT	DESCRIPTION	RESET TYPE	ROL
F170	Thermal L Feedback	The inductance core of a module has overheated.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F171	Filter LC Current	The current through filter C does not correspond to the theoretical current.	Manual	Power Electronics staff
F172	C LC Filter	The calculated capacity does not correspond to the theoretical capacity of the inverter.	Manual	Power Electronics staff
F173	CTs measure	The inverter has detected a current error higher than the configured threshold between the current transformers and the current summation of the modules.	Manual	Power Electronics staff
F174	AC circuit breaker fb	Inconsistency between the expected state of the circuit breaker and the reading of the feedback.	Manual	Power Electronics staff
F175	Overcurrent C	An overcurrent has been detected in filter C.	Manual	Power Electronics staff
F176	Multi. Overcurr Events	Overcurrent control in VISMA.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F177	Wrong Filt C Config.	A configuration error in the C filter has been detected.	Manual	Power Electronics staff
F185	Disco. Cfilt Hall sensor	In the C filter of an inverter with HW1.4 or higher, it has been detected that one or more Hall current sensors are disconnected or malfunctioning.	Manual	Power Electronics staff
F186	Overtemp. SKID cabinet	Overheating detected in the MV Skid control cabinet.	Auto-reset by measurement	N/A
F187	SKID trans overtemp.	The MV Skid transformer has exceeded the maximum temperature.	Auto-reset by measurement	N/A
F190	Combined F-V	One of the adjusted protection thresholds has been exceeded.	Auto-reset by measurement	N/A
F197	HW Overvoltage	An overvoltage has been reached on the AC low side, higher than the hardware overvoltage limit of the inverter.	Auto-reset by measurement	N/A
F198	HW Undervoltage	An undervoltage has been reached on the AC low side, lower than the hardware undervoltage limit of the inverter.	Auto-reset by measurement	N/A
F200	Overvoltage Lim DC	The DC voltage measurement of the module is below the average of the main control board.	Manual	Power Electronics staff
F202	Taux overtemperature	Overheating has been detected in the auxiliary transformer.	Auto-reset by measurement	N/A
F203	MV overtemperature	One of the PT100 of the transformer has exceeded the set temperature.	Auto-reset by measurement	N/A
F204	Fan impulsion MT	Loss of communication with inverter fans.	Auto-reset by measurement	N/A
F205	Fan DU zone		Auto-reset by measurement	N/A
F206	Fan zone filter LC		Auto-reset by measurement	N/A
F207	Fan zone modules		Auto-reset by measurement	N/A
F208	MV cri overtemperature	The critical temperature threshold of the MV transformer has been exceeded.	Manual	Power Electronics staff
F210	uC-FPGA Comms	Communication between the MCU and the FPGA of the main control board has failed.	Auto-reset by measurement	N/A
F211	Intern Isolation	An insulation resistance value below the configured fault threshold has been detected.	Auto-reset by measurement	N/A

CODE	FAULT	DESCRIPTION	RESET TYPE	ROL
F212	Extern Isolation	The insulation monitor has measured a resistance value below the configured fault threshold.	Auto-reset by measurement	N/A
F213	Negative DU current	The DU current is below the negative current threshold value.	Manual	Power Electronics staff
F214	IMD Comms	Failure of fiber optic communications with the isolation monitor device.	Auto-reset by measurement	N/A
F215	Timeout GFDI	Close or open order has been sent to the GFDI and does not match the action.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F216	Timeout IMI Meas	The time set for an isolation measurement has expired.	Auto-reset (5 retries; 5 sec between retries)	Power Electronics staff
F217	Critical Temp LC filter	The temperature of the LC filter is above the configured limit.	Auto-reset by measurement	N/A
F219	Overvoltage Lim AC	Overvoltage has been detected in the AC voltage limiter.	Manual	Power Electronics staff
F220	PT100 induc no connected	One or more of the PT100 of the choke is not connected to one of its terminals.	Auto-reset by measurement	N/A
F221	PT100 traf not connected	One or more of the PT100 of the transformer is not connected to one of its terminals.	Auto-reset by measurement	N/A
F222	NTC adm temp not connect	Intake temperature NTC resistor is not connected.	Auto-reset by measurement	N/A
F223	NTC LC filter not connec	The filter temperature NTC resistor is not connected.	Auto-reset by measurement	N/A
F224	Ind overtemperature	The temperature of one of the module inductances is higher than the configured one.	Auto-reset by measurement	N/A
F225	Ind cri overtemperature	The temperature of one of the module inductances is higher than the one set as critical.	Manual	Power Electronics staff
F226	IMI Hardware	The hardware device is not compatible with the number of modules in the inverter.	Manual	Power Electronics staff
F227	Fault Unknown	The DSP or the FPGA have reported a fault unknown by the microcontroller.	Manual	Power Electronics staff
F228	Fan setpoint	One of the fans is not operating at the configured speed setpoint.	Auto-reset by measurement	N/A
F229	Timeout IMI Channel	Some of the IMD channels have not been able to give a correct insulation measurement.	Auto-reset (Unlimited retries; 5 sec between retries)	Power Electronics staff
F230	Fan type not correct DU	Incorrect fan type.	Auto-reset by measurement	N/A
F231	Fan type not correct MT		Auto-reset by measurement	N/A
F232	Fan type not correct MOD		Auto-reset by measurement	N/A
F234	Thermal Bus Plus	The digital input connected to the Bus Plus cabinet thermal has been activated.	Manual	Power Electronics staff
F235	T/H sensors comms	Communications with the temperature and humidity sensors of all the inverter's boards have failed.	Auto-reset by measurement	N/A
F236	DU sensor comms lost	Communications with the temperature and humidity sensors of the two DUs of the unit or, in the case of MPCS, with the temperature/humidity sensor of its DU have failed.	Manual	Power Electronics staff

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CODE	FAULT	DESCRIPTION	RESET TYPE	ROL
F237	Negative union fuse	Fault associated with the blowing of one of the union fuses of the negative busbars of the DC cabinet (PV field input). It only applies to inverters with Bus Plus and negative grounded pole.	Manual	Power Electronics staff
F238	AEMO comms lost	Communications failure between PPC / SCADA and the inverter related to AEMO standard.	Manual	Qualified customer
F240	Du BP overcurrent	The sum of currents measured in the DU channels exceeds the threshold and the inverter is not switching.	Manual	Power Electronics staff
F241	Delta _T max DU	The temperature increase in the DU is higher than a set value. **Note:** from version 4.1.0, this fault is deprecated and it has been replaced by W241.	Manual	Power Electronics staff
F242	Delta_T LC	The temperature increase in the LC filter area is higher than a set value.	Manual	Power Electronics staff
F243	Fan saturated setpoint	The setpoint of 100% has been exceeded in one of the fans.	Manual	Power Electronics staff
F244	T LC Silent mode	The ambient temperature is above 25 °C, with the ventilation of the LC filter area set to [G15.1.6-Behavior] = Stop.	Auto-reset by measurement	N/A
F245	Over temp silent mode	When silent mode is activated, if the ambient temperature exceeds 50 °C, the inverter will stop and remain in fault mode until the temperature drops below the hysteresis defined in [G15.1.7-Threshold].	Auto-reset by measurement	N/A
F246	Clean filters	The maximum number of consecutive days (configured in [SG5.2.1-Filter Fault]) with the [W194: Module Filter] warning active, without cleaning or replacing the filters in one or more modules has been exceeded.	Manual	Power Electronics staff
F249	Low Temp. KNAN	Fault associated with the KNAN transformer due to an oil temperature reading below -10 °C.	Auto-reset by measurement	N/A
F252	RFI OverCur Persistent	The inverter has read a current for an RFI greater than the configured threshold ([G5.13.5-Max RFI I protection]) or greater than 50A.	Auto-reset by measurement	N/A
F253	Imbalance V	There is a voltage unbalance between busbars greater than the value set in the protection.	Auto-reset by measurement	N/A
F255	BAT Open detection	A sudden opening of the battery has been detected.	Auto-reset by measurement	N/A
F257	Soft-Charge AC Timeout	The busbar did not complete the AC soft-charging process within the expected time.	Auto-reset by measurement	N/A

3.4. Warnings

The table below summarizes all possible inverter warnings and their descriptions. The status and history of each notification is displayed using the visualization group [SV7.2: Warnings].

CODE	WARNING	DESCRIPTION
W1	Input high voltage	The input voltage is reaching a dangerous level; the value is above the value protection set up.
W2	Low voltage input	The input voltage is reaching a dangerous level; the value is below the value protection set up.
W3	High frecuency	The input frequency is reaching a dangerous level; the value is above the value protection set up.
W4	Low frecuency	The input frequency is reaching a dangerous level; the value is below the value protection set up.
W5	No start conditions	The AC conditions for soft bus charging and the AC and DC start conditions are not fulfilled.
W7	P limit	The set limit of the inverter's active power has been reached.
W8	Q limit	The set limit of the inverter's reactive power has been reached.
W9	Is limit	Limit Is has been reached. The modules are operating at maximum current.
W10	Id limit	The Id current limit has been reached. The message code indicates the cause of the limit: (1): Limit due to AC. (2): Limit due to DC.
W11	Derating Vdc	A derating has been activated due to overvoltage DC bus voltage.
W12	Derating thermal IGBT	A derating has been activated due to overheating of IGBTs.
W13	Derating thermal adm	A derating has been activated due to high admission temperature.
W14	Low P	The active power is lower than the configured minimum AC power.
W15	Low Vdc	The DC voltage is lower than the configured minimum DC voltage.
W18	Tracker output low Vac	The output voltage of the module in trackers mode is lower than the configured value.
W19	LVRT	Grid support by low voltage (voltage sags). The inverter is working using the LVRT control mode.
W20	HVRT	Grid support by high voltage (voltage sags). The inverter is working using the HVRT control mode.
W21	Curve P(f)	The power of the inverter is reduced due to a high frequency by running the Power Per Frequency (PPF) algorithm.
W22	Curve Q(V)	Grid support by means of reactive power curve as a function of voltage.
W23	Curve P(V)	Grid support by means of active power curve as a function of voltage.
W24	DU close night	DU will close at night without PV voltage in 30 s. If the ECON mode is available, DU will be closed during the night after an LVRT/OVRT event.
W25	Close AC without premag.	If DU opens after an LVRT/OVRT event (ECON mode available), MV switchgear will close without premagnetization in 30 s, provided that the parameter [SG1.8.7-En. Shutdown premag] is disabled.
W27	Overvoltage limit AC	The AC overvoltage protector has been activated.
W28	Derating freq temp IGBT	A frequency derating has been activated due to overheating of IGBTs.
W29	DGPT2 Temperature alarm	MV Skid transformer temperature alarm signal has been activated.
W30	Overtemperature DU	DU overtemperature limitation has been activated.
W31	MV Autocalibration	The inverter is self-calibrating the MV voltage.
W32	MV Premagnetization	The inverter is pre-magnetizing the MV transformer.
W33	Econ Mode	The inverter is operating in ECON mode.
W34	Night Conditions	The inverter has detected that the conditions for night mode are met.
W35	Autonight Night Day	The inverter is in the transition from night to day.

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CODE	WARNING	DESCRIPTION
W36	Autonight Day Night	The inverter is in the transition from day to night.
W37	Night Mode	The inverter has the night mode enabled.
W38	Opening DU	The inverter has detected conditions for opening the DU and it will open it after 30 s.
W39	DU Open	The inverter has opened the DU due to some algorithm or event. There is a discrepancy between the state of the DU and the state of the Virtual DC Selector.
W40	Opening AC	The inverter has detected conditions for opening AC and it will open it after 15 s.
W41	AC Open	The inverter has opened the AC voltage due to some algorithm or event. There is a discrepancy between the state of the AC voltage and the state of the Virtual AC Selector.
W42	AC Disconnect	AC disconnection conditions are being met, and the inverter will fail after the configured time has elapsed.
W43	Air conditioning	The conditions for air conditioning for a HEMK inverter with MV Skid and UPS are not fulfilled. **Note**: this warning is not available from version 5.4.0 onwards.
W44	Active heating enabled	The active heating system is running.
W45	No modules active heating	No modules are available during heating. There is a fault or they are disabled.
W46	Filter LC Current Unbal	An unbalanced current has been detected, which has exceeded a configured threshold during a configured time, in the monitoring of the fundamental frequency current of the grid flowing through the LC filter capacitors during normal operation of the inverters.
W47	C LC Filter	A loss of capacity is diagnosed during operation of the inverter according to the estimation of the C capacity of the LC filter made as a function of the current and voltage measurement.
W48	Current Balancing Mod	The control algorithm to reduce the unbalance of direct sequence currents between modules when the inverter works in open loop is running.
W50	CTs measure	The inverter has detected a current error higher than the configured threshold between the current CTs (current transformers) and the current summation of the modules. **Note:** from version 4.10.0, this warning also indicates that the phase sequency in the current measure is incorrect.
W51	Speed limit	The configured speed limit when the inverter is working in voltage type (VISMA) has been reached.
W52	VAC limit	The configured AC voltage limit when the inverter is working in voltage type (VISMA) has been reached.
W53	NO_SYNC_FAC	AC voltage has been detected when the inverter is configured for VSI control type and FAC control subtype (the MV switchgear must be open and there must be no LV voltage).
W54	Vdc/Vac limit	The algorithm that limits power (P and Q) of the inverter by Vdc/Vac ratio is running.
W55	AC Circ. Breaker Open	The AC circuit breaker has opened upon a network event and will not close until the network meets the voltage and frequency specifications set in subgroup [G2.3: Reconnection] for the delay time specified in the same subgroup.
W56	Silent mode derating	A power reduction is being applied to maintain the noise level set it [G15.1.2-Silent mode level].
W57	No meas Vac FPGA	It has been detected that the AC voltage measurement of the module FPGA is not correctly wired or is not wired at all.
W58	Limit Q per VAC	The algorithm limiting the inductive or capacitive reactive power has been activated because the grid voltage has taken a value equal to the enabled voltage protections plus/minus the configured margin. The reference protection is the input threshold of the OVRT/LVRT algorithm and, if not enabled, will be the least restrictive over/undervoltage network protection. If neither of these two protections is enabled, the functionality is disabled. The configuration of this functionality can be found in group [G3.1: P/Q] and the effective values can be found in group [SV1.8: Limits].
W59	Incorrect filter C cfg.	The measured value of the filter capacitance does not match the one corresponding to the filter type selected in SG10.5.0.
W60	Discon. Cfilt Hall sensor	In the C filter of an inverter with HW1.3 or lower, it has been detected that one or more Hall current sensors are disconnected or malfunctioning.
W61	Act Balancing Triggering	Active balancing algorithm has been activated due to the voltage difference between busbars.
W65	Overcurrent AC	The AC current measurement in one of the modules has exceeded the protection threshold.
W66	Mod. Bus DC Unbalanced	There is a voltage unbalance (the threshold is configurable) between the high and low side of the DC bus.

CODE	WARNING	DESCRIPTION
W69	Overvoltage DC (P)	Overvoltage (>830V) on the positive side of the DC bus.
W70	Overvoltage DC (N)	Overvoltage (>830V) on the negative side of the DC bus.
W71	Desaturation R1(H)	A desaturation has been triggered in one of the IGBT (inners or outers) of the power module due to an overcurrent in the IGBT or faults caused by a loss of power to the gate drive controlling the IGBT.
W72	Desaturation R2(H)	
W73	Desaturation R2(L)	
W74	Desaturation R3(L)	
W75	Desaturation S1(H)	
W76	Desaturation S2(H)	
W77	Desaturation S2(L)	
W78	Desaturation S3(L)	
W79	Desaturation T1(H)	
W80	Desaturation T2(H)	
W81	Desaturation T2(L)	
W82	Desaturation T3(L)	
W83	Multiple desaturation	Several of the IGBT have reported desaturation at the same time.
W84	Comunications	There is a communication frame lost between the FPGA of the main control board and module board.
W85	Time out soft charge	A module is in soft charge mode and the voltage has not reached the set value within the defined time.
W86	Feedback breaker	The DC switch disconnector is not closing/opening when required.
W90	Tº IGBTS HW	The temperature measurement of an IGBT has exceeded the maximum permissible threshold (126 ºC) or an IGBT temperature sensor has failed.
W95	Source 15V	It has been detected that the voltage of the 24Vdc and 15Vdc sources has dropped.
W98	Idc derivation	The module has measured an average current greater than that set in the DC current sensor when the module is stopped.
W99	Differential Iac stop	The module has measured an AC current greater than that configured in one of the AC sensors when the module is stopped.
W102	Unbalanced current	A module output current unbalance greater than a configurable threshold has been detected within the defined time.
W103	IGBTs temperature	The temperature of one or more IGBT has exceeded the configured maximum permissible value.
W104	PCB temperature	A temperature has been reached on the PCB of one of the modules higher than the fault threshold configured.
W106	Unbal/Max Vbus	There is a voltage unbalance (>200V) between the high and low side of the DC bus.
W107	Iac not reached	The output current has not exceeded at least 50% of the reference current provided that the reference current has exceeded the defined trigger threshold too.
W108	Overcurrent Idc	Although fault 69 has been triggered, the voltage is still unstable, when it reaches 875V the warning will be triggered.
W109	Vdc module low	The module has detected a very low DC voltage, which is insufficient to start and continue operation.
W111	Fan module	The fan of the module communicates a fault.
W112	Emergency OCAC	The module has detected an AC overcurrent that has been sustained for more than 500us when the IGBTs stop switching.
W113	Checksum comms	Four frames in a row with invalid CRC have been detected.
W114	Sensed current R	The module's phase R current sensor is not measuring correctly; there is an offset error.
W115	Sensed current S	The module's phase S current sensor is not measuring correctly; there is an offset error.

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CODE	WARNING	DESCRIPTION
W116	Sensed current T	The module's phase T current sensor is not measuring correctly; there is an offset error.
W117	Stage HW2.0	A GEN3 HW2.0 module control board has been detected in a GEN3 inverter with a previous hardware version (HW1.4 or lower).
W118	Disable mod. SC	The module has been disabled by DSP due to a fault during soft charging or the module has failed too many times.
W119	Desaturation R1(L)	A desaturation has been triggered in one of the IGBT (inner or outer) of the power module due to an overcurrent in the IGBT or faults caused by a loss of power to the gate drive controlling the IGBT.
W120	Desaturation R3(H)	
W121	Desaturation S1(L)	
W122	Desaturation S3(H)	
W123	Desaturation T1(L)	
W124	Desaturation T3(H)	
W125	Diff Iac mod run	The zero sequence current calculation is higher than the set value for more than 500us.
W128	Spring loaded	The spring of the DC switch disconnector is not charged when it should be.
W129	Isolation Module	An insulation fault has been detected in one of the power modules.
W130	Softcharge Feedback	Soft charge feedback not received (applies only MPCS).
W131	Exceeded Faults limit	The module has given more than 'N' faults in the configured time and the DSP has blocked it.
W132	VDC HW Overvoltage	The module has detected an overvoltage on the entire module bus. The fault threshold is set to 1550V total bus voltage.
W139	Mod. low Vdc recover	While the module is running, a drop of its total bus voltage below the average threshold set in [G5.10.15-Low module Vdc] has been detected and, within 500 µs, the voltage has recovered above the upper threshold ([G5.10.15-Module Low Vdc] + [G5.10.15a-Low module Vdc High]).
W151	Lockable module	One or more modules has been interlocked / disabled due to a fault configured in the list of interlocking faults.
W152	PPC comms	Power limitation has been activated due to the parameters set in the communication timeout with the PPC.
W156	High resistor isolation	The insulation monitor has detected an insulation resistance higher than the permissible insulation resistance.
W158	General Isolation	The inverter has measured an earth resistance lower than the configured threshold (DU is close).
W159	Intern Isolation	An insulation resistance value below the configured fault threshold has been detected (either the channel is AC and the IGBTs are not switching or the DU is open).
W160	Extern Isolation	The insulation monitor has measured a resistance value below the configured fault threshold (the channel is DC and the DU is open).
W161	Fan impulsion MT	Loss of communications with impulsion fans (MV cabinet).
W162	Fan DU zone	Loss of communications with the impulsion fans of the clean zone (DC cabinet).
W163	Fan zone filter LC	Loss of communications with the sinusoidal filter impulsion fans (LC filter cabinet).
W164	Fan zone modules	Loss of communications with the impulsion fans of the power stages (modules cabinets).
W165	Fan motor status	The fault bit has been detected internally in the fan (EBM, Tesoer or Delta).
W166	Temp sensor read	Communication between the microcontroller (MCU) and the temperature sensor of the main control board has failed.
W167	AntiPID ON	The Anti-PID algorithm is running, as long as the conditions described in [G4.13] are fulfilled.
W170	Overvoltage DC	DC bus overvoltage detected.
W171	No RTC	Communication with the RTC is failing.
W172	LC filter Temperature	The temperature read by the sensor in the LC filter is higher than the configured threshold in [G5.4.5.1-Wrng Temp. Threshold].
W173	IMI Disabled Channels	Channels of Insulation Monitoring Device are disabled.

CODE	WARNING	DESCRIPTION
W174	Comms Med Temp	A communication fault has been detected with one of the Bus Plus temperature measurement boards. The warning instance allows identifying which measurement board is affected. The detailed communication status of each board can be checked through the parameters: [SV1.17.1.1-Comms status] (board 1) and [SV1.17.2.1-Comms status] (board 2).
W175	BUSPLUS NTC problem	It has been detected that one of the NTC temperature sensors measuring the temperatures in the DC switch disconnectors (Bus Plus Advanced) is not connected.
W176	DC source fuse	The fuse on busbar 1 of the DC power supply has blown.
W178	Blown fuse	One of the DC cabinet active channel fuses has blown.
W179	FTP_OPEN	The FTP session is open, no information is being written to the SD card.
W180	SD problem	One of the created logs cannot be opened.
W181	Aux transf overvoltaje	The digital overvoltage signal on the auxiliary transformer has tripped.
W182	Therma Bus Plus	The digital input connected to the Bus Plus thermal circuit breaker has been activated.
W183	Internal comms	Fault of internal communications between uP and DSP when the frame loss rate is 10%.
W184	MV transformer Temp	The temperature read by any of the PT100 of the MV Skid transformer connected to the central control exceeds the threshold configured in [G5.4.3.7-MV warning threshold] for HEMK with motorised MV Skid.
W185	IMI not calibrated	Insulation Monitoring Device is not calibrated.
W186	Supply low	The input voltage (24V) of the main control board is lower than 22 V and some peripherals may not have enough voltage to work properly.
W187	DU Delta Temp Threshold	The difference between the temperature/humidity measured by the master DU board and the measured by the slave is higher than the value configured in [G5.10.18-DU Master Slave DifT]. It will not be triggered when the system is a MPCS.
W188	DU sensor lost comms	Communications with the humidity/temperature sensor of one of the DU boards (master or slave) have been lost. It will not be triggered when the system is a MPCS.
W189	Cold start ongoing	The cold start algorithm is running.
W190	AEMO comms lost	Communications failure between PPC / SCADA and the inverter related to AEMO standard.
W191	Antipid wrong V	If a voltage 50V higher than the initial voltage has not been reached within 10 seconds after activation of the AntiPID, it is necessary to stop the AntiPID.
W192	Antipid V not reached	If a voltage of 700-800V has not been reached within 60 seconds after activation of the AntiPID, it is necessary to stop the AntiPID.
W193	Antipid retries exceed	Three attempts have been made to start the antiPID and it has failed to run.
W194	Module filter	In at least one of the modules, the difference between the temperature of the IGBT with the highest temperature and the ambient temperature has exceeded the threshold set in [SG5.1.6-Ativation threshold]. This module must be serviced. **Note: this warning is disabled.**
W195	Protec. vent inductance	The inductance temperature has reached the setting threshold, and the fan speed of the power stages has been increased to provide extra cooling to the inductance.
W196	Fan Incorrect Frames	The number of erroneous frames from the fan has exceeded the configured limit during the monitoring time specified in [G11.10: Error frames].
W197	Modbus traffic exceed	It indicates that the configured minimum time between requests is not being respected, and therefore excessive Modbus traffic is being generated. Requests made in a shorter time than the time configured in [G6.2.14-Min time between req.] will be discarded.
W199	DC Source	The DCDC power supply is supplied from busbar 3; there is voltage on busbar 1, but it is insufficient to supply the system.
W241	Delta T max DU	Excessive temperature difference between modules and ambient.
W252	RFI OverCur Persistent	The current measured by the RFI is greater than the configured threshold.
W253	Imbalance V	The difference in voltage between the busbars is greater than the value set in the protection.
W255	BAT open detection	A sudden opening of the battery has been detected.
W257	Soft-Charge AC Timeout	The busbar has not completed the soft charge from AC in the expected time.

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CODE	WARNING	DESCRIPTION
W303	RFI OverCurrent	The current measured by the RFI is reaching a dangerous level. A current reduction shall be applied.
W304	Imbalance P	There is a difference between the power active (P) setpoints of the busbars, which is reaching a dangerous level. The active power limitation shall be activated.
W305	Limit VDC BESS	The BESS Vdc voltage is outside the configured limits. A power reduction will be applied.
W306	CCL Limit	The inverter is limiting by charge current.
W307	DCL Limit	The inverter is limiting by discharge current.
W308	HW Limit	The difference between the maximum and minimum current reference is higher than the configured threshold. A limitation shall be applied.
W309	Busbar Disable	A busbar has been disabled by the parameters of the group G2.
W310	Module Disable	The modules of an enabled busbar are internally disabled via the configuration parameters (SG).
W311	Low Vdc	Busbar has detected stop condition by low Vdc voltage.
W312	High Vdc	Busbar has detected stop condition by high Vdc voltage.
W313	Unbalanced Vdc	Busbar has detected stop condition due to voltage unbalance between busbars.

4. OPERATING INSTRUCTIONS

4

4.1. Inverter, grid, and battery controls

To configure the control of the inverter, the grid, and the batteries, go to the parameter group **[G8: Control]**. To configure the DC voltage and current limits for both charging and discharging the batteries connected to the different busbars, go to the parameter group **[G3.7: Batteries]**.

4.2. Protections and trip limits for voltage and frequency

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The description of the inverter protection settings (protection range, default values, and protection response times) can be found in the parameter group **[G5: Protections]**. The field-adjustable trip limits for voltage and frequency (voltage and frequency adjustment range, trip time, and default parameters) are specifically found in the parameter group **[G5.1: Voltage-frequency]**.

The following table shows the values of measurement accuracy of inverters up to HW1.3[1].

	STEADY-STATE MEASUREMENTS			TRANSIENT MEASUREMENT		
PARAMETER	**MIN. MEASUREMENT ACCURACY**	**MEASUREMENT WINDOW**	**RANGE**	**MIN. MEASUREMENT ACCURACY**	**MEASUREMENT WINDOW**	**RANGE**
Voltage, RMS	±1 % V_{nom}	10 cycles	0.5 p.u. to 1.2 p.u.	±2 % V_{nom}	5 cycles	0.0 p.u. to 1.4 p.u.
Frequency [2]	10 mHz	60 cycles	50 Hz to 66 Hz	100 mHz	5 cycles	50 Hz to 66 Hz
Active Power	< ± 2 %	10 cycles	0.2 p.u.< P < 1.0 p.u.	Not required	N/A	N/A
Reactive Power	< ± 5 %	10 cycles	0.2 p.u.< Q < 1.0 p.u.	Not required	N/A	N/A
Time	1 % of measured duration	N/A	5 s to 600 s	< 1 cycle	N/A	10 ms to 5 s

For inverters up to HW1.4[1], the measurement accuracy values are as follows:

	STEADY-STATE MEASUREMENTS			TRANSIENT MEASUREMENT		
PARAMETER	**MIN. MEASUREMENT ACCURACY**	**MEASUREMENT WINDOW**	**RANGE**	**MIN. MEASUREMENT ACCURACY**	**MEASUREMENT WINDOW**	**RANGE**
Voltage, RMS	± 1 % V_{nom}	10 cycles	0,5 p.u. to 1,2 p.u.	± 2 % V_{nom}	5 cycles	0,0 p.u. to 1,4 p.u.
Frequency [2]	10 mHz	60 cycles	40 Hz to 66 Hz	± 10 mHz	6 cycles	40 Hz to 66 Hz
Current, RMS	± 2,5 % I_{nom}	10 cycles	0,2 p.u. to 1,2 p.u.	± 10 %	1 cycle	0,1 p.u. to 1,4 p.u.
Active Power	± 5 % S_{nom}	10 cycles	0,2 p.u.< P < 1,0 p.u.	Not required	N/A	N/A
Reactive Power	± 2 % S_{nom}	10 cycles	0,2 p.u.< Q < 1,0 p.u.	Not required	N/A	N/A
Time	1 % of measured duration	N/A	5 s to 600 s	2 cycles	N/A	10 ms to 5 s

[1] Measurement accuracy requirements specified in this table are applicable for voltage THD < 2,5% and individual voltage harmonics <1,5%.
[2] Accuracy requirements for frequency are applicable only when the fundamental voltage is greater than 30% of the nominal voltage.

5. VISUALIZATION AND STATUS PARAMETERS

5

It is possible to check at any time the inverter status and information related to its dynamic parameters. All parameters to visualize and monitor the operation status of the inverter are grouped and listed in the menu “Visualization”. They will be described in this section.

5.1. Group SV1: Internal visualization

This group allows the visualization of the internal state of the system. It shows the status of the inverter, the starting conditions, the status of the AC and DC switchgear, the status of the digital inputs and outputs, temperatures of the electronic boards that make up the inverter and the temperatures of the inlet, filter and MV transformer, as well as the limits and power of the inverter. On the other hand, it shows the inverter status, whether there are start conditions, drive select information and parameters related to medium voltage control.

5.1.1. Subgroup SV1.1: Status

SV1.1.0-Busbar Status

The subgroup SV1.1.0.x indicates the status of the busbars when the product is an **PCS/MPCS inverter**. Depending on the internally configured number of busbars, the respective parameters are displayed.

Note: The subgroup SV1.1.0.x is displayed if [SV1.3.1.1-Inverter type = PCS/MPCS].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.1.0.1-Busbar 1 status = No valid state</td><td>NONE</td><td rowspan="4">Shows the status of the busbars from DC cabinet according to the following table:
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>No valid state</td><td>Indicates an invalid busbar status.</td></tr>
<tr><td>1</td><td>Stop state</td><td>The busbar status indicates that the busbar is stopped.</td></tr>
<tr><td>2</td><td>Stop soft charge</td><td>The busbar status indicates soft charging.</td></tr>
<tr><td>3</td><td>ready state</td><td>The busbar status indicates that the busbar is prepared or ready.</td></tr>
<tr><td>4</td><td>AC ready state</td><td>The busbar status indicates that the AC part of the inverter is prepared or ready. The DC part is closed.</td></tr>
<tr><td>5</td><td>Pre Run</td><td>Indicates that the busbar is starting.</td></tr>
<tr><td>6</td><td>Run</td><td>Indicates that the busbar status is started.</td></tr>
<tr><td>7</td><td>Busbar discharge status</td><td>The busbar status is in the process of discharging.</td></tr>
<tr><td>8</td><td>Waiting for Vdc start cond</td><td>Indicates that the busbar starting, and voltage conditions are not reached.</td></tr>
<tr><td>9</td><td>Wait Unbalance Vdc start cond</td><td>Indicates that the busbar starting, and unbalance voltage conditions are not reached.</td></tr>
<tr><td>10</td><td>Waiting for start conditions</td><td>Indicates that the busbar is ready to start but starting conditions are not reached.</td></tr>
<tr><td>22</td><td>Rest state</td><td>Starts with this status after power up.</td></tr>
<tr><td>55</td><td>Busbar init state</td><td>Indicates the initialization of the busbar.</td></tr>
<tr><td>60</td><td>No modules</td><td>No modules are operating.</td></tr>
<tr><td>61</td><td>Busbar stopped: low voltage</td><td>The busbar has stopped due to undervoltage shutdown conditions.</td></tr>
<tr><td>62</td><td>Busbar stopped: high voltage</td><td>The busbar has stopped due to high voltage shutdown conditions.</td></tr>
<tr><td>63</td><td>Busbar stop: unbalanc voltage</td><td>The busbar has stopped due to unbalance voltage shutdown conditions.</td></tr>
<tr><td>252</td><td>RFI Overcurrent Persistent</td><td>Indicates a persistent RFI overcurrent on the busbar.</td></tr>
<tr><td>253</td><td>Voltage difference exceeded</td><td>Indicates an exceeded voltage difference on the busbar.</td></tr>
<tr><td>254</td><td>VDC not reached</td><td>Indicates a Vdc voltage not reached on the busbar.</td></tr>
<tr><td>255</td><td>Battery open detection</td><td>Indicates open battery detection on the busbar.</td></tr>
</table>
Note: The different options of these parameters will be displayed if [SG10.1.14-Number of busbar] ≥ n, being ‘n’ the number of the busbar. For more information about the distribution of the busbars in the DC cabinet, please consult the Hardware and Installation Manual of the inverters.</td></tr>
<tr><td>SV1.1.0.2-Busbar 2 status = No valid state</td><td>NONE</td></tr>
<tr><td>SV1.1.0.3-Busbar 3 status = No valid state</td><td>NONE</td></tr>
<tr><td>SV1.1.0.4-Busbar 4 status = No valid state</td><td>NONE</td></tr>
<tr><td>SV1.1.1-Current fault = NO FLT</td><td>NONE</td><td>Shows the current fault code in case there is any. To consult fault codes and actions to be taken in case any of them occur, please refer to section 2.
The coding of the fault is given by a decimal number. After making all the required conversions, first to hexadecimal and then to binary, a sixteen bits register is generated. This register defines the code of the fault and in which module has it occurred. From right to left, the ten first bits (LSB) define the code of the fault and the following six bits (MSB) represent the module.
Note: if the MSB represents 0, it means that is an inverter fault.
Example:
Decimal Modbus value 1127 = Binary value 0000010001100111➔MSB = 1; LSB =103➔ Module number 1 has triggered [F103: IGBTs temperature].
Note: if there was more than one fault at the same moment, they will be shown cyclically, one after another, until they are resolved.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.1.2-Current fault module = 0</td><td>NONE</td><td>When a module fault results in a system fault, this parameter shows the number of the module that has produced the original fault. If several modules have produced the fault, it will show the last module that reported it.</td></tr>
<tr><td>SV1.1.3-Current warning = NO WRN</td><td>NONE</td><td>Shows the current warning code in case there is any. The interpretation of the parameter warnings follows the same procedure as described in [SV1.1.1-Current fault].</td></tr>
<tr><td>SV1.1.4-Current status = Power Up</td><td>NONE</td><td>Shows the current status of the inverter.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>OPTION</th><th>STATUS</th></tr>
<tr><td>0</td><td>Power Up</td><td>9</td><td>Ready</td></tr>
<tr><td>1</td><td>Init</td><td>10</td><td>Wait</td></tr>
<tr><td>2</td><td>OFF</td><td>11</td><td>Pre ON</td></tr>
<tr><td>3</td><td>Standby DC</td><td>12</td><td>ON</td></tr>
<tr><td>4</td><td>Precharge DC</td><td>13</td><td>Stopping</td></tr>
<tr><td>5</td><td>Softcharge DC</td><td>14</td><td>Ready AC</td></tr>
<tr><td>6</td><td>Ready DC</td><td>15</td><td>Diagnostic AC</td></tr>
<tr><td>7</td><td>Standby AC</td><td>16</td><td>Discharge</td></tr>
<tr><td>8</td><td>Blackstart</td><td>17</td><td>Fault</td></tr>
</table></td></tr>
<tr><td>SV1.1.5-Power Source = 0.00V</td><td>V</td><td>Shows the voltage measured on the supply and auxiliary services transformer cabinet.</td></tr>
</table>

5.1.2. Subgroup SV1.2: Start conditions

This subgroup shows the start and stop conditions of the inverter busbars.

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer].

SV1.2.0a-Busbar Start Cond

This subgroup allows to check the busbar conditions and its start delay. Consult the *Hardware and Installation Manual* for more information about the busbar distribution.

Note: The different options and parameters of this subgroup will be displayed if [SG10.1.14-Number of busbar ≥ n], being 'n' the number of the busbar.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.2.0a.1-Start VDC = 0000</td><td>NONE</td><td>Allows to check if the DC voltage start condition is fulfilled.
<table>
<tr><th>BIT</th><th>STATUS</th></tr>
<tr><td>0</td><td>Busbar 1</td></tr>
<tr><td>1</td><td>Busbar 2</td></tr>
<tr><td>2</td><td>Busbar 3</td></tr>
<tr><td>3</td><td>Busbar 4</td></tr>
</table></td></tr>
<tr><td>SV1.2.0a.2-Start VDC Unbal = 0000</td><td>NONE</td><td>Allows to check if the busbar unbalance start condition is fulfilled.
<table>
<tr><th>BIT</th><th>STATUS</th></tr>
<tr><td>0</td><td>Busbar 1</td></tr>
<tr><td>1</td><td>Busbar 2</td></tr>
<tr><td>2</td><td>Busbar 3</td></tr>
<tr><td>3</td><td>Busbar 4</td></tr>
</table></td></tr>
<tr><td>SV1.2.0a.3-VDC Delay 1 = 0 s</td><td>s</td><td>Allows to check the delay counter of DC voltage start conditions from busbar 1.</td></tr>
<tr><td>SV1.2.0a.4-VDC Delay 2 = 0 s</td><td>s</td><td>Allows to check the delay counter of DC voltage start conditions from busbar 2.</td></tr>
<tr><td>SV1.2.0a.5-VDC Delay 3 = 0 s</td><td>s</td><td>Allows to check the delay counter of DC voltage start conditions from busbar 3.</td></tr>
</table>

SCREEN	UNITS	DESCRIPTION
SV1.2.0a.6-VDC Delay 4 = 0 s	s	Allows to check the delay counter of DC voltage start conditions from busbar 4.
SV1.2.0a.7-Unbalan Delay 1 = 0 s	s	Allows to check the delay counter of the unbalance start conditions from busbar 1.
SV1.2.0a.8-Unbalan Delay 2 = 0 s	s	Allows to check the delay counter of the unbalance start conditions from busbar 2.
SV1.2.0a.9-Unbalan Delay 3 = 0 s	s	Allows to check the delay counter of the unbalance start conditions from busbar 3.
SV1.2.0a.10-Unbalan Delay 4 = 0 s	s	Allows to check the delay counter of the unbalance start conditions from busbar 4.

SV1.2.0b-Busbar Stop Cond

This subgroup allows to check the stop conditions of the busbar and its delay. Consult the *Hardware and Installation Manual* for more information about the busbar distribution.

Note: The different options and parameters of this subgroup will be displayed if [SG10.1.14-Number of busbar ≥ n], being 'n' the number of the busbar.

EN

SCREEN	UNITS	DESCRIPTION
SV1.2.0b.1-Stop VDC = 0000	NONE	Allows to check if the DC voltage stop condition is fulfilled. BIT 0: Busbar 1 BIT 1: Busbar 2 BIT 2: Busbar 3 BIT 3: Busbar 4
SV1.2.0b.2-Stop VDC Unbal = 0000	NONE	Allows to check if the busbar unbalance stop condition is fulfilled. BIT 0: Busbar 1 BIT 1: Busbar 2 BIT 2: Busbar 3 BIT 3: Busbar 4
SV1.2.0b.3-VDC Delay 1 = 0 s	s	Allows to check the delay counter of DC voltage stop conditions from busbar 1.
SV1.2.0b.4-VDC Delay 2 = 0 s	s	Allows to check the delay counter of DC voltage stop conditions from busbar 2.
SV1.2.0b.5-VDC Delay 3 = 0 s	s	Allows to check the delay counter of DC voltage stop conditions from busbar 3.
SV1.2.0b.6-VDC Delay 4 = 0 s	s	Allows to check the delay counter of DC voltage stop conditions from busbar 4.
SV1.2.0b.7-Unbalan Delay 1 = 0 s	s	Allows to check the delay counter of the unbalance stop conditions from busbar 1.
SV1.2.0b.8-Unbalan Delay 2 = 0 s	s	Allows to check the delay counter of the unbalance stop conditions from busbar 2.
SV1.2.0b.9-Unbalan Delay 3 = 0 s	s	Allows to check the delay counter of the unbalance stop conditions from busbar 3.
SV1.2.0b.10-Unbalan Delay 4 = 0 s	s	Allows to check the delay counter of the unbalance stop conditions from busbar 4.
SV1.2.1-Start conditions all = No	NONE	Shows if there are start conditions.
SV1.2.2-Start conditions V = No	NONE	Shows if there are AC voltage start conditions.
SV1.2.2b-Seq. Start cond. = Yes	NONE	Shows whether there are sequence start conditions in the network. If this sequence matches the one set in VISMA mode in [G8.1.2.3-Freq Sequence], Yes will be displayed. **Note**: this parameter is shown if [SG10.1.19-Visma enable] = Enabled.

SCREEN	UNITS	DESCRIPTION
SV1.2.3-Start conditions f = No	NONE	Shows if there are frequency start conditions.
SV1.2.4-Start conditions Vdc = No	NONE	Shows if there are DC voltage start conditions.
SV1.2.5-Start cond MV cell = No	NONE	Shows if there are AC MV cell start conditions (circuit breaker and disconnector closed, earth open). **Note:** This parameter only applies to HEM inverters.
SV1.2.6-Start conditions P = No	NONE	Shows if there are power start conditions.
SV1.2.7-Start Cond. FPGA Vac = No	NONE	Shows the FPGA AC voltage start condition. When it is correctly wired, this parameter will be shown as Yes. It will be hidden if the overcurrent control is disabled or if VISMA is not enabled in the drive select. **Note:** this parameter is shown if [SG10.1.19-Visma enable] = Yes or [G8.2.1.11.0-Enable] = Yes.
SV1.2.10-Vdc min = 0 V	V	Shows the minimum DC voltage applied by the DSP during the inverter startup, based on the ambient temperature. **Note**: this parameter is shown if [G1.3-User] = Power Electronics staff and [SG10.1.4-Inverter type] ≠ PCS/MPCS.

5.1.3. Subgroup SV1.3: Drive select

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SV1.3.1: General

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.3.1.1-Inverter type = Solar inverter</td><td>NONE</td><td>Shows the application of the inverter.
<table>
<tr><th>OPT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Solar inverter</td><td>Shows when the product is a solar inverter (HEM or HEMK).</td></tr>
<tr><td>1</td><td>Basic BUSPLUS</td><td>Shows when the inverter has the optional Bus Plus basic. This allows to connect up to 2 1200 kW DC/DC or 4 525 kW DC/DC.</td></tr>
<tr><td>2</td><td>Advanced BUSPLUS</td><td>Shows when the inverter has the optional Bus Plus advanced. This allows to connect up to 4 1200 kW DC/DC or 8 525 kW DC/DC.</td></tr>
<tr><td>3</td><td>PCS/MPCS</td><td>Shows when the product is a battery inverter (PCSK/MPCKS).</td></tr>
</table></td></tr>
<tr><td>SV1.3.1.1a-Bus Plus = [SG10.1.5a-Bus Plus]</td><td>NONE</td><td>Shows the DC/DC converter power connected to Bus Plus terminals.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>No</td><td>No Bus Plus.</td></tr>
<tr><td>1 x 525 KW</td><td>1 DC/DC converter of 525 kW.</td></tr>
<tr><td>2 x 525 KW</td><td>2 DC/DC converter of 525 kW.</td></tr>
<tr><td>3 x 525 KW</td><td>3 DC/DC converter of 525 kW.</td></tr>
<tr><td>4 x 525 KW</td><td>4 DC/DC converter of 525 kW.</td></tr>
<tr><td>5 x 525 KW</td><td>5 DC/DC converter of 525 kW.</td></tr>
<tr><td>6 x 525 KW</td><td>6 DC/DC converter of 525 kW.</td></tr>
<tr><td>7 x 525 KW</td><td>7 DC/DC converter of 525 kW.</td></tr>
<tr><td>8 x 525 KW</td><td>8 DC/DC converter of 525 kW.</td></tr>
<tr><td>1 x 1200 KW</td><td>1 DC/DC converter of 1200 kW.</td></tr>
<tr><td>2 x 1200 KW</td><td>2 DC/DC converter of 1200 kW.</td></tr>
<tr><td>3 x 1200 KW</td><td>3 DC/DC converter of 1200 kW.</td></tr>
<tr><td>4 x 1200 KW</td><td>4 DC/DC converter of 1200 kW.</td></tr>
</table>
Note: This parameter is only displayed if [SV1.3.1.1-Inverter type = Solar inverter].</td></tr>
</table>

SCREEN	UNITS	DESCRIPTION
SV1.3.1.2-Nominal AC voltage = [SG10.1.6- Nominal Vac] V	V	Shows the rated AC voltage.
SV1.3.1.3-Nominal S = [SG10.1.11-Nominal S] kVA	kVA	Shows the rated apparent power of each power stage.
SV1.3.1.4-Number of modules = [SG10.1.12-Number of modules]	NONE	Shows the number of the modules.
SV1.3.1.5-Number of MPPs = [SG10.1.14-Number of busbar]	NONE	Shows the number of MPPs.
SV1.3.1.6-Nominal bus voltage = 1500 V	V	Shows the rated DC voltage.
SV1.3.1.7-Max AC current = [SG10.1.9-Iac max] A	A	Shows the AC current limit in each power stage.
SV1.3.1.8-Max DC current = [SG10.1.10-Idc max] A	A	Shows the DC current limit in each power stage.
SV1.3.1.10-MV Skid = [SG10.7.3-MV Skid]	NONE	Shows if the inverter (only HEMK or PCSK) is connected to a Power Electronics' MV Skid.
SV1.3.1.11-Nominal frequency = [SG10.1.7-Nominal Frequency] Hz	Hz	Shows the grid frequency.

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SV1.3.2: Protections

SCREEN	UNITS	DESCRIPTION
SV1.3.2.1-Overcurr prot = [SG10.2.1-Overcurrent threshold] A	A	Shows the overcurrent protection threshold.

SV1.3.3: Limits

SCREEN	UNITS	DESCRIPTION
SV1.3.3.1-Inductive I limit = [SG10.3.1-Inductive I limit] A	A	Shows the inductive reactive current limit.
SV1.3.3.2-Capacitive I limit = [SG10.3.2-Capacitive I limit] A	A	Shows the capacitive reactive current limit.
SV1.3.3.3-Max P limit = [SG10.3.3-Max P limit] %	%	Shows the maximum P value.
SV1.3.3.4-Max Q limit = [SG10.3.4-Max Q limit] %	%	Shows the maximum Q value.
SV1.3.3.5-Max S limit = [SG10.3.5-Max S limit] %	%	Shows the maximum S value.
SV1.3.3.6-VRT reactive I = [SG10.3.6-VRT reactive I] A	A	Shows the maximum reactive current per module (P-Q curve per module) that the system can provide during the LVRT / HVRT events.
SV1.3.3.7-Inductive cosphi = [SG10.3.7-Inductive cosphi]	NONE	Displays the value of the inductive limit of the inverter for 1 p.u. AC voltage.
SV1.3.3.8-Capacitive cosphi = [SG10.3.8-Capacitive cosphi]	NONE	Displays the value of the capacitive limit of the inverter for 1 p.u. AC voltage.

SV1.3.4: Transformer

These parameters apply to HEM inverters and PCSM battery inverters [SG10.1.3-Product type = HEM].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.3.4.1-Cell type = V</td><td>NONE</td><td>Shows the cell protection type.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>V</td><td>Protection cell (V).</td></tr>
<tr><td>1</td><td>LV</td><td>Input + Protection cell (LV).</td></tr>
<tr><td>2</td><td>LVL</td><td>Input + Protection + Output cell (LVL).</td></tr>
</table>
Note: this parameter will be also shown if [SG10.7.3-MV Skid = Yes, Motorized] & [SG10.1.3-Product type = HEMK].</td></tr>
<tr><td>SV1.3.4.2-MV Voltage = 0 V</td><td>V</td><td>Shows the MV transformer voltage.
Note: this parameter will be also shown if [SG10.7.3-MV Skid = Yes, Motorized] & [SG10.1.3-Product type = HEMK].</td></tr>
<tr><td>SV1.3.4.3-Pn = 0 kW</td><td>kW</td><td>Shows the MV transformer active power.</td></tr>
<tr><td>SV1.3.4.4-Xcc = 0.00 %</td><td>kW</td><td>Shows the MV transformer short-circuit reactance.</td></tr>
<tr><td>SV1.3.4.5-Pfe = 0.00 kW</td><td>kW</td><td>Shows the MV transformer no load losses.</td></tr>
<tr><td>SV1.3.4.6-Pcu = 0.0 kW</td><td>kW</td><td>Shows the MV transformer copper losses at full-load current.</td></tr>
<tr><td>SV1.3.4.7-Test Frequency = 0 Hz</td><td>Hz</td><td>Shows the MV transformer frequency.</td></tr>
<tr><td>SV1.3.4.8-Open current = 0.000 %</td><td>%</td><td>Shows the MV transformer open current.</td></tr>
</table>

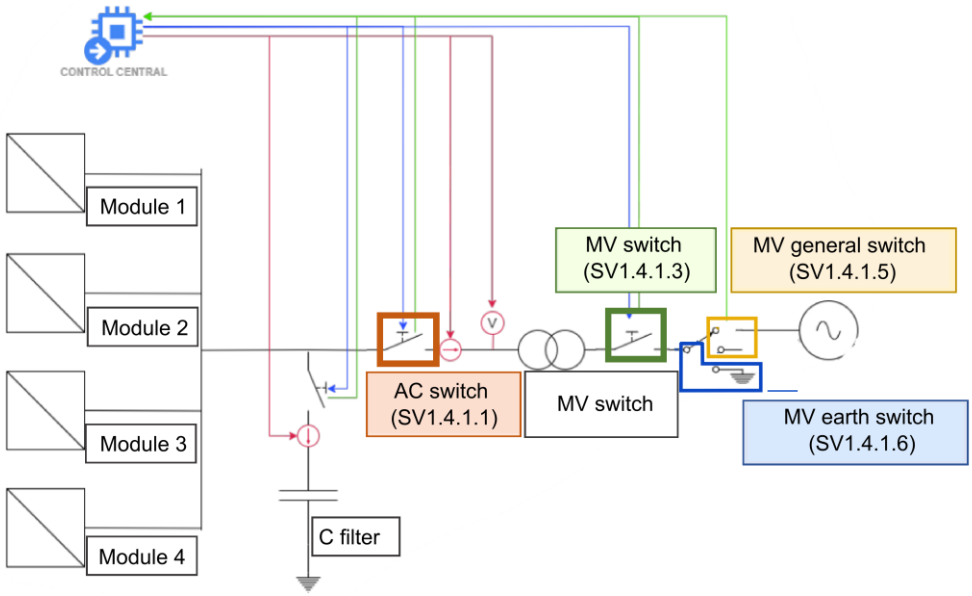
SV1.3.5: Filter

SCREEN	UNITS	DESCRIPTION
SV1.3.5.1-R (mOhm) = [SG10.5.1-R (mOhm)] mOhms	mOhms	Shows the LC filter resistance.
SV1.3.5.2-L (mH) = [SG10.5.2-L (mH)] mH	mH	Shows the modular inductance.
SV1.3.5.3-C (uF) = [SG10.5.3-C (uF)] µF	µF	Shows the LC filter capacitance.
SV1.3.5.4-Lc (uH) = [SG10.5.4-Lc (uH)] uH	uH	Shows the LC filter inductance.

5.1.4. Subgroup SV1.4: Equipment

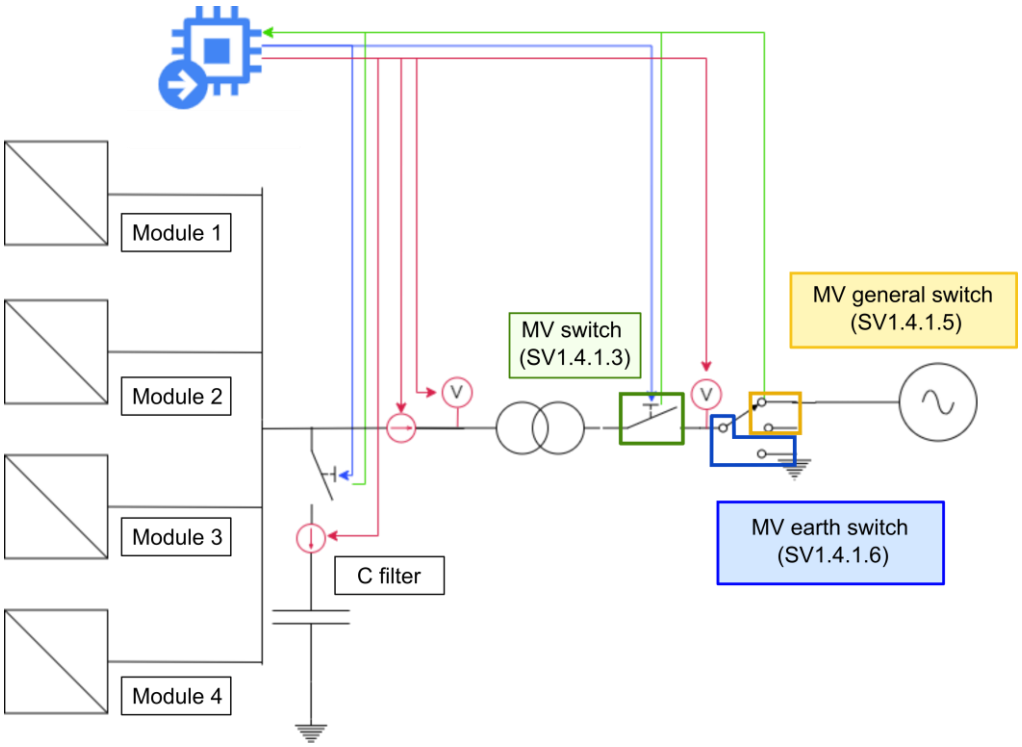
SV1.4.1: AC equipment

This subgroup shows the status of the different switch disconnectors that make up the AC part of the inverter. To visualize where each of the switch disconnectors referred to in this group is located, below is an example of a **HEMK** inverter in combination with a motorized **MV Skid.**



AC-side disconnectors of HEMK inverters with motorized MV Skid

On the other hand, the following picture shows an example of a **HEM** inverter.



AC-side disconnectors of HEM inverters

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer].

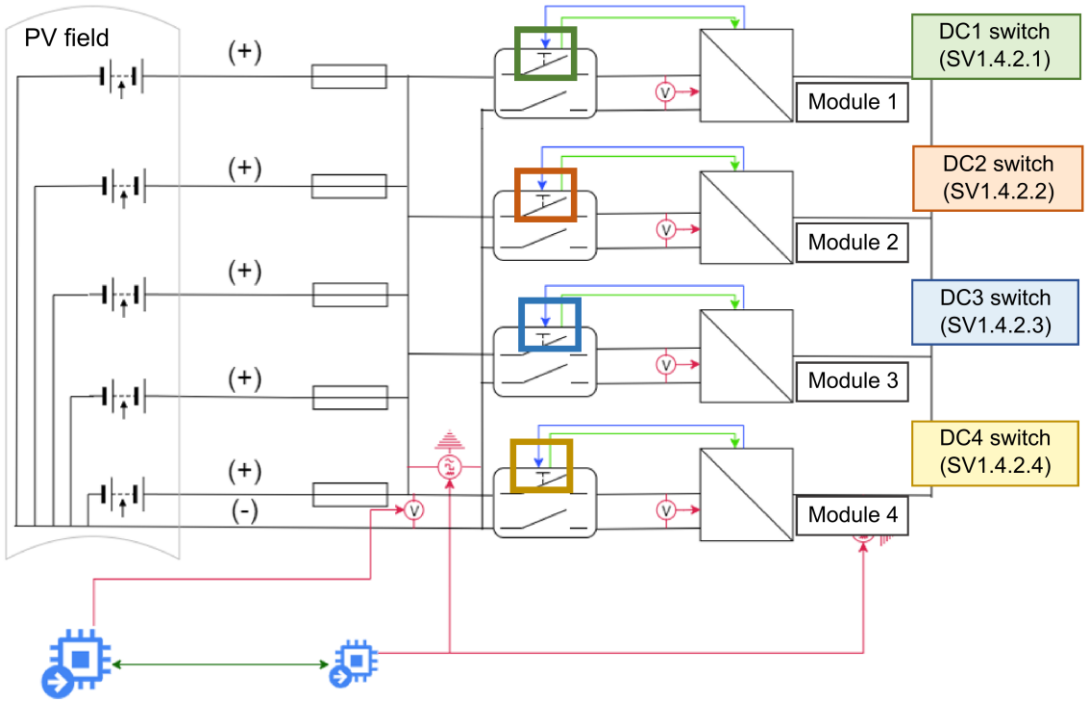
<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.4.1.1-LV AC switch status = Open</td><td>NONE</td><td>Shows the status of the AC circuit breaker (status of its digital output). This circuit breaker has three states: open, closed, and unknown. Unknown status will be displayed when it is not possible to determine in which state the circuit breaker is in.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Open</td><td>The circuit breaker is open (the digital output of the circuit breaker is open).</td></tr>
<tr><td>1</td><td>Closed</td><td>The circuit breaker is closed (the digital output of the circuit breaker is closed).</td></tr>
<tr><td>2</td><td>Unknown</td><td>The command given to the circuit breaker and the status of the corresponding digital output do not match.</td></tr>
</table>
Note: This parameter is displayed if [SG10.1.3 Product type] = HEMK.</td></tr>
<tr><td>SV1.4.1.2-AC spring status = Unknown</td><td>NONE</td><td>Shows the status of the AC spring circuit breaker (status of its digital output). The AC circuit breaker of the inverter is motorized. It has an associated spring action which allows the circuit breaker to close. In this way, when the circuit breaker is commanded to close, the spring must be charged first. Once the circuit breaker is closed, the spring is unloaded. This parameter displays the status of the spring.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Discharged</td><td>Shows when the AC spring circuit breaker is discharged.</td></tr>
<tr><td>1</td><td>Charged</td><td>Shows when the AC spring circuit breaker is charged.</td></tr>
<tr><td>2</td><td>Unknown</td><td>Shows when the AC spring circuit breaker is unknown.</td></tr>
</table>
Note: This parameter is only displayed if [SG10.1.3 Product type] = HEMK.</td></tr>
<tr><td>SV1.4.1.3-MV circ brkr status = Unknown</td><td>NONE</td><td>Shows the status of the circuit breaker of the MV switchgear (status of its digital output). This circuit breaker has three states: open, closed, and unknown. Unknown status will be displayed when it is not possible to determine in which state the circuit breaker is in.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Open</td><td>The circuit breaker is open (the digital output that actuates the circuit breaker is open).</td></tr>
<tr><td>1</td><td>Closed</td><td>The circuit breaker is closed (the digital output that the disconnector is closed).</td></tr>
<tr><td>2</td><td>Unknown</td><td>The command given to the circuit breaker and the status of the corresponding digital output do not match.</td></tr>
</table>
Note: This parameter is only displayed if [SV1.3.1.10-MV Skid ≠ No] and [SG10.1.3-Product type = HEMK] or [SG10.1.3-Product type = HEM].</td></tr>
<tr><td>SV1.4.1.4-MV spring status = Unknown</td><td>NONE</td><td>Shows the status of the MV spring circuit breaker (status of its digital output. The MV circuit breaker of the inverter is motorized. It has an associated spring action which allows the circuit breaker to close. In this way, when the circuit breaker is commanded to close, the spring must be charged first. Once the circuit breaker is closed, the spring is unloaded. This parameter displays the status of the spring.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Discharged</td><td>Shows when the MV spring circuit breaker is discharged.</td></tr>
<tr><td>1</td><td>Charged</td><td>Shows when the MV spring circuit breaker is charged.</td></tr>
<tr><td>2</td><td>Unknown</td><td>Shows when the MV spring circuit breaker is unknown.</td></tr>
</table>
Note: This parameter is only displayed if [SV1.3.1.10-MV Skid ≠ No] and [SG10.1.3-Product type = HEMK] or [SG10.1.3-Product type = HEM].</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.4.1.5-MV swtch discn stat = Unknown</td><td>NONE</td><td>Shows the status of the switch disconnector of the MV switchgear (status of its digital output). This switch disconnector has three states: open, closed, and unknown. Unknown status will be displayed when it is not possible to determine in which state the switch disconnector is in.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Open</td><td>The switch disconnector is open (the digital output of the switch disconnector is open).</td></tr>
<tr><td>1</td><td>Closed</td><td>The switch disconnector is closed (the digital output of the disconnector is closed).</td></tr>
<tr><td>2</td><td>Unknown</td><td>The command given to the switch disconnector and the status of the corresponding digital output do not match.</td></tr>
</table>
Note: This parameter is only displayed if [SV1.3.1.10-MV Skid ≠ No] and [SG10.1.3-Product type = HEMK] or [SG10.1.3-Product type = HEM].</td></tr>
<tr><td>SV1.4.1.6-MV earth switch sta = Unknown</td><td>NONE</td><td>Shows the status of the earthing switch of the MV switchgear (status of its digital output). This switch disconnector has three states: open, closed, and unknown. Unknown status will be displayed when it is not possible to determine in which state the switch disconnector is in.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Open</td><td>The earthing switch disconnector is open (the digital output of the earthing switch disconnector is open).</td></tr>
<tr><td>1</td><td>Closed</td><td>The earthing switch disconnector is closed (the digital output of the disconnector is closed).</td></tr>
<tr><td>2</td><td>Unknown</td><td>The command given to the earthing switch disconnector and the status of the corresponding digital output do not match.</td></tr>
</table>
Note: This parameter is only displayed if [SV1.3.1.10-MV Skid ≠ No] and [SG10.1.3-Product type = HEMK] or [SG10.1.3-Product type = HEM].</td></tr>
</table>

EN

SV1.4.2: DC equipment

This subgroup shows the status of the different switch disconnectors that make up the DC part of the inverter. To visualize where each of the switch disconnectors referred to in this group is located, an example of a HEM inverter with insulated poles is shown below. This case is similar to the rest of the product range. The following image shows an example of the DC part of the inverter**. Note that this part of the inverter may vary according to the project specifications.**



DC-side disconnectors of HEM inverters with isolated poles

Note: Parameters of this subgroup are displayed if [SG10.1.12-Number of modules ≥ 'n'] being 'n' the number of the module.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.4.2.1-DC 1 switch status = Unknown</td><td>NONE</td><td>Shows the status of the DC switch disconnector of module 1 of the inverter (status of its digital output). It has three states: open, closed, and unknown. Unknown status will be displayed when it is not possible to determine in which state the switch disconnector is in.
Note: This parameter is displayed if it has been configured internally.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Open</td><td>The switch disconnector is open (the digital output of the switch disconnector is open).</td></tr>
<tr><td>1</td><td>Closed</td><td>The switch disconnector is closed (the digital output of the disconnector is closed).</td></tr>
<tr><td>2</td><td>Unknown</td><td>The command given to the switch disconnector and the status of the corresponding digital output do not match.</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.4.2.2-DC 2 switch status = Unknown</td><td>NONE</td><td>Shows the status of the DC switch disconnector of module 2 of the inverter (status of its digital output). It has three states: open, closed, and unknown. Unknown status will be displayed when it is not possible to determine in which state the switch disconnector is in.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Open</td><td>The switch disconnector is open (the digital output of the switch disconnector is open).</td></tr>
<tr><td>1</td><td>Closed</td><td>The switch disconnector is closed (the digital output of the disconnector is closed).</td></tr>
<tr><td>2</td><td>Unknown</td><td>The command given to the switch disconnector and the status of the corresponding digital output do not match.</td></tr>
</table></td></tr>
<tr><td>SV1.4.2.3-DC 3 switch status = Unknown</td><td>NONE</td><td>Shows the status of the DC switch disconnector of module 3 of the inverter (status of its digital output). It has three states: open, closed, and unknown. Unknown status will be displayed when it is not possible to determine in which state the switch disconnector is in.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Open</td><td>The switch disconnector is open (the digital output of the switch disconnector is open).</td></tr>
<tr><td>1</td><td>Closed</td><td>The switch disconnector is closed (the digital output of the disconnector is closed).</td></tr>
<tr><td>2</td><td>Unknown</td><td>The command given to the switch disconnector and the status of the corresponding digital output do not match.</td></tr>
</table></td></tr>
<tr><td>SV1.4.2.4-DC 4 switch status = Unknown</td><td>NONE</td><td>Shows the status of the DC switch disconnector of module 4 of the inverter (status of its digital output). It has three states: open, closed, and unknown. Unknown status will be displayed when it is not possible to determine in which state the switch disconnector is in.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Open</td><td>The switch disconnector is open (the digital output of the switch disconnector is open).</td></tr>
<tr><td>1</td><td>Closed</td><td>The switch disconnector is closed (the digital output of the disconnector is closed).</td></tr>
<tr><td>2</td><td>Unknown</td><td>The command given to the switch disconnector and the status of the corresponding digital output do not match.</td></tr>
</table></td></tr>
<tr><td>SV1.4.2.5-DC 1 spring status = Unknown</td><td>NONE</td><td>Shows the status of the DC busbar 1 switch disconnector spring (status of its digital output). The switch disconnector and DC busbar belong to module 1. The DC switch disconnector of the inverter is motorized. It has an associated spring action, which allows the switch disconnector to close. In this way, when the switch disconnector is commanded to close, the spring must be charged first. Once the switch disconnector is closed, the spring is unloaded. This parameter displays the status of the spring.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Discharged</td><td>Shows when DC busbar 1 spring switch disconnector is discharged.</td></tr>
<tr><td>1</td><td>Charged</td><td>Shows when DC busbar 1 spring switch disconnector is charged.</td></tr>
<tr><td>2</td><td>Unknown</td><td>Shows when the DC busbar 1 spring switch disconnector is unknown.</td></tr>
</table></td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.4.2.6-DC 2 spring status = Unknown</td><td>NONE</td><td>Shows the status of the DC busbar 2 spring switch disconnector (status of its digital output). The switch disconnector and DC busbar belong to module 2. The DC switch disconnector of the inverter is motorized. It has an associated spring action, which allows the switch disconnector to close. In this way, when the switch disconnector is commanded to close, the spring must be charged first. Once the switch disconnector is closed, the spring is unloaded. This parameter displays the status of the spring.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Discharged</td><td>Shows when DC busbar 2 spring switch disconnector is discharged.</td></tr>
<tr><td>1</td><td>Charged</td><td>Shows when DC busbar 2 spring switch disconnector is charged.</td></tr>
<tr><td>2</td><td>Unknown</td><td>Shows when the DC busbar 2 spring switch disconnector is unknown.</td></tr>
</table></td></tr>
<tr><td>SV1.4.2.7-DC 3 spring status = Unknown</td><td>NONE</td><td>Shows the status of the DC busbar 3 spring switch disconnector (status of its digital output). The switch disconnector and DC busbar belong to module 3. The DC switch disconnector of the inverter is motorized. It has an associated spring action, which allows the switch disconnector to close. In this way, when the switch disconnector is commanded to close, the spring must be charged first. Once the switch disconnector is closed, the spring is unloaded. This parameter displays the status of the spring.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Discharged</td><td>Shows when DC busbar 3 spring switch disconnector is discharged.</td></tr>
<tr><td>1</td><td>Charged</td><td>Shows when DC busbar 3 spring switch disconnector is charged.</td></tr>
<tr><td>2</td><td>Unknown</td><td>Shows when the DC busbar 3 spring switch disconnector is unknown.</td></tr>
</table></td></tr>
<tr><td>SV1.4.2.8-DC 4 spring status = Unknown</td><td>NONE</td><td>Shows the status of the DC busbar 4 spring switch disconnector (status of its digital output). The switch disconnector and DC busbar belong to module 4. The DC switch disconnector of the inverter is motorized. It has an associated spring action, which allows the switch disconnector to close. In this way, when the switch disconnector is commanded to close, the spring must be charged first. Once the switch disconnector is closed, the spring is unloaded. This parameter displays the status of the spring.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Discharged</td><td>Shows when DC busbar 4 spring switch disconnector is discharged.</td></tr>
<tr><td>1</td><td>Charged</td><td>Shows when DC busbar 4 spring switch disconnector is charged.</td></tr>
<tr><td>2</td><td>Unknown</td><td>Shows when the DC busbar 4 spring switch disconnector is unknown.</td></tr>
</table></td></tr>
</table>

5.1.5. Subgroup SV1.5: Digital I/O status

This group shows information about the status of the digital inputs and outputs.

SV1.5.1: DSP-uC central

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.5.1.1-uC Digital inputs = 0000000000000000</td><td>NONE</td><td>Each bit (from right to left) represents a digital input, as described in the table below.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Loto DC</td><td>Displays the status of the LOTO DC.</td></tr>
<tr><td>1</td><td>Loto AC</td><td>Displays the status of the LOTO AC.</td></tr>
<tr><td>2</td><td>Retro Aux 1</td><td>Shows the status of the auxiliary contactor 1 feedback.</td></tr>
<tr><td>3</td><td>Circuit breaker loaded</td><td>Displays if the circuit breaker is loaded.</td></tr>
<tr><td>4</td><td>Retro Circuit Breaker Open</td><td>Displays if the circuit breaker is open.</td></tr>
<tr><td>5</td><td>Retro Circuit Breaker Closed</td><td>Displays if the circuit breaker is closed.</td></tr>
<tr><td>6</td><td>Retro Transf SSAA</td><td>Shows the status of the transformer thermal contact.</td></tr>
<tr><td>7</td><td>Retro DCDC Int</td><td>Shows the status of the DC/DC converter.</td></tr>
<tr><td>8</td><td>Retro Aux 2</td><td>Displays status of LC contactor feedback.</td></tr>
<tr><td>9</td><td>Selector Key</td><td>Shows the status of selector key.</td></tr>
<tr><td>10</td><td>Emergency Button</td><td>Shows the status of emergency button.</td></tr>
<tr><td>11</td><td>IP Default</td><td>Shows the IP default.</td></tr>
<tr><td>12</td><td>Config 2</td><td>Shows the status config 2 is ok.</td></tr>
<tr><td>13</td><td>ISO Aux 1</td><td>Displays the status of isolation monitoring device.</td></tr>
<tr><td>14</td><td>GFDI Fault</td><td>Displays if the GFDI fuse status is not ok.</td></tr>
<tr><td>15</td><td>ISO Aux 2</td><td>Displays the status of isolation monitoring device.</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.2-uC Digital inputs 2 = 0000000000000000</td><td>NONE</td><td>Each bit (from right to left) represents a digital input, as described in the table below.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Retro Inductance Overvoltage</td><td>Feedback overvoltage inductance.</td></tr>
</table></td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.5.1.3-uC Digital outputs = 0000000000000000</td><td>NONE</td><td>Each bit (from right to left) represents a digital output, as described in the table below.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Act Aux 1</td><td>Reserved.</td></tr>
<tr><td>1</td><td>Act Aux 2</td><td>Reserved.</td></tr>
<tr><td>2</td><td>Act Aux 3</td><td>Reserved.</td></tr>
<tr><td>3</td><td>Led Fault Wrn</td><td>Displays the output status of the Alarm / Fault LED output.</td></tr>
<tr><td>4</td><td>Led Green</td><td>Shows the output status of the RUN LED output.</td></tr>
<tr><td>5</td><td>Close AC Circuit Breaker</td><td>Shows the output status AC circuit breaker.</td></tr>
<tr><td>6</td><td>Open UVT Circuit Breaker</td><td>Shows the output status UVT circuit breaker.</td></tr>
<tr><td>7</td><td>Open SHT Circuit Breaker</td><td>Shows the output status SHT circuit breaker.</td></tr>
<tr><td>8</td><td>Act Aux 4</td><td>Reserved.</td></tr>
<tr><td>9</td><td>DCDC Int Stop</td><td>Shows the output status DCDC int stop.</td></tr>
<tr><td>10</td><td>Act Aux 5</td><td>Reserved.</td></tr>
<tr><td>11</td><td>Act Fault</td><td>Reserved.</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.4-Status error = 000</td><td>NONE</td><td>Displays information on the following errors:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Softcharge contactor fault</td><td>Displays the error status of the soft charge contactor fault (0:OK, 1: NOT OK).</td></tr>
<tr><td>1</td><td>Trackers contactor fault</td><td>Displays the error status of the trackers mode contactor fault (0:OK, 1: NOT OK).</td></tr>
<tr><td>3</td><td>LC Contactor fault</td><td>Shows the error status of the LC contactor fault (0:OK, 1: NOT OK).</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.5-Status DSP RJ13 = 000000</td><td>NONE</td><td>Shows input and output status of connector RJ13 of the main control board. Each bit (from right to left) represents digital input / output status of connector RJ13 of main control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DO open cell</td><td>Digital output to control the opening of the MV switchgear.</td></tr>
<tr><td>1</td><td>DO close cell</td><td>Digital output to control the closing of the MV switchgear.</td></tr>
<tr><td>2</td><td>DO MV out aux</td><td>Not applicable to this product version and earlier.</td></tr>
<tr><td>3</td><td>DI breaker open</td><td>Feedback indicating that the protection cell is open</td></tr>
<tr><td>4</td><td>DI breaker close</td><td>Feedback indicating that the protection cell is closed</td></tr>
<tr><td>5</td><td>DI spring loaded</td><td>Feedback indicating that the protection cell spring is loaded</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.5.1.6-Status DSP RJ14 = 0000000</td><td>NONE</td><td>Shows input and output status of connector RJ14 of the main control board. Each bit (from right to left) represents digital input / output status of connector RJ14 of main control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DI disconnector open</td><td>Feedback indicating that the disconnector is open.</td></tr>
<tr><td>1</td><td>DI Disconnector close</td><td>Feedback indicating that the disconnector is closed.</td></tr>
<tr><td>2</td><td>DI Disconnector ground close</td><td>Feedback indicating that the disconnector is grounded.</td></tr>
<tr><td>3</td><td>DI Disconnector ground open</td><td>Feedback indicating that the disconnector is not grounded.</td></tr>
<tr><td>4</td><td>DI Feedback shoot aux</td><td>Not applicable to this product version and earlier.</td></tr>
<tr><td>5</td><td>DI Pressure gas sf6 cell</td><td>Feedback indicating that the pressure in the MV switchgear transformer is too low.</td></tr>
<tr><td>6</td><td>DI Watchdog[3] relay</td><td>MV switchgear watchdog feedback.</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.7-Status DSP RJ15 = 0000000</td><td>NONE</td><td>Shows input and output status of connector RJ15 of the main control board. Each bit (from right to left) represents digital input / output status of connector RJ15 of main control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DI CL2 disc. Ground close</td><td>Line cell 2 disconnector earthing switch feedback status is closed.</td></tr>
<tr><td>1</td><td>DI CL2 disconnector close</td><td>Line cell 2 disconnector feedback status is closed.</td></tr>
<tr><td>2</td><td>DI CL2 disconnector open</td><td>Line cell 2 disconnector feedback switch status is open.</td></tr>
<tr><td>3</td><td>DI CL1 discon. ground open</td><td>Line cell 1 disconnector earthing switch feedback status is open.</td></tr>
<tr><td>4</td><td>DI CL1 discon. ground close</td><td>Line cell 1 earthing switch disconnector feedback status is closed.</td></tr>
<tr><td>5</td><td>DI CL1 disconnector close</td><td>Line cell 1 switch disconnector feedback status is closed.</td></tr>
<tr><td>6</td><td>DI CL1 disconnector open</td><td>Line cell disconnector 1 switch disconnector feedback status is open.</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.8-Status DSP RJ16 = 0000000</td><td>NONE</td><td>Shows input and output status of connector RJ16 of the main control board. Each bit (from right to left) ≥represents digital input / output status of connector RJ16 of main control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DI CL2 discon. ground open</td><td>Line cell 2 earthing switch feedback status is open.</td></tr>
<tr><td>1</td><td>DI DGPT2 temperature fault</td><td>Signal from the MV Skid MV transformer indicating temperature fault.</td></tr>
<tr><td>2</td><td>DI DGPT2 temperature warning</td><td>Signal from the MV Skid MV transformer indicating temperature warning.</td></tr>
<tr><td>3</td><td>DI DGPT2 pressure fault</td><td>Signal from the MV Skid MV transformer indicating pressure failure.</td></tr>
<tr><td>4</td><td>DI DGPT2 level fault</td><td>Signal from the MV Skid MV transformer indicating oil or gas level failure.</td></tr>
<tr><td>5</td><td>DI MV aux 1</td><td>Not applicable to this product version and earlier.</td></tr>
<tr><td>6</td><td>DI MV aux 2</td><td>Not applicable to this product version and earlier.</td></tr>
</table></td></tr>
</table>

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[3] A watchdog is a peripheral that periodically checks that the state of the microcontroller is correct. If it is not, it restarts it.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.5.1.9-Status DSP RJ17 = 000000</td><td>NONE</td><td>Each bit (from right to left) represents digital input / output status of connector RJ17 of main control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DO activation contactor 1[4]</td><td>Command to activation contactor 1.</td></tr>
<tr><td>1</td><td>DO activation contactor 2[4]</td><td>Command to activation contactor 2.</td></tr>
<tr><td>2</td><td>DO activation contactor 3[4]</td><td>Command to activation contactor 3.</td></tr>
<tr><td>3</td><td>DI feedback contactor 1</td><td>Feedback that the contactor 1 has been activated.</td></tr>
<tr><td>4</td><td>DI feedback contactor 2</td><td>Feedback that the contactor 2 has been activated.</td></tr>
<tr><td>5</td><td>DI feedback contactor 3</td><td>Feedback that the contactor 3 has been activated.</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.10-Status DSP RJ18 = 000000</td><td>NONE</td><td>Each bit (from right to left) represents digital input / output status of connector RJ18 of main control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DO activation contactor 4[4]</td><td>Command to activation contactor 4.</td></tr>
<tr><td>1</td><td>DO activation aux 1</td><td>Command to activation sdp AC.</td></tr>
<tr><td>2</td><td>DO activation aux 2</td><td>Command to activation inductance.</td></tr>
<tr><td>3</td><td>DI feedback contactor 4</td><td>Feedback that the contactor 4 has been activated.</td></tr>
<tr><td>4</td><td>DI feedback sdp AC</td><td>Feedback that the sdp AC has been activated.</td></tr>
<tr><td>5</td><td>DI feedback inductance</td><td>Feedback that the inductance has been activated.</td></tr>
</table></td></tr>
</table>

[4] These contactors connect the capacitor C of the LCL filter to the output of the inverter.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.5.1.11-Status DSP RJ20 = 000000</td><td>NONE</td><td>Shows input and output status of connector RJ20 of the main control board. Each bit (from right to left) represents digital input / output status of connector RJ20 of main control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved for future use</td><td>-</td></tr>
<tr><td>1</td><td>Internal fault</td><td>The internal fault bit indicates the status of the digital output whether it is activated or deactivated. The inverter activates this digital output when a special fault is detected. It remains active until a manual reset is performed.
Note: This applies to DCDC / MPCS.</td></tr>
<tr><td>2</td><td>Reserved for future use</td><td>-</td></tr>
<tr><td>3</td><td>E-stop</td><td>The bit corresponding to E-Stop indicates the status of the digital input whether it is activated or deactivated. When the digital input "E-STOP Customer" is activated, the device activates fault F166: E-Stop. This fault indicates that the customer has opened the safety chain via the following terminals:
- One activates the E-STOP digital input.
- The other terminal also activates the E-STOP digital input but in turn opens the safety chain of the device if the customer signal is de-energized.
This applies to the solar and storage ranges (M and K types).</td></tr>
<tr><td>4</td><td>External fault</td><td>The external fault bit indicates the status of the digital input whether it is activated or deactivated. Corresponds to the customer remote emergency button. When this digital input is activated, the inverter triggers fault F55: External fault/OVGR.
This digital input is the communication via the customer to the inverter to command a shutdown due to an external inverter fault.</td></tr>
<tr><td>5</td><td>Reserved for future use</td><td>-</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.12-Status DSP RJ21 = 0</td><td>NONE</td><td>Each bit (from right to left) represents digital input / output status of connector RJ21 of main control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DO activation fault</td><td>Command to activation fault.</td></tr>
</table></td></tr>
<tr><td>SV1.5.1.13-Status DSP RJ22 = 0</td><td>NONE</td><td>Each bit (from right to left) represents digital input / output status of connector RJ22 of main control board.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DI emergency button 2</td><td>Feedback of the activation of emergency stop button 2.</td></tr>
</table></td></tr>
</table>

SV1.5.2: FPGA module interf

Note: These parameters are displayed if [G1.3-User ≥ Power Electronics staff] and [SG10.1.12-Number of modules ≥ n], being "n" the number of the module.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.5.2.1-Mod 1 Digital In = 000000</td><td>NONE</td><td>Each bit (from right to left) shows information about the following digital inputs from the FPGA of module 1.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>di_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>di_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>di_softcharge</td><td>Softcharge feedback.</td></tr>
<tr><td>3</td><td>dis_charge</td><td>Feedback that the DC switch disconnector spring is charged.</td></tr>
<tr><td>4</td><td>dis_open</td><td>Feedback that the DC switch disconnector is open.</td></tr>
<tr><td>5</td><td>dis_close</td><td>Feedback that the DC switch disconnector is closed.</td></tr>
</table></td></tr>
<tr><td>SV1.5.2.2-Mod 1 Digital Out = 000000</td><td>NONE</td><td>Each bit (from right to left) shows information about the following digital outputs from the FPGA of module 1:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>do_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>do_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>do_aux_1</td><td>Softcharge activation.</td></tr>
<tr><td>3</td><td>do_close</td><td>Command to close DC switch disconnector.</td></tr>
<tr><td>4</td><td>do_open</td><td>Command to open DC switch disconnector.</td></tr>
<tr><td>5</td><td>do_charge</td><td>Command to charge spring of DC switch disconnector.</td></tr>
</table></td></tr>
<tr><td>SV1.5.2.3-Mod 2 Digital In = 000000</td><td>NONE</td><td>Each bit (from right to left) shows information about the following digital inputs from the FPGA of module 2.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>di_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>di_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>di_softcharge</td><td>Softcharge feedback.</td></tr>
<tr><td>3</td><td>dis_charge</td><td>Feedback that the DC switch disconnector spring is charged.</td></tr>
<tr><td>4</td><td>dis_open</td><td>Feedback that the DC switch disconnector is open.</td></tr>
<tr><td>5</td><td>dis_close</td><td>Feedback that the DC switch disconnector is closed.</td></tr>
</table></td></tr>
<tr><td>SV1.5.2.4-Mod 2 Digital Out = 000000</td><td>NONE</td><td>Each bit (from right to left) shows information about the following digital outputs from the FPGA of module 2:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>do_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>do_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>do_aux_1</td><td>Softcharge activation.</td></tr>
<tr><td>3</td><td>do_close</td><td>Command to close DC switch disconnector.</td></tr>
<tr><td>4</td><td>do_open</td><td>Command to open DC switch disconnector.</td></tr>
<tr><td>5</td><td>do_charge</td><td>Command to charge spring of DC switch disconnector.</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.5.2.5-Mod 3 Digital In = 000000</td><td>NONE</td><td>Each bit (from right to left) shows information about the following digital inputs from the FPGA of module 3.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>di_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>di_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>di_softcharge</td><td>Softcharge feedback.</td></tr>
<tr><td>3</td><td>dis_charge</td><td>Feedback that the DC switch disconnector spring is charged.</td></tr>
<tr><td>4</td><td>dis_open</td><td>Feedback that the DC switch disconnector is open.</td></tr>
<tr><td>5</td><td>dis_close</td><td>Feedback that the DC switch disconnector is closed.</td></tr>
</table></td></tr>
<tr><td>SV1.5.2.6-Mod 3 Digital Out = 000000</td><td>NONE</td><td>Each bit (from right to left) shows information about the following digital outputs from the FPGA of module 3:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>do_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>do_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>do_aux_1</td><td>Softcharge activation.</td></tr>
<tr><td>3</td><td>do_close</td><td>Command to close DC switch disconnector.</td></tr>
<tr><td>4</td><td>do_open</td><td>Command to open DC switch disconnector.</td></tr>
<tr><td>5</td><td>do_charge</td><td>Command to charge spring of DC switch disconnector.</td></tr>
</table></td></tr>
<tr><td>SV1.5.2.7-Mod 4 Digital In = 000000</td><td>NONE</td><td>Each bit (from right to left) shows information about the following digital inputs from the FPGA of module 4.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>di_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>di_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>di_softcharge</td><td>Softcharge feedback.</td></tr>
<tr><td>3</td><td>dis_charge</td><td>Feedback that the DC switch disconnector spring is charged.</td></tr>
<tr><td>4</td><td>dis_open</td><td>Feedback that the DC switch disconnector is open.</td></tr>
<tr><td>5</td><td>dis_close</td><td>Feedback that the DC switch disconnector is closed.</td></tr>
</table></td></tr>
<tr><td>SV1.5.2.8-Mod 4 Digital Out = 000000</td><td>NONE</td><td>Each bit (from right to left) shows information about the following digital outputs from the FPGA of module 4:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>do_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>do_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>do_aux_1</td><td>Softcharge activation.</td></tr>
<tr><td>3</td><td>do_close</td><td>Command to close DC switch disconnector.</td></tr>
<tr><td>4</td><td>do_open</td><td>Command to open DC switch disconnector.</td></tr>
<tr><td>5</td><td>do_charge</td><td>Command to charge spring of DC switch disconnector.</td></tr>
</table></td></tr>
</table>

EN

SV1.5.3-MV control interface

This group is not available from version 5.0.0.

Note: These parameters are displayed if [SG10.1.3-Product type = HEM]. Applies to HEM / MPCSM inverters.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.5.3.1 Digital Inputs = 00000000000000</td><td>NONE</td><td>Each bit shows information about the following digital inputs from the MV switchgear.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>3</td><td>MV Cell protection open.</td><td>Displays the status of the digital inputs of the MV cell protection open (0: Open, 1: Closed).</td></tr>
<tr><td>4</td><td>MV Cell protection close.</td><td>Displays the status of the digital inputs of the MV cell protection close (0: Open, 1: Closed).</td></tr>
<tr><td>6</td><td>MV switch gear open.</td><td>Displays the status of the digital inputs of the MV switchgear open (0: Open, 1: Closed).</td></tr>
<tr><td>7</td><td>MV switch gear close.</td><td>Displays the status of the digital inputs of the MV switchgear close (0: Open, 1: Closed).</td></tr>
<tr><td>8</td><td>MV Earth disconnector close.</td><td>Displays the status of the digital inputs of the MV earth disconnector close (0: Open, 1: Closed).</td></tr>
<tr><td>9</td><td>MV Earth disconnector open.</td><td>Displays the status of the digital inputs of the MV earth disconnector open (0: Open, 1: Closed).</td></tr>
<tr><td>11</td><td>MV FANs feedback.</td><td>It shows the status of the digital inputs of the MV fans feedback (0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV1.5.3.2 Digital Outputs = 00000</td><td>NONE</td><td>Each bit shows information about the following digital outputs from the MV switchgear.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>MV switch gear close.</td><td>Displays the status of the digital outputs of the MV switchgear close (0: Open, 1: Closed).</td></tr>
<tr><td>1</td><td>MV Cell protection trip coil.</td><td>Displays the status of the digital outputs MV switchgear protection trip coil (0: Open, 1: Closed).</td></tr>
<tr><td>2</td><td>MV Trafo fans contactor.</td><td>Displays the status of the digital outputs of the MV transformer fans contactor (0: Open, 1: Closed).</td></tr>
</table></td></tr>
<tr><td>SV1.5.3.3 Status error = 000</td><td>NONE</td><td>Each bit shows information about the status error from the MV switchgear.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>MV switch gear fault.</td><td>Shows the fault status of the circuit breaker in the MV circuit breaker failure (0: OK, 1: NOT OK).</td></tr>
<tr><td>1</td><td>MV trip coil.</td><td>Shows the fault status of the MV cell tripping coil (0: OK, 1: NOT OK).</td></tr>
<tr><td>2</td><td>Trafo fan contactor fault</td><td>Displays the failure status of the transformer fan contactor (0: OK, 1: NOT OK).</td></tr>
</table>
Note: This parameter is displayed if [G1.3-User ≥ Commissioning].</td></tr>
</table>

5.1.6. Subgroup SV1.6: Environment

The parameters of this subgroup show the values of temperature and humidity of the sensors located in several areas of the inverter.

Applies to HEM/PCSM and HEMK/PCSK.

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer].

SCREEN	UNITS	DESCRIPTION
SV1.6.1-Internal T° = 35 ºC	ºC	Shows the main control cabinet internal temperature.
SV1.6.2-Max modules T° = 0.00 ºC	ºC	Shows the maximum internal temperature of four modules.
SV1.6.3-DU Internal T° = 35 ºC	ºC	The system can have two different DC cabinet boards. This parameter shows the internal temperature of the main control board of the first DC cabinet board connected to the second one by optical fiber.
SV1.6.3b-DU 2 Internal T° = 35 ºC	ºC	The system can have two different DC cabinet boards. This parameter shows the internal temperature of the main control board of the second DC cabinet board connected to the first one by optical fiber.
SV1.6.4-Admission T° = 0 ºC	ºC	Shows the temperature measured by the sensor located at the air inlet in the AC module.
SV1.6.5-Admission T° Avg = 0 ºC	ºC	Shows the average temperature measured by the NTC sensor located in the AC cabinet. **Note**: Only users with role Developer can see and modify this parameter.
SV1.6.6-Wrong Sampl T° Adm = 0	ºC	Shows if the sample of the admission temperature is out of range.
SV1.6.7-Internal humidity = 50 %	ºC	Shows the main control board humidity.
SV1.6.8-MV PT100 (1) = 0 ºC	ºC	Shows the measured temperature by the PT100 located in the MV transformer cabinet. **Note:** This parameter is only displayed if [SG10.1.3 Product type = HEM] or [SG10.1.3 Product type = HEMK] and [SG10.7.5-Skid MV Transf PT100 = Yes].
SV1.6.1.9-MV PT100 (2) = 0 ºC	ºC	Shows the measured temperature by the PT100 located in the MV transformer cabinet. **Note:** This parameter is only displayed if [SG10.1.3 Product type = HEM] or [SG10.1.3 Product type = HEMK] and [SG10.7.5-Skid MV Transf PT100 = Yes].
SV1.6.10-MV PT100 (3) = 0 ºC	ºC	Shows the measured temperature by the PT100 located in the MV transformer cabinet. **Note:** This parameter is only displayed if [SG10.1.3 Product type = HEM] or [SG10.1.3 Product type = HEMK] and [SG10.7.5-Skid MV Transf PT100 = Yes].
SV1.6.11-Ind PT100 (1) = 0 ºC	ºC	Shows the measured temperature by the PT100 located in the LC filter.
SV1.6.12-Ind PT100 (2) = 0 ºC	ºC	Shows the measured temperature by the PT100 located in the LC filter.
SV1.6.13-Ind PT100 (3) = 0 ºC	ºC	Shows the measured temperature by the PT100 located in the LC filter.
SV1.6.14-Ind PT100 (4) = 0 ºC	ºC	Shows the measured temperature by the PT100 located in the LC filter.
SV1.6.15-LCL filter Temp = 0 ºC	ºC	Shows the LC filter temperature measured by the NTC sensor connected in the J34 pin of the main control board.

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5.1.7. Subgroup SV1.7: Inverter Control

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.7.1-P control = 000</td><td>NONE</td><td>Indicates the current P control mode. There are different control modes available:
<table>
<tr><th>BIT</th><th>CONTROL MODE</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>MPPt</td><td>The inverter is working using the MPPT mode.</td></tr>
<tr><td>1</td><td>Id</td><td>The inverter is working using the active current mode.</td></tr>
<tr><td>2</td><td>P</td><td>The inverter is working using the active power mode (solar application).</td></tr>
<tr><td>3</td><td>Vdc</td><td>The inverter is working using voltage control mode.</td></tr>
<tr><td>7</td><td>PAC</td><td>The inverter is working using the active power mode (MPCS application).</td></tr>
<tr><td>8</td><td>Multi PAC</td><td>The inverter is working using the active power mode per busbar (MPCS application).</td></tr>
<tr><td>10</td><td>LVRT</td><td>The inverter is working using the LVRT control mode.</td></tr>
<tr><td>11</td><td>HVRT</td><td>The inverter is working using the HVRT control mode.</td></tr>
<tr><td>12</td><td>NIGHT</td><td>The inverter is working using the voltage control mode (Bus Plus night mode application).</td></tr>
<tr><td>13</td><td>NIGHT MOD</td><td>The inverter is working using the voltage control mode (night mode, solar or MPCS application).</td></tr>
<tr><td>20</td><td>VISMA_PAC</td><td>The inverter operates in VISMA control mode regulating AC reactive power.</td></tr>
<tr><td>21</td><td>VISMA_FREQ</td><td>The inverter is operating in VISMA control mode regulating the grid frequency.</td></tr>
<tr><td>22</td><td>VISMA_LVRT</td><td>Inverter is operating in VISMA_PAC/VISMA_FREQ control mode and has switched control to LVRT.</td></tr>
<tr><td>23</td><td>VISMA_HVRT</td><td>The inverter is operating in VISMA_PAC/VISMA_FREQ control mode and has switched control to HVRT.</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.7.2-Q control = 000</td><td>NONE</td><td>Indicates the current Q control mode. There are different control modes available:
<table>
<tr><th>BIT</th><th>CONTROL MODE</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>CPHI</td><td>The inverter is working using the cos phi mode.</td></tr>
<tr><td>1</td><td>Q</td><td>The inverter is working using the reactive power control mode (solar application).</td></tr>
<tr><td>2</td><td>Iq</td><td>The inverter is working using the reactive current control mode.</td></tr>
<tr><td>3</td><td>Cos Phi P</td><td>The inverter is working using the cos phi(P) control mode.</td></tr>
<tr><td>7</td><td>QAC</td><td>The inverter is working using the reactive power control mode (MPCS application).</td></tr>
<tr><td>10</td><td>LVRT</td><td>The inverter is working using the LVRT control mode.</td></tr>
<tr><td>11</td><td>HVRT</td><td>The inverter is working using the HVRT control mode.</td></tr>
<tr><td>12</td><td>NIGHT</td><td>The inverter is working using the reactive power control mode (night mode).</td></tr>
<tr><td>20</td><td>FAC</td><td>The inverter is working using the voltage source Vac closed loop control mode.</td></tr>
<tr><td>21</td><td>VD</td><td>The inverter is working using the voltage source open loop control mode.</td></tr>
<tr><td>22</td><td>VISMA_VAC</td><td>The inverter operates with the VISMA Vac voltage source in closed loop control mode.</td></tr>
<tr><td>23</td><td>VISMA_Q</td><td>The inverter operates in VISMA control mode regulating AC reactive power.</td></tr>
<tr><td>24</td><td>VISMA_IQ</td><td>The inverter operates in VISMA control mode regulating AC reactive current.</td></tr>
<tr><td>25</td><td>VISMA_LVRT</td><td>The inverter is operating in VISMA_Q/VISMA_IQ control mode and has switched control to LVRT.</td></tr>
<tr><td>26</td><td>VISMA_HVRT</td><td>The inverter is operating in VISMA_Q/VISMA_IQ control mode and has switched control to HVRT.</td></tr>
</table></td></tr>
</table>

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5.1.8. Subgroup SV1.8: Limits

Note: this subgroup is displayed if [G1.3-User ≥ Qualified customer].

SV1.8.1: Deratings

SCREEN	UNITS	DESCRIPTION
SV1.8.1.1-Derating global = 0.0 %	%	Shows the limit of the most restrictive module maximum current between the Vdc IGBTs temp and admission temp deratings.
SV1.8.1.2-Derating Vdc = 0.0 %	%	Shows the limit of the maximum module current due to overvoltage Vdc.
SV1.8.1.3-Derating Temp IGBTs = 0.0 %	%	Shows the limit of the maximum module current due to overheating of IGBTs.
SV1.8.1.4-Derating admiss T = 0.0 %	%	Shows the limit of the maximum module current due to admission temp.
SV1.8.1.5-Derating frequency = 0.0 Hz	Hz	Shows the limit of the switching frequency of the IGBTs due to overheating of the IGBTs.
SV1.8.1.6-Silent m limit = 0.0 %	%	Displays the calculated derating limit to be applied during silent mode. **Note**: this parameter is displayed if [G15.1.1-Enable silent mode] = Enabled.
SV1.8.1.7-Silent m feedback = 0.0 %	%	Displays the actual derating limit applied during silent mode. **Note**: this parameter is displayed if [G15.1.1-Enable silent mode] = Enabled.

SV1.8.2-Idc Limits

Note: this subgroup is displayed if [G1.3-User ≥ Power Electronics staff] and if [SG10.1.4-Inverter type = PCS/MPCS].

SCREEN	UNITS	DESCRIPTION
SV1.8.2.1-Busbar1 lim dischar = 0.00 %	%	Shows the load and discharge limits being applied on busbar 1. **Note:** These parameters are displayed if [SG10.1.14-Number of busbar ≥ 1].
SV1.8.2.2-Busbar1 lim charge = 0.00 %	%	
SV1.8.2.3-Busbar2 lim dischar = 0.00 %	%	Shows the load and discharge limits being applied on busbar 2. **Note:** These parameters are displayed if [SG10.1.14-Number of busbar ≥ 2].
SV1.8.2.4-Busbar2 lim charge = 0.00 %	%	
SV1.8.2.5-Busbar3 lim dischar = 0.00 %	%	Shows the load and discharge limits being applied on busbar 3. **Note:** These parameters are displayed if [SG10.1.14-Number of busbar ≥ 3].
SV1.8.2.6-Busbar3 lim charge = 0.00 %	%	
SV1.8.2.7-Busbar4 lim dischar = 0.00 %	%	Shows the load and discharge limits being applied on busbar 4. **Note:** These parameters are displayed if [SG10.1.14-Number of busbar ≥ 4].
SV1.8.2.8-Busbar4 lim charge = 0.00 %	%	

SV1.8.3: Self limitation

This subgroup shows information about voltage limitation.

Note: this subgroup is displayed if [G1.3-User] ≥ Qualified customer.

SCREEN	UNITS	DESCRIPTION
SV1.8.3.1-Vac overV enable =	%	Displays the grid voltage threshold for activating the reactive power limitation due to overvoltage. When the measured voltage exceeds the value specified in this parameter, the capacitive reactive power will be limited. **Note**: this parameter is displayed if [G3.1.5.1-Overvoltage margin] ≠ disabled. In versions prior to 5.2.0, this parameter was called [SV1.8.3.1-Overvoltage Vac lim].
SV1.8.3.1b-Vac overV. deact = 0.0 %	%	Displays the grid voltage threshold for deactivating the reactive power limitation due to overvoltage. When the measured voltage drops below the value defined in this parameter, if the delay ([G3.1.5.1b-OverV output delay]): - is enabled, the counter ([SV1.8.3.1c-OverV deac counter]) will start to deactivate the capacitive reactive power limitation. - is disabled, the limitation will be deactivated immediately. **Note**: this parameter is displayed if [G3.1.5.1-Overvoltage margin] ≠ disabled.
SV1.8.3.1c-OverV deac counter = 0 ms	ms	Displays the time during which the voltage conditions for deactivating the capacitive reactive power limitation, defined in [SV1.8.3.1b-Vac overV. deact], have been maintained. The counter is reset to 0 if the conditions are no longer met or if the limitation is not active. When the counter value reaches the delay configured in [G3.1.5.1b-OverV output delay], the limitation is deactivated. **Note:** this parameter is displayed if [G1.3-User] ≥ Power Electronics staff and [G3.1.5.1-Overvoltage margin] ≠ disabled.
SV1.8.3.2-Q capacitive limit = 0.0 %	%	Displays the capacitive power limit imposed by the inverter self-limitation in case of too high AC voltage measurement. The activation threshold applied for this limit can be found in parameter [SV1.8.3.1-Vac overV enable]. **Note**: this parameter is displayed if [G3.1.5.1-Overvoltage margin] ≠ disabled.
SV1.8.3.3-Vac underV enable = 0.0 %	%	Displays the grid voltage threshold used for activating the reactive power limitation due to undervoltage. When the measured voltage is below the value specified in this parameter, the inductive reactive power will be limited. **Note**: this parameter is displayed if [G3.1.5.2-Undervoltage margin] ≠ disabled. In versions prior to 5.2.0, this parameter was called [SV1.8.3.3-Undervoltage Vac limit].
SV1.8.3.3b-Vac underV. deact = 0.0 %	%	Displays the grid voltage threshold for deactivating the reactive power limitation due to undervoltage. When the measured voltage rises above the value defined in this parameter, if the delay ([G3.1.5.2b-UnderV output delay]): - is enabled, the counter ([SV1.8.3.3c-SubV deact counter]) will start to deactivate the inductive reactive power limitation. - is disabled, the limitation will be deactivated immediately. **Note**: this parameter is displayed if [G3.1.5.2-Undervoltage margin] ≠ disabled.
SV1.8.3.3c-SubV deact counter = 0 ms	ms	Displays the time during which the voltage conditions for deactivating the inductive reactive power limitation, defined in [SV1.8.3.3b-Vac underV. deact], have been maintained. The counter is reset to 0 if the conditions are no longer met or if the limitation is not active. When the counter value reaches the delay configured in [G3.1.5.2b-UnderV output delay], the limitation is deactivated. **Note:** this parameter is displayed if [G1.3-User] ≥ Power Electronics staff and [G3.1.5.2-Undervoltage margin] ≠ disabled.
SV1.8.3.4-Q inductiva limit = 0.0 %	%	Displays the inductive power limit imposed by the inverter's self-limitation in case of too low AC voltage measurement. The activation threshold applied for this limit can be found in parameter [SV1.8.3.3-Vac underV enable]. **Note**: this parameter is displayed if [G3.1.5.2-Undervoltage margin] ≠ disabled.

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5.1.9. Subgroup SV1.9: Available power

This subgroup shows the apparent, active and reactive power of the inverter considering the deratings that are running at any time, as well as the available power stages (module FPGAs) both in the inverter and in the busbar.

The available power is calculated as the value of the most restrictive current limit multiplied by the voltage and displayed as a percentage. This subgroup considers the available modules and busbars, as well as the power limits in operation (deratings, DU limit, SVR, global limits, module limits, etc.). It does not take into account the power reduction algorithms in operation (PV, Pf, etc).

SV1.9.1-Equipment

SCREEN	UNITS	DESCRIPTION
SV1.9.1.1-P discharge = 0.00 %	%	Shows the available active discharge power calculated in the DSP. It is displayed with positive sign.
SV1.9.1.2-P charge = 0.00 %	%	Shows the active power at load available calculated in the DSP. It is displayed with a negative sign.
SV1.9.1.3-Q = 0.00 %	%	Shows the available reactive power calculated in the DSP. It is displayed in absolute value.
SV1.9.1.4-S = 0.00 %	%	Shows the available apparent power calculated in the DSP, between 0 and the apparent power limit set in [G3.1.4-S Limit].

SV1.9.2- Busbars

Note: Parameters in this subgroup are displayed if [SV1.3.1.1-Inverter type = PCS/MPCS] and [SG10.1.14- Number of busbar ≥ n], being 'n' the number of the busbar.

SCREEN	UNITS	DESCRIPTION
SV1.9.2.1-P discharge emb n1 = 0.00 %	%	Shows the active power in discharge available calculated in the DSP at busbar 1. It is displayed with positive sign.
SV1.9.2.2-P charge busbar n1 = 0.00 %	%	Shows the active power at load available calculated in the DSP at busbar 1. It is displayed with a negative sign.
SV1.9.2.3-Q busbar n1 = 0.00 %	%	Shows the available reactive power calculated in the DSP at busbar 1. It is displayed in absolute value.
SV1.9.2.4-P discharge emb n2 = 0.00 %	%	Shows the active power in discharge available calculated in the DSP at busbar 2. It is displayed with positive sign.
SV1.9.2.5-P charge busbar n2 = 0.00 %	%	Shows the active power at load available calculated in the DSP at busbar 2. It is displayed with a negative sign.
SV1.9.2.6-Q busbar n2 = 0.00 %	%	Shows the available reactive power calculated in the DSP at busbar 2. It is displayed in absolute value.
SV1.9.2.7-P discharge emb n3 = 0.00 %	%	Shows the active power in discharge available calculated in the DSP at busbar 3. It is displayed with positive sign.
SV1.9.2.8-P charge busbar n3 = 0.00 %	%	Shows the active power at load available calculated in the DSP at busbar 3. It is displayed with a negative sign.
SV1.9.2.9-Q busbar n3 = 0.00 %	%	Shows the available reactive power calculated in the DSP at busbar 3. It is displayed in absolute value.
SV1.9.2.10-P discharge emb n4 = 0.00 %	%	Shows the active power in discharge available calculated in the DSP at busbar 4. It is displayed with positive sign.
SV1.9.2.11-P charge bus n4 = 0.00 %	%	Shows the active power at load available calculated in the DSP at busbar 4. It is displayed with a negative sign.
SV1.9.2.12-Q busbar n4 = 0.00 %	%	Shows the available reactive power calculated in the DSP at busbar 4. It is displayed in absolute value.

5.1.10. Subgroup SV1.10: C LC Filter

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV1.10.1-Total C = 0 µF	µF	Shows the filter LC total capacitance measured in run mode. This mode is shown in parameter [SV12.x.1-Status], being 'x' the number of the module.
SV1.10.1b-C filter autodetec = 0.00%	%	Shows, as percentage, the value of the estimated capacity based on the theoretical capacity as configured in SG10.5.0
SV1.10.2-Bank1 C = 0 µF	µF	Shows the filter LC bank1 capacitance measured in AC auto diagnosis status. The status of the inverter is shown in parameter [SV1.1.4-Current status]. **Note:** This parameter is displayed if [SG10.5.5-Benchs Number C ≥ 1].
SV1.10.3-Bank2 C = 0 µF	µF	Shows the filter LC bank2 capacitance measured in AC auto diagnosis status. The status of the inverter is shown in parameter [SV1.1.4-Current status]. **Note:** This parameter is displayed if [SG10.5.5-Benchs Number C ≥ 2].
SV1.10.4-Bank3 C = 0 µF	µF	Shows the filter LC bank3 capacitance measured in AC auto diagnosis status. The status of the inverter is shown in parameter [SV1.1.4-Current status]. **Note:** This parameter is displayed if [SG10.5.5-Benchs Number C ≥ 3].
SV1.10.5-Bank4 C = 0 µF	µF	Shows the filter LC bank4 capacitance measured in AC auto diagnosis status. The status of the inverter is shown in parameter [SV1.1.4-Current status]. **Note:** This parameter is displayed if [SG10.5.5-Benchs Number C ≥ 4].
SV1.10.6-HW filter version = HW version 1.3	NONE	Shows the hardware version of the C filter that has been detected.

5.1.11. Subgroup SV1.11: MPCS Busbar Setpoint

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV1.11.1-Busbar1 P ref = 0.00 %	%	Shows the MPCS busbar 1 active power setpoint.
SV1.11.2-Busbar2 P ref = 0.00 %	%	Shows the MPCS busbar 2 active power setpoint. **Note:** This parameter is displayed if [SV1.3.1.1-Inverter type = PCS/MPCS] and [SG10.1.14-Number of busbar ≥ 2].
SV1.11.3-Busbar3 P ref = 0.00 %	%	Shows the MPCS busbar 3 active power setpoint. **Note:** This parameter is displayed if [SV1.3.1.1-Inverter type = PCS/MPCS] and [SG10.1.14-Number of busbar ≥ 3].
SV1.11.4-Busbar4 P ref = 0.00 %	%	Shows the MPCS busbar 4 active power setpoint. **Note:** This parameter is displayed if [SV1.3.1.1-Inverter type = PCS/MPCS] and [SG10.1.14-Number of busbar ≥ 4].
SV1.11.5-Busbars Q ref = 0.00 %	%	Shows the MPCS busbars reactive power setpoint.
SV1.11.6-sharing P ref = 0.00 %	%	Shows the sharing of the MPCS active power (P) setpoint among the different DC busbars. **Note**: this parameter is shown if [SG10.1.19-Visma enable] = Enabled.
SV1.11.7-sharing Q ref = 0.00 %	%	Shows the sharing of the MPCS reactive power (Q) setpoint among the different DC busbars. **Note**: this parameter is shown if [SG10.1.19-Visma enable] = Enabled.

EN

5.1.12. Subgroup SV1.12: Instant power

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer].

SCREEN	UNITS	DESCRIPTION
SV1.12.1-Pinst = 0 kW	kW	Shows the instantaneous active power at the output of the inverter in AC. For HEM / PCSM inverters correspond to the high voltage, for HEMK / PCSK corresponds to low voltage.
SV1.12.2-Qinst = 0 kVA	kVA	Shows the instantaneous reactive power at the output of the inverter in AC. For HEM / PCSM inverters it corresponds to the high voltage, for HEMK / PCSK inverters it corresponds to low voltage.

5.1.13. Subgroup SV1.13: Start Delay

This subgroup shows the delays on start.

SCREEN	UNITS	DESCRIPTION
SV1.13.1-Start DLY Grid Cond = 0 s	s	Shows the start delay of the inverter due to grid conditions.
SV1.13.2-Start Delay = 0 s	s	Shows the start delay of the inverter. This delay occurs after the delay due to grid conditions which is showed in [SV1.13.1-Start DLY Grid Cond].
SV1.13.3-Start DLY Seq Req =0 s	s	Shows the start delay of the inverter for the sequential reconnection.

5.1.14. Subgroup SV1.14: Stop Cond.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV1.14.1-Applied Pmin = 0.00 %	%	Minimum power value applied in the minimum P-stop condition.

5.1.15. Subgroup SV1.15: Fans setpoint

This subgroup allows to display the assigned and corrected speed setpoints for each of the fans installed in the inverter. Refer to section "G11: Fans" for more information on fan configuration.

Note: this subgroup is displayed if [G1.3-User] ≥ Qualified customer.

SCREEN	UNITS	DESCRIPTION
SV1.15.1-DU 1 assigned = 0 %	%	Display of the setpoint assigned to fan DU 1. This setpoint does not take into account the correction when the fan is Delta.
SV1.15.1a-DU 1 corrected = 0 %	%	Display of the corrected setpoint applied to the DU 1 fan when the fan is Delta. **Note**: this parameter is displayed if [G11.2.1-DU Fan 1 Model] = Delta.
SV1.15.2-DU 2 assigned= 0 %	%	Display of the setpoint assigned to the fan DU 2. This setpoint does not take into account the correction when the fan is Delta.
SV1.15.2a-DU 2 corrected= 0 %	%	Display of the corrected setpoint applied to the fan DU 2 when the fan is Delta.. **Note**: this parameter is displayed if [G11.2.2-DU Fan 2 Model] = Delta.
SV1.15.3-Module 1 assigned = 0 %	%	Display of the setpoint assigned to the fan of module 1.

SCREEN	UNITS	DESCRIPTION
SV1.15.4-Module 2 assigned = 0 %	%	Display of the setpoint assigned to the fan of module 2.
SV1.15.5-Module 3 assigned = 0 %	%	Display of the setpoint assigned to the fan of module 3.
SV1.15.6-Module 4 assigned = 0 %	%	Display of the setpoint assigned to the fan of module 4.
SV1.15.7-LC filter assigned = 0 %	%	Display of the setpoint assigned to the fan of filter LC.
SV1.15.8-MV 1 assigned = 0 %	%	Display of the setpoint assigned to the fan MT1. This setpoint does not take into account the correction when the fan is Delta. Only available for HEM / PCSM inverter.
SV1.15.8a-MV 1 corrected = 0 %	%	Display of the corrected setpoint applied to the fan MT1 when the fan is Delta. Only available for HEM / PCSM inverter. **Note**: this parameter is displayed if [G11.2.8-MT Fan 1 Model] = Delta and [SV11.8.16-Fan delta model] = L9210.
SV1.15.9-MV 2 assigned = 0 %	%	Display of the setpoint assigned to the fan MT2. This setpoint does not take into account the correction when the fan is Delta. Only available for HEM / PCSM inverter.
SV1.15.9a-MV 2 corrected = 0 %	%	Display of the corrected setpoint applied to the fan MT2 when the fan is Delta. Only available for HEM / PCSM inverter. **Note**: this parameter is displayed if [G11.2.9-MT Fan 2 Model] = Delta and [SV11.9.16-Fan delta model] = L9210.
SV1.15.10-MV 3 assigned = 0 %	%	Display of the setpoint assigned applied to the fan MT3. This setpoint does not take into account the correction when the fan is Delta. Only available for HEM / PCSM inverter.
SV1.15.10a-MV 3 corrected = 0 %	%	Display of the corrected setpoint applied to the fan MT3 when the fan is Delta. Only available for HEM / PCSM inverter. **Note**: this parameter is displayed if [G11.2.10-MT Fan 3 Model] = Delta and [SV11.10.16-Fan delta model] = L9210.
SV1.15.11-MV 4 assigned = 0 %	%	Display of the corrected setpoint applied to the fan MT4. This setpoint does not take into account the correction when the fan is Delta. Only available for HEM / PCSM inverter.
SV1.15.11a-MV 4 corrected = 0 %	%	Display of the corrected setpoint applied to the fan MT4 when the fan is Delta. Only available for HEM / PCSM inverter. **Note**: this parameter is displayed if [G11.2.11-MT Fan 4 Model] = Delta and [SV11.11.16-Fan delta model] = L9210.

5.1.16. Subgroup SV1.16: AC Circuit Breaker

Note: this subgroup is displayed if [G1.3-User] ≥ Admin.

SCREEN	UNITS	DESCRIPTION
SV1.16.1-Cir Break res count =	s	Displays the time during which the voltage and frequency network conditions have met the constraints specified in parameter group [G2.3: Reconnection] after a grid failure that has caused the AC circuit breaker to open. Once the value specified in parameter [G2.3.6-C.Breaker reset delay] is reached, it recloses the AC circuit breaker. **Note**: this parameter is displayed if any of the AC circuit breaker opening parameters is set to Enabled ([G5.1.1.13-Open AC Circ. Break], [G5.1.2.13-Open AC Circ. Break], [G5.1.3.12-Open AC Circ. Break] o [G5.1.4.12-Open AC Circ. Break]).

5.1.17. Subgroup SV1.17: Bus Plus temp

This group allows to display the temperatures associated with the Bus Plus measurement boards, as well as the status of communications with those boards.

Note: this subgroup is displayed if [G1.3-User] ≥ Qualified customer and [SG10.7.8-Temp board] = Enabled.

Subgroup SV1.17.1: Temp board 1

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.17.1.0-ID board = Off</td><td>NONE</td><td>Allows deactivating temperature measurement board 1 or configuring its ID:
<table><tr><th>BIT</th><th>STATUS</th></tr><tr><td>99</td><td>Off.</td></tr><tr><td>100 to 115</td><td>ID board.</td></tr></table>Note: the assigned ID must match the position of the DIP switches on temperature measurement board 1. ID configuration is performed through the position of the switches on the board itself.</td></tr>
<tr><td>SV1.17.1.1-Comms status = Disabled</td><td>NONE</td><td>Displays the communication status with temperature measurement board 1.
<table><tr><th>BIT</th><th>STATUS</th></tr><tr><td>0</td><td>Disabled.</td></tr><tr><td>1</td><td>OK.</td></tr><tr><td>2</td><td>Error.</td></tr></table>Note: this parameter is shown if [SV1.17.1.0-ID board] ≥ 100.</td></tr>
<tr><td>SV1.17.1.2-NTC1 Temperature = 0 ºC</td><td>°C</td><td rowspan="9">These parameters show the temperature readings for each of the plates on DC switch disconnectors No. 1 and No. 2:
<table><tr><th>SENSOR</th><th>DC SWITCH DISCONNECTOR NO.</th><th>PLATE NO.</th></tr><tr><td>NTC1</td><td rowspan="4">1</td><td>1</td></tr><tr><td>NTC2</td><td>2</td></tr><tr><td>NTC3</td><td>3</td></tr><tr><td>NTC4</td><td>4</td></tr><tr><td>NTC5</td><td rowspan="4">2</td><td>1</td></tr><tr><td>NTC6</td><td>2</td></tr><tr><td>NTC7</td><td>3</td></tr><tr><td>NTC8</td><td>4</td></tr></table>They also indicate whether there are any issues with the NTC temperature sensors:
<table><tr><th>BIT</th><th>STATUS</th></tr><tr><td>151</td><td>NTC not connectected.</td></tr><tr><td>152</td><td>NTC short circuit.</td></tr></table>Note: this parameter is shown if [SV1.17.1.0-ID board] ≥ 100.</td></tr>
<tr><td>SV1.17.1.3-NTC2 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.1.4-NTC3 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.1.5-NTC4 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.1.6-NTC5 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.1.7-NTC6 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.1.8-NTC7 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.1.9-NTC8 Temperature = 0 ºC</td><td>°C</td></tr>
</table>

Subgroup SV1.17.2: Temp board 2

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV1.17.2.0-ID board = Off</td><td>NONE</td><td>Allows deactivating temperature measurement board 2 or configuring its ID:
<table><tr><th>BIT</th><th>STATUS</th></tr><tr><td>99</td><td>Off.</td></tr><tr><td>100 to 115</td><td>ID board.</td></tr></table>Note: the assigned ID must match the position of the DIP switches on temperature measurement board 2. ID configuration is performed through the position of the switches on the board itself.</td></tr>
<tr><td>SV1.17.2.1-Comms status = Disabled</td><td>NONE</td><td>Displays the communication status with temperature measurement board 2.
<table><tr><th>BIT</th><th>STATUS</th></tr><tr><td>0</td><td>Disabled.</td></tr><tr><td>1</td><td>OK.</td></tr><tr><td>2</td><td>Error.</td></tr></table>Note: this parameter is shown if [SV1.17.2.0-ID board] ≥ 100.</td></tr>
<tr><td>SV1.17.2.2-NTC1 Temperature = 0 ºC</td><td>°C</td><td rowspan="9">These parameters show the temperature readings for each of the plates on DC switch disconnectors No. 3 and No. 4:
<table><tr><th>SENSOR</th><th>DC SWITCH DISCONNECTOR NO.</th><th>PLATE NO.</th></tr><tr><td>NTC1</td><td rowspan="4">3</td><td>1</td></tr><tr><td>NTC2</td><td>2</td></tr><tr><td>NTC3</td><td>3</td></tr><tr><td>NTC4</td><td>4</td></tr><tr><td>NTC5</td><td rowspan="4">4</td><td>1</td></tr><tr><td>NTC6</td><td>2</td></tr><tr><td>NTC7</td><td>3</td></tr><tr><td>NTC8</td><td>4</td></tr></table>They also indicate whether there are any issues with the NTC temperature sensors:
<table><tr><th>BIT</th><th>STATUS</th></tr><tr><td>151</td><td>NTC not connectected.</td></tr><tr><td>152</td><td>NTC short circuit.</td></tr></table>Note: this parameter is shown if [SV1.17.2.0-ID board] ≥ 100.</td></tr>
<tr><td>SV1.17.2.3-NTC2 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.2.4-NTC3 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.2.5-NTC4 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.2.6-NTC5 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.2.7-NTC6 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.2.8-NTC7 Temperature = 0 ºC</td><td>°C</td></tr>
<tr><td>SV1.17.2.9-NTC8 Temperature = 0 ºC</td><td>°C</td></tr>
</table>

5.1.18. Subgroup SV1.18: Discharge Conditions

This subgroup shows whether the conditions for transitioning to the Discharge status have been met and the time remaining until the discharge actually begins.

SCREEN	UNITS	DESCRIPTION
SV1.18.1-Discharge Conditions= No	NONE	Indicates whether the necessary conditions for the inverter to transition to the Discharge status are met. Transitions to the Discharge status can only occur from the Init or Off statuses. This parameter is deactivated once the discharge is complete.
SV1.18.2-Remaining Delay = 10 s	s	Shows a 10-second countdown from the moment the conditions for transitioning to the Discharge status are met until the discharge actually begins.

5.2. Group SV2: DC source

This group shows the information corresponding to the electrical measurements (power, voltage and current) of the inverter.

Note: This group is displayed if [SG10.1.14-Number of busbar ≥ n] being 'n' the number of the busbar.

SCREEN	UNITS	DESCRIPTION
SV2.1-Input power 1 = 0 kW	kW	Shows the DC input power of the inverter busbar 1.
SV2.2-Input power 2 = 0 kW	kW	Shows the DC input power of the inverter busbar 2.
SV2.3-Input power 3 = 0 kW	kW	Shows the DC input power of the inverter busbar 3.
SV2.4-Input power 4 = 0 kW	kW	Shows the DC input power of the inverter busbar 4.
SV2.5-I DC 1 = 0 A	A	Shows the DC current of the inverter busbar 1.
SV2.6-I DC 2 = 0 A	A	Shows the DC current of the inverter busbar 2.
SV2.7-I DC 3 = 0 A	A	Shows the DC current of the inverter busbar 3.
SV2.8-I DC 4 = 0 A	A	Shows the DC current of the inverter busbar 4.
SV2.9-Dc Voltage extern 1 = 0.0 V	V	These parameters show the external DC bus voltage. This is, the voltage measured on the external PV/BESS connection busbar (upstream of the DC switch disconnector). Depending on the inverter, more or fewer measurements will be available: **HEM/K:** Only DC voltage bus 1 **PCSM/K:** - Frame 4 – 1 busbar: Only Dc Voltage extern 1. - Frame 4 – 2 busbars: Dc Voltage extern 1 = Busbar 1; Dc Voltage extern 3 = Busbar 2. - Frame 4 – 4 busbars: One DC voltage per busbar. - Frame 3 – 1 busbar: Only Dc Voltage extern 1. - Frame 3 – 3 busbars: One DC voltage per busbar. - Frame 2 – 1 busbar - Only Dc Voltage extern 1. - Frame 2 – 2 busbars: one DC voltage per busbar.
SV2.10-Dc Voltage extern 2 = 0.0 V	V	
SV2.11-Dc Voltage extern 3 = 0.0 V	V	
SV2.12-Dc Voltage extern 4 = 0.0 V	V	
SV2.13-DC volt last start 1 = 0.0 V	V	Displays the last voltage of the PV field before starting.
SV2.14-DC volt last start 2 = 0.0 V		
SV2.15-DC volt last start 3 = 0.0 V		
SV2.16-DC volt last start 4 = 0.0 V		
SV2.17-PV Power = 0 kW	kW	Displays the power at the DC cabinet calculated with the current measured by the DU shunt and the field voltage. **Note:** This parameter is displayed if [SG10.1.4-Inverter type ≠ PCS/MPCS].
SV2.18-PV DC Current = 0 A	A	Displays the sum of all the current measured by the DC cabinet shunt and the field voltage. The sum is as measured by the shunt it is not protected against negative measurement in order to detect field current. **Note:** This parameter is displayed if [SG10.1.4-Inverter type ≠ PCS/MPCS].

5.3. Group SV3: Grid visualization

This group allows the visualization of voltage and current per phase, as well as the power injected into the network, power factor and network frequency.

5.3.1. Subgroup SV3.1: Grid MV

Note: This subgroup is only shown for [G3.1-User] ≥ Qualified customer and only applies to HEM/PCSM and HEMK/PCSK with MV Skid configured in [SG10.1.3-Product type] = HEM or [SG10.1.3-Product type] = HEMK and [SG10.7.4-Volt Meas MV Skid] = Yes.

SCREEN	UNITS	DESCRIPTION
SV3.1.1-Grid voltage RS MV = 0.0 kV	kV	Shows the medium voltage phase voltage (RS).
SV3.1.2-Grid voltage ST MV = 0.0 kV	kV	Shows the medium voltage phase voltage (ST).
SV3.1.3-Grid voltage TR MV = 0.0 kV	kV	Shows the medium voltage phase voltage (TR).
SV3.1.4-Grid current I1 MV = 0.0 A	A	Shows the medium voltage phase current (I1).
SV3.1.5-Grid current I2 MV = 0.0 A	A	Shows the medium voltage phase current (I2).
SV3.1.6-Grid current I3 MV = 0.0 A	A	Shows the medium voltage phase current (I3).
SV3.1.7-P MV = 0 kW	kW	Shows the medium voltage active power.
SV3.1.8-Q MV = 0 kVAr	kVAr	Shows the medium voltage reactive power.
SV3.1.9-S MV = 0 kVA	kVA	Shows the medium voltage apparent power.
SV3.1.10-Cos phi MV = 0.000	NONE	Shows the medium voltage cos phi.

5.3.2. Subgroup SV3.2: Grid LV

SCREEN	UNITS	DESCRIPTION
SV3.2.1-Grid voltage RS LV = 0.0 V	V	Shows the low voltage phase voltage (RS).
SV3.2.2-Grid voltage ST LV = 0.0 V	V	Shows the low voltage phase voltage (ST).
SV3.2.3-Grid voltage TR LV = 0.0 V	V	Shows the low voltage phase voltage (TR).
SV3.2.4-Grid current I1 LV = 0.0 A	A	Shows the low voltage phase current R (I1).
SV3.2.5-Grid current I2 LV = 0.0 A	A	Shows the low voltage phase current S (I2).
SV3.2.6-Grid current I3 LV = 0.0 A	A	Shows the low voltage phase current T (I3).
SV3.2.7-P LV = 0 kW	kW	Shows the low voltage active power.
SV3.2.8-Q LV = 0 kVAr	kVAr	Shows the low voltage reactive power.
SV3.2.9-S LV = 0 kVA	kVA	Shows the low voltage apparent power.
SV3.2.10-Cos phi LV = 0.000	NONE	Shows the low voltage cosine of phi.
SV3.2.11-Grid sequence = Positive sequence	NONE	Indicates the sign (positive or negative) of the grid sequence. BIT 0: Positive sequence. BIT 1: Negative sequence.
SV3.2.12-Grid freq = 0.000 Hz	Hz	Shows the absolute value of the grid frequency.
SV3.2.13-Vac SD = 0.00 %	%	Shows the direct voltage sequence measurement, as a percentage of the product's rated voltage.
SV3.2.14-Vac SI = 0.00 %	%	Shows the inverse voltage sequence measurement, as a percentage of the product's rated voltage.

5.3.3. Subgroup SV3.3: C Filter

This subgroup shows information about the LCL filter measures.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV3.3.1-RS LV Grid Voltage = 0.0 V	V	Shows the low voltage phase voltage (RS).
SV3.3.2-ST LV Grid Voltage = 0.0 V	V	Shows the low voltage phase voltage (ST).
SV3.3.3-TR LV Grid Voltage = 0.0 V	V	Shows the low voltage phase voltage (TR).
SV3.3.4-C Ir Current = 0.00 A	A	Shows the RMS (Root Mean Square) Ir current of the filter.
SV3.3.5-C Is Current = 0.00 A	A	Shows the RMS (Root Mean Square) Is current of the filter.
SV3.3.6-C It Current = 0.00 A	A	Shows the RMS (Root Mean Square) It current of the filter.
SV3.3.7-Fund Ir Current = 0.00 A	A	Shows the RMS value of the Ir fundamental component of the filter.
SV3.3.8-Fund Is Current = 0.00 A	A	Shows the RMS value of the Is fundamental component of the filter.
SV3.3.9-Fund It Current = 0.00 A	A	Shows the RMS value of the It fundamental component of the filter.

5.3.4. Subgroup SV3.4: Presen. of Voltage

This subgroup shows the presence of voltage on the DSP measurement channels.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV3.4.1-Vrs LV = 0	NONE	Shows the presence of voltage on the RS measurement channel of the DSP.
SV3.4.2-Vst LV = 0	NONE	Shows the presence of voltage on the ST measurement channel of the DSP.
SV3.4.3-Vtr LV = 0	NONE	Shows the presence of voltage on the TR measurement channel of the DSP.
SV3.4.4-Vr HV = 0	NONE	Shows the presence of voltage on the R measurement channel of the DSP.
SV3.4.5-Vs HV = 0	NONE	Shows the presence of voltage on the S measurement channel of the DSP.
SV3.4.6-Vt HV = 0	NONE	Shows the presence of voltage on the T measurement channel of the DSP.

5.4. Group SV4: Algorithms

This group shows information about the algorithms run by the inverter.

Note: This group is displayed if [G1.3-User ≥ Qualified customer].

5.4.1. Subgroup SV4.1: MPPt

This algorithm searches for the maximum power point of the photovoltaic arrays connected to the inverter. These parameters only apply to HEM inverters.

Note: This subgroup is displayed if [G1.3-User ≥Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV4.1.1-Status = Off	NONE	Shows the result of the MPPT algorithm (disable, starting or activated). OPTION / STATUS / DESCRIPTION 0 / Off / Algorithm disabled or not running 1 / Init / Algorithm is starting, it is bringing the field voltage to the initial working point. 2 / ON / Algorithm activated, once the initial point (Init) is reached, turn ON and MPPT starts.
SV4.1.2-Irradiation = Still	NONE	Shows the MPPt algorithm irradiance. OPTION / STATUS / DESCRIPTION 0 / Still / Indicates constant irradiation. 1 / Rise / Indicates increasing irradiation. 2 / Fall / Indicates decreasing irradiation.
SV4.1.3-Delta irradiation = 0.0 %	%	Shows the MPPT algorithm irradiance delta.
SV4.1.4-Current action = 0.0 V	V	Shows the current action that is carrying out the algorithm.
SV4.1.5-Current reference = 0.0 V	V	Shows the current reference that is using the algorithm.

5.4.2. Subgroup SV4.2: Curve Q(P)

This subgroup shows the Q(P) algorithm status configured in "G4.12: Q(P) Curve" group.

SCREEN	UNITS	DESCRIPTION
SV4.2.1-Q(P) status = Disabled	NONE	Shows the zone of the Q(P) Curve algorithm that is operating. OPTION / DESCRIPTION / OPTION / DESCRIPTION 0 / Disabled. / 5 / Zone 0 charge. 1 / Zone 0 discharge. / 6 / Zone 1 charge. 2 / Zone 1 discharge. / 7 / Zone 2 charge. 3 / Zone 2 discharge. / 8 / Zone 3 charge. 4 / Zone 3 discharge. / - / -
SV4.2.2-Q(P) P grid = 0.00 %	%	Corresponds to the power used in algorithm Q(P) to interpolate.
SV4.2.3-Q(P) Q ref = 0.00 %	%	Shows the algorithm output power.

EN

5.4.3. Subgroup SV4.3: Curve Q(V)

This subgroup shows the Q(V) algorithm configured in subgroup G4.5: Curve Q(V).

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.3.1-Q(V) status = Off</td><td>NONE</td><td>Shows the result of the Curve Q(V) algorithm.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>FUNCTION</th></tr>
<tr><td>0</td><td>Off</td><td>The algorithm is disabled. To enable it, activate in [G4.5.1- Enable].</td></tr>
<tr><td>1</td><td>Dead Band</td><td>The algorithm is enabled but not activated.</td></tr>
<tr><td>2</td><td>Q ind. Ramp up</td><td>The algorithm is active by increasing the inductive reactive power.</td></tr>
<tr><td>3</td><td>Q ind. Ramp down</td><td>The algorithm is active by decreasing the inductive reactive power.</td></tr>
<tr><td>4</td><td>Q cap. Ramp up</td><td>The algorithm is active by increasing the capacitive reactive power.</td></tr>
<tr><td>5</td><td>Q cap. Ramp down</td><td>The algorithm is active by decreasing the capacitive reactive power.</td></tr>
</table></td></tr>
<tr><td>SV4.3.2-Q(V) reference = 0 %</td><td>%</td><td>Shows the result of the Q calculation.</td></tr>
<tr><td>SV4.3.3-Vac = 0.00 %</td><td>%</td><td>Shows the average Vac measured.</td></tr>
</table>

5.4.4. Subgroup SV4.4: Anti-Island

These parameters display information related to the anti-island algorithm.

Note: These parameters are displayed if [G1.3-User ≥ Power Electronics staff] and [G5.3.1-Mode ≥ Active].

SCREEN	UNITS	DESCRIPTION
SV4.4.1-Dev. F passive (Hz/s) = 0.0	Hz/s	Displays the frequency deviation, in Hz/s, for passive anti-island algorithm.
SV4.4.2-Dev. F active = 0.0 Hz	Hz	Displays the frequency deviation, in Hz, for active anti-island algorithm.

5.4.5. Subgroup SV4.5: P(f)

This subgroup allows the visualization of the P(f) algorithm configured in G4.7: P(f) Curve.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.5.1-P(f) status = Off</td><td>NONE</td><td>Shows the P(f) algorithm status.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>FUNCTION</th></tr>
<tr><td>0</td><td>Deactivated</td><td>Algorithm deactivated. Enabled in parameter [G4.7.1- Reference Power].</td></tr>
<tr><td>1</td><td>Dead Band</td><td>Algorithm enabled, but the conditions for execution have not been provided. It is in the dead band</td></tr>
<tr><td>2</td><td>High Frequency</td><td>High frequency algorithm active, active power reduction or consumption in progress.</td></tr>
<tr><td>3</td><td>High Frequency Hyst</td><td>High frequency algorithm active, active power increase or injection in progress.</td></tr>
<tr><td>4</td><td>Low Frequency</td><td>Low frequency algorithm active, active power reduction or consumption in progress.</td></tr>
<tr><td>5</td><td>Low Frequency Hyst</td><td>Low frequency algorithm active, active power boosting or injection in progress.</td></tr>
</table></td></tr>
<tr><td>SV4.5.2-P(f) frozen Pref = 0.00 %</td><td>%</td><td>Shows the frozen power reference.</td></tr>
<tr><td>SV4.5.3-P(f) calc P limit = 0.00 %</td><td>%</td><td>Shows the calculated P limit.</td></tr>
<tr><td>SV4.5.4-P(f) Calc. P. Ref = 0.00 %</td><td>%</td><td>Shows the calculated active power reference.</td></tr>
<tr><td>SV4.5.5-P(f) Input Frequency = 0.00 %</td><td>%</td><td>Shows the input frequency.</td></tr>
</table>

5.4.6. Subgroup SV4.6: Batteries V Intern

This subgroup shows the internal busbar voltages estimation. Applies to **PCS/MPCS** inverters.

Note: These parameters are displayed if [SV1.3.1.1-Inverter type = PCS/MPCS] and [SG10.1.14-Number of busbar ≥ n] being 'n' the number of the busbar.

SCREEN	UNITS	DESCRIPTION
SV4.6.1-VDC Battery 1 = 0 V	V	Shows the internal Vdc of battery busbar 1.
SV4.6.2-VDC Battery 2 = 0 V	V	Shows the internal Vdc of battery busbar 2.
SV4.6.3-VDC Battery 3 = 0 V	V	Shows the internal Vdc of battery busbar 3.
SV4.6.4-VDC Battery 4 = 0 V	V	Shows the internal Vdc of battery busbar 4.

5.4.7. Subgroup SV4.7: Internal Control Vars

This subgroup shows internal bits for debug.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV4.7.1-Debug Logic 1 = 0	NONE	Shows the internal debug logic 1.
SV4.7.2-Debug Logic 2 = 0	NONE	Shows the internal debug logic 2.
SV4.7.3-Debug Logic 3 = 0	NONE	Shows the internal debug logic 3.
SV4.7.4-Debug Logic 4 = 0	NONE	Shows the internal debug logic 4.

5.4.8. Subgroup SV4.10: P(V) Curve

This subgroup allows the visualization of the P(V) algorithm configured in G4.6: P(V) Curve.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.10.1-Status = Disabled</td><td>NONE</td><td>Shows P(V) status according to enumeration.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>FUNCTION</th></tr>
<tr><td>0</td><td>Disabled</td><td>Algorithm disabled. To enable, go to parameter [G4.6.1 Enable].</td></tr>
<tr><td>1</td><td>Off</td><td>The conditions for running the algorithm have not been met. It is in the dead zone.</td></tr>
<tr><td>2</td><td>Limiting P</td><td>The conditions for running the algorithm have been given. The inverter is limiting the active power as a function of voltage.</td></tr>
<tr><td>3</td><td>Recovering</td><td>The power recovery stage is being executed. The algorithm is exiting.</td></tr>
<tr><td>4</td><td>Finishing</td><td>5 ms state with no utility. Each exit from the algorithm is occurring.</td></tr>
</table></td></tr>
<tr><td>SV4.10.2-P Limit = 0.0 %</td><td>%</td><td>Shows current P limit.</td></tr>
<tr><td>SV4.10.3-P Reference = 0.0 %</td><td>%</td><td>P reference in PCS.</td></tr>
<tr><td>SV4.10.4-Avg Vac = 0 V</td><td>V</td><td>Voltage average of three phase.</td></tr>
</table>

5.4.9. Subgroup SV4.11: SVR

This subgroup allows to visualize the status of the Suppression of Voltage Rise (SVR) algorithm. This algorithm limits an uncontrolled increase in voltage to the grid.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.11.1-SVR Status = Off</td><td>NONE</td><td>Shows SVR algorithm status.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Off</td><td>SVR algorithm not active.</td></tr>
<tr><td>1</td><td>Q injection</td><td>The threshold V1 (minimum grid voltage threshold at which the SVR algorithm is activated) has been exceeded. Inductive reactive power starts to be injected with the configured gradient.</td></tr>
<tr><td>2</td><td>P Reduction</td><td>Active power reduction to zero starts following the configured ramp. This phase starts when the threshold V2 (configured threshold referring to the grid voltage) has been exceeded.</td></tr>
<tr><td>3</td><td>P recovery</td><td>Active power recovery starts following the configured ramp. This phase starts when the voltage is below the configured threshold V2 minus the configured hysteresis V2. The configured cosine of phi is maintained.</td></tr>
<tr><td>4</td><td>Q recovery</td><td>Reactive power recovery starts following the configured ramp. This phase starts when the mains voltage is below the configured threshold V1 minus the configured hysteresis V1.</td></tr>
</table></td></tr>
<tr><td>SV4.11.2-P ref frozen = 0.0 %</td><td>%</td><td>Id value before SVR algorithm activation.</td></tr>
<tr><td>SV4.11.3-Q ref frozen = 0.0 %</td><td>%</td><td>Id value before SVR algorithm activation.</td></tr>
<tr><td>SV4.11.4-P lim calculated = 0.0 %</td><td>%</td><td>P limit calculated by SVR algorithm.</td></tr>
<tr><td>SV4.11.5-Q lim calculated = 0.0 %</td><td>%</td><td>Signed Q limit calculated by SVR algorithm.</td></tr>
<tr><td>SV4.11.6-Vac average measured = 0 V</td><td>V</td><td>Vac monitored by SVR algorithm.</td></tr>
</table>

EN

5.4.10. Subgroup SV4.12: DC Autodiagnostic

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.12.1-Status DC Autodiag = Init</td><td>NONE</td><td>Indicates the internal states of the DC diagnosis.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Init</td><td>Corresponds to the variable initialization and first state of the Finite State Machine (FSM).</td></tr>
<tr><td>1</td><td>Soft Charge Modules</td><td>A soft charge is carried out separately for each module up to 300 V. Once the internal bus has been charged, the discharge command is given.</td></tr>
<tr><td>2</td><td>DC Isolation Measure</td><td>Checks that the insulation of the batteries has been measured at least once. All insulation measurements must be correct in order to be able to initiate a switchover.
This option only applies to PCS inverter.</td></tr>
<tr><td>3</td><td>Soft Charge and close DC</td><td>The bus of the inverter is loaded until it reaches the level of the external bus. Then, the DC disconnectors are closed.</td></tr>
<tr><td>4</td><td>Close AC breaker</td><td>In the case of HEMK or PCSK inverter connected to a MV Skid, it indicates that the AC circuit breaker is closed.
In HEM and MPCSM, closes AC directly.</td></tr>
<tr><td>5</td><td>Filter diagnosis</td><td>Indicates the filter diagnosis.</td></tr>
<tr><td>6</td><td>IMI pre AC</td><td>AC voltage is generated to measure insulation. If GFDI is not available, the parameters are set to generate an AC voltage to measure insulation in the following logic state: 60 Hz, positive sequence, open control loop and a 200 V reference.</td></tr>
<tr><td>7</td><td>DC isolation meas from AC</td><td>The generated AC voltage is maintained, and the insulation measurement is awaited.
A DC isolation measurement is made but measuring in AC, as it is easier to measure the PV field isolation with this methodology when the PV field voltage is not stable.</td></tr>
<tr><td>8</td><td>Complete</td><td>Indicates that the diagnosis has been completed.</td></tr>
<tr><td>9</td><td>Fault</td><td>This option does not apply to this product version.</td></tr>
</table></td></tr>
</table>

5.4.11. Subgroup SV4.13: AC Autodiagnostic

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.13.1-Status AC Autodiag = Init</td><td>NONE</td><td>Indicates the internal states of the AC diagnosis.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Init</td><td>Corresponds to the variable initialization and first state of the Finite State Machine (FSM).</td></tr>
<tr><td>1</td><td>AC Isolation Measure</td><td>Corresponds to the first isolation measurement that is carried out without switching.</td></tr>
<tr><td>2</td><td>Filter diagnosis</td><td>A passive diagnosis of the capacitance of each capacitor bank is carried out.</td></tr>
<tr><td>3</td><td>Soft charge</td><td>The active module is soft charged without closing. Disabling the closing of all modules prevents them from closing.
Once the module is in the standby state of its soft charge, the soft charge command is removed, and the control is configured to switch in open loop.</td></tr>
<tr><td>4</td><td>Module switching</td><td>Indicates that module switchover has been started.</td></tr>
<tr><td>5</td><td>Isolation measure</td><td>The insulation measurement is carried out with the IGBTs closed.</td></tr>
<tr><td>6</td><td>Stop Isolation measure</td><td>The module has stopped. If not finished, selects a new module and return to “Filter diagnosis”.</td></tr>
<tr><td>7</td><td>Complete</td><td>Indicates that the diagnosis has been completed.</td></tr>
<tr><td>8</td><td>Fault</td><td>Indicates the state reached due to failure at some specific point in the diagnostics, or if the main logic of the DSP has failed. All the variables of this logic are initialized and the order to stop will be given, transitioning to the "Init" state.</td></tr>
</table>
</td></tr>
</table>

EN

5.4.12. Subgroup SV4.14: Premagnetization

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff] and if [SG10.1.3 Product type is HEM] or [SG10.1.3 Product type is HEMK] and [SG10.7.3 MV Skid ≠ No].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.14.1-Premag status = Premag off</td><td>NONE</td><td>Shows the internal status of premagnetization.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Premag off</td><td>Premagnetization is not in progress. Initial or idle state. The system is idle and checks whether the conditions for starting are met.</td></tr>
<tr><td>1</td><td>Premag check cell status</td><td>The status of the MV switchgear is checked. The available modules are checked and selected to prepare the pre-magnetization process.</td></tr>
<tr><td>2</td><td>Premag check frequency MV</td><td>It ensures that the MV frequency is within the allowed ranges and the MV PLL is synchronized.</td></tr>
<tr><td>3</td><td>Premag check mod Vac MV</td><td>MV voltage magnitude and DC bus conditions are checked to ensure that they are within operating limits.</td></tr>
<tr><td>4</td><td>Premag close C filter</td><td>The capacitor filter is closed to prepare the system for voltage generation. The AC circuit breaker of the HEMK inverters with MV Skid is also closed.</td></tr>
<tr><td>5</td><td>Premag start modules</td><td>Premagnetization is starting. The selected modules are started and their correct operation is ensured before proceeding. In this state the Finite State Machine (FSM) is bifurcated depending on whether a premagnetization or a hard close is performed.</td></tr>
<tr><td>6</td><td>Premag black start</td><td>Initial AC voltage is generated for a black start, providing an initial reference.</td></tr>
<tr><td>7</td><td>Premag check Vac to close</td><td>The generated voltage is checked (magnitude, frequency, angle) and, if it is correct, the circuit breaker of the MV switchgear is closed.</td></tr>
<tr><td>8</td><td>Premag close MV</td><td>It is verified that the circuit breaker of the MV switchgear has been properly closed, the modules are stopped and the control is disabled.</td></tr>
<tr><td>9</td><td>Premag done</td><td>Premagnetization is finished. It is checked that the medium voltage does not deviate from the low voltage; if this is the case, an autocalibration of the MV measurement is triggered.</td></tr>
<tr><td>10</td><td>Premag fault</td><td>Premagnetization has failed. The system is prepared to retry pre-magnetization or stop.</td></tr>
<tr><td>11</td><td>Precharge</td><td>Precharging of the modules by AC before closing the circuit breaker of the MV switchgear to avoid performing the maneuver without internal voltage in the power stages.</td></tr>
<tr><td>12</td><td>Hard Close</td><td>The MV switchgear is closed directly without performing the standard premagnetization. Before closing, the inverter stops switching.</td></tr>
</table></td></tr>
<tr><td>SV4.14.2-Vac MV = 0.00 %</td><td>%</td><td>The medium voltage in the premagnetization is displayed.</td></tr>
<tr><td>SV4.14.3-Vac LV = 0.00 %</td><td>%</td><td>The low voltage in the premagnetization is displayed.</td></tr>
<tr><td>SV4.14.4-Freq MV = 0.00 Hz</td><td>Hz</td><td>The medium voltage frequency in the premagnetization is displayed.</td></tr>
<tr><td>SV4.14.5-Freq LV = 0.00 Hz</td><td>Hz</td><td>The low voltage frequency in the premagnetization is displayed.</td></tr>
</table>

5.4.13. Subgroup SV4.15: OVRT

This subgroup allows to visualize the OVRT algorithm status configured in G4.4: OVRT.

Note: this subgroup is displayed if [G1.3-User ≥ Power Electronics staff] and [G4.4.1.1-Enable ≠ Disabled].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.15.1-State = OFF</td><td>NONE</td><td>Shows the status of the OVRT algorithm.
<table>
<tr><th>OPTION</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>OFF</td><td>Algorithm is deactivated or not detected.</td></tr>
<tr><td>1</td><td>ON Derive</td><td>Algorithm is detected and the voltage differential (dV) is in direct or inverse sequence and activated.</td></tr>
<tr><td>2</td><td>ON</td><td>Algorithm is detected and activated.</td></tr>
<tr><td>3</td><td>Fault Avoid</td><td>Overcurrent detected and it is not OVRT.</td></tr>
<tr><td>4</td><td>Fast Current Limit Avoid</td><td>An overcurrent has been detected by the fast current limit.</td></tr>
<tr><td>5</td><td>Fault Avoid Out</td><td>Not used.</td></tr>
<tr><td>6</td><td>Momentary Cessation</td><td>The second level used in some standards has been reached.</td></tr>
<tr><td>7</td><td>Pre Out</td><td>Not used.</td></tr>
<tr><td>8</td><td>Out</td><td>Indicates the necessary conditions to end the execution of the OVRT algorithm.</td></tr>
<tr><td>9</td><td>Recovery</td><td>The dip is over, the recovery process continues.</td></tr>
<tr><td>10</td><td>End</td><td>Algorithm has finished.</td></tr>
</table></td></tr>
<tr><td>SV4.15.2-Number = 0</td><td>NONE</td><td>Shows the number of overvoltage conditions detected.</td></tr>
<tr><td>SV4.15.3-Depth DS Last = 0.00 %</td><td>%</td><td>Shows the last OVRT depth direct sequence.</td></tr>
<tr><td>SV4.15.4-Depth IS Last = 0.00 %</td><td>%</td><td>Shows the last OVRT depth inverse sequence.</td></tr>
<tr><td>SV4.15.5-Depth VSRT Max Last = 0.00 %</td><td>%</td><td>Shows the RMS of the maximum time composite voltage at the last event.</td></tr>
<tr><td>SV4.15.6-V Increment = 0.00 %</td><td>%</td><td>Shows the voltage OVRT increment.</td></tr>
<tr><td>SV4.15.7-Last Duration = 0 ms</td><td>ms</td><td>Shows the duration of the last OVRT.</td></tr>
<tr><td>SV4.15.8-V threshold active = 0.00 %</td><td>%</td><td>Shows the threshold voltage to activate OVRT algorithm.
Note: If [G4.4.1.2b-k V Threshold ≠ 0.000], take this into account for the threshold voltage.</td></tr>
</table>

EN

5.4.14. Subgroup SV4.16: LVRT

This subgroup allows to visualize the LVRT algorithm status configured in G4.3: LVRT.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff] and [G4.3.1.1-Enable ≠ Disabled].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.16.1-State = OFF</td><td>NONE</td><td>Shows the status of the LVRT algorithm.
<table>
<tr><th>OPT.</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>OFF</td><td>Algorithm is deactivated or not detected.</td></tr>
<tr><td>1</td><td>ON Derive</td><td>Algorithm is detected and the voltage differential (dV) is in direct or inverse sequence and activated.</td></tr>
<tr><td>2</td><td>ON</td><td>Algorithm is detected and activated.</td></tr>
<tr><td>3</td><td>Fault Avoid</td><td>Overcurrent detected and it is not LVRT.</td></tr>
<tr><td>4</td><td>Fast Current Limit Avoid</td><td>An overcurrent has been detected by the fast current limit.</td></tr>
<tr><td>5</td><td>Fault Avoid Out</td><td>Not used.</td></tr>
<tr><td>6</td><td>Momentary Cessation</td><td>The second level used in some standards has been reached.</td></tr>
<tr><td>7</td><td>Pre Out</td><td>Not used.</td></tr>
<tr><td>8</td><td>Out</td><td>Indicates the necessary conditions to end the execution of the LVRT algorithm.</td></tr>
<tr><td>9</td><td>Recovery</td><td>The dip is over, the recovery process continues.</td></tr>
<tr><td>10</td><td>End</td><td>Algorithm has finished.</td></tr>
</table></td></tr>
<tr><td>SV4.16.2-Number = 0</td><td>NONE</td><td>Shows the number of dips detected since power supply is detected by the control board. If power supply is removed, the counter will restart.</td></tr>
<tr><td>SV4.16.3-Depth DS Last = 0.00 %</td><td>%</td><td>Shows the last LVRT depth direct sequence.</td></tr>
<tr><td>SV4.16.4-Depth IS Last = 0.00 %</td><td>%</td><td>Shows the last LVRT depth inverse sequence.</td></tr>
<tr><td>SV4.16.5-Depth VRST Min Last = 0.00 %</td><td>%</td><td>Shows the last VRST.</td></tr>
<tr><td>SV4.16.6-V Increment = 0.00 %</td><td>%</td><td>Shows the voltage LVRT increment.</td></tr>
<tr><td>SV4.16.7-Last Duration = 0 ms</td><td>NONE</td><td>Shows the duration of the last LVRT.</td></tr>
<tr><td>SV4.16.8-Vdrop number = 0</td><td>NONE</td><td>Number of times a voltage drop has occurred.</td></tr>
<tr><td>SV4.16.9-V threshold active = 0.00 %</td><td>%</td><td>Shows the threshold voltage to activate LVRT algorithm.
Note: If [G4.3.1.2b-k V Threshold ≠ 0.000], take this into account for the threshold voltage.</td></tr>
</table>

5.4.15. Subgroup SV4.17: Modulation

This subgroup allows to visualize the internal details for modulation status.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.17.1-Active Mod. = Single</td><td>NONE</td><td>Allows to change the modulation type for laboratory testing.
Note: The modulation type pre-set to "Single" must not be changed.</td></tr>
<tr><td>SV4.17.2-Modules in DS = 00000000</td><td>NONE</td><td>Displays the 8-bit mask with the modules working in double signal.
First module is the bit located in the right place and the last module, the bit located in the left place.
<table>
<tr><th>BIT</th><th>STATUS</th><th>BIT</th><th>STATUS</th></tr>
<tr><td>0</td><td>Module 1</td><td>5</td><td>Not available</td></tr>
<tr><td>1</td><td>Module 2</td><td>6</td><td>Not available</td></tr>
<tr><td>2</td><td>Module 3</td><td>7</td><td>Not available</td></tr>
<tr><td>3</td><td>Module 4</td><td>8</td><td>Not available</td></tr>
<tr><td>4</td><td>Not available</td><td>9</td><td>Not available</td></tr>
</table>
Note: These options are displayed if [SG10.1.12-Number of modules < n] being ‘n’ the module number.</td></tr>
<tr><td>SV4.17.3-Nº DS Activations = 0</td><td>NONE</td><td>Shows how many times the modulation has been changed to double signal.</td></tr>
<tr><td>SV4.17.4-Fault counter = 0</td><td>NONE</td><td>Shows the value of the counter of failure time in double signal modulation.</td></tr>
</table>

5.4.16. Subgroup SV4.18: FAC

This subgroup allows to visualize the internal references for VSI control type and FAC control subtype.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV4.18.1-Vd Direct sequence = 0.00 %	%	Shows the direct voltage sequence reference on the D axis, as a percentage of the product’s rated voltage.
SV4.18.2-Vq Direct sequence = 0.00 %	%	Shows the direct voltage sequence reference on the Q axis, as a percentage of the product’s rated voltage.
SV4.18.3-Vd Inverse sequence = 0.00 %	%	Shows the inverse voltage sequence reference on the D axis, as a percentage of the product’s rated voltage.
SV4.18.2-Vq Inverse sequence = 0.00 %	%	Shows the inverse voltage sequence reference on the Q axis, as a percentage of the product’s rated voltage.

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5.4.17. Subgroup SV4.19: VISMA

This subgroup allows to visualize the internal references for VSI control type and VISMA control subtype.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV4.19.1-P ref = 0.00 %	%	Shows the power reference for VSI control type and VISMA control subtype.
SV4.19.2-Damping = 0.00 %	%	Shows the damping action for VSI control type and VISMA control subtype.
SV4.19.3-P fb = 0.00 %	%	Shows the power calculation used as feedback for VISMA machine.
SV4.19.4-Freq ref = 0.00 %	%	Shows the frequency reference for VSI control type and VISMA control subtype.
SV4.19.5-Freq rotor = 0.00 %	%	Shows the rotor frequency of VISMA machine.
SV4.19.6-Acceleration = 0.00 %	%	Shows the rotor acceleration of VISMA machine.
SV4.19.7-Vac ref = 0.00 %	%	Shows the AC voltage reference for VSI control type and VISMA control subtype.
SV4.19.8-Vac fb = 0.00 %	%	Shows the voltage used as feedback of VISMA machine.
SV4.19.9-Iq ref = 0.00 %	%	Shows the reactive current reference for VSI control type and VISMA control subtype.
SV4.19.10-Iq fb = 0.00 %	%	Shows the reactive current used as feedback of VISMA machine.
SV4.19.11-Vd = 0.00 %	%	Shows the modulation index used.
SV4.19.12-Damping freq = 0.00 Hz	Hz	Shows the frequency used for damping.

5.4.18. Subgroup SV4.20: Limit Vdc/Vac

This subgroup allows to visualize the value of the parameters of the algorithm that limits P and Q by Vdc/Vac ratio.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV4.20.1-Index modulation = 0.00 %	%	Shows the most restrictive modulation index.
SV4.20.2-Output counter = 0 ms	ms	Shows the output counter of the algorithm.
SV4.20.3-P limit = 0.00 %	%	Shows the P limit value of the algorithm.
SV4.20.4-Q limit = 0.00 %	%	Shows the Q limit value of the algorithm.

5.4.19. Subgroup SV4.21: AC Precharge

This subgroup allows the AC precharge statuses to be displayed.

Note: this subgroup is displayed if [G1.3-User] ≥ Power Electronics staff.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.21.1-Precharge status = INIT</td><td>NONE</td><td>Displays the status of the AC precharge DSP algorithm. This algorithm performs a DC by AC bus charging when it detects that there is a module with the bus unloaded and cannot perform the softcharge. It is activated during premagnetization, before connecting the inverter to the grid.

Note: this parameter applies only to HEM and HEMK inverters and in any control mode.
<table><tr><th>BIT</th><th>STATUS</th><th>BIT</th><th>STATUS</th></tr><tr><td>0</td><td>INIT</td><td>4</td><td>ON</td></tr><tr><td>1</td><td>OFF</td><td>5</td><td>STOP</td></tr><tr><td>2</td><td>CHECK AC</td><td>6</td><td>DONE</td></tr><tr><td>3</td><td>PRE ON</td><td>7</td><td>FAULT</td></tr></table></td></tr>
<tr><td>SV4.21.2-Active counter = 0</td><td>NONE</td><td>It shows the active counter of the algorithm at the moment.</td></tr>
</table>

EN

5.4.20. Subgroup SV4.22: P (ROCOF)

This subgroup displays the status of the control algorithm that enables the inverter to respond to rapid changes in grid frequency (ROCOF – Rate of Change of Frequency), in accordance with the requirements of the Lithuanian grid code.

Note: this subgroup is displayed if [G1.3-User] ≥ Qualified customer.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV4.22.1-Status = Dead Band</td><td>NONE</td><td>Displays the current status of the P-control algorithm based on the rate of change of the grid frequency (ROCOF).
<table><tr><th>OPT.</th><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>Dead Band</td><td>Indicates that the ROCOF is within the configured deadband. In this status, the inverter does not apply any control action to the P associated with the ROCOF function.</td></tr><tr><td>1</td><td>Zone P max</td><td>Indicates that the ROCOF exceeds the upper threshold of the deadband. In this status, the inverter applies a linear response by increasing the P delivered to the grid, in accordance with the configured response points, relative to the power setpoint.</td></tr><tr><td>2</td><td>Zone P min</td><td>Indicates that the ROCOF exceeds the lower threshold of the dead band. In this status, the inverter applies a linear response by reducing the P delivered to the grid, in accordance with the configured response points, relative to the power setpoint.</td></tr></table></td></tr>
<tr><td>SV4.22.2-ROCOF (Hz/s) = 0.00</td><td>NONE</td><td>Displays the instantaneous value of the rate of change of grid frequency (ROCOF) used as input to the P-control algorithm.</td></tr>
<tr><td>SV4.22.3-P reference = 0.0 %</td><td>%</td><td>Displays the P reference generated by the ROCOF algorithm. This reference is added to the P references commanded by the [G8: Control] group, contributing to the inverter's final P setpoint.</td></tr>
</table>

5.4.21. Subgroup SV4.23: Active Balance

This subgroup displays the parameter values associated with the active balance control mode between busbars, in which the inverter control gains are assigned externally.

Note: this subgroup is displayed if [G1.3-User] ≥ Power Electronics staff, [SG10.1.4-Inverter type] = PCS/MPCS, and [SG10.1.14-Number of busbar] ≥ 2.

SCREEN	UNITS	DESCRIPTION
SV4.23.1-P lim discharge = 0.00 %	%	Displays the discharge limit applied to the internal active power references (Pac) during operation in active balance mode between busbars. In this software version, the limit applies only in VISMA mode.
SV4.23.2-P lim charge = 0.00 %	%	Displays the charge limit applied to the internal active power references (Pac) during operation in active balance mode between busbars. In this software version, the limit applies only in VISMA mode.
SV4.23.3-Pac ref global = 0.00 %	%	Global active power reference used by the active balance algorithm, without applying individual limits per busbar.
SV4.23.4-Pac ref BESS 1 = 0.00 %	%	Active power reference used for load sharing between busbars, once the configured limits have been applied. It corresponds to Busbar 1.
SV4.23.5-Pac ref BESS 2 = 0.00 %	%	Active power reference used for load sharing between busbars, once the configured limits have been applied. It corresponds to Busbar 2.
SV4.23.6-Pac ref BESS 3 = 0.00 %	%	Active power reference used for load sharing between busbars, once the configured limits have been applied. It corresponds to Busbar 3. **Note:** this parameter is shown if [SG10.1.14-Number of busbar] ≥ 3.
SV4.23.7-Pac ref BESS 4 = 0.00 %	%	Active power reference used for load sharing between busbars, once the configured limits have been applied. It corresponds to Busbar 4. **Note:** this parameter is shown if [SG10.1.14-Number of busbar] ≥ 4.

5.5. Group SV5: Counters

The counters are started when the IGBTs of the power stages start switching. They stop when the IGBTs stop switching, opening the AC and DC part of the inverter.

This group shows the active and reactive energy history, as well as the operating time, of each power stage and the total energy of the inverter.

To read data from several modules at a time, use Modbus communication and refer to "Energy visualization parameters" section.

5.5.1. Subgroup SV5.1: Active energy

SCREEN	UNITS	DESCRIPTION
SV5.1.1-Select module = Module 1	NONE	Shows the information related to the active energy of the selected module.
SV5.1.2-Daily energy = 0 kWh	kWh	Shows the daily active energy of the selected module.
SV5.1.3-Yesterday energy = 0 kWh	kWh	Shows yesterday's active energy of the selected module.
SV5.1.4-Month energy = 0 kWh	kWh	Shows the current month active energy of the selected module.
SV5.1.5-Last month energy = 0 kWh	kWh	Shows the last month active energy of the selected module.
SV5.1.6-Total energy = 0 kWh	kWh	Shows the total active energy of the selected module.
SV5.1.7-Consumed daily energy = 0 kWh	kWh	Shows the daily active energy consumed of the selected module. **Note:** This parameter is displayed if [SV5.1.1-Select module = All].
SV5.1.8-Consumed total energy = 0 kWh	kWh	Shows the total active energy consumed of the selected module. **Note:** This parameter is displayed if [SV5.1.1-Select module = All].

5.5.2. Subgroup SV5.2: Reactive energy

SCREEN	UNITS	DESCRIPTION
SV5.2.1-Select module = Module 1	NONE	Shows the information related to the reactive energy of the selected module.
SV5.2.2-Daily reactive = 0 kVArh	kVArh	Shows the daily reactive energy of the selected module.
SV5.2.3-Yesterday reactive = 0 kVArh	kVArh	Shows the yesterday reactive energy of the selected module.
SV5.2.4-Month reactive = 0 kVArh	kVArh	Shows the current month's reactive energy of the selected module.
SV5.2.5-Last month reactive = 0 kVArh	kVArh	Shows the last month's reactive energy of the selected module.
SV5.2.6-Total reactive = 0 kVArh	kVArh	Shows the total reactive energy of the selected module.

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5.5.3. Subgroup SV5.3: Run time counters

SCREEN	UNITS	DESCRIPTION
SV5.3.1-Total run time years = 0	NONE	Displays the total run time years counter.
SV5.3.2-Total run time days = 0	NONE	Displays the total run time days counter.
SV5.3.3-Total run time hours = 0	NONE	Displays the total run time hours counter.
SV5.3.4-Partial run time year = 0	NONE	Displays the partial run time years counter.
SV5.3.5-Partial run time days = 0	NONE	Displays the partial run time days counter.
SV5.3.6-Partial run time hour = 0	NONE	Displays the partial run time hours counter.

5.5.4. Subgroup SV5.4: Grid connections

Note: These parameters must be configured in subgroup SG9.3: Rst grid conn counters.

SCREEN	UNITS	DESCRIPTION
SV5.4.1-Total grid conns = 0	NONE	Displays the total number of grid connections. **Note:** elevated privileges are required to reset the grid counter.
SV5.4.2-Partial grid conns = 0	NONE	Displays the partial number of grid connections. **Note:** elevated privileges are required to reset the grid counter.

5.5.5. Subgroup SV5.5: MV cell counters

Note: These parameters are displayed if [SG10.1.3-Product type] = HEMK and [SG10.7.3-MV Skid] = Yes, Motorized] or [SG10.1.3-Product type] = HEM.

SCREEN	UNITS	DESCRIPTION
SV5.5.1-No premag closures = 0	NONE	Shows the number of MV switchgear closures without premagnetization, including forced closures as specified in [G2.7.1.1-MV cell force contro] and physical closures.
SV5.5.1a-Manual cmd closures = 0	NONE	Shows the number of manual closures of the MV switchgear initiated from the control system (without premagnetization). These closures are executed using the MV switchgear control override commands: - [G2.7.1.1-MV cell force contro] = Communications and - [G2.7.1.2-MV cell switch actio] = Closed. The counter is incremented when, following a closing command sent by the DSP, the change in state from open to closed is detected. The counter value persists after a power interruption and can be reset using the parameter [SG9.4.1a-Rst manual closure]. **Note**: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV5.5.1b-Uncommanded closures = 0	NONE	Shows the number of manual closures of the MV switchgear not commanded by the control system (without premagnetization). These closures are caused by physical or external actions on the MV switchgear, without a closure command being issued by the DSP. The counter increments based on the change in state from open to closed detected in the switchgear. The counter value persists after a power interruption and can be reset using the parameter [SG9.4.1b-Rst uncommanded clos]. **Note**: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV5.5.2-Premag closures = 0	NONE	Shows the number of MV switchgear closures with premagnetization, including closures with AC pre-charge and premagnetization logic.
SV5.5.3-Loaded openings = 0	NONE	Shows the number of MV switchgear openings being the switchgear loaded.
SV5.5.4-Not loaded openings = 0	NONE	Shows the number of MV switchgear openings being the switchgear not loaded.

5.6. Group SV6: Comms visualization

This group shows information about the communications status in the inverter components. Inverters are composed of several components or devices that communicate with each other to provide proper operation.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

5.6.1. Subgroup SV6.2: Central Board

This subgroup allows to visualize the internal communications status at the main control board.

SV6.2.1: uP-DSP

This group shows the status of the communication between the uP and DSP devices of the main control board. It is important to underscore that this communication can be done using the slow SPI bus and the fast SPI bus. This group shows the status of both.

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SCREEN	UNITS	DESCRIPTION
SV6.2.1.1-DSP Fast Status = Comms Error	NONE	Shows the communication quality between uP and DSP (fast SPI bus). OPTION / STATUS / DESCRIPTION 0 / Disabled / Show the communication is disabled 1 / Comms OK / Show the communication quality is ok 2 / Comms Error / Show the communication quality is not ok
SV6.2.1.2-DSP Slow Status = Comms Error	NONE	Shows the communication quality between uP and DSP (low SPI bus). OPTION / STATUS / DESCRIPTION 0 / Disabled / Show the communication is disabled 1 / Comms OK / Show the communication quality is ok 2 / Comms Error / Show the communication quality is not ok
SV6.2.1.3-Fast Frame Error = 0	NONE	Counter that displays number of incorrect frames in a time window of 500 consecutive frames.
SV6.2.1.4-Fast Frame Err Int = 0	NONE	Counter that displays the number of incorrect frames due to error in the frame itself in a time window of 500 consecutive frames.
SV6.2.1.5-Fast Frame Err Drv = 0	NONE	Counter that displays the number of incorrect frames due to reception failure by uP in a time window of 500 consecutive frames.
SV6.2.1.6-Fast Frame Err Tout = 0	NONE	Counter that displays the number of incorrect frames due to timeout in communication establishment between the two devices in a time window of 500 consecutive frames.
SV6.2.1.7-Fast Frame Success = 0	NONE	Counter that displays the number of correct frames in a time window of 500 consecutive frames.
SV6.2.1.8-Fast Frame Total = 0	NONE	Counter up to 500 frames and then reset.
SV6.2.1.9-Fast Err/Total = 0 %	%	Error rate of 500 frames successive.
SV6.2.1.10-Slow Frame Error = 0	NONE	Counter that displays the number of incorrect frames in a time window of 125 consecutive frames.
SV6.2.1.11-Slow Frame Integ = 0	NONE	Counter that displays number of incorrect frames due to error in the frame itself in a time window of 125 consecutive frames.
SV6.2.1.12-Slow Frame Driver = 0	NONE	Counter that displays the number of incorrect frames due to reception failure by uP in a time window of 125 consecutive frames.

SCREEN	UNITS	DESCRIPTION
SV6.2.1.13-Slow Frame Tout = 0	NONE	Counter that displays the number of incorrect frames due to timeout in communication establishment between the two devices in a time window of 125 consecutive frames.
SV6.2.1.14-Slow Frame Success = 0	NONE	Counter that displays the number of correct frames in a time window of 125 consecutive frames.
SV6.2.1.15-Slow Frame Total = 0	NONE	Counter up to 125 frames and then reset.
SV6.2.1.16-Slow Err/Total = 0 %	%	Error rate of 125 successive frames.

SV6.2.2: Ethernet

SCREEN	UNITS	DESCRIPTION
SV6.2.2.1-Reset counters = 0	NONE	Shows the reset counters.
SV6.2.2.2-Total Processed = 0.00 %	%	Shows the process.
SV6.2.2.3-Socket1 Status = Offline	NONE	Shows the socket 1 status. OPTION / STATUS 0 Offline 1 Online
SV6.2.2.4-Socket1 IP = 0.0.0.0	NONE	Shows the socket 1 IP.
SV6.2.2.5-Socket1 Packets = 0	NONE	Indicates the number of packets sent by socket 1 of the central control Up.
SV6.2.2.6-Socket1 Processed = 0	NONE	Indicates the number of successfully processed packets with respect to those sent by socket 1 of the central control uP.
SV6.2.2.7-Socket2 Status = Offline	NONE	Shows the socket 2 status. OPTION / STATUS 0 Offline 1 Online
SV6.2.2.8-Socket2 IP = 0.0.0.0	NONE	Shows the socket 2 IP.
SV6.2.2.9-Socket2 Packets = 0	NONE	Indicates the number of packets sent by socket 2 of the central control Up.
SV6.2.2.10-Socket2 Processed = 0	NONE	Indicates the number of successfully processed packets with respect to those sent by socket 2 of the central control uP.
SV6.2.2.11-Socket3 Status = Offline	NONE	Shows the socket 3 status. OPTION / STATUS 0 Offline 1 Online
SV6.2.2.12-Socket3 IP = 0.0.0.0	NONE	Shows the socket 3 IP.
SV6.2.2.13-Socket3 Packets = 0	NONE	Indicates the number of packets sent by socket 3 of the central control Up.
SV6.2.2.14-Socket3 Processed = 0	NONE	Indicates the number of successfully processed packets with respect to those sent by socket 3 of the central control uP.
SV6.2.2.15-Socket4 Status = Offline	NONE	Shows the socket 4 status. OPTION / STATUS 0 Offline 1 Online
SV6.2.2.16-Socket4 IP = 0.0.0.0	NONE	Shows the socket 4 IP.
SV6.2.2.17-Socket4 Packets = 0	NONE	Indicates the number of packets sent by socket 4 of the central control Up.

SCREEN	UNITS	DESCRIPTION
SV6.2.2.18-Socket4 Processed = 0	NONE	Indicates the number of successfully processed packets with respect to those sent by socket 4 of the central control uP.
SV6.2.2.19-Socket5 Status = Offline	NONE	Shows the socket 5 status. OPTION / STATUS 0 Offline 1 Online
SV6.2.2.20-Socket5 IP = 0.0.0.0	NONE	Shows the socket 5 IP.
SV6.2.2.21-Socket5 Packets = 0	NONE	Indicates the number of packets sent by socket 5 of the central control Up.
SV6.2.2.22-Socket5 Processed = 0	NONE	Indicates the number of successfully processed packets with respect to those sent by socket 5 of the central control uP.
SV6.2.2.23-Socket6 Status = Offline	NONE	Shows the socket 6 status. OPTION / STATUS 0 Offline 1 Online
SV6.2.2.24-Socket6 IP = 0.0.0.0	NONE	Shows the socket 6 IP address.
SV6.2.2.25-Socket6 Packets = 0	NONE	Indicates the number of packets sent by socket 6 of the central control Up.
SV6.2.2.26-Socket6 Processed = 0	NONE	Indicates the number of successfully processed packets with respect to those sent by socket 6 of the central control uP.

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SV6.2.3: uP-IMI

SCREEN	UNITS	DESCRIPTION
SV6.2.3.1-Successfull Frames = 0	NONE	Indicates the number of frames that have been successfully processed between the uP of the main control board and the isolation monitoring device.
SV6.2.3.2-Error Frames = 0	NONE	Indicates the number of erroneous frames between the uP of the main control board and the isolation monitoring device.

SV6.2.4: uP-FPGA

This group shows the status of the communication between the uP of the main control board and the central FPGA.

SCREEN	UNITS	DESCRIPTION
SV6.2.4.1-Successful frames = 0	NONE	Indicates the number of frames that have been successfully processed between the uP of the main control board and the central FPGA.
SV6.2.4.2-Error Frames = 0	NONE	Indicates the number of erroneous frames between the uP of the main control board and the central FPGA.
SV6.2.4.3-Status = Disabled	NONE	Shows the communications status between the uP of the main control board and the central FPGA.

OPTION	STATUS	DESCRIPTION
0	Disabled	There is no communication between the uP of the main control board and the central FPGA.
1	Comms OK	Communication between the uP of the main control board and the central FPGA is correct.
2	Timeout	The time allowed to establish communication between the uP of the main control board and the central FPGA has expired. It triggers F10.
3	Error CRC	It is not possible to establish communication between the uP of the main control board and the central FPGA due to an error in the frame.
4	Error Driver	It is not possible to establish communication between the uP of the main control board and the central FPGA due to a failure in the reception of the frame.

SV6.2.4.4-Frames/sec OK = 0	NONE	Counter that displays the number of correct frames per second between the two devices.
SV6.2.4.5-Frames/sec Timeout = 0	NONE	Counter that displays the number of frames that have failed per timeout per second between both devices.
SV6.2.4.6-Frames/sec CRC = 0	NONE	Counter that displays the number of frames that have failed due to CRC (communications frame error) per second between both devices.
SV6.2.4.7-Frames/sec Driver = 0	NONE	Counter that displays the number of frames that have failed due to an error in the reception of the frame shared between uP and central FPGA.

SV6.2.5: Fans

This group of parameters allows displaying both the frames that have been correctly processed and the erroneous ones between the uP of the central control board and each of the fans. These frame counters can be manually reset by writing a zero. To know about the ventilation layout, please consult "SV11: Fans" section.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

- **SV6.2.5.1: Fan 1**

SCREEN	UNITS	DESCRIPTION
SV6.2.5.1.1-Fan 1 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 1.
SV6.2.5.1.2-Fan 1 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control and fan 1.

- **SV6.2.5.2: Fan 2**

SCREEN	UNITS	DESCRIPTION
SV6.2.5.2.1-Fan 2 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 2.
SV6.2.5.2.2-Fan 2 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control and fan 2.

- **SV6.2.5.3: Fan 3**

SCREEN	UNITS	DESCRIPTION
SV6.2.5.3.1-Fan 3 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 3.
SV6.2.5.3.2-Fan 3 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control and fan 3.

- **SV6.2.5.4: Fan 4**

SCREEN	UNITS	DESCRIPTION
SV6.2.5.4.1-Fan 4 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 4.
SV6.2.5.4.2-Fan 4 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control and fan 4.

- **SV6.2.5.5: Fan 5**

SCREEN	UNITS	DESCRIPTION
SV6.2.5.5.1-Fan 5 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 5.
SV6.2.5.5.2-Fan 5 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control and fan 5.

- **SV6.2.5.6: Fan 6**

SCREEN	UNITS	DESCRIPTION
SV6.2.5.6.1-Fan 6 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 6.
SV6.2.5.6.2-Fan 6 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control and fan 6.

- **SV6.2.5.7: Fan 7**

SCREEN	UNITS	DESCRIPTION
SV6.2.5.7.1-Fan 7 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 7.
SV6.2.5.7.2-Fan 7 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control and fan 7.

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- **SV6.2.5.8: Fan 8**

SCREEN	UNITS	DESCRIPTION
SV6.2.5.8.1-Fan 8 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 8.
SV6.2.5.8.2-Fan 8 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control and fan 8.

- **SV6.2.5.8: Fan 9**

SCREEN	UNITS	DESCRIPTION
SV6.2.5.9.1-Fan 9 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 9.
SV6.2.5.9.2-Fan 9 Err Frames = 0	NONE	Indicates the number of failed frames between uP central control and fan 9.

- **SV6.2.5.8: Fan 10**

SCREEN	UNITS	DESCRIPTION
SV6.2.5.10.1-Fan 10 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 10.
SV6.2.5.10.2-Fan 10 Er Frames = 0	NONE	Indicates the number of failed frames between uP central control and fan 10.

- **SV6.2.5.8: Fan 11**

SCREEN	UNITS	DESCRIPTION
SV6.2.5.11.1-Fan 11 Ok Frames = 0	NONE	Indicates the number of frames that have been successfully processed between uP central control and fan 11.
SV6.2.5.11.2-Fan 11 Er Frames = 0	NONE	Indicates the number of failed frames between uP central control and fan 11.

SV6.2.6: Central FPGA-DU

SCREEN	UNITS	DESCRIPTION
SV6.2.6.1- Successful frames = 0	NONE	Indicates the number of frames that have been successfully processed between central FPGA and DU.
SV6.2.6.2-Error frames = 0	NONE	Number of error frames shared between the DU and the central FPGA.

SV6.2.7-FPGA Central-Module

SCREEN	UNITS	DESCRIPTION
SV6.2.7.1-Cent->M1 frame OK = 0	NONE	Displays how many frames sent from the central FPGA to the module 1 FPGA are correct. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 1].
SV6.2.7.2-Cent->M1 Er frame = 0	NONE	Displays how many frames sent from the central FPGA to the module 1 FPGA are wrong. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 1].
SV6.2.7.3-M1->Cent frame OK = 0	NONE	Displays how many frames sent from module 1 FPGA to the central FPGA are correct. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 1].

SCREEN	UNITS	DESCRIPTION
SV6.2.7.4-M1->Cent Er frame = 0	NONE	Displays how many frames sent from module 1 FPGA to the central FPGA are wrong. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 1].
SV6.2.7.5-Cent->M2 frame OK = 0	NONE	Displays how many frames sent from the central FPGA to the module 2 FPGA are correct. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 2].
SV6.2.7.6-Cent->M2 Er frame = 0	NONE	Displays how many frames sent from the central FPGA to the module 2 FPGA are wrong. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 2].
SV6.2.7.7-M2->Cent frame OK = 0	NONE	Displays how many frames sent from module 2 FPGA to the central FPGA are correct. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 2].
SV6.2.7.8-M2->Cent Er frame = 0	NONE	Displays how many frames sent from module 2 FPGA to the central FPGA are wrong. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 2].
SV6.2.7.9-Cent->M3 frame OK = 0	NONE	Displays how many frames sent from the central FPGA to the module 3 FPGA are correct. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 3].
SV6.2.7.10-Cent->M3 Er frame = 0	NONE	Displays how many frames sent from the central FPGA to the module 3 FPGA are wrong. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 3].
SV6.2.7.11-M3->Cent frame OK = 0	NONE	Displays how many frames sent from module 3 FPGA to the central FPGA are correct. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 3].
SV6.2.7.12-M3->Cent Er frame = 0	NONE	Displays how many frames sent from module 3 FPGA to the central FPGA are wrong. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 3].
SV6.2.7.13-Cent->M4 frame OK = 0	NONE	Displays how many frames sent from the central FPGA to the module 4 FPGA are correct. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 4].
SV6.2.7.14-Cent->M4 Er frame = 0	NONE	Displays how many frames sent from the central FPGA to the module 4 FPGA are wrong. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 4].
SV6.2.7.15-M4->Cent frame OK = 0	NONE	Displays how many frames sent from module 4 FPGA to the central FPGA are correct. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 4].
SV6.2.7.16-M4->Cent Er frame = 0	NONE	Displays how many frames sent from module 4 FPGA to the central FPGA are wrong. **Note:** This parameter is displayed if [SG10.1.12-Number of modules ≥ 4].

SV6.2.8-DSP-Central FPGA

SCREEN	UNITS	DESCRIPTION
SV6.2.8.1-Successful frames = 0	NONE	Shows the number of successful frames shared between the central FPGA and DSP.
SV6.2.8.2-Error frames = 0	NONE	Shows the number of error frames shared between the central FPGA and DSP.

SV6.2.9-uP bus i2c

This group shows the data related to the temperature and humidity sensor of the main control board that are monitoring in the I2c bus.

SCREEN	UNITS	DESCRIPTION
SV6.2.9.1-Sens T/H Errors I2C = 0	NONE	Shows the number of errors of the temperature and humidity sensor due to I2C communication faults.
SV6.2.9.2-Sensor T/H Sensor Type = Unknown	NONE	Shows the name of the temperature and humidity sensor of the main control board. OPT. / STATUS / DESCRIPTION 0 / Unknown / Temperature and humidity sensor unknown. 1 / Sensor 7020 / 7020 temperature and humidity sensor. 2 / Sensor HTU2ID / HTU2ID temperature and humidity sensor.
SV6.2.9.3-Sensor T/H Current Temperature = 35 ºC	ºC	Shows the registered temperature by the temperature and humidity sensor.
SV6.2.9.4-Sensor T/H Current Humidity = 50 %	%	Shows the percentage of humidity registered by the temperature and humidity sensor.

SV6.2.10: BusPlus temp

This subgroup allows to display the communication frame counters for the temperature measurement boards for Bus Plus.

Note: this subgroup is displayed if [SG10.7.8-Temp. board] = Enabled.

- **SV6.2.10.1: Board 1**

SCREEN	UNITS	DESCRIPTION
SV6.2.10.1.1-B1 frames ok = 0	NONE	Shows the counter for OK frames received from measurement board 1. It is possible to reset the counter by entering the value 0.
SV6.2.10.1.2-B1 frames error = 0	NONE	Shows the frame error counter for measurement board 1. It is possible to reset the counter by entering the value 0.

- **SV6.2.10.2: Board 2**

SCREEN	UNITS	DESCRIPTION
SV6.2.10.2.1-B2 frames ok = 0	NONE	Shows the counter for OK frames received from measurement board 2. It is possible to reset the counter by entering the value 0.
SV6.2.10.2.2-B2 frames error = 0	NONE	Shows the frame error counter for measurement board 2. It is possible to reset the counter by entering the value 0.

SV6.2.11: PTP

This subgroup shows the diagnostic parameters for time synchronization via the PTP (Precision Time Protocol) between the plant's PTP Master and the inverter's central control board. Synchronization is performed through the PSAMS, which acts as a gateway between the PTP Master and the inverter. These parameters make it possible to assess the status of communications with the master and to verify that synchronization remains active.

Note: for information on selecting the time synchronization source and the PSAMS PTP status, refer to section I.4.5 of Annex I: PSAMS.

SCREEN	UNITS	DESCRIPTION
SV6.2.11.1-PTP Seconds = 0 s	s	Shows the value of the inverter's clock once synchronized with the PTP Master, in Unix Epoch Time format (seconds elapsed since 1 January 1970, UTC). If, when converting the value to date and time, it matches the plant's actual time, this confirms that the inverter's clock is correctly synchronized with the master. As this is an unsigned 32-bit value, it is transmitted in two 16-bit registers: most significant word (MSW, 50289) and least significant word (LSW, 50290), which must be read together.
SV6.2.11.2-Recv Sync Seq ID = 0	NONE	Counter associated with the synchronization frames received from the master. Its value must increase continuously during normal operation: if it increases, the inverter is correctly receiving the synchronization information from the master; if it remains static, this indicates that reception has been interrupted and, therefore, a problem in the communication with the master.
SV6.2.11.3-Sent DReq Seq ID = 0	NONE	Counter associated with the requests that the inverter sends to the master to measure the network delay. Its continuous increase confirms that the inverter is actively participating in the delay measurement, a step required to compensate for it and achieve the demanded synchronization accuracy. It is interpreted together with [SV6.2.11.4-Recv DResp Seq ID].
SV6.2.11.4-Recv DResp Seq ID = 0	NONE	Counter associated with the master's responses to the delay measurement requests sent by the inverter. Under normal conditions its value must be one unit lower than that of [SV6.2.11.3-Sent DReq Seq ID] (if [SV6.2.11.3-Sent DReq Seq ID] = N, then [SV6.2.11.4-Recv DResp Seq ID] = N − 1), since the response is counted after the request. Maintaining this N / N − 1 relationship confirms that the exchange is completed correctly and that the network delay compensation is operational.

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5.7. Group SV7: Faults and warnings

This group allows to visualize the history of faults, warnings and events of the inverter, as well as the real time status of the faults and warnings through a series of parameters, in which bit by bit represents whether the fault and/or warning is active or inactive.

5.7.1. Subgroup SV7.1: Faults

SV7.1.1: Faults status

Parameters within this group show the status of each fault (0 = Not active; X = Active) in groups of 16 faults (the lowest number being the bit to the upmost right).

SCREEN	UNITS	DESCRIPTION
SV7.1.1.1-Fault 1 to 16 = 0000000000000000	NONE	Displays the status of faults 1 to 16.
SV7.1.1.2-Fault 17 to 32 = 0000000000000000	NONE	Displays the status of faults 17 to 32.
SV7.1.1.3-Fault 33 to 48 = 0000000000000000	NONE	Displays the status of faults 33 to 48.

SCREEN	UNITS	DESCRIPTION
SV7.1.1.4-Fault 49 to 64 = 0000000000000000	NONE	Displays the status of faults 49 to 64.
SV7.1.1.5-Fault 65 to 80 = 0000000000000000	NONE	Displays the status of faults 65 to 80.
SV7.1.1.6-Fault 81 to 96 = 0000000000000000	NONE	Displays the status of faults 81 to 96.
SV7.1.1.7-Fault 97 to 112 = 0000000000000000	NONE	Displays the status of faults 97 to 112.
SV7.1.1.8-Fault 113 to 128 = 0000000000000000	NONE	Displays the status of faults 113 to 128.
SV7.1.1.9-Fault 129 to 144 = 0000000000000000	NONE	Displays the status of faults 129 to 144.
SV7.1.1.10-Fault 145 to 160 = 0000000000000000	NONE	Displays the status of faults 145 to 160.
SV7.1.1.11-Fault 161 to 176 = 0000000000000000	NONE	Displays the status of faults 161 to 176.
SV7.1.1.12-Fault 177 to 192 = 0000000000000000	NONE	Displays the status of faults 177 to 192.
SV7.1.1.13-Fault 193 to 208 = 0000000000000000	NONE	Displays the status of faults 193 to 208.
SV7.1.1.14-Fault 209 to 224 = 0000000000000000	NONE	Displays the status of faults 209 to 224.
SV7.1.1.15-Fault 225 to 240 = 0000000000000000	NONE	Displays the status of faults 225 to 240.
SV7.1.1.16-Fault 241 to 256 = 0000000000000000	NONE	Displays the status of faults 241 to 256.
SV7.1.1.17-Fault 257 to 272 = 0000000000000000	NONE	Displays the status of faults 257 to 272.
SV7.1.1.18-Fault 273 to 288 = 0000000000000000	NONE	Displays the status of faults 273 to 288.
SV7.1.1.19-Fault 289 to 304 = 0000000000000000	NONE	Displays the status of faults 289 to 304.
SV7.1.1.20-Fault 305 to 320 = 0000000000000000	NONE	Displays the status of faults 305 to 320.
SV7.1.1.21-Fault 321 to 336 = 0000000000000000	NONE	Displays the status of faults 321 to 336.
SV7.1.1.22-Fault 337 to 352 = 0000000000000000	NONE	Displays the status of faults 337 to 352.
SV7.1.1.23-Fault 353 to 368 = 0000000000000000	NONE	Displays the status of faults 353 to 368.
SV7.1.1.24-Fault 369 to 384 = 0000000000000000	NONE	Displays the status of faults 369 to 384.
SV7.1.1.25-Fault 385 to 400 = 0000000000000000	NONE	Displays the status of faults 385 to 400.
SV7.1.1.26-Fault 401 to 416 = 0000000000000000	NONE	Displays the status of faults 401 to 416.
SV7.1.1.27-Fault 417 to 432 = 0000000000000000	NONE	Displays the status of faults 417 to 432.
SV7.1.1.28-Fault 433 to 448 = 0000000000000000	NONE	Displays the status of faults 433 to 448.
SV7.1.1.29-Fault 449 to 464 = 0000000000000000	NONE	Displays the status of faults 449 to 464.
SV7.1.1.30-Fault 465 to 480 = 0000000000000000	NONE	Displays the status of faults 465 to 480.
SV7.1.1.31-Fault 481 to 496 = 0000000000000000	NONE	Displays the status of faults 481 to 496.
SV7.1.1.32-Fault 497 to 512 = 0000000000000000	NONE	Displays the status of faults 497 to 512.

SV7.1.2: Faults history

This subgroup allows to visualize the list of the last ten faults of the system in chronological order. The parameters that register the date contain two Modbus addresses corresponding to the date and time of the fault history register.

The following is an example of how the date and time registers for each fault are interpreted internally.

The parameters return decimal numbers recording the date and time of the fault occurred. This information must be transformed into binary and then converted back to decimal. The process is as follows. Example:

- Date: 5/9/2037
  - Day (5): it is transformed into binary by shifting 11 bits to the left obtaining 0010 1000 0000 0000 0000. The decimal value is 10240.
  - Month (9): transformed into binary by shifting 7 bits to the left to obtain 0100 1000 0000. The decimal value is 1152.
  - Year (2037): 2000 must be subtracted from the year. The decimal value is 37.

Once the decimal values of the date have been obtained, they are added together to obtain 11429. This value is read by Modbus. For the time the process is similar:

- Time: 16:11
  - Time (16): it is transformed into binary by shifting 8 bits to the left, obtaining 0001 0000 0000 0000 0000. The decimal value is 4096.
  - Minutes (11): The decimal value is 11.

When adding the decimals that have come out of the hour and minute conversion, the value is 4107. This value is read by Modbus.

The parameters corresponding to this subgroup are shown below.

SCREEN	DESCRIPTION	FUNCTION
SV7.1.2.1-Last fault = No faults	Register number 1 of fault history.	Displays a list of the last ten inverter faults, in chronological order. The most recent fault is shown first and stored. Whenever a new fault occurs, it will be placed at the top of the list and the oldest fault messages will drop a position, deleting the last fault message. The inverter is reset by pressing the STOP-RESET button on the display or using an external reset if it is available. Certain faults may rearm automatically using the Auto Reset.
SV7.1.2.2a-Date = 01/01/2000 00:00	Date and time of the register number 1 of fault history.	
SV7.1.2.2c-Counter = 0	Shows the number of times the fault corresponding to fault history register 1 occurred on the same day.	
SV7.1.2.3-Ninth fault = No faults	Register number 2 of fault history.	
SV7.1.2.4a-Date = 01/01/2000 00:00	Date and time of the register number 2 of fault history.	
SV7.1.2.2c-Counter = 0	Shows the number of times the fault corresponding to fault history register 2 occurred on the same day.	
SV7.1.2.5-Eighth fault = No faults	Register number 3 of fault history.	
SV7.1.2.6a-Date = 01/01/2000 00:00	Date and time of the register number 3 of fault history.	
SV7.1.2.6c-Counter = 0	Shows the number of times the fault corresponding to fault history register 3 occurred on the same day.	
SV7.1.2.7-Seventh fault = No faults	Register number 4 of fault history.	
SV7.1.2.8a-Date = 01/01/2000 00:00	Date and time of the register number 4 of fault history.	

SCREEN	DESCRIPTION	FUNCTION
SV7.1.2.8c-Counter = 0	Shows the number of times the fault corresponding to fault history register 4 occurred on the same day.	
SV7.1.2.9-Sixth fault = No faults	Register number 5 of fault history.	
SV7.1.2.10a-Date = 01/01/2000 00:00	Date and time of the register number 5 of fault history.	
SV7.1.2.10c-Counter = 0	Shows the number of times the fault corresponding to fault history register 5 occurred on the same day.	
SV7.1.2.11-Fifth fault = No faults	Register number 6 of fault history.	
SV7.1.2.12a-Date = 01/01/2000 00:00	Date and time of the register number 6 of fault history.	
SV7.1.2.12c-Counter = 0	Shows the number of times the fault corresponding to fault history register 6 occurred on the same day.	
SV7.1.2.13-Fourth fault = No faults	Register number 7 of fault history.	
SV7.1.2.14a-Date = 01/01/2000 00:00	Date and time of the register number 7 of fault history.	
SV7.1.2.14c-Counter = 0	Shows the number of times the fault corresponding to fault history register 7 occurred on the same day.	
SV7.1.2.15-Third fault = No faults	Register number 8 of fault history.	
SV7.1.2.16a-Date = 01/01/2000 00:00	Date and time of the register number 8 of fault history.	
SV7.1.2.16c-Counter = 0	Shows the number of times the fault corresponding to fault history register 8 occurred on the same day.	
SV7.1.2.17-Second fault = No faults	Register number 9 of fault history.	
SV7.1.2.18a-Date = 01/01/2000 00:00	Date and time of the register number 9 of fault history.	
SV7.1.2.18c-Counter = 0	Shows the number of times the fault corresponding to fault history register 9 occurred on the same day.	
SV7.1.2.19-First fault = No faults	Register number 10 of fault history.	
SV7.1.2.20a-Date = 01/01/2000 00:00	Date and time of the register number 10 of fault history.	
SV7.1.2.20c-Counter = 0	Shows the number of times the fault corresponding to fault history register 10 occurred on the same day.	

SV7.1.3: Faults type

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.1.3.0-Fault type Select = Abrupt stop</td><td>NONE</td><td>Shows the type of fault to get the information.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table>Note: Status Abrupt stop and Controlled stop are only displayed when there is a module fault.</td></tr>
<tr><td>SV7.1.3.1-Select page = 1</td><td>NONE</td><td>Shows the page of the faults to be consulted.</td></tr>
<tr><td>SV7.1.3.2a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.2b-Fault type = Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.3a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.3b-Fault type =Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.4a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.1.3.4b-Fault type =Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.5a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.5b-Fault type =Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.6a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.6b-Fault type =Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.7a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.7b-Fault type =Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.8a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.1.3.8b-Fault type =Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.9a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.9b-Fault type =Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.10a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.10b-Fault type = Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.11a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.11b-Fault type = Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.12a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.1.3.12b-Fault type = Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.13a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.13b-Fault type =Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.14a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.14b-Fault type = Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.15a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.15b-Fault type = Special</td><td>NONE</td><td>Shows the type of the fault.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td></tr>
<tr><td>SV7.1.3.16a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.1.3.16b-Fault type = Special</td><td>NONE</td><td>Shows the type of the fault.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr>
<tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr>
<tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr>
<tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr>
<tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr>
</table></td></tr>
<tr><td>SV7.1.3.17a-Fault number = No faults</td><td>NONE</td><td>Shows the number of the fault.</td></tr>
<tr><td>SV7.1.3.17b-Fault type = Special</td><td>NONE</td><td>Shows the type of the fault.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr>
<tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr>
<tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr>
<tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr>
<tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr>
</table></td></tr>
</table>

EN

5.7.2. Subgroup SV7.2: Warnings

SV7.2.1: Warning status

Parameters within this group show the status of each warning (0 = Not active; X = Active) in groups of 16 warnings (being the lowest number the bit to the upmost right).

SCREEN	UNITS	DESCRIPTION
SV7.2.1.1-Warning 1 to 16 = 0000000000000000	NONE	Displays the status of warnings 1 to 16.
SV7.2.1.2-Warning 17 to 32 = 0000000000000000	NONE	Displays the status of warnings 17 to 32.
SV7.2.1.3-Warning 33 to 48 = 0000000000000000	NONE	Displays the status of warnings 33 to 48.
SV7.2.1.4-Warning 49 to 64 = 0000000000000000	NONE	Displays the status of warnings 49 to 64.
SV7.2.1.5-Warning 65 to 80 = 0000000000000000	NONE	Displays the status of warnings 65 to 80.
SV7.2.1.6-Warning 81 to 96 = 0000000000000000	NONE	Displays the status of warnings 81 to 96.
SV7.2.1.7-Warning 97 to 112 = 0000000000000000	NONE	Displays the status of warnings 97 to 112.
SV7.2.1.8-Warning 113 to 128 = 0000000000000000	NONE	Displays the status of warnings 113 to 128.
SV7.2.1.9-Warning 129 to 144 = 0000000000000000	NONE	Displays the status of warnings 129 to 144.
SV7.2.1.10-Warning 145 to 160 = 0000000000000000	NONE	Displays the status of warnings 145 to 160.
SV7.2.1.11-Warning 161 to 176 = 0000000000000000	NONE	Displays the status of warnings 161 to 176.
SV7.2.1.12-Warning 177 to 192 = 0000000000000000	NONE	Displays the status of warnings 177 to 192.

SCREEN	UNITS	DESCRIPTION
SV7.2.1.13-Warning 193 to 208 = 0000000000000000	NONE	Displays the status of warnings 193 to 208.
SV7.2.1.14-Warning 209 to 224 = 0000000000000000	NONE	Displays the status of warnings 209 to 224.
SV7.2.1.15-Warning 225 to 240 = 0000000000000000	NONE	Displays the status of warnings 225 to 240.
SV7.2.1.16-Warning 241 to 256 = 0000000000000000	NONE	Displays the status of warnings 241 to 256.
SV7.2.1.17-Warning 257 to 272 = 0000000000000000	NONE	Displays the status of warnings 257 to 272.
SV7.2.1.18-Warning 273 to 288 = 0000000000000000	NONE	Displays the status of warnings 273 to 288.
SV7.2.1.19-Warning 289 to 304 = 0000000000000000	NONE	Displays the status of warnings 289 to 304.
SV7.2.1.20-Warning 305 to 320 = 0000000000000000	NONE	Displays the status of warnings 305 to 320.
SV7.2.1.21-Warning 321 to 336 = 0000000000000000	NONE	Displays the status of warnings 321 to 336.
SV7.2.1.22-Warning 337 to 352 = 0000000000000000	NONE	Displays the status of warnings 337 to 352.
SV7.2.1.23-Warning 353 to 368 = 0000000000000000	NONE	Displays the status of warnings 353 to 368.
SV7.2.1.24-Warning 369 to 384 = 0000000000000000	NONE	Displays the status of warnings 369 to 384.
SV7.2.1.25-Warning 385 to 400 = 0000000000000000	NONE	Displays the status of warnings 385 to 400.
SV7.2.1.26-Warning 401 to 416 = 0000000000000000	NONE	Displays the status of warnings 401 to 416.
SV7.2.1.27-Warning 417 to 432 = 0000000000000000	NONE	Displays the status of warnings 417 to 432.
SV7.2.1.28-Warning 433 to 448 = 0000000000000000	NONE	Displays the status of warnings 433 to 448.
SV7.2.1.29-Warning 449 to 464 = 0000000000000000	NONE	Displays the status of warnings 449 to 464.
SV7.2.1.30-Warning 465 to 480 = 0000000000000000	NONE	Displays the status of warnings 465 to 480.
SV7.2.1.31-Warning 481 to 496 = 0000000000000000	NONE	Displays the status of warnings 481 to 496.
SV7.2.1.32-Warning 497 to 512 = 0000000000000000	NONE	Displays the status of warnings 497 to 512.

SV7.2.1a: Warning private status

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.2.1.1a-Warn. 1 to 16 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 1 to 16.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W1: Input high voltage</td><td>8</td><td>W9: Is limit</td></tr>
<tr><td>1</td><td>W2: Low voltage input</td><td>9</td><td>W10: Id limit</td></tr>
<tr><td>2</td><td>W3: High frecuency</td><td>10</td><td>W11: Derating Vdc</td></tr>
<tr><td>3</td><td>W4: Low frecuency</td><td>11</td><td>W12: Derating thermal IGBT</td></tr>
<tr><td>4</td><td>W5: No start conditions</td><td>12</td><td>W13: Derating thermal adm</td></tr>
<tr><td>5</td><td>Reserved.</td><td>13</td><td>W14: Low P</td></tr>
<tr><td>6</td><td>W7: P limit</td><td>14</td><td>W15: Low Vdc</td></tr>
<tr><td>7</td><td>W8: Q limit</td><td>15</td><td>Reserved.</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.2.1.2a-Warn. 17 to 32 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 17 to 32.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved.</td><td>8</td><td>W25: Close AC without premag.</td></tr>
<tr><td>1</td><td>W18: Tracker output low Vac</td><td>9</td><td>Reserved.</td></tr>
<tr><td>2</td><td>W19: LVRT</td><td>10</td><td>W27: Overvoltage limit AC</td></tr>
<tr><td>3</td><td>W20: HVRT</td><td>11</td><td>W28: Derating freq temp IGBT</td></tr>
<tr><td>4</td><td>W21: Curve P(f)</td><td>12</td><td>W29: DGPT2 Temperature alarm</td></tr>
<tr><td>5</td><td>W22: Curve Q(V)</td><td>13</td><td>W30: Overtemperature DU</td></tr>
<tr><td>6</td><td>W23: Curve P(V)</td><td>14</td><td>W31: MV Autocalibration</td></tr>
<tr><td>7</td><td>W24: DU close night</td><td>15</td><td>W32: MV Premagnetization</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.3a-Warn. 33 to 48 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 33 to 48.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W33: ECON Mode</td><td>8</td><td>W41: AC Open</td></tr>
<tr><td>1</td><td>W34: Night Conditions</td><td>9</td><td>W42:AC Disconnect</td></tr>
<tr><td>2</td><td>W35: Autonight Night Day</td><td>10</td><td>W43: Air conditioning Note: not available from version 5.4.0 onwards.</td></tr>
<tr><td>3</td><td>W36: Autonight Day Night</td><td>11</td><td>W44:Active heating enabled</td></tr>
<tr><td>4</td><td>W37: Night Mode</td><td>12</td><td>W45:No modules active heating</td></tr>
<tr><td>5</td><td>W38: Opening DU</td><td>13</td><td>W46:Filter LC Current Unbal</td></tr>
<tr><td>6</td><td>W39: DU Open</td><td>14</td><td>W47: C LC Filter</td></tr>
<tr><td>7</td><td>W40: Opening AC</td><td>15</td><td>W48: Current Balancing Mod</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.4a-Warn. 49 to 64 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 49 to 64.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved</td><td>8</td><td>W57: No meas Vac FPGA</td></tr>
<tr><td>1</td><td>W50: CTs measure</td><td>9</td><td>W58: Limit Q per VAC</td></tr>
<tr><td>2</td><td>W51: Speed limit</td><td>10</td><td>W59: Incorrect filter C cfg.</td></tr>
<tr><td>3</td><td>W52: VAC limit</td><td>11</td><td>W60: Discon. Cfilt Hall sensor</td></tr>
<tr><td>4</td><td>W53: NO_SYNC_FAC</td><td>12</td><td>W61: Act Balancing Triggering</td></tr>
<tr><td>5</td><td>W54: Vdc/Vac limit</td><td>13</td><td>Reserved</td></tr>
<tr><td>6</td><td>W55: AC Circ. Breaker Open</td><td>14</td><td>Reserved</td></tr>
<tr><td>7</td><td>W56: Silent mode derating</td><td>15</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.5a-Warn. 65 to 80 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 65 to 80.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W65: Overcurrent AC</td><td>8</td><td>W73:Desaturation R2(L)</td></tr>
<tr><td>1</td><td>W66: Mod. Bus DC Unbalanced</td><td>9</td><td>W74:Desaturation R3(L)</td></tr>
<tr><td>2</td><td>Reserved.</td><td>10</td><td>W75:Desaturation S1(H)</td></tr>
<tr><td>3</td><td>Reserved</td><td>11</td><td>W76:Desaturation S2(H)</td></tr>
<tr><td>4</td><td>W69: Overvoltage DC (P)</td><td>12</td><td>W77:Desaturation S2(L)</td></tr>
<tr><td>5</td><td>W70: Overvoltage DC (N)</td><td>13</td><td>W78:Desaturation S3(L)</td></tr>
<tr><td>6</td><td>W71:Desaturation R1(H)</td><td>14</td><td>W79:Desaturation T1(H)</td></tr>
<tr><td>7</td><td>W72:Desaturation R2(H)</td><td>15</td><td>W80:Desaturation T2(H)</td></tr>
</table></td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.2.1.6a-Warn. 81 to 96 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 81 to 96.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W81:Desaturation T2(L)</td><td>8</td><td>Reserved</td></tr>
<tr><td>1</td><td>W82:Desaturation T3(L)</td><td>9</td><td>W90: Tª IGBTS HW</td></tr>
<tr><td>2</td><td>W83:Multiple desaturation</td><td>10</td><td>Reserved.</td></tr>
<tr><td>3</td><td>W84:Comunications</td><td>11</td><td>Reserved</td></tr>
<tr><td>4</td><td>W85:Time out soft charge</td><td>12</td><td>Reserved.</td></tr>
<tr><td>5</td><td>W86:Feedback breaker</td><td>13</td><td>Reserved.</td></tr>
<tr><td>6</td><td>Reserved.</td><td>14</td><td>W95: Source 15V</td></tr>
<tr><td>7</td><td>Reserved</td><td>15</td><td>Reserved.</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.7a-Warn. 97 to 112 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 97 to 112.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved</td><td>8</td><td>Reserved</td></tr>
<tr><td>1</td><td>W98: Idc derivation</td><td>9</td><td>W106: Unbal/Max Vbus</td></tr>
<tr><td>2</td><td>W99: Differential Iac stop</td><td>10</td><td>W107: Iac not reached</td></tr>
<tr><td>3</td><td>Reserved</td><td>11</td><td>W108: Overcurrent Idc</td></tr>
<tr><td>4</td><td>Reserved.</td><td>12</td><td>W109: Vdc module low</td></tr>
<tr><td>5</td><td>W102: Unbalanced current</td><td>13</td><td>Reserved</td></tr>
<tr><td>6</td><td>W103: IGBTs temperature</td><td>14</td><td>W111: Fan module</td></tr>
<tr><td>7</td><td>W104:PCB temperature</td><td>15</td><td>W112: Emergency OCAC</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.8a-Warn. 113 to 128 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 113 to 128.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W113: Checksum comms</td><td>8</td><td>W121: Desaturation S1(L)</td></tr>
<tr><td>1</td><td>W114: Sensed current R</td><td>9</td><td>W122: Desaturation S3(H)</td></tr>
<tr><td>2</td><td>W115: Sensed current S</td><td>10</td><td>W123: Desaturation T1(L)</td></tr>
<tr><td>3</td><td>W116: Sensed current T</td><td>11</td><td>W124: Dessaturation T3(H)</td></tr>
<tr><td>4</td><td>W117: Stage HW2.0</td><td>12</td><td>W125: Diff Iac mod run</td></tr>
<tr><td>5</td><td>W118: Disable mod. SC</td><td>13</td><td>Reserved.</td></tr>
<tr><td>6</td><td>W119: Desaturation R1(L)</td><td>14</td><td>Reserved.</td></tr>
<tr><td>7</td><td>W120: Desaturation R3(H)</td><td>15</td><td>W128: Spring loaded</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.9a-Warn. 129 to 144 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 129 to 144.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W129: Isolation Module</td><td>8</td><td>Reserved</td></tr>
<tr><td>1</td><td>W130: Softcharge Feedback</td><td>9</td><td>Reserved</td></tr>
<tr><td>2</td><td>W131:Exceeded Faults limit</td><td>10</td><td>Reserved</td></tr>
<tr><td>3</td><td>W132: VDC HW Overvoltage</td><td>11</td><td>Reserved</td></tr>
<tr><td>4</td><td>Reserved.</td><td>12</td><td>Reserved</td></tr>
<tr><td>5</td><td>Reserved</td><td>13</td><td>Reserved</td></tr>
<tr><td>6</td><td>Reserved</td><td>14</td><td>Reserved</td></tr>
<tr><td>7</td><td>Reserved</td><td>15</td><td>Reserved</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.2.1.10a-Warn. 145 to 160 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 145 to 160.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved</td><td>8</td><td>Reserved.</td></tr>
<tr><td>1</td><td>Reserved</td><td>9</td><td>Reserved</td></tr>
<tr><td>2</td><td>Reserved</td><td>10</td><td>Reserved.</td></tr>
<tr><td>3</td><td>Reserved</td><td>11</td><td>W156: High resistor isolation</td></tr>
<tr><td>4</td><td>Reserved</td><td>12</td><td>Reserved.</td></tr>
<tr><td>5</td><td>Reserved.</td><td>13</td><td>W158: General Isolation</td></tr>
<tr><td>6</td><td>W151: Lockable module</td><td>14</td><td>W159: Intern Isolation</td></tr>
<tr><td>7</td><td>W152: PPC comms</td><td>15</td><td>W160: Extern Isolation</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.11a-Warn. 161 to 176 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 161 to 176.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W161: Fan impulsion MT</td><td>8</td><td>Reserved.</td></tr>
<tr><td>1</td><td>W162: Fan DU zone</td><td>9</td><td>W170: Overvoltage DC</td></tr>
<tr><td>2</td><td>W163: Fan zone filter LC</td><td>10</td><td>W171: No RTC</td></tr>
<tr><td>3</td><td>W164: Fan zone modules</td><td>11</td><td>W172: LC filter Temperature</td></tr>
<tr><td>4</td><td>W165: Fan motor status</td><td>12</td><td>W173: IMI Disabled Channels</td></tr>
<tr><td>5</td><td>W166: Temp sensor read</td><td>13</td><td>W174: Med Temp Com</td></tr>
<tr><td>6</td><td>W167:AntiPID ON</td><td>14</td><td>W175: BUSPLUS NTC problem</td></tr>
<tr><td>7</td><td>Reserved.</td><td>15</td><td>W176: DC source fuse</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.12a-Warn. 177 to 192 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 177 to 192.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved.</td><td>8</td><td>W185: IMI not calibrated</td></tr>
<tr><td>1</td><td>W178:Blown fuse</td><td>9</td><td>W186: Supply low</td></tr>
<tr><td>2</td><td>W179:FTP_OPEN</td><td>10</td><td>W187: DU Delta Temp Threshold</td></tr>
<tr><td>3</td><td>W180:SD problem</td><td>11</td><td>W188: DU sensor lost comms</td></tr>
<tr><td>4</td><td>W181: Aux transf overvoltaje</td><td>12</td><td>W189: Cold start ongoing</td></tr>
<tr><td>5</td><td>W182: Therma Bus Plus</td><td>13</td><td>W190: AEMO comms lost</td></tr>
<tr><td>6</td><td>W183: Internal comms</td><td>14</td><td>W191:Antipid wrong V</td></tr>
<tr><td>7</td><td>W184: MV transformer Temp</td><td>15</td><td>W192:Antipid V not reached</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.13a-Warn. 193 to 208 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 193 to 208.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W193: Antipid retries exceed</td><td>8</td><td>Reserved</td></tr>
<tr><td>1</td><td>Reserved</td><td>9</td><td>Reserved</td></tr>
<tr><td>2</td><td>W195: Protec. vent inductance</td><td>10</td><td>Reserved</td></tr>
<tr><td>3</td><td>W196: Fan Incorrect Frames</td><td>11</td><td>Reserved</td></tr>
<tr><td>4</td><td>W197: Modbus traffic exceed</td><td>12</td><td>Reserved</td></tr>
<tr><td>5</td><td>Reserved</td><td>13</td><td>Reserved</td></tr>
<tr><td>6</td><td>W199: DC Source</td><td>14</td><td>Reserved</td></tr>
<tr><td>7</td><td>Reserved</td><td>15</td><td>Reserved</td></tr>
</table></td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.2.1.14a-Warn. 209 to 224 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 209 to 224.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved</td><td>8</td><td>Reserved</td></tr>
<tr><td>1</td><td>Reserved</td><td>9</td><td>Reserved</td></tr>
<tr><td>2</td><td>Reserved</td><td>10</td><td>Reserved</td></tr>
<tr><td>3</td><td>Reserved</td><td>11</td><td>Reserved</td></tr>
<tr><td>4</td><td>Reserved</td><td>12</td><td>Reserved</td></tr>
<tr><td>5</td><td>Reserved</td><td>13</td><td>Reserved</td></tr>
<tr><td>6</td><td>Reserved</td><td>14</td><td>Reserved</td></tr>
<tr><td>7</td><td>Reserved</td><td>15</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.15a-Warn. 225 to 240 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 225 to 240.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved</td><td>8</td><td>Reserved</td></tr>
<tr><td>1</td><td>Reserved</td><td>9</td><td>Reserved</td></tr>
<tr><td>2</td><td>Reserved</td><td>10</td><td>Reserved</td></tr>
<tr><td>3</td><td>Reserved</td><td>11</td><td>Reserved</td></tr>
<tr><td>4</td><td>Reserved</td><td>12</td><td>Reserved</td></tr>
<tr><td>5</td><td>Reserved</td><td>13</td><td>Reserved</td></tr>
<tr><td>6</td><td>Reserved</td><td>14</td><td>Reserved</td></tr>
<tr><td>7</td><td>Reserved</td><td>15</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.16a-Warn. 241 to 256 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 241 to 256.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W241: Delta T max DU</td><td>8</td><td>Reserved</td></tr>
<tr><td>1</td><td>Reserved</td><td>9</td><td>Reserved</td></tr>
<tr><td>2</td><td>Reserved</td><td>10</td><td>Reserved</td></tr>
<tr><td>3</td><td>Reserved</td><td>11</td><td>W252: RFI OverCur Persistent</td></tr>
<tr><td>4</td><td>Reserved</td><td>12</td><td>W253: Imbalance V</td></tr>
<tr><td>5</td><td>Reserved</td><td>13</td><td>Reserved.</td></tr>
<tr><td>6</td><td>Reserved</td><td>14</td><td>W255:BAT open detection</td></tr>
<tr><td>7</td><td>Reserved</td><td>15</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.17a-Warn. 257 to 272 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 257 to 272.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W257: Soft-Charge AC Timeout</td><td>8</td><td>Reserved</td></tr>
<tr><td>1</td><td>Reserved</td><td>9</td><td>Reserved</td></tr>
<tr><td>2</td><td>Reserved</td><td>10</td><td>Reserved</td></tr>
<tr><td>3</td><td>Reserved</td><td>11</td><td>Reserved</td></tr>
<tr><td>4</td><td>Reserved</td><td>12</td><td>Reserved</td></tr>
<tr><td>5</td><td>Reserved</td><td>13</td><td>Reserved</td></tr>
<tr><td>6</td><td>Reserved</td><td>14</td><td>Reserved</td></tr>
<tr><td>7</td><td>Reserved</td><td>15</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.18a-Warn. 273 to 288 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 273 to 288.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved</td><td>8</td><td>Reserved</td></tr>
<tr><td>1</td><td>Reserved</td><td>9</td><td>Reserved</td></tr>
<tr><td>2</td><td>Reserved</td><td>10</td><td>Reserved</td></tr>
<tr><td>3</td><td>Reserved</td><td>11</td><td>Reserved</td></tr>
<tr><td>4</td><td>Reserved</td><td>12</td><td>Reserved</td></tr>
<tr><td>5</td><td>Reserved</td><td>13</td><td>Reserved</td></tr>
<tr><td>6</td><td>Reserved</td><td>14</td><td>Reserved</td></tr>
<tr><td>7</td><td>Reserved</td><td>15</td><td>Reserved</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV7.2.1.19a-Warn. 289 to 304 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 289 to 304.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Reserved</td><td>8</td><td>Reserved</td></tr>
<tr><td>1</td><td>Reserved</td><td>9</td><td>Reserved</td></tr>
<tr><td>2</td><td>Reserved</td><td>10</td><td>Reserved</td></tr>
<tr><td>3</td><td>Reserved</td><td>11</td><td>Reserved</td></tr>
<tr><td>4</td><td>Reserved</td><td>12</td><td>Reserved</td></tr>
<tr><td>5</td><td>Reserved</td><td>13</td><td>Reserved</td></tr>
<tr><td>6</td><td>Reserved</td><td>14</td><td>W303: RFI OverCurrent</td></tr>
<tr><td>7</td><td>Reserved</td><td>15</td><td>W304: Imbalance P</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.20a-Warn. 305 to 320 = 0000000000000000</td><td>NONE</td><td>Displays the warning private status 305 to 320.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>W305: Limit VDC BESS</td><td>8</td><td>W313: Unbalanced Vdc</td></tr>
<tr><td>1</td><td>W306: CCL Limit</td><td>9</td><td>Reserved</td></tr>
<tr><td>2</td><td>W307: DCL Limit</td><td>10</td><td>Reserved</td></tr>
<tr><td>3</td><td>W308: HW Limit</td><td>11</td><td>Reserved</td></tr>
<tr><td>4</td><td>W309: Busbar Disable</td><td>12</td><td>Reserved</td></tr>
<tr><td>5</td><td>W310: Module Disable</td><td>13</td><td>Reserved</td></tr>
<tr><td>6</td><td>W311: Low Vdc</td><td>14</td><td>Reserved</td></tr>
<tr><td>7</td><td>W312: High Vdc</td><td>15</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>SV7.2.1.21a-Warn. 321 to 336 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
<tr><td>SV7.2.1.22a-Warn. 337 to 352 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
<tr><td>SV7.2.1.23a-Warn. 353 to 368 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
<tr><td>SV7.2.1.24a-Warn. 369 to 384 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
<tr><td>SV7.2.1.25a-Warn. 385 to 400 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
<tr><td>SV7.2.1.26a-Warn. 401 to 416 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
<tr><td>SV7.2.1.27a-Warn. 417 to 432 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
<tr><td>SV7.2.1.28a-Warn. 433 to 448 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
<tr><td>SV7.2.1.29a-Warn. 449 to 464 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
<tr><td>SV7.2.1.30a-Warn. 465 to 480 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
<tr><td>SV7.2.1.31a-Warn. 481 to 496 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
<tr><td>SV7.2.1.32a-Warn. 497 to 512 = 0000000000000000</td><td>NONE</td><td>Reserved.</td></tr>
</table>

EN

SV7.2.2: Warnings history

Note: See section "SV7.1.2: Faults history" for more information about the internal interpretation procedure of dates and times for warning history.

SCREEN	DESCRIPTION	FUNCTION
SV7.2.2.1-Last warning = 0	Register number 1 of the warning history.	Displays a list of the last ten warnings, in chronological order. When a warning occurs, it will be displayed on the status bar flashing with the current state. It will remain being displayed until the problem is fixed. The most recent warning is shown first and stored. Whenever a new warning occurs, it will be placed at the top of the list and the oldest warning messages will drop a position, deleting the last warning message.
SV7.2.2.2-Date = 01/01/2000 00:00	Date and time of the register number 1 of warning history.	
SV7.2.2.3-Ninth warning = 0	Register number 2 of the warning history.	
SV7.2.2.4-Date = 01/01/2000 00:00	Date and time of the register number 2 of warning history.	
SV7.2.2.5-Eighth warning = 0	Register number 3 of the warning history.	
SV7.2.2.6-Date = 01/01/2000 00:00	Date and time of the register number 3 of warning history.	
SV7.2.2.7-Seventh warning = 0	Register number 4 of the warning history.	
SV7.2.2.8-Date = 01/01/2000 00:00	Date and time of the register number 4 of warning history.	
SV7.2.2.9-Sixth warning = 0	Register number 5 of the warning history.	
SV7.2.2.10-Date = 01/01/2000 00:00	Date and time of the register number 5 of warning history.	
SV7.2.2.11-Fifth warning = 0	Register number 6 of the warning history.	
SV7.2.2.12-Date = 01/01/2000 00:00	Date and time of the register number 6 of warning history.	
SV7.2.2.13-Fourth warning = 0	Register number 7 of the warning history.	
SV7.2.2.14-Date = 01/01/2000 00:00	Date and time of the register number 7 of warning history.	
SV7.2.2.15-Third warning = 0	Register number 8 of the warning history.	
SV7.2.2.16-Date = 01/01/2000 00:00	Date and time of the register number 8 of warning history.	
SV7.2.2.17-Second warning = 0	Register number 9 of the warning history.	
SV7.2.2.18-Date = 01/01/2000 00:00	Date and time of the register number 9 of warning history.	
SV7.2.2.19-First warning = 0	Register number 10 of the warning history.	
SV7.2.2.20-Date = 01/01/2000 00:00	Date and time of the register number 10 of warning history.	

SV7.2.3: Private warn history

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	DESCRIPTION	FUNCTION
SV7.2.3.1-Last warning = 0	Register number 1 of the warning history.	Displays a list of the last ten private warnings, in chronological order. When a private warning occurs, it will be displayed on the status bar flashing with the current state. It will remain being displayed until the problem is fixed. The most recent private warning is shown first and stored. Whenever a new private warning occurs, it will be placed at the top of the list and the oldest private warning messages will drop a position deleting the last warning message.
SV7.2.3.2-Date = 01/01/2000 00:00	Date and time of the register number 1 of warning history.	
SV7.2.3.3-Ninth warning = 0	Register number 2 of the warning history.	
SV7.2.3.4-Date = 01/01/2000 00:00	Date and time of the register number 2 of warning history.	
SV7.2.3.5-Eighth warning = 0	Register number 3 of the warning history.	
SV7.2.3.6-Date = 01/01/2000 00:00	Date and time of the register number 3 of warning history.	
SV7.2.3.7-Seventh warning = 0	Register number 4 of the warning history.	
SV7.2.3.8-Date = 01/01/2000 00:00	Date and time of the register number 4 of warning history.	
SV7.2.3.9-Sixth warning = 0	Register number 5 of the warning history.	
SV7.2.3.10-Date = 01/01/2000 00:00	Date and time of the register number 5 of warning history.	
SV7.2.3.11-Fifth warning = 0	Register number 6 of the warning history.	
SV7.2.3.12-Date = 01/01/2000 00:00	Date and time of the register number 6 of warning history.	
SV7.2.3.13-Fourth warning = 0	Register number 7 of the warning history.	
SV7.2.3.14-Date = 01/01/2000 00:00	Date and time of the register number 7 of warning history.	
SV7.2.3.15-Third warning = 0	Register number 8 of the warning history.	
SV7.2.3.16-Date = 01/01/2000 00:00	Date and time of the register number 8 of warning history.	
SV7.2.3.17-Second warning = 0	Register number 9 of the warning history.	
SV7.2.3.18-Date = 01/01/2000 00:00	Date and time of the register number 9 of warning history.	
SV7.2.3.19-First warning = 0	Register number 10 of the warning history.	
SV7.2.3.20-Date = 01/01/2000 00:00	Date and time of the register number 10 of warning history.	

EN

5.7.3. Subgroup SV7.3: Events

SV7.3.1: Events status

Parameters within this group show the status of each event (0 = Not active; X = Active) in groups of 16 events (being the lowest number the bit to the upmost right).

SCREEN	UNITS	DESCRIPTION
SV7.3.1.1-Event 1 to 16 = 0000000000000000	NONE	Displays the status of events 1 to 16.
SV7.3.1.2-Event 17 to 32 = 0000000000000000	NONE	Displays the status of events 17 to 32.
SV7.3.1.3-Events 33 to 48 = 0000000000000000	NONE	Displays the status of events 33 to 48.
SV7.3.1.4-Event 49 to 64 = 0000000000000000	NONE	Displays the status of events 49 to 64.

SV7.3.2: Events history

Note: See section "SV7.1.2: Faults history" for more information about the internal interpretation procedure of dates and times for events history.

SCREEN	DESCRIPTION	FUNCTION
SV7.3.2.1-Last event = No events	Register number 1 of the event history.	Displays a list of the last ten events, in chronological order. When an event occurs, it will be displayed on the status bar flashing with the current state. It will remain being displayed until the problem is fixed. The most recent event is shown first and stored. Whenever a new event occurs, it will be placed at the top of the list and the oldest event messages will drop a position deleting the last event message. See section "Events" for more information on possible events and their codes.
SV7.3.2.2-Date = 01/01/2000 00:00	Date and time of the register number 1 of event history.	
SV7.3.2.3-Ninth event = No events	Register number 2 of the event history.	
SV7.3.2.4-Date = 01/01/2000 00:00	Date and time of the register number 2 of event history.	
SV7.3.2.5-Eighth event = No events	Register number 3 of the event history.	
SV7.3.2.6-Date = 01/01/2000 00:00	Date and time of the register number 3 of event history.	
SV7.3.2.7-Seventh event = No events	Register number 4 of the event history.	
SV7.3.2.8-Date = 01/01/2000 00:00	Date and time of the register number 4 of event history.	
SV7.3.2.9-Sixth event = No events	Register number 5 of the event history.	
SV7.3.2.10-Date = 01/01/2000 00:00	Date and time of the register number 5 of event history.	
SV7.3.2.11-Fifth event = No events	Register number 6 of the event history.	
SV7.3.2.12-Date = 01/01/2000 00:00	Date and time of the register number 6 of event history.	
SV7.3.2.13-Fourth event = No events	Register number 7 of the event history.	
SV7.3.2.14-Date = 01/01/2000 00:00	Date and time of the register number 7 of event history.	
SV7.3.2.15-Third event = No events	Register number 8 of the event history.	
SV7.3.2.16-Date = 01/01/2000 00:00	Date and time of the register number 8 of event history.	
SV7.3.2.17-Second event = No events	Register number 9 of the event history.	
SV7.3.2.18-Date = 01/01/2000 00:00	Date and time of the register number 9 of event history.	
SV7.3.2.19-First event = No events	Register number 10 of the event history.	
SV7.3.2.20-Date = 01/01/2000 00:00	Date and time of the register number 10 of event history.	

5.7.4. Subgroup SV7.4: Module events

This subgroup allows to monitor the behaviour of the power stage of the inverter. For that, it is required to save the last ten events of each module FPGA in a similar way to the previous subgroups SV7.1.2: Faults history, SV7.2.2: Warnings history or SV7.3.2: Events history. Depending on the modules configured internally, this subgroup shows the respective lists.

Notes:

- This subgroup is displayed if [G1.3-User ≥ Power Electronics staff] and if [SV1.3.1.4-Number of modules ≥ n], being 'n' the number of the module.
- See section "SV7.1.2: Faults history" for more information about the internal interpretation procedure of dates and times.

SV7.4.1-Module 1 events

SCREEN	DESCRIPTION	FUNCTION
SV7.4.1.1-Mod 1 last event = Invalid	Register number 1 of the event history.	Displays a list of module status messages, in chronological order. The most recent event is shown first and stored. Whenever a new event occurs, it will be placed at the top of the list and the oldest event messages will drop a position deleting the last event message. See section "Module status messages" for more information on the statuses and their codes.
SV7.4.1.2-Date = 01/01/2000 00:00	Date and time of the register number 1 of event history.	
SV7.4.1.3-Mod 1 nineth event = Invalid	Register number 2 of the event history.	
SV7.4.1.4-Date = 01/01/2000 00:00	Date and time of the register number 2 of event history.	
SV7.4.1.5-Mod 1 eight event = Invalid	Register number 3 of the event history.	
SV7.4.1.6-Date = 01/01/2000 00:00	Date and time of the register number 3 of event history.	
SV7.4.1.7-Mod 1 seventh event = Invalid	Register number 4 of the event history.	
SV7.4.1.8-Date = 01/01/2000 00:00	Date and time of the register number 4 of event history.	
SV7.4.1.9-Mod 1 sixth event = Invalid	Register number 5 of the event history.	
SV7.4.1.10-Date = 01/01/2000 00:00	Date and time of the register number 5 of event history.	
SV7.4.1.11-Mod 1 fifth event = Invalid	Register number 6 of the event history.	
SV7.4.1.12-Date = 01/01/2000 00:00	Date and time of the register number 6 of event history.	
SV7.4.1.13-Mod 1 fourth event = Invalid	Register number 7 of the event history.	
SV7.4.1.14-Date = 01/01/2000 00:00	Date and time of the register number 7 of event history.	
SV7.4.1.15-Mod 1 third event = Invalid	Register number 8 of the event history.	
SV7.4.1.16-Date = 01/01/2000 00:00	Date and time of the register number 8 of event history.	
SV7.4.1.17-Mod 1 second event = Invalid	Register number 9 of the event history.	
SV7.4.1.18-Date = 01/01/2000 00:00	Date and time of the register number 9 of event history.	
SV7.4.1.19-Mod 1 first event = Invalid	Register number 10 of the event history.	
SV7.4.1.20-Date = 01/01/2000 00:00	Date and time of the register number 10 of event history.	

EN

SV7.4.2-Module 2 events

SCREEN	DESCRIPTION	FUNCTION
SV7.4.2.1-Mod 2 last event = Invalid	Register number 1 of the event history.	Displays a list of module status messages, in chronological order. The most recent event is shown first and stored. Whenever a new event occurs, it will be placed at the top of the list and the oldest event messages will drop a position deleting the last event message. See section "Module status messages" for more information on the statuses and their codes.
SV7.4.2.2-Date = 01/01/2000 00:00	Date and time of the register number 1 of event history.	
SV7.4.2.3-Mod 2 nineth event = Invalid	Register number 2 of the event history.	
SV7.4.2.4-Date = 01/01/2000 00:00	Date and time of the register number 2 of event history.	
SV7.4.2.5-Mod 2 eight event = Invalid	Register number 3 of the event history.	
SV7.4.2.6-Date = 01/01/2000 00:00	Date and time of the register number 3 of event history.	
SV7.4.2.7-Mod 2 seventh event = Invalid	Register number 4 of the event history.	
SV7.4.2.8-Date = 01/01/2000 00:00	Date and time of the register number 4 of event history.	
SV7.4.2.9-Mod 2 sixth event = Invalid	Register number 5 of the event history.	
SV7.4.2.10-Date = 01/01/2000 00:00	Date and time of the register number 5 of event history.	
SV7.4.2.11-Mod 2 fifth event = Invalid	Register number 6 of the event history.	
SV7.4.2.12-Date = 01/01/2000 00:00	Date and time of the register number 6 of event history.	
SV7.4.2.13-Mod 2 fourth event = Invalid	Register number 7 of the event history.	
SV7.4.2.14-Date = 01/01/2000 00:00	Date and time of the register number 7 of event history.	
SV7.4.2.15-Mod 2 third event = Invalid	Register number 8 of the event history.	
SV7.4.2.16-Date = 01/01/2000 00:00	Date and time of the register number 8 of event history.	
SV7.4.2.17-Mod 2 second event = Invalid	Register number 9 of the event history.	
SV7.4.2.18-Date = 01/01/2000 00:00	Date and time of the register number 9 of event history.	
SV7.4.2.19-Mod 2 first event = Invalid	Register number 10 of the event history.	
SV7.4.2.20-Date = 01/01/2000 00:00	Date and time of the register number 10 of event history.	

SV7.4.2-Module 3 events

SCREEN	DESCRIPTION	FUNCTION
SV7.4.3.1-Mod 3 last event = Invalid	Register number 1 of the event history.	Displays a list of module status messages, in chronological order. The most recent event is shown first and stored. Whenever a new event occurs, it will be placed at the top of the list and the oldest event messages will drop a position deleting the last event message. See section "Module status messages" for more information on the statuses and their codes.
SV7.4.3.2-Date = 01/01/2000 00:00	Date and time of the register number 1 of event history.	
SV7.4.3.3-Mod 3 nineth event = Invalid	Register number 2 of the event history.	
SV7.4.3.4-Date = 01/01/2000 00:00	Date and time of the register number 2 of event history.	
SV7.4.3.5-Mod 3 eight event = Invalid	Register number 3 of the event history.	
SV7.4.3.6-Date = 01/01/2000 00:00	Date and time of the register number 3 of event history.	
SV7.4.3.7-Mod 3 seventh event = Invalid	Register number 4 of the event history.	
SV7.4.3.8-Date = 01/01/2000 00:00	Date and time of the register number 4 of event history.	
SV7.4.3.9-Mod 3 sixth event = Invalid	Register number 5 of the event history.	
SV7.4.3.10-Date = 01/01/2000 00:00	Date and time of the register number 5 of event history.	
SV7.4.3.11-Mod 3 fifth event = Invalid	Register number 6 of the event history.	
SV7.4.3.12-Date = 01/01/2000 00:00	Date and time of the register number 6 of event history.	
SV7.4.3.13-Mod 3 fourth event = Invalid	Register number 7 of the event history.	
SV7.4.3.14-Date = 01/01/2000 00:00	Date and time of the register number 7 of event history.	
SV7.4.3.15-Mod 3 third event = Invalid	Register number 8 of the event history.	
SV7.4.3.16-Date = 01/01/2000 00:00	Date and time of the register number 8 of event history.	
SV7.4.3.17-Mod 3 second event = Invalid	Register number 9 of the event history.	
SV7.4.3.18-Date = 01/01/2000 00:00	Date and time of the register number 9 of event history.	
SV7.4.3.19-Mod 3 first event = Invalid	Register number 10 of the event history.	
SV7.4.3.20-Date = 01/01/2000 00:00	Date and time of the register number 10 of event history.	

EN

SV7.4.4-Module 4 events

SCREEN	DESCRIPTION	FUNCTION
SV7.4.4.1-Mod 4 last event = Invalid	Register number 1 of the event history.	Displays a list of module status messages, in chronological order. The most recent event is shown first and stored. Whenever a new event occurs, it will be placed at the top of the list and the oldest event messages will drop a position deleting the last event message. See section "Module status messages" for more information on the statuses and their codes.
SV7.4.4.2-Date = 01/01/2000 00:00	Date and time of the register number 1 of event history.	
SV7.4.4.3-Mod 4 nineth event = Invalid	Register number 2 of the event history.	
SV7.4.4.4-Date = 01/01/2000 00:00	Date and time of the register number 2 of event history.	
SV7.4.4.5-Mod 4 eight event = Invalid	Register number 3 of the event history.	
SV7.4.4.6-Date = 01/01/2000 00:00	Date and time of the register number 3 of event history.	
SV7.4.4.7-Mod 4 seventh event = Invalid	Register number 4 of the event history.	
SV7.4.4.8-Date = 01/01/2000 00:00	Date and time of the register number 4 of event history.	
SV7.4.4.9-Mod 4 sixth event = Invalid	Register number 5 of the event history.	
SV7.4.4.10-Date = 01/01/2000 00:00	Date and time of the register number 5 of event history.	
SV7.4.4.11-Mod 4 fifth event = Invalid	Register number 6 of the event history.	
SV7.4.4.12-Date = 01/01/2000 00:00	Date and time of the register number 6 of event history.	
SV7.4.4.13-Mod 4 fourth event = Invalid	Register number 7 of the event history.	
SV7.4.4.14-Date = 01/01/2000 00:00	Date and time of the register number 7 of event history.	
SV7.4.4.15-Mod 4 third event = Invalid	Register number 8 of the event history.	
SV7.4.4.16-Date = 01/01/2000 00:00	Date and time of the register number 8 of event history.	
SV7.4.4.17-Mod 4 second event = Invalid	Register number 9 of the event history.	
SV7.4.4.18-Date = 01/01/2000 00:00	Date and time of the register number 9 of event history.	
SV7.4.4.19-Mod 4 first event = Invalid	Register number 10 of the event history.	
SV7.4.4.20-Date = 01/01/2000 00:00	Date and time of the register number 10 of event history.	

5.8. Group SV9: IMD

This group allows the visualization of the insulation monitoring device (IMD) information. This meter is able to measure the earth resistance of the inverter. If this is too low, the inverter is internally configured to go to a non-risk situation until the technical staff can perform a shunt to earth. There are two types of insulation testers: two-channel (one channel to measure AC and one channel to measure DC) and five-channel (one channel to measure AC and 4 channels to measure DC).

Normally, for PCSM/K inverters, the five-channel option is used. For HEM/K inverters, the two-channel option is used.

Note: This group is displayed if [G1.3-User ≥ Qualified customer].

5.8.1. Subgroup SV9.1: General

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV9.1.0-Comms Status = Disabled	NONE	Shows the communication status of uC with isolation monitoring device. **STATUS** / **DESCRIPTION** Disabled: IMI not enabled. It may not communicate within the first 25 seconds of powering up the electronics. In this case, check that parameter [SG4.21-IMI] is enabled. Comms OK: IMI communications are correct. Timeout: A timeout has been exceeded when trying to establish communications with the IMI. Communications error. Error CRC: Communications error with the IMI. Invalid modbus code: An attempt is made to communicate with the IMI with an unsupported Modbus code (Modbus protocol).
SV9.1.1-Mode = STOP	NONE	Indicates if the isolation monitoring device is measuring. **STATUS** / **DESCRIPTION** STOP: IMI is not measuring. CONTINUOS: IMI is measuring continuously.
SV9.1.2-Request Channel = AC	NONE	Indicates which channel of the isolation monitoring device has been ordered to be measured. **STATUS** / **DESCRIPTION** AC: IMI requests to measure AC channel. DC1: IMI requests to measure DC1 channel. DC2: IMI requests to measure DC2 channel. DC3: IMI requests to measure DC3 channel. DC4: IMI requests to measure DC4 channel.
SV9.1.3-Current Channel = AC	NONE	Indicates which channel of the isolation monitoring device is being requested to be measured. **STATUS** / **DESCRIPTION** AC: IMI requests to measure AC channel. DC1: IMI requests to measure DC1 channel. DC2: IMI requests to measure DC2 channel. DC3: IMI requests to measure DC3 channel. DC4: IMI requests to measure DC4 channel.
SV9.1.4-Measure time = 0 s	s	Indicates the measurement time the channel is taking to measure.
SV9.1.5-Autodiagnos Remaining = 0 min	min	Indicates the time remaining to launch the next auto diagnosis.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV9.1.6-General State = IDLE</td><td>NONE</td><td>Shows the current status of isolation monitoring device.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>IDLE</td><td>Isolation monitoring device is stopped. Its algorithm is not running.</td></tr>
<tr><td>MEASURE</td><td>Isolation monitoring device is measuring on one of its channels.</td></tr>
<tr><td>AUTODIAGNOSIS</td><td>Isolation monitoring device is running the auto diagnosis algorithm.</td></tr>
<tr><td>ERROR</td><td>Isolation monitoring device has reported an internal fault, if this status appears, it will be reported as F49.</td></tr>
</table></td></tr>
<tr><td>SV9.1.7-Measure State = PULSE POS</td><td>NONE</td><td>Indicates the measurement status.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>PULSE POS</td><td>A positive pulse is being measured.</td></tr>
<tr><td>PULSE NEG</td><td>A negative pulse is being measured.</td></tr>
</table></td></tr>
<tr><td>SV9.1.8-Autodiagnosis State = STEP 1</td><td>NONE</td><td>Indicates the status of the auto diagnosis process when the auto diagnosis algorithm is executed. Indicates at which step of the auto diagnosis process it is.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>STEP 1</td><td>The auto diagnosis process is in Step 1.</td></tr>
<tr><td>STEP 2</td><td>The auto diagnosis process is in Step 2.</td></tr>
<tr><td>STEP 3</td><td>The auto diagnosis process is in Step 3.</td></tr>
<tr><td>OK</td><td>The internal diagnosis of the isolation monitoring device is correct. The main control board has not fault.</td></tr>
<tr><td>ERROR</td><td>The internal diagnosis of the isolation monitoring device has failed. The main control board does not work properly.</td></tr>
</table></td></tr>
<tr><td>SV9.1.9-Digital Inputs = 00000000</td><td>NONE</td><td>Indicates bit by bit the status of the digital inputs that are connected to the isolation monitoring device main control board. The order is as follows:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Busplus Feedback</td><td>Status of the digital output feedback that grounds the Bus Plus contactor.</td></tr>
<tr><td>1</td><td>DC overvoltage</td><td>Status of the DC voltage limiter connected to the inverter. It trips [F200: Overvoltage Lim DC] fault when an overvoltage is present.</td></tr>
<tr><td>2</td><td>GFDI blown fuse</td><td>GFDI feedback indicating that one of the PV negative pole (Bus Plus) fuses has blown.</td></tr>
<tr><td>3</td><td>Smoke Detector</td><td>Status of the smoke detector connected to the IMI. It is activated when dangerous smoke levels are detected in the system.</td></tr>
<tr><td>4</td><td>Reserved 3.</td><td>Not used.</td></tr>
<tr><td>5</td><td>GFDI Shoot.</td><td>GFDI feedback indicating that the GFDI has tripped.</td></tr>
<tr><td>6</td><td>GFDI Closed.</td><td>GFDI feedback indicating that it has closed.</td></tr>
<tr><td>7</td><td>GFDI Open</td><td>GFDI feedback indicating that it is open.</td></tr>
</table></td></tr>
<tr><td>SV9.1.10-Digital Outputs 1 = 0000000000000</td><td>NONE</td><td>Indicates bit by bit the status of the digital outputs that are connected to the isolation monitoring device PCB. The order is as follows:
<table>
<tr><th>BIT</th><th>STATUS</th><th>BIT</th><th>STATUS</th></tr>
<tr><td>0</td><td>Measuring relay 1</td><td>7</td><td>Reset measurement 2</td></tr>
<tr><td>1</td><td>Measuring relay 2</td><td>8</td><td>Relay test 1</td></tr>
<tr><td>2</td><td>Positive pulse</td><td>9</td><td>Relay test 2</td></tr>
<tr><td>3</td><td>Negative pulse</td><td>10</td><td>J29 Output</td></tr>
<tr><td>4</td><td>Measure gain 1</td><td>11</td><td>GFDI Close</td></tr>
<tr><td>5</td><td>Measure gain 2</td><td>12</td><td>GFDI Open/Close Antipid</td></tr>
<tr><td>6</td><td>Reset measurement 1</td><td></td><td></td></tr>
</table></td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV9.1.11-Digital Outputs 2 = 00000000</td><td>NONE</td><td>Indicates bit by bit the status of the digital outputs that are connected to the isolation monitoring device main control board. The order is as follows:
<table>
<tr><th>BIT</th><th>STATUS</th><th>BIT</th><th>STATUS</th></tr>
<tr><td>0</td><td>Channel 1</td><td>4</td><td>Channel 5</td></tr>
<tr><td>1</td><td>Channel 2</td><td>5</td><td>Channel Fault</td></tr>
<tr><td>2</td><td>Channel 3</td><td>6</td><td>Reserved 2</td></tr>
<tr><td>3</td><td>Channel 4</td><td>7</td><td>AntiPID / Cont Busplus</td></tr>
</table></td></tr>
</table>

5.8.2. Subgroup SV9.2: Channels

Note: Parameters within this subgroup are only displayed if [G9.1.1-Enable = Yes] and [G9.2.X.1-Enable = Yes].

EN

SV9.2.1: Channel AC

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV9.2.1.1-Status = 0000</td><td>NONE</td><td>Indicates the channel AC status of the isolation monitoring device.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Error</td><td>Shows if the channel is in isolation fault.</td></tr>
<tr><td>1</td><td>Warning</td><td>Shows if the channel is in isolation warning.</td></tr>
<tr><td>2</td><td>Rmax</td><td>Shows if the channel exceeds maximum 2 Mohms.</td></tr>
<tr><td>3</td><td>Ready</td><td>The channel cool down time is set to ready. This time defines when a channel must pass without measuring for its associated resistor card to cool down to the same temperature it had heated while measuring.</td></tr>
</table></td></tr>
<tr><td>SV9.2.1.2-N Measures = 0</td><td>NONE</td><td>Defines the number of measurements the channel has performed since the last isolation monitoring device reset.</td></tr>
<tr><td>SV9.2.1.3-Resistor = 0 Ohms</td><td>Ohms</td><td>Measured resistance.</td></tr>
<tr><td>SV9.2.1.4-Last measure time = 0 s</td><td>s</td><td>Shows the time taken to measure the last measurement.</td></tr>
<tr><td>SV9.2.1.5-Time to ready = 0 s</td><td>s</td><td>Shows time left to cool down and retake a measurement.</td></tr>
<tr><td>SV9.2.1.6-T. aval. no measure = 0 min</td><td>min</td><td>Shows the time that the IMD is taking to get a new valid measure.</td></tr>
<tr><td>SV9.2.1.7-Res no filtered = 0 Ohms</td><td>Ohms</td><td>Indicates the resistance value without applying the AC channel's RP filter.</td></tr>
</table>

SV9.2.2: Channel DC 1

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV9.2.2.1-Status = 0000</td><td>NONE</td><td>Indicates the channel DC1 status of the isolation monitoring device.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Error</td><td>Shows if the channel is in isolation fault.</td></tr>
<tr><td>1</td><td>Warning</td><td>Shows if the channel is in isolation warning.</td></tr>
<tr><td>2</td><td>Rmax</td><td>Shows if the channel exceeds maximum 2 Mohms.</td></tr>
<tr><td>3</td><td>Ready</td><td>The channel cool down time is set to ready. This time defines when a channel must pass without measuring for its associated resistor card to cool down to the same temperature it had heated while measuring.</td></tr>
</table></td></tr>
</table>

SCREEN	UNITS	DESCRIPTION
SV9.2.2.2-N Measures = 0	NONE	Defines the number of measurements the channel has performed since the last isolation monitoring device reset.
SV9.2.2.3-Resistor = 0 Ohms	Ohms	Measured resistance.
SV9.2.2.4-Last measure time = 0 s	s	Shows the time taken to measure the last measurement.
SV9.2.2.5-Time to ready = 0 s	s	Shows time left to cool down and retake a measurement.
SV9.2.2.6-T. aval. no measure = 0 min	min	Shows the time that the IMD is taking to get a new valid measure.
SV9.2.2.7-Res no filtered = 0 Ohms	Ohms	Indicates the resistance value without applying the RP filter of channel DC 1.

SV9.2.3: Channel DC 2

SCREEN	UNITS	DESCRIPTION
SV9.2.3.1-Status = 0000	NONE	Indicates the channel DC 2 status of the isolation monitoring device. BIT / STATUS / DESCRIPTION 0 / Error / Shows if the channel is in isolation fault. 1 / Warning / Shows if the channel is in isolation warning. 2 / Rmax / Shows if the channel exceeds maximum 2 Mohms. 3 / Ready / The channel cool down time is set to ready. This time defines when a channel must pass without measuring for its associated resistor card to cool down to the same temperature it had heated while measuring.
SV9.2.3.2-N Measures = 0	NONE	Defines the number of measurements the channel has performed since the last isolation monitoring device reset.
SV9.2.3.3-Resistor = 0 Ohms	Ohms	Measured resistance.
SV9.2.3.4-Last measure time = 0 s	s	Shows the time taken to measure the last measurement.
SV9.2.3.5-Time to ready = 0 s	s	Shows time left to cool down and retake a measurement.
SV9.2.3.6-T. aval. no measure = 0 min	min	Shows the time that the IMD is taking to get a new valid measure.
SV9.2.3.7-Res no filtered = 0 Ohms	Ohms	Indicates the resistance value without applying the RP filter of channel DC 2.

SV9.2.4: Channel DC 3

SCREEN	UNITS	DESCRIPTION
SV9.2.4.1-Status = 0000	NONE	Indicates the channel DC 3 status of the isolation monitoring device. BIT / STATUS / DESCRIPTION 0 / Error / Shows if the channel is in isolation fault. 1 / Warning / Shows if the channel is in isolation warning. 2 / Rmax / Shows if the channel exceeds maximum 2 Mohms. 3 / Ready / The channel cool down time is set to ready. This time defines when a channel must pass without measuring for its associated resistor card to cool down to the same temperature it had heated while measuring.
SV9.2.4.2-N Measures = 0	NONE	Defines the number of measurements the channel has performed since the last isolation monitoring device reset.
SV9.2.4.3-Resistor = 0 Ohms	Ohms	Measured resistance.

SCREEN	UNITS	DESCRIPTION
SV9.2.4.4-Last measure time = 0 s	s	Shows the time taken to measure the last measurement.
SV9.2.4.5-Time to ready = 0 s	s	Shows time left to cool down and retake a measurement.
SV9.2.4.6-T. aval. no measure = 0 min	min	Shows the time that the IMD is taking to get a new valid measure.
SV9.2.4.7-Res no filtered = 0 Ohms	Ohms	Indicates the resistance value without applying the RP filter of channel DC 3.

SV9.2.5: Channel DC 4

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV9.2.5.1-Status = 0000</td><td>NONE</td><td>Indicates the channel DC 4 status of the isolation monitoring device.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Error</td><td>Shows if the channel is in isolation fault.</td></tr>
<tr><td>1</td><td>Warning</td><td>Shows if the channel is in isolation warning.</td></tr>
<tr><td>2</td><td>Rmax</td><td>Shows if the exceeds maximum 2 Mohms.</td></tr>
<tr><td>3</td><td>Ready</td><td>The channel cooling time is set to ready. This time defines when a channel must pass without measuring for its associated resistor card to cool down to the same temperature it had heated while measuring.</td></tr>
</table></td></tr>
<tr><td>SV9.2.5.2-N Measures = 0</td><td>NONE</td><td>Defines the number of measurements the channel has performed since the last isolation monitoring device reset.</td></tr>
<tr><td>SV9.2.5.3-Resistor = 0 Ohms</td><td>Ohms</td><td>Measured resistance.</td></tr>
<tr><td>SV9.2.5.4-Last measure time = 0 s</td><td>s</td><td>Shows the time taken to measure the last measurement.</td></tr>
<tr><td>SV9.2.5.5-Time to ready = 0 s</td><td>s</td><td>Shows time left to cool down and retake a measurement.</td></tr>
<tr><td>SV9.2.5.6-T. aval. no measure = 0 min</td><td>min</td><td>Shows the time that the IMD is taking to get a new valid measure.</td></tr>
<tr><td>SV9.2.5.7-Res no filtered = 0 Ohms</td><td>Ohms</td><td>Indicates the resistance value without applying the RP filter of channel DC 4.</td></tr>
</table>

5.8.3. Subgroup SV9.3: Maneuver

Note: Parameters within this subgroup are only displayed if [G9.1.1-Enable = Yes].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV9.3.1-Maneuver State = Normal Operation</td><td>NONE</td><td>Indicates the maneuver status of isolation monitoring devices.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Normal operation</td><td>Consecutively measures the isolation on each of the enabled channels.</td></tr>
<tr><td>GFDI maneuvre</td><td>Shows the performs GFDI maneuver when the inverter is earthed.</td></tr>
<tr><td>Diagnosis maneuvre</td><td>Shows when the inverter performs the diagnosis of the inverter.</td></tr>
<tr><td>Manual maneuvre</td><td>Indicates when the isolation monitoring device is working in manual mode.</td></tr>
</table>
Note: This parameter is displayed if [G9.1.1-Enable = Yes].</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV9.3.2-GFDI Substate = Open GFDI</td><td>NONE</td><td>Shows the GFDI maneuver status.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Open GFDI</td><td>Proceed to open the GFDI.</td></tr><tr><td>Measure GFDI</td><td>The insulation is measured when the GFDI is open.</td></tr><tr><td>Close GFDI</td><td>Proceed to close the GFDI.</td></tr><tr><td>Measure GFDI closed</td><td>Proceed to measure insulation with GFDI closed.</td></tr></table></td></tr>
<tr><td>SV9.3.3-Diagnosis Substate = Waiting Request</td><td>NONE</td><td>Shows the status during the diagnostic process of the inverter.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Waiting Request</td><td>Shows the isolation monitoring device stays in STOP, waiting for orders from the main control board.</td></tr><tr><td>Measuring DC</td><td>Shows when isolation monitoring device proceeds to measure all available DCs in DC. Only available for MPCS</td></tr><tr><td>Measuring AC and DC</td><td>Shows when isolation monitoring device proceeds to measure AC with DC closed and switching.</td></tr><tr><td>Measuring AC</td><td>Shows when isolation monitoring device proceeds to measure AC with DC open and not switching.</td></tr><tr><td>Measuring Modules</td><td>Shows when isolation monitoring device proceeds to measure AC with DC open and switching, module by module.</td></tr></table>Note: This parameter is displayed if [G9.1.1-Enable = Yes] and [SV9.3.1-Maneuver State = Diagnosis Maneuver].</td></tr>
<tr><td>SV9.3.4-Measuring Channel = AC</td><td>NONE</td><td>Indicates the channel is measuring.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>AC</td><td>IMI requests to measure AC channel.</td></tr><tr><td>DC1</td><td>IMI requests to measure DC1 channel.</td></tr><tr><td>DC2</td><td>IMI requests to measure DC2 channel.</td></tr><tr><td>DC3</td><td>IMI requests to measure DC3 channel.</td></tr><tr><td>DC4</td><td>IMI requests to measure DC4 channel.</td></tr><tr><td>No measuring</td><td>IMI is not measuring.</td></tr></table>Note: This parameter is displayed if [G9.1.1-Enable = Yes].</td></tr>
<tr><td>SV9.3.5-GFDI Output = Open</td><td>NONE</td><td>Indicates the GFDI status, open or closed.
Note: this parameter is shown if [SG10.6.1-DU Type] = GFDI + IMD.</td></tr>
<tr><td>SV9.3.6-GFDI Feedback = Open</td><td>NONE</td><td>Indicates the GFDI feedback status.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Open</td><td>The GFDI feedback is open.</td></tr><tr><td>Closed</td><td>The GFDI feedback is closed.</td></tr><tr><td>Shooted</td><td>The GFDI output indicating overcurrent trip is activated.</td></tr><tr><td>Undeterminated</td><td>GFDI feedback signals both open and closed, or neither</td></tr></table>Note: this parameter is shown if [SG10.6.1-DU Type] = GFDI + IMD.</td></tr>
<tr><td>SV9.3.7-GFDI Daily Maneuver = Pending</td><td>NONE</td><td>Shows if the GFDI maneuver of the day is done.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Pending</td><td>The GFDI maneuver has not yet been performed. It is done once a day.</td></tr><tr><td>Done</td><td>The GFDI maneuver has been performed.</td></tr></table></td></tr>
</table>

5.8.4. Subgroup SV9.4: Advanced

Notes:

- This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].
- Parameters within this subgroup are only displayed if [G9.1.1-Enable = Yes].

SCREEN	UNITS	DESCRIPTION
SV9.4.1-RP1 Not Calibrated = 0 Ohms	Ohms	Shows the result of resistance obtained when measuring positive pulse.
SV9.4.2-RP2 Not Calibrated = 0 Ohms	Ohms	Shows the result of resistance obtained when measuring negative pulse.
SV9.4.3-RP1 Calibrated = 0 Ohms	Ohms	Shows the result of resistance Rp1 measured after calibration.
SV9.4.4-RP2 Calibrated = 0 Ohms	Ohms	Shows the result of resistance Rp2 measured after calibration.
SV9.4.5-Predicted Iterations = 0	NONE	Shows the predicted iterations to perform a correct isolation measurement.
SV9.4.6-ADC Meas 1 = 0	NONE	Shows the measurement received by the ADC isolation monitoring device.
SV9.4.7-ADC Meas 2 = 0	NONE	Shows the measurement received by the ADC isolation monitoring device.
SV9.4.8-ADC V48P = 0	NONE	Shows the measurement received by the ADC isolation monitoring device.
SV9.4.9-ADC V48N = 0	NONE	Shows the measurement received by the ADC isolation monitoring device.
SV9.4.10-ADC Earth = 0	NONE	Shows the measurement received by the ADC isolation monitoring device.
SV9.4.11-Calibration Status = Not calibrated	NONE	Indicates the calibration status of IMI. The IMI should be calibrated in fabric, in order to work properly. **STATUS** / **DESCRIPTION** Not calibrated: The IMI is not calibrated. Point 1 calibrated: IMI device calibration points. In the calibration process, whereby the resistance measurements are adjusted to optimize the IMI's insulation resistance measurement algorithm, it first calibrates with a low value resistance (point 1) and then calibrates another point with a high value resistance (point 2), so that it can then interpolate between the two points. If either point has not been calibrated, it cannot be guaranteed that it can measure correctly. Point 2 calibrated Calibrated: The IMI is calibrated. If the IMI is not calibrated, then warning "W185: IMI not calibrated" will be triggered.
SV9.4.12-ADC Semibus Pos = 0	V	Corresponds to the semi bus positive to earth ADC (Analog to Digital Converter) from IMI which is the device that measures this data. This parameter allows the voltage to be monitored.

EN

5.9. Group SV10: DU

This group allows the visualization of the currents passing through each of the enabled channels of the DC cabinet, as well as the current passing through the DC and AC RFI filters. Refer to the *Hardware and Installation Manual* for more information on channel layout in the DC cabinet.

Note: This group is displayed if [SG10.6.1-DU Type = Enabled].

5.9.1. Subgroup SV10.1: String Block A

SCREEN	UNITS	DESCRIPTION
SV10.1.1-Channel 1 = 0.0 A	A	Shows the current measured on channels 1 to 10 of the DC cabinet.
SV10.1.2-Channel 2 = 0.0 A	A	
SV10.1.3-Channel 3 = 0.0 A	A	
SV10.1.4-Channel 4 = 0.0 A	A	
SV10.1.5-Channel 5 = 0.0 A	A	
SV10.1.6-Channel 6 = 0.0 A	A	
SV10.1.7-Channel 7 = 0.0 A	A	
SV10.1.8-Channel 8 = 0.0 A	A	
SV10.1.9-Channel 9 = 0.0 A	A	
SV10.1.10-Channel 10 = 0.0 A	A	

5.9.2. Subgroup SV10.2: String Block B

SCREEN	UNITS	DESCRIPTION
SV10.2.1-Channel 11 = 0.0 A	A	Shows the current measured on channels 11 to 20 of the DC cabinet.
SV10.2.2-Channel 12 = 0.0 A	A	
SV10.2.3-Channel 13 = 0.0 A	A	
SV10.2.4-Channel 14 = 0.0 A	A	
SV10.2.5-Channel 15 = 0.0 A	A	
SV10.2.6-Channel 16 = 0.0 A	A	
SV10.2.7-Channel 17 = 0.0 A	A	
SV10.2.8-Channel 18 = 0.0 A	A	
SV10.2.9-Channel 19 = 0.0 A	A	
SV10.2.10-Channel 20 = 0.0 A	A	

5.9.3. Subgroup SV10.3: String Block C

SCREEN	UNITS	DESCRIPTION
SV10.3.1-Channel 21 = 0.0 A	A	Shows the current measured on channels 21 to 30 of the DC cabinet.
SV10.3.2-Channel 22 = 0.0 A	A	
SV10.3.3-Channel 23 = 0.0 A	A	
SV10.3.4-Channel 24 = 0.0 A	A	
SV10.3.5-Channel 25 = 0.0 A	A	
SV10.3.6-Channel 26 = 0.0 A	A	
SV10.3.7-Channel 27 = 0.0 A	A	
SV10.3.8-Channel 28 = 0.0 A	A	
SV10.3.9-Channel 29 = 0.0 A	A	
SV10.3.10-Channel 30 = 0.0 A	A	

EN

5.9.4. Subgroup SV10.4: String Block D

SCREEN	UNITS	DESCRIPTION
SV10.4.1-Channel 31 = 0.0 A	A	Shows the current measured on channels 31 to 40 of the DC cabinet.
SV10.4.2-Channel 32 = 0.0 A	A	
SV10.4.3-Channel 33 = 0.0 A	A	
SV10.4.4-Channel 34 = 0.0 A	A	
SV10.4.5-Channel 35 = 0.0 A	A	
SV10.4.6-Channel 36 = 0.0 A	A	
SV10.4.7-Channel 37 = 0.0 A	A	
SV10.4.8-Channel 38 = 0.0 A	A	
SV10.4.9-Channel 39 = 0.0 A	A	
SV10.4.10-Channel 40 = 0.0 A	A	

5.9.5. Subgroup SV10.5: Temperature DU master

SCREEN	UNITS	DESCRIPTION
SV10.5-Temperature DU master = 0 ºC	ºC	Shows the master DC cabinet board temperature.

5.9.6. Subgroup SV10.6: Temperature DU slave

SCREEN	UNITS	DESCRIPTION
SV10.6-Temperature DU slave = 0 ºC	ºC	Shows the slave DC cabinet board temperature. **Note:** This parameter is shown if [SG10.1.4-Inverter type] ≠ PCS/MPCS.

5.9.7. Subgroup SV10.7: RFI current 1

SCREEN	UNITS	DESCRIPTION
SV10.7-RFI current 1 = 0 A	A	Shows the current through RFI 1 measured in DC.

5.9.8. Subgroup SV10.8: RFI current 2

SCREEN	UNITS	DESCRIPTION
SV10.8-RFI current 2 = 0 A	A	Shows the current through RFI 2 measured in DC.

5.9.9. Subgroup SV10.9: RFI current 3

SCREEN	UNITS	DESCRIPTION
SV10.9-RFI current 3 = 0 A	A	Shows the current through RFI 3 measured in DC.

5.9.10. Subgroup SV10.10: RFI current 4

SCREEN	UNITS	DESCRIPTION
SV10.10-RFI current 4 = 0 A	A	Shows the current through RFI 4 measured in DC.

5.10.Group SV11: Fans

This group allows to visualize the information provided by each of the fans installed in the inverters for monitoring purposes. Consult section "G11: Fans" for more information about the fan's configuration.

Note: This group is displayed if [G1.3-User] ≥ Qualified customer.

5.10.1. Subgroup SV11.1: Fan DU 1

Note: these parameters are only displayed if [G11.11-Avaliables Fan Nº] ≥ 1.

SV11.1.0: Fan Monitor DU 1

This subgroup shows in real time what is happening in the DU 1 fan speed monitoring module.

SCREEN	UNITS	DESCRIPTION
SV11.1.0.1-Fan State = Not Monitoring	NONE	Shows the fan status according to the following table: **STATUS** / **DESCRIPTION** Not Monitoring: The status of the fan is not being monitored. This is the usual state before the fan starts. Ramp Transition: Fan in ramp transition (accelerating or decelerating) taking into account the transition period as a function of the difference for the times defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Periodic Check: Periodic monitoring of the fan (when it is not in ramp transition). Fan fault: Fan on fault due to not reaching the set speed (F228).
SV11.1.0.2-Curr. Setp. RPM = 0 rpm	rpm	Shows current setpoint in rpm.
SV11.1.0.3-Err. Margin RPM = 0 rpm	rpm	Shows the maximum permissible margin of error of the fan. It is updated each time there is a new setpoint independently if this triggers the transition state. It is different as a function of the fan type. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.1.0.4-Current Speed = 0 rpm	rpm	Shows the fan current speed in rpm.
SV11.1.0.5-Max Speed = 0 rpm	rpm	Shows the fan maximum speed in rpm. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.1.0.6-Ramp Time = 0 s	s	Shows the calculated ramp time for the change of setpoint, this is, the time that it will take the fan to reach the commanded speed. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.1.0.7-Elap. ramp time = 0 s	s	Shows the elapsed ramp time. This value is incremented until the value set in parameter [SV11.1.0.6-Ramp Time] is reached. During this time, the fan will be in state "Ramp Transition" and when the calculated ramp time is reached, the fan status will switch to "Periodic Check". **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.1.0.8-Deviation = 0 %	%	Shows the deviation error configured in [G11.14-Setpoint error deviation]. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.1.0.9-Error counter = 0 s	s	Shows the time during which the current speed is not withing the admissible setpoint margin. If the value of [G11.15-Setpoint error delay][5] is reached, the fan will trigger fault F228. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.1.0.10-Dev. Error Delay= 0 s	s	This parameter shows the maximum time that can elapse without reaching the setpoint before changing to state "Fan fault". This value is set in [G11.15-Setpoint error delay]. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.

[5] Also shown in SV11.1.0.10.

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.1.0.11-Ramp UP Hyster = 0 rpm</td><td>rpm</td><td rowspan="2">These parameters show the calculated hysteresis for the fan ramp-up [SV11.1.0.11] and ramp-down [SV11.1.0.12].

The hysteresis is calculated as a function of the maximum fan speed and the ramp-up time defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Its minimum possible value is 2% of the maximum speed (this value has been chosen as the minimum hysteresis value because in the fan curves the minimum change is 1%).
Note: these parameters are shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.1.0.12-Ramp DOWN Hyster = 0 rpm</td><td>rpm</td></tr>
<tr><td>SV11.1.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the fan DU 1.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Fan DU1 is disabled.</td></tr><tr><td>OK fan</td><td>Communication between the uP and the fan DU1 is correct.</td></tr><tr><td>Comms Error</td><td>uP is unable to communicate with the fan DU1.</td></tr><tr><td>Error</td><td>Fan DU1 is in error.</td></tr></table></td></tr>
<tr><td>SV11.1.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.1.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan.
Note: This parameter is only shown if [G11.2.1-DU Fan 1 Model] = EBM-Papst.
It is not displayed in Power Electronics remote management tool.</td></tr>
<tr><td>SV11.1.4-Fan zone = DU zone</td><td>NONE</td><td>Shows the zone assigned to the fan DU 1.</td></tr>
<tr><td>SV11.1.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>Unknown request</td></tr><tr><td>Check ID</td><td>The ID of the fan with which the inverter is communicating is checked for correctness.</td></tr><tr><td>Configuration 1</td><td rowspan="7">The inverter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Inverter sends a reset to the communicating fan.</td></tr><tr><td>Configuration 2</td></tr><tr><td>Configuration 3</td></tr><tr><td>Configuration 4</td></tr><tr><td>Configuration 5</td></tr><tr><td>Configuration 6</td></tr><tr><td>Configuration 7</td></tr><tr><td>Read serial number</td><td>The inverter reads the serial number of the fan.</td></tr><tr><td>Reset</td><td>The inverter sends a reset to the communicating fan.</td></tr><tr><td>Get status</td><td>The inverter asks the fan to indicate its internal status to the inverter.</td></tr><tr><td>Get speed</td><td>The inverter asks the fan to get the current fan speed.</td></tr><tr><td>Get temperature</td><td>The inverter asks the fan to get the internal temperature of the fan.</td></tr><tr><td>Get power</td><td>The inverter asks the fan to get the current power consumption of the fan.</td></tr><tr><td>Set setpoint</td><td>The inverter sets the speed setpoint configured in the G11 to the fan.</td></tr><tr><td>Special request</td><td>Special request.</td></tr><tr><td>Get references</td><td>The inverter uses the reference values (voltage and current) from the fan to calculate the power consumption.
Note: this applies only to EBM-Papst fans.</td></tr></table>Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.1.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.1.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.1.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.1.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>RS485 communication fault with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.1-DU Fan 1 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.1.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fan fault.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.1-DU Fan 1 Model] = Tesoer.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.1.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronics temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>High line voltage.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>The shedding function is active.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.1-DU Fan 1 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.1.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL out of control.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.1-DU Fan 1 Model] = Tesoer.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.1.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the DELTA fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DC bus average overcurrent</td><td>Average DC bus overcurrent.</td></tr>
<tr><td>1</td><td>DC bus average overvoltage</td><td>Average DC bus overvoltage.</td></tr>
<tr><td>2</td><td>DC bus average undervoltage</td><td>Average DC bus undervoltage.</td></tr>
<tr><td>3</td><td>EE ove protection temperature</td><td>Fan overtemperature protection.</td></tr>
<tr><td>4</td><td>Fan lock</td><td>Fan lock.</td></tr>
<tr><td>5</td><td>AC lose phase</td><td>AC phase loss.</td></tr>
<tr><td>6</td><td>Fan reverse run</td><td>Fan reverse.</td></tr>
<tr><td>7</td><td>Hall signal abnormal</td><td>Abnormal hall signal.</td></tr>
<tr><td>8</td><td>EEPROM read/write fail</td><td>EEPROM read/write fault.</td></tr>
<tr><td>9</td><td>DC bus peak over current</td><td>DC bus peak overcurrent.</td></tr>
<tr><td>10</td><td>AC bus over voltage</td><td>AC bus overvoltage.</td></tr>
<tr><td>11</td><td>AC bus under voltage</td><td>AC bus undervoltage.</td></tr>
<tr><td>12</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>13</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.1-DU Fan 1 Model] = Delta.</td></tr>
<tr><td>SV11.1.11-Temperature motor = 0 ºC</td><td>ºC</td><td>Shows the temperature of the fan motor.
Note: this parameter is shown if [G11.2.1-DU Fan 1 Model] ≠ Delta and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.1.12-Temperature module = 0 ºC</td><td>ºC</td><td>Shows the temperature of the module motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.1.13-Temp. electronic = 0 ºC</td><td>ºC</td><td>Shows the temperature of the electronics of the motor.
Note: this parameter is shown if [G11.2.1-DU Fan 1 Model] = EBM-Papst and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.1.14-Current power = 0 W</td><td>W</td><td>Shows the current active power of the motor.</td></tr>
<tr><td>SV11.1.15-Fan type = [G11.2.1-DU Fan 1 Model]</td><td>NONE</td><td>Shows the fan type configured in [G11.2.1-DU Fan 1 Model].
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>EBM-Papst</td><td>EBM-Papst fan supplier.</td></tr>
<tr><td>Tesoer</td><td>Tesoer fan supplier.</td></tr>
<tr><td>Delta</td><td>Delta fan supplier.</td></tr>
</table></td></tr>
<tr><td>SV11.1.16-Fan delta model = Unknown</td><td>NONE</td><td>Indicates which Delta fan model is configured.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>The found Delta fan does not correspond to a valid fan.</td></tr>
<tr><td>L9210</td><td>The found Delta fan corresponds to the L9210 reference.</td></tr>
<tr><td>L9587</td><td>The found Delta fan corresponds to the L9587 reference.</td></tr>
<tr><td>L9169</td><td>The found Delta fan corresponds to the L9169 reference.</td></tr>
</table>
Notes:
– This parameter is displayed if [G11.2.1-DU Fan 1 Model] = Delta.
– If two different types of Delta are installed in the same zone, some of these faults will be triggered: F230, F231 and F232.</td></tr>
</table>

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5.10.2. Subgroup SV11.2: Fan DU 2

Note: these parameters are only displayed if [G11.11-Avaliables Fan Nº] ≥ 2.

SV11.2.0: Fan Monitor DU 2

This subgroup shows in real time what is happening in the DU 2 fan speed monitoring module.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.2.0.1-Fan State = Not Monitoring</td><td>NONE</td><td>Shows the fan status according to the following table:
<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Not Monitoring</td><td>The status of the fan is not being monitored.
This is the usual state before the fan starts.</td></tr><tr><td>Ramp Transition</td><td>Fan in ramp transition (accelerating or decelerating) taking into account the transition period as a function of the difference for the times defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time].</td></tr><tr><td>Periodic Check</td><td>Periodic monitoring of the fan (when it is not in ramp transition).</td></tr><tr><td>Fan fault</td><td>Fan on fault due to not reaching the set speed (F228).</td></tr></table></td></tr>
<tr><td>SV11.2.0.2-Curr. Setp. RPM = 0 rpm</td><td>rpm</td><td>Shows current setpoint in rpm.</td></tr>
<tr><td>SV11.2.0.3-Err. Margin RPM = 0 rpm</td><td>rpm</td><td>Shows the maximum permissible margin of error of the fan. It is updated each time there is a new setpoint independently if this triggers the transition state. It is different as a function of the fan type.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.0.4-Current Speed = 0 rpm</td><td>rpm</td><td>Shows the fan current speed in rpm.</td></tr>
<tr><td>SV11.2.0.5-Max Speed = 0 rpm</td><td>rpm</td><td>Shows the fan maximum speed in rpm.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.0.6-Ramp Time = 0 s</td><td>s</td><td>Shows the calculated ramp time for the change of setpoint, this is, the time that it will take the fan to reach the commanded speed.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.0.7-Elap. ramp time = 0 s</td><td>s</td><td>Shows the elapsed ramp time. This value is incremented until the value set in parameter [SV11.2.0.6-Ramp Time] is reached. During this time, the fan will be in state “Ramp Transition” and when the calculated ramp time is reached, the fan status will switch to “Periodic Check”.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.0.8-Deviation = 0 %</td><td>%</td><td>Shows the deviation error configured in [G11.14-Setpoint error deviation].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.0.9-Error counter = 0 s</td><td>s</td><td>Shows the time during which the current speed is not withing the admissible setpoint margin. If the value of [G11.15-Setpoint error delay][6] is reached, the fan will trigger fault F228.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.0.10-Dev. Error Delay= 0 s</td><td>s</td><td>This parameter shows the maximum time that can elapse without reaching the setpoint before changing to state “Fan fault”. This value is set in [G11.15-Setpoint error delay].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.0.11-Ramp UP Hyster = 0 rpm</td><td>rpm</td><td rowspan="2">These parameters show the calculated hysteresis for the fan ramp-up [SV11.2.0.11] and ramp-down [SV11.2.0.12].

The hysteresis is calculated as a function of the maximum fan speed and the ramp-up time defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Its minimum possible value is 2% of the maximum speed (this value has been chosen as the minimum hysteresis value because in the fan curves the minimum change is 1%).
Note: these parameters are shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.0.12-Ramp DOWN Hyster = 0 rpm</td><td>rpm</td></tr>
</table>

[6] Also shown in SV11.2.0.10.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.2.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the DU2 fan.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Disabled</td><td>Fan DU2 is disabled.</td></tr>
<tr><td>OK fan</td><td>Communication between the uP and the fan DU2 is correct.</td></tr>
<tr><td>Comms Error</td><td>uP is unable to communicate with the fan DU2.</td></tr>
<tr><td>Error</td><td>Fan DU2 is in error.</td></tr>
</table></td></tr>
<tr><td>SV11.2.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.2.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan.
Note: This parameter is only shown if [G11.2.2-DU Fan 2 Model] = EBM-Papst.
It is not displayed in Power Electronics remote management tool.</td></tr>
<tr><td>SV11.2.4-Fan zone = DU zone</td><td>NONE</td><td>Shows the zone assigned to the fan DU2.</td></tr>
<tr><td>SV11.2.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>Unknown request.</td></tr>
<tr><td>Check ID</td><td>The ID of the fan with which the inverter is communicating is checked for correctness.</td></tr>
<tr><td>Configuration 1</td><td rowspan="7">The inverter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Inverter sends a reset to the communicating fan.</td></tr>
<tr><td>Configuration 2</td></tr>
<tr><td>Configuration 3</td></tr>
<tr><td>Configuration 4</td></tr>
<tr><td>Configuration 5</td></tr>
<tr><td>Configuration 6</td></tr>
<tr><td>Configuration 7</td></tr>
<tr><td>Read serial number</td><td>The inverter reads the serial number of the fan.</td></tr>
<tr><td>Reset</td><td>The inverter sends a reset to the communicating fan.</td></tr>
<tr><td>Get status</td><td>The inverter asks the fan to indicate its internal status to the inverter.</td></tr>
<tr><td>Get speed</td><td>The inverter asks the fan to get the current fan speed.</td></tr>
<tr><td>Get temperature</td><td>The inverter asks the fan to get the internal temperature of the fan.</td></tr>
<tr><td>Get power</td><td>The inverter asks the fan to get the current power consumption of the fan.</td></tr>
<tr><td>Set setpoint</td><td>The inverter sets the speed setpoint configured in the G11 to the fan.</td></tr>
<tr><td>Special request</td><td>Special request.</td></tr>
<tr><td>Get references</td><td>The inverter uses the reference values (voltage and current) from the fan to calculate the power consumption.
Note: this applies only to EBM-Papst fans.</td></tr>
</table>
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.2.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.2.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>RS485 communication fault with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.2-DU Fan 2 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.2.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fan fault.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.2-DU Fan 2 Model] = Tesoer.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.2.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronics temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>High line voltage.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>The shedding function is active.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.2-DU Fan 2 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.2.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL out of control.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.2-DU Fan 2 Model] = Tesoer.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.2.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the DELTA fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DC bus average overcurrent</td><td>Average DC bus overcurrent.</td></tr>
<tr><td>1</td><td>DC bus average overvoltage</td><td>Average DC bus overvoltage.</td></tr>
<tr><td>2</td><td>DC bus average undervoltage</td><td>Average DC bus undervoltage.</td></tr>
<tr><td>3</td><td>EE ove protection temperature</td><td>Fan overtemperature protection.</td></tr>
<tr><td>4</td><td>Fan lock</td><td>Fan lock.</td></tr>
<tr><td>5</td><td>AC lose phase</td><td>AC phase loss.</td></tr>
<tr><td>6</td><td>Fan reverse run</td><td>Fan reverse.</td></tr>
<tr><td>7</td><td>Hall signal abnormal</td><td>Abnormal hall signal.</td></tr>
<tr><td>8</td><td>EEPROM read/write fail</td><td>EEPROM read/write fault.</td></tr>
<tr><td>9</td><td>DC bus peak over current</td><td>DC bus peak overcurrent.</td></tr>
<tr><td>10</td><td>AC bus over voltage</td><td>AC bus overvoltage.</td></tr>
<tr><td>11</td><td>AC bus under voltage</td><td>AC bus undervoltage.</td></tr>
<tr><td>12</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>13</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.2-DU Fan 2 Model] = Delta.</td></tr>
<tr><td>SV11.2.11-Temperature motor = 0 ºC</td><td>ºC</td><td>Shows the temperature of the fan motor.
Note: this parameter is displayed if [G11.2.2-DU Fan 2 Model] ≠ Delta and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.12-Temperature module = 0 ºC</td><td>ºC</td><td>Shows the temperature of the module motor.
Note: this parameter is displayed if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.13-Temp. electronic = 0 ºC</td><td>ºC</td><td>Shows the temperature of the electronics of the motor.
Note: This parameter is only displayed if [G11.2.2-DU Fan 2 Model] = EBM-Papst and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.2.14-Current power = 0 W</td><td>W</td><td>Shows the current active power of the motor.</td></tr>
<tr><td>SV11.2.15-Fan type = [G11.2.2-DU Fan 2 Model]</td><td>NONE</td><td>Shows the fan type configured in [G11.2.2-DU Fan 2 Model].
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>EBM-Papst</td><td>EBM-Papst fan supplier.</td></tr>
<tr><td>Tesoer</td><td>Tesoer fan supplier.</td></tr>
<tr><td>Delta</td><td>Delta fan supplier.</td></tr>
</table></td></tr>
<tr><td>SV11.2.16-Fan delta model = Unknown</td><td>NONE</td><td>Indicates which Delta fan model is configured.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>The found Delta fan does not correspond to a valid fan.</td></tr>
<tr><td>L9210</td><td>The found Delta fan corresponds to the L9210 reference.</td></tr>
<tr><td>L9587</td><td>The found Delta fan corresponds to the L9587 reference.</td></tr>
<tr><td>L9169</td><td>The found Delta fan corresponds to the L9169 reference.</td></tr>
</table>
Notes:
– This parameter is displayed if [G11.2.2-DU Fan 2 Model] = Delta.
– If two different types of Delta are installed in the same zone, some of these faults will be triggered: F230, F231 and F232.</td></tr>
</table>

5.10.3. Subgroup SV11.3: Fan Mod. 1

Note: these parameters are only displayed if [G11.11-Avaliables Fan Nº] ≥ 3.

SV11.3.0: Fan Monitor Mod. 1

This subgroup shows in real time what is happening in the module 1 fan speed monitoring module.

SCREEN	UNITS	DESCRIPTION
SV11.3.0.1-Fan State = Not Monitoring	NONE	Shows the fan status according to the following table: **STATUS** / **DESCRIPTION** Not Monitoring: The status of the fan is not being monitored. This is the usual state before the fan starts. Ramp Transition: Fan in ramp transition (accelerating or decelerating) taking into account the transition period as a function of the difference for the times defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Periodic Check: Periodic monitoring of the fan (when it is not in ramp transition). Fan fault: Fan on fault due to not reaching the set speed (F228).
SV11.3.0.2-Curr. Setp. RPM = 0 rpm	rpm	Shows current setpoint in rpm.
SV11.3.0.3-Err. Margin RPM = 0 rpm	rpm	Shows the maximum permissible margin of error of the fan. It is updated each time there is a new setpoint independently if this triggers the transition state. It is different as a function of the fan type. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.3.0.4-Current Speed = 0 rpm	rpm	Shows the fan current speed in rpm.
SV11.3.0.5-Max Speed = 0 rpm	rpm	Shows the fan maximum speed in rpm. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.3.0.6-Ramp Time = 0 s	s	Shows the calculated ramp time for the change of setpoint, this is, the time that it will take the fan to reach the commanded speed. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.3.0.7-Elap. ramp time = 0 s	s	Shows the elapsed ramp time. This value is incremented until the value set in parameter [SV11.3.0.6-Ramp Time] is reached. During this time, the fan will be in state "Ramp Transition" and when the calculated ramp time is reached, the fan status will switch to "Periodic Check". **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.3.0.8-Deviation = 0 %	%	Shows the deviation error configured in [G11.14-Setpoint error deviation]. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.3.0.9-Error counter = 0 s	s	Shows the time during which the current speed is not withing the admissible setpoint margin. If the value of [G11.15-Setpoint error delay][7] is reached, the fan will trigger fault F228. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.3.0.10-Dev. Error Delay= 0 s	s	This parameter shows the maximum time that can elapse without reaching the setpoint before changing to state "Fan fault". This value is set in [G11.15-Setpoint error delay]. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.3.0.11-Ramp UP Hyster = 0 rpm	rpm	These parameters show the calculated hysteresis for the fan ramp-up [SV11.3.0.11] and ramp-down [SV11.3.0.12]. The hysteresis is calculated as a function of the maximum fan speed and the ramp-up time defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Its minimum possible value is 2% of the maximum speed (this value has been chosen as the minimum hysteresis value because in the fan curves the minimum change is 1%). **Note:** these parameters are shown if [G1.3-User] ≥ Power Electronics staff.
SV11.3.0.12-Ramp DOWN Hyster = 0 rpm	rpm	

[7] Also shown in SV11.3.0.10.

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.3.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the module 1 fan.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Disabled</td><td>Fan module 1 is disabled.</td></tr>
<tr><td>OK fan</td><td>Communication between the uP and the fan module 1 is correct.</td></tr>
<tr><td>Comms Error</td><td>uP is unable to communicate with the fan module 1.</td></tr>
<tr><td>Error</td><td>Fan module 1 is in error.</td></tr>
</table></td></tr>
<tr><td>SV11.3.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.3.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan.
Note: This parameter is only shown if [G11.2.3-Module Fan 1 Model] = EBM-Papst. It is not displayed in Power Electronics remote management tool.</td></tr>
<tr><td>SV11.3.4-Fan zone = Modules zone</td><td>NONE</td><td>Shows the zone assigned to the module 1 fan.</td></tr>
<tr><td>SV11.3.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>Unknown request.</td></tr>
<tr><td>Check ID</td><td>The ID of the fan with which the inverter is communicating is checked for correctness.</td></tr>
<tr><td>Configuration 1</td><td rowspan="7">The inverter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Inverter sends a reset to the communicating fan.</td></tr>
<tr><td>Configuration 2</td></tr>
<tr><td>Configuration 3</td></tr>
<tr><td>Configuration 4</td></tr>
<tr><td>Configuration 5</td></tr>
<tr><td>Configuration 6</td></tr>
<tr><td>Configuration 7</td></tr>
<tr><td>Read serial number</td><td>The inverter reads the serial number of the fan.</td></tr>
<tr><td>Reset</td><td>The inverter sends a reset to the communicating fan.</td></tr>
<tr><td>Get status</td><td>The inverter asks the fan to indicate its internal status to the inverter.</td></tr>
<tr><td>Get speed</td><td>The inverter asks the fan to get the current fan speed.</td></tr>
<tr><td>Get temperature</td><td>The inverter asks the fan to get the internal temperature of the fan.</td></tr>
<tr><td>Get power</td><td>The inverter asks the fan to get the current power consumption of the fan.</td></tr>
<tr><td>Set setpoint</td><td>The inverter sets the speed setpoint configured in the G11 to the fan.</td></tr>
<tr><td>Special request</td><td>Special request.</td></tr>
<tr><td>Get references</td><td>The inverter uses the reference values (voltage and current) from the fan to calculate the power consumption.
Note: this applies only to EBM-Papst fans.</td></tr>
</table>
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.3.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.3.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.3.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.3.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>RS485 communication fault with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.3-Module Fan 1 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.3.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fan fault.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.3-Module Fan 1 Model] = Tesoer.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.3.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronics temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>High line voltage.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>The shedding function is active.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.3-Module Fan 1 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.3.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL out of control.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.3-Module Fan 1 Model] = Tesoer.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.3.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the DELTA fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DC bus average overcurrent</td><td>Average DC bus overcurrent.</td></tr>
<tr><td>1</td><td>DC bus average overvoltage</td><td>Average DC bus overvoltage.</td></tr>
<tr><td>2</td><td>DC bus average undervoltage</td><td>Average DC bus undervoltage.</td></tr>
<tr><td>3</td><td>EE ove protection temperature</td><td>Fan overtemperature protection.</td></tr>
<tr><td>4</td><td>Fan lock</td><td>Fan lock.</td></tr>
<tr><td>5</td><td>AC lose phase</td><td>AC phase loss.</td></tr>
<tr><td>6</td><td>Fan reverse run</td><td>Fan reverse.</td></tr>
<tr><td>7</td><td>Hall signal abnormal</td><td>Abnormal hall signal.</td></tr>
<tr><td>8</td><td>EEPROM read/write fail</td><td>EEPROM read/write fault.</td></tr>
<tr><td>9</td><td>DC bus peak over current</td><td>DC bus peak overcurrent.</td></tr>
<tr><td>10</td><td>AC bus over voltage</td><td>AC bus overvoltage.</td></tr>
<tr><td>11</td><td>AC bus under voltage</td><td>AC bus undervoltage.</td></tr>
<tr><td>12</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>13</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.3-Module Fan 1 Model] = Delta.</td></tr>
<tr><td>SV11.3.11-Temperature motor = 0 ºC</td><td>ºC</td><td>Shows the temperature of the fan motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.3.12-Temperature module = 0 ºC</td><td>ºC</td><td>Shows the temperature of the module motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.3.13-Temp. electronic = 0 ºC</td><td>ºC</td><td>Shows the temperature of the electronics of the motor.
Note: This parameter is only displayed if [G11.2.3-Module Fan 1 Model] ≠ Tesoer and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.3.14-Current power = 0 W</td><td>W</td><td>Shows the current active power of the motor.</td></tr>
<tr><td>SV11.3.15-Fan type = [G11.2.3-Module Fan 1 Model]</td><td>NONE</td><td>Shows the fan type configured in [G11.2.3-Module Fan 1 Model].
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>EBM-Papst</td><td>EBM-Papst fan supplier.</td></tr>
<tr><td>Tesoer</td><td>Tesoer fan supplier.</td></tr>
<tr><td>Delta</td><td>Delta fan supplier.</td></tr>
</table></td></tr>
<tr><td>SV11.3.16-Fan delta model = Unknown</td><td>NONE</td><td>Indicates which Delta fan model is configured.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>The found Delta fan does not correspond to a valid fan.</td></tr>
<tr><td>L9210</td><td>The found Delta fan corresponds to the L9210 reference.</td></tr>
<tr><td>L9587</td><td>The found Delta fan corresponds to the L9587 reference.</td></tr>
<tr><td>L9169</td><td>The found Delta fan corresponds to the L9169 reference.</td></tr>
</table>
Notes:
– This parameter is displayed if [G11.2.3-Module Fan 1 Model] = Delta.
– If two different types of Delta are installed in the same zone, some of these faults will be triggered: F230, F231 and F232.</td></tr>
</table>

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5.10.4. Subgroup SV11.4: Fan Mod. 2

Note: these parameters are only displayed if [G11.11-Avaliables Fan Nº] ≥ 4.

SV11.4.0: Fan Monitor Mod. 2

This subgroup shows in real time what is happening in the module 2 fan speed monitoring module.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.4.0.1-Fan State = Not Monitoring</td><td>NONE</td><td>Shows the fan status according to the following table:
<table><tr><th>STATE</th><th>DESCRIPTION</th></tr><tr><td>Not Monitoring</td><td>The state of the fan is not being monitored. This is the usual state before the fan starts.</td></tr><tr><td>Ramp Transition</td><td>Fan in ramp transition (accelerating or decelerating) taking into account the transition period as a function of the difference for the times defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time].</td></tr><tr><td>Periodic Check</td><td>Periodic monitoring of the fan (when it is not in ramp transition).</td></tr><tr><td>Fan fault</td><td>Fan on fault due to not reaching the set speed (F228).</td></tr></table></td></tr>
<tr><td>SV11.4.0.2-Curr. Setp. RPM = 0 rpm</td><td>rpm</td><td>Shows current setpoint in rpm.</td></tr>
<tr><td>SV11.4.0.3-Err. Margin RPM = 0 rpm</td><td>rpm</td><td>Shows the maximum permissible margin of error of the fan. It is updated each time there is a new setpoint independently if this triggers the transition state. It is different as a function of the fan type.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.0.4-Current Speed = 0 rpm</td><td>rpm</td><td>Shows the fan current speed in rpm.</td></tr>
<tr><td>SV11.4.0.5-Max Speed = 0 rpm</td><td>rpm</td><td>Shows the fan maximum speed in rpm.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.0.6-Ramp Time = 0 s</td><td>s</td><td>Shows the calculated ramp time for the change of setpoint, this is, the time that it will take the fan to reach the commanded speed.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.0.7-Elap. ramp time = 0 s</td><td>s</td><td>Shows the elapsed ramp time. This value is incremented until the value set in parameter [SV11.4.0.6-Ramp Time] is reached. During this time, the fan will be in state “Ramp Transition” and when the calculated ramp time is reached, the fan status will switch to “Periodic Check”.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.0.8-Deviation = 0 %</td><td>%</td><td>Shows the deviation error configured in [G11.14-Setpoint error deviation].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.0.9-Error counter = 0 s</td><td>s</td><td>Shows the time during which the current speed is not withing the admissible setpoint margin. If the value of [G11.15-Setpoint error delay][8] is reached, the fan will trigger fault F228.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.0.10-Dev. Error Delay= 0 s</td><td>s</td><td>This parameter shows the maximum time that can elapse without reaching the setpoint before changing to state “Fan fault”. This value is set in [G11.15-Setpoint error delay].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.0.11-Ramp UP Hyster = 0 rpm</td><td>rpm</td><td rowspan="2">These parameters show the calculated hysteresis for the fan ramp-up [SV11.4.0.11] and ramp-down [SV11.4.0.12].

The hysteresis is calculated as a function of the maximum fan speed and the ramp-up time defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Its minimum possible value is 2% of the maximum speed (this value has been chosen as the minimum hysteresis value because in the fan curves the minimum change is 1%).
Note: these parameters are shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.0.12-Ramp DOWN Hyster = 0 rpm</td><td>rpm</td></tr>
</table>

[8] Also shown in SV11.4.0.10.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.4.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the module 2 fan.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Disabled</td><td>Fan module 2 is disabled.</td></tr>
<tr><td>OK fan</td><td>Communication between the uP and the fan module 2 is correct.</td></tr>
<tr><td>Comms Error</td><td>uP is unable to communicate with the fan module 2.</td></tr>
<tr><td>Error</td><td>Fan module 2 is in error.</td></tr>
</table></td></tr>
<tr><td>SV11.4.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.4.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan.
Note: This parameter is only shown if [G11.2.4-Module Fan 2 Model] = EBM-Papst. It is not displayed in Power Electronics remote management tool.</td></tr>
<tr><td>SV11.4.4-Fan zone = Modules zone</td><td>NONE</td><td>Shows the zone assigned to the module 2 fan.</td></tr>
<tr><td>SV11.4.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>Unknown request.</td></tr>
<tr><td>Check ID</td><td>The ID of the fan with which the inverter is communicating is checked for correctness.</td></tr>
<tr><td>Configuration 1</td><td rowspan="7">The inverter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Inverter sends a reset to the communicating fan.</td></tr>
<tr><td>Configuration 2</td></tr>
<tr><td>Configuration 3</td></tr>
<tr><td>Configuration 4</td></tr>
<tr><td>Configuration 5</td></tr>
<tr><td>Configuration 6</td></tr>
<tr><td>Configuration 7</td></tr>
<tr><td>Read serial number</td><td>The inverter reads the serial number of the fan.</td></tr>
<tr><td>Reset</td><td>The inverter sends a reset to the communicating fan.</td></tr>
<tr><td>Get status</td><td>The inverter asks the fan to indicate its internal status to the inverter.</td></tr>
<tr><td>Get speed</td><td>The inverter asks the fan to get the current fan speed.</td></tr>
<tr><td>Get temperature</td><td>The inverter asks the fan to get the internal temperature of the fan.</td></tr>
<tr><td>Get power</td><td>The inverter asks the fan to get the current power consumption of the fan.</td></tr>
<tr><td>Set setpoint</td><td>The inverter sets the speed setpoint configured in the G11 to the fan.</td></tr>
<tr><td>Special request</td><td>Special request.</td></tr>
<tr><td>Get references</td><td>The inverter uses the reference values (voltage and current) from the fan to calculate the power consumption.
Note: this applies only to EBM-Papst fans.</td></tr>
</table>
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.4.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.4.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>RS485 communication fault with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.4-Module Fan 2 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.4.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fan fault.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.4-Module Fan 2 Model] = Tesoer.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.4.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronics temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>High line voltage.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>The shedding function is active.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.4-Module Fan 2 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.4.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL out of control.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.4-Module Fan 2 Model] = Tesoer.</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.4.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the DELTA fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DC bus average overcurrent</td><td>Average DC bus overcurrent.</td></tr>
<tr><td>1</td><td>DC bus average overvoltage</td><td>Average DC bus overvoltage.</td></tr>
<tr><td>2</td><td>DC bus average undervoltage</td><td>Average DC bus undervoltage.</td></tr>
<tr><td>3</td><td>EE ove protection temperature</td><td>Fan overtemperature protection.</td></tr>
<tr><td>4</td><td>Fan lock</td><td>Fan lock.</td></tr>
<tr><td>5</td><td>AC lose phase</td><td>AC phase loss.</td></tr>
<tr><td>6</td><td>Fan reverse run</td><td>Fan reverse.</td></tr>
<tr><td>7</td><td>Hall signal abnormal</td><td>Abnormal hall signal.</td></tr>
<tr><td>8</td><td>EEPROM read/write fail</td><td>EEPROM read/write fault.</td></tr>
<tr><td>9</td><td>DC bus peak over current</td><td>DC bus peak overcurrent.</td></tr>
<tr><td>10</td><td>AC bus over voltage</td><td>AC bus overvoltage.</td></tr>
<tr><td>11</td><td>AC bus under voltage</td><td>AC bus undervoltage.</td></tr>
<tr><td>12</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>13</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: this parameter is only displayed if [G11.2.4-Module Fan 2 Model] = Delta.</td></tr>
<tr><td>SV11.4.11-Temperature motor = 0 °C</td><td>°C</td><td>Shows the temperature of the fan motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.12-Temperature module = 0 °C</td><td>°C</td><td>Shows the temperature of the module motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.13-Temp. electronic = 0 °C</td><td>°C</td><td>Shows the temperature of the electronics of the motor.
Note: This parameter is only shown if [G11.2.4-Module Fan 2 Model] ≠ Tesoer and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.4.14-Current power = 0 W</td><td>W</td><td>Shows the current active power of the motor.</td></tr>
<tr><td>SV11.4.15-Fan type = [G11.2.4-Module Fan 2 Model]</td><td>NONE</td><td>Shows the fan type configured in [G11.2.4-Module Fan 2 Model].
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>EBM-Papst</td><td>EBM-Papst fan supplier.</td></tr>
<tr><td>Tesoer</td><td>Tesoer fan supplier.</td></tr>
<tr><td>Delta</td><td>Delta fan supplier.</td></tr>
</table></td></tr>
<tr><td>SV11.4.16-Fan delta model = Unknown</td><td>NONE</td><td>Indicates which Delta fan model is configured.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>The found Delta fan does not correspond to a valid fan.</td></tr>
<tr><td>L9210</td><td>The found Delta fan corresponds to the L9210 reference.</td></tr>
<tr><td>L9587</td><td>The found Delta fan corresponds to the L9587 reference.</td></tr>
<tr><td>L9169</td><td>The found Delta fan corresponds to the L9169 reference.</td></tr>
</table>
Notes:
– This parameter is displayed if [G11.2.4-Module Fan 2 Model] = Delta.
– If two different types of Delta are installed in the same zone, some of these faults will be triggered: F230, F231 and F232.</td></tr>
</table>

5.10.5. Subgroup SV11.5: Fan Mod. 3

Note: these parameters are only displayed if [G11.11-Avaliables Fan Nº] ≥ 5.

SV11.5.0: Fan Monitor Mod. 3

This subgroup shows in real time what is happening in the module 3 fan speed monitoring module.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.5.0.1-Fan State = Not Monitoring</td><td>NONE</td><td>Shows the fan status according to the following table:
<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Not Monitoring</td><td>The status of the fan is not being monitored.
This is the usual state before the fan starts.</td></tr><tr><td>Ramp Transition</td><td>Fan in ramp transition (accelerating or decelerating) taking into account the transition period as a function of the difference for the times defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time].</td></tr><tr><td>Periodic Check</td><td>Periodic monitoring of the fan (when it is not in ramp transition).</td></tr><tr><td>Fan fault</td><td>Fan on fault due to not reaching the set speed (F228).</td></tr></table></td></tr>
<tr><td>SV11.5.0.2-Curr. Setp. RPM = 0 rpm</td><td>rpm</td><td>Shows current setpoint in rpm.</td></tr>
<tr><td>SV11.5.0.3-Err. Margin RPM = 0 rpm</td><td>rpm</td><td>Shows the maximum permissible margin of error of the fan. It is updated each time there is a new setpoint independently if this triggers the transition state. It is different as a function of the fan type.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.0.4-Current Speed = 0 rpm</td><td>rpm</td><td>Shows the fan current speed in rpm.</td></tr>
<tr><td>SV11.5.0.5-Max Speed = 0 rpm</td><td>rpm</td><td>Shows the fan maximum speed in rpm.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.0.6-Ramp Time = 0 s</td><td>s</td><td>Shows the calculated ramp time for the change of setpoint, this is, the time that it will take the fan to reach the commanded speed.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.0.7-Elap. ramp time = 0 s</td><td>s</td><td>Shows the elapsed ramp time. This value is incremented until the value set in parameter [SV11.5.0.6-Ramp Time] is reached. During this time, the fan will be in state “Ramp Transition” and when the calculated ramp time is reached, the fan status will switch to “Periodic Check”.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.0.8-Deviation = 0 %</td><td>%</td><td>Shows the deviation error configured in [G11.14-Setpoint error deviation].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.0.9-Error counter = 0 s</td><td>s</td><td>Shows the time during which the current speed is not withing the admissible setpoint margin. If the value of [G11.15-Setpoint error delay][9] is reached, the fan will trigger fault F228.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.0.10-Dev. Error Delay= 0 s</td><td>s</td><td>This parameter shows the maximum time that can elapse without reaching the setpoint before changing to state “Fan fault”. This value is set in [G11.15-Setpoint error delay].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.0.11-Ramp UP Hyster = 0 rpm</td><td>rpm</td><td rowspan="2">These parameters show the calculated hysteresis for the fan ramp-up [SV11.5.0.11] and ramp-down [SV11.5.0.12].

The hysteresis is calculated as a function of the maximum fan speed and the ramp-up time defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Its minimum possible value is 2% of the maximum speed (this value has been chosen as the minimum hysteresis value because in the fan curves the minimum change is 1%).
Note: these parameters are shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.0.12-Ramp DOWN Hyster = 0 rpm</td><td>rpm</td></tr>
</table>

[9] Also shown in SV11.5.0.10.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.5.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the module 3 fan.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Fan module 3 is disabled.</td></tr><tr><td>OK fan</td><td>Communication between the uP and the fan module 3 is correct.</td></tr><tr><td>Comms Error</td><td>uP is unable to communicate with the fan module 3.</td></tr><tr><td>Error</td><td>Fan module 3 is in error.</td></tr></table></td></tr>
<tr><td>SV11.5.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.5.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan.
Note: This parameter is only shown if [G11.2.5-Module Fan 3 Model] = EBM-Papst. It is not displayed in Power Electronics remote management tool.</td></tr>
<tr><td>SV11.5.4-Fan zone = Modules zone</td><td>NONE</td><td>Shows the zone assigned to the module 3 fan.</td></tr>
<tr><td>SV11.5.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>Unknown request.</td></tr><tr><td>Check ID</td><td>The ID of the fan with which the inverter is communicating is checked for correctness.</td></tr><tr><td>Configuration 1</td><td rowspan="7">The inverter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Inverter sends a reset to the communicating fan.</td></tr><tr><td>Configuration 2</td></tr><tr><td>Configuration 3</td></tr><tr><td>Configuration 4</td></tr><tr><td>Configuration 5</td></tr><tr><td>Configuration 6</td></tr><tr><td>Configuration 7</td></tr><tr><td>Read serial number</td><td>The inverter reads the serial number of the fan.</td></tr><tr><td>Reset</td><td>The inverter sends a reset to the communicating fan.</td></tr><tr><td>Get status</td><td>The inverter asks the fan to indicate its internal status to the inverter.</td></tr><tr><td>Get speed</td><td>The inverter asks the fan to get the current fan speed.</td></tr><tr><td>Get temperature</td><td>The inverter asks the fan to get the internal temperature of the fan.</td></tr><tr><td>Get power</td><td>The inverter asks the fan to get the current power consumption of the fan.</td></tr><tr><td>Set setpoint</td><td>The inverter sets the speed setpoint configured in the G11 to the fan.</td></tr><tr><td>Special request</td><td>Special request.</td></tr><tr><td>Get references</td><td>The inverter uses the reference values (voltage and current) from the fan to calculate the power consumption.
Note: this applies only to EBM-Papst fans.</td></tr></table>Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.5.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.5.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>RS485 communication fault with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.5-Module Fan 3 Model] = EBM-Papst</td></tr>
<tr><td>SV11.5.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fan fault.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.5-Module Fan 3 Model] = Tesoer.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.5.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronics temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>High line voltage.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>The shedding function is active.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.5-Module Fan 3 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.5.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL out of control.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.5-Module Fan 3 Model] = Tesoer.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.5.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the DELTA fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DC bus average overcurrent</td><td>Average DC bus overcurrent.</td></tr>
<tr><td>1</td><td>DC bus average overvoltage</td><td>Average DC bus overvoltage.</td></tr>
<tr><td>2</td><td>DC bus average undervoltage</td><td>Average DC bus undervoltage.</td></tr>
<tr><td>3</td><td>EE ove protection temperature</td><td>Fan overtemperature protection.</td></tr>
<tr><td>4</td><td>Fan lock</td><td>Fan lock.</td></tr>
<tr><td>5</td><td>AC lose phase</td><td>AC phase loss.</td></tr>
<tr><td>6</td><td>Fan reverse run</td><td>Fan reverse.</td></tr>
<tr><td>7</td><td>Hall signal abnormal</td><td>Abnormal hall signal.</td></tr>
<tr><td>8</td><td>EEPROM read/write fail</td><td>EEPROM read/write fault.</td></tr>
<tr><td>9</td><td>DC bus peak over current</td><td>DC bus peak overcurrent.</td></tr>
<tr><td>10</td><td>AC bus over voltage</td><td>AC bus overvoltage.</td></tr>
<tr><td>11</td><td>AC bus under voltage</td><td>AC bus undervoltage.</td></tr>
<tr><td>12</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>13</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.5-Module Fan 3 Model] = Delta.</td></tr>
<tr><td>SV11.5.11-Temperature motor = 0 °C</td><td>°C</td><td>Shows the temperature of the fan motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.12-Temperature module = 0 °C</td><td>°C</td><td>Shows the temperature of the module motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.13-Temp. electronic = 0 °C</td><td>°C</td><td>Shows the temperature of the electronics of the motor.
Note: This parameter is only displayed if [G11.2.5-Module Fan 3 Model] ≠ Tesoer and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.5.14-Current power = 0 W</td><td>W</td><td>Shows the current active power of the motor.</td></tr>
<tr><td>SV11.5.15-Fan type = [G11.2.5-Module Fan 3 Model]</td><td>NONE</td><td>Shows the fan type configured in [G11.2.5-Module Fan 3 Model].
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>EBM-Papst</td><td>EBM-Papst fan supplier.</td></tr>
<tr><td>Tesoer</td><td>Tesoer fan supplier.</td></tr>
<tr><td>Delta</td><td>Delta fan supplier.</td></tr>
</table></td></tr>
<tr><td>SV11.5.16-Fan delta model = Unknown</td><td>NONE</td><td>Indicates which Delta fan model is configured.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>The found Delta fan does not correspond to a valid fan.</td></tr>
<tr><td>L9210</td><td>The found Delta fan corresponds to the L9210 reference.</td></tr>
<tr><td>L9587</td><td>The found Delta fan corresponds to the L9587 reference.</td></tr>
<tr><td>L9169</td><td>The found Delta fan corresponds to the L9169 reference.</td></tr>
</table>
Notes:
– This parameter is displayed if [G11.2.5-Module Fan 3 Model] = Delta.
– If two different types of Delta are installed in the same zone, some of these faults will be triggered: F230, F231 and F232.</td></tr>
</table>

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5.10.6. Subgroup SV11.6: Fan Mod. 4

Note: these parameters are only displayed if [G11.11-Avaliables Fan Nº] ≥ 6.

SV11.6.0: Fan Monitor Mod. 4

This subgroup shows in real time what is happening in the module 4 fan speed monitoring module.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.6.0.1-Fan State = Not Monitoring</td><td>NONE</td><td>Shows the fan status according to the following table:
<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Not Monitoring</td><td>The status of the fan is not being monitored.
This is the usual state before the fan starts.</td></tr><tr><td>Ramp Transition</td><td>Fan in ramp transition (accelerating or decelerating) taking into account the transition period as a function of the difference for the times defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time].</td></tr><tr><td>Periodic Check</td><td>Periodic monitoring of the fan (when it is not in ramp transition).</td></tr><tr><td>Fan fault</td><td>Fan on fault due to not reaching the set speed (F228).</td></tr></table></td></tr>
<tr><td>SV11.6.0.2-Curr. Setp. RPM = 0 rpm</td><td>rpm</td><td>Shows current setpoint in rpm.</td></tr>
<tr><td>SV11.6.0.3-Err. Margin RPM = 0 rpm</td><td>rpm</td><td>Shows the maximum permissible margin of error of the fan. It is updated each time there is a new setpoint independently if this triggers the transition state. It is different as a function of the fan type.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.0.4-Current Speed = 0 rpm</td><td>rpm</td><td>Shows the fan current speed in rpm.</td></tr>
<tr><td>SV11.6.0.5-Max Speed = 0 rpm</td><td>rpm</td><td>Shows the fan maximum speed in rpm.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.0.6-Ramp Time = 0 s</td><td>s</td><td>Shows the calculated ramp time for the change of setpoint, this is, the time that it will take the fan to reach the commanded speed.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.0.7-Elap. ramp time = 0 s</td><td>s</td><td>Shows the elapsed ramp time. This value is incremented until the value set in parameter [SV11.6.0.6-Ramp Time] is reached. During this time, the fan will be in state “Ramp Transition” and when the calculated ramp time is reached, the fan status will switch to “Periodic Check”.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.0.8-Deviation = 0 %</td><td>%</td><td>Shows the deviation error configured in [G11.14-Setpoint error deviation].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.0.9-Error counter = 0 s</td><td>s</td><td>Shows the time during which the current speed is not withing the admissible setpoint margin. If the value of [G11.15-Setpoint error delay][10] is reached, the fan will trigger fault F228.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.0.10-Dev. Error Delay= 0 s</td><td>s</td><td>This parameter shows the maximum time that can elapse without reaching the setpoint before changing to state “Fan fault”. This value is set in [G11.15-Setpoint error delay].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.0.11-Ramp UP Hyster = 0 rpm</td><td>rpm</td><td rowspan="2">These parameters show the calculated hysteresis for the fan ramp-up [SV11.6.0.11] and ramp-down [SV11.6.0.12].

The hysteresis is calculated as a function of the maximum fan speed and the ramp-up time defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Its minimum possible value is 2% of the maximum speed (this value has been chosen as the minimum hysteresis value because in the fan curves the minimum change is 1%).
Note: these parameters are shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.0.12-Ramp DOWN Hyster = 0 rpm</td><td>rpm</td></tr>
</table>

[10] Also shown in SV11.6.0.10.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.6.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the module 4 fan.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Fan module 4 is disabled.</td></tr><tr><td>OK fan</td><td>Communication between the uP and the fan module 4 is correct.</td></tr><tr><td>Comms Error</td><td>uP is unable to communicate with the fan module 4.</td></tr><tr><td>Error</td><td>Fan module 4 is in error.</td></tr></table></td></tr>
<tr><td>SV11.6.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.6.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan.
Note: This parameter is only shown if [G11.2.6-Module Fan 4 Model] = EBM-Papst. It is not displayed in Power Electronics remote management tool.</td></tr>
<tr><td>SV11.6.4-Fan zone = Modules zone</td><td>NONE</td><td>Shows the zone assigned to the module 4 fan.</td></tr>
<tr><td>SV11.6.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>Unknown request.</td></tr><tr><td>Check ID</td><td>The ID of the fan with which the inverter is communicating is checked for correctness.</td></tr><tr><td>Configuration 1</td><td rowspan="8">The inverter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Inverter sends a reset to the communicating fan.</td></tr><tr><td>Configuration 2</td></tr><tr><td>Configuration 3</td></tr><tr><td>Configuration 4</td></tr><tr><td>Configuration 5</td></tr><tr><td>Configuration 6</td></tr><tr><td>Configuration 7</td></tr><tr><td>Read serial number</td><td>The inverter reads the serial number of the fan.</td></tr><tr><td>Reset</td><td>The inverter sends a reset to the communicating fan.</td></tr><tr><td>Get status</td><td>The inverter asks the fan to indicate its internal status to the inverter.</td></tr><tr><td>Get speed</td><td>The inverter asks the fan to get the current fan speed.</td></tr><tr><td>Get temperature</td><td>The inverter asks the fan to get the internal temperature of the fan.</td></tr><tr><td>Get power</td><td>The inverter asks the fan to get the current power consumption of the fan.</td></tr><tr><td>Set setpoint</td><td>The inverter sets the speed setpoint configured in the G11 to the fan.</td></tr><tr><td>Special request</td><td>Special request.</td></tr><tr><td>Get references</td><td>The inverter uses the reference values (voltage and current) from the fan to calculate the power consumption.
Note: this applies only to EBM-Papst fans.</td></tr></table>Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.6.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.6.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>RS485 communication fault with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.6-Module Fan 4 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.6.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fan fault.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.6-Module Fan 4 Model] = Tesoer.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.6.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.

<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronics temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>High line voltage.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>The shedding function is active.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.6-Module Fan 4 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.6.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.

<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL out of control.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.6-Module Fan 4 Model] = Tesoer.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.6.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the DELTA fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DC bus average overcurrent</td><td>Average DC bus overcurrent.</td></tr>
<tr><td>1</td><td>DC bus average overvoltage</td><td>Average DC bus overvoltage.</td></tr>
<tr><td>2</td><td>DC bus average undervoltage</td><td>Average DC bus undervoltage.</td></tr>
<tr><td>3</td><td>EE ove protection temperature</td><td>Fan overtemperature protection.</td></tr>
<tr><td>4</td><td>Fan lock</td><td>Fan lock.</td></tr>
<tr><td>5</td><td>AC lose phase</td><td>AC phase loss.</td></tr>
<tr><td>6</td><td>Fan reverse run</td><td>Fan reverse.</td></tr>
<tr><td>7</td><td>Hall signal abnormal</td><td>Abnormal hall signal.</td></tr>
<tr><td>8</td><td>EEPROM read/write fail</td><td>EEPROM read/write fault.</td></tr>
<tr><td>9</td><td>DC bus peak over current</td><td>DC bus peak overcurrent.</td></tr>
<tr><td>10</td><td>AC bus over voltage</td><td>AC bus overvoltage.</td></tr>
<tr><td>11</td><td>AC bus under voltage</td><td>AC bus undervoltage.</td></tr>
<tr><td>12</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>13</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.6-Module Fan 4 Model] = Delta.</td></tr>
<tr><td>SV11.6.11-Temperature motor = 0 ºC</td><td>ºC</td><td>Shows the temperature of the fan motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.12-Temperature module = 0 ºC</td><td>ºC</td><td>Shows the temperature of the module motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.13-Temp. electronic = 0 ºC</td><td>ºC</td><td>Shows the temperature of the electronics of the motor.
Note: This parameter is only displayed if [G11.2.6-Module Fan 4 Model] ≠Tesoer and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.6.14-Current power = 0 W</td><td>W</td><td>Shows the current active power of the motor.</td></tr>
<tr><td>SV11.6.15-Fan type = [G11.2.6-Module Fan 4 Model]</td><td>NONE</td><td>Shows the fan type configured in [G11.2.6-Module Fan 4 Model].
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>EBM-Papst</td><td>EBM-Papst fan supplier.</td></tr>
<tr><td>Tesoer</td><td>Tesoer fan supplier.</td></tr>
<tr><td>Delta</td><td>Delta fan supplier.</td></tr>
</table></td></tr>
<tr><td>SV11.6.16-Fan delta model = Unknown</td><td>NONE</td><td>Indicates which Delta fan model is configured.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>The found Delta fan does not correspond to a valid fan.</td></tr>
<tr><td>L9210</td><td>The found Delta fan corresponds to the L9210 reference.</td></tr>
<tr><td>L9587</td><td>The found Delta fan corresponds to the L9587 reference.</td></tr>
<tr><td>L9169</td><td>The found Delta fan corresponds to the L9169 reference.</td></tr>
</table>
Notes:
– This parameter is displayed if [G11.2.6-Module Fan 4 Model] = Delta.
– If two different types of Delta are installed in the same zone, some of these faults will be triggered: F230, F231 and F232.</td></tr>
</table>

5.10.7. Subgroup SV11.7: Fan LC

Note: these parameters are only displayed if [G11.11-Avaliables Fan Nº] ≥ 7.

SV11.7.0: Fan Monitor LC

This subgroup shows in real time what is happening in the LC filter fan speed monitoring module.

SCREEN	UNITS	DESCRIPTION
SV11.7.0.1-Fan State = Not Monitoring	NONE	Shows the fan status according to the following table: **STATUS** / **DESCRIPTION** Not Monitoring: The status of the fan is not being monitored. This is the usual state before the fan starts. Ramp Transition: Fan in ramp transition (accelerating or decelerating) taking into account the transition period as a function of the difference for the times defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Periodic Check: Periodic monitoring of the fan (when it is not in ramp transition). Fan fault: Fan on fault due to not reaching the set speed (F228).
SV11.7.0.2-Curr. Setp. RPM = 0 rpm	rpm	Shows current setpoint in rpm.
SV11.7.0.3-Err. Margin RPM = 0 rpm	rpm	Shows the maximum permissible margin of error of the fan. It is updated each time there is a new setpoint independently if this triggers the transition state. It is different as a function of the fan type. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.7.0.4-Current Speed = 0 rpm	rpm	Shows the fan current speed in rpm.
SV11.7.0.5-Max Speed = 0 rpm	rpm	Shows the fan maximum speed in rpm. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.7.0.6-Ramp Time = 0 s	s	Shows the calculated ramp time for the change of setpoint, this is, the time that it will take the fan to reach the commanded speed. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.7.0.7-Elap. ramp time = 0 s	s	Shows the elapsed ramp time. This value is increased until the value set in parameter [SV11.7.0.6-Ramp Time] is reached. During this time, the fan will be in state "Ramp Transition" and when the calculated ramp time is reached, the fan status will switch to "Periodic Check". **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.7.0.8-Deviation = 0 %	%	Shows the deviation error configured in [G11.14-Setpoint error deviation] **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.7.0.9-Error counter = 0 s	s	Shows the time during which the current speed is not withing the admissible setpoint margin. If the value of [G11.15-Setpoint error delay][11] is reached, the fan will trigger fault F228. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.7.0.10-Dev. Error Delay= 0 s	s	This parameter shows the maximum time that can elapse without reaching the setpoint before changing to state "Fan fault". This value is set in [G11.15-Setpoint error delay]. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.7.0.11-Ramp UP Hyster = 0 rpm	rpm	These parameters show the calculated hysteresis for the fan ramp-up [SV11.7.0.11] and ramp-down [SV11.7.0.12]. The hysteresis is calculated as a function of the maximum fan speed and the ramp-up time defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Its minimum possible value is 2% of the maximum speed (this value has been chosen as the minimum hysteresis value because in the fan curves the minimum change is 1%). **Note:** these parameters are shown if [G1.3-User] ≥ Power Electronics staff.
SV11.7.0.12-Ramp DOWN Hyster = 0 rpm	rpm	

[11] Also shown in SV11.7.0.10.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.7.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the LC filter fan.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Fan LC filter is disabled.</td></tr><tr><td>OK fan</td><td>Communication between the uP and the fan LC filter is correct.</td></tr><tr><td>Comms Error</td><td>uP is unable to communicate with the fan LC filter.</td></tr><tr><td>Error</td><td>Fan LC filter is in error.</td></tr></table></td></tr>
<tr><td>SV11.7.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.7.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan.
Note: This parameter is only shown if [G11.2.7-Filter LC Fan Model] = EBM-Papst. It is not displayed in Power Electronics remote management tool.</td></tr>
<tr><td>SV11.7.4-Fan zone = LC filter zone</td><td>NONE</td><td>Shows the zone assigned to the LC filter fan.</td></tr>
<tr><td>SV11.7.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>Unknown request.</td></tr><tr><td>Check ID</td><td>The ID of the fan with which the inverter is communicating is checked for correctness.</td></tr><tr><td>Configuration 1</td><td rowspan="7">The inverter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Inverter sends a reset to the communicating fan.</td></tr><tr><td>Configuration 2</td></tr><tr><td>Configuration 3</td></tr><tr><td>Configuration 4</td></tr><tr><td>Configuration 5</td></tr><tr><td>Configuration 6</td></tr><tr><td>Configuration 7</td></tr><tr><td>Read serial number</td><td>The inverter reads the serial number of the fan.</td></tr><tr><td>Reset</td><td>The inverter sends a reset to the communicating fan.</td></tr><tr><td>Get status</td><td>The inverter asks the fan to indicate its internal status to the inverter.</td></tr><tr><td>Get speed</td><td>The inverter asks the fan to get the current fan speed.</td></tr><tr><td>Get temperature</td><td>The inverter asks the fan to get the internal temperature of the fan.</td></tr><tr><td>Get power</td><td>The inverter asks the fan to get the current power consumption of the fan.</td></tr><tr><td>Set setpoint</td><td>The inverter sets the speed setpoint configured in the G11 to the fan.</td></tr><tr><td>Special request</td><td>Special request.</td></tr><tr><td>Get references</td><td>The inverter uses the reference values (voltage and current) from the fan to calculate the power consumption.
Note: this applies only to EBM-Papst fans.</td></tr></table>Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.7.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.7.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.7.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.7.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>RS485 communication fault with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.7-Filter LC Fan Model] = EBM-Papst.</td></tr>
<tr><td>SV11.7.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fan fault.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.7-Filter LC Fan Model] = Tesoer.</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.7.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronics temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>High line voltage.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>The shedding function is active.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.7-Filter LC Fan Model] = EBM-Papst.</td></tr>
<tr><td>SV11.7.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL out of control.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.7-Filter LC Fan Model] = Tesoer.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.7.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the DELTA fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DC bus average overcurrent</td><td>Average DC bus overcurrent.</td></tr>
<tr><td>1</td><td>DC bus average overvoltage</td><td>Average DC bus overvoltage.</td></tr>
<tr><td>2</td><td>DC bus average undervoltage</td><td>Average DC bus undervoltage.</td></tr>
<tr><td>3</td><td>EE ove protection temperature</td><td>Fan overtemperature protection.</td></tr>
<tr><td>4</td><td>Fan lock</td><td>Fan lock.</td></tr>
<tr><td>5</td><td>AC lose phase</td><td>AC phase loss.</td></tr>
<tr><td>6</td><td>Fan reverse run</td><td>Fan reverse.</td></tr>
<tr><td>7</td><td>Hall signal abnormal</td><td>Abnormal hall signal.</td></tr>
<tr><td>8</td><td>EEPROM read/write fail</td><td>EEPROM read/write fault.</td></tr>
<tr><td>9</td><td>DC bus peak over current</td><td>DC bus peak overcurrent.</td></tr>
<tr><td>10</td><td>AC bus over voltage</td><td>AC bus overvoltage.</td></tr>
<tr><td>11</td><td>AC bus under voltage</td><td>AC bus undervoltage.</td></tr>
<tr><td>12</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>13</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: this parameter is only displayed if [G11.2.7-Filter LC Fan Model] = Delta.</td></tr>
<tr><td>SV11.7.11-Temperature motor = 0 ºC</td><td>ºC</td><td>Shows the temperature of the fan motor.
Note: this parameter is only displayed if [G11.2.7-Filter LC Fan Model] ≠ Delta and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.7.12-Temperature module = 0 ºC</td><td>ºC</td><td>Shows the temperature of the module motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.7.13-Temp. electronic = 0 ºC</td><td>ºC</td><td>Shows the temperature of the electronics of the motor.
Note: This parameter is only displayed if [G11.2.7-Filter LC Fan Model] = EBM-Papst and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.7.14-Current power = 0 W</td><td>W</td><td>Shows the current active power of the motor.</td></tr>
<tr><td>SV11.7.15-Fan type = [G11.2.7-Filter LC Fan Model]</td><td>NONE</td><td>Shows the fan type configured in [G11.2.7-Filter LC Fan Model].
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>EBM-Papst</td><td>EBM-Papst fan supplier.</td></tr>
<tr><td>Tesoer</td><td>Tesoer fan supplier.</td></tr>
<tr><td>Delta</td><td>Delta fan supplier.</td></tr>
</table></td></tr>
<tr><td>SV11.7.16-Fan delta model = Unknown</td><td>NONE</td><td>Indicates which Delta fan model is configured.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>The found Delta fan does not correspond to a valid fan.</td></tr>
<tr><td>L9210</td><td>The found Delta fan corresponds to the L9210 reference.</td></tr>
<tr><td>L9587</td><td>The found Delta fan corresponds to the L9587 reference.</td></tr>
<tr><td>L9169</td><td>The found Delta fan corresponds to the L9169 reference.</td></tr>
</table>
Notes:
– This parameter is displayed if [G11.2.7-Filter LC Fan Model] = Delta.
– If two different types of Delta are installed in the same zone, some of these faults will be triggered: F230, F231 and F232.</td></tr>
<tr><td>SV11.7.17-Followed setpoint = LC filter</td><td>NONE</td><td>Shows the setpoint being followed by the filter LC fan according to the following table:
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>LC filter</td><td>LC filter setpoint is followed.</td></tr>
<tr><td>Modules</td><td>Modules setpoint is followed.</td></tr>
</table>
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

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5.10.8. Subgroup SV11.8: Fan MT 1

Note: these parameters are only displayed if [G11.11-Avaliables Fan Nº] ≥ 8.

SV11.8.0: Fan Monitor MT 1

This subgroup shows in real time what is happening in the MT1 fan speed monitoring module.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.8.0.1-Fan State = Not Monitoring</td><td>NONE</td><td>Shows the fan status according to the following table:
<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Not Monitoring</td><td>The status of the fan is not being monitored.
This is the usual state before the fan starts.</td></tr><tr><td>Ramp Transition</td><td>Fan in ramp transition (accelerating or decelerating) taking into account the transition period as a function of the difference for the times defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time].</td></tr><tr><td>Periodic Check</td><td>Periodic monitoring of the fan (when it is not in ramp transition).</td></tr><tr><td>Fan fault</td><td>Fan on fault due to not reaching the set speed (F228).</td></tr></table></td></tr>
<tr><td>SV11.8.0.2-Curr. Setp. RPM = 0 rpm</td><td>rpm</td><td>Shows current setpoint in rpm.</td></tr>
<tr><td>SV11.8.0.3-Err. Margin RPM = 0 rpm</td><td>rpm</td><td>Shows the maximum permissible margin of error of the fan. It is updated each time there is a new setpoint independently if this triggers the transition state. It is different as a function of the fan type.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.0.4-Current Speed = 0 rpm</td><td>rpm</td><td>Shows the fan current speed in rpm.</td></tr>
<tr><td>SV11.8.0.5-Max Speed = 0 rpm</td><td>rpm</td><td>Shows the fan maximum speed in rpm.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.0.6-Ramp Time = 0 s</td><td>s</td><td>Shows the calculated ramp time for the change of setpoint, this is, the time that it will take the fan to reach the commanded speed.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.0.7-Elap. ramp time = 0 s</td><td>s</td><td>Shows the elapsed ramp time. This value is incremented until the value set in parameter [SV11.8.0.6-Ramp Time] is reached. During this time, the fan will be in state “Ramp Transition” and when the calculated ramp time is reached, the fan status will switch to “Periodic Check”.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.0.8-Deviation = 0 %</td><td>%</td><td>Shows the deviation error configured in [G11.14-Setpoint error deviation].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.0.9-Error counter = 0 s</td><td>s</td><td>Shows the time during which the current speed is not withing the admissible setpoint margin. If the value of [G11.15-Setpoint error delay][12] is reached, the fan will trigger fault F228.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.0.10-Dev. Error Delay= 0 s</td><td>s</td><td>This parameter shows the maximum time that can elapse without reaching the setpoint before changing to state “Fan fault”. This value is set in [G11.15-Setpoint error delay].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.0.11-Ramp UP Hyster = 0 rpm</td><td>rpm</td><td rowspan="2">These parameters show the calculated hysteresis for the fan ramp-up [SV11.8.0.11] and ramp-down [SV11.8.0.12].

The hysteresis is calculated as a function of the maximum fan speed and the ramp-up time defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Its minimum possible value is 2% of the maximum speed (this value has been chosen as the minimum hysteresis value because in the fan curves the minimum change is 1%).
Note: these parameters are shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.0.12-Ramp DOWN Hyster = 0 rpm</td><td>rpm</td></tr>
</table>

[12] Also shown in SV11.8.0.10.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.8.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the MT1 fan.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Disabled</td><td>Fan MT1 is disabled.</td></tr>
<tr><td>OK fan</td><td>Communication between the uP and the fan MT1 is correct.</td></tr>
<tr><td>Comms Error</td><td>uP is unable to communicate with the fan MT1.</td></tr>
<tr><td>Error</td><td>Fan MT1 is in error.</td></tr>
</table></td></tr>
<tr><td>SV11.8.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.8.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan MT1.
Note: This parameter is only shown if [G11.2.8-MT Fan 1 Model] = EBM-Papst. It is not displayed in Power Electronics remote management tool.</td></tr>
<tr><td>SV11.8.4-Fan zone = MT zone</td><td>NONE</td><td>Shows the zone assigned to the MT1 fan.</td></tr>
<tr><td>SV11.8.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>Unknown request.</td></tr>
<tr><td>Check ID</td><td>The ID of the fan with which the inverter is communicating is checked for correctness.</td></tr>
<tr><td>Configuration 1</td><td rowspan="7">The inverter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Inverter sends a reset to the communicating fan.</td></tr>
<tr><td>Configuration 2</td></tr>
<tr><td>Configuration 3</td></tr>
<tr><td>Configuration 4</td></tr>
<tr><td>Configuration 5</td></tr>
<tr><td>Configuration 6</td></tr>
<tr><td>Configuration 7</td></tr>
<tr><td>Read serial number</td><td>The inverter reads the serial number of the fan.</td></tr>
<tr><td>Reset</td><td>The inverter sends a reset to the communicating fan.</td></tr>
<tr><td>Get status</td><td>The inverter asks the fan to indicate its internal status to the inverter.</td></tr>
<tr><td>Get speed</td><td>The inverter asks the fan to get the current fan speed.</td></tr>
<tr><td>Get temperature</td><td>The inverter asks the fan to get the internal temperature of the fan.</td></tr>
<tr><td>Get power</td><td>The inverter asks the fan to get the current power consumption of the fan.</td></tr>
<tr><td>Set setpoint</td><td>The inverter sets the speed setpoint configured in the G11 to the fan.</td></tr>
<tr><td>Special request</td><td>Special request.</td></tr>
<tr><td>Get references</td><td>The inverter uses the reference values (voltage and current) from the fan to calculate the power consumption.
Note: this applies only to EBM-Papst fans.</td></tr>
</table>
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.8.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.8.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>RS485 communication fault with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.8-MT Fan 1 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.8.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fan fault.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.8-MT Fan 1 Model] = Tesoer.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.8.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronics temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>High line voltage.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>The shedding function is active.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.8-MT Fan 1 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.8.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL out of control.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.8-MT Fan 1 Model] = Tesoer.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.8.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the DELTA fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DC bus average overcurrent</td><td>Average DC bus overcurrent.</td></tr>
<tr><td>1</td><td>DC bus average overvoltage</td><td>Average DC bus overvoltage.</td></tr>
<tr><td>2</td><td>DC bus average undervoltage</td><td>Average DC bus undervoltage.</td></tr>
<tr><td>3</td><td>EE ove protection temperature</td><td>Fan overtemperature protection.</td></tr>
<tr><td>4</td><td>Fan lock</td><td>Fan lock.</td></tr>
<tr><td>5</td><td>AC lose phase</td><td>AC phase loss.</td></tr>
<tr><td>6</td><td>Fan reverse run</td><td>Fan reverse.</td></tr>
<tr><td>7</td><td>Hall signal abnormal</td><td>Abnormal hall signal.</td></tr>
<tr><td>8</td><td>EEPROM read/write fail</td><td>EEPROM read/write fault.</td></tr>
<tr><td>9</td><td>DC bus peak over current</td><td>DC bus peak overcurrent.</td></tr>
<tr><td>10</td><td>AC bus over voltage</td><td>AC bus overvoltage.</td></tr>
<tr><td>11</td><td>AC bus under voltage</td><td>AC bus undervoltage.</td></tr>
<tr><td>12</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>13</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.8-MT Fan 1 Model] = Delta.</td></tr>
<tr><td>SV11.8.11-Temperature motor = 0 ºC</td><td>ºC</td><td>Shows the temperature of the fan motor.
Note: this parameter is only displayed if [G11.2.8-MT Fan 1 Model] ≠ Delta and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.12-Temperature module = 0 ºC</td><td>ºC</td><td>Shows the temperature of the module motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.13-Temp. electronic = 0 ºC</td><td>ºC</td><td>Shows the temperature of the electronics of the motor.
Note: This parameter is only displayed if [G11.2.8-MT Fan 1 Model] = EBM-Papst and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.8.14-Current power = 0 W</td><td>W</td><td>Shows the current active power of the motor.</td></tr>
<tr><td>SV11.8.15-Fan type = [G11.2.8-MT Fan 1 Model]</td><td>NONE</td><td>Shows the fan type configured in [G11.2.8-MT Fan 1 Model].
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>EBM-Papst</td><td>EBM-Papst fan supplier.</td></tr>
<tr><td>Tesoer</td><td>Tesoer fan supplier.</td></tr>
<tr><td>Delta</td><td>Delta fan supplier.</td></tr>
</table></td></tr>
<tr><td>SV11.8.16-Fan delta model = Unknown</td><td>NONE</td><td>Indicates which Delta fan model is configured.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>The found Delta fan does not correspond to a valid fan.</td></tr>
<tr><td>L9210</td><td>The found Delta fan corresponds to the L9210 reference.</td></tr>
<tr><td>L9587</td><td>The found Delta fan corresponds to the L9587 reference.</td></tr>
<tr><td>L9169</td><td>The found Delta fan corresponds to the L9169 reference.</td></tr>
</table>
Notes:
– This parameter is displayed if [G11.2.8-MT Fan 1 Model] = Delta.
– If two different types of Delta are installed in the same zone, some of these faults will be triggered: F230, F231 and F232.</td></tr>
</table>

5.10.9. Subgroup SV11.9: Fan MT 2

Note: these parameters are only displayed if [G11.11-Avaliables Fan Nº] ≥ 9.

SV11.9.0: Fan Monitor MT 2

This subgroup shows in real time what is happening in the MT2 fan speed monitoring module.

SCREEN	UNITS	DESCRIPTION
SV11.9.0.1-Fan State = Not Monitoring	NONE	Shows the fan status according to the following table: **STATUS** / **DESCRIPTION** Not Monitoring: The status of the fan is not being monitored. This is the usual state before the fan starts. Ramp Transition: Fan in ramp transition (accelerating or decelerating) taking into account the transition period as a function of the difference for the times defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Periodic Check: Periodic monitoring of the fan (when it is not in ramp transition). Fan fault: Fan on fault due to not reaching the set speed (F228).
SV11.9.0.2-Curr. Setp. RPM = 0 rpm	rpm	Shows current setpoint in rpm.
SV11.9.0.3-Err. Margin RPM = 0 rpm	rpm	Shows the maximum permissible margin of error of the fan. It is updated each time there is a new setpoint independently if this triggers the transition state. It is different as a function of the fan type. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.9.0.4-Current Speed = 0 rpm	rpm	Shows the fan current speed in rpm.
SV11.9.0.5-Max Speed = 0 rpm	rpm	Shows the fan maximum speed in rpm. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.9.0.6-Ramp Time = 0 s	s	Shows the calculated ramp time for the change of setpoint, this is, the time that it will take the fan to reach the commanded speed. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.9.0.7-Elap. ramp time = 0 s	s	Shows the elapsed ramp time. This value is incremented until the value set in parameter [SV11.9.0.6-Ramp Time] is reached. During this time, the fan will be in state "Ramp Transition" and when the calculated ramp time is reached, the fan status will switch to "Periodic Check". **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.9.0.8-Deviation = 0 %	%	Shows the deviation error configured in [G11.14-Setpoint error deviation] **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.9.0.9-Error counter = 0 s	s	Shows the time during which the current speed is not withing the admissible setpoint margin. If the value of [G11.15-Setpoint error delay][13] is reached, the fan will trigger fault F228. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.9.0.10-Dev. Error Delay= 0 s	s	This parameter shows the maximum time that can elapse without reaching the setpoint before changing to state "Fan fault". This value is set in [G11.15-Setpoint error delay]. **Note:** this parameter is shown if [G1.3-User] ≥ Power Electronics staff.
SV11.9.0.11-Ramp UP Hyster = 0 rpm	rpm	These parameters show the calculated hysteresis for the fan ramp-up [SV11.9.0.11] and ramp-down [SV11.9.0.12]. The hysteresis is calculated as a function of the maximum fan speed and the ramp-up time defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Its minimum possible value is 2% of the maximum speed (this value has been chosen as the minimum hysteresis value because in the fan curves the minimum change is 1%). **Note:** these parameters are shown if [G1.3-User] ≥ Power Electronics staff.
SV11.9.0.12-Ramp DOWN Hyster = 0 rpm	rpm	

[13] Also shown in SV11.9.0.10.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.9.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the MT2 fan.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Disabled</td><td>Fan MT2 is disabled.</td></tr>
<tr><td>OK fan</td><td>Communication between the uP and the fan MT2 is correct.</td></tr>
<tr><td>Comms Error</td><td>uP is unable to communicate with the fan MT2.</td></tr>
<tr><td>Error</td><td>Fan MT2 is in error.</td></tr>
</table></td></tr>
<tr><td>SV11.9.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.9.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan MT2.
Note: This parameter is only shown if [G11.2.9-MT Fan 2 Model] = EBM-Papst. It is not displayed in Power Electronics remote management tool.</td></tr>
<tr><td>SV11.9.4-Fan zone = MT zone</td><td>NONE</td><td>Shows the zone assigned to the MT2 fan.</td></tr>
<tr><td>SV11.9.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>Unknown request.</td></tr>
<tr><td>Check ID</td><td>The ID of the fan with which the inverter is communicating is checked for correctness.</td></tr>
<tr><td>Configuration 1</td><td rowspan="8">The inverter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Inverter sends a reset to the communicating fan.</td></tr>
<tr><td>Configuration 2</td></tr>
<tr><td>Configuration 3</td></tr>
<tr><td>Configuration 4</td></tr>
<tr><td>Configuration 5</td></tr>
<tr><td>Configuration 6</td></tr>
<tr><td>Configuration 7</td></tr>
<tr><td>Read serial number</td><td>The inverter reads the serial number of the fan</td></tr>
<tr><td>Reset</td><td>The inverter sends a reset to the communicating fan</td></tr>
<tr><td>Get status</td><td>The inverter asks the fan to indicate its internal status to the inverter</td></tr>
<tr><td>Get speed</td><td>The inverter asks the fan to get the current fan speed.</td></tr>
<tr><td>Get temperature</td><td>The inverter asks the fan to get the internal temperature of the fan.</td></tr>
<tr><td>Get power</td><td>The inverter asks the fan to get the current power consumption of the fan.</td></tr>
<tr><td>Set setpoint</td><td>The inverter sets the speed setpoint configured in the G11 to the fan.</td></tr>
<tr><td>Special request</td><td>Special request.</td></tr>
<tr><td>Get references</td><td>The inverter uses the reference values (voltage and current) from the fan to calculate the power consumption.
Note: this applies only to EBM-Papst fans.</td></tr>
</table>
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.9.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.9.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.9.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.9.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>RS485 communication fault with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.9-MT Fan 2 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.9.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fan fault.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.9-MT Fan 2 Model] = Tesoer.</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.9.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronics temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>High line voltage.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>The shedding function is active.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.9-MT Fan 2 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.9.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL out of control.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.9-MT Fan 2 Model] = Tesoer.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.9.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the DELTA fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DC bus average overcurrent</td><td>Average DC bus overcurrent.</td></tr>
<tr><td>1</td><td>DC bus average overvoltage</td><td>Average DC bus overvoltage.</td></tr>
<tr><td>2</td><td>DC bus average undervoltage</td><td>Average DC bus undervoltage.</td></tr>
<tr><td>3</td><td>EE ove protection temperature</td><td>Fan overtemperature protection.</td></tr>
<tr><td>4</td><td>Fan lock</td><td>Fan lock.</td></tr>
<tr><td>5</td><td>AC lose phase</td><td>AC phase loss.</td></tr>
<tr><td>6</td><td>Fan reverse run</td><td>Fan reverse.</td></tr>
<tr><td>7</td><td>Hall signal abnormal</td><td>Abnormal hall signal.</td></tr>
<tr><td>8</td><td>EEPROM read/write fail</td><td>EEPROM read/write fault.</td></tr>
<tr><td>9</td><td>DC bus peak over current</td><td>DC bus peak overcurrent.</td></tr>
<tr><td>10</td><td>AC bus over voltage</td><td>AC bus overvoltage.</td></tr>
<tr><td>11</td><td>AC bus under voltage</td><td>AC bus undervoltage.</td></tr>
<tr><td>12</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>13</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.9-MT Fan 2 Model] = Delta.</td></tr>
<tr><td>SV11.9.11-Temperature motor = 0 ºC</td><td>ºC</td><td>Shows the temperature of the fan motor.
Note: This parameter is only displayed if [G11.2.9-MT Fan 2 Model] ≠ Delta and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.9.12-Temperature module = 0 ºC</td><td>ºC</td><td>Shows the temperature of the module motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.9.13-Temp. electronic = 0 ºC</td><td>ºC</td><td>Shows the temperature of the electronics of the motor.
Note: This parameter is only displayed if [G11.2.9-MT Fan 2 Model] = EBM-Papst and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.9.14-Current power = 0 W</td><td>W</td><td>Shows the current active power of the motor.</td></tr>
<tr><td>SV11.9.15-Fan type = [G11.2.9-MT Fan 2 Model]</td><td>NONE</td><td>Shows the fan type configured in [G11.2.9-MT Fan 2 Model].
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>EBM-Papst</td><td>EBM-Papst fan supplier.</td></tr>
<tr><td>Tesoer</td><td>Tesoer fan supplier.</td></tr>
<tr><td>Delta</td><td>Delta fan supplier.</td></tr>
</table></td></tr>
<tr><td>SV11.9.16-Fan delta model = Unknown</td><td>NONE</td><td>Indicates which Delta fan model is configured.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>The found Delta fan does not correspond to a valid fan.</td></tr>
<tr><td>L9210</td><td>The found Delta fan corresponds to the L9210 reference.</td></tr>
<tr><td>L9587</td><td>The found Delta fan corresponds to the L9587 reference.</td></tr>
<tr><td>L9169</td><td>The found Delta fan corresponds to the L9169 reference.</td></tr>
</table>
Notes:
– This parameter is displayed if [G11.2.9-MT Fan 2 Model] = Delta.
– If two different types of Delta are installed in the same zone, some of these faults will be triggered: F230, F231 and F232.</td></tr>
</table>

EN

5.10.10. Subgroup SV11.10: Fan MT 3

Note: these parameters are only displayed if [G11.11-Avaliables Fan Nº] ≥ 10.

SV11.10.0: Fan Monitor MT 3

This subgroup shows in real time what is happening in the MT3 fan speed monitoring module.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.10.0.1-Fan State = Not Monitoring</td><td>NONE</td><td>Shows the fan status according to the following table:
<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Not Monitoring</td><td>The status of the fan is not being monitored. This is the usual state before the fan starts.</td></tr><tr><td>Ramp Transition</td><td>Fan in ramp transition (accelerating or decelerating) taking into account the transition period as a function of the difference for the times defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time].</td></tr><tr><td>Periodic Check</td><td>Periodic monitoring of the fan (when it is not in ramp transition).</td></tr><tr><td>Fan fault</td><td>Fan on fault due to not reaching the set speed (F228).</td></tr></table></td></tr>
<tr><td>SV11.10.0.2-Curr. Setp. RPM = 0 rpm</td><td>rpm</td><td>Shows current setpoint in rpm.</td></tr>
<tr><td>SV11.10.0.3-Err. Margin RPM = 0 rpm</td><td>rpm</td><td>Shows the maximum permissible margin of error of the fan. It is updated each time there is a new setpoint independently if this triggers the transition state. It is different as a function of the fan type.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.0.4-Current Speed = 0 rpm</td><td>rpm</td><td>Shows the fan current speed in rpm.</td></tr>
<tr><td>SV11.10.0.5-Max Speed = 0 rpm</td><td>rpm</td><td>Shows the fan maximum speed in rpm.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.0.6-Ramp Time = 0 s</td><td>s</td><td>Shows the calculated ramp time for the change of setpoint, this is, the time that it will take the fan to reach the commanded speed.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.0.7-Elap. ramp time = 0 s</td><td>s</td><td>Shows the elapsed ramp time. This value is incremented until the value set in parameter [SV11.10.0.6-Ramp Time] is reached. During this time, the fan will be in state “Ramp Transition” and when the calculated ramp time is reached, the fan status will switch to “Periodic Check”.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.0.8-Deviation = 0 %</td><td>%</td><td>Shows the deviation error configured in [G11.14-Setpoint error deviation]
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.0.9-Error counter = 0 s</td><td>s</td><td>Shows the time during which the current speed is not withing the admissible setpoint margin. If the value of [G11.15-Setpoint error delay][14] is reached, the fan will trigger fault F228.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.0.10-Dev. Error Delay= 0 s</td><td>s</td><td>This parameter shows the maximum time that can elapse without reaching the setpoint before changing to state “Fan fault”. This value is set in [G11.15-Setpoint error delay].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.0.11-Ramp UP Hyster = 0 rpm</td><td>rpm</td><td rowspan="2">These parameters show the calculated hysteresis for the fan ramp-up [SV11.10.0.11] and ramp-down [SV11.10.0.12].

The hysteresis is calculated as a function of the maximum fan speed and the ramp-up time defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Its minimum possible value is 2% of the maximum speed (this value has been chosen as the minimum hysteresis value because in the fan curves the minimum change is 1%).
Note: these parameters are shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.0.12-Ramp DOWN Hyster = 0 rpm</td><td>rpm</td></tr>
</table>

[14] Also shown in SV11.10.0.10.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.10.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the MT3 fan.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Fan MT3 is disabled.</td></tr><tr><td>OK fan</td><td>Communication between the uP and the fan MT3 is correct.</td></tr><tr><td>Comms Error</td><td>uP is unable to communicate with the fan MT3.</td></tr><tr><td>Error</td><td>Fan MT3 is in error.</td></tr></table></td></tr>
<tr><td>SV11.10.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.10.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan MT3.
Note: This parameter is only shown if [G11.2.10-MT Fan 3 Model] = EBM-Papst. It is not displayed in Power Electronics remote management tool.</td></tr>
<tr><td>SV11.10.4-Fan zone = MT zone</td><td>NONE</td><td>Shows the zone assigned to the MT3 fan.</td></tr>
<tr><td>SV11.10.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>Unknown request.</td></tr><tr><td>Check ID</td><td>The ID of the fan with which the inverter is communicating is checked for correctness.</td></tr><tr><td>Configuration 1</td><td rowspan="7">The inverter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Inverter sends a reset to the communicating fan.</td></tr><tr><td>Configuration 2</td></tr><tr><td>Configuration 3</td></tr><tr><td>Configuration 4</td></tr><tr><td>Configuration 5</td></tr><tr><td>Configuration 6</td></tr><tr><td>Configuration 7</td></tr><tr><td>Read serial number</td><td>The inverter reads the serial number of the fan.</td></tr><tr><td>Reset</td><td>The inverter sends a reset to the communicating fan.</td></tr><tr><td>Get status</td><td>The inverter asks the fan to indicate its internal status to the inverter.</td></tr><tr><td>Get speed</td><td>The inverter asks the fan to get the current fan speed.</td></tr><tr><td>Get temperature</td><td>The inverter asks the fan to get the internal temperature of the fan.</td></tr><tr><td>Get power</td><td>The inverter asks the fan to get the current power consumption of the fan.</td></tr><tr><td>Set setpoint</td><td>The inverter sets the speed setpoint configured in the G11 to the fan.</td></tr><tr><td>Special request</td><td>Special request.</td></tr><tr><td>Get references</td><td>The inverter uses the reference values (voltage and current) from the fan to calculate the power consumption.
Note: this applies only to EBM-Papst fans.</td></tr></table>Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.10.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.10.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>RS485 communication fault with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.10-MT Fan 3 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.10.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fan fault.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.10-MT Fan 3 Model] = Tesoer.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.10.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronics temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input..</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>High line voltage.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>The shedding function is active.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.10-MT Fan 3 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.10.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL out of control.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.10-MT Fan 3 Model] = Tesoer.</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.10.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the DELTA fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DC bus average overcurrent</td><td>Average DC bus overcurrent.</td></tr>
<tr><td>1</td><td>DC bus average overvoltage</td><td>Average DC bus overvoltage.</td></tr>
<tr><td>2</td><td>DC bus average undervoltage</td><td>Average DC bus undervoltage.</td></tr>
<tr><td>3</td><td>EE ove protection temperature</td><td>Fan overtemperature protection.</td></tr>
<tr><td>4</td><td>Fan lock</td><td>Fan lock.</td></tr>
<tr><td>5</td><td>AC lose phase</td><td>AC phase loss.</td></tr>
<tr><td>6</td><td>Fan reverse run</td><td>Fan reverse.</td></tr>
<tr><td>7</td><td>Hall signal abnormal</td><td>Abnormal hall signal.</td></tr>
<tr><td>8</td><td>EEPROM read/write fail</td><td>EEPROM read/write fault.</td></tr>
<tr><td>9</td><td>DC bus peak over current</td><td>DC bus peak overcurrent.</td></tr>
<tr><td>10</td><td>AC bus over voltage</td><td>AC bus overvoltage.</td></tr>
<tr><td>11</td><td>AC bus under voltage</td><td>AC bus undervoltage.</td></tr>
<tr><td>12</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>13</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.10-MT Fan 3 Model] = Delta.</td></tr>
<tr><td>SV11.10.11-Temperature motor = 0 ºC</td><td>ºC</td><td>Shows the temperature of the fan motor.
Note: this parameter is only displayed if [G11.2.10-MT Fan 3 Model] ≠ Delta and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.12-Temperature module = 0 ºC</td><td>ºC</td><td>Shows the temperature of the module motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.13-Temp. electronic = 0 ºC</td><td>ºC</td><td>Shows the temperature of the electronics of the motor.
Note: This parameter is only displayed if [G11.2.10-MT Fan 3 Model] = EBM-Papst and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.10.14-Current power = 0 W</td><td>W</td><td>Shows the current active power of the motor.</td></tr>
<tr><td>SV11.10.15-Fan type = [G11.2.10-MT Fan 3 Model]</td><td>NONE</td><td>Shows the fan type configured in [G11.2.10-MT Fan 3 Model].
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>EBM-Papst</td><td>EBM-Papst fan supplier.</td></tr>
<tr><td>Tesoer</td><td>Tesoer fan supplier.</td></tr>
<tr><td>Delta</td><td>Delta fan supplier.</td></tr>
</table></td></tr>
<tr><td>SV11.10.16-Fan delta model = Unknown</td><td>NONE</td><td>Indicates which Delta fan model is configured.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>The found Delta fan does not correspond to a valid fan.</td></tr>
<tr><td>L9210</td><td>The found Delta fan corresponds to the L9210 reference.</td></tr>
<tr><td>L9587</td><td>The found Delta fan corresponds to the L9587 reference.</td></tr>
<tr><td>L9169</td><td>The found Delta fan corresponds to the L9169 reference.</td></tr>
</table>
Notes:
– This parameter is displayed if [G11.2.10-MT Fan 3 Model] = Delta.
– If two different types of Delta are installed in the same zone, some of these faults will be triggered: F230, F231 and F232.</td></tr>
</table>

5.10.11. Subgroup SV11.11: Fan MT 4

Note: these parameters are only displayed if [G11.11-Avaliables Fan Nº] ≥ 11.

SV11.11.0: Fan Monitor MT 4

This subgroup shows in real time what is happening in the MT4 fan speed monitoring module.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.11.0.1-Fan State = Not Monitoring</td><td>NONE</td><td>Shows the fan status according to the following table:
<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Not Monitoring</td><td>The status of the fan is not being monitored. This is the usual state before the fan starts.</td></tr><tr><td>Ramp Transition</td><td>Fan in ramp transition (accelerating or decelerating) taking into account the transition period as a function of the difference for the times defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time].</td></tr><tr><td>Periodic Check</td><td>Periodic monitoring of the fan (when it is not in ramp transition).</td></tr><tr><td>Fan fault</td><td>Fan on fault due to not reaching the set speed (F228).</td></tr></table></td></tr>
<tr><td>SV11.11.0.2-Curr. Setp. RPM = 0 rpm</td><td>rpm</td><td>Shows current setpoint in rpm.</td></tr>
<tr><td>SV11.11.0.3-Err. Margin RPM = 0 rpm</td><td>rpm</td><td>Shows the maximum permissible margin of error of the fan. It is updated each time there is a new setpoint independently if this triggers the transition state. It is different as a function of the fan type.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.0.4-Current Speed = 0 rpm</td><td>rpm</td><td>Shows the fan current speed in rpm.</td></tr>
<tr><td>SV11.11.0.5-Max Speed = 0 rpm</td><td>rpm</td><td>Shows the fan maximum speed in rpm.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.0.6-Ramp Time = 0 s</td><td>s</td><td>Shows the calculated ramp time for the change of setpoint, this is, the time that it will take the fan to reach the commanded speed.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.0.7-Elap. ramp time = 0 s</td><td>s</td><td>Shows the elapsed ramp time. This value is incremented until the value set in parameter [SV11.11.0.6-Ramp Time] is reached. During this time, the fan will be in state “Ramp Transition” and when the calculated ramp time is reached, the fan status will switch to “Periodic Check”.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.0.8-Deviation = 0 %</td><td>%</td><td>Shows the deviation error configured in [G11.14-Setpoint error deviation].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.0.9-Error counter = 0 s</td><td>s</td><td>Shows the time during which the current speed is not withing the admissible setpoint margin. If the value of [G11.15-Setpoint error delay][15] is reached, the fan will trigger fault F228.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.0.10-Dev. Error Delay= 0 s</td><td>s</td><td>This parameter shows the maximum time that can elapse without reaching the setpoint before changing to state “Fan fault”. This value is set in [G11.15-Setpoint error delay].
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.0.11-Ramp UP Hyster = 0 rpm</td><td>rpm</td><td rowspan="2">These parameters show the calculated hysteresis for the fan ramp-up [SV11.11.0.11] and ramp-down [SV11.11.0.12].

The hysteresis is calculated as a function of the maximum fan speed and the ramp-up time defined in [G11.7.9-Ramp-up time] and [G11.7.10-Ramp-down time]. Its minimum possible value is 2% of the maximum speed (this value has been chosen as the minimum hysteresis value because in the fan curves the minimum change is 1%).
Note: these parameters are shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.0.12-Ramp DOWN Hyster = 0 rpm</td><td>rpm</td></tr>
</table>

[15] Also shown in SV11.11.0.10.

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.11.1-State comms = Disabled</td><td>NONE</td><td>Shows the status of communications between the µP and the MT4 fan.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Fan MT4 is disabled.</td></tr><tr><td>OK fan</td><td>Communication between the uP and the fan MT4 is correct.</td></tr><tr><td>Comms Error</td><td>uP is unable to communicate with the fan MT4.</td></tr><tr><td>Error</td><td>Fan MT4 is in error.</td></tr></table></td></tr>
<tr><td>SV11.11.2-Error = 0x00 Hex</td><td>NONE</td><td>Shows the type of error of the fan.</td></tr>
<tr><td>SV11.11.3-Nserial Final =</td><td>NONE</td><td>Shows the serial number of the fan MT4.
Note: This parameter is only displayed if [G11.2.11-MT Fan 4 Model] = EBM-Papst. It is not displayed in Power Electronics remote management tool.</td></tr>
<tr><td>SV11.11.4-Fan zone = MT zone</td><td>NONE</td><td>Shows the zone assigned to the fan MT4.</td></tr>
<tr><td>SV11.11.5-Current Request = Unknown</td><td>NONE</td><td>Shows what action is requested to the fan.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Unknown</td><td>Unknown request.</td></tr><tr><td>Check ID</td><td>The ID of the fan with which the inverter is communicating is checked for correctness.</td></tr><tr><td>Configuration 1</td><td rowspan="7">The inverter is setting the internal fan configuration.
Inverter reads the serial number of the fan.
Inverter sends a reset to the communicating fan.</td></tr><tr><td>Configuration 2</td></tr><tr><td>Configuration 3</td></tr><tr><td>Configuration 4</td></tr><tr><td>Configuration 5</td></tr><tr><td>Configuration 6</td></tr><tr><td>Configuration 7</td></tr><tr><td>Read serial number</td><td>The inverter reads the serial number of the fan.</td></tr><tr><td>Reset</td><td>The inverter sends a reset to the communicating fan.</td></tr><tr><td>Get status</td><td>The inverter asks the fan to indicate its internal status to the inverter.</td></tr><tr><td>Get speed</td><td>The inverter asks the fan to get the current fan speed.</td></tr><tr><td>Get temperature</td><td>The inverter asks the fan to get the internal temperature of the fan.</td></tr><tr><td>Get power</td><td>The inverter asks the fan to get the current power consumption of the fan.</td></tr><tr><td>Set setpoint</td><td>The inverter sets the speed setpoint configured in the G11 to the fan.</td></tr><tr><td>Special request</td><td>Special request.</td></tr><tr><td>Get references</td><td>The inverter uses the reference values (voltage and current) from the fan to calculate the power consumption.
Note: this applies only to EBM-Papst fans.</td></tr></table>Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.6-Id modbus slave = 0</td><td>NONE</td><td>Shows the ID of the fan.</td></tr>
<tr><td>SV11.11.7-Speed = 0</td><td>NONE</td><td>Shows the current speed of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.8-Rotation = Counter clockwise</td><td>NONE</td><td>Shows the rotation direction of the fan.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.11.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Phase failure</td><td>Fault in one of the fan phases.</td></tr>
<tr><td>1</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>2</td><td>output stage overheating</td><td>Overheating in the fan output.</td></tr>
<tr><td>3</td><td>Communication error</td><td>RS485 communication fault with the fan.</td></tr>
<tr><td>4</td><td>General error</td><td>Fan general fault.</td></tr>
<tr><td>5</td><td>Motor overheating</td><td>Motor overheating of the fan.</td></tr>
<tr><td>6</td><td>Hall sensor error</td><td>Hall sensor fault.</td></tr>
<tr><td>7</td><td>Motor blocked</td><td>Motor blocked.</td></tr>
<tr><td>8</td><td>Speed limit exceeded</td><td>The maximum fan speed limit has been exceeded.</td></tr>
<tr><td>9</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>10</td><td>Rotor calibration error</td><td>Fan rotor calibration.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>DC link undervoltage</td><td>DC undervoltage in the fan.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>15</td><td>Bit unused</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.11-MT Fan 4 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.11.9-Motor status = 0000000000000000</td><td>NONE</td><td>Shows the motor status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Power on</td><td>Fan on.</td></tr>
<tr><td>1</td><td>Stopped</td><td>Fan stopped.</td></tr>
<tr><td>2</td><td>Measuring offset current</td><td>Measuring offset of fan current.</td></tr>
<tr><td>3</td><td>Charging boot strap capacitate</td><td>Charging fan capacity.</td></tr>
<tr><td>4</td><td>Motor running</td><td>Fan motor is running.</td></tr>
<tr><td>5</td><td>Fault</td><td>Fan fault.</td></tr>
<tr><td>6</td><td>Catch spin</td><td>Fan rotation mode.</td></tr>
<tr><td>7</td><td>Parking</td><td>Fan is stopping.</td></tr>
<tr><td>8</td><td>Open loop acceleration</td><td>Fan is in the acceleration phase following an open loop.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.11-MT Fan 4 Model] = Tesoer.</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.11.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the EBM PAPST fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Current limit engaged</td><td>The fan is reaching its current limit.</td></tr>
<tr><td>1</td><td>Line impedance too high</td><td>Fan has too high impedance.</td></tr>
<tr><td>2</td><td>Power limitation engaged</td><td>The fan is reaching its power limit.</td></tr>
<tr><td>3</td><td>Output stage temperature high</td><td>Fan output temperature is too high.</td></tr>
<tr><td>4</td><td>Motor temperature high</td><td>Motor fan temperature is too high.</td></tr>
<tr><td>5</td><td>Temp inside electronics high</td><td>Electronics temperature is too high.</td></tr>
<tr><td>6</td><td>DC-link voltage low</td><td>DC voltage fan is too low.</td></tr>
<tr><td>7</td><td>Brake operation</td><td>Braking operation is running.</td></tr>
<tr><td>8</td><td>Calibration of rotor sensor</td><td>Calibration phase of rotor sensor.</td></tr>
<tr><td>9</td><td>Speed is less than limit</td><td>Fan speed is lower than limit.</td></tr>
<tr><td>10</td><td>Wire break analog input/PWM</td><td>Break connection of analog input.</td></tr>
<tr><td>11</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>12</td><td>High voltage</td><td>High voltage.</td></tr>
<tr><td>13</td><td>Bit unused</td><td>Not used.</td></tr>
<tr><td>14</td><td>Line voltage high</td><td>High line voltage.</td></tr>
<tr><td>15</td><td>Shedding function active</td><td>The shedding function is active.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.11-MT Fan 4 Model] = EBM-Papst.</td></tr>
<tr><td>SV11.11.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the TESOER fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Motor gatekill fault</td><td>Motor gatekill fault.</td></tr>
<tr><td>1</td><td>DC bus critical over V critic</td><td>DC bus critical voltage level.</td></tr>
<tr><td>2</td><td>DC bus overvoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>3</td><td>DC bus undervoltage fault</td><td>DC bus overvoltage fault.</td></tr>
<tr><td>4</td><td>PLL fault</td><td>PLL out of control.</td></tr>
<tr><td>5</td><td>Zero speed fault</td><td>Zero speed fault.</td></tr>
<tr><td>6</td><td>Over temperature fault</td><td>Over temperature fault.</td></tr>
<tr><td>7</td><td>Rotor lock fault</td><td>The fan motor is locked.</td></tr>
<tr><td>8</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>9</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>10</td><td>MCE execution fault</td><td>MCE execution fault.</td></tr>
<tr><td>11</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>12</td><td>Parameter load fault</td><td>Parameter load fault.</td></tr>
<tr><td>13</td><td>UART link break fault</td><td>UART communication link is broken.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.11-MT Fan 4 Model] = Tesoer.</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV11.11.10-Warnings = 0000000000000000</td><td>NONE</td><td>Shows the warning status of the DELTA fan.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>DC bus average overcurrent</td><td>Average DC bus overcurrent.</td></tr>
<tr><td>1</td><td>DC bus average overvoltage</td><td>Average DC bus overvoltage.</td></tr>
<tr><td>2</td><td>DC bus average undervoltage</td><td>Average DC bus undervoltage.</td></tr>
<tr><td>3</td><td>EE ove protection temperature</td><td>Fan overtemperature protection.</td></tr>
<tr><td>4</td><td>Fan lock</td><td>Fan lock.</td></tr>
<tr><td>5</td><td>AC lose phase</td><td>AC phase loss.</td></tr>
<tr><td>6</td><td>Fan reverse run</td><td>Fan reverse.</td></tr>
<tr><td>7</td><td>Hall signal abnormal</td><td>Abnormal hall signal.</td></tr>
<tr><td>8</td><td>EEPROM read/write fail</td><td>EEPROM read/write fault.</td></tr>
<tr><td>9</td><td>DC bus peak over current</td><td>DC bus peak overcurrent.</td></tr>
<tr><td>10</td><td>AC bus over voltage</td><td>AC bus overvoltage.</td></tr>
<tr><td>11</td><td>AC bus under voltage</td><td>AC bus undervoltage.</td></tr>
<tr><td>12</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>13</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>14</td><td>Reserved</td><td>Not used.</td></tr>
<tr><td>15</td><td>Reserved</td><td>Not used.</td></tr>
</table>
Note: This parameter is only displayed if [G11.2.11-MT Fan 4 Model].= Delta.</td></tr>
<tr><td>SV11.11.11-Temperature motor = 0 ºC</td><td>ºC</td><td>Shows the temperature of the fan motor.
Note: This parameter is only displayed if [G11.2.11-MT Fan 4 Model] ≠ Delta and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.12-Temperature module = 0 ºC</td><td>ºC</td><td>Shows the temperature of the module motor.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.13-Temp. electronic = 0 ºC</td><td>ºC</td><td>Shows the temperature of the electronics of the motor.
Note: This parameter is only displayed if [G11.2.11-MT Fan 4 Model] = EBM-Papst and [G1.3-User] ≥ Power Electronics staff.</td></tr>
<tr><td>SV11.11.14-Current power = 0 W</td><td>W</td><td>Shows the current active power of the motor.</td></tr>
<tr><td>SV11.11.15-Fan type = [G11.2.11-MT Fan 4 Model]</td><td>NONE</td><td>Shows the fan type configured in [G11.2.11-MT Fan 4 Model].
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>EBM-Papst</td><td>EBM-Papst fan supplier.</td></tr>
<tr><td>Tesoer</td><td>Tesoer fan supplier.</td></tr>
<tr><td>Delta</td><td>Delta fan supplier.</td></tr>
</table></td></tr>
<tr><td>SV11.11.16-Fan delta model = Unknown</td><td>NONE</td><td>Indicates which Delta fan model is configured.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Unknown</td><td>The found Delta fan does not correspond to a valid fan.</td></tr>
<tr><td>L9210</td><td>The found Delta fan corresponds to the L9210 reference.</td></tr>
<tr><td>L9587</td><td>The found Delta fan corresponds to the L9587 reference.</td></tr>
<tr><td>L9169</td><td>The found Delta fan corresponds to the L9169 reference.</td></tr>
</table>
Notes:
– This parameter is displayed if [G11.2.11-MT Fan 4 Model] = Delta.
– If two different types of Delta are installed in the same zone, some of these faults will be triggered: F230, F231 and F232.</td></tr>
</table>

EN

5.11.Group SV12: Modules Information

This group allows the visualization of the status of the power modules of the inverter. The subgroup [SV12.0: General Information] shows general information about the power modules. From subgroup [SV12.1: Module 1] onwards, specific information on the individual modules is displayed. There are as many groups as there are internally configured modules.

Note: This group is displayed if [SG10.1.12-Number of modules ≥ 'n'] being 'n' the number of the module.

5.11.1. Subgroup SV12.0: General information

SCREEN	UNITS	DESCRIPTION
SV12.0.1-Running modules = 0000000000	NONE	Shows which modules are currently working.
SV12.0.2-Num running modules = 0	NONE	Shows the number of the modules that are currently working.
SV12.0.3-Next starting module = Module 1	NONE	Shows which module is the next to start working.
SV12.0.4-Modules start stop = 0000000000	NONE	Modules start stop order.
SV12.0.5-Modules enable reg = 0000000000	NONE	Modules enable register.

5.11.2. Subgroup SV12.1: Module 1

SV12.1.1-Status

SCREEN	UNITS	DESCRIPTION
SV12.1.1-Status = Invalid	NONE	Shows the status of the module. See section "Module status messages" for more information on the statuses.

SV12.1.2-Electrical measures

SCREEN	UNITS	DESCRIPTION
SV12.1.2.1-Current R = 0.0 A	A	Shows the phase current.
SV12.1.2.2-Current S = 0.0 A	A	Shows the phase current.
SV12.1.2.3-Current T = 0.0 A	A	Shows the phase current.
SV12.1.2.4-DC Current = 0.0 A	A	Shows the DC current of the module.
SV12.1.2.5-DC Voltage P = 0 V	V	Shows the P DC voltage of the module.
SV12.1.2.6-DC Voltage N = 0 V	V	Shows the N DC voltage of the module.
SV12.1.2.7-DC Voltage = 0 V	V	Shows the DC voltage of the module.
SV12.1.2.8-P = 0.0 kW	kW	Shows the active power of the module.
SV12.1.2.9-Q = 0.0 kVAr	kVAr	Shows the reactive power of the module.
SV12.1.2.10-AC Vrs = 0 V	V	Shows the RS AC voltage of the module.
SV12.1.2.11-AC Vst = 0 V	V	Shows the ST AC voltage of the module.
SV12.1.2.12-AC Vtr = 0 V	V	Shows the TR AC voltage of the module.

SV12.1.3-Temp. Measures

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.1.3.1-Ambient Temp = 0 ºC	ºC	Shows the ambient temperature that is measuring the module.
SV12.1.3.2-Max temperature = 0.00 ºC	ºC	Shows the maximum temperature of the module IGBT.
SV12.1.3.3-Temperature R1 = 0.00 ºC	ºC	Shows the temperature of the IGBT in each of its phase's R, S and T of module 1.
SV12.1.3.4-Temperature R2 = 0.00 ºC	ºC	
SV12.1.3.5-Temperature R3 = 0.00 ºC	ºC	
SV12.1.3.6-Temperature S1 = 0.00 ºC	ºC	
SV12.1.3.7-Temperature S2 = 0.00 ºC	ºC	
SV12.1.3.8-Temperature S3 = 0.00 ºC	ºC	
SV12.1.3.9-Temperature T1 = 0.00 ºC	ºC	
SV12.1.3.10-Temperature T2 = 0.00 ºC	ºC	
SV12.1.3.11-Temperature T3 = 0.00 ºC	ºC	

SV12.1.4-Versions

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.1.4.1-HW version = 0.0.0-0	NONE	Shows the HW version.
SV12.1.4.2-SW version = 0.0.0-0	NONE	Shows the SW version.

SV12.1.5-Debug

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.1.5.1-Debug 0 = 0	NONE	Used by developer.
SV12.1.5.2-Debug 1 = 0	NONE	
SV12.1.5.3-Debug 2 = 0	NONE	
SV12.1.5.4-Debug 3 = 0	NONE	
SV12.1.5.5-Debug 4 = 0	NONE	
SV12.1.5.6-Debug 5 = 0	NONE	
SV12.1.5.7-Debug 6 = 0	NONE	
SV12.1.5.8-Debug 7 = 0	NONE	

EN

SV12.1.6-Inputs/Outputs

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.1.6.1-Digital Inputs = 000000	NONE	Each bit shows information about the following digital inputs from the FPGA of module 1. BIT / STATUS / DESCRIPTION 0 / di_softstart_3 / Not used. 1 / di_aux_2 / Not used. 2 / di_softcharge / Soft charged feedback. 3 / dis_charge / Feedback that the DC spring is charged. 4 / dis_open / Feedback that the circuit breaker DC is open. 5 / dis_close / Feedback that the circuit breaker DC is closed.
SV12.1.6.2-Digital Outputs = 000000	NONE	Each bit shows information about the following digital outputs from the FPGA of module 1: BIT / STATUS / DESCRIPTION 0 / do_softstart_3 / Not used. 1 / do_aux_2 / Not used. 2 / do_aux_1 / Activation soft charged. 3 / do_close / Command to close DC circuit breaker. 4 / do_open / Command to open DC circuit breaker. 5 / do_charge / Command to charge spring.

SV12.1.7-Humidity

SCREEN	UNITS	DESCRIPTION
SV12.1.7-Humidity = 50 %	NONE	Shows the humidity of the module control board measured internally.

5.11.3. Subgroup SV12.2: Module 2

SV12.2.1-Status

SCREEN	UNITS	DESCRIPTION
SV12.2.1-Status = Invalid	NONE	Shows the status of the module. See section "Module status messages" for more information on the statuses.

SV12.2.2-Electrical Measures

SCREEN	UNITS	DESCRIPTION
SV12.2.2.1-Current R = 0.0 A	A	Shows the phase current.
SV12.2.2.2-Current S = 0.0 A	A	Shows the phase current.
SV12.2.2.3-Current T = 0.0 A	A	Shows the phase current.
SV12.2.2.4-DC Current = 0.0 A	A	Shows the DC current of the module.
SV12.2.2.5-DC Voltage P = 0 V	V	Shows the positive DC voltage of the module.

SCREEN	UNITS	DESCRIPTION
SV12.2.2.6-DC Voltage N = 0 V	V	Shows the negative DC voltage of the module.
SV12.2.2.7-DC Voltage = 0 V	V	Shows the DC voltage of the module.
SV12.2.2.8-P = 0.0 kW	kW	Shows the active power of the module.
SV12.2.2.9-Q = 0.0 kVAr	kVAr	Shows the reactive power of the module.
SV12.2.2.10-AC Vrs = 0 V	V	Shows the RS AC voltage of the module.
SV12.2.2.11-AC Vst = 0 V	V	Shows the ST AC voltage of the module.
SV12.2.2.12-AC Vtr = 0.0 V	V	Shows the TR AC voltage of the module.

SV12.2.3-Temp. Measures

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.2.3.1-Ambient Temp = 0 ºC	ºC	Shows the ambient temperature that is measuring the module.
SV12.2.3.2-Max temperature = 0.00 ºC	ºC	Shows the maximum temperature of the module IGBT.
SV12.2.3.3-Temperature R1 = 0.00 ºC	ºC	Shows the temperature of the IGBT in each of its phases R, S and T of module 2.
SV12.2.3.4-Temperature R2 = 0.00 ºC	ºC	
SV12.2.3.5-Temperature R3 = 0.00 ºC	ºC	
SV12.2.3.6-Temperature S1 = 0.00 ºC	ºC	
SV12.2.3.7-Temperature S2 = 0.00 ºC	ºC	
SV12.2.3.8-Temperature S3 = 0.00 ºC	ºC	
SV12.2.3.9-Temperature T1 = 0.00 ºC	ºC	
SV12.2.3.10-Temperature T2 = 0.00 ºC	ºC	
SV12.2.3.11-Temperature T3 = 0.00 ºC	ºC	

SV12.2.4-Versions

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.2.4.1-HW version = 0.0.0-0	NONE	Shows the HW version.
SV12.2.4.2-SW version = 0.0.0-0	NONE	Shows the SW version.

EN

SV12.2.5-Debug

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.2.5.1-Debug 0 = 0	NONE	Used by developer.
SV12.2.5.2-Debug 1 = 0	NONE	
SV12.2.5.3-Debug 2 = 0	NONE	
SV12.2.5.4-Debug 3 = 0	NONE	
SV12.2.5.5-Debug 4 = 0	NONE	
SV12.2.5.6-Debug 5 = 0	NONE	
SV12.2.5.7-Debug 6 = 0	NONE	
SV12.2.5.8-Debug 7 = 0	NONE	

SV12.2.6-Inputs/Outputs

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.2.6.1-Digital Inputs = 000000	NONE	Each bit shows information about the following digital inputs from the FPGA of module 2.
SV12.2.6.2-Digital Outputs = 000000	NONE	Each bit shows information about the following digital outputs from the FPGA of module 2:

BIT	STATUS	DESCRIPTION
0	di_softstart_3	Not used.
1	di_aux_2	Not used.
2	di_softcharge	Soft charged feedback.
3	dis_charge	Feedback that the DC spring is charged.
4	dis_open	Feedback that the circuit breaker DC is open.
5	dis_close	Feedback that the circuit breaker DC is closed.

BIT	STATUS	DESCRIPTION
0	do_softstart_3	Not used.
1	do_aux_2	Not used.
2	do_aux_1	Activation soft charged.
3	do_close	Command to close DC circuit breaker.
4	do_open	Command to open DC circuit breaker.
5	do_charge	Command to charge spring.

SV12.2.7-Humidity

SCREEN	UNITS	DESCRIPTION
SV12.2.7-Humidity = 50 %	NONE	Shows the humidity of the module control board measured internally.

5.11.4. Subgroup SV12.3: Module 3

SV12.3.1-Status

SCREEN	UNITS	DESCRIPTION
SV12.3.1-Status = Invalid	NONE	Shows the status of the module. See section "Module status messages" for more information on the statuses.

SV12.3.2-Electrical Measures

SCREEN	UNITS	DESCRIPTION
SV12.3.2.1-Current R = 0.0 A	A	Shows the phase current.
SV12.3.2.2-Current S = 0.0 A	A	Shows the phase current.
SV12.3.2.3-Current T = 0.0 A	A	Shows the phase current.
SV12.3.2.4-DC Current = 0.0 A	A	Shows the DC current of the module.
SV12.3.2.5-DC Voltage P = 0 V	V	Shows the P DC voltage of the module.
SV12.3.2.6-DC Voltage N = 0 V	V	Shows the N DC voltage of the module.
SV12.3.2.7-DC Voltage = 0 V	V	Shows the DC voltage of the module.
SV12.3.2.8-P = 0.0 kW	kW	Shows the active power of the module.
SV12.3.2.9-Q = 0.0 kVAr	kVAr	Shows the reactive power of the module.
SV12.3.2.10-AC Vrs = 0 V	V	Shows the RS AC voltage of the module.
SV12.3.2.11-AC Vst = 0 V	V	Shows the ST AC voltage of the module.
SV12.3.2.12-AC Vtr = 0 V	V	Shows the TR AC voltage of the module.

SV12.3.3-Temp. Measures

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.3.3.1-Ambient Temp = 0 ºC	ºC	Shows the ambient temperature that is measuring the module.
SV12.3.3.2-Max temperature = 0.00 ºC	ºC	Shows the maximum temperature of the module IGBT.
SV12.3.3.3-Temperature R1 = 0.00 ºC	ºC	Shows the temperature of the IGBT in each of its phases R, S and T of module 3.
SV12.3.3.4-Temperature R2 = 0.00 ºC	ºC	
SV12.3.3.5-Temperature R3 = 0.00 ºC	ºC	
SV12.3.3.6-Temperature S1 = 0.00 ºC	ºC	
SV12.3.3.7-Temperature S2 = 0.00 ºC	ºC	
SV12.3.3.8-Temperature S3 = 0.00 ºC	ºC	
SV12.3.3.9-Temperature T1 = 0.00 ºC	ºC	
SV12.3.3.10-Temperature T2 = 0.00 ºC	ºC	
SV12.3.3.11-Temperature T3 = 0.00 ºC	ºC	

EN

SV12.3.4-Versions

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.3.4.1-HW version = 0.0.0-0	NONE	Shows the HW version.
SV12.3.4.2-SW version = 0.0.0-0	NONE	Shows the SW version.

SV12.3.5-Debug

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.3.5.1-Debug 0 = 0	NONE	Used by developer.
SV12.3.5.2-Debug 1 = 0	NONE	
SV12.3.5.3-Debug 2 = 0	NONE	
SV12.3.5.4-Debug 3 = 0	NONE	
SV12.3.5.5-Debug 4 = 0	NONE	
SV12.3.5.6-Debug 5 = 0	NONE	
SV12.3.5.7-Debug 6 = 0	NONE	
SV12.3.5.8-Debug 7 = 0	NONE	

SV12.3.6-Inputs/Outputs

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV12.3.6.1-Digital Inputs = 000000</td><td>NONE</td><td>Each bit shows information about the following digital inputs from the FPGA of module 3.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>di_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>di_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>di_softcharge</td><td>Soft charged feedback.</td></tr>
<tr><td>3</td><td>dis_charge</td><td>Feedback that the DC spring is charged.</td></tr>
<tr><td>4</td><td>dis_open</td><td>Feedback that the circuit breaker DC is open.</td></tr>
<tr><td>5</td><td>dis_close</td><td>Feedback that the circuit breaker DC is closed.</td></tr>
</table></td></tr>
<tr><td>SV12.3.6.2-Digital Outputs = 000000</td><td>NONE</td><td>Each bit shows information about the following digital outputs from the FPGA of module 3:
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>do_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>do_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>do_aux_1</td><td>Activation soft charged.</td></tr>
<tr><td>3</td><td>do_close</td><td>Command to close DC circuit breaker.</td></tr>
<tr><td>4</td><td>do_open</td><td>Command to open DC circuit breaker.</td></tr>
<tr><td>5</td><td>do_charge</td><td>Command to charge spring.</td></tr>
</table></td></tr>
</table>

SV12.3.7-Humidity

SCREEN	UNITS	DESCRIPTION
SV12.3.7-Humidity = 50 %	NONE	Shows the humidity of the module control board measured internally.

5.11.5. Subgroup SV12.4: Module 4

SV12.4.1-Status

SCREEN	UNITS	DESCRIPTION
SV12.4.1-Status = Invalid	NONE	Shows the status of the module. See section "Module status messages" for more information on the statuses.

SV12.4.2-Electrical Measures

SCREEN	UNITS	DESCRIPTION
SV12.4.2.1-Current R = 0.0 A	A	Shows the phase current.
SV12.4.2.2-Current S = 0.0 A	A	Shows the phase current.
SV12.4.2.3-Current T = 0.0 A	A	Shows the phase current.
SV12.4.2.4-DC Current = 0.0 A	A	Shows the DC current of the module.
SV12.4.2.5-DC Voltage P = 0 V	V	Shows the P DC voltage of the module.
SV12.4.2.6-DC Voltage N = 0 V	V	Shows the N DC voltage of the module.
SV12.4.2.7-DC Voltage = 0 V	V	Shows the DC voltage of the module.
SV12.4.2.8-P = 0.0 kW	kW	Shows the active power of the module.
SV12.4.2.9-Q = 0.0 kVAr	kVAr	Shows the reactive power of the module.
SV12.4.2.10-AC Vrs = 0 V	V	Shows the RS AC voltage of the module.
SV12.4.2.11-AC Vst = 0 V	V	Shows the ST AC voltage of the module.
SV12.4.2.12-AC Vtr = 0 V	V	Shows the TR AC voltage of the module.

EN

SV12.4.3-Temp. Measures

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.4.3.1-Ambient Temp = 0 ºC	ºC	Shows the ambient temperature that is measuring the module.
SV12.4.3.2-Max temperature = 0.00 ºC	ºC	Shows the maximum temperature of the module IGBT.
SV12.4.3.3-Temperature R1 = 0.00 ºC	ºC	Shows the temperature of the IGBT in each of its phases R, S and T of module 4.
SV12.4.3.4-Temperature R2 = 0.00 ºC	ºC	
SV12.4.3.5-Temperature R3 = 0.00 ºC	ºC	
SV12.4.3.6-Temperature S1 = 0.00 ºC	ºC	
SV12.4.3.7-Temperature S2 = 0.00 ºC	ºC	
SV12.4.3.8-Temperature S3 = 0.00 ºC	ºC	
SV12.4.3.9-Temperature T1 = 0.00 ºC	ºC	
SV12.4.3.10-Temperature T2 = 0.00 ºC	ºC	
SV12.4.3.11-Temperature T3 = 0.00 ºC	ºC	

SV12.4.4-Versions

SCREEN	UNITS	DESCRIPTION
SV12.4.4.1-HW version = 0.0.0-0	NONE	Shows the HW version.
SV12.4.4.2-SW version = 0.0.0-0	NONE	Shows the HW version.

SV12.4.5-Debug

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	UNITS	DESCRIPTION
SV12.4.5.1-Debug 0 = 0	NONE	Used by developer.
SV12.4.5.2-Debug 1 = 0	NONE	
SV12.4.5.3-Debug 2 = 0	NONE	
SV12.4.5.4-Debug 3 = 0	NONE	
SV12.4.5.5-Debug 4 = 0	NONE	
SV12.4.5.6-Debug 5 = 0	NONE	
SV12.4.5.7-Debug 6 = 0	NONE	
SV12.4.5.8-Debug 7 = 0	NONE	

SV12.4.6-Inputs/Outputs

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

<table>
<tr><th>SCREEN</th><th>UNITS</th><th>DESCRIPTION</th></tr>
<tr><td>SV12.4.6.1-Digital Inputs = 000000</td><td>NONE</td><td>Each bit shows information about the following digital inputs from the FPGA of module 4.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>di_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>di_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>di_softcharge</td><td>Soft charge feedback.</td></tr>
<tr><td>3</td><td>dis_charge</td><td>Feedback that the DC spring is charged.</td></tr>
<tr><td>4</td><td>dis_open</td><td>Feedback that the DC switch disconnector is open.</td></tr>
<tr><td>5</td><td>dis_close</td><td>Feedback that the DC switch disconnector is closed.</td></tr>
</table></td></tr>
<tr><td>SV12.4.6.2-Digital Outputs = 000000</td><td>NONE</td><td>Each bit shows information about the following digital outputs from the FPGA of module 4.
<table>
<tr><th>BIT</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>do_softstart_3</td><td>Not used.</td></tr>
<tr><td>1</td><td>do_aux_2</td><td>Not used.</td></tr>
<tr><td>2</td><td>do_aux_1</td><td>Soft charge activation.</td></tr>
<tr><td>3</td><td>do_close</td><td>Command to close DC switch disconnector.</td></tr>
<tr><td>4</td><td>do_open</td><td>Command to open DC switch disconnector.</td></tr>
<tr><td>5</td><td>do_charge</td><td>Command to charge spring.</td></tr>
</table></td></tr>
</table>

SV12.4. 7-Humidity

SCREEN	UNITS	DESCRIPTION
SV12.4.7-Humidity = 50 %	NONE	Shows the humidity of the module control board measured internally.

EN

6.PROGRAMMING PARAMETERS

6

Programming parameters allow setting and modifying the inverter operating conditions and general parameters.

NOTICE

Depending on their role, the user will have read and write or read only permissions.

6.1. Group G1: General

In this group, user can enter the password to change the user mode and adjust date, time and UTC offset.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G1.1-Init parameters = No	No Yes	Initializes all parameters when is set to "Yes". **OPTION** / **DESCRIPTION** No: Not initialize. Yes: Initializes programming group parameters with the following exceptions: • G11: Fans • G6: Communications • G1.6-UTC offset • G9: IMD • G10: DU • G14: Custom Modbus **Note:** This parameter is only displayed if [G1.3-User ≥ Power Electronics staff].	No
G1.2-Access code = 0	0 to 65535	Shows the current user mode access code.	Yes
G1.3-User = Customer	Customer Commissioning Qualified customer Power Electronics staff Developer	Shows the current user mode: **OPTION** / **DESCRIPTION** Customer: Basic customer access. Commissioning: Not available. Qualified customer (Admin): Advanced customer access. Power Electronics staff: Power Electronics staff access. Developer: Developer access.	Yes
G1.5-Date/time = 01/01/2016 0:00:00	01/01/2000 0:00:00 to 31/12/2099 23:59:59	Sets current date and time. Format DD/MM/YYYY H:M:S **Note**: see section "SV7.1.2: Faults history" for more information on the internal interpretation procedure of the date and time.	Yes
G1.6 -UTC offset = 0 min	-720 to 840 min	Sets the difference, in minutes, from UTC (Universal Time Coordinated) for a particular place, if needed.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G1.7-Inverter name = Poste1	32 characters	Displays the inverter's name. This parameter is set up with a string of 32 characters. **Note:** Only users with role higher than Admin can modify this parameter.	Yes
G1.8-Facility name = Planta1	32 characters	Displays the name of the facility to identify the location. This parameter is set up with a string of 32 characters. **Note:** Only users with role higher than Admin can modify this parameter.	Yes
G1.9-Serial number = 0	18 characters	Shows the serial number of the inverter. This parameter is set up with a string of 18 characters. **Notes:** • Only users with role higher than Admin can modify this parameter by entering the password correctly. • To modify this parameter, the password in [SG1.1.29-Serial Number pass] must be entered first.	Yes

EN

6.2. Group G2: Running conditions

This group covers all the definition concerning running, stop and reconnection conditions of the inverter.

6.2.1. Subgroup G2.1: Start

Parameters to start the inverter can be set in this subgroup.

Notes:

- This subgroup is displayed if [G1.3-User ≥ Qualified customer].
- Only users with a role higher than Admin (Qualified customer) can modify these parameters, except G2.1.1 and G2.1.8 which requires a role equal or higher to Admin.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.1.1-Vdc low = Vac nominal (drive select) * 1.4142*1.2	Vac nominal (drive select) * 1.4142 to 1500 V	Allows setting the minimum DC voltage to start the inverter. **Note:** This parameter is not shown in STATCOM.	Yes
G2.1.2-Vac low = 95.0%	80.0 to 100.0 %	Allows setting the minimum permissible AC voltage to start the inverter.	Yes
G2.1.3-Vac high = 105.0%	100.0 to 120.0 %	Allows setting the maximum permissible AC voltage to start the inverter.	Yes
G2.1.4-F low = 59.3 Hz	45.0 to 60.0 Hz	Allows setting the minimum permissible frequency to start the inverter.	Yes
G2.1.5-F high = 60.5 Hz	50.0 to 65.0 Hz	Allows setting the maximum permissible frequency to start the inverter.	Yes
G2.1.6-Start delay = 300 s	0 to 600 s	Allows setting the delay applied since the AC grid reconnection conditions of voltage and frequency are met.	Yes

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G2.1.7-Start mode = Sequential</td><td>Sequential
Centralised</td><td>Configures the start mode of the inverter.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Sequential</td><td>The modules start sequentially. The start-up ramp of the inverter will be staggered at the conditions established on [G2.4. Modules connection] and following the gradient limitation parameters adjusted in [G4.2.1-Start].</td></tr><tr><td>Centralised</td><td>The inverter starts setting all enabled modules to zero power and increases power following the reconnect gradient adjusted in [G4.2.1 Start]. All modules will be started at the same time in their "Pre ON" state.
In MPCS, all modules are forced to start, by symmetry. The inverter will be kept in “Pre ON” until all modules are started.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G2.1.8-Start delay Extra = 5 s</td><td>0 to 6000 s</td><td>Allows configuring a pure start delay to product start.
This delay shall be given after the network conditions delay is met and the start command is received.</td><td>Yes</td></tr>
<tr><td>G2.1.9-Vdc constant [V/ºC] = 4.00 V</td><td>Disabled = 0.00;
0.01 to 5.00 V</td><td>Allows to adjust the inverter's DC startup voltage, set as the Vpv (photovoltaic array voltage) based on the ambient temperature, using linear compensation relative to a reference temperature of 25 °C.
The startup voltage is calculated as:
$V_{startup} = V_{pv} - V_{dc}constant * (T_{amb} - 25\ ºC)$
Example:
If the photovoltaic array voltage (Vpv) is 1100 Vdc (at 25 °C) and the ambient temperature (Tamb) is 50 °C, and assuming a temperature coefficient of 2 V/°C, the startup photovoltaic voltage (Vstartup) will be 1050 Vdc.
Note: this parameter is shown if [G1.3-User] ≥ Power Electronics staff and [SG10.1.4-Inverter type] ≠ PCS/MPCS.</td><td>Yes</td></tr>
</table>

6.2.2. Subgroup G2.2: Stop

Notes:

- This subgroup is displayed if [G1.3-User ≥ Qualified customer].
- Only users with a role higher than Admin can modify these parameters, except [G2.2.2-Min P] which requires a role equal or higher to Admin.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.2.1-Min P enable = AC	No AC DC	Allows enabling or disabling the stop of the inverter due to minimum active power value defined in [G2.2.2-Min P].	Yes
G2.2.2-Min P = 1.2 %	0.1 to 100.0 %	Allows setting the minimum DC power in order to stop the inverter.	Yes
G2.2.3-P min delay = 420 s	1 to 600 s	Allows setting the delay to stop the inverter when [G2.2.2-Min P] reaches its defined value.	Yes
G2.2.4-Min Vdc enable = Yes	No Yes	Allows enabling or disabling the stop of the inverter due to minimum V_{dc} value defined in [G2.2.5-Min Vdc].	Yes
G2.2.5-Min Vdc = 0 V	(Vac nominal (drive select) * 1.4142) -20V to (Vac nominal (drive select) * 1.4142) -20V	Allow configuring the automatic inverter shutdown depending on the minimum DC voltage. If voltage drops below the value of [G2.2.5-Min Vdc], and the timeout set in [G2.2.6-Min Vdc delay], the inverter will stop.	Yes
G2.2.6-Min Vdc delay = 5 s	1 to 300 s		Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.2.7-Re start delay = 300.0 s	0.0 to 600.0 s	Defines the minimum time after which the inverter will start again if the conditions are fulfilled.	Yes
G2.2.8-V_P-min = Disabled	Disabled = 1501; √2*[SG10.1.6-Nominal Vac] to 1500 V	Allows adding a new condition (Vdc is less than Vdcmin) before stopping the inverter if it receives a Plim setpoint lower than Pmin. The inverter will stop if [G2.2.1-Min P enable] is enabled, the DC power is less than the threshold configured in [G2.2.2-Min P] and the Vdc is less than the threshold configured in this parameter G2.2.8 during the time configured in [G2.2.3-P min delay]. In order to account for the set delay in [G2.2.3-P min delay], the next conditions (G2.2.2 and G2.2.8) must be fulfilled at the same time. If AC or DC enabled, it evaluates the power at the mains or at the DU. **Note:** The minimum value of the parameter is set internally.	Yes

6.2.3. Subgroup G2.3: Reconnection

Notes:

- This subgroup is displayed if [G1.3-User] ≥ Qualified customer.
- Parameters [G2.3.1-Vac low] to [G2.3.5-Delay grid fault] can only be modified by users with a role higher than Admin (qualified customer).

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.3.1-Vac low = 95.0 %	80.0 to 100.0 %	Defines the minimum AC voltage to reconnect the inverter.	Yes
G2.3.2-Vac high = 105.0 %	100.0 to 120.0 %	Defines the maximum AC voltage to reconnect the inverter.	Yes
G2.3.3-F low = 59.3 Hz	46.0 to 61.0 Hz	Defines the minimum allowed frequency to reconnect the inverter.	Yes
G2.3.4-F high = 60.5 Hz	49.0 to 63.0 Hz	Defines the maximum allowed frequency to reconnect the inverter.	Yes
G2.3.5-Delay grid fault = 300 s	0 to 600 s	Defines the time after which the inverter will start again if the conditions are fulfilled.	Yes
G2.3.6-C.Breaker reset delay = 600 s	0 to 3000 s	It allows setting the time that must elapse for the fault conditions to disappear after the AC circuit breaker opens due to a grid failure and closes automatically. **Note**: it is displayed if any of the AC circuit breaker opening parameters is set to Enabled ([G5.1.1.13-Open AC Circ. Break], [G5.1.2.13-Open AC Circ. Break], [G5.1.3.12-Open AC Circ. Break] or [G5.1.4.12- Open AC Circ. Break]).	Yes

6.2.4. Subgroup G2.4: Modules connection

In this group, conditions for connecting modules can be adjusted.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.4.1-Activation level = 30 %	0 to 100 %	Allows setting the module activation power level.	Yes
G2.4.2-Deactivation level = 30 %	0 to 100 %	Allows setting the module deactivation power level.	Yes
G2.4.3-Activation time = 1 s	1 to 300 s	Allows setting the module activation time after detecting the activation power level set on [G2.4.1-Activation level].	Yes
G2.4.4-Deactivation time = 10 s	1 to 300 s	Allows setting the module deactivation time after detecting the deactivation power level set on [G2.4.2-Deactivation level].	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.4.5-Available Q backup = 0 %	0 to 100 %	This parameter indicates the value of reactive power that the inverter should be able to provide with a quick response. Depending on this threshold, the number of the modules that need to be started in order to satisfy the reactive demand is calculated.	No

6.2.5. Subgroup G2.5: Night detection

This group allows setting the parameters to enable and disable the night mode.

The night functions can only be executed once the night has been detected. Similarly, these functions forcefully cease to operate when the day is detected.

Night is detected when the voltage measured in panels is lower than the inverter disconnection voltage minus a configurable threshold [G2.5.1-Night threshold].

Day is detected when the voltage measured in panels is greater than the starting voltage configured in the inverter [G2.1.1-Vdc low] minus a configurable threshold [G2.5.2-Day threshold].

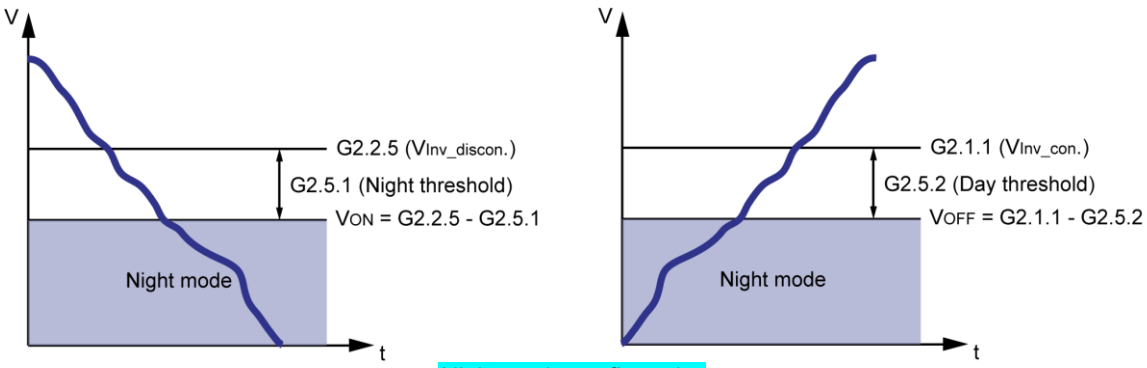
Night detection is indicated on the display by the “NIGHT” status.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.5.1-Night threshold = 50 V	5 to 400 V	Allows setting the Night/Heating mode input voltage as a hysteresis regarding the DC disconnection voltage set on [G2.2.5-Min Vdc].	Yes
G2.5.2-Day threshold = 50 V	5 to 400 V	Allows setting the Night/Heating mode output voltage as a hysteresis regarding the DC connection voltage set on [G2.1.1-Vdc low].	Yes
G2.5.3-Day delay = 900 s	0 to 3600 s	To switch to NIGHT MODE, the inverter stops the power and opens DU. The PV voltage rises to the open circuit level. The PV voltage in open circuit could be higher than [G2.5.2-Day threshold] and therefore the logic could lead us back to the RUN state). This parameter allows the inverter to wait for this delay time before returning to the RUN state.	Yes

When the “NIGHT” mode is activated, it is possible to enable the heating and the night mode. If both modes are enabled, the night mode will always be prioritized.

The following figures illustrates the night mode configuration:



Night mode configuration

6.2.6. Subgroup G2.6: Selectors

G2.6.1-Selectors Maneuver

Note: Only users with role equal or higher than Admin (qualified customer) can modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.6.1.1-General Selector = Off	Off On	Allows to configure the selector in order to start working with the inverter. **Note**: This parameter is displayed if [SG1.1.11-Selector config ≥ Remote].	Yes
G2.6.1.2-DC Virtual Sel conf = Selector	Selector Comms	Configures the type of selector (communications or selector).	Yes
G2.6.1.2a-DC Virtual Selector = Off	Off On	Enables / disables the virtual selector for DC. **Note:** This parameter is only displayed if [G2.6.1.2-DC Virtual Sel conf = Comms].	Yes
G2.6.1.3-AC Virtual Sel conf = Selector	Selector Comms	Configures the type of selector (communications or selector).	Yes
G2.6.1.3a-AC Virtual Selector = Off	Off On	Enables / disables the virtual selector for AC. **Note:** This parameter is only displayed if [G2.6.1.3-AC Virtual Sel conf = Comms].	Yes

G2.6.2-Ready Mode

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.6.2-Ready Mode = AC/DC Ready	AC/DC Ready DC Ready AC Ready	Configures the logical state to go in the inverter when given the "ready" command. **OPTION** / **DESCRIPTION** AC/DC Ready – The inverter is connected to AC and DC. DC Ready – The inverter is connected to DC. AC Ready – The inverter is connected to AC.	Yes

6.2.7. Subgroup G2.7: MV Cell

This subgroup allows to manage remotely the MV switchgear of HEM/PCSM or HEMK/PCSK with MV Skid.

G2.7.1-MV Cell Control

Notes:

- This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with a role equal to or higher than Admin (qualified customer) can modify these parameters.
- Applies to [SG10.7.3-MV Skid = Yes, Motorized] and [SG10.1.3-Product type = HEMK] or [SG10.1.3-Product type = HEM].

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G2.7.1.1-MV cell force control = Virtual Selector AC</td><td>Communications
Virtual Selector AC
Independent</td><td>Allows the configuration of the control mode of the MV switchgear when the inverter is a HEM/PCSM or HEMK/PCSK with motorised MV Skid.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Communications</td><td>This option allows the manual control of the MV switchgear by communications. The MV switchgear closes via parameter [G2.7.1.2-MV cell switch actio].</td></tr><tr><td>Virtual Selector AC</td><td>This option manages the MV switchgear automatically by the inverter itself, through the AC virtual selector configured in [G2.6.1.3a-AC Virtual Selector].</td></tr><tr><td>Independent</td><td>This option allows the MT switchgear to be closed independently from the inverter's main logic. Closing is performed without premagnetization by applying the sequential reconnection delay, provided that the medium voltage is above 55% and the line cells are closed.
Note: it is displayed only if [SG1.8.7-En. Shutdown premag] = Yes, [SG10.1.3-Product type] = HEMK, [SG10.7.3-MV Skid] = Yes, Motorized and [SG10.7.4-Volt Meas MV Skid] = Yes.</td></tr></table></td><td>No</td></tr>
<tr><td>G2.7.1.2-MV cell switch actio = Idle</td><td>Idle
Open
Closed</td><td>Enables the closing and/or opening manoeuvre of the MV switchgear, provided that the disconnector spring is charged. This parameter can only be used when [G2.7.1.1-MV cell force contro] is configured as "Communications".
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Idle</td><td>No action is commanded to the MT switchgear, it is idle.</td></tr><tr><td>Open</td><td>Commands the action of opening the MV switchgear disconnector.</td></tr><tr><td>Closed</td><td>Commands the action of closing the MV switchgear disconnector.</td></tr></table></td><td>No</td></tr>
</table>

G2.7.2-Sequential Reconnec

This functionality has the objective of reducing the inrush current produced by the simultaneous connection of all inverters that compound a solar plant. The sequential connection of the plant inverters is managed by the firmware (or logical support) and can be both a remote connection or an automatic connection.

Before starting any maneuver, it is necessary to check that the disconnectors of the line and protection switchgear of the inverter are closed, and the motorized disconnector is open. To do this, the status of these disconnectors must be checked on the digital inputs RJ13, RJ14, RJ15 and RJ16 (subgroup SV1.5: Digital I/O status).

Applies to HEMK/PCSK and MV/TWIN Skid inverter with motorized switchgear or HEM/PCSM.

Notes:

- Only users with a role higher than Admin (qualified customer) can modify this subgroup except G2.7.2.1 which requires a role equal or higher than Admin.
- This subgroup is displayed if [SV1.3.1.10-MV Skid = Yes. Motorized] and [SG10.1.3-Product type = HEMK] or [SG10.1.3-Product type = HEM].

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.7.2.1-Enable = Disabled	Disabled Enabled	Allows to enable or disable the algorithm. **Note**: This parameter can be read for all roles.	Yes
G2.7.2.2-Condition = Remote	Remote Automatic	Configures if the order to close the MV switchgear is triggered by an external device such as PPC or SCADA (Remote) or by the inverter (Automatic). **OPTION** / **DESCRIPTION** Remote: The SCADA or PPC commands the closing action of the MV switchgear. The transformers will be energized consecutively depending on the delay configured [G2.7.2.4-Delay time]. Automatic: The system connects to grid when the MV grid voltage is stablished. **Note**: This parameter is displayed if [G2.7.2.1-Enable = Disabled].	Yes
G2.7.2.3-Delay Type = Automatic	Manual Automatic	Configures the type of delay that operates during the sequential reconnection algorithm. **OPTION** / **DESCRIPTION** Manual: The delay is configured in [G2.7.2.4-Delay Time]. Each inverter should have a different delay. Automatic: The delay is calculated from the first octet of the inverter IP multiplied by the delay configured in [G2.7.2.4-Delay Time]. An example is shown below: 1. The IP of the inverter is 192.168.204.2. 2. The delay set in the [G2.7.2.4-Delay Time] parameter is 15 s. 3. The delay applied in this inverter will be 2*15 s = 30 s. The IP of the inverters of the plant should be a fix IP and must start in 1. **Notes:** - This maneuver is available for inverters HEM and HEMK with MV Skid motorized. - This parameter is displayed if [G2.7.2.1-Enable = Disabled].	Yes
G2.7.2.4-Delay Time = 10 s	1 to 65535 s	Configures the delay to close the MV switchgear after the appropriate conditions are met. **Note:** This parameter is displayed if [G2.7.2.1-Enable = Disabled].	Yes

6.2.8. Subgroup G2.8: Busbars Configuration

This subgroup allows to enable or disable the configuration of busbars of the inverter. The DSP implements a logic that allows busbars to be started, stopped, and reconnected according to the parameterization. When a busbar or module is enabled or disabled, it is ramped. If the ramps are disabled or a higher ramp is configured, there is a gradient which limits the operation: power down by 1000 %/s, power up by 2000 %/s.

In any of these operations, compliance with the hardware limit between busbars is mandatory. When a limit is set, the running busbar ramps down its power to allow the second busbar to start and together they return to the references. However, an abrupt shutdown of one module due to faults can cause the hardware limit to be breached in the other busbar module. Reference changes to correct this situation will follow the active gradients.

Applies to [SG10.1.4-Inverter type = PCS/MPCS].

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with a role equal or higher than Admin can modify these parameters.

G2.8.0-Individual Cfg

This subgroup allows to configure which module FPGAs and which busbar are active using the parameters described in this subgroup. When it is enabled or disabled the module FPGA or busbar, parameters [G2.8.1-Enable Busbars] and [G2.8.2-Enable Modules] are updated to consider this modification. In other words, subgroup G2.8.0 and parameters G2.8.1 and G2.8.2 are linked.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.8.0.1-Enable Busbar1 = Enabled	Disabled Enabled	Enables or disables the different busbars of the inverter. **Note:** These parameters are displayed if [SG10.1.14-Number of busbar ≥ n], being 'n', the number of the busbar.	Yes
G2.8.0.2-Enable Busbar2 = Enabled	Disabled Enabled		Yes
G2.8.0.3-Enable Busbar3 = Enabled	Disabled Enabled		Yes
G2.8.0.4-Enable Busbar4 = Enabled	Disabled Enabled		Yes
G2.8.0.5-Enable Module1 = Enabled	Disabled Enabled	Enables or disables the power modules of the inverter. **Note:** These parameters are displayed if [SG10.1.12-Number of modules = 4] and [SG10.1.14-Number of busbar = 2].	Yes
G2.8.0.6-Enable Module2 = Enabled	Disabled Enabled		Yes
G2.8.0.7-Enable Module3 = Enabled	Disabled Enabled		Yes
G2.8.0.8-Enable Module4 = Enabled	Disabled Enabled		Yes

G2.8.1- Enable Busbars and G2.8.2-Enable Modules

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G2.8.1-Enable Busbars = XXXX</td><td>0000 to XXXX</td><td>Enables the different busbars of the inverter. Each bit corresponds to a busbar as shown in the following table, being 0=0, X=1. The first busbar is the bit on the right side; the last busbar is the bit on the left side.
<table><tr><th>BIT</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>Busbar 1</td></tr><tr><td>1</td><td>Busbar 2</td></tr><tr><td>2</td><td>Busbar 3</td></tr><tr><td>3</td><td>Busbar 4</td></tr></table>Busbar activation is possible via SCADA. When a SCADA writes to a Modbus register, the binary-decimal columns of the table show the conversion. If the SCADA wants to enable a busbar, it must write in register G2.8.1 the corresponding value in binary or in decimal.
In the last case, the SCADA will write a decimal number from 0 to 15 corresponding to the busbar activation.
An example is shown in the following table to know which busbars are active:
<table><tr><th>Binary</th><th>Busbar 4</th><th>Busbar 3</th><th>Busbar 2</th><th>Busbar 1</th><th>Decimal number</th></tr><tr><td>0000</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0001</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0010</td><td>0</td><td>0</td><td>1</td><td>0</td><td>2</td></tr><tr><td>0011</td><td>0</td><td>0</td><td>1</td><td>1</td><td>3</td></tr><tr><td>0100</td><td>0</td><td>1</td><td>0</td><td>0</td><td>4</td></tr><tr><td>0101</td><td>0</td><td>1</td><td>0</td><td>1</td><td>5</td></tr></table>The leftmost bit is the busbar 4. The decimal number 15 corresponds to the activation of all busbars, being its binary 1111. The conversion formula is as follows
DECIMAL = Bit EMB1 + Bit EMB2 * 2 + Bit EMB3 * 4 + Bit EMB4 * 8
Note: Busbars depend on internal configuration. It is only displayed if [SV1.3.1.1-Inverter type = PCS/MPCS].</td><td>Yes</td></tr>
<tr><td>G2.8.2-Enable Modules = 00XXXXXXXX</td><td>0000000000 to 00XXXXXXXX</td><td>Parameter which each bit corresponds to a module FPGA. Enables to work with the different power stages. First module is the bit located in the right place. Last module the bit located in the left place.
<table><tr><th>BIT</th><th>STATUS</th><th>BIT</th><th>STATUS</th></tr><tr><td>0</td><td>Module 1</td><td>6</td><td>Not available</td></tr><tr><td>1</td><td>Module 2</td><td>7</td><td>Not available</td></tr><tr><td>2</td><td>Module 3</td><td>8</td><td>Not available</td></tr><tr><td>3</td><td>Module 4</td><td>9</td><td>Not available</td></tr><tr><td>4</td><td>Not available</td><td>--</td><td>-</td></tr></table>Notes:
• This parameter is displayed if [SG10.1.12-Number of modules = 4] and [SG10.1.14-Number of busbar = 2].
• These options are displayed if [SG10.1.12-Number of modules ≥ ‘n’] being ‘n’ the number of the module.</td><td>Yes</td></tr>
<tr><td>G2.8.3-DCINPUT Thres = 650 V</td><td>Disabled = 0;
1 to 1500 V</td><td>Allows configuring the voltage threshold of busbar 1 at which the DC-DC power supply is switched. When the voltage measured on busbar 1 drops below this value, the system switches the power supply to the electronics to busbar 3, ensuring that the inverter starts up and operates in PCS mode.
Note: this parameter is shown if [SG10.1.4-Inverter type] = PCS/MPCS</td><td>Yes</td></tr>
<tr><td>G2.8.4-DCINPUT Status = Busbar 1</td><td>Disabled
Initializating
Busbar 1
Busbar 3</td><td>Allows to visualize the status of the DCDC power supply control, showing which busbar is powering the inverter electronics based on the measured voltage and the configured threshold.
Note:
• This parameter is read-only (RO).
• This parameter is shown if [SG10.1.4-Inverter type] = PCS/MPCS or [G2.8.3-DCINPUT threshold] ≠ Disabled.</td><td>Yes</td></tr>
</table>

EN

6.2.9. Subgroup G2.9: Busbar Start

This subgroup controls the starting conditions of the internally configured busbars. In case they are not fulfilled, the warning, the warning “W5: No start conditions”, will trip.

The delay configured in parameter [G2.9.1-Delay] will start counting from when the start conditions are met.

Applies to [SG10.1.4-Inverter type = PCS/MPCS].

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with a role higher than Admin (qualified customer) can modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.9.1-Delay = 5 s	0 to 600 s	Allows to configure the time to wait after complying conditions to start busbar in order to command the order.	Yes
G2.9.2-Vdc min = 1050 V	[G3.7.1.1-Min Vdc] to 1425 V	Allows to configure the minimum DC voltage that allows to start the busbar.	Yes
G2.9.3-Vdc max = 1500 V	581 to 1530 V	Allows to configure the maximum DC voltage that allows to start the busbar.	Yes
G2.9.4-Unbalanced = 75 V	0 to 85 V	Allows to configure the maximum DC voltage difference that allows to start the busbar.	Yes

6.2.10. Subgroup G2.10: Busbar Stop

This subgroup controls the stop conditions when the inverter is in the ON state and the busbar is running.

When any stop condition is met, the busbar will stop when the time set in [G2.10.1-Delay] expires.

In the event that the inverter stops, the busbar shutdown conditions will be reset.

If the busbar stops, the stop conditions will be reset when the busbar stops. In this case, the reclosing conditions configured in subgroup “G2.11: Busbar Reconnection” will be activated.

Applies to [SG10.1.4-Inverter type = PCS/MPCS].

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with a role higher than Admin (qualified customer) can modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.10.1-Delay = 2 s	0 to 600 s	Allows to configure the time to wait after complying conditions to stop busbar in order to command the order.	Yes
G2.10.2-Vdc min = 1000 V	363 to 1500 V	Allows to configure the minimum DC voltage that allows to stop the busbar.	Yes
G2.10.3-Vdc max = 1500 V	[G2.10.2-Vdc min] to 1530 V	Allows to configure the maximum DC voltage that allows to stop the busbar.	Yes
G2.10.4-Unbalanced = 100 V	0 to 120 V	Allows to configure the maximum DC voltage difference that allows to stop the busbar.	Yes

6.2.11. Subgroup G2.11: Busbar Reconnection

This subgroup controls the busbar reconnection conditions after the stop configured in subgroup [G2.10: Busbar Stop].

Only the reconnection conditions are monitored when the inverter is in the ON state and the busbar has stopped due to stop conditions. If the busbar has stopped because it has been disabled, the reconnection conditions will not be activated.

If the inverter is not in the ON state, the reconnection conditions will be reset.

When the reconnection conditions are activated and met, the busbar will be reset after the start delay set in [G2.11.1-Delay] has elapsed.

Applies to [SG10.1.4-Inverter type = PCS/MPCS].

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with a role higher than Admin can modify these parameters.

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.11.1-Delay = 10 s	10 to 600 s	Allows to configure the waiting time after the conditions are met to reconnect the busbar in order to command the order.	Yes
G2.11.2-Vdc min = 1425 V	382 to 1425 V	Allows to configure the minimum DC voltage that allows to reconnect the busbar.	Yes
G2.11.3-Vdc max = 1500 V	582 to 1530 V	Allows to configure the maximum DC voltage that allows to reconnect the busbar.	Yes
G2.11.4-Unbalanced = 50 V	0 to 75 V	Allows to configure the maximum DC voltage difference that allows to reconnect the busbar.	Yes

6.2.12. Subgroup G2.12: Softcharge mode

Note: This parameter is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.12-Softcharge mode = Internal DC central	External DC Internal DC central	Defines soft charge mode that inverter uses. There are two modes: External DC (value = 2) only for debug purposes and Internal DC central (value =4) used by default.	Yes

6.2.13. Subgroup G2.13: Softcharge offset

Note: This parameter is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G2.13-Softcharge offset = 50 V	Off = 0; 1 to 150 V	Indicates the difference between internal and external voltage before closing DC contactor.	Yes

6.3. Group G3: Limits

This group is intended for the configuration of the limits of the inverter, both power and deratings implemented in the inverter, currents and batteries.

6.3.1. Subgroup G3.1: P/Q

Notes: Only users with a role equal or higher than Admin (qualified customer) can modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.1.1-P limit Discharge = 100 %	0.00 to 100.00 %	Sets the P limit for discharge (current flow to photovoltaic or battery to grid).	Yes
G3.1.1b-P Charge limit = -100.00 %	-100.00 to 0.00 %	Sets the P Limit for Charge (current flow is from grid to DC Load: busbar or battery). **Note:** It is displayed if [SV1.3.1.1-Inverter type ≥ Basic BUSPLUS].	Yes
G3.1.2-Q limit = 100.00 %	0.00 to 150.00 %	Sets the reactive power limit configuration.	Yes
G3.1.3-P/Q priority = Q	P Q	Defines which reference, P or Q, takes priority if the maximum apparent power (S) is reached.	Yes
G3.1.4-S limit = 100.00 %	0.00 to 200.00 %	Configures the apparent power limit.	Yes

G3.1.5-Self limitation

This subgroup is used to configure the self-limiting voltage protection.

Notes: this subgroup is only displayed if [G1.3-User] ≥ Qualified Customer.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.1.5.1-Overvoltage margin = 5.0 %	Disabled = 0.0; 0.1 to 5.0 %	It allows configuring the margin used for the activation of the capacitive reactive power limitation in the event of a voltage peak is detected in the network. When the measured voltage exceeds the value of the enabled overvoltage protection minus the margin, the capacitive power will be limited. The reference protection is the OVRT algorithm input threshold and, if it is not enabled, it will be the least restrictive overvoltage network protection. If neither of these two protections is enabled, the functionality is disabled. The applied limits can be found in group [SV1.8.3: Self limitation]. **Note**: it is shown if the OVRT algorithm is not disabled [G4.4.1.1-Enable] ≠ Disabled or [G5.1.2.1-Enable] > 0.	Yes
G3.1.5.1a-Overvoltage hyst. = Disabled %	Disabled = 0.0; 0.1 to 5.0 %	Allows to configure the threshold for deactivating the reactive power limitation due to grid overvoltage. **Note**: it is shown if the OVRT algorithm is not disabled [G4.4.1.1-Enable] ≠ Disabled or [G5.1.2.1-Enable] > 0.	Yes
G3.1.5.1b-OverV output delay = Disabled ms	Disabled = 0; 1 to 65535 ms	Allows to configure the delay for deactivating reactive power limitation due to grid overvoltage. **Note**: it is shown if [G3.1.5.1a-Overvoltage hyst] ≠ Disabled.	Yes
G3.1.5.2-Undervoltage margin = 1.0 %	Disabled = 0.0; 0.1 to 5.0 %	It allows configuring the margin used for the activation of the inductive reactive power limitation in the event of a voltage drop detected in the network. When the measured voltage is lower than the value of the enabled undervoltage protection plus the margin, the inductive power will be limited. The reference protection is the LVRT algorithm input threshold and, if it is not enabled, it will be the least restrictive undervoltage network protection. If neither of these two protections is enabled, the functionality is disabled. The applied limits can be found in group [SV1.8.3: Self limitation]. **Note**: it is shown if the LVRT algorithm is not disabled [G4.3.1.1-Enable] ≠ Disabled or [G5.1.1.1-Enable] > 0.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.1.5.2a-Undervoltage hyst. = Disabled %	Disabled = 0.0; 0.1 to 5.0 %	Allows to configure the threshold for deactivating the reactive power limitation due to grid undervoltage. **Note**: it is shown if the LVRT algorithm is not disabled [G4.3.1.1-Enable] ≠ Disabled or [G5.1.1.1-Enable] > 0.	Yes
G3.1.5.2b-UnderV output delay = Disabled ms	Disabled = 0; 1 to 65535 ms	Allows to configure the delay for deactivating reactive power limitation due to grid undervoltage. **Note**: it is shown if [G3.1.5.2a-Undervoltage hyst] ≠ Disabled.	Yes

6.3.2. Subgroup G3.2: Derating V

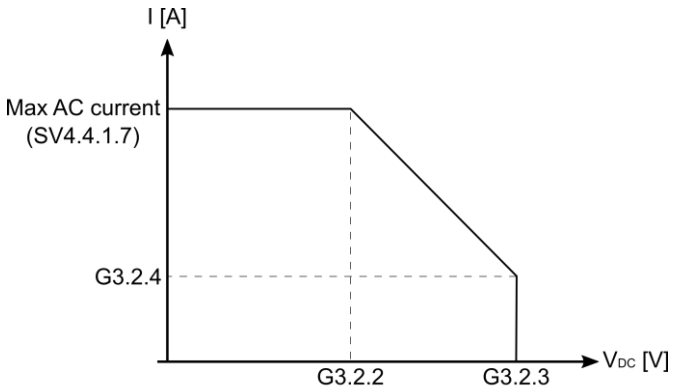
Parameters to configure the derating by voltage function.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.2.1-Enable = Yes	No Yes	If this parameter is set to Yes, when the voltage reaches the value set in [G3.2.2-Start Vdc], derating will be start. At G3.2.3, current will be equal to the value set in [G3.2.4-Imin (Vdc max)]. If voltage keeps rising and overcomes 1500 V, fault F2 (HW Vbus) will be triggered, and the inverter will stop. **OPTION** / **DESCRIPTION** No – The derating V is disabled. Yes – The derating V is enabled.	Yes
G3.2.2-Start Vdc = 1500 V	0 to [G3.2.3-Max Vdc] V	Allows setting the DC voltage derating start expressed in volts. **Note:** This parameter is displayed if [G3.2.1-Enable = Yes].	Yes
G3.2.3-Max Vdc = 1600 V	[G3.2.2-Start Vdc] to 1600 V	Allows setting the DC maximum voltage expressed in volts. **Note:** This parameter is displayed if [G3.2.1-Enable = Yes].	Yes
G3.2.4-Imin (Vdc max) = 0 A	0 to 920 A	Allows setting the current at the maximum DC voltage defined in [G3.2.3-Max Vdc]. **Note:** This parameter is displayed if [G3.2.1-Enable = Yes].	Yes

The following figure illustrates the voltage derating function:



Voltage derating function

6.3.3. Subgroup G3.3: Derating temperature

In this subgroup, the parameters used to configure the module thermal derating function are included.

The intake temperature derating limits the inverter capacity as a function of the measured ambient temperature, with the aim of protecting internal components and extending their service life.

From software version 5.0.0 onwards, the limit applied to the module is calculated considering two constraints:

- **Module current derating (Der Ipwm):** result of the intake temperature derating curve configured in this subgroup.
- **Apparent power-based derating (Der S in Ipwm):** current limit equivalent to the apparent power limit defined by parameter [SG10.3.5-Max S Limit], taking into account the AC control voltage.

The final applied limit is the most restrictive of both:

$$I_{limit} = \min(\text{Der Ipwm}, \text{Der S in Ipwm})$$

In this way, the inverter limits the effective power delivered to the system according to the thermal constraints.

- Under AC overvoltage conditions (Vac > 1.0 p.u.), the apparent power-based limitation may be more restrictive than the current-based limitation, preventing power overdelivery.
- Under nominal voltage or undervoltage conditions (Vac ≤ 1.0 p.u.), the limit is determined by the module current derating.

The apparent power-based derating is only applied when the following conditions are simultaneously met:

- Vac > 1.0 p.u.
- [SG10.3.5-Max S Limit] = 100%

In all other conditions, or when [SG10.3.5-Max S Limit] ≥ 110%, the thermal derating behavior is equivalent to that of previous versions.

Notes:

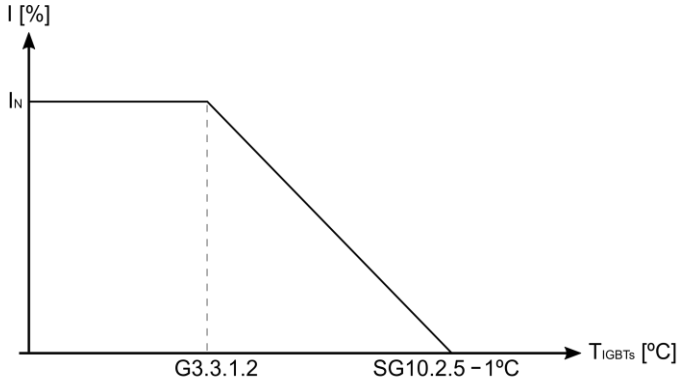
- This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].
- The diagrams shown in this subgroup represent the behavior of the intake temperature derating. Under certain AC overvoltage conditions, the limit may be additionally constrained as described above.

G3.3.1-Derating disc Tª IGBT

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G3.3.1.1-Enable = Yes</td><td>No
Yes</td><td>If temperature derating is enabled, when the IGBT temperature reaches the value set in [G3.3.1.2-Initial Tª IGBTs], derating will start.
If the temperature keeps rising and reaches the protection threshold value [SG10.2.2-Igbts disc temp thre], fault F40 (AC mod overtemperature) will be triggered.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>No</td><td>The temperature derating is disabled.</td></tr>
<tr><td>Yes</td><td>The temperature derating is enabled.</td></tr>
</table></td><td>No</td></tr>
<tr><td>G3.3.1.2-Initial Tª IGBTs = 98 ºC</td><td>0 to 150 ºC</td><td>Defines the starting point for the derating by temperature of the IGBTs when the inverter is in discharge.
Note: This parameter is displayed if [G3.3.1.1-Enable = Yes].</td><td>Yes</td></tr>
</table>

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.3.1.3- TªIGBTsDeviationMax = Temp max igbts discharge deviation activation - Range def ºC	5 to 25 ºC; Disabled = 26 ºC	Allows setting the maximum admissible difference between the highest IGBT temperature and the average of the highest IGBT temperatures per module, so if that difference is not exceeded the derating starts according to the highest temperature of all operating modules, and if exceeded, begins according to the highest temperature that is not out of range. **Note:** This parameter is displayed if [G3.3.1.1-Enable = Yes].	Yes

The following figure illustrates the derating function depending on IGBT temperature:

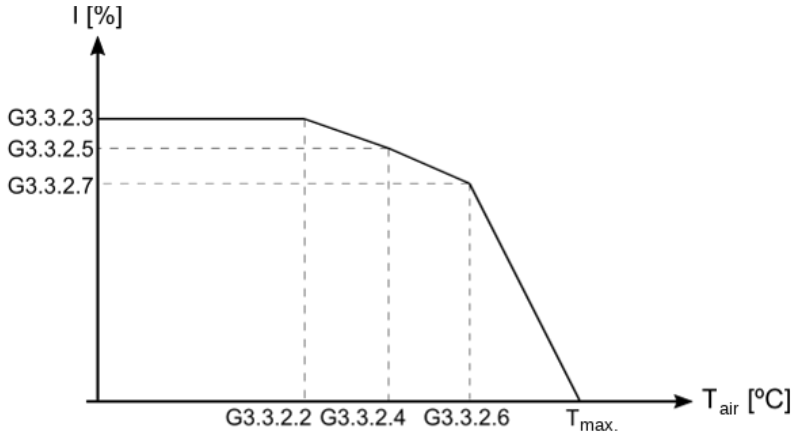


Derating function depending on IGBT temperature

G3.3.2- Derating Tª adm disc

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.3.2.1-Enable = Yes	No Yes	If the derating by intake temperature is enabled, the configuration of 3 current and 3 temperature points is enabled, both for charging and discharging. In this way, the inverter adjusts the output current according to its intake temperature. OPTION / DESCRIPTION No: The derating by intake temperature is disabled. Yes: The derating by intake temperature is enabled. **Note**: this parameter is automatically set to "Yes" if [SG10.7.9-Cooling type] = KNAN.	Yes
G3.3.2.2-Adm. T1 discharge = 40 ºC	-35 to 40 ºC	Allows setting the current derating intake temperature in discharge for the first temperature point. **Note:** This parameter is displayed if [G3.3.2.1-Enable = Yes].	Yes
G3.3.2.3-Adm. I1 discharge = 100.00 %	[G3.3.2.5-Adm. I2 discharge] to 100.00 %	Allows setting the current derating intake temperature in discharge for the first current point. **Note:** This parameter is displayed if [G3.3.2.1-Enable = Yes].	Yes
G3.3.2.4-Adm. T2 discharge = 50 ºC	[G3.3.2.2-Adm. T1 discharge] to 50 ºC	Allows setting the current derating intake temperature in discharge for the second temperature point. **Note:** This parameter is displayed if [G3.3.2.1-Enable = Yes].	Yes
G3.3.2.5-Adm. I2 discharge = 93.00 %	[G3.3.2.7-Adm. I3 discharge] to 100.00 %	Allows setting the current derating intake temperature in discharge for the second current point. **Note:** This parameter is displayed if [G3.3.2.1-Enable] = Yes and is automatically set to 86 % if [SG10.7.9-Cooling type] = KNAN.	Yes
G3.3.2.6-Adm. T3 discharge = 53 ºC	[G3.3.2.4-Adm. T2 discharge] to 55 ºC	Allows setting the current derating intake temperature in discharge for the third temperature point. **Note:** This parameter is displayed if [G3.3.2.1-Enable = Yes].	Yes
G3.3.2.7-Adm. I3 discharge = 86.00 %	0.00 to 86.36 %	Allows setting the current derating intake temperature in discharge for the third current point. **Note:** This parameter is displayed if [G3.3.2.1-Enable = Yes].	Yes

The following figure illustrates the derating function depending on air intake temperature, which depends on the inverter configuration:



Derating function depending on air intake temperature

G3.3.3- Derating char Tª IGBT

Note: These parameters are only displayed if [SG10.1.4-Inverter type ≥ Basic BUSPLUS].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.3.3.1-Enable = Yes	No Yes	Maximum module AC current derating function by IGBT temperature is allowed. OPTION: No – Current reduction with temperature is disabled. Yes – Current reduction with temperature is enabled.	Yes
G3.3.3.2-Initial Tª IGBTs = 94 ºC	0 to 150 ºC	Defines the starting point for the derating by temperature of the IGBTs when the inverter is in charge. **Note:** This parameter is only displayed if [G3.3.3.1-Enable = Yes].	Yes
G3.3.3.3-TªIGBTsDeviationMax = Temp max igbts charge deviation activation - Range def ºC	5 to 25 ºC; Disabled = 26 ºC	Allows to establish the maximum admissible difference between the highest temperature of the IGBTs and the average of the highest temperatures of the IGBTs per module in load mode. **Note:** This parameter is only displayed if [G3.3.3.1-Enable = Yes].	Yes

G3.3.4- Derating T adm charge

Note: These parameters are only displayed if [SV1.3.1.1-Inverter type ≥ Basic BUSPLUS].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.3.4.1-Enable = Yes	No Yes	Allows the configuration of the maximum module current derating function by temperature. OPTION: No – The intake temperature-based derating is disabled. Yes – The intake temperature-based derating is enabled. **Note**: this parameter is automatically set to "Yes" if [SG10.7.9-Cooling type] = KNAN.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.3.4.2-Admission T1 charge = 40 ºC	-35 to [G3.3.2.2-Adm. T1 discharge] ºC	Sets the intake temperature for the first derating point. **Note:** This parameter is only displayed if [G3.3.4.1-Enable = Yes].	Yes
G3.3.4.3-Admission I1 charge = 100.00 %	0.00 to [G3.3.2.3-Adm. I1 discharge] %	Sets the percentage of the rated current for the first derating point. **Note:** This parameter is only displayed if [G3.3.4.1-Enable = Yes].	Yes
G3.3.4.4-Admission T2 charge = 50 ºC	[G3.3.4.2-Admission T1 charge] to [G3.3.2.4-Adm. T2 discharge] ºC	Sets the intake temperature for the second derating point. **Note:** This parameter is only displayed if [G3.3.4.1-Enable = Yes].	Yes
G3.3.4.5-Admission I2 charge = 93.00 %	0.00 to -Min between admission I1 charge and admission I2 discharge %	Sets the percentage of the rated current for the second derating point. **Note:** This parameter is only displayed if [G3.3.4.1-Enable = Yes] and is automatically set to 86 % if [SG10.7.9-Cooling type] = KNAN.	Yes
G3.3.4.6-Admission T3 charge = 53 ºC	[G3.3.4.4-Admission T2 charge] to [G3.3.2.6-Adm. T3 discharge] ºC	Sets the intake temperature for the third derating point. **Note:** This parameter is only displayed if [G3.3.4.1-Enable = Yes].	Yes
G3.3.4.7-Admission I3 charge = 86.00 %	0.00 to -Min between admission I2 charge and admission I3 discharge %	Sets the percentage of the rated current for the third derating point. **Note:** This parameter is only displayed if [G3.3.4.1-Enable = Yes].	Yes
G3.3.4.8-Admission T4 charge = 60 ºC	0 to 60 ºC	Sets the intake temperature for the maximum derating point. **Note:** This parameter is only displayed if [G3.3.4.1-Enable = Yes].	Yes

G3.3.5- Frequency derating Tª

This group allows the parameters configuring for frequency derating. The frequency derating allows to regulate the IGBTs switching frequency so that their operating temperature is reduced.

Note: Only users with role Developer can modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.3.5.1-Enable = No	No Yes	The power stage must apply a frequency derating on the carrier signal in order to reduce the overheating of the semiconductors (IGBTs) themselves while maintaining the maximum output current possible. Frequency derating is applied to all stages, and not only to the one containing the IGBT that has reached the "initial TªIGBTs" temperature. • A frequency derating will be applied when the thermal limit is reached until the minimum frequency is reached following a ramp. The initial IGBTs temperature coincides with the nominal frequency point and the maximum temperature with the set minimum frequency. • When the minimum frequency has been reached, current derating by IGBT temperature is applied while maintaining the minimum frequency. OPTION: No — The frequency derating is disabled. OPTION: Yes — The frequency derating is enabled. **Note:** This parameter can only be read and not modified.	Yes
G3.3.5.2-Disc initial T IGBTs = 90 ºC	90 to [G3.3.5.3-Disc final T IGBTs] ºC	It allows setting initial point of frequency derating by IGBTs when the inverter is in discharge.	Yes
G3.3.5.3-Disc final T IGBTs = 90 ºC	[G3.3.5.2-Disc initial T IGBTs] to 65535 ºC	It allows setting initial point of frequency derating by IGBTs when the inverter is in discharge.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.3.5.4-Char initial T IGBTs = 86 ºC	86 to [G3.3.5.5-Charge final T IGBTs] ºC	It allows setting initial point of frequency derating by IGBTs when the inverter is in charge.	Yes
G3.3.5.5-Charge final T IGBTs = 86 ºC	[G3.3.5.4-Char initial T IGBTs] to 65535 ºC	It allows setting initial point of frequency derating by IGBTs when the inverter is in charge.	Yes
G3.3.5.6-Frequency = 2580 Hz	2000 to 3000 Hz	Defines the minimum frequency during derating. **Note:** This parameter is only displayed if [G3.3.5.1-Enable = Yes].	Yes

6.3.4. Subgroup G3.4: Current limits

In this group, the limits for current can be configured.

Note: This subgroup is only displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.4.1-I max (dis)charge DC = 1200 A	0 to 65535 A	Sets the value to configure DC charge / discharge.	Yes
G3.4.2-Reactive current VRT = 100.00 %	0.00 to 100.00 %	Defines the reactive current limit LVRT or OVRT.	Yes
G3.4.3-Direct current VRT = 100.00 %	0.00 to 100.00 %	Defines the reactive current limit during LVRT or OVRT events when the system is working with direct sequence.	Yes
G3.4.4-Inverse current VRT = 100.00 %	0.00 to 100.00 %	Defines the reactive current limit during LVRT or OVRT events when the system is working with inverse sequence.	Yes

6.3.5. Subgroup G3.5: DU

Projects with Bus Plus functionality can lead to the installation of a large photovoltaic power. This can cause overheating on the DC side where the inverter cannot control the power of the DC/DC converter which is managed by the SCADA. Basically, the inverter will limit its own power (not the power of the DC/DC converter) by reading the current of the DC cabinet and the inlet temperature in order not to exceed the above limits.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff] and [SV1.3.1.1-Inverter type ≠ PCS/MPCS].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.5.1-Enable = Yes	No Yes	Enables the DC current limitation.	Yes
G3.5.2-DC switch limit 40ºC = -DC switch current limit 40 ºC default A	0 to 2500 A	Allows setting the current limit of the DC switch and the DC busbar when the temperature inside the DC cabinet is 40 ºC. **Note:** This parameter is only displayed if [G3.5.1-Enable = Yes]. Not available since version 5.1.0.	Yes
G3.5.3-DC switch limit 50ºC = -DC switch current limit 50 ºC default A	0 to 2500 A	Allows setting the current limit of the DC switch and the DC busbar when the temperature inside the DC cabinet is 50 ºC. **Note:** This parameter is only displayed if [G3.5.1-Enable = Yes]. Not available since version 5.1.0.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.5.4-DC switch limit 60ºC = -DC switch current limit 60 ºC default A	0 to 2500 A	Allows setting the current limit of the DC switch and the DC busbar when the temperature inside the DC cabinet is 60 ºC. **Note:** This parameter is only displayed if [G3.5.1-Enable = Yes]. Not available since version 5.1.0.	Yes
G3.5.5-Current variation = 10 %	0 to 100 %	Allows adjusting the percentage of the nominal current of the inverter that will be reduced for each ºC that the temperature of 60 ºC is exceeded. **Note:** This parameter is only displayed if [G3.5.1-Enable = Yes]. Not available since version 5.1.0.	Yes
G3.5.6-Wait time = 30 s	0 to 600 s	Allows adjusting the time the inverter will wait between two current variations (configured in the previous parameter). **Note:** This parameter is only displayed if [G3.5.1-Enable = Yes]. Not available since version 5.1.0.	Yes

6.3.6. Subgroup G3.6: Reference Differences

G3.6.1-P Difference[1]

Notes:

- This subgroup is displayed if [G1.3-User ≥ Qualified customer] and if [SG10.1.4-Inverter type ≥ Basic BUSPLUS].
- Only users with role higher than Admin (qualified customer) can modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.6.1.1-Enable P Limit = Yes	No Yes	Enables or disables the active power limits of the busbar.	Yes
G3.6.1.2-Busbar P threshold = 30 %	1 to 40 %	Sets the maximum difference of the active power between of the busbar. **Note:** This parameter is displayed if [G3.6.1.1-Enable P Limit = Yes].	Yes

6.3.7. Subgroup G3.7: Batteries

Note: This subgroup is displayed if [SG10.1.4-Inverter type ≥ Basic BUSPLUS].

G3.7.1-DC voltage

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with role higher than Admin can modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.7.1.1-Vdc min =1050 V	381 to 1425 V	Minimum voltage for battery discharge. If the battery voltage reaches this value, the inverter will regulate the current so as not to reduce the battery voltage any further.	Yes
G3.7.1.2-Vdc max = 1500 V	481 to 1530 V	Maximum voltage for battery charging. If the battery voltage reaches this value, the inverter will regulate the current so as not to increase the battery voltage any further.	Yes

[1] Consult Power Electronics for limits.

G3.7.2-DC current

Notes:

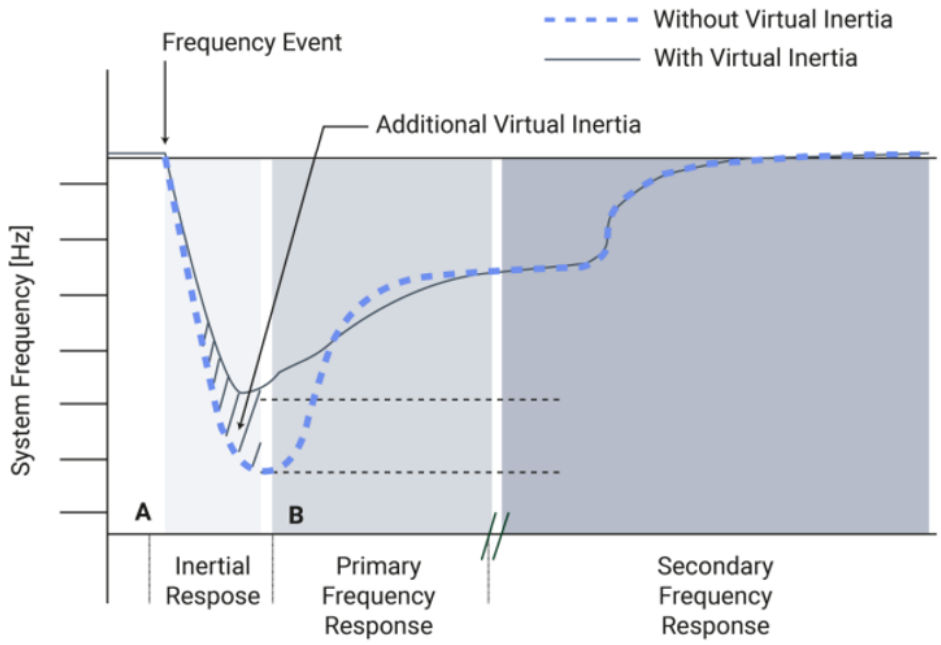
- Only users with role equal or higher than Admin can modify these parameters.
- It is displayed if [SG10.1.14-Number of busbar ≥ n], being 'n' the number of the busbar.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.7.2.1-Bus bar 1 DC dischar = (4600 / [SG10.1.14-Number of busbar]) A	0 to (5000 / [SG10.1.14-Number of busbar]) A	Sets the Discharge Current Limit (DCL), which is the DC current limit for the discharge of the battery connected to busbar 1.	Yes
G3.7.2.2-Bus bar 1 DC charge = (4600 / [SG10.1.14-Number of busbar]) A	0 to (5000 / [SG10.1.14-Number of busbar]) A	Sets the Charge Current Limit (CCL), which is the DC current limit for the charge of the battery connected to busbar 1.	Yes
G3.7.2.3-Bus bar 2 DC dischar = (4600 / [SG10.1.14-Number of busbar]) A	0 to (5000 / [SG10.1.14-Number of busbar]) A	Sets the Discharge Current Limit (DCL), which is the DC current limit for the discharge of the battery connected to busbar 2.	Yes
G3.7.2.4-Bus bar 2 DC charge = (4600 / [SG10.1.14-Number of busbar]) A	0 to (5000 / [SG10.1.14-Number of busbar]) A	Sets the Charge Current Limit (CCL), which is the DC current limit for the charge of the battery connected to busbar 2.	Yes
G3.7.2.5-Bus bar 3 DC dischar = (4600 / [SG10.1.14-Number of busbar]) A	0 to (5000 / [SG10.1.14-Number of busbar]) A	Sets the Discharge Current Limit (DCL), which is the DC current limit for the discharge of the battery connected to busbar 3.	Yes
G3.7.2.6-Bus bar 3 DC charge = (4600 / [SG10.1.14-Number of busbar]) A	0 to (5000 / [SG10.1.14-Number of busbar]) A	Sets the Charge Current Limit (CCL), which is the DC current limit for the charge of the battery connected to busbar 3.	Yes
G3.7.2.7-Bus bar 4 DC dischar = (4600 / [SG10.1.14-Number of busbar]) A	0 to (5000 / [SG10.1.14-Number of busbar]) A	Sets the Discharge Current Limit (DCL), which is the DC current limit for the discharge of the battery connected to busbar 4.	Yes
G3.7.2.8-Bus bar 4 DC charge = (4600 / [SG10.1.14-Number of busbar]) A	0 to (5000 / [SG10.1.14-Number of busbar]) A	Sets the Charge Current Limit (CCL), which is the DC current limit for the charge of the battery connected to busbar 4.	Yes

6.3.8. Subgroup G3.8: VISMA limits

Power load and discharge variations on the grid generate disturbances in the frequency that conventional synchronous generators compensate for with their mechanical inertia.

In grids with a high percentage of renewable energy, generators lack this mechanical inertia. For this purpose, the Visma algorithm has been implemented, which is a new type of control that simulates the behavior of a traditional electromechanical synchronous generator, behaving like a voltage source. The aim of this virtual inertia is to operate in grid-forming mode, stabilizing the grid frequency.



VISMA algorithm operation under frequency variations

Parameters within this group allow configuring the limits in voltage mode.

Note: This subgroup will be displayed if [G1.3-User ≥ Power Electronics staff] and [SG10.1.19-Visma enable = Enabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.8.1-Visma Freq max = 120.0 %	100.0 to 200.0 %	Allows to configure the maximum frequency limit that limits the active power of the inverter so as not to exceed the set frequencies in [G8.1.2.2-Freq increment]. If the frequency approaches the limit, it can be used to detect an Off Grid situation, but in that case, it may fail or change the control type/subtype/mode of the inverter. **Note:** This parameter applies if [G8.0.2-Control subtype = VISMA] & [G8.0.7-Control Theta = SWING EQ].	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G3.8.2-Visma Freq min = 70.0 %	0.1 to 100.0 %	Allows to configure the minimum frequency limit that limits the active power of the inverter so as not to exceed the set frequencies [G8.1.2.2-Freq increment]. If the frequency approaches the limit, it can be used to detect an Off Grid situation, but in that case, it may fail or change the control type/subtype/mode of the inverter. **Note:** This parameter is displayed if [G8.0.2-Control subtype = VISMA] & [G8.0.7-Control Theta = SWING EQ].	Yes
G3.8.3-Visma Vac max = 115.0 %	105.0 to 140.0 %	Allows to configure the maximum AC voltage limit that limits the reactive power of the inverter so as not to exceed the set voltages [G8.1.2.5-Vac Reference]. If there is an Off Grid situation, the inverter would limit the voltage within a range, avoiding a fault triggering. **Note:** This parameter is displayed if [G8.0.2-Control subtype = VISMA] & [G8.0.6-Q VSI control = Q].	Yes
G3.8.4-Visma Vac min = 70.0 %	0.1 to 95.0 %	Allows to configure the minimum AC voltage limit that limits the reactive power of the inverter so as not to exceed the set voltages [G8.1.2.5-Vac Reference]. If there is an Off Grid situation, the inverter would limit the voltage within a range, avoiding a fault triggering. **Note:** This parameter is displayed if [G8.0.2-Control subtype = VISMA] & [G8.0.6-Q VSI control = Q].	Yes
G3.8.5-Visma Is start = 30.00 %	0.10 to 80.00 %	Allows to configure the current limit available during a Blackstart start for VSI control type. **Note:** this parameter is displayed if [SG1.10.1-Black Start = Enabled] and [G8.0.5-P VSI control = Freq VISMA] and [G8.0.6-Q VSI control = Vac].	Yes
G3.8.6-Visma Is dynamic = 100.0 %	0.1 to 100.0 %	Allows to configure the dynamic apparent current limit of the inverter for VSI control type. **Note:** this parameter is displayed if [G8.0.2-Control subtype] = VISMA.	Yes
G3.8.7-Visma Id dynamic = 100.0 %	0.1 to 100.0 %	Allows to configure the dynamic active current limit of the inverter for VSI control type. **Note:** this parameter is displayed if [G8.0.2-Control subtype] = VISMA.	Yes
G3.8.8-Visma Iq dynamic = 100.0 %	0.1 to 100.0 %	Allows to configure the dynamic reactive current limit of the inverter for VSI control type. **Note:** this parameter is displayed if [G8.0.2-Control subtype] = VISMA.	Yes
G3.8.9-Visma Is LVRT = 100.0%	0.1 to 100.0%	Defines the dynamic current limit Is with respect to the nominal current applied in LVRT during VISMA control mode. **Note:** this parameter is displayed if [G8.0.2-Control subtype] = VISMA and [G4.3.1.1-Enable] = Disabled.	Yes
G3.8.10-Visma Id LVRT = 100.0%	0.1 to 100.0%	Defines the dynamic current limit Id with respect to the nominal current applied in LVRT during VISMA control mode. **Note:** this parameter is displayed if [G8.0.2-Control subtype] = VISMA and [G4.3.1.1-Enable] = Disabled.	Yes
G3.8.11-Visma Iq LVRT = 100.0%	0.1 to 100.0%	Defines the dynamic current limit Iq with respect to the nominal current applied in LVRT during VISMA control mode. **Note:** this parameter is displayed if [G8.0.2-Control subtype] = VISMA and [G4.3.1.1-Enable] = Disabled.	Yes
G3.8.12-Visma Is OVRT = 100.0%	0.1 to 100.0%	Defines the dynamic current limit Is with respect to the nominal current applied in OVRT during VISMA control mode. **Note:** this parameter is displayed if [G8.0.2-Control subtype] = VISMA and [G4.4.1.1-Enable] = Disabled.	Yes
G3.8.13-Visma Id OVRT = 100.0%	0.1 to 100.0%	Defines the dynamic current limit Id with respect to the nominal current applied in OVRT during VISMA control mode. **Note:** this parameter is displayed if [G8.0.2-Control subtype] = VISMA and [G4.4.1.1-Enable] = Disabled.	Yes
G3.8.14-Visma Iq OVRT = 100.0%	0.1 to 100.0%	Defines the dynamic current limit Iq with respect to the nominal current applied in OVRT during VISMA control mode. **Note:** this parameter is displayed if [G8.0.2-Control subtype] = VISMA and [G4.4.1.1-Enable] = Disabled.	Yes

6.4. Group G4: Algorithms

This group includes all the parameters that allow the execution of the different algorithms that help to ensure optimum performance of the inverter.

6.4.1. Subgroup G4.1: MPPt

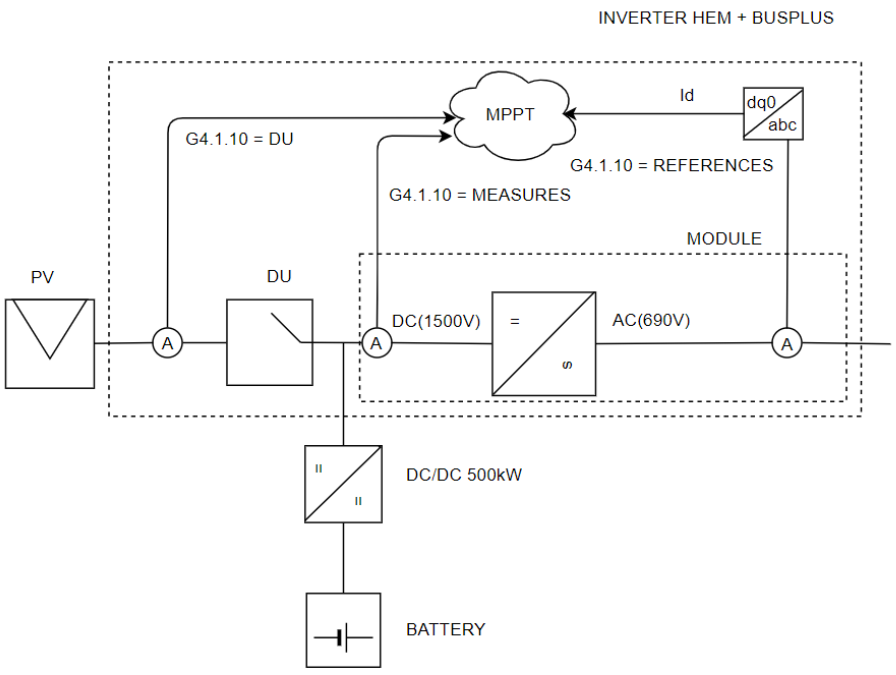
The MPPt (Maximum Power Point Tracking) algorithm searches constantly for the point of maximum power, which helps maximize the power generated by the panels and contributes to battery maintenance.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.1.1-Min Vdc = 0 V	0 to 65535 V	Defines the minimum voltage to start the MPPT algorithm.	Yes
G4.1.2-Max Vdc = 0 V	0 to 65535 V	Defines the maximum voltage to start the MPPT algorithm.	Yes
G4.1.3-Start voltage = 85.0 %	0.0 to 95.0 %	Defines the multiplication factor applied to not loaded charge voltage to start finding MPPT.	Yes
G4.1.4-Sample time = 3.0 s	0.1 to 10.0 s	Defines the time interval between executions of the MPPT algorithm.	Yes
G4.1.5-Max delta Vdc = 10 V	1 to 20 V	Defines the maximum voltage variations resulting from the MPPT algorithm.	Yes
G4.1.6-Delta Vdc = 2.5 V	0.1 to 10.0 V	Increases or decreases the voltage variation (increase or decrease) resulting of the MPPT algorithm.	Yes
G4.1.7-Delta P = 10.00 %	0.01 to 25.00 %	Defines the power MPP constant threshold.	Yes
G4.1.8-Delta Irr = 2.50 %	0.01 to 25.00 %	Defines the irradiation threshold to detect irradiation changes.	Yes
G4.1.9-Vac Hyst = 2 %	0 to 25 %	Minimum AC voltage value that the MPPt algorithm can set. This parameter enables controlling the minimum MPPt voltage as a percentage of the real AC voltage.	Yes
G4.1.10-Mode = References	Measures References DU	Allows to stablish the reference modes: **OPTION** / **DESCRIPTION** Measures: MPPT is calculated from Vdc measures and the sum of modules Idc. References: MPPT is calculated from MPPT reference and Id reference. DU: MPPT is calculated from the DC current measures. If the Bus Plus option is activated in parameter [SG10.1.4-Inverter type] = Solar inverter, it is advisable to set the parameter to DU.	Yes

EN

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.1.11-Negative Power = 0.0 %</td><td>0.0 to 100.0 %</td><td>Parameter to allow charging batteries from the grid depending on the inverter type [SG10.1.4-Type inverter] and following MPPT algorithm.
Usually, the inverters regulate DC Voltage bus according to the MPPt algorithm.
There are different options depending on the inverter type and permission to charge grid batteries:
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>0-Standard inverter. A solar inverter is usually not able to consume power from grid.
If this is the case, G4.1.11 must be set to 0%.</td></tr>
<tr><td>0</td><td>0-Bus Plus inverter type with no permissions to charge batteries. In the case of a Bus Plus inverter type with no permission to get power from grid to charge the batteries, G4.1.11 must be set to 0%.</td></tr>
<tr><td>Nx550 / Sbase</td><td>Bus Plus inverter type with permissions to charge batteries.
In the case of a Bus Plus inverter type, the DCDC external converter could be requested to charge batteries when there is not enough PV power.
In that case, this parameter allows to get power from grid to charge the batteries.
If this is accepted by the electric grid operator (or grid rules), then parameter G4.1.11 can be configured differently than 0%.
The parameter setting must be in relation to the power of the number of DCDCs (N) installed and what the network operator allows for charging the batteries.
In order to set this parameter, note that base power (Sbase) of the inverter is calculated by SG10.1.7 Nº modules x SG10.1.6 Nominal S module (kVA).</td></tr>
</table></td><td>Yes</td></tr>
</table>



MPPT algorithm in HEM inverters with Bus Plus

6.4.2. Subgroup G4.2: Gradients

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with role equal or higher than Admin (qualified customer) can see and modify these parameters.

G4.2.1-Start

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.2.1.1-Start grad P enable = Enabled	Disabled Enabled	Allows to enable or disable the start active power gradient. When this option is enabled, the power is injected gradually.	Yes
G4.2.1.2-Start gradient P = 50.0 %/s	0.10 to 600.00 %/s	Allows adjusting the speed at which the active power changes during inverter startup.	Yes
G4.2.1.3-Start grad Q enable = Enabled	Disabled Enabled	Allows to enable or disable the start reactive power gradient.	Yes
G4.2.1.4-Start gradient Q = 50.0 %/s	0.10 to 600.00 %/s	Allows adjusting the speed at which the reactive power changes during inverter startup.	Yes
G4.2.1.5- Start grad V Enable = Yes	No Yes	Allows enabling or disabling the start voltage gradient in voltage control type. **Note**: this parameter will be displayed if [SG10.1.19-Visma enable = Enabled].	Yes
G4.2.1.6- Start Gradient V = 10.00 %s	0.10 to 600.00 %/s	Allows adjusting the start voltage gradient in voltage control type. **Note**: this parameter will be displayed if [SG10.1.19-Visma enable = Enabled].	Yes

G4.2.2-Upward

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.2.2.1-Upward grad P enable = Enabled	Disabled Enabled	Allows enabling or disabling the active power ramp-up applied during normal inverter operation.	Yes
G4.2.2.2-Upward gradient P = 50.0 %/s	0.1 to 3000.0 %/s	Allows configuring the active power ramp-up gradient, that is, the maximum rate at which active power can increase.	Yes
G4.2.2.3-Upward grad Q enable = Enabled	Disabled Enabled	Allows enabling or disabling the reactive power ramp-up applied during normal inverter operation. The inverter will increase its output power.	Yes
G4.2.2.4-Upward gradient Q = 50.0 %/s	0.1 to 3000.0 %/s	Allows configuring the reactive power ramp-up gradient, i.e. the maximum rate at which reactive power can increase.	Yes
G4.2.2.5- Upward grad V enable = Yes	No Yes	Allows enabling or disabling the voltage ramp-up applied while the inverter is operating in voltage control mode. **Note**: this parameter will be displayed if [SG10.1.19-Visma enable = Enabled].	Yes
G4.2.2.6- Upward grad V = 10.0 %s	0.1 to 3000.0 %/s	Allows configuring the voltage ramp-up gradient in voltage control mode. **Note**: this parameter will be displayed if [SG10.1.19-Visma enable = Enabled].	Yes
G4.2.2.7- Upw grad freq enab = Yes	No Yes	Allows enabling or disabling the frequency ramp-up in voltage control mode. **Note**: this parameter will be displayed if [SG10.1.19-Visma enable = Enabled].	Yes
G4.2.2.8- Upward grad freq = 10.0 %s	0.1 to 3000.0 %/s	Allows configuring the start frequency upward gradient in voltage control mode. **Note**: this parameter will be displayed if [SG10.1.19-Visma enable = Enabled].	Yes
G4.2.2.9-Up grad sh P ena = Disabled	Disabled Enabled	Allows enabling or disabling the active power sharing ramp-up (P sharing) during normal inverter operation. **Note**: This parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes

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SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.2.2.10-Up grad sh P = 500.0 %/s	0.1 to 3000.0 %/s	Allows configuring the active power sharing ramp-up gradient (P sharing), that is, the maximum rate at which the sharing increases. **Note**: This parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes
G4.2.2.11-Up grad sh Q en = Disabled	Disabled Enabled	Allows enabling or disabling the reactive power sharing ramp-up (Q sharing) during normal inverter operation. **Note**: This parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes
G4.2.2.12-Up grad sh Q = 500.0 %/s	0.1 to 3000.0 %/s	Allows configuring the reactive power sharing ramp-up gradient (Q sharing), that is, the maximum rate at which reactive power sharing increases. **Note:** This parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes

G4.2.3-Downward

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.2.3.1-Down grad P enable = Enabled	Disabled Enabled	Allows enabling or disabling the active power ramp-down applied during normal inverter operation.	Yes
G4.2.3.2-Downward gradient P = 50.0 %/s	0.1 to 3000.0 %/s	Allows configuring the active power ramp-down gradient, which defines the maximum rate at which the active power can decrease.	Yes
G4.2.3.3-Down grad Q enable = Enabled	Disabled Enabled	Allows enabling or disabling the reactive power ramp-down during normal inverter operation.	Yes
G4.2.3.4-Downward gradient Q = 50.0 %/s	0.1 to 3000.0 %/s	Allows configuring the gradient of the reactive power ramp-down, which defines the maximum rate at which the reactive power can decrease.	Yes
G4.2.3.5- Down grad V enab = Yes	No Yes	Allows enabling or disabling the voltage ramp-down applied while the inverter is operating in voltage control mode. **Note**: this parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes
G4.2.3.6- Downward grad V = 10.0 %s	0.1 to 3000.0 %/s	Allows configuring the voltage ramp-down gradient, which defines the maximum rate at which voltage can decrease in voltage control mode **Note**: this parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes
G4.2.3.7- Down grad freq enab = Yes	No Yes	Allows enabling or disabling the frequency ramp-down in voltage control mode. **Note**: this parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes
G4.2.3.8- Downward grad freq = 10.0 %s	0.1 to 3000.0 %/s	Allows configuring the frequency ramp-down gradient, which defines the maximum rate at which frequency can decrease in voltage control mode. **Note**: this parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes
G4.2.3.9-Dw grad sh P ena = Disabled	Disabled Enabled	Allows enabling or disabling the active power sharing ramp-down (P sharing) applied during normal inverter operation. **Note**: this parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes
G4.2.3.10-Dw grad sh P = 500.0 %/s	0.1 to 3000.0 %/s	Allows configuring the active power sharing ramp-down gradient (P sharing), which defines the maximum rate at which active power sharing decreases. **Note**: this parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes
G4.2.3.11-Dw grad sh Q en = Disabled	Disabled Enabled	Allows enabling or disabling the reactive power sharing ramp-down (Q sharing) during normal inverter operation. **Note**: this parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes
G4.2.3.12-Dw grad sh Q = 500.0 %/s	0.1 to 3000.0 %/s	Allows configuring the reactive power sharing ramp-down gradient (Q sharing), which defines the maximum rate at which reactive power sharing decreases. **Note**: this parameter is displayed if [SG10.1.19-Visma enable] = Enabled.	Yes

G4.2.4-Stop

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.2.4.1-Stop grad P enable = Enabled	Disabled Enabled	Allows enabling or disabling the stop active power gradient. The inverter will reduce its output power until it reaches 0.	Yes
G4.2.4.2-Stop gradient P = 50.0 %/s	0.1 to 3000.0 %/s	Allows configuring the speed at which the active power changes during inverter stop.	Yes
G4.2.4.3-Stop grad Q enable = Enabled	Disabled Enabled	Allows enabling or disabling the stop reactive power gradient. The inverter will reduce its exit power until it reaches 0.	Yes
G4.2.4.4-Stop gradient Q = 50.0 %/s	0.1 to 3000.0 %/s	Allow configuring the speed at which the reactive power changes during inverter stop.	Yes

G4.2.5-AEMO

This subgroup allows the configuration of the specific AEMO gradients applied when the inverter operates under the conditions defined by that standard, such as in the event of a loss of communication with the PPC/SCADA.

Note: parameters within this group can only be modified by users with role Power Electronics staff or higher.

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SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.2.5.1- P upward gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the active power P ramp-up gradient applied during operation in standard AEMO mode.	Yes
G4.2.5.2- Q upward gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the reactive power Q ramp-up gradient applied during operation in standard AEMO mode.	Yes
G4.2.5.3- P downward gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the active power P ramp-down gradient applied during operation in standard AEMO mode.	Yes
G4.2.5.4- Q downward gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the reactive power Q ramp-down gradient applied during operation in standard AEMO mode.	Yes
G4.2.5.5- Alt P up gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the active power P ramp-up gradient applied during operation in alternative AEMO mode.	Yes
G4.2.5.6- Alt Q up gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the reactive power Q ramp-up gradient applied during operation in alternative AEMO mode.	Yes
G4.2.5.7- Alt P down gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the active power P ramp-down gradient applied during operation in alternative AEMO mode.	Yes
G4.2.5.8- Alt Q down gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the reactive power Q ramp-down gradient applied during operation in alternative AEMO mode.	Yes
G4.2.5.9-Sh P up gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the active power sharing ramp-up gradient (P sharing) applied during operation in standard AEMO mode.	Yes
G4.2.5.10-Sh Q up gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the reactive power sharing ramp-up gradient (Q sharing) applied during operation in standard AEMO mode.	Yes
G4.2.5.11-Sh P dw gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the active power sharing ramp-down gradient (P sharing) applied during operation in standard AEMO mode.	Yes
G4.2.5.12-Sh Q dw gradient = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the reactive power sharing ramp-down gradient (Q sharing) applied during operation in standard AEMO mode.	Yes
G4.2.5.13-Alt sh P up gradien = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the active power sharing ramp-up gradient (P sharing) applied during operation in alternative AEMO mode.	Yes
G4.2.5.14-Alt sh Q up gradien = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the reactive power sharing ramp-up gradient (Q sharing) applied during operation in alternative AEMO mode.	Yes
G4.2.5.15-Alt sh P dw gradien = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the active power sharing ramp-down gradient (P sharing) applied during operation in alternative AEMO mode.	Yes
G4.2.5.16-Alt sh Q dw gradien = 1.43 %/s	0.35 to 655.00 %/s	Allows configuring the reactive power sharing ramp-down gradient (Q sharing) applied during operation in alternative AEMO mode.	Yes

G4.2.6: VDE

This subgroup allows the reactive power to be regulated according to VDE standards.

Note: this subgroup is displayed if [G1.3-User] ≥ Power Electronics staff.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.2.6.1-Enable Q Response = Disabled	Disabled Enabled	Allows to activate the specific behavior according to VDE standards on reactive power (Q) regulation so that the generated Q resembles the response of a first order system.	Yes
G4.2.6.2-3TAU Q Response = 10 s	1 to 180 s	Allows configuring the 3TAU constant of the first order system for reactive power (Q) regulation according to VDE standards.	Yes

6.4.3. Subgroup G4.3: LVRT

The LVRT (Low Voltage Ride Through) algorithm helps controlling the behavior of the inverter in case of an AC voltage dip, to prevent it from stopping due to that voltage drop.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

G4.3.1-LVRT General

This subgroup allows the configuration of the conditions for voltage event detection and input to the LVRT algorithm.

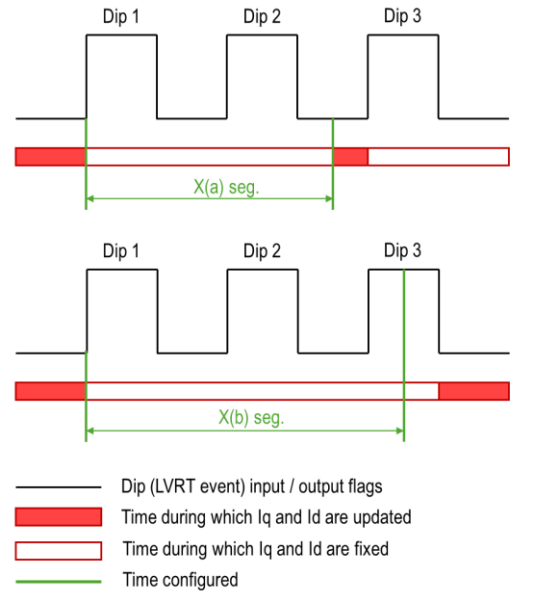
Note: Parameters within this subgroup from [G4.3.1.2-Threshold] to [G4.3.1.9-Freezing time refs] will be shown if [G4.3.1.1-Enable] ≠ Disabled.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.3.1.1-Enable = Mode 2	Disabled Mode 1 Mode 2 Mode 3	Enables or disables the LVRT algorithm and defines the voltage measurement to be used to determine the input to the LVRT algorithm in a voltage event. **OPTION** / **DESCRIPTION** Disabled: LVRT algorithm is disabled. Mode 1: Enable low voltage ride-through. The reference voltage corresponds with the direct sequence voltage. Mode 2: Enable low voltage ride-through. The reference voltage corresponds with the lowest value of phase-to-phase voltages. Mode 3: Enable the algorithm. The LVRT algorithm takes into account the estimated voltage in the POI to access or not to LVRT. **Note:** If this parameter is set to Disabled, the rest of parameters within this group up to up to [G4.3.1.9-Freezing time refs] will not be shown.	Yes
G4.3.1.2-Threshold = 85.0 %	0.0 to 95.0 %	Sets the threshold for the LVRT algorithm. If the AC voltage drops under this value, then the LVRT algorithm will be activated.	Yes
G4.3.1.2a-Inverse threshold = 0.0 %	0.0 to 50.0 %	Allows to configure the reverse voltage level that triggers reverse sequence current injection when LVRT algorithm is activated. When [G4.3.1.2a-Inverse threshold] = 0.0 %, the parameter is disabled.	Yes
G4.3.1.2b-k V Threshold = 0.000	0.000 to 0.500	Allows to configure a constant which will multiply the Iq value injected at the moment and the result will be added to the threshold configured for the activation of the LVRT algorithm.	Yes

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.3.1.3-Moment Cessation = Disabled</td><td>Disabled = 100.1;
1.0 to 100.0 %</td><td>It configures a nominal voltage threshold below which the inverter does not inject power to the grid anymore (P=0- Q=0). When the AC voltage increases its value above the system threshold then the standard LVRT algorithm will start working again.</td><td>Yes</td></tr>
<tr><td>G4.3.1.4-Configuration = Is prev + kdVdir</td><td>kdV
Iq prev + kdVdir
Is prev + kdVdir
Proportional
Advanced
kdV + Iq offset</td><td>Defines the system behavior in case of an LVRT event.

<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>kdV</td><td>The response is proportional to voltage variation.</td></tr><tr><td>Iq prev + kdVdir</td><td>The response is proportional to the direct sequence voltage variation + previous reactive current.</td></tr><tr><td>Is prev + kdVdir</td><td>The response is proportional to the direct sequence voltage variation + previous apparent current.</td></tr><tr><td>Proportional</td><td>The response tries to maintain the pre-event direct sequence active and reactive power and adds reactive current in direct and reverse sequence proportional to the voltage variation, limiting both currents equally (proportional). Prioritizes the injection of Q over P.</td></tr><tr><td>Advanced</td><td>Allows to configure a customized behavior. When this option is selected, the parameters [Injection mode Id] and [Injection mode Iq] will be displayed.</td></tr><tr><td>kdV + Iq offset</td><td>The response is proportional to voltage variation + offset reactive current.
Note: this option is not available from version 4.20.0.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G4.3.1.5-P Control Mode = Current</td><td>Current
Power</td><td>Allow selecting the P control mode:
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Current</td><td>Control by current.</td></tr><tr><td>Power</td><td>Control by power.</td></tr></table>Note: This parameter will only be shown if [G4.3.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
<tr><td>G4.3.1.6-Q Control Mode = Current</td><td>Current
Power</td><td>Allow selecting the Q control mode:
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Current</td><td>Control by current.</td></tr><tr><td>Power</td><td>Control by power.</td></tr></table>Note: This parameter will only be shown if [G4.3.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
<tr><td>G4.3.1.7-P/Q Priority = Q</td><td>P
Q</td><td>This parameter allows deciding which magnitude P or Q will take priority to keep S constant at a maximum value. P selects active power priority and Q selects reactive power priority.
Note: This parameter will only be shown if [G4.3.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
<tr><td>G4.3.1.8-Sequence Priority = Direct</td><td>Direct
Inverse
Proportional</td><td>Defines the LVRT sequence priority:
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Direct</td><td>Direct sequence in current injection takes priority.</td></tr><tr><td>Inverse</td><td>Inverse sequence in current injection takes priority.</td></tr></table>Note: This parameter will only be shown if [G4.3.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.3.1.9-Freezing time refs = 1.0 s</td><td>Disabled = 0.0;
0.1 to 3.0 s</td><td>Allows to configure the time during which the Iq and Id values prior to a LVRT are saved.
When an LVRT event is initiated, the inverter takes Iq and Id values that were being used at that time prior to the activation of the LVRT algorithm, so that:
a) if two or more voltage drops (dips) occur within the configured time, the values used will be the input values of the first LVRT event that occurred, provided that this time has not elapsed.
b) if the time elapses, but the dip does not end, the values will remain fixed until the dip ends, and meanwhile, the inverter will regulate on them.
Illustrative example:
 
</td><td>Yes</td></tr>
<tr><td>G4.3.1.10-Control VSI P = No alternative</td><td>No alternative
Pac
Freq</td><td>Allows to configure a change of the VISMA control mode within LVRT with respect to the control mode configured in [G8.1.2-VSI control].
Note: this parameter will be shown when [SG10.1.19- Visma enable] = Enabled.</td><td>Yes</td></tr>
<tr><td>G4.3.1.11-Control VSI Q = No alternative</td><td>No alternative
Vac
Qac</td><td>Allows to configure a change of the VISMA control mode within LVRT with respect to the control mode configured in [G8.1.2-VSI control].
Note: this parameter will be shown when [SG10.1.19- Visma enable] = Enabled.</td><td>Yes</td></tr>
<tr><td>G4.3.1.1b-Alt. LVRT Enable = Mode 2</td><td>Disabled
Mode 1
Mode 2
Mode 3</td><td>Allows to set the alternative configuration for LVRT enablement.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Disabled</td><td>LVRT algorithm is disabled with alternative configuration.</td></tr>
<tr><td>Mode 1</td><td>Enable low voltage ride-through with alternative configuration. The reference voltage corresponds with the direct sequence voltage.</td></tr>
<tr><td>Mode 2</td><td>Enable low voltage ride-through with alternative configuration. The reference voltage corresponds with the lowest value of phase-to-phase voltages.</td></tr>
<tr><td>Mode 3</td><td>Enable the algorithm with alternative configuration. The LVRT algorithm takes into account the estimated voltage in the POI to access or not to LVRT.</td></tr>
</table>
Note: this parameter is enabled in [G8.2.3-Enable Alt. Config.].</td><td>Yes</td></tr>
</table>

G4.3.2-LVRT Config

This subgroup allows to configure the current injection characteristics when inside the LVRT algorithm. The accessible parameters depend on [G4.3.1.4-Configuration].

Note: Parameters within this subgroup will be shown if [G4.3.1.1-Enable ≠ Disabled].

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.3.2.1-Control = Mode 1</td><td>Mode 1
Mode 2
Mode 3</td><td>Allows to define the voltage measurement to be used to control the response of the LVRT algorithm once inside the voltage event.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Mode 1</td><td>Enable algorithm. The reference voltage corresponds to the direct sequence voltage.</td></tr><tr><td>Mode 2</td><td>Enable algorithm. The reference voltage corresponds to the lowest value of the phase-to-phase voltages.</td></tr><tr><td>Mode 3</td><td>Enable algorithm. The LVRT algorithm takes into account the estimated voltage at the POI to access LVRT or not.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G4.3.2.2-Vset = Disabled</td><td>[G4.3.1.2-Threshold] to 120.1 %
Disabled = 120.2</td><td>Allows setting the amount of Vset based power injection in the LVRT algorithm. The Vset parameter is a value that will be used to calculate the deltaV used in the LVRT algorithm. If the deltaV calculation is disabled, it is done with respect to the voltage prior to the event.</td><td>Yes</td></tr>
<tr><td>G4.3.2.2a-Inverse Vset = 0.0 %</td><td>0.0 to [G4.3.1.2a-Inverse threshold] %</td><td>Allows to configure the reference voltage level which is used by the LVRT inverse control algorithm.</td><td>Yes</td></tr>
<tr><td>G4.3.2.3-Id Injection Mode = Constant</td><td>X Prev
Constant
MPPt</td><td>Allows selecting the Id injection mode.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>X Prev</td><td>Injects the previous value, where x = P or I depending on the previously set P control.</td></tr><tr><td>Constant</td><td>Injects a constant value.</td></tr><tr><td>MPPt</td><td>Uses the MPPt algorithm for injection.</td></tr></table>Note: This parameter will only be shown if [G4.3.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
<tr><td>G4.3.2.4-Iq DIR Injection Mod = X prev + kdVdir</td><td>kdVdir
X prev + kdVdir
Constant
kdV + Iq offset</td><td>Allows selecting the Iq direct injection mode.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>kdVdir</td><td>Injects a value proportional to the depth of the voltage dip.</td></tr><tr><td>X prev + kdVdir</td><td>Injects the above value, where X depends on the configuration set.</td></tr><tr><td>Constant</td><td>Injects a constant value.</td></tr><tr><td>kdV + Iq offset</td><td>The response is proportional to voltage variation + offset reactive current.</td></tr></table>Note: This parameter will only be shown if [G4.3.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.3.2.5-Iq INV Injection Mod = Power Stabilizer</td><td>KdVInv
Power Stabilizer
Constant</td><td>Allows selecting the Iq Inverse injection mode.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>KdVInv</td><td>Injects a value proportional to the inverse voltage sequence during the voltage dip.</td></tr><tr><td>Power Stabilizer</td><td>Injects reverse current in order to keep the active power constant in dips with reverse voltage sequence.
Note: this option is not available.</td></tr><tr><td>Constant</td><td>Injects a constant value of reverse Iq.</td></tr></table>Note: This parameter will only be shown if [G4.3.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
<tr><td>G4.3.2.6-Ref. Id DIR = 0.0 %</td><td>0.0 to 100.0 %</td><td>Sets the reference for current Iq in direct sequence during LVRT operation.
Note: This parameter will be shown if [G4.3.2.3-Id Injection Mode] = Constant.</td><td>Yes</td></tr>
<tr><td>G4.3.2.7-K DIR = 2.0</td><td>0.0 to 10.0</td><td>Sets the voltage proportional multiplication factor K (dV) for direct sequence during LVRT operation.
Note: This parameter will be shown if [G4.3.2.4-Iq DIR Injection] ≠ Constant.</td><td>Yes</td></tr>
<tr><td>G4.3.2.8-Ref. Iq DIR = 100.0 %</td><td>0.0 to 100.0 %</td><td>Sets the reference for current Iq in direct sequence during LVRT operation.
Note: This parameter will be shown if [G4.3.2.4-Iq DIR Injection] = Constant.</td><td>Yes</td></tr>
<tr><td>G4.3.2.9-K INV = 0.0</td><td>0.0 to 10.0</td><td>Sets the voltage proportional multiplication factor K (dV) in inverse sequence during LVRT operation.
Note: This parameter will be shown if [G4.3.2.5-Iq INV Injection] = kdVInv.</td><td>Yes</td></tr>
<tr><td>G4.3.2.10-Ref. Iq INV = 0.0 %</td><td>0.0 to 100.0 %</td><td>Sets the reference for current Iq in inverse sequence during LVRT operation.
Note: This parameter will be shown if [G4.3.2.5-Iq INV Injection] = Constant.</td><td>Yes</td></tr>
<tr><td>G4.3.2.11-KPS = 100.0 %</td><td>0.0 to 100.0 %</td><td>Allows configuring the proportional gain of the power stabilizer.
Note: This parameter will be shown if [G4.3.2.5-Iq INV Injection] = Power Stabilizer.</td><td>Yes</td></tr>
<tr><td>G4.3.2.12-KdV limit dir seq = Disabled</td><td>0.0 to 200 %;
Disabled = 200.1</td><td>Defines the current limit to estimate the kdV of LVRT event in direct sequence.
Note: This parameter is displayed if [G4.3.2.4-Iq DIR Injection Mod] ≠ Constant.</td><td>Yes</td></tr>
<tr><td>G4.3.2.13-KdV limit inv seq = Disabled</td><td>0.0 to 200 %;
Disabled = 200.1</td><td>Defines the current limit to estimate the kdV of LVRT event in inverse sequence.
Note: This parameter is displayed if [G4.3.2.5-Iq INV Injection Mod] = kdVInv.</td><td>Yes</td></tr>
<tr><td>G4.3.2.14-Iq offset = 0.0 %</td><td>0.0 to 100.0 %</td><td>Allows to configure Iq offset which will be added to the Q reference in an LVRT event, in order to account for the inductive loads between the inverter and POI. This offset must be positive (capacitive).</td><td>Yes</td></tr>
<tr><td>G4.3.2.15-Kref overcurr. VSI = 0.0 %</td><td>0.0 to 100.0 %</td><td>Allows to configure the gain applied to the injection current prior to activating the overcurrent control.
Note: this parameter will be shown when [SG10.1.19- Visma enable] = Enabled.</td><td>Yes</td></tr>
<tr><td>G4.3.2.16-T. slow filt. Vin = 0.02 s</td><td>0.00 to 1.00 s</td><td>It allows to set the time constant of the slow filter applied to the voltage used for reference generation in LVRT. It is only used when the value of the fast filter and that of the slow filter are similar, i.e. stationary.</td><td>Yes</td></tr>
</table>

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.3.2.17-Hist. Slow/Fast Vin = 5 %	0 to 100 %	It allows to set the value used to determine when to change the input voltage filtering of the LVRT algorithm. When the slow filter is being used and the difference between the result of the slow filter and the fast filter is greater than this value, the result of the fast filter starts to be used for reference generation (in LVRT).	Yes
G4.3.2.18-T. fast filt. Vin = 0.02 s	0.00 to 1.00 s	It allows to set the time constant of the fast filter applied to the voltage used for reference generation in LVRT. It is only used when the value of the fast filter and the slow filter diverge, i.e. in transients.	Yes
G4.3.2.19-Hist. Fast/Slow Vin = 5 %	0 to 100 %	It allows to set the value used to determine when to change the input voltage filtering of the LVRT algorithm. When the fast filter is being used and the difference between the result of the slow filter and the fast filter is less than this value, the result of the slow filter is started to be used for reference generation (in LVRT).	Yes

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G4.3.3-LVRT Exit

This subgroup defines the exit conditions of the LVRT algorithm.

Note: Parameters within this subgroup from [G4.3.3.1-Threshold Delay] to [G4.3.3.8-IQ Recover Ramp] are shown if [G4.3.1.1-Enable] ≠ Disabled.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.3.3.1-Threshold Delay = Infinite	0.005 to 60.000 s Infinite = 60.001;	Sets a delay at the output of the LVRT algorithm. When set to 60001 the time is infinite.	Yes
G4.3.3.2-Hysteresis Delay = 0.005 s	0.005 to 60.000 s	Sets a delay to the output of the LVRT algorithm if the hysteresis condition of parameter [G4.3.3.3-Hysteresis] is reached.	Yes
G4.3.3.3-Hysteresis = 5 %	1 to 15 %	Defines the hysteresis to obtain the output voltage value (voltage at which there is no dip). If the voltage is below the threshold, it will be considered as a voltage dip and will end when the voltage is "threshold" + "hysteresis".	Yes
G4.3.3.4-P Recover = 100.0 %	0.0 to 100.0 %	Sets the active power output of LVRT recovery. This is the P power value that the algorithm will apply on dip recovery before normal control starts acting again.	Yes
G4.3.3.5-Q Recover = 100.0 %	0.0 to 100.0 %	Sets the reactive power output of LVRT recovery. This is the Q power value that the algorithm will apply on dip recovery before normal control starts acting again.	Yes
G4.3.3.6a-P Recover Time = Disabled	Disabled = 0.0; 0.1 to 60.0 s	Time during which, after an LVRT event, active power references remain frozen.	Yes
G4.3.3.6b-Q Recover Time = Disabled	Disabled = 0.0; 0.1 to 60.0 s	Time during which, after an LVRT event, reactive power references remain frozen.	Yes
G4.3.3.7-ID Recover Ramp = Disabled	Disabled = 19; 20 to 3000 %/s	Recovery ramp at the LVRT exit. If the parameter is configured as “Disabled” the ramp will be disabled. Otherwise, the set value will determine the slope (percentage per second) of P.	Yes
G4.3.3.8-IQ Recover Ramp = Disabled	Disabled = 19; 20 to 3000 %/s	Recovery ramp at the LVRT exit. If the parameter is configured as “Disabled” the ramp will be disabled. Otherwise, the set value will determine the slope (percentage per second) of Q.	Yes

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.3.3.9-Exit Criteria = Voltage</td><td>Voltage
Voltage and Timeout
Voltage or Timeout</td><td rowspan="2">The [G4.3.3.9-Exit Criteria] parameter allows to define the conditions that determine when a LVRT event ends depending on the selected option.<table><tr><th>[G4.3.3.9-Exit Criteria]</th><th>DESCRIPTION</th></tr><tr><td>Voltage</td><td>The LVRT event ends when the AC voltage at the inverter terminals returns to within the specified exit conditions. No time-based criteria apply.</td></tr><tr><td>Voltage and Timeout</td><td>The LVRT event ends only when the AC voltage meets the exit conditions and the time set in [G4.3.3.10-Exit Timeout] (minimum time) has elapsed.</td></tr><tr><td>Voltage or Timeout</td><td>The LVRT event ends when the first of the following conditions is met:
- the AC voltage meets the exit conditions, or
- the time set in [G4.3.3.10-Exit Timeout] (maximum time) is reached.</td></tr></table>Note: the [G4.3.3.10-Exit Timeout] parameter is displayed only if [G4.3.3.9-Exit Criteria] ≠ Voltage.</td><td>Yes</td></tr>
<tr><td>G4.3.3.10-Exit Timeout = Disabled s</td><td>Disabled = 0.0;
0.1 to 30.0 s</td><td>Yes</td></tr>
<tr><td>G4.3.3.11-T. Blocking new Ref = Disabled s</td><td>Disabled = 0.0;
0.1 to 100.0 s</td><td>This parameter allows to define an interval (measured from the start of the LVRT event) during which, if the LVRT event is time-based, the P and Q setpoints are maintained at their pre-event values, preventing new setpoints from being applied.
This does not apply when the event ends due to voltage recovery.
This applies to both threshold-based dips and voltage drift-based dips.
Note: this parameter is displayed if [G4.3.3.9-Exit Critera] = Voltage or Timeout or [G4.3.4.4-Exit Delay] ≠ Disabled.</td><td>Yes</td></tr>
<tr><td>G4.3.3.12-Calm T avoid OVRT = 100 ms</td><td>0 to 1000 ms</td><td>This parameter allows to set a quiet period following the occurrence of an LVRT event, during which the OVRT event input is blocked, preventing immediate transitions between events during voltage recovery.
This period is applied immediately after the interval configured in [SG1.2.a6-Re-entry block Time].
Note: this parameter is displayed if [G4.4.1.1-Enable] ≠ Disabled.</td><td>Yes</td></tr>
</table>

G4.3.4-LVRT Voltage Derive

This subgroup allows to configure the detection, timing, and recovery criteria for the LVRT algorithm performance in case of voltage drift.

Note: Parameters within this subgroup will be shown if [G4.3.1.1-Enable ≠ Disabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.3.4.1-Enable = Disabled	Disabled Enabled	Enables or disables the input and output to the LVRT algorithm due to voltage drift.	Yes
G4.3.4.2-DIR dV Threshold = Disabled	0.00 to 25.00 %; Disabled = 25.01	This parameter defines the forward voltage variation threshold from which the dip algorithm will be activated.	Yes
G4.3.4.2a-RMS dV Threshold = Disabled %	0.00 to 25.00 %; Disabled = 25.01	This parameter defines the rapid variation threshold used as the input criterion for the voltage drift-based LVRT algorithm. Detection is performed on a per-phase basis by evaluating the variation in the RMS voltage value.	Yes
G4.3.4.3-INV dV Threshold = Disabled	0.00 to 25.00 %; Disabled = 25.01	This parameter defines the inverse voltage variation threshold from which the dip algorithm will be activate.	Yes
G4.3.4.4-Exit Delay = Disabled	Disabled = 0.0; 0.1 to 30.0 s	Sets a delay in seconds at the output of the LVRT algorithm when drift entered.	Yes
G4.3.4.5-Operation Ramp = 10 %/s	1 to 1000%/s; Disabled = 1001	Operation ramp. Input output and operation when the inverter is operating in LVRT and the event that caused it is the drift algorithm.	Yes

G4.3.5- LVRT Fast Exit

This subgroup allows to set the configuration to exit the LVRT algorithm quickly.

Note: Parameters within this subgroup will be shown if [G4.3.1.1-Enable ≠ Disabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.3.5.1-Enable = Disabled	Disabled Enabled	This parameter allows quickly enable or disable the LVRT output mode.	Yes
G4.3.5.2-Delay = 60 ms	30 to 100 ms	Once the inverter enters LVRT mode, a timer is started, and the timer has to exceed [G4.2.4.1-Stop grad P enable] to start the algorithm for exiting LVRT quickly.	Yes
G4.3.5.3-Confirmation Time = 10 ms	10 to 100 ms	Defines the confirmation time to be passed by the inverter before switching from LVRT to RUN.	Yes
G4.3.5.4-Retries Number = 2	1 to 10	During the PREEXIT substation if any of the output conditions are still met it will go to the LVRT and the counter in this parameter will increase.	Yes
G4.3.5.5-Exit Threshd 1 (u/s) = 120	10 to 200	Once the fast exit algorithm is started then the dv / dt and Voltage threshold to exit conditions are checked. When one of these conditions is met then the algorithm goes to a substation called PREEXIT where it has to spend more time than Confirmation Time to bring the inverter from LVRT to RUN".	Yes
G4.3.5.6-Exit Threshold 2 = 120 %	100 to 140 %	Once the fast exit algorithm is started then the dv / dt and Voltage threshold to exit conditions are checked. When one of these conditions is met then the algorithm goes to a substation called PREEXIT where it has to spend more time than Confirmation Time to bring the inverter from LVRT to RUN".	Yes

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G4.3.6- LVRT Dynamic Improv

This subgroup allows to configure dynamic improvements to the LVRT algorithm.

Note: Parameters within this subgroup will be shown if [G4.3.1.1-Enable ≠ Disabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.3.6.1-Dynamic Extra DIR = 0.0 %	-100.0 to 100.0 %	Allows setting a time over- or under-reference to improve dynamic performance (direct sequence).	Yes
G4.3.6.2-Dynamic Extra INV = 0.0 %	-100.0 to 100.0 %	Allows setting a time over- or under-reference to improve dynamic performance (inverse sequence).	Yes
G4.3.6.3-Dyn Ramp DIR (%/ms) = 0.0	0.0 to 1000.0	Sets the recovery ramp to over- or under-reference time (direct sequence).	Yes
G4.3.6.4-Dyn Ramp INV (%/ms) = 0.0	0.0 to 1000.0	Sets the recovery ramp to over- or under-reference time (inverse sequence).	Yes

6.4.4. Subgroup G4.4: OVRT

The OVRT (Over Voltage Ride Through) algorithm is similar to the LVRT algorithm but used in case there is a voltage peak. It helps maintaining inverter operation stable.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

G4.4.1-OVRT General

This subgroup allows the configuration of the conditions for voltage event detection and input to the OVRT algorithm.

Note: Parameters within this subgroup from [G4.4.1.2-Threshold] to [G4.4.1.9-Freezing time refs] will be shown if [G4.4.1.1-Enable] ≠ Disabled.

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.4.1.1-Enable = Disabled</td><td>Disabled
Mode 1
Mode 2
Mode 3</td><td>Enables or disables the OVRT algorithm and defines the voltage measurement to be used to determine the OVRT algorithm input to a voltage event.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>OVRT algorithm is disabled.</td></tr><tr><td>Mode 1</td><td>Enable high voltage ride-through algorithm. The reference voltage corresponds with the direct sequence voltage.</td></tr><tr><td>Mode 2</td><td>Enable high voltage ride-through algorithm. The reference voltage corresponds with the highest value of phase-to-phase voltages.</td></tr><tr><td>Mode 3</td><td>Enable algorithm. The OVRT algorithm takes into account the estimated voltage at the POI to access OVRT or not.</td></tr></table>Note: If this parameter is set to Disabled, the rest of parameters within this group up to [G4.4.1.9-Freezing time refs] will not be shown.</td><td>Yes</td></tr>
<tr><td>G4.4.1.2-Threshold = 110.0 %</td><td>105.0 to 128.0 %</td><td>Sets the threshold for the OVRT algorithm. If the voltage reaches a higher value, the OVRT algorithm will be executed.</td><td>Yes</td></tr>
<tr><td>G4.4.1.2a-Inverse threshold = 0.0 %</td><td>0.0 to 50.0 %</td><td>Allows to configure the reverse voltage level that triggers reverse sequence current injection when OVRT algorithm is activated.</td><td>Yes</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.4.1.2b-k V Threshold = 0.000</td><td>0.000 to 0.500</td><td>Allows to configure a constant which will multiply the Iq value injected at the moment and the result will be added to the threshold configured for the activation of the OVRT algorithm.</td><td>Yes</td></tr>
<tr><td>G4.4.1.3-Moment Cessation = Disabled</td><td>100.0 to 120.0%;
Disabled = 120.1</td><td>This parameter allows to define a threshold in % of the nominal voltage above which the inverter stops feeding power into the grid (P=0 and Q=0).
When the voltage drops below the threshold, the inverter continues with the configured dip algorithm.</td><td>Yes</td></tr>
<tr><td>G4.4.1.4-Configuration = Iq prev + kdVdir</td><td>kdV
Iq prev + kdVdir
Is prev + kdVdir
Proportional
Advanced
kdV + Iq offset</td><td>Defines the system behavior in case of an OVRT event.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>kdV</td><td>The response is proportional to the direct sequence voltage variation.</td></tr>
<tr><td>Iq prev + kdVdir</td><td>The response is proportional to the direct sequence voltage variation + previous reactive current.</td></tr>
<tr><td>Is prev + kdVdir</td><td>The response is proportional to the direct sequence voltage variation + previous apparent current.</td></tr>
<tr><td>Proportional</td><td>The response tries to maintain the pre-event direct sequence active and reactive power and adds reactive current in direct and reverse sequence proportional to the voltage variation, limiting both currents equally (proportional). Prioritizes the injection of Q over P.</td></tr>
<tr><td>Advanced</td><td>Allows to configure a customized behavior. When this option is selected, the parameters [Injection mode Id] and [Injection mode Iq] will be displayed.</td></tr>
<tr><td>kdV + Iq offset</td><td>The response is proportional to voltage variation + offset reactive current.
Note: this option is not available from version 4.20.0.</td></tr>
</table></td><td>Yes</td></tr>
<tr><td>G4.4.1.5-P Control Mode = Current</td><td>Current
Power</td><td>Allow selecting the P control mode:
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Current</td><td>Control by current.</td></tr>
<tr><td>Power</td><td>Control by power.</td></tr>
</table>
Note: This parameter will only be shown if [G4.4.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
<tr><td>G4.4.1.6-Q Control Mode = Current</td><td>Current
Power</td><td>Allow selecting the Q control mode:
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Current</td><td>Control by current.</td></tr>
<tr><td>Power</td><td>Control by power.</td></tr>
</table>
Note: This parameter will only be shown if [G4.4.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
<tr><td>G4.4.1.7-P/Q Priority = Q</td><td>P
Q</td><td>This parameter allows to select which magnitude P or Q will take priority to keep S constant at a maximum value. P selects active power priority and Q selects reactive power priority.
Note: This parameter will only be shown if [G4.4.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.4.1.8-Sequence Priority = Direct</td><td>Direct
Inverse
Proportional</td><td>Defines the OVRT sequence priority:<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Direct</td><td>Direct sequence in current injection takes priority.</td></tr><tr><td>Inverse</td><td>Inverse sequence in current injection takes priority.</td></tr></table>Note: This parameter will only be shown if [G4.4.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
<tr><td>G4.4.1.9-Freezing time refs = 1.0 s</td><td>Disabled = 0.0;
0.1 to 3.0 s</td><td>Allows to configure the time during which the Iq and Id values prior to an OVRT are saved.</td><td>Yes</td></tr>
<tr><td>G4.3.1.10-Control VSI P = No alternative</td><td>No alternative
Pac
Freq</td><td>Allows to configure a change of the VISMA control mode within OVRT with respect to the control mode configured in [G8.1.2-VSI control].
Note: this parameter will be shown when [SG10.1.19- Visma enable] = Enabled.</td><td>Yes</td></tr>
<tr><td>G4.3.1.11-Control VSI Q = No alternative</td><td>No alternative
Vac
Qac</td><td>Allows to configure a change of the VISMA control mode within OVRT with respect to the control mode configured in [G8.1.2-VSI control].
Note: this parameter will be shown when [SG10.1.19- Visma enable] = Enabled.</td><td>Yes</td></tr>
<tr><td>G4.4.1.1b-Alt. OVRT Enable = Disabled</td><td>Disabled
Mode 1
Mode 2
Mode 3</td><td>Allows to set the alternative configuration for HVRT/OVRT enablement.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>OVRT algorithm is disabled with alternative configuration.</td></tr><tr><td>Mode 1</td><td>Enable high voltage ride-through algorithm with alternative configuration. The reference voltage corresponds with the direct sequence voltage.</td></tr><tr><td>Mode 2</td><td>Enable high voltage ride-through algorithm with alternative configuration. The reference voltage corresponds with the highest value of phase-to-phase voltages.</td></tr><tr><td>Mode 3</td><td>Enable algorithm with alternative configuration. The OVRT algorithm takes into account the estimated voltage at the POI to access OVRT or not.</td></tr></table>Note: this parameter is enabled in [G8.2.3-Enable Alt. Config.].</td><td>Yes</td></tr>
</table>

G4.4.2-OVRT Config

This subgroup allows to configure the current injection characteristics when inside the OVRT algorithm.

Note: Parameters within this subgroup will not be shown if [G4.4.1.1-Enable ≠ Disabled].

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.4.2.1-Control = Mode 1</td><td>Mode 1
Mode 2
Mode 3</td><td>Allows to define the voltage measurement that will be used to control the response of the OVRT algorithm once inside the voltage event.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Mode 1</td><td>Enable algorithm. The reference voltage corresponds to the direct sequence voltage.</td></tr><tr><td>Mode 2</td><td>Enable algorithm. The reference voltage corresponds to the lowest value of the phase-to-phase voltages.</td></tr><tr><td>Mode 3</td><td>Enable algorithm. The OVRT algorithm takes into account the estimated voltage at the POI to access OVRT or not.</td></tr></table></td><td>Yes</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.4.2.2-Vset = Disabled</td><td>[G4.3.1.2-Threshold] to 120.1%
Disabled = 120.2</td><td>Allows setting the amount of Vset based power injection in the HVRT algorithm. The Vset parameter is a value that will be used to calculate the deltaV used in the OVRT algorithm. If the deltaV calculation is disabled, it is done with respect to the voltage prior to the event.</td><td>Yes</td></tr>
<tr><td>G4.4.2.2a-Inverse Vset = 0.0 %</td><td>0.0 to [G4.4.1.2a-Inverse threshold] %</td><td>Allows to configure the reference voltage level which is used by the OVRT inverse control algorithm.
Note: this parameter will be displayed if [G4.4.1.1-Enable] ≠ Disabled.</td><td>Yes</td></tr>
<tr><td>G4.4.2.3-Id Injection Mode = Constant</td><td>X Prev
Constant
MPPt</td><td>Allows selecting the Id injection mode.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>X Prev</td><td>Injects the above value, where x = P or I depending on the previously set P control.</td></tr><tr><td>Constant</td><td>Injects a constant value.</td></tr><tr><td>MPPt</td><td>Uses the MPPt algorithm for injection.</td></tr></table>Note: This parameter will only be shown if [G4.4.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
<tr><td>G4.4.2.4-Iq DIR Injection Mod = X prev + kdVdir</td><td>kdVdir
X prev + kdVdir
Constant
kdV + Iq offset</td><td>Allows selecting the Iq direct injection mode.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>KdVdir</td><td>Injects a value proportional to the overvoltage.</td></tr><tr><td>X prev + kdVdir</td><td>Injects the previous value, where X depends on the set configuration.</td></tr><tr><td>Constant</td><td>Injects a constant value.</td></tr><tr><td>kdV + Iq offset</td><td>The response is proportional to voltage variation + offset reactive current.</td></tr></table>Note: This parameter will only be shown if [G4.4.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
<tr><td>G4.4.2.5-Iq INV Injection Mod = Power Stabilizer</td><td>KdVInv
Power Stabilizer
Constant</td><td>Allows selecting the Iq Inverse injection mode.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>KdVInv</td><td>Injects a value proportional to the reverse voltage sequence during the overvoltage.</td></tr><tr><td>Power Stabilizer</td><td>Injects reverse current in order to keep the active power constant in a reverse voltage sequence overvoltage.
Note: this option is not available.</td></tr><tr><td>Constant</td><td>Injects a constant value of reverse Iq.</td></tr></table>Note: This parameter will only be shown if [G4.4.1.4-Configuration] = Advanced.</td><td>Yes</td></tr>
<tr><td>G4.4.2.6-Ref. Id DIR = 0.0 %</td><td>0.0 to 100.0 %</td><td>Sets the reference for current Iq in direct sequence during OVRT operation.
Note: This parameter will be shown if [G4.4.2.3-Id Injection Mode] = Constant.</td><td>Yes</td></tr>
<tr><td>G4.4.2.7-K DIR = 2.0</td><td>0.0 to 10.0</td><td>Sets the voltage proportional multiplication factor K (dV) for direct sequence during OVRT operation.
Note: This parameter will be shown if [G4.4.2.4-Iq DIR Injection Mod] ≠ Constant.</td><td>Yes</td></tr>
<tr><td>G4.4.2.8-Ref. Iq DIR = 100.0 %</td><td>0.0 to 100.0 %</td><td>Sets the reference for current Iq in direct sequence during OVRT operation.
Note: This parameter will be shown if [G4.4.2.4-Iq DIR Injection Mod] = Constant.</td><td>Yes</td></tr>
</table>

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SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.4.2.9-K INV = 0.0	0.0 to 10.0	Sets the voltage proportional multiplication factor K (dV) in inverse sequence during OVRT operation. **Note:** This parameter will be shown if [G4.4.2.5-Iq INV Injection Mod] = kdVInv.	Yes
G4.4.2.10-Ref. Iq INV = 0.0 %	0.0 to 100.0 %	Sets the reference for current Iq in inverse sequence during OVRT operation. **Note:** This parameter will be shown if [G4.4.2.5-Iq INV Injection Mod] = Constant.	Yes
G4.4.2.11-KPS = 100.0 %	0.0 to 100.0 %	Allows configuring the proportional gain of the power stabilizer. **Note:** This parameter will be shown if [G4.4.2.5-Iq INV Injection Mod] = Power Stabilizer.	Yes
G4.4.2.12-KdV limit dir seq = Disabled	0.0 to 200 %; Disabled = 200.1	Defines the current limit to estimate the kdV in OVRT event and direct sequence. **Note:** This parameter is displayed if [G4.4.2.4-Iq DIR Injection Mod] = Constant.	Yes
G4.4.2.13-KdV limit inv seq = Disabled	0.0 to 200 %; Disabled = 200.1	Defines the current limit to estimate the kdV in OVRT event and inverse sequence. **Note:** This parameter is displayed if [G4.4.2.5-Iq INV Injection Mod] = kdVInv.	Yes
G4.4.2.14-Iq offset = 0.0 %	0.0 to 100.0 %	Allows to configure Iq offset which will be added to the Q reference in an OVRT event, in order to account for the inductive loads between the inverter and POI. This offset must be positive (capacitive).	Yes
G4.4.2.15-Kref overcurr. VSI = 0.0 %	0.0 to100.0 %	Allows to configure the gain applied to the injection current prior to activating the overcurrent control. **Note**: this parameter will be shown when [SG10.1.19- Visma enable] = Enabled.	Yes
G4.4.2.16-T. slow filt. Vin = 0.02 s	0.00 to 1.00 s	It allows to set the time constant of the slow filter applied to the voltage used for reference generation in OVRT. It is only used when the value of the fast filter and that of the slow filter are similar, i.e. stationary.	Yes
G4.4.2.17-Hist. Slow/Fast Vin = 5 %	0 to 100 %	It allows to set the value used to determine when to change the input voltage filtering of the OVRT algorithm. When the slow filter is being used and the difference between the result of the slow filter and the fast filter is greater than this value, the result of the fast filter starts to be used for reference generation (in OVRT).	Yes
G4.4.2.18-T. fast filt. Vin = 0.02 s	0.00 to 1.00 s	It allows to set the time constant of the fast filter applied to the voltage used for reference generation in OVRT. Only used when the value of the fast filter and the slow filter diverge, i.e. in transients.	Yes
G4.4.2.19-Hist. Fast/Slow Vin = 5 %	0 to 100 %	It allows to set the value used to determine when to change the input voltage filtering of the OVRT algorithm. When the fast filter is being used and the difference between the result of the slow filter and the fast filter is less than this value, the result of the slow filter is started to be used for reference generation (in OVRT).	Yes

G4.4.3-OVRT Exit

This subgroup defines the output conditions of the OVRT algorithm.

Note: Parameters within this subgroup will not be shown if [G4.4.1.1-Enable ≠ Disabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.4.3.1-Threshold Delay = Infinite	0.005 to 60.000 s Infinite = 60.001;	Sets a delay at the output of the OVRT algorithm. When set to 60001 the time is infinite.	Yes
G4.4.3.2-Hysteresis Delay = 0.005 s	0.005 to 60.000 s	Sets a delay to the OVRT algorithm output if the hysteresis condition [G4.4.3.3-Hysteresis] is met.	Yes
G4.4.3.3-Hysteresis = 2 %	1 to 15 %	Defines the hysteresis to obtain the output voltage value (voltage at which there is no overvoltage). If the voltage is above the threshold, it will be considered as an overvoltage and will end when the voltage is "threshold" - "hysteresis".	Yes
G4.4.3.4-P Recover = 100.0 %	0.0 to 100.0 %	Sets the active power output of OVRT recovery. This is the P power value that the algorithm will apply on dip recovery before normal control starts acting again.	Yes
G4.4.3.5-Q Recover = 100.0 %	0.0 to 100.0 %	Sets the reactive power output of OVRT recovery. This is, the Q power value that the algorithm will apply on dip recovery before normal control starts acting again,	Yes
G4.4.3.6a-P Recover Time = Disabled	Disabled = 0.0; 0.1 to 60.0 s	Time during which, after an HVRT event, active power references remain frozen.	Yes
G4.4.3.6b-Q Recover Time = 0.0 s	Disabled = 0.0; 0.1 to 60.0 s	Time during which, after an OVRT event, reactive power references remain frozen.	Yes
G4.4.3.7-ID Recover Ramp = Disabled	Disabled = 19; 20 to 3000 %/s	Recovery ramp at the OVRT exit. If the parameter is configured as “Disabled” the ramp will be disabled. Otherwise, the set value will determine the slope (percentage per second) of P.	Yes
G4.4.3.8-IQ Recover Ramp = Disabled	Disabled = 19; 20 to 3000 %/s	Recovery ramp at the OVRT exit. If the parameter is configured as “Disabled” the ramp will be disabled. Otherwise, the set value will determine the slope (percentage per second) of Q.	Yes
G4.4.3.9-Exit Criteria = Voltage	Voltage Voltage and Timeout Voltage or Timeout	The [G4.4.3.9-Exit Criteria] parameter allows to define the conditions that determine when a OVRT event ends depending on the selected option. **[G4.4.3.9-Exit Criteria]** / **DESCRIPTION** Voltage: The OVRT event ends when the AC voltage at the inverter terminals returns to within the specified exit conditions. No time-based criteria apply. Voltage and Timeout: The OVRT event ends only when the AC voltage meets the exit conditions and the time set in [G4.4.3.10-Exit Timeout] (minimum time) has elapsed. Voltage or Timeout: The OVRT event ends when the first of the following conditions is met: - the AC voltage meets the exit conditions, or - the time set in [G4.4.3.10-Exit Timeout] (maximum time) is reached. **Note:** the [G4.4.3.10-Exit Timeout] parameter is displayed only if [G4.4.3.9-Exit Criteria] ≠ Voltage.	Yes
G4.4.3.10-Exit Timeout = Disabled s	Disabled = 0.0; 0.1 to 30.0 s		Yes

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SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.4.3.11-T. Blocking new Ref = Disabled s	Disabled = 0.0; 0.1 to 100.0 s	This parameter allows to define an interval (measured from the start of the OVRT event) during which, if the OVRT event is time-based, the P and Q setpoints are maintained at their pre-event values, preventing new setpoints from being applied. This does not apply when the event ends due to voltage recovery. This applies to both threshold-based dips and voltage drift-based dips. **Note**: this parameter is displayed if [G4.4.3.9-Exit Critera] = Voltage or Timeout or [G4.4.4.4-Exit Delay] ≠ Disabled.	Yes
G4.4.3.12-Calm T avoid LVRT = 100 ms	0 to 1000 ms	This parameter allows to set a quiet period following the occurrence of an OVRT event, during which the LVRT event input is blocked, preventing immediate transitions between events during voltage recovery. This period is applied immediately after the interval configured in [SG1.2.a6-Re-entry block Time]. **Note**: this parameter is displayed if [G4.3.1.1-Enable] ≠ Disabled.	Yes

G4.4.4- HVRT Voltage Derive

This subgroup allows to configure the performance of the OVRT algorithm in case of voltage drift.

Note: Parameters within this subgroup will not be shown if [G4.4.1.1-Enable ≠ Disabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.4.4.1-Enable = Disabled	Disabled Enabled	Enables or disables input and output to the OVRT algorithm due to voltage drift.	Yes
G4.4.4.2-Dir dV Threshold = Disabled	0.00 to 25.00 %; Disabled = 25.01	This parameter defines the forward voltage variation threshold from which the dip algorithm will be activated.	Yes
G4.4.4.2a-RMS dV Threshold = Disabled %	0.00 to 25.00 %; Disabled = 25.01	This parameter defines the rapid variation threshold used as the input criterion for the voltage drift-based OVRT algorithm. Detection is performed on a per-phase basis by evaluating the variation in the RMS voltage value.	Yes
G4.4.4.3-Inv dV Threshold = Disabled	0.00 to 25.00 %; Disabled = 25.01	This parameter defines the inverse voltage variation threshold from which the dip algorithm will be activated.	Yes
G4.4.4.4-Exit Delay = Disabled	Disabled = 0.0; 0.1 to 30.0 s	Sets a delay in seconds at the output of the HVRT algorithm when drift entered.	Yes
G4.4.4.5-Operation Ramp = 10 %/s	1 to 1000 %/s; Disabled = 1001	Operation ramp. Input, output and operation when the inverter is operating in HVRT and the event that caused it is the drift algorithm.	Yes

G4.4.5-OVRT Fast Exit

This subgroup allows to adjust the settings to exit the OVRT algorithm quickly.

Note: Parameters within this subgroup will not be shown if [G4.4.1.1-Enable ≠ Disabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.4.5.1-Enable = Disabled	Disabled Enabled	This parameter allows to quickly enable or disable the OVRT output mode.	Yes
G4.4.5.2-Delay = 60 ms	30 to 100 ms	Once the inverter enters LVRT mode, a timer is started, and the timer has to exceed G4.2.4.1 to start the algorithm for exiting LVRT quickly.	Yes
G4.4.5.3-Confirmation Time = 10 ms	10 to 100 ms	Defines the confirmation time to be passed by the inverter before switching from OVRT to RUN.	Yes
G4.4.5.4-Retries Number = 2	1 to 10	During the PREEXIT substation if any of the output conditions are still met it will go to the OVRT and the counter in this parameter will increase.	Yes
G4.4.5.5-Exit Threshd 1 (u/s) = 120	10 to 200	Once the fast exit algorithm is started then the dv / dt and Voltage threshold to exit conditions are checked. When one of these conditions is met then the algorithm goes to a substation called PREEXIT where it has to spend more time than Confirmation Time to bring the inverter from OVRT to RUN".	Yes
G4.4.5.6-Exit Threshold 2 = 120 %	100 to 140 %	Once the fast exit algorithm is started then the dv / dt and Voltage threshold to exit conditions are checked. When one of these conditions is met then the algorithm goes to a substation called PREEXIT where it has to spend more time than Confirmation Time to bring the inverter from OVRT to RUN".	Yes

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G4.4.6- OVRT Dynamic Improv

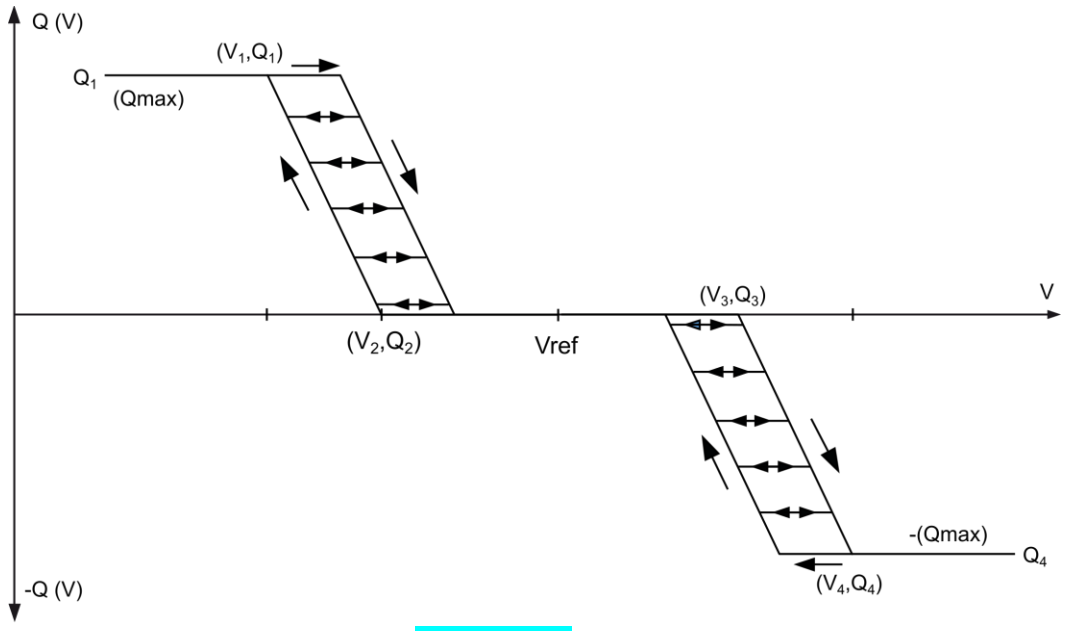
This subgroup allows to configure dynamic enhancements to the OVRT algorithm.

Note: Parameters within this subgroup will not be shown if [G4.4.1.1-Enable ≠ Disabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.4.6.1-Dynamic Extra DIR = 0.0 %	-100.0 to 100.0 %	Allows setting a time over- or under-reference to improve dynamic performance (direct sequence).	Yes
G4.4.6.2-Dynamic Extra INV = 0.0 %	-100.0 to 100.0 %	Allows setting a time over- or under-reference to improve dynamic performance (inverse sequence).	Yes
G4.4.6.3-Dyn Ramp DIR (%/ms) = 0.0	0.0 to 1000.0	Sets the recovery ramp to over- or under-reference time (direct sequence).	Yes
G4.4.6.4-Dyn Ramp INV (%/ms) = 0.0	0.0 to 1000.0	Sets the recovery ramp to over- or under-reference time (inverse sequence).	Yes

6.4.5. Subgroup G4.5: Curve Q(V)

This subgroup shows the configuration of the Q(V) algorithm. The following graph shows the curve points configured in the parameters of this subgroup.



Q(V) algorithm

Notes:

- This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with role higher than Admin (qualified customer) can modify these parameters.
- Parameters within this subgroup will be shown if [G4.5.1-Enable = Yes].

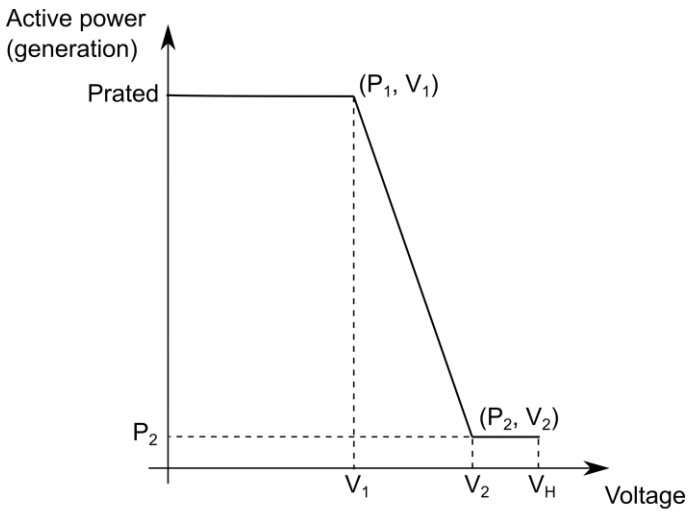
SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.5.1-Enable = No	No Yes	Enables or disables algorithm Q(v).	Yes
G4.5.2-Vref = 100.00 %	90.00 to 110.00 %	Defines the reference voltage of algorithm Q(v).	Yes
G4.5.3-V1 = 90.00 %	0.00 to [G4.5.4-V2] %	Sets the voltage from which the reactive capacitive power injection is calculated to inject the power configured in Q1 [G4.5.7-Q1].	Yes
G4.5.4-V2 = 92.00 %	0.00 to 655.35 %	Sets the voltage from which the reactive capacitive power injection is calculated to inject the power configured in Q2 [G4.5.8-Q2].	Yes
G4.5.5-V3 = 108.00 %	0.00 to 140.00 %	Sets the voltage from which the reactive capacitive power injection is calculated to inject the power configured in Q3 [G4.5.9-Q3].	Yes
G4.5.6-V4 = 110.00 %	[G4.5.5-V3] to 140.00 %	Sets the voltage from which the reactive capacitive power injection is calculated to inject the power configured in Q4 [G4.5.10-Q4].	Yes
G4.5.7-Q1 = 48.43 %	[G4.5.8-Q2] to 250.00 %	Configures the reactive power when the voltage is equal to V1 [G4.5.3-V1].	Yes
G4.5.8-Q2 = 0.00 %	0.00 to 100.00 %	Configures the reactive power when the voltage is equal to V2 [G4.5.4- V2].	Yes
G4.5.9-Q3 = 0.00 %	-100.00 to 0.00 %	Configures the reactive power when the voltage is equal to V3 [G4.5.5- V3].	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.5.10-Q4 = -48.43 %	-250.00 to [G4.5.9-Q3] %	Configures the reactive power when the voltage is equal to V4 [G4.5.6- V4].	Yes
G4.5.11-V hysteresis = 1.0 %	0.0 to 10.0 %	Adjusts the voltage hysteresis of the V/Q control mode.	Yes
G4.5.12-P activation = 20.00 %	0.00 to 100.00 %	Adjusts the activation active power of the V/Q control mode.	Yes
G4.5.13-P desactivation = 5.00 %	0.00 to 100.00 %	Adjusts the deactivation active power of the V/Q control mode.	Yes
G4.5.14-Response delay = 0.3 s	0.3 to 10.0 s	Defines the waiting time before activating the V/Q control mode when the activation conditions are accomplished.	Yes
G4.5.15-Response time = 5.0 s	0.0 to 90.0 s	Adjusts the response time (seconds) of the V/Q control mode. The response time is defined as the time that may take to arrive to 90 % of the reference once the modification has been made.	Yes
G4.5.16-Minimum cos phi = Disabled	Disabled = 0.000; 0.001 a 0.950	Allows configuring the minimum power factor for the Q(V) algorithm in order to limit both inductive and capacitive reactive power. **Note**: this parameter is shown if [G1.3-User] ≥ Emp. Power Electronics and [G4.5.1-Enable] = Yes.	Yes

EN

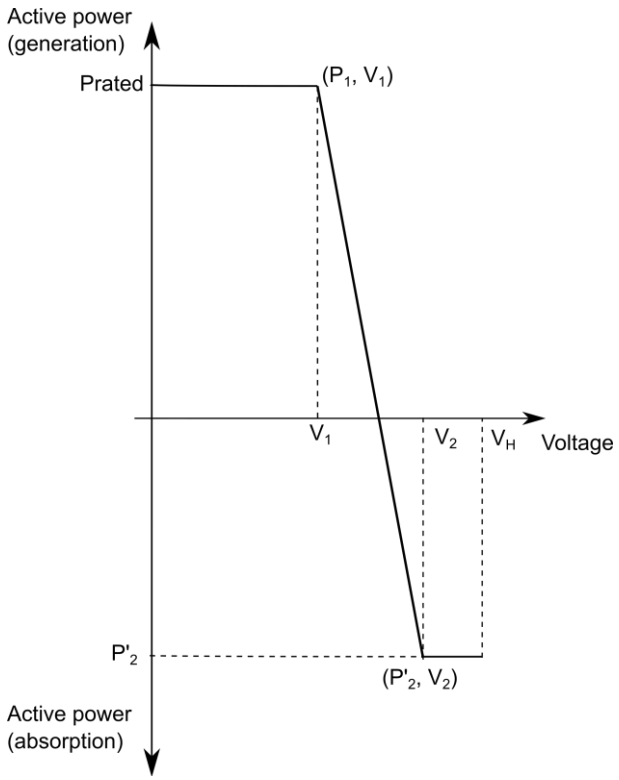
6.4.6. Subgroup G4.6: P(V) Curve

This subgroup allows the configuration of the P(V) algorithm. In case of HEM/K inverters, the P(V) curve is represented as follows.



P(V) algorithm in HEMK without Bus Plus inverters

In the case of HEM/K inverters with Bus Plus basic or advanced, the P(V) curve is represented as follows.



P(V) algorithm in HEMK with Bus Plus inverters

Notes:

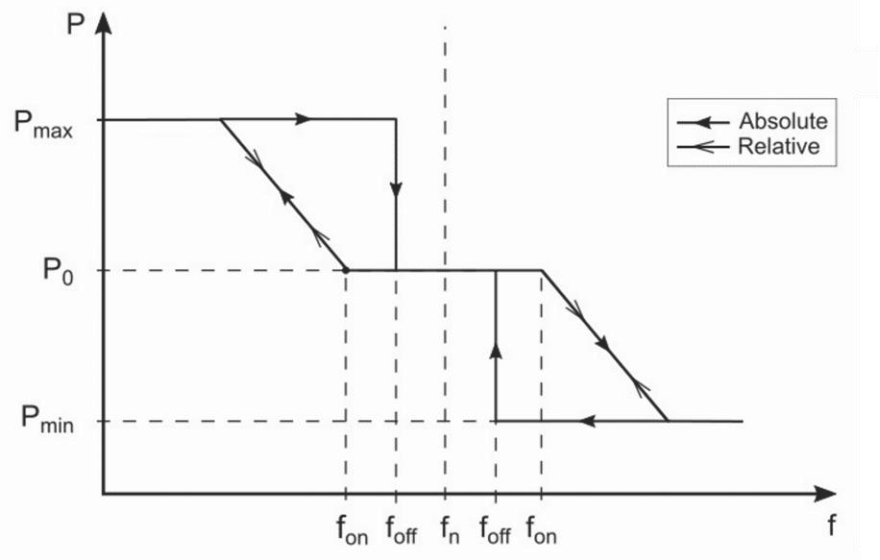
- This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with role equal or higher than Admin (qualified customer) can modify these parameters.
- Parameters within this subgroup will be shown if [G4.6.1-Enable = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.6.1-Enable = No	No Yes	Allows the system to configure the P(V) Curve.	Yes
G4.6.2-V1 = 110.00 %	100.00 to 140.00 %	Sets the V1 P(V) point.	Yes
G4.6.3-V2 = 120.00 %	[G4.6.2-V1] to 140.00 %	Sets the V2 P(V) point.	Yes
G4.6.4-P1 = 100.00 %	0.00 to 100.00 %	Sets the P1 P(V) point.	Yes
G4.6.5-P2 = 0.00 %	-100.00 to [G4.6.4-P1] %	Sets the P2 P(V) point.	Yes
G4.6.6-Hysteresis = 0.0 %	0.0 to 10.0 %	Sets the hysteresis to recover the active power with voltage reduction.	Yes
G4.6.7-Exit Delay = 60.0 s	0.0 to 90.0 s	Sets the hysteresis delay of the P/V curve. If the hysteresis value [G4.6.6-Hysteresis] is exceeded, then this delay will be applied as the trigger to start working with the P/V algorithm.	Yes
G4.6.8-Response time = 10.0 s	0.0 to 90.0 s	Adjusts the response time of the P/V algorithm. The response time is defined as the time that the system may take to achieve 90% of the reference once the modification has been made.	Yes
G4.6.9-Reference power = Relative	Relative Nominal	Defines the reference power mode for the P(V) event: Relative (power preview to event) or Nominal (Nominal power).	Yes
G4.6.10-Recovery mode = Absolut	Absolut Relative	Defines the recover mode after the P/V event. Absolut hold more restrictive power all event. Relative hold power during hysteresis and recover in ramp after hysteresis.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.6.11-Enable 1st order = No	No Yes	Allows enabling or disabling the active power generation response of the P(V) algorithm with dynamics equivalent to a first-order system in response to changes in the P setpoint.	No
G4.6.12-3TAU P response = 3 s	1 to 180 s	Allows configuring the time constant (TAU) of the first-order response of the active power generation of the P(V) algorithm, when this functionality is enabled.	Yes

6.4.7. Subgroup G4.7: P(f) Curve

This subgroup allows the configuration of the P(f) algorithm. The following graph represents the algorithm curve.



P(f) algorithm

Notes: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with role higher than Admin (qualified customer) can modify these parameters.

G4.7.1-Common Parameters

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.7.1.1-Power Reference = Nominal	Relative Nominal	Defines the reduction / increase percentage by Hz that will be applied to the power at the moment the activation frequency (relative option) or the nominal power (nominal option) is reached.	Yes
G4.7.1.2-Recovery Mode = Relative	Absolut Relative	Defines the recover mode of the P/F algorithm. OPTION / DESCRIPTION Absolut: The power limitation obtained when the frequency recover process begins is maintained until the deactivation frequency is reached. Relative: Applies an adjustable hysteresis.	Yes
G4.7.1.3-Gradient = Disabled	Disabled = 0.00 0.01 to 655.00 %/s	Adjusts the gradient that limits the power variation when the P/F algorithm is working.	Yes
G4.7.1.4-Response Time = 5.00 s	0.00 to 5.00 s	Adjusts the response time of the P/V algorithm. The response time is defined as the time that the system may take to achieve the 90 % of the reference once the modification has been made.	Yes
G4.7.1.5-Change Chrg Dischrg = Disabled	Disabled Enabled	Allows to enable the passage from charge to discharge and vice versa in case of frequency event. **Note:** This parameter is displayed if [SG10.1.4-Inverter type = PCS/MPCS].	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.7.1.6-Enable Freeze Pref = Disabled	Disabled Enabled	Enable or disable freezing of the reference of active power (P) in case of frequency event. **Note:** This parameter is displayed if [SG10.1.4-Inverter type = PCS/MPCS].	Yes
G4.7.1.7-VDE Output P %/min = 10	1 to 100	Allows to set the gradient of P to be applied when the frequency is reset after a P(f) event (VDE standard specific). **Note**: This parameter is displayed if [G1.3-User] ≥ Power Electronics staff.	Yes
G4.7.1.8-VDE Output P Time = Disabled min	Disabled = 0; 1 to 60 min	Allows to set the time during which the P gradient set in [G4.7.1.7-VDE output P %/min] (VDE standard specific) will be applied. **Note**: This parameter is displayed if [G1.3-User] ≥ Power Electronics staff.	Yes
G4.7.1.9-Entry Delay = Disabled s	Disabled = 0.0; 0.1 to 2.0 s	Allows configuring an input delay for the P(f) algorithm so that the algorithm action is carried out once the configured time has elapsed since the triggering condition occurs.	Yes

G4.7.2-Over-Frequency

Parameters within this subgroup will be shown if [G4.7.2.1-Enable = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.7.2.1-Enable = Disabled	Disabled Enabled	Defines the function activation frequency. Depending on this frequency the active power will be reduced as it is configured.	Yes
G4.7.2.2-Freq. Activation = 60.300 Hz	50.000 to 65.000 Hz	Defines the function activation frequency. Depending on this frequency the active power will be reduced as it is configured.	Yes
G4.7.2.3-Hysteresis = 0.0 Hz	0.0 to 5.0 Hz	Sets the hysteresis value to recover the active power with a frequency reduction. The frequency to stop the regulation is defined as $F_{off} = F_{on}$ – Hysteresis.	Yes
G4.7.2.4-Statism = 4.000 %	0.001 to 20.000 %	Defines the frequency variation for each 100% load variation: statism (%) = 100/ (fn(Hz)*rate(pu/Hz).	Yes
G4.7.2.5-Final Frequency = 65.000 Hz	50.000 to 65.000 Hz	Sets the final frequency value of the first constant power section of the over frequencies range - Ireland Code.	Yes
G4.7.2.6- Zero Frequency = 65.000 Hz	50.000 to 65.000 Hz	Sets the final frequency value of the second constant power section. It is equal to zero on the over frequencies range - Ireland Code.	Yes
G4.7.2.7-P final = -100.0 %	-250.0 to 250.0 %	Sets the first constant power section (over frequencies) - Ireland Code.	Yes
G4.7.2.8-P zero = -100.0 %	-250.0 to 250.0 %	Sets the second constant power section (over frequencies) - Ireland Code.	Yes
G4.7.2.9-Delay Hys Exit = 0.0 s	0.0 to 600.0 s	Sets the delay time that the system waits after the desired frequency is reached in order to exit the algorithm - Italy Code. **Note:** This parameter is only displayed if [G4.7.1.2-Recovery Mode = Absolute] and [G4.7.2.1-Enable = Yes].	Yes
G4.7.2.10-Enable Recover Grad = 25 %	Slow Fast Operation Disabled	Allows to: • Configure always the slow gradient. • Configure always the fast gradient. • Configure a dual gradient depending on the power decrease forced by the over frequency event. • Configure any exit gradient.	Yes
G4.7.2.11-Slow Gradient %/min = 5	1 to 600	Defines the power increase that will be applied at the exit of the algorithm when the active power decrease is higher than the P delta configured in the previous parameter. **Note:** This parameter is only displayed if ([G4.7.2.10-Enable Recover Grad = Disabled] or [G4.7.2.10-Enable Recover Grad = Slow] or [G4.7.2.10-Enable Recover Grad = 1 to 50]).	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.7.2.12-Fast Gradient %/min = 20	1 to 600	Defines the power increase that will be applied at the exit of the algorithm when the active power decrease is lower than the P delta configured in the previous parameter. **Note:** This parameter is only displayed if ([G4.7.2.10-Enable Recover Grad = Disabled] or [G4.7.2.10-Enable Recover Grad = Fast] or [G4.7.2.10-Enable Recover Grad = 1 to 50]).	Yes

G4.7.3–Under-Frequency

Parameters within this subgroup will be shown if [G4.7.3.1-Enable = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.7.3.1-Enable = Disabled	Disabled Enabled	Enables / disables the frequency regulation. Active power is increased depending on the frequency value.	Yes
G4.7.3.2-Freq. Activation = 59.700 Hz	45.000 to 65.000 Hz	Defines the activation frequency of the function. That means that the power starts increasing when the frequency has reached the value configured in this parameter.	Yes
G4.7.3.3-Hysteresis = 0.0 Hz	0.0 to 5.0 Hz	Defines the hysteresis value to recover the active power with a frequency reduction. The frequency to stop the regulation is defined as $F_{off} = F_{on}$ + Hysteresis.	Yes
G4.7.3.4-Statism = 4.000 %	0.001 to 20.000 %	Defines the frequency variation for each 100% load variation: statism (%) = 100/ (fn(Hz)*slope(pu/Hz).	Yes
G4.7.3.5-Power Reserve = 0.0 %	0.0 to 10.0 %	Defines the power stock that the inverter creates with regard to the nominal energy.	Yes
G4.7.3.6-Initial Frequency = 45.000 Hz	45.000 to 60.000 Hz	Defines the initial frequency value of the first constant power section of the under frequencies range - Ireland Code.	Yes
G4.7.3.7-Final Frequency = 45.000 Hz	45.000 to 60.000 Hz	Defines the final frequency value of the first constant power section of the under frequencies range - Ireland Code.	Yes
G4.7.3.8-P initial = 100.0 %	0.0 to 250.0 %	Defines the first constant power section (under frequencies) - Ireland Code.	Yes
G4.7.3.9-P final = 100.0 %	0.0 to 250.0 %	Defines the second constant power section (under frequencies) - Ireland Code.	Yes

G4.7.4–Active Pow. Ranges

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.7.4.1-P1 = 0.0 %	-25.0 to 25.0 %	Sets the minimum active power within the dead band. Allows to carry out a symmetrical algorithm.	Yes
G4.7.4.2-P2 = 100.0 %	0.0 to 250.0 %	Sets the maximum positive limit.	Yes
G4.7.4.3-P3 = -100.0 %	-250.0 to 0.0 %	Sets the maximum negative limit.	Yes

6.4.8. Subgroup G4.8: Econ mode

In order to save energy during periods when the PV field or battery is not required to interact with the grid, the **HEM / PCSM** inverters have an operating mode called Econ Mode.

In this mode, the inverter disconnects from the grid by opening the MV switchgear and thus avoids the no-load consumption of the main MV transformer.

The general feeding for the system may be provided by an external transformer or by external supply provided by the customer.

By its own characteristics, ECON mode is not compatible with other night modes as heating or Night mode (Vdc bus regulation).

ECON mode is an optional within the inverter and must be configured internally, [SG10.7.1-Econmode].

Notes:

- This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with a role higher than Admin (qualified customer) can modify these parameters except G4.8.1 which requires at least role equal to Admin.
- Parameters within this subgroup will be shown if [G4.8.1-Enable = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.8.1-Enable = No	No Yes	Allows the SCADA to activate the Econ Mode algorithm during the night. **Note:** This function depends on (Enabled = YES and in the Drive Select inverter type = HEM).	Yes
G4.8.2-Priority = No	No Yes	Sets the priority of Econ Mode overnight modes (night mode, active heating and DC disconnection). There are two situations depending on the selected priority: • YES: Deactivates night modes active heating (internal or external) and DC night opening. • NO: The activation of night mode has priority over Econ Mode. This parameter can only be changed with the MV switchgear closed in order to avoid several openings at night. Otherwise, a warning will be triggered.	Yes

6.4.9. Subgroup G4.9: SVR

This subgroup allows to configure the Suppression of Voltage Rise (SVR) algorithm. This algorithm limits an uncontrolled increase in voltage to the grid.

Notes:

- This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with role higher than Admin (qualified customer) can modify parameters within this subgroup.
- Parameters within this subgroup will be shown if [G4.9.1-Enable = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.9.1-Enable = No	No Yes	Allows to enable / disable the SVR algorithm.	Yes
G4.9.2-Threshold V1 = 105.0 %	100.0 to 120.0 %	Sets the lowest threshold to activate the SVR algorithm. If the grid voltage exceeds this value, the inverter will start to inject reactive power with the ramp set and limited in the following parameters.	Yes
G4.9.3-Hysteresis V1 = 2.0 %	0.5 to 10.0 %	When the mains voltage is less than the threshold V1 minus the hysteresis (Threshold V1 - Hysteresis V1) the inductive reactive power injection will be reduced.	Yes
G4.9.4-Injection ramp Q = 0.3 %/s	0.1 to 200.0 %/s	Adjusts the injection ramp Q when threshold V1 is exceeded.	Yes
G4.9.5-Recovery ramp Q = 1.7 %/s	0.1 to 200.0 %/s	Adjusts the recovery ramp Q when voltage is below Threshold V1 - Hysteresis V1.	Yes
G4.9.6-Limit cos phi = 0.85	0.80 to 1.00	Sets the power factor limit value for Q injection.	Yes
G4.9.7-Threshold V2 = 107.0 %	[G4.9.2-Threshold V1] to 120.0 %	Sets the second threshold from which P will be reduced to zero with the ramp configured and limited in these parameters.	Yes
G4.9.8-Hysteresis V2 = 2.0 %	0.5 to 10.0 %	When the main voltage is less than the threshold V2 minus the hysteresis (Threshold V2 - Hysteresis V2) the inductive active power injection will be reduced.	Yes
G4.9.9-Reduction ramp P = 0.3 %/s	0.1 to 200.0 %/s	Adjusts the reduction ramp P when threshold V2 is exceeded.	Yes
G4.9.10-Recovery ramp P = 1.7 %/s	0.1 to 200.0 %/s	Adjusts the recovery ramp P when voltage is below Threshold V2 - Hysteresis V2. Limit Cos phi must continue to be maintained.	Yes

EN

6.4.10. Subgroup G4.10: Night mode

This subgroup allows to configure the night mode algorithm, both solar inverters and PCS inverters.

Notes:

- This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with role higher than Admin (qualified customer) can modify the parameters G4.10.1 and G4.10.3.
- Parameters within this subgroup will be shown if [G4.10.1-Enable = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.10.1-Enable = No	No Yes	This parameter allows enabling or disabling the night mode algorithm. OPTION: No — The night mode is disabled. OPTION: Yes — Enables night mode option. This means that a smooth transition (without stopping the inverter) to NIGHT MODE will be done when conditions are fulfilled. **Note:** Even if this mode is selected, if the start command is removed, the inverter will not be able to start.	Yes
G4.10.2-Q reference = 0.0 %	-100.0 to 100.0 %	Sets the reactive power that the system must apply at night mode.	Yes
G4.10.3-Vbus reference = 1120 V	500 to 1500 V	Allows setting the bus voltage reference for the night mode.	Yes
G4.10.4-Activate = No	No Yes	Allows to activate the night mode if the algorithm is enabled in [G4.10.1-Enable] parameter.	Yes
G4.10.5-Force = No	No Yes	Allows night mode to be forced to run the algorithm.	Yes

6.4.11. Subgroup G4.11: Active balancing

It is an automatic power scaling algorithm between multiple battery banks. This strategy consists of keeping the battery banks in the same operating point (equal voltage and SOC).

Applies to PCS / MPCS inverters with at least 2 busbars ([SG10.1.4-Inverter type = PCS/MPCS] and [SG10.1.14-Number of busbar] ≥ 2).

Note: this subgroup is displayed if [G1.3-User] ≥ Qualified customer.

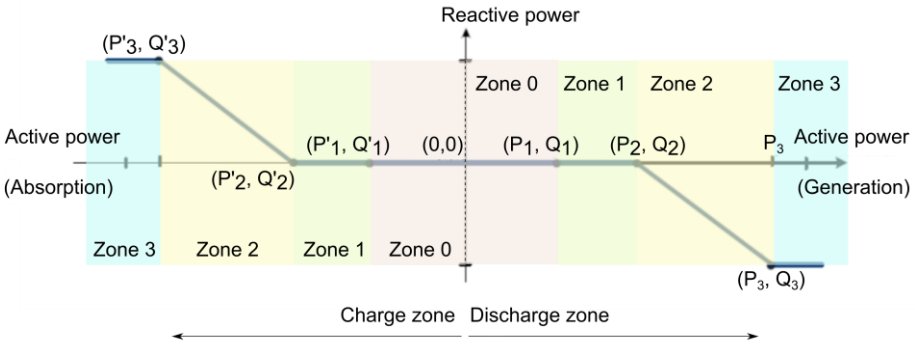
SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.11.1-Enable = No	No Yes	Allows enabling (Yes) or disabling (No) the Active balancing algorithm.	Yes
G4.11.2-Activation = No	No Yes	Once the algorithm is enabled, this parameter allows the activation of the algorithm. **Note:** an internal configuration is required for this parameter to be shown (it is shown if [G4.11.1-Enable] = Yes).	Yes
G4.11.3-Gain = 2.00	1.00 to 100.00	Allows to set the value required to calculate shifting factor and adjust the power to be diverted between battery banks. The bigger the gain, the faster the adjustment. If the gain is 0, then there will be no action. The maximum gain will depend on the maximum shifting admissible by Multi PCSK. **Note:** this parameter is shown if [G4.11.1-Enable] = Yes.	Yes
G4.11.4-V activation threshol = 5 V	0 to 10 V	Threshold of the bank voltage difference from which the algorithm is activated. If the voltage difference is less than this value, the algorithm does not activate. **Note:** this parameter is shown if [G4.11.1-Enable] = Yes.	Yes
G4.11.4b-SOC activation th = 1.0 %	0.0 to 10.0 %	This allows to set the State of Charge (SOC) difference threshold at which the Active Balancing algorithm is activated when the activation criterion is based on SOC. **Note**: this parameter is displayed if [G4.11.5-Balance mode] = SOC.	No
G4.11.5-Balance mode = Vdc	Vdc SOC Extern	Allows to select the criteria used to calculate the gain of the Active Balancing algorithm. **OPTION** / **DESCRIPTION** Vdc: Balancing is based on the DC voltage of the battery banks. SOC: Balancing is based on the state of charge (SOC). Extern: Balancing is based on an external SOC. The gain calculation is performed externally to the inverter and applied directly to the configured values.	No
G4.11.6-SOC BESS 1 = 0.00 %	0.00 to 120.00 %	This allows to configure the variable that registers the SOC value for the busbar 1. This value must be entered externally and is not calculated internally by the inverter. **Note**: this parameter is displayed if [G4.11.5-Balance mode] = SOC.	Yes
G4.11.7-SOC BESS 2 = 0.00 %	0.00 to 120.00 %	This allows to configure the variable that registers the SOC value for the busbar 2. This value must be entered externally and is not calculated internally by the inverter. **Note**: this parameter is displayed if [G4.11.5-Balance mode] = SOC.	Yes
G4.11.8-SOC BESS 3 = 0.00 %	0.00 to 120.00 %	This allows to configure the variable that registers the SOC value for the busbar 3. This value must be entered externally and is not calculated internally by the inverter. **Note**: this parameter is displayed if [G4.11.5-Balance mode] = SOC and [SG10.1.14-Number of busbar] ≥ 3.	Yes
G4.11.9-SOC BESS 4 = 0.00 %	0.00 to 120.00 %	This allows to configure the variable that registers the SOC value for the busbar 4. This value must be entered externally and is not calculated internally by the inverter. **Note**: this parameter is displayed if [G4.11.5-Balance mode] = SOC and [SG10.1.14-Number of busbar] ≥ 4.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.11.10-K BESS 1 = 0.00 %	-100.00 to 100.00 %	This allows to configure the gain of the Active Balancing algorithm assigned to Busbar 1 for the distribution of active power (Pac) in multi-busbar configurations. It applies to VISMA and CSI modes.	Yes
G4.11.11-K BESS 2 = 0.00 %	-100.00 to 100.00 %	This allows to configure the gain of the Active Balancing algorithm assigned to Busbar 2 for the distribution of active power (Pac) in multi-busbar configurations. It applies to VISMA and CSI modes.	Yes
G4.11.12-K BESS 3 = 0.00 %	-100.00 to 100.00 %	This allows to configure the gain of the Active Balancing algorithm assigned to Busbar 3 for the distribution of active power (Pac) in multi-busbar configurations. It applies to VISMA and CSI modes.	Yes
G4.11.13-K BESS 4 = 0.00 %	-100.00 to 100.00 %	This allows to configure the gain of the Active Balancing algorithm assigned to Busbar 4 for the distribution of active power (Pac) in multi-busbar configurations. It applies to VISMA and CSI modes.	Yes

6.4.12. Subgroup G4.12: Q(P) Curve

The Q(P) algorithm controls the reactive power injected or absorbed by the inverter as a function of active power. This control is defined by a curve composed of several active power (P) points and their corresponding reactive power (Q) values, distinguishing between charging and discharging regions.

The Q(P) curve is defined from a central point P0 and three additional active power points in each direction (charging and discharging), allowing the inverter response to be adjusted for different active power levels. The control is applied linearly between the configured points, with no delay included, as shown in the following figure.



Q(P) algorithm

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with role higher than Admin (qualified customer) can modify the parameters.

G4.12.1-Common Parameters

Allows the configuration of the common parameters of the Q(P) charge/discharge algorithm.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.12.1.1-Q0 = 0.00 %	-100.00 to 100.00 %	Allows configuring the Q value when P = 0. It is the starting point of the Q(P) curve.	No
G4.12.1.1a-P0 = 0.00 %	-100.00 to 100.00 %	Allows configuring the active power value of point P0 of the Q(P) curve, corresponding to the origin of the curve. This parameter allows shifting the central point of the Q(P) curve along the active power axis, enabling the configuration of curves with different separation between charging and discharging regions.	No
G4.12.1.2-Gradient = Disabled	Disabled = 0.0; 0.1 to 3000.0 %/s	Allows configuring the algorithm output ramp.	Yes

G4.12.2-Discharge

This subgroup allows the Q(P) discharge algorithm configuration.

Note: Parameters within this subgroup will be shown if [G4.12.2.1-Discharge Enable] = Enabled.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.12.2.1-Discharge Enable = Disabled	Disabled Enabled	Allows enabling the algorithm in discharge area.	Yes
G4.12.2.2-Discharge P1 = 20.00 %	0.00 to 100.00 %	Allows configuring the value of the first active power point in the discharging region of the Q(P) curve. This parameter defines the P value from which Q control in discharging begins. Its minimum configurable value is determined by parameter [G4.12.1.1a-P0], preventing overlap between the charging and discharging operating regions.	No
G4.12.2.3-Discharge P2 = 50.00 %	0.00 to 100.00 %	Allows configuring the value of the second active power point in the discharging region of the Q(P) curve. This parameter defines an intermediate point at which Q is adjusted as a function of P during discharging.	No
G4.12.2.4-Discharge P3 = 100.00 %	0.00 to 100.00 %	Allows configuring the value of the third active power point in the discharging region of the Q(P) curve. This parameter defines the maximum discharging P value, above which the configured Q value remains constant.	No
G4.12.2.5-Discharge Q1 = 0.00 %	-100.00 to 100.00 %	Allows configuring the value of the first reactive power point to be applied in the discharging region of the Q(P) curve, based on P.	No
G4.12.2.6-Discharge Q2 = 0.00 %	-100.00 to 100.00 %	Allows configuring the value of the second reactive power point to be applied in the discharging region of the Q(P) curve, based on P.	No
G4.12.2.7-Discharge Q3 = -44.00 %	-100.00 to 100.00 %	Allows configuring the value of the third reactive power point to be applied in the discharging region of the Q(P) curve, based on P.	No

G4.12.3-Charge

Note: These parameters are displayed if [SG10.1.4-Inverter type ≥ Basic BUSPLUS] and [G4.12.3.1-Charge Enable = Enabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.12.3.1-Charge Enable = Disabled	Disabled Enabled	Enables the algorithm in the charge area.	Yes
G4.12.3.2-Charge P1 = -20.00 %	-100.00 to 0.00 %	Allows configuring the value of the first active power point in the charging region of the Q(P) curve. This parameter defines the P value at which Q control in charging starts. Its maximum configurable value is limited by parameter [G4.12.1.1a-P0], preventing overlap between charging and discharging regions.	No
G4.12.3.3-Charge P2 = - 50.00 %	-100.00 to 0.00 %	Allows configuring the value of the second active power point in the charging region of the Q(P) curve. This parameter defines an intermediate point at which Q is adjusted as a function of P during charging.	No
G4.12.3.4-Charge P3 = - 100.00 %	-100.00 to 0.00 %	Allows configuring the value of the third active power point in the charging region of the Q(P) curve. This parameter defines the maximum P value in charging, above which the configured Q value remains constant.	No
G4.12.3.5-Charge Q1 = 0.00 %	-100.00 to 100.00 %	Allows configuring the value of the first reactive power point to be applied in the Q1 charging region of the Q(P) curve, based on P.	No
G4.12.3.6-Charge Q2 = 0.00 %	-100.00 to 100.00 %	Allows configuring the value of the second reactive power point to be applied in the Q2 charging region of the Q(P) curve, based on P.	No
G4.12.3.7-Charge Q3 = 44.00 %	-100.00 to 100.00 %	Allows configuring the value of the third reactive power point to be applied in the Q3 charging region of the Q(P) curve, based on P.	No

6.4.13. Subgroup G4.13: AntiPID

The PID (Potential Induce Degradation) corresponds to the power degradation that the solar panels suffer, when the inverter is configured as isolated poles, due to eddy current. If the effect of degradation has already started, it is possible to restraint it or even revert it. For this purpose, an external hardware is connected to the inverter in such a way that, via a trigger signal from the inverter, it is possible to activate or deactivate the Anti-PID hardware.

This optional can be included in the inverter using extra electronics. The main control of the Anti-PID is performed by a couple of digital outputs located over the IMI board device and is configured in [SG10.7.2-Anti-PID]. It is only compatible when [SG10.6.1-DU Type = IMD]. As per its own specifications this device is not compatible when:

- [SG10.6.1- DU Type] = GFDI+IMD.
- [SG10.1.4-Inverter type] = Basic or Advanced BUSPLUS.

This functionality is compatible with the night mode, active heating and also the Econ Mode functionality.

The activation sequence is carried out when the system switches from day to night and the PV voltage between positive and negative is less than 50V. The deactivation sequence is carried out when the system switches from night to day.

Two warnings are related to the Anti-PID:

- W191: Antipid wrong V. It is triggered while the activation algorithm is being carried out, and the voltage between DC and earth is too low. This means, when the voltage is below the measure of the voltage before closing relays of the Anti-PID plus 50V.
- W192: Antipid V not reached. It is triggered when the voltage between DC and earth is not between 700 and 800V, once the Anti-PID is already activated.

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. These parameters are only displayed if [SG10.7.2-Anti-PID = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.13.1-Activated = No	No Yes	Allows to activate the Anti-PID algorithm. When the above conditions are met, the algorithm will be started. **Note:** Only users with role equal or higher than Admin can modify this parameter.	Yes
G4.13.2-Time After Night = 30 s	0 to 60 s	Indicates the time elapsed since nighttime conditions are reached. **Note:** Only users with role higher than Admin can modify this parameter.	Yes
G4.13.3-PV voltage threshold = 50 V	10 to 50 V	Allows to configure the PV voltage below which the Anti-PID algorithm may start.	Yes
G4.13.4-Start time delay = 7 min	0 to 15 min	Allows to configure the delay to start the Anti-PID algorithm once the conditions are met.	Yes
G4.13.5-AntiPID Reset = No	No Yes	Allows to reboot the Anti-PID algorithm. This algorithm includes some protections against wrong behavior due to external causes. Thus, in case it has been tried to start the Anti-PID by three times but it is not possible to finish the activation sequence, the system does not try it anymore until the next day. This parameter allows to reset the Anti-PID in this situation and it also allows to erase the warning “W193:Antipid retries exceed”.	Yes

6.4.14. Subgroup G4.14: Active heating

The system should be provided by an active heating system that takes advantage of the power or heat losses generated by the power stage to carry out the heating based on the temperature and humidity.

When the conditions to activate the active heating are fulfilled, the night mode will start forcing Q reference to a configurable value. The ventilation system of DC cabinet will also start.

The active heating can be activated because of the temperature and humidity of the central control, module FPGA or DU boards are above or below the configured thresholds. In other words, when:

- The temperature is below a minimum configurable threshold.
- The temperature is equal or above the configurable threshold and simultaneously, the humidity is superior to the configurable threshold.

The activation of the active heating functionality also depends on the maximum temperature of the IGBTs, as the switching of the module FPGA will stop if this temperature is above the configurable threshold and starts when it is below a configurable threshold.

The ventilation system of DC cabinet and the active heating will remain activated for the period of time that the power stage stops switching, in order to cool the IGBTs and the system heating to continue.

The night mode is a priority compared with the active heating. Likewise, the active heating is a priority compared with the Econ Mode functionality.

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with role higher than Admin (qualified customer) can modify the parameters.

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G4.14.1-Active heating activa = Disabled</td><td>Disabled
Automatic</td><td>Allows to set the active heating mode configured.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>The active heating system is disabled.</td></tr><tr><td>Automatic</td><td>The active heating system is configured as automatic mode.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G4.14.2-Q reference = 0.0 %</td><td>0.0 to 30.0 %</td><td>Allows to set the reactive power reference for the active heating mode.</td><td>Yes</td></tr>
<tr><td>G4.14.3-Vbus reference = 1120 V</td><td>400 to 1500 V</td><td>Allows to set the bus voltage reference for the active heating mode.
Note: This parameter is displayed if [G4.14.1-Active heating activa ≠ Disabled].</td><td>Yes</td></tr>
<tr><td>G4.14.4-Setpoint DU fans = 30 %</td><td>0 to 100 %</td><td>Sets the setpoint of the DU zone to the setpoint of this parameter when active heating is activated.
Note: This parameter is displayed if [G4.14.1-Active heating activa ≠ Disabled].</td><td>Yes</td></tr>
<tr><td>G4.14.5-Activation temp = 5 ºC</td><td>-30 to 47 ºC</td><td rowspan="2">Allows to set the temperature below which active heating will be activated regardless of the humidity value and above which active heating will start if and only if the relative humidity exceeds its activation threshold.
Note: These parameters are displayed if [G4.14.1-Active heating activa ≠ Disabled].</td><td>Yes</td></tr>
<tr><td>G4.14.6-Deactivation temp = 8 ºC</td><td>0 to 50 ºC</td><td>Yes</td></tr>
<tr><td>G4.14.7-Activation humidity = 65 %</td><td>0 to 100 %</td><td>Allows to set the humidity above which active heating will be activated if the activation temperature condition is met at the same time.
Note: This parameter is displayed if [G4.14.1-Active heating activa ≠ Disabled].</td><td>Yes</td></tr>
</table>

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.14.8-Deactivation humidity = 50 %	0 to 100 %	Allows to set the humidity below which active heating will be deactivated if the temperature is higher than the activation temperature. **Note:** This parameter is displayed if [G4.14.1-Active heating activa ≠ Disabled].	Yes
G4.14.9-Deactiv temp IGBTS = 90 ºC	0 to 120 ºC	Allows to set the IGBTs temperature above which the IGBTs switching will be deactivated. Ventilation and active heating will remain activated. **Note:** This parameter is displayed if [G4.14.1-Active heating activa ≠ Disabled].	Yes
G4.14.10-Activ temp IGBTS = 60 ºC	0 to 120 ºC	Allows to set the IGBTs temperature below which the IGBTs switching will be activated. Ventilation and active heating will remain activated. **Note:** This parameter is displayed if [G4.14.1-Active heating activa ≠ Disabled].	Yes

6.4.15. Subgroup G4.15: Cold start

EN

The inverter that may be installed in places with extreme climate conditions (this is, in places that can reach temperatures of around -25ºC), should include a heating kit that increases the temperature inside the system to avoid damages of its components.

This way, when the temperature measured by the admission NTC and by the power stage is less than -20ºC, the system needs to carry out a special start process, called cold start.

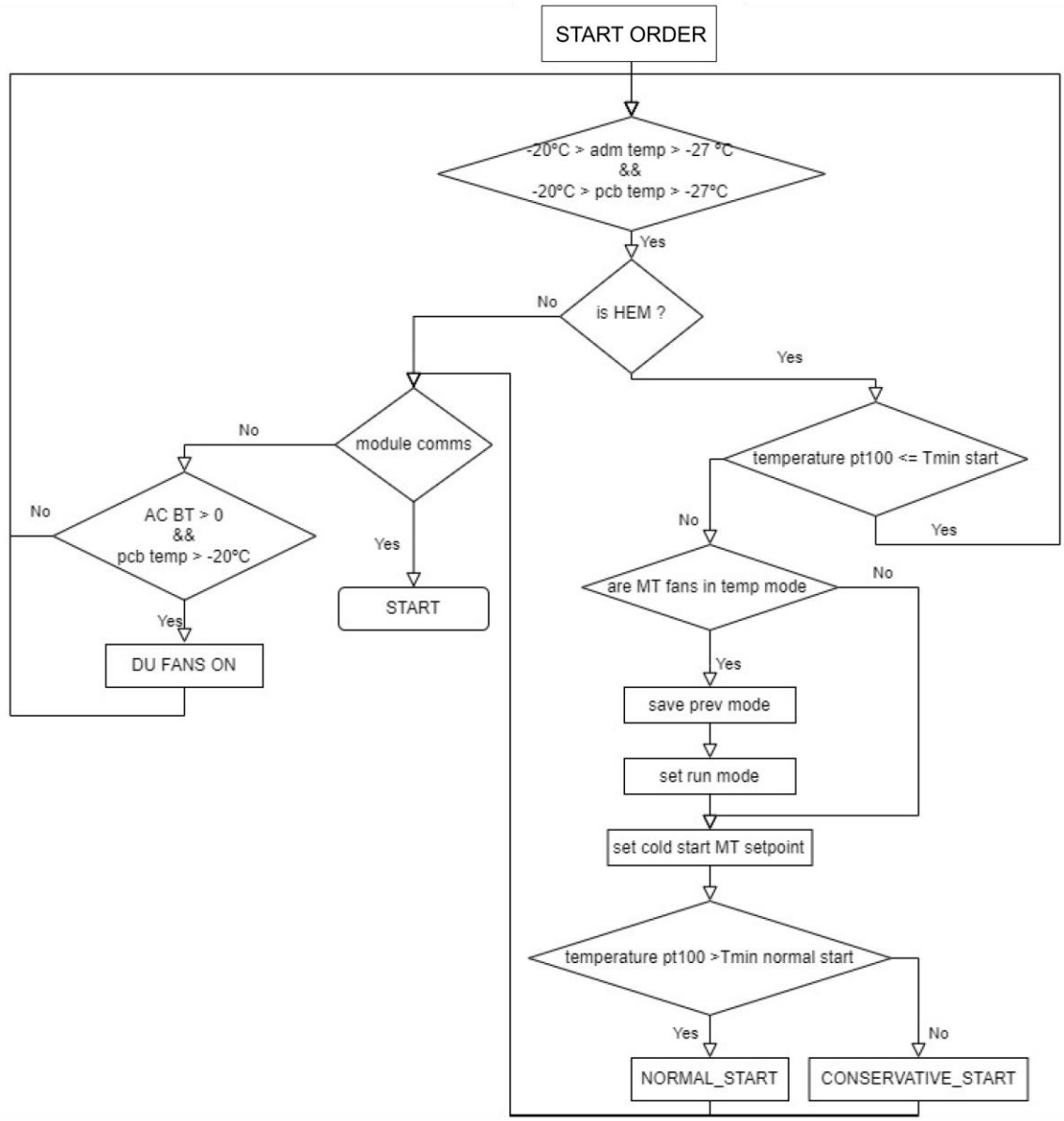
The next sequence is handled when the conditions of cold start are met: If no communication with the power stage can be established, but there is AC low voltage, the ventilation of DC cabinet starts being the speed setpoint as configured in parameter [G4.15.2 -DU fans speed].

At this point, it is necessary to differentiate if the inverter is type “M” or “K”. If it is type “M”, before checking the communication with the power stage, it is necessary to oversee the status of the MV zone. That is, if one or more of the MV transformer PT100 measure more than the threshold configured in [G4.15.5-Min start temp] and less than the threshold configured in [G4.15.6-Min normal start temp], then it is necessary to carry out a conservative start limiting the apparent power to the configured in [G4.15.4-S power limit].

When the temperature reaches a higher value above the configured in the [G4.15.6-Min normal start temp], then the system can carry out a normal start.

If the temperature measured in the MV transformer zone is less than the configured in the [G4.15.5-Min start temp], then the fans of this zone should start. The speed setpoint is the configured in [G4.15.7-MV fans speed].

The next diagram sums up the sequence implemented regarding the cold start.



Sequence applied during cold start

Note: Parameters in this subgroup can only be seen and edited by Power Electronics staff o Developer login roles.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.15.1-DU fans temp = -20 ºC	-35 to 50 ºC	Defines the temperature at which the fans from the DC cabinet will start running in case no communication with the power stage.	No
G4.15.2-DU fans speed = 30 %	0 to 100 %	Defines the speed setpoint of the fans from the DC cabinet in case there is no communication with the power stage.	No
G4.15.3-DU fans stop delay = 3600 s	0 to 3600 s	Configures the fans from the DC cabinet to stop delay at the end of cold start.	No
G4.15.4-S power limit = 50 %	0 to 100 %	Defines the power limit when the inverter starts being the temperature below -25ºC (measured by the PT100 of the MV transformer) until the temperature reaches the configured in [G4.15.5-Min start temp]. **Note:** This parameter is displayed if [SG10.1.3-Product type = HEM].	No
G4.15.5-Min start temp = -34 ºC	-34 to 50 ºC	Sets the minimum temperature below which the inverter will ignore the start or connection order. **Note:** This parameter is displayed if [SG10.1.3-Product type = HEM].	No
G4.15.6-Min normal start temp = -10 ºC	-25 to 50 ºC	Sets the minimum temperature above which the inverter will start in normal operation. **Note:** This parameter is displayed if [SG10.1.3-Product type = HEM].	No
G4.15.7-MV fans speed = 80 %	0 to 100 %	Defines the speed setpoint of the fans from MV transformer at cold start. **Note:** This parameter is displayed if [SG10.1.3-Product type = HEM].	No

EN

6.4.16. Subgroup G4.16: Standby mode

This group includes the configuration parameters for Standby mode, in which the IGBTs stop switching when the P and Q setpoints are 0%.

Note: this subgroup is displayed if [G1.3-User] ≥ Qualified customer.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.16.1-Activation = Disabled	Disabled Enabled	Enables Standby mode.	Yes
G4.16.2-Activation delay = 1.0 s	0.0 to 300.0 s	Allows the timeout to be set from when the P and Q references at the inverter output are 0% until the IGBTs stop switching. **Note**: the minimum value and default value will be 60.0 s if [G4.16.4-Filter status] = Offline.	No
G4.16.3-Deactivation delay = 0.1 s	0.0 to 30.0 s	Allows the timeout to exit Standby mode to be set.	No
G4.16.4-Filter status = Online	Offline Online	Allows the status of filter C during Standby mode to be configured. **OPTION** / **DESCRIPTION** Offline: Filter C is disconnected from the grid during Standby mode. Online: Filter C remains connected to the grid during Standby mode.	No
G4.16.5-Fans = Yes	No Yes	Whenever int]. Allows the inverter ventilation to be configured based on the behavior of the selected filter when Standby mode is activated. **Note:** this parameter is only shown if [G1.3-User] ≥ Power Electronics staff and if [G4.16.4-Filter status] = Online.	No

6.4.17. Subgroup G4.17: P(ROCOF)

This subgroup allows to configure the control algorithm that enables the inverter to respond to rapid changes in grid frequency (ROCOF), in accordance with the requirements of the Lithuanian grid code. The response is achieved through dynamic adjustments to active power (P), providing a frequency support function based on synthetic inertia.

Note: this subgroup is displayed if [G1.3-User] ≥ Qualified customer. The parameters from [G4.17.2-ROCOF db neg (Hz/s)] to [G4.17.11-En Hysteresis] are displayed if [G4.17.1-Enable] = Yes.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G4.17.1-Enable = No	No Yes	This option allows to enable or disable the ROCOF-based active power control algorithm. When enabled, the inverter can adjust the active power based on the rate of change of the grid frequency.	Yes
G4.17.2-ROCOF db neg (Hz/s) = -0.25	-5.00 to 0.00	This allows to set the negative ROCOF value at which the inverter exits the dead band and begins to apply active power modulation in response to a rapid drop in grid frequency.	Yes
G4.17.3-ROCOF P max (Hz/s) = -0.75	-10.00 to 0.00	This allows to set the negative ROCOF value at which the inverter reaches the maximum positive active power variation. Above this value, the power variation remains constant within the limit defined by parameter [G4.17.4-ROCOF dP max].	Yes
G4.17.4-ROCOF dP max = 100 %	0 to 200 %	This allows to set the maximum positive deviation in active power that the ROCOF algorithm can apply relative to the power setpoint.	Yes
G4.17.5-ROCOF db pos (Hz/s) = 0.25	0.00 to 5.00	This allows to set the positive ROCOF value at which the inverter exits the dead band and begins to apply active power modulation in response to a rapid increase in grid frequency.	Yes
G4.17.6-ROCOF P min (Hz/s) = 0.75	0.00 to 10.00	This allows to set the positive ROCOF value at which the inverter reaches the maximum negative active power variation. Above this value, the power variation remains constant at the limit defined by parameter [G4.17.7-ROCOF dP min].	Yes
G4.17.7-ROCOF dP min = -100 %	-200 to 0 %	This setting allows to configure the maximum negative deviation in active power that the ROCOF algorithm can apply relative to the power setpoint.	Yes
G4.17.8-Freezing time dP = 10 s	0 to 100 s	Once the ROCOF algorithm is activated, this parameter allows to set the duration during which the inverter maintains the maximum active power variation at a constant level, regardless of subsequent changes in the ROCOF.	Yes
G4.17.9-Test mode frequency = Disabled Hz	0 to 100 Hz; Disabled = 101 Hz	This allows to set the input frequency used by the algorithm when test mode is enabled, enabling verification of the ROCOF control's behavior without relying on the actual grid frequency.	Yes
G4.17.10-Test mod grad(Hz/s) = 1.00	0.00 to 10.00	This allows to define the frequency variation gradient applied in test mode, which is used to simulate a controlled rate of frequency change (ROCOF).	Yes
G4.17.11-En Hysteresis = 0.10 s	Disabled = 0.00; 0.01 to 2.00 s	It allows to set a time hysteresis to prevent repetitive re-entrancy of the ROCOF algorithm, ensuring stability in the case of minor fluctuations in the rate of frequency change.	Yes

6.5. Group G5: Protections

This group provides access to all the parameters for configuring, enabling or disabling, and/or setting limits for the various protections available, as well as defining the behavior of all the inverter's faults and warnings.

The high and low voltage protections (G5.X) save the values of the low measured AC voltage or the one estimated in the POI according to the value set in parameter [SG1.2.26-max time cons LVRT].

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Except for the parameters [G5.1.x.13-Open AC Circ. Break], [G5.17.3-Enable Q limit DS] and [G5.17.4-Input grad. Q limit] and the subgroup [G5.12: GFDI], only users with a role higher than Admin (qualified customer) can modify these parameters.

6.5.1. Subgroup G5.1: Voltage frequency

G5.1.1-Low AC voltage

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G5.1.1.1-Enable = XXXX0</td><td>00000 to XXXXX</td><td rowspan="11">There are five low voltage protection levels. The following table shows, from right to left, the bits associated with each protection.
<table><tr><th>BIT</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>Very fast protection 2.</td></tr><tr><td>1</td><td>Very fast protection.</td></tr><tr><td>2</td><td>Fast protection 2.</td></tr><tr><td>3</td><td>Fast protection.</td></tr><tr><td>4</td><td>Slow protection.</td></tr></table>To enable each protection, set the corresponding bit and adjust the voltage values and delay as desired (parameters [G5.1.1.2-Slow protection] to [G5.1.1.11-Dly vry fst 2 pro]).
• Protection levels can be set from 0.0 % up to 100.0 %. 100.0 % corresponds with the rated voltage[2].
• The slow protection delay can be set from 0.0 s up to 6550.0 s.
• Fast protections delay can be set from 0.00 s up to 655.00 s.
• Very fast protections delay range varies between 0.000 and 65.500 s.
It is possible to enable/disable each protection level individually in [G5.1.1.1-Enable]. Default values, on the other hand, depend on the selected normative.</td><td>Yes</td></tr>
<tr><td>G5.1.1.2-Slow protection = 85.0 %</td><td>0.0 to 100.0 %</td><td>Yes</td></tr>
<tr><td>G5.1.1.3-Delay slow prot = 10.0 s</td><td>0.0 to 6550.0 s</td><td>Yes</td></tr>
<tr><td>G5.1.1.4-Fast protection = 65.0 %</td><td>0.0 to 100.0 %</td><td>Yes</td></tr>
<tr><td>G5.1.1.5-Delay fast prot = 5.00 s</td><td>0.00 to 655.00 s</td><td>Yes</td></tr>
<tr><td>G5.1.1.6-Fast 2 protection = 45.0 %</td><td>0.0 to 100.0 %</td><td>Yes</td></tr>
<tr><td>G5.1.1.7-Delay fast 2 prot = 2.00 s</td><td>0.00 to 655.00 s</td><td>Yes</td></tr>
<tr><td>G5.1.1.8-Very fast protection = 20.0 %</td><td>0.0 to 100.0 %</td><td>Yes</td></tr>
<tr><td>G5.1.1.9-Delay very fast prot = 1.000 s</td><td>0.000 to 65.000 s</td><td>Yes</td></tr>
<tr><td>G5.1.1.10-Very fast 2 prot = 0.0 %</td><td>0.0 to 100.0 %</td><td>Yes</td></tr>
<tr><td>G5.1.1.11-Dly vry fst 2 pro = 1.000 s</td><td>0.000 to 65.000 s</td><td>Yes</td></tr>
<tr><td>G5.1.1.1b-Interpolation = No</td><td>No
Yes</td><td>Whenever interpolation of low voltage protections is enabled ([G5.1.1.1b] = YES), the inverter will calculate, in real time, the time to trigger protections depending on the current voltage and the curves/points configured in parameters [G5.1.1.2] to [G5.1.1.11].
To enable interpolation of high voltage protections, go to parameter [G5.1.2.1b].
Note: this parameter only applies for Egypt grid code and it is only shown if [G1.3-User] ≥ Power Electronics staff.</td><td>Yes</td></tr>
<tr><td>G5.1.1.12-Enable HW protec = Yes</td><td>No
Yes</td><td>Enable HW undervoltage protection.
Note: this parameter is only shown if [G1.3-User] ≥ Power Electronics staff.</td><td>Yes</td></tr>
</table>

[2] The ranges indicated correspond to the software of the inverter. The validated hardware ranges must be respected in the process of parameterisation of the inverter. Consult Power Electronics for more information.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.1.13-Open AC Circ. Break = Disabled	Disabled Enabled	Allows to configure the opening of the AC circuit breaker in the event of an AC undervoltage fault. **Note:** this parameter is only shown for HEMK inverters ([SG10.1.3-Tipo product] = HEMK).	Yes

G5.1.2- High AC voltage

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.2.1-Enable = XX000	00000 to XXXXX	There are five high voltage protection levels. The following table shows, from right to left, the bits associated with each protection. BIT 0 – Very fast protection 2. BIT 1 – Very fast protection. BIT 2 – Fast protection 2. BIT 3 – Fast protection. BIT 4 – Slow protection. To enable each protection, set the corresponding bit and adjust the voltage values and delay as desired (parameters [G5.1.2.1-Enable] to [G5.1.2.11-Dly vry fast 2)]. • Protection levels can be set from 100.0 % up to 140.0%. 100.0 % corresponds with the rated voltage[3]. • The slow protection delay can be set from 0.0 s up to 6550.0 s. • Fast protections delay can be set from 0.00 s up to 655.00 s. • Very fast protections delay range varies between 0.000 and 65.500 s. It is possible to enable/disable each protection level individually in [G5.1.2.1-Enable]. Default values, on the other hand, depend on the selected normative.	Yes
G5.1.2.2-Slow protection = 115.0 %	100.0 to 140.0 %		Yes
G5.1.2.3-Delay slow prot = 2.0 s	0.0 to 6550.0 s		Yes
G5.1.2.4-Fast protection = 125.0 %	100.0 to 140.0 %		Yes
G5.1.2.5-Delay fast prot = 0.10 s	0.00 to 655.00 s		Yes
G5.1.2.6-Fast 2 protection = 140.0 %	100.0 to 140.0 %		Yes
G5.1.2.7-Delay fast 2 prot = 0.10 s	0.00 to 655.00 s		Yes
G5.1.2.8-Very fast protection = 140.0 %	100.0 to 140.0 %		Yes
G5.1.2.9-Delay very fast prot = 0.100 s	0.000 to 65.500 s		Yes
G5.1.2.10-Very fast 2 prot = 140.0 %	100.0 to 140.0 %		Yes
G5.1.2.11-Dly vry fast 2 prot = 0.100 s	0.000 to 65.500 s		Yes
G5.1.2.1b-Interpolation = No	No Yes	Whenever interpolation of high voltage protections is enabled ([G5.1.2.1b] = YES), the inverter will calculate, in real time, the time to trigger protections depending on the current voltage and the curves/points configured in parameters [G5.1.2.2] to [G5.1.2.11]. To enable interpolation of low voltage protections, go to parameter [G5.1.1.1b]. **Note**: This parameter only applies for Egypt grid code and it is only shown if [G1.3-User] ≥ Power Electronics staff.	Yes
G5.1.2.12-Enable HW protec = Yes	No Yes	Enable HW overvoltage protection. **Note:** this parameter is only shown if [G1.3-User] ≥ Power Electronics staff.	Yes
G5.1.2.13-Open AC Circ. Break = Disabled	Disabled Enabled	Allows to configure the opening of the AC circuit breaker in the event of an AC overvoltage fault. **Note:** this parameter is only shown for HEMK inverters ([SG10.1.3-Tipo product] = HEMK).	Yes

[3] The ranges indicated correspond to the software of the inverter. The validated hardware ranges must be respected in the process of parameterisation of the inverter. Consult Power Electronics for more information.

G5.1.3-Low AC freq

Note: these parameters can only be modified by users with role higher than Admin, except for [G5.1.3.12-Open AC Circ. Break].

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G5.1.3.1-Enable = XXX00</td><td>00000 to XXXXX</td><td rowspan="11">There are five low frequency protection levels. The following table shows, from right to left, the bits associated with each protection.
<table><tr><th>BIT</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>Very fast protection 2.</td></tr><tr><td>1</td><td>Very fast protection.</td></tr><tr><td>2</td><td>Fast protection.</td></tr><tr><td>3</td><td>Slow protection 2.</td></tr><tr><td>4</td><td>Slow protection.</td></tr></table>To enable each protection, set the corresponding bit and adjust the frequency values and delay as desired (parameters [G5.1.3.1-Enable] to [G5.1.3.11-Dly vry fast 2]).
• Protection levels can be set from 45.0 Hz up to 60.0 Hz[4].
• The slow protection delay can be set from 0.0 s up to 6550.0 s.
• Fast protections delay can be set from 0.00 s up to 655.00 s.
• Very fast protections delay range varies between 0.000 and 65.500 s.
It is possible to enable/disable each protection level individually in [G5.1.3.1-Enable]. Default values, on the other hand, depend on the selected normative.</td><td>Yes</td></tr>
<tr><td>G5.1.3.2-Slow protection = 58.0 Hz</td><td>45.0 to 60.0 Hz</td><td>Yes</td></tr>
<tr><td>G5.1.3.3-Delay slow prot = 300.0 s</td><td>0.0 to 6550.0 s</td><td>Yes</td></tr>
<tr><td>G5.1.3.4-Slow 2 protection = 56.5 Hz</td><td>45.0 to 60.0 Hz</td><td>Yes</td></tr>
<tr><td>G5.1.3.5-Delay slow 2 prot = 1.0 s</td><td>0.0 to 6550.0 s</td><td>Yes</td></tr>
<tr><td>G5.1.3.6-Fast protection = 55.0 Hz</td><td>45.0 to 60.0 Hz</td><td>Yes</td></tr>
<tr><td>G5.1.3.7-Delay fast prot = 0.10 s</td><td>0.00 to 655.00 s</td><td>Yes</td></tr>
<tr><td>G5.1.3.8-Very fast protection = 45.0 Hz</td><td>45.0 to 60.0 Hz</td><td>Yes</td></tr>
<tr><td>G5.1.3.9-Delay very fast prot = 0.100 s</td><td>0.000 to 65.500 s</td><td>Yes</td></tr>
<tr><td>G5.1.3.10-Very fast 2 prot = 45.0 Hz</td><td>45.0 to 60.0 Hz</td><td>Yes</td></tr>
<tr><td>G5.1.3.11-Dly vry fast 2 prot = 0.100 s</td><td>0.000 to 65.500 s</td><td>Yes</td></tr>
<tr><td>G5.1.3.12-Open AC Circ. Break = Disabled</td><td>Disabled
Enabled</td><td>Allows to configure the opening of the AC circuit breaker in the event of an underfrequency fault.
Note: this parameter is only shown for HEMK inverters ([SG10.1.3-Tipo product] = HEMK).</td><td>Yes</td></tr>
</table>

EN

[4] The ranges indicated correspond to the software of the inverter. The validated hardware ranges must be respected in the process of parameterisation of the inverter. Consult Power Electronics for more information.

G5.1.4-High AC freq

Note: these parameters can only be modified by users with role higher than Admin, except for [G5.1.4.12-Open AC Circ. Break].

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G5.1.4.1-Enable = XXX00</td><td>00000 to XXXXX</td><td rowspan="11">There are five high frequency protection levels. The following table shows, from right to left, the bits associated with each protection.
<table><tr><th>BIT</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>Very fast protection 2.</td></tr><tr><td>1</td><td>Very fast protection.</td></tr><tr><td>2</td><td>Fast protection.</td></tr><tr><td>3</td><td>Slow protection 2.</td></tr><tr><td>4</td><td>Slow protection.</td></tr></table>To enable each protection, set the corresponding bit and adjust the frequency values and delay as desired (parameters [G5.1.4.1-Enable] to [G5.1.4.11-Dly vry fast 2]).
• Protection levels can be set from 50.0 Hz up to 67.0 Hz[5].
• Slow protection delays can be set from 0. 0s up to 6550.0 s.
• The fast protection delay can be set from 0.00 s up to 655.00 s.
• Very fast protections delay range varies between 0.000 and 65.500 s.
It is possible to enable/disable each protection level individually in [G5.1.4.1-Enable]. Default values, on the other hand, depend on the selected normative.</td><td>Yes</td></tr>
<tr><td>G5.1.4.2-Slow protection = 62.0 Hz</td><td>50.0 to 67.0 Hz</td><td>Yes</td></tr>
<tr><td>G5.1.4.3-Delay slow prot = 300.0 s</td><td>0.0 to 6550.0 s</td><td>Yes</td></tr>
<tr><td>G5.1.4.4-Slow 2 protection = 63.5 Hz</td><td>50.0 to 67.0 Hz</td><td>Yes</td></tr>
<tr><td>G5.1.4.5-Delay slow 2 prot = 1.0 s</td><td>0.00 to 655.00 s</td><td>Yes</td></tr>
<tr><td>G5.1.4.6-Fast protection = 65.0 Hz</td><td>50.0 to 67.0 Hz</td><td>Yes</td></tr>
<tr><td>G5.1.4.7-Delay fast prot = 0.10 s</td><td>0.00 to 655.00 s</td><td>Yes</td></tr>
<tr><td>G5.1.4.8-Very fast protection = 67.0 Hz</td><td>50.0 to 67.0 Hz</td><td>Yes</td></tr>
<tr><td>G5.1.4.9-Delay very fast prot = 0.100 s</td><td>0.000 to 65.500 s</td><td>Yes</td></tr>
<tr><td>G5.1.4.10-Very fast 2 prot = 67.0 Hz</td><td>50.0 to 67.0 Hz</td><td>Yes</td></tr>
<tr><td>G5.1.4.11-Dly vry fast 2 prot = 0.100 s</td><td>0.000 to 65.500 s</td><td>Yes</td></tr>
<tr><td>G5.1.4.12-Open AC Circ. Break = Disabled</td><td>Disabled
Enabled</td><td>Allows to configure the opening of the AC circuit breaker in the event of an overfrequency fault.
Note: this parameter is only shown for HEMK inverters ([SG10.1.3-Tipo product] = HEMK).</td><td>Yes</td></tr>
</table>

[5] The ranges indicated correspond to the software of the inverter. The validated hardware ranges must be respected in the process of parameterisation of the inverter. Consult Power Electronics for more information.

G5.1.4b- Volt/freq Combined

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.4b-Enable combo protec = 0000	0000 to XXXX	Enables/disables combined high-frequency/high-voltage protection. BIT 0: Enable Hf-Hv. BIT 1: Enable Hf-Lv. BIT 2: Enable Lf-Lv. BIT 3: Enable Lf-Lv. A frequency value and voltage limits are configured. If the inverter exceeds these limits, during the time configured for them, then the inverter will go into fault [F190: Combined F-V].	Yes
G5.1.4b.1.2-Comb V/F Hf-Hv V1 = 120.0 %	100.0 to 140.0 %	If the voltage is above V1 value and the frequency is above f1 for the time t1, the fault [F190: Combined F-V] will be triggered.	Yes
G5.1.4b.1.3-Comb V/F Hf-Hv f1 = 62.00 Hz	50.00 to 67.00 Hz		Yes
G5.1.4b.1.4-Comb V/F Hf-Hv t1 = 1.0 s	0.0 to 1800.0 s		Yes
G5.1.4b.2.2-Comb V/F Hf-Lv V2 = 80.0 %	0.0 to 100.0 %	If the voltage is below V2 value and the frequency is above f2 for the time t2, the fault [F190: Combined F-V] will be triggered.	Yes
G5.1.4b.2.3-Comb V/F Hf-Lv f2 = 62.00 Hz	50.00 to 67.00 Hz		Yes
G5.1.4b.2.4-Comb V/F Hf-Lv t2 = 1.0 s	0.0 to 1800.0 s		Yes
G5.1.4b.3.2-Comb V/F Lf-Hv V3 = 120.0 %	100.0 to 140.0 %	If the voltage is above V3 value and the frequency is below f3 for the time t3, the fault [F190: Combined F-V] will be triggered.	Yes
G5.1.4b.3.3-Comb V/F Lf-Hv f3 = 58.00 Hz	45.00 to 60.00 Hz		Yes
G5.1.4b.3.4-Comb V/F Lf-Hv t3 = 1.0 s	0.0 to 1800.0 s		Yes
G5.1.4b.4.2-Comb V/F Lf-Lv V4 = 80.0 %	0.0 to 100.0 %	If the voltage is below V4 value and the frequency is below f4 for the time t4, the fault [F190: Combined F-V] will be triggered.	Yes
G5.1.4b.4.3-Comb V/F Lf-Lv f4 = 58.00 Hz	45.00 to 60.00 Hz		Yes
G5.1.4b.4.4-Comb V/F Lf-Lv t4 = 1.0 s	0.0 to 1800.0 s		Yes

EN

G5.1.5-Protect V trigger

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.5-Protect V trigger = Worst	Worst All	Allows defining when voltage protections must be enabled. It applies both to the preexisting protections and the interpolation protections. **OPTION** / **DESCRIPTION** Worst: Whenever voltage in any phase (RS, ST or TR) overcomes the threshold, the protection will be triggered: - High voltage faults are checked against the maximum voltage. - Low voltage faults are checked against the minimum voltage. All: The protection will be triggered when voltage in all phases (RS, ST and TR) has overcome the threshold: - High voltage faults are checked against the minimum voltage. - Low voltage faults are checked against the maximum voltage.	Yes

G5.1.6-Protect V location

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.6-Protect V location = Inverter	Inverter POI Both	Allows determining where the measure for voltage protections (overvoltage and low voltages) will be taken in the medium voltage side. **OPTION** / **DESCRIPTION** Inverter: Voltages are measured at the inverter terminals. POI: Voltages are estimated/calculated from inverter voltage, current and line impedance. Both: Any voltage that overcomes the limits will trigger the protections	Yes

G5.1.7- Transient high Vac

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.7.1- Enable = 000	000 to XXX	Allows protection to be enabled with the time voltages directly for times of less than one mains cycle. **BIT** / **STATUS** 0: Enable protection 1. 1: Enable protection 2. 2: Enable protection 3.	No
G5.1.7.2- Curve Point 1 = 130.0 %	100.0 to 200.0 %	Set first point 1 of transient Vac voltage curve.	No
G5.1.7.3- Delay 1 = 0.500 ms	0.000 to 2.000 ms	Delay for the first point 1 on the transient Vac voltage curve.	No
G5.1.7.4- Curve Point 2 = 140.0 %	100.0 to 200.0 %	Set first point 2 of transient Vac voltage curve.	No
G5.1.7.5- Delay 2 = 0.200 ms	0.000 to 2.000 ms	Delay for the first point 2 on the transient Vac voltage curve.	No

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.7.6- Curve Point 3 = 150.0 %	100.0 to 200.0 %	Set first point 3 of transient Vac voltage curve.	No
G5.1.7.7- Delay 3 = 0.100 ms	0.000 to 2.000 ms	Delay for the first point 3 on the transient Vac voltage curve.	No

G5.1.8- AC Disconnection

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.8.1-Enable = Enabled	Disabled Enabled	Allows enable disable disconnection protection.	No
G5.1.8.2-Threshold = 60.00 %	0.00 to 100.00 %	Threshold at which the disconnection condition starts counting.	No
G5.1.8.3-Delay = 80 ms	0 to 1000 ms	Indicates the delay to AC disconnection if the conditions are met.	No

G5.1.9-Frequency in VRT

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.1.9.1-Recovery time = 0 ms	0 to 500 ms	Allows to configure the time of the reactivation of frequency protection after VRT event. During this event, frequency protections are managed using time delays to prevent unintended disconnections. Once the configured time has elapsed, the protections return to normal operation in accordance with the connection requirements established by MISO standard.	Yes

EN

6.5.2. Subgroup G5.2: Quality

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

G5.2.1 – Unbalanced

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.2.1.1-Enable =XX	00 to XX	Enables current/voltage unbalance. BIT 0: Current unbalance. BIT 1: Voltage unbalance.	Yes
G5.2.1.2-Unbalance I = 30 %	1 to 100 %	Current unbalance threshold.	Yes
G5.2.1.3-Unbalance I min = 30 %	0 to 100 %	Adjusts minimal current to apply protection.	Yes
G5.2.1.4-Delay unbalance I = 5 s	1 to 60 s	Delay to report current unbalance.	Yes
G5.2.1.5-Unbalance V = 25 %	1 to 100 %	Voltage unbalance threshold.	Yes
G5.2.1.6-Delay unbalance V = 5 s	1 to 60 s	Delay to report voltage unbalance.	Yes
G5.2.1.7-Module Homopolar I = Disabled	Disabled = 0; 1 to 100 %	Current homopolar threshold for FPGA.	Yes
G5.2.1.8-DU I unbal threshold = 25 %	Disabled = 4; 5 to 100 %	DU current unbalance threshold.	Yes
G5.2.1.9-Unbalance Halfbus Mo = 200 V	0 to 301 V	Enables the protection of half bus power stage unbalance voltage.	Yes
G5.2.1.10-Delay Unbal Halfbus = 10 s	0 to 60 s	Indicates the delay in the unbalance modules.	Yes

G5.2.2 – THDi Protection

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.2.2.1-THDi Prot Config = Warning	Disabled Warning Fault	Allows to configure the THDi protection.	Yes
G5.2.2.2-THDi Threshold = 5.0 %	0.0 to 100.0 %	Defines the threshold for harmonic distortion (THDi).	Yes

G5.2.3 – CTs

This subgroup allows configuring the fault due to non-functional CTs.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.2.3.1-Enable CTs prot = Warning	Disabled Warning Fault	Allows to configure the behavior of the non-functional CTs fault. OPT. / MODE / DESCRIPTION 0 / Disabled / No action shall be taken. 1 / Warning / If the event occurs. 2 / Fault / If the event occurs. **Note:** The default value depends on the value configured in the parameter [SG10.1.19-Visma enable]: SG10.1.19 / Default value of G5.2.3.1 Disabled / Warning Enabled (*) / Fault (*) In this case, the only option available will be "Fault" (the other options will not be shown).	Yes
G5.2.3.2-Unbalance threshold = 25.0 %	0.0 to 100.0 %	Allows setting the level of the fault due to non-functional CTs.	Yes

EN

6.5.3. Subgroup G5.3: Anti islanding

An anti-island condition occurs when the inverter keeps injecting on the grid although it does not have voltage because it comes into resonance with some load. This is dangerous because all the grids should be without current but there is a section (island) where current is flowing without being controlled.

Currently, there are two modes: active and passive.

In active mode, the inverter causes a disturbance in the grid (slightly distorts the current waveform). If the inverter is connected to the grid frequency cannot vary, but, if there is an island, the frequency will vary due to distortion. Parameters used in this mode are G5.3.2 to G5.3.4.

In passive mode, the inverter only checks frequency variation during a time lapse. If the frequency variation ($\Delta freq$) overcomes a threshold [G5.3.5] during the time lapse set in [G5.3.6], the anti-island protection will be triggered.

$$\Delta freq = current\ frequency - previous\ frequency$$

Simultaneous mode is a combination of active and passive modes.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.3.1-Mode = Off	Off Active Passive Simultaneous	Enables to select the anti-islanding modes: OPTION / DESCRIPTION Off / Disable anti-islanding mode. Active / Enable anti-islanding mode. Passive / Enable ROCOF anti-islanding. Simultaneous / Active and passive modes operate at the same time.	Yes
G5.3.2-Drift = 0.00 %	0.00 to 5.00 %	Constant drift added in the frequency. **Note:** Parameter available if [G5.3.1 Mode] = Active or Simultaneous.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.3.3-K error = 0.5 %	0.0 to 50.0 %	Drift added depending on the frequency error differential. **Note:** Parameter available if [G5.3.1 Mode] = Active or Simultaneous.	Yes
G5.3.4-Fault level (º) = 5	1 to 60	Threshold for triggering the anti-islanding protection. **Note:** Parameter available if [G5.3.1 Mode] = Active or Simultaneous.	Yes
G5.3.5-ROCOF (Hz/s) = 4.0	1.0 to 50.0	Constant drift added in the frequency. **Note:** Parameter available if [G5.3.1 Mode] = Passive or Simultaneous.	Yes
G5.3.5d-ROCOF fine (Hz/s) = 0.0	0.0 to 2.0	If the value set in this parameter is overcome, anti-island will be activated. **Note:** Parameter available if [G5.3.1 Mode] = Passive or Simultaneous.	Yes
G5.3.6-Fault delay = 250 ms	150 to 1000 ms	Configure a delay before triggering an anti-islanding fault when ROCOF mode is active. **Note:** Parameter available if [G5.3.1 Mode] = Passive or Simultaneous.	Yes
G5.3.6d-ROCOF delay fine = 0 ms	0 to 1000 ms	Passive anti islanding delay correction. **Note:** Parameter available if [G5.3.1 Mode] = Passive or Simultaneous.	Yes

6.5.4. Subgroup G5.4: Temperature

Parameters to configure the temperature protections for IGBTs, AC module and the medium voltage transformer. Please notice protections for the transformer (G5.4.3.2-MVTransformrThreshld to G5.4.3.5-MVTransTcriticThld) only apply to HEM inverters.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

G5.4.1- Power stage

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.1.1-IGBTs T fault enable = Enabled	Disabled Enabled	Allows to enable / disable the IGBTs temperature protection.	Yes
G5.4.1.2-IGBTs temp thres flt = [SG10.2.2-Igbts disc temp thre] ºC	20 to 150 ºC	Sets the IGBT temperature protection threshold applied when the inverter is in discharge.	Yes
G5.4.1.3-IGBTsTempHysteresis = 10 ºC	0 to 50 ºC	Configures the hysteresis of the IGBT temperature, below which the fault is cleared.	Yes

G5.4.2- PCB

G5.4.2.1-Module

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.2.1.1-Mod T fault enable = Enabled	Disabled Enabled	Allows to enable / disable the internal module main control board temperature protection.	Yes
G5.4.2.1.2-ModuleTempThresFlt = 75 ºC	20 to 87 ºC	Sets the module main control board protection threshold.	Yes
G5.4.2.1.3-ModulTempHysteresi = 10 ºC	0 to 50 ºC	Configures the hysteresis of the module main control board temperature, below which the fault is cleared.	Yes

G5.4.2.2-Central

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.2.2.1-Main control board = Disabled	20 to 85 ºC; Disabled = 86 ºC	Temperature protection threshold for the P2G control board. When the threshold is exceeded, Fault 40 is displayed.	Yes
G5.4.2.2.2-CentTempHysteresis = 10 ºC	0 to 50 ºC	Configures the hysteresis of the control central board temperature, below which the fault is cleared.	Yes

G5.4.2.3-DU

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.2.3.1-DU tem fault thres = 70 ºC	20 to 85 ºC; Disabled = 86 ºC	Sets the temperature threshold for DC cabinet board temperature protection. If the temperature measured by the DC cabinet board is higher than this threshold, F159: Overtemperature DU, will be activated. The default value depends on the inverter type configured in the parameter [SG10.1.4-Inverter type]. SG10.1.4 / Default value of G5.4.2.3.1: Solar inverter: 70 ºC Basic BUSPLUS: 70 ºC Advanced BUSPLUS: 70 ºC PCS/MPCS: 80 ºC	Yes
G5.4.2.3.2-DU temp hys = 10 ºC	0 to 50 ºC	Sets the temperature hysteresis for DC cabinet board temperature protection. If the temperature of the board measured by the DC cabinet is below the threshold configured in [G5.4.2.3.1-DU tem fault thres] minus the hysteresis configured in this parameter, then the [F159: Overtemperature DU] will be reset.	Yes

EN

G5.4.3- Transformer

The MV transformer from the HEM inverter has temperature protection which is based in temperature probes located inside the transformer's columns (windings). Additionally, this protection is allowed when inverter is a HEMK type and it has a connected PT-100 probe within MV Skid transformer. In this case the inverter will have an associated MV switchgear that can be motorized or not.

Note: These parameters are only displayed if [SG10.1.3-Product type = HEM] or ([SG10.1.3-Product type = HEMK] and [SG10.7.3-MV Skid ≥ Yes, Not Motorized] and [SG10.7.5-Skid MV Transf PT100 = Yes]).

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G5.4.3.1-Trafo T fault enable = Disabled</td><td>Disabled
Enabled</td><td>Allows to enable / disable the MV transformer temperature protection.
Note: this parameter is read-only (RO).</td><td>Yes</td></tr>
<tr><td>G5.4.3.2-MVTransformrThreshld = 105 ºC</td><td>120 to 180 ºC</td><td>Sets the overtemperature protection threshold for the MV transformer. Depending on the product settings this threshold will have a different value; for the HEMK inverter, this depends on the configuration of parameter [SG10.7.9-Cooling Type]:
<table>
<tr><th>SG10.1.3</th><th>SG10.7.9</th><th>G5.4.3.2 default value</th></tr>
<tr><td>HEM</td><td>-</td><td>120 ºC</td></tr>
<tr><td rowspan="2">HEMK</td><td>ONAN</td><td>100 ºC</td></tr>
<tr><td>KNAN</td><td>110 ºC</td></tr>
</table></td><td>Yes</td></tr>
<tr><td>G5.4.3.3-MV transformr delay = 120 s</td><td>0 to 300 s</td><td>Sets the delay (in seconds) for the MV transformer overtemperature protection.</td><td>Yes</td></tr>
<tr><td>G5.4.3.4-Trafo crash T enable = Disabled</td><td>Disabled
Enabled</td><td>Allows to enable / disable the MV transformer critical temperature protection.
Note: this parameter is read-only (RO).</td><td>Yes</td></tr>
<tr><td>G5.4.3.5-MVTransTcriticThld = 130 ºC</td><td>130 to 160 ºC</td><td>Defines the minimum temperature at which the fault F208: Critical temperature of the MV transformer will be triggered. Depending on settings this threshold will get different value; for the HEMK inverter, this depends on the configuration of parameter [SG10.7.9-Cooling Type]:
<table>
<tr><th>SG10.1.3</th><th>SG10.7.9</th><th>G5.4.3.5 default value</th></tr>
<tr><td>HEM</td><td>-</td><td>120 ºC</td></tr>
<tr><td rowspan="2">HEMK</td><td>ONAN</td><td>113 ºC</td></tr>
<tr><td>KNAN</td><td>125 ºC</td></tr>
</table></td><td>Yes</td></tr>
<tr><td>G5.4.3.6-MV Temp Hysteresis = 5 ºC</td><td>0 to 50 ºC</td><td>Defines the hysteresis required to clear the fault automatically.</td><td>Yes</td></tr>
<tr><td>G5.4.3.7-MV warning threshold = 90 ºC</td><td>0 to 150 ºC</td><td>Allow to configure the warning threshold of the MV transformer temperature for the HEMK with motorised MV Skid. In other words, when the temperature read by any of the PT100 of the transformer connected to the central control exceeds the threshold configured in this parameter, the warning [W184: MV transformer temperature] will be triggered. The default value depends on the configuration of parameter [SG10.7.9-Cooling Type]:
<table>
<tr><th>SG10.1.3</th><th>SG10.7.9</th><th>G5.4.3.7 default value</th></tr>
<tr><td rowspan="2">HEMK</td><td>ONAN</td><td>90 ºC</td></tr>
<tr><td>KNAN</td><td>100 ºC</td></tr>
</table></td><td>Yes</td></tr>
</table>

G5.4.4- Inductance

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.4.1-Ind T fault enable = Enabled	Enabled	Allows to enable the inductance temperature protection. **Note**: this parameter is read-only (RO).	Yes
G5.4.4.2-Ind Threshold = 140 ºC	120 to 150 ºC	Defines the inductance protection threshold.	Yes
G5.4.4.3-Ind delay = 120 s	0 to 300 s	Defines the time that the fault waits until it is triggered.	Yes
G5.4.4.4-Ind crash T enable = Enabled	Disabled Enabled	Allows to enable / disable the inductance critical temperature protection. **Note:** this parameter is read-only (RO).	Yes
G5.4.4.5-IndTcriticThld = 150 ºC	130 to 160 ºC	Defines the inductance critical protection threshold.	Yes
G5.4.4.6-Ind Temp Hysteresis = 10 ºC	0 to 50 ºC	Defines the hysteresis required to clear the fault automatically.	Yes

EN

G5.4.5- LC filter

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.5.1-Wrng Temp. Threshold = **68** ºC	40 to [G5.4.5.3-Critical T Threshold] ºC	Adjusts the warning threshold. When the temperature read by the sensor is above this threshold W172 is triggered.	Yes
G5.4.5.2-Temp fault delay = 120 s	0 to 300 s	Sets the delay (in seconds) for the LC filter overtemperature protection.	Yes
G5.4.5.3-Critical T Threshold = 70 ºC	[G5.4.5.1-Wrng Temp. Threshold][6] to 75 ºC	Adjusts the fault threshold. When the temperature read by the sensor is above the configurable threshold during a configurable time [G5.4.5.2-Temp fault delay] F217 is triggered.	Yes
G5.4.5.4-Delta T LC = 25 ºC	0 to 50 ºC; Disabled = 51 ºC;	LC filter overtemperature protection. When the difference between the temperature measured in the cabinet [SV1.6.15-LCL filter Temp] and the ambient temperature measured at the air inlet [SV1.6.4-Admission Tº] is above this threshold F242 is triggered.	No

G5.4.6- DU

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.6.1-Max increase DU temp = 18 ºC	0 to 50 ºC	DU overtemperature protection. When the difference between the temperature measured in one of the modules [SV12.x.3.1- Ambient Temp] and the ambient temperature measured at the air inlet [SV1.6.4-Admission Tº] is above this threshold F241 is triggered. **Note:** from version 4.1.0, F241 is deprecated and it has been replaced by W241.	Yes

[6] The minimum value is 65 °C, unless [G5.4.5.1-Wrng Temp. Threshold] is higher. In this case, [G5.4.5.1-Wrng Temp. Threshold] will be the minimum value.

G5.4.7: Inductance ventilation

This group allows to set the threshold and gain of the ventilation protection for the inductance.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.7.1-Threshold activation = 125 ºC	100 to 150 ºC	Allows to set the temperature threshold measured in the hottest PT100 sensor of the inductance above which the increase in the ventilation setpoint of the module zone is activated.	Yes
G5.4.7.2-Gain [%/ºC] = 2	0 to 10	Allows to set the constant that relates the increase of the fan speed to the temperature increase in the inductance above the threshold.	Yes

G5.4.8-Skid Transf Admission

This group allows configuring the MV SKID transformer intake temperature fault (F187).

Note: this subgroup is displayed if [SG10.7.3-MT Skid] ≠ No, [SG10.1.3-Product Type] = HEMK, and [SG10.7.9-Cooling Type] = KNAN.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.8.1-Transf T fault enabl = Enable	Disabled Enabled	Allows enabling or disabling the reporting of fault F187	Yes
G5.4.8.2-Transf T fault thres = 50 ºC	0 to 80 ºC	Allows configuring the threshold above which, if the intake temperature exceeds it, fault F187 is reported	Yes

G5.4.9-Skid Cabinet

This group allows configuring the MV SKID cabinet overtemperature fault (F186).

Note: this subgroup is displayed if [SG10.7.3-MT Skid] ≠ No and [SG10.1.3-Product Type] = HEMK.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.4.9.1-Cabinet fault enable = Enabled	Disabled Enabled	Allows enabling or disabling the reporting of fault F186. When enabled, if the MV SKID cabinet temperature exceeds 60 °C, digital input J16.6 is activated and fault F186 is reported.	Yes
G5.4.9.2-Cabinet fault reset = 5 min	0 to 30 min	Allows configuring the delay time between the loss of the high level at digital input J16.6 and the automatic reset of fault F186.	Yes

6.5.5. Subgroup G5.5: I/O Protections

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.5.1-SF6 MT cell = Warning	Warning Fault	Sets SF MV cell as warning or fault. **Note:** This parameter is only displayed if [SV1.3.1.10-MV Skid ≠ No] and [SG10.1.3-Product type = HEMK] or [SG10.1.3-Product type = HEM].	No
G5.5.2-Feedback AC breaker = Warning	Warning Fault	Sets feedback breaker AC as warning or fault. **Note:** This parameter is displayed if [SV1.3.1.10-MV Skid ≠ No] and [SG10.1.3-Product type = HEMK].	No

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.5.3-Feedback MT cell = Warning	Warning Fault	Sets feedback MV cell as warning or fault. **Note**: This parameter is only displayed if [SG10.1.3-Product type = HEMK] or [SG10.1.3-Product type = HEM].	No
G5.5.4-Feedback breaker DC = Warning	Warning Fault	Sets feedback breaker DC as warning or fault.	No
G5.5.5-Over VDC Fault Type = Warning	Warning Fault	Sets overvoltage DC as warning or fault.	No
G5.5.6-Over VAC Fault Type = Warning	Warning Fault	Sets overvoltage AC as warning or fault.	No
G5.5.7-Thermal Bus Plus = Warning	Warning Fault Disabled	Sets the type of protection of the Bus Plus thermal circuit breaker (warning or fault). This circuit breaker is a digital input connected to the microcontroller unit (MCU) and is only available when the system has Bus Plus. If it is configured as warning, then the warning "W182: Therma Bus Plus" will be triggered. If it is configured as fault, then the "F234: Thermal Bus Plus", will be triggered. If it is configured as disable, then the protection will not be triggered under any circumstances. **Note:** This parameter is displayed if [SG10.1.4-Inverter type = Advanced BUSPLUS].	No

EN

6.5.6. Subgroup G5.6: DU

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.6.1-DUtotal neg cur = Disabled	0 to 1000 A; Disabled = 1001 A	Allows to protect the system against DC current injection to the PV panels when an unexpected stop of the inverter has taken place in the situation of a system composed by DCDC connected to the Bus Plus.	No
G5.6.2-DUtotal neg cur del = 2 s	0 to 10 s	Defines the delay to trigger the negative current injection protection.	No
G5.6.3-DU overcurrent = Off	Off = 0.0; 0.1 to 600.0 A	Allows setting a current value for the DC channels. If this value is overcome, a fault will be triggered.	Yes
G5.6.4-DU Inv current prot = -350.0 A	Off = -350.1; -350.0 to -19.0 A	In a photovoltaic field sometimes inverse or negative currents can appear; they can be read through the current transformers (in each channel) that are in the DC cabinet (DU). This parameter defines the inverter protections based on these currents. When the negative current of a channel in the DC cabinet exceeds [G5.6.3-DU overcurrent], the inverter stops (but does not open the DU). The protection is activated if there are DC current sensors. **Note**: In the case of PCS, the fault will be disabled.	Yes

6.5.7. Subgroup G5.7: Faults/Warnings config

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

G5.7.1- Faults

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G5.7.1.0-Time to Zero State = 3600 s</td><td>Off = 0;
1 to 10800 s</td><td>Allows the time to attempt to clear all fault counters and set the inverter's auto reset mechanism to zero state only applies if the inverter is not faulted at the time of expiration.</td><td>Yes</td></tr>
<tr><td>G5.7.1.1-Fault selected = F1: Watchdog</td><td>Check complete list in tab Faults</td><td>Allows fault selected to configure.</td><td>Yes</td></tr>
<tr><td>G5.7.1.2-Fault type = Special</td><td>Special
DU + Abrupt stop
DU + Controlled stop
Abrupt stop
Controlled stop</td><td>Allows to select the type of fault in case of special fault, abrupt or controlled stop.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the switch disconnector and isolate the inverter (opens DC and AC low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs.</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G5.7.1.3-Type of clean = Manual</td><td>Manual
Autoreset
By measured</td><td>Allows to configure whether the type of fault clearing selected is manual, auto reset or by measure. The “by measure” option is given by the internal configuration of the fault.
Note: This parameter is displayed if [G5.7.1.2-Fault type ≥ DU+Abrupt stop].</td><td>Yes</td></tr>
<tr><td>G5.7.1.4-Number of retries = 1</td><td>1
2
5
10
20
Infinite</td><td>Allows number of retries of auto reset.
Note: This parameter is displayed if [G5.7.1.2-Fault type ≥ DU+Abrupt stop=2] and [G5.7.1.3-Type of clean ≠ Manual].</td><td>Yes</td></tr>
<tr><td>G5.7.1.5-Time between retries = 1 s</td><td>1
2
5
10
20</td><td>Allows time between retries of auto reset.
Note: This parameter is displayed if [G5.7.1.2-Fault type ≥ DU+Abrupt stop=2] and [G5.7.1.3-Type of clean ≠ Manual].</td><td>Yes</td></tr>
<tr><td>G5.7.1.6-User Manual Reset = Customer</td><td>Customer
Comissioning
Qualified customer
Power Electronics staff
Developer</td><td>User Role that is allowed to perform a reset of the current fault.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Customer</td><td>Resetting the fault can be done by a customer role or higher.</td></tr><tr><td>Comissioning</td><td>Resetting the fault can be done by a commissioning role or higher.</td></tr><tr><td>Qualified customer</td><td>Resetting the fault can be done by a qualified customer role or higher.</td></tr><tr><td>Power Electronics staff</td><td>Resetting the fault can be done by a PEE Staff role or higher.</td></tr><tr><td>Developer</td><td>Resetting the fault can be done by a developer role.</td></tr></table></td><td>Yes</td></tr>
</table>

G5.7.2- Warnings

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.7.2.1-Warning selected = W1:Input high voltage	Check complete list in tab Warnings	Allows to configure the warning modules selected.	Yes
G5.7.2.2-Source = UP	UP DSP Modulo	Sets the warning source.	Yes
G5.7.2.3-Private = No	No Yes	Allows the set if warning is private or not.	Yes

G5.7.3- Clean History

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.7.3.1-Faults = No	No Yes	Allows the clean faults history.	Yes
G5.7.3.2-Warnings = No	No Yes	Allows the clean warning history.	Yes

EN

6.5.8. Subgroup G5.8: Faults module config

The modules have their fault status and depending on how they are configured within this group, they may behave as an inverter fault, so that the inverter will stop. In this case the module fault code will coincide with the inverter fault code.

See section "Module status messages" and “Faults” for more information on the statuses and their codes.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.8.1-Fault selected = F65: Overcurrent AC	Check complete list in tab Faults	Allows to configure the faults modules selected.	Yes
G5.8.2-Fault action = Normal	Normal Lockable Isolated DC Device	Sets the actions. **OPTION** / **DESCRIPTION** Normal: Treat as a non-special ruling unless it is configured as a special ruling. Lockable: Not isolates the module by not opening the DC module DC disconnect switch. A warning will be displayed until the inverter is checked. Isolated DC: Isolates the module by not opening the DC module DC disconnect switch. Device: The fault is escalated to inverter fault and the behavior after the fault will be configured in the inverter fault list from the following: Special fault, DU open fault with ramp stop or DU open fault with abrupt stop.	Yes

6.5.9. Subgroup G5.9: Faults busbar config

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.9.1-Fault selected = F252:RFI OverCur Persistent	Check complete list in tab Faults	Allows to configure the faults busbar selected.	Yes
G5.9.2-Fault action = Normal	Normal Isolated DC Device	Sets the actions. OPTION / DESCRIPTION Normal: Treat as a non-special ruling unless it is configured as a special ruling. Isolated DC: Isolates the module by not opening the DC module DC disconnect switch. Device: The fault is escalated to inverter fault and the behavior after the fault will be configured in the inverter fault list from the following: Special fault, DU open fault with ramp stop or DU open fault with abrupt stop.	Yes

6.5.10. Subgroup G5.10: Internal protections

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.10.3-Unbal bus enable = XX	00 to XX	Enables / disables the protection because of bus unbalance (triggered at 200 V) or half bus maximum voltage (triggered at 800 V). BIT / STATUS 0: Vmax half bus. 1: Bus unbalance.	Yes
G5.10.4-Unbal bus threshold = 275 V	150 to 300 V	Sets the voltage threshold to trigger the respective fault.	Yes
G5.10.5-Delay unbalance bus = 3 s	1 to 10 s	Sets the time that the fault waits until it is triggered. **Note**: this parameter is not available from version 2.2.0.	Yes
G5.10.6-V max half bus = 850 V	300 to 875 V	Sets the voltage threshold to trigger the respective fault.	Yes
G5.10.7-Delay V max half bus = 3 s	1 to 10 s	Sets the time that the fault waits until it is triggered. **Note**: this parameter is not available from version 2.2.0.	Yes
G5.10.8-Unb Pac Pdc threshold = 30 %	10 to 100 %; Off = 101	Sets the power threshold to trigger the respective fault. **Note:** Only users with role Developer can see and modify this parameter.	Yes
G5.10.9-Deriv. Idc Threshold = 100 A	50 to 250 A	Sets the current threshold to trigger the respective fault.	Yes
G5.10.10-Deriv. Iac threshold = 250 A	50 to 500 A	Sets the current threshold to trigger the respective fault.	Yes
G5.10.11-Smoke detector DU = No	No Yes	The default value of this parameter will be used to Low module Vdc protection. When the AC voltage configured in the drive select is lower than 250 V, the protection will be disabled.	Yes
G5.10.13-Thr Activ Flt Idref = 100 A	50 to 400 A	This parameter sets the threshold to activate fault / warning F107 / W107: Iac not reached. When the output current does not reach, as minimum, 50% of the reference current, having overcome the threshold set in G5.10.13, system will display a fault or warning 107.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.10.14-Idc overcurrent = Off A	Off = 9; 10 to 1200 A	Sets the threshold for DC overcurrent detection. If this parameter is enabled, and its value overcome, fault F67: Mod. DC overcurrent will be triggered.	Yes
G5.10.15-Low module Vdc =650 V	Off = 49; 50 to 950 V	The default value of this parameter is used to protect the inverter against voltage changing from fixed to variable depending on the following equation. DEF (G5.10.15) = √(2)*Vac_configured-300V. Fault F109: Mod. Low Vdc will be triggered when these changes occur. When the voltage configured internally is lower than 250 V, the Vdc undervoltage protection will be disabled.	Yes
G5.10.15a-Low mod Vdc High = 50 V	0 to 200 V	Enables / disables the protection because of bus unbalance (triggered at 200 V) or half bus maximum voltage (triggered at 800 V). **Note:** Only users with role Developer can modify this parameter.	Yes
G5.10.15b-Low mod Vdc Low = 100 V	0 to 200 V	Sets the voltage threshold to trigger the respective fault. **Note:** Only users with role Developer can modify this parameter.	Yes
G5.10.16-Low DSP Vdc = 300 V	20 to 500 V	Sets the time that the fault waits until it is triggered.	Yes
G5.10.17-Low DSP Vdc delay = 3.0 ms	1.0 to 100.0 ms; Off = 100.1	Sets the voltage threshold to trigger the respective fault.	Yes
G5.10.18-DU MasterSlave DifT = 5 ºC	0 to 10 ºC	Allow to configure the maximum temperature that can vary from the temperature measured by the master and the temperature measured by the slave (temperature of the board). If both temperatures differ the configured in the [G5.10.18-DU MasterSlave DifT], then [W187: DU Delta Temp Threshold] will be triggered. Note that this warning will not be triggered when the system is a MPCS, because MPCS only includes one DU board. **Note:** This parameter is displayed if [SV1.3.1.1-Inverter type] ≠ PCS/MPCS.	Yes

EN

6.5.11. Subgroup G5.12: GFDI

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.12.0-Reset mode GFDI Fault = Manual	Manual Automatic	Allows to configure manual or automatic reset of fault [F41: GFDI].	Yes
G5.12.1-Reset GFDI Fault = No	No Yes	Allows the customer to launch a GFDI maneuver in DC to reset the GFDI fault. **Note:** this parameter is displayed if [G5.12.0-Reset mode GFDI Fault] = Manual.	Yes
G5.12.2-Remaining Attempts = 2	0 to 65535	Indicates the number of remaining attempts available to try to reset the GFDI fault. **Note:** Only users with a role higher than Admin can modify this parameter.	Yes

6.5.12. Subgroup G5.13: Unbalanced Busbar

Note: This subgroup is displayed if [G1.3-User] ≥ Power Electronics staff.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.13.1-Enable V Unbalance = Enabled	Disabled Enabled	Enables or disables busbar voltage protections.	No
G5.13.2-Busbar V Threshold = 100 V	0 to 110 V	Sets the maximum busbar voltage threshold. **Note**: This parameter is only displayed if [G5.13.1-Enable V Unbalance] = Enabled.	Yes
G5.13.3-RFI I enable = Disabled	Disabled Enabled	Enables or disables the RFI current protection.	No
G5.13.4-RFI-I-meas-method = Measured current	Measured current Calculated current Most restrictive	Enables to configure busbar current protections: OPTION: Measured current — DESCRIPTION: Uses the measured current as a parameter for I RFI protection. OPTION: Calculated current — DESCRIPTION: Uses the calculated current as a parameter for I RFI protection. OPTION: Most restrictive — DESCRIPTION: Uses the most restrictive current (measured or calculated) as a parameter for I RFI protection. **Note**: This parameter is only displayed if [G5.13.3-RFI I enable] = Enabled.	No
G5.13.5-Max RFI I protection = 25 A	0 to 35 A	Sets the maximum RFI current to trigger the protection (F252 triggers). **Note**: This parameter is only displayed if [G5.13.3-RFI I enable] = Enabled.	Yes
G5.13.6-Max Time RFI Fault = 30 s	0 to 60 s	Sets the maximum measured time over which F252 is triggered. **Note**: This parameter is only displayed if [G5.13.3-RFI I enable] = Enabled.	Yes

6.5.13. Subgroup G5.15: Batteries opening

This subgroup allows to fit the protection regarding the premature opening of the batteries. It must be not adjusted by the customer, otherwise “F255: BAT open detection” will be triggered. Applies to PCS / MPCS inverters [SV1.3.1.1-Inverter type = PCS/MPCS].

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.15.1-Vdc High Threshold = 50 V	25 to 100 V	Corresponds to the detection level of the batteries opening because of high voltage. This level is differential with regard to the batteries high voltage stop level.	Yes
G5.15.2-Vdc High Delay = 10 ms	3 to 20 ms	Corresponds to the batteries opening detection delay because of high voltage.	Yes
G5.15.3-Vdc Low Threshold = 50 V	25 to 100 V	Corresponds to the batteries opening detection level because of low voltage. This level is differential with regard to the batteries low voltage stop level.	Yes
G5.15.4-Vdc Low Delay = 10 ms	3 to 20 ms	Corresponds to the batteries opening detection delay because of low voltage.	Yes
G5.15.5-Fast increment Thres = 100 V	25 to 1000 V	Corresponds to a variation in the voltage measurement. If the inverter is in the ON state and a sudden variation in the measurement is detected being above the threshold configured in [G5.15.5- Fast increment Thres], the battery status will be considered as open and will trigger fault F255: BAT open detection. If the inverter is in shutdown process and there is a sudden variation of the voltage measurement being above the threshold configurable in [G5.15.6-Threshold Incr. stopped], the batteries are considered to have been opened in shutdown state and will trigger fault F255: BAT open detection. Parameter [G5.15.7-Stop Incr. Delay] indicates the time in which the bus discharge event occurs.	Yes
G5.15.6-Stop increment Thres = 250 V	25 to 1000 V		Yes
G5.15.7-Delay increment Stop = 25 s	10 to 3000 s		Yes

6.5.14. Subgroup G5.16: LC Filter

This subgroup allows to configure the LCL filter protection.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

G5.16.1-LC Filter Unbal I

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.16.1.1-Protection type = Disabled	Disabled Warning Fault	Configures the type of protection associated with the monitoring of the fundamental frequency current of the grid flowing through the LC filter capacitors during normal operation of the inverters. **OPTION** / **DESCRIPTION** Disabled: This option disables the fault and warning configured. Warning: If an unbalance of currents is detected, and this unbalance exceeds a configurable threshold in [G5.16.1.3-Fault Threshold] during a configurable time in [G5.16.1.4-Fault Delay], F171 or W46 will be activated, depending on how the protection is configured. Fault: (same as Warning)	Yes
G5.16.1.2-Reference Current = Calculated	Calculated Calibrated	Allows to configure the reference current according to the following table: **OPTION** / **DESCRIPTION** Calculated: Calculated reference from the measured values of frequency, Vac and C configured in the Drive Select. Calibrated: Calibrated reference during routine / field testing. **Note:** This option does not apply to this product version.	Yes
G5.16.1.3-Fault Threshold = 25 %	0 to 100 %	Corresponds to the LC filter current unbalance fault threshold.	Yes
G5.16.1.4-Fault Delay = 10 s	0 to 3600 s	Corresponds to the LC filter current unbalance delay.	Yes

G5.16.2-Estim. Capacitance

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.16.2.1-Protection type = Disabled	Disabled Warning Fault	Configure the type of protection associated with the functionality by which an estimation of the C capacity of the LC filter can be made as a function of the current and voltage measurement. **OPTION** / **DESCRIPTION** Disabled: This option disables the fault and warning configured. Warning: When a loss of capacity is diagnosed during operation of the inverter, this parameter configures a signal by means of the warning W47: C LC Filter. Fault: When a loss of capacity is diagnosed during operation of the inverter, this parameter configures a signal by means of the fault F172: C LC Filter, which will carry out an abrupt stop and will be reset manually by default.	Yes
G5.16.2.2-Ref. Capacitance = Calculated	Calculated Calibrated	Allows to configure the reference capacitance according to the following table: **OPTION** / **DESCRIPTION** Calculated: It is based on the C value set in the drive select. Calibrated: Corresponds to the value calibrated during routine field testing.	Yes
G5.16.2.3-Fault Threshold = 25 %	0 to 100 %	Corresponds to the LC filter capacitance fault threshold.	Yes
G5.16.2.4-Fault Delay = 10 s	0 to 3600 s	Corresponds to the LC filter capacitance delay.	Yes

EN

G5.16.3: Autodetection LC

This subgroup allows setting parameters related to the auto-detection of the LC filter type.

The filter capacitance is estimated and calculated with the theoretical value it should have according to the filter type configured in SG10.5.0.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.16.3.1-Enable = Yes	No Yes	Enables or disables the protection that detects LC filter misconfiguration. If any of the thresholds are exceeded, a warning or fault will be triggered (depending on the setting of G5.16.3.4). **Note:** this parameter can only be seen and modified by users with role Developer.	Yes
G5.16.3.2-Fail threshold bank = 15%	10 to 50%	Configures the detection threshold per capacitor bank. That is, the admissible difference between the expected capacitance and the calculated capacitance within the same bank. **Note:** this parameter is shown if [G5.16.3.1-Enable] = Yes.	Yes
G5.16.3.3-Fail threshold assm = 20%	10 to 50%	Allows to set the detection threshold of the capacitor assembly. That is, the admissible difference between the expected capacitance and the calculated capacitance of the total capacitor assembly. **Note:** this parameter is shown if [G5.16.3.1-Enable] = Yes.	Yes
G5.16.3.4-Protection type = Fault	No Yes	Allows configuring whether a fault or warning will be triggered in case a problem is encountered during filter autodetection. **Note:** this parameter is shown if [G5.16.3.1-Enable] = Yes.	Yes

G5.16.4-CFilter Disc. Hall

This subgroup allows configuring the detection of disconnection or failure of the filter C current Hall sensor, as well as the generation of the corresponding warning or fault.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.16.4.1-Protection Enable = Yes	No Yes	Enables or disables detection of the Hall sensor for the C filter when it is disconnected or malfunctioning. When enabled, the inverter generates the fault [F185: Disco. Cfilt Hall sensor] or the warning [W60: Discon. Cfilt Hall sensor], depending on the filter's hardware version ([SV1.10.6-HW filter version]).	Yes

6.5.15. Subgroup G5.17: *Double-Signal*

Parameters for the configuration of the maximum time allowed in double-signal modulation.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.17.1-Enable DS = Enabled	Disabled Enabled	Enables or disables double signal modulation protection. **Note :** this parameter can only be modified by users with role Developer.	Yes
G5.17.2-Down Grad. DS = 10%/s	1 to 100%/s	Allows to set the value of the down gradient of the double modulation counter. This gradient is defined in % of the up gradient which is fixed at 1s (1 count/s). Example: A 50% down gradient means that while in single-signal the failure counter will be decremented at half the rate of the up gradient (0.5 counts/s).	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G5.17.3-Enable Q limit DS = Enabled	Disabled Enabled	This allows to enable or disable the reactive power limit Q during double-signal modulation operation.	Yes
G5.17.4-Input grad. Q limit = 10.00 %/s	0.10 to 100.00 %/s	This allows you to configure the input ramp to the reactive power limit Q when the inverter is operating in double-signal modulation. **Note**: this parameter is displayed if [G5.17.3-Enable Q limit DS] = Enabled.	Yes

6.6. Group G6: Communications

This group allows the configuration of the Modbus, Ethernet and FTP communication properties of the inverter, as well as the definition of the communications with the PPC / external SCADA.

EN

6.6.1. Subgroup G6.1: Modbus

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

G6.1.1-Master 1

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.1.1.1-Modbus baudrate = 19200 bps	9600 bps 19200 bps 57600 bps 115200 bps 230400 bps 460800 bps 921600 bps	Adjusts the baud rate (bits per second) of Master 1.	Yes
G6.1.1.2-Modbus parity = None	None Odd Even	Adjusts the parity of Master 1.	Yes
G6.1.1.3-Modbus stop bits = 1 bit	1 bit 2 bits	Adjusts the stop bits number (per byte) of Master 1.	Yes
G6.1.1.4-Modbus error CRC = 0	0 to 65535	Shows the number of frames that have been lost of Master 1. **Note**: it allows to manually reset the counter by typing a zero.	Yes
G6.1.1.5-Modbus OK frames = 0	0 to 65535	Shows the number of successful frames that have been sent of Master 1. **Note**: it allows to manually reset the counter by typing a zero.	Yes
G6.1.1.6-Modbus recover = No	No Yes	These parameters should only be used by Developers.	Yes
G6.1.1.7-Modbus Timeout = 0.6 s	0.2 to 5.0 s		Yes
G6.1.1.8-Period = 0.250 s	0.010 to 1.000 s		Yes

G6.1.2-Master 2

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.1.2.1-Modbus baudrate = 19200 bps	9600 bps 19200 bps 57600 bps 115200 bps 230400 bps 460800 bps 921600 bps	Adjusts the baud rate (bits per second) of Master 2.	Yes
G6.1.2.2-Modbus parity = None	None Odd Even	Adjusts the parity of Master 2.	Yes
G6.1.2.3-Modbus stop bits = 1 bit	1 bit 2 bits	Adjusts the stop bits number (per byte) of Master 2.	Yes
G6.1.2.4-Modbus error CRC = 0	0 to 65535	Shows the number of frames that have been lost of Master 2. **Note**: it allows to manually reset the counter by typing a zero.	Yes
G6.1.2.5-Modbus OK frames = 0	0 to 65535	Shows the number of successful frames that have been sent of Master 2. **Note**: it allows to manually reset the counter by typing a zero.	Yes
G6.1.2.6-Modbus recover = No	No Yes	These parameters should only be used by Developers.	Yes
G6.1.2.7-Modbus Timeout 2 = 0.6 s	0.2 to 5.0 s		Yes
G6.1.2.8-Period = 0.250 s	0.010 to 1.000 s		Yes

G6.1.3-Master 3

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.1.3.1-Modbus baudrate = 115200 bps	9600 bps 19200 bps 57600 bps 115200 bps 230400 bps 460800 bps 921600 bps	Adjusts the baud rate (bits per second) of Master 3.	Yes
G6.1.3.2-Modbus parity = None	None Odd Even	Adjusts the parity of Master 3.	Yes
G6.1.3.3-Modbus stop bits = 1 bit	1 bit 2 bits	Adjusts the stop bits number (per byte) of Master 3.	Yes
G6.1.3.4-Modbus error CRC = 0	0 to 65535	Shows the number of frames that have been lost of Master 3. **Note**: it allows to manually reset the counter by typing a zero.	Yes
G6.1.3.5-Modbus OK frames = 0	0 to 65535	Shows the number of successful frames that have been sent of Master 3. **Note**: it allows to manually reset the counter by typing a zero.	Yes
G6.1.3.6-Modbus recover = No	No Yes	These parameters should only be used by Developers.	Yes
G6.1.3.7-Modbus Timeout 3 = 0.6 s	0.2 to 5.0 s		Yes
G6.1.3.8-Period = 0.100 s	0.010 to 1.000 s		Yes

6.6.2. Subgroup G6.2: Ethernet

Note: Only users with a role equal or higher than Admin (qualified customer) can modify these parameters.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.2.1-Automatic IP = No	No Yes	Allows selecting the IP assignment method of the module: **OPTION** / **DESCRIPTION** No: The inverter will take the IP address, subnet mask and gateway set by the user. Yes: DHCP protocol is enabled, and the network server will automatically set the IP address, subnet mask and gateway.	Yes
G6.2.2-Assigned IP = 0.0.0.0	0.0.0.0 to 255.255.255.255	Shows the IP address assigned to the module when DHCP is active.	Yes
G6.2.3-Assigned subnet = 0.0.0.0	0.0.0.0 to 255.255.255.255	Shows the subnet mask address assigned to the module when DHCP is active.	Yes
G6.2.4-Assigned gateway = 0.0.0.0	0.0.0.0 to 255.255.255.255	Shows the gateway address assigned to the module.	Yes
G6.2.5-IP address = 192.168.204.202	1.0.0.1 to 255.255.255.254	Sets the IP address in the user's local network, according to the guidelines given by the network Admin. IP address format is: A.B.C.D. To adjust this address, introduce a value in each of the four parameters that compose it.	Yes
G6.2.6-Subnet Mask = 255.255.255.0	1.0.0.0 to 255.255.255.255	Sets the subnet mask address in the user's local network, according to the guidelines given by the network Admin. Subnet mask address format is: A.B.C.D. To adjust this address, introduce a value in each of the four parameters that compose it.	Yes
G6.2.7-Gateway = 0.0.0.0	0.0.0.0 to 255.255.255.254	Sets the gateway address in the user's local network so that the inverter is able to communicate to an external network. This address must follow the guidelines given by the network Admin. Gateway's address format is: A.B.C.D. To adjust this address, introduce a value in each of the four parameters that compose it.	Yes
G6.2.8-MAC address = 21-1B-78-81-E9-44	00-00-00-00-00-00 to FF-FF-FF-FF-FF-FE	Set the MAC address. This address is unique and exclusive, and it is related with the LAN / inverter board. It must be given by Power Electronics. MAC address format is: A.B.C.D.E.F. To adjust this address, introduce a value in each of the six parameters that compose it. **Note:** this parameter is displayed if [G1.3-User ≥ Power Electronics staff].	Yes
G6.2.9-Primary DNS = 8.8.8.8	1.0.0.1 to 255.255.255.254	Sets the IP to connect to the remote log.	Yes
G6.2.10-Secondary DNS = 8.8.4.4	1.0.0.1 to 255.255.255.254	Sets the IP to connect to the remote log.	Yes
G6.2.11-Enable UDP response = Enabled	Disabled Enabled	Allows to enable or disable the response when there is a UDP request. **Note:** this parameter is displayed if [G1.3-User] ≥ Power Electronics staff.	Yes
G6.2.12-Number sockets by IP = 6	1 to 6	Allows to define the maximum number of connections available for the same IP. **Note**: this parameter is displayed if [G1.3-User] ≥ Emp. Power Electronics.	Yes
G6.2.13-IP address golden = 0.0.0.0	0.0.0.0 to 255.255.255.255	Allows the configuration of an IP address to which the device will always allow connection, regardless of which sockets are occupied. The IP address format is: A.B.C.D. To set this address, enter a value for each of the four parameters that comprise it. **Note**: this parameter is displayed if [G1.3-User] ≥ Emp. Power Electronics.	Yes
G6.2.14-Min time between req. = Off ms	Off = 0; 1 to 100 ms	Allows to configure the minimum time required between Modbus requests. **Note:** this parameter is displayed if [G1.3-User] ≥ Power Electronics staff.	Yes
G6.2.15-Request type filter = Write	Write Write and read	Allows to select between filtering write request or read/write request. **Note:** this parameter is displayed if [G1.3-User] ≥ Power Electronics staff.	Yes

EN

6.6.3. Subgroup G6.4: FTP Guest

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with a role equal to or higher than Admin (qualified customer) can modify these parameters except G6.4.1 which requires a role higher than Admin.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.4.1-Enable FTP Guest = Yes	No Yes	Enables or disables the FTP Guest functionality.	Yes
G6.4.2-Guest Username = guest	10 characters	Sets the type of user: User 1.	Yes
G6.4.3-Guest Password = guest	10 characters	Configures password for user 1.	Yes

6.6.4. Subgroup G6.5: PPC

This subgroup is developed to configure the behavior of the system against communication lost events between PPC/SCADA and how the inverter behaves when that event comes up.

PPC or SCADA sends periodical commands to the inverter related to P and/or Q setpoints according to the control mode within the plant.

To establish communication with the inverter, the SCADA or PPC (Power Plant Controller) must first initiate communication via the PPC Heartbeat variable[7]. To do this, the SCADA or PPC must write in this variable a value other than 0 (valid range 0 to 65535) and update the value regularly. Each time a signal is received from the PPC Heartbeat variable, the parameter [G6.5.9- PPC Heartbeat Reg], a read-only variable, is updated.

If the SCADA or PPC does not update the value in the PPC Heartbeat variable and a time greater than that indicated by parameter [G6.5.10-HB lost time] elapses, the active and reactive power reference is modified to those configured in parameters [G6.5.3-Home P ref] and [G6.5.4-Home Q ref]. The gradients with which the inverter works to reach these references are those configured in parameters [G6.5.5-P rec ramp] and [G6.5.6-Q rec ramp].

If before the time set in [G6.5.10-HB lost time] expires, communications are re-established, the inverter will go to the references indicated by the SCADA or PPC with the recovery ramps [G6.5.5-P rec ramp] and [G6.5.6-Q rec ramp]. When the new references indicated by the SCADA or PPC are reached, operation is resumed with the initially configured operating values.

If the time indicated by parameter [G6.5.10-HB lost time] has expired and the time indicated by [G6.5.1-Home refs timeout] has also expired, then the P and Q references are modified to those configured in parameters [G6.5.3-Home P ref] and [G6.5.4-Home Q ref] according to the type of P and Q control that is configured at that time, and the [W152: PPC comms] warning is triggered. If communications are re-established before the loss of communications time expires, proceed in the same way as in the previous paragraph.

[7] See section "Remote control functions" to access the Modbus register of the variable.

On the other hand, if the three times indicated in parameters [G6.5.10-HB lost time], [G6.5.1-Home refs timeout] and [G6.5.2-Comm timeout] expire, fault [F51: PPC comms] is triggered. At this point, the inverter stops. The inverter is configured to work with the operation references defined by the SCADA or PPC. Once the fault has been resolved, the inverter will go to the references indicated by means of the recovery ramps [G6.5.5-P rec ramp] and [G6.5.6-Q rec ramp].

If the PPC or SCADA writes a zero to the PPC Heartbeat variable, the system is disabled. This step is equivalent to setting [G6.5.1-Home refs timeout] and [G6.5.2-Comm timeout] to zero.

Note: This subgroup is displayed if [G1.3-User ≥ Qualified customer]. Only users with a role equal to or higher than Admin (qualified customer) can modify these parameters.

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G6.5.1-Home refs timeout = 4 s</td><td>Off = 0
1 to 60 s</td><td rowspan="2">If no communication is detected between the Power Plant Controller and the inverter for a longer time than the timeout set in [G6.5.1-Home refs timeout] and/or [G6.5.10-HB lost time], all references will be set to [G6.5.3-Home P ref] and parameter [G6.5.4-Home Q ref].
Parameter [G6.5.2-Comm timeout], on the other hand, defines the maximum time communications can be down before triggering a PPC communications fault (F51: PPC comms).
Notes:
-[G6.5.1] is only displayed if [G8.0.1-Control type ≠ VSI] or [G6.5.12-Ramp selector ≠ Aemo].
-[G6.5.2] is only displayed if [G6.5.12-Ramp selector ≠ Aemo].</td><td>Yes</td></tr>
<tr><td>G6.5.2-Comm timeout = 60 s</td><td>Off = 0
1 to 1200 s</td><td>Yes</td></tr>
<tr><td>G6.5.3-Home P ref = 0.00 %</td><td>0.00 to 100.00 %</td><td>Active power setpoint value when the time set in [G6.5.1-Home refs timeout] and/or [G6.5.10-HB lost time] has elapsed.
Note: This parameter is only displayed if [G6.5.12-Ramp selector ≠ Aemo].</td><td>Yes</td></tr>
<tr><td>G6.5.4-Home Q ref = 0.00 %</td><td>-100.00 to 100.00 %</td><td>Reactive power setpoint when the time set in [G6.5.1-Home refs timeout] and/or [G6.5.10-HB lost time] has elapsed.
Note: This parameter is only displayed if [G6.5.12-Ramp selector ≠ Aemo].</td><td>Yes</td></tr>
<tr><td>G6.5.5-P rec ramp = 10.0 %/s</td><td>0.1 to 3000.0 %/s</td><td rowspan="2">Defines the active and reactive power gradients with which the inverter operates under inverter loss of communications conditions (F51: PPC communications).
Note: This parameter is only displayed if [G6.5.12-Ramp selector = Recovery Ramps].</td><td>Yes</td></tr>
<tr><td>G6.5.6-Q rec ramp = 10.0 %/s</td><td>0.1 to 3000.0 %/s</td><td>Yes</td></tr>
<tr><td>G6.5.7-IP filter enable = No</td><td>No
Yes</td><td>Defines the PPC IP filter.</td><td>Yes</td></tr>
<tr><td>G6.5.8-IP address = 192.168.0.150</td><td>1.0.0.1 to 255.255.255.254</td><td>Defines the PPC IP address.</td><td>Yes</td></tr>
<tr><td>G6.5.9-PPC Heartbeat Reg = 0</td><td>0 to 65535</td><td>Defines the PPC heartbeat register. Allows to verify if the communication with the PPC / SCADA is working properly.
Note: this parameter is read-only (RO).</td><td>Yes</td></tr>
<tr><td>G6.5.10-HB lost time = 1.0 s</td><td>0.1 to 5.0 s</td><td>Defines the time after which the communication loss algorithm starts to operate.
Note: This parameter is displayed if [G6.5.12-Ramp selector ≠ Aemo].</td><td>Yes</td></tr>
<tr><td>G6.5.11-Comms state = First heartbeat</td><td>First heartbeat
Communicating
No Communicating</td><td>Enables to set the PPC communication status:
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>First heartbeat</td><td>PPC is waiting to receive the first heartbeat.</td></tr><tr><td>Communicating</td><td>PPC is communicating.</td></tr><tr><td>No Communicating</td><td>PPC is not communicating.</td></tr></table>Note: This parameter is only displayed if [G6.5.12-Ramp selector ≠ Aemo] and is read-only (RO).</td><td>Yes</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G6.5.12-Ramp selector = No Ramps</td><td>No Ramps
Recovery Ramps
Aemo</td><td>Allows to configure how the inverter reacts when the communication between the system and the PPC/SCADA is lost.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>No Ramps</td><td>If this option is set, then no ramps will be applied. The transition between the last reference received before the communication loss event and the first reference received after the communication is recovered is carried out using the operation gradients configured in:
G4.2.2.1-Upward grad P enable.
G4.2.2.2-Upward gradient P.
G4.2.2.3-Upward grad Q enable.
G4.2.2.4-Upward gradient Q.</td></tr><tr><td>Recovery Ramps</td><td>If this option is set, then the recovery ramps algorithm will be applied. This means the transition between references received before and after the communication lost are the configured in:
G6.5.5-P rec ramp.
G6.5.6-Q rec ramp.</td></tr><tr><td>Aemo</td><td>Not applicable to this product version.</td></tr></table>Note: [G6.5.12-Ramp selector = No Ramps] if [G8.0.1-Control type = VSI].</td><td>No</td></tr>
<tr><td>G6.5.13-P ramp state = Deactivated</td><td>Deactivated
Activated
Running</td><td>Shows the status of the PPC active power ramp.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Deactivated</td><td>The ramp is deactivated.</td></tr><tr><td>Activated</td><td>The ramp is activated.</td></tr><tr><td>Running</td><td>The ramp is being applied.</td></tr></table>Notes:
-This parameter is displayed if [SG10.1.4-Inverter type = PCS/MPCS].
- This parameter is only displayed if [G6.5.12-Ramp selector ≠ NO RAMPS].
- This parameter is read-only (RO).</td><td>Yes</td></tr>
<tr><td>G6.5.14-Q ramp state = Deactivated</td><td>Deactivated
Activated
Running</td><td>Shows the status of the PPC reactive power ramp.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Deactivated</td><td>The ramp is deactivated.</td></tr><tr><td>Activated</td><td>The ramp is activated.</td></tr><tr><td>Running</td><td>The ramp is being applied.</td></tr></table>Notes:
- This parameter is displayed if [SG10.1.4-Inverter type = PCS/MPCS].
- This parameter is only displayed if [G6.5.12-Ramp selector ≠ NO RAMPS].
- This parameter is read-only (RO).</td><td>Yes</td></tr>
<tr><td>G6.5.15-P cross offset = 1.00 %</td><td>0.10 to 10.00 %</td><td rowspan="2">Allows the system to ramp between two P/Q references in order to avoid grid disturbances. The recovery ramping algorithm is initiated when a loss of communication event occurs with the PPC or SCADA. The recovery ramping algorithm needs a stop condition to return to the operating gradients.
[G6.5.15-P cross offset] parameter allows the measured active power of the PPC and the reference to cross within this offset to indicate when the ramp ends.
[G6.5.16-Q cross offset] parameter allows the measured reactive power of the PPC and the reference to cross within this offset to indicate when the ramp ends.
Note: This parameter is only displayed if [G6.5.12-Ramp selector ≠ Aemo].</td><td>Yes</td></tr>
<tr><td>G6.5.16-Q cross offset = 1.00 %</td><td>0.10 to 10.00 %</td><td>Yes</td></tr>
<tr><td>G6.5.17-Bbar1 P ramp state = Deactivated</td><td>Deactivated
Activated
Running</td><td>Shows the status of the PPC busbar 1 active power ramp. It applies to PCS/MPCS inverter.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Deactivated</td><td>The ramp is deactivated.</td></tr><tr><td>Activated</td><td>The ramp is activated.</td></tr><tr><td>Running</td><td>The ramp is being applied.</td></tr></table>Notes:
- This parameter is displayed if [SG10.1.4-Inverter type = PCS/MPCS].
- This parameter is only displayed if [G6.5.12-Ramp selector ≠ NO RAMPS].
- This parameter is read-only (RO).</td><td>Yes</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G6.5.18-Bbar2 P ramp state = Deactivated</td><td>Deactivated
Activated
Running</td><td>Shows the status of the PPC busbar 2 active power ramp. It applies to PCS/MPCS inverter.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Deactivated</td><td>The ramp is deactivated.</td></tr><tr><td>Activated</td><td>The ramp is activated.</td></tr><tr><td>Running</td><td>The ramp is being applied.</td></tr></table>Notes:
- This parameter is displayed if [SG10.1.4-Inverter type = PCS/MPCS] and [SG10.1.14-Number of busbar ≥ 2].
- This parameter is only displayed if [G6.5.12-Ramp selector ≠ NO RAMPS].
- This parameter is read-only (RO).</td><td>Yes</td></tr>
<tr><td>G6.5.19-Bbar3 P ramp state = Deactivated</td><td>Deactivated
Activated
Running</td><td>Shows the status of the PPC busbar 3 active power ramp. It applies to PCS/MPCS inverter.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Deactivated</td><td>The ramp is deactivated.</td></tr><tr><td>Activated</td><td>The ramp is activated.</td></tr><tr><td>Running</td><td>The ramp is being applied.</td></tr></table>Notes:
- This parameter is displayed if [SG10.1.4-Inverter type = PCS/MPCS] and [SG10.1.14-Number of busbar ≥ 3].
- This parameter is only displayed if [G6.5.12-Ramp selector ≠ NO RAMPS].
- This parameter is read-only (RO).</td><td>Yes</td></tr>
<tr><td>G6.5.20-Bbar4 P ramp state = Deactivated</td><td>Deactivated
Activated
Running</td><td>Shows the status of the PPC busbar 4 active power ramp. It applies to PCS/MPCS inverter.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Deactivated</td><td>The ramp is deactivated.</td></tr><tr><td>Activated</td><td>The ramp is activated.</td></tr><tr><td>Running</td><td>The ramp is being applied.</td></tr></table>Notes:
- This parameter is displayed if [SG10.1.4-Inverter type = PCS/MPCS] and [SG10.1.14-Number of busbar ≥ 4].
- This parameter is displayed if [G6.5.12-Ramp selector ≠ NO RAMPS].
- This parameter is read-only (RO).</td><td>Yes</td></tr>
<tr><td>G6.5.20a-Sharing P ramp state = Deactivated</td><td>Deactivated
Activated
Running</td><td>Displays the status of the active power sharing (P_sharing) ramp (configurable in [G8.2.1.6.5-P]) during an AEMO / VISMA event.
Notes:
-This parameter is displayed if [SG10.1.4-Inverter type] = PCS/MPCS.
-This parameter is displayed if [SG10.1.19-Visma enable] = Enabled.
-This parameter is displayed if [G6.5.12-Ramp selector] ≠ No Ramps.
-This parameter is read-only (RO).</td><td>Yes</td></tr>
<tr><td>G6.5.21-Bbars Q ramp state = Deactivated</td><td>Deactivated
Activated
Running</td><td>Shows the status of the PPC busbars reactive power ramp. It applies to PCS/MPCS inverter.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Deactivated</td><td>The ramp is deactivated.</td></tr><tr><td>Activated</td><td>The ramp is activated.</td></tr><tr><td>Running</td><td>The ramp is being applied.</td></tr></table>Notes:
- This parameter is displayed if [SG10.1.4-Inverter type = PCS/MPCS].
- This parameter is displayed if [G6.5.12-Ramp selector ≠ NO RAMPS].
- This parameter is read-only (RO).</td><td>Yes</td></tr>
<tr><td>G6.5.21a-Sharing Q ramp state = Deactivated</td><td>Deactivated
Activated
Running</td><td>Displays the status of the reactive power sharing (Q_sharing) ramp (configurable in [G8.2.1.6.6-Q]) during an AEMO / VISMA event.
Notes:
-This parameter is displayed if [SG10.1.4-Inverter type] = PCS/MPCS.
-This parameter is displayed if [SG10.1.19-Visma enable] = Enabled.
-This parameter is displayed if [G6.5.12-Ramp selector≠ No Ramps.
-This parameter is read-only (RO).</td><td>Yes</td></tr>
</table>

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.5.22-AEMO detect loss = 500 ms	1 to 5000[8] ms	Allows to configure the maximum time to detect a loss of communication with SCADA or PPC according to AEMO standard configuration. **Notes:** -This parameter is only displayed if [G6.5.12-Ramp selector = AEMO]. -Only users with a role higher than Admin can modify this parameter.	Yes
G6.5.22a-AEMO alt detect Loss = 500 ms	1 to 5000 ms	Allows to configure the maximum time to detect a loss of communication with SCADA or PPC according to AEMO alternative configuration. **Notes:** - This parameter is displayed if [G6.5.12-Ramp selector] = AEMO and [SG10.1.19-Visma enable] = Enabled. - Only users with a role higher than Admin can modify this parameter.	Yes
G6.5.23-AEMO timeout 1st loss = 30 s	1[9] to 60 s	Allows to configure the time in the first loss according to AEMO standard configuration, after which set the value of [G6.5.3-Home P ref] or [G6.5.4-Home Q ref] depending on the control mode of the inverter. **Notes:** -These parameters are displayed if [G6.5.12-Ramp selector = AEMO]. -Only users with a role higher than Admin can modify these parameters. -The parameter with integer value (default value = 30 s) is hidden from Power Electronics remote management tool from version 3.3.2.5, but its Modbus address (41965) can be accessed through SCADA.	Yes
G6.5.23-AEMO timeout 1st loss = 30.000 s	1.000[10] to 60.000 s		Yes
G6.5.23a-AEMO alt timeout 1st = 3600 s	5 to 3600 s	Allows to configure the time in the first loss according to AEMO alternative configuration, after which set the value of [G6.5.3-Home P ref] or [G6.5.4-Home Q ref] depending on the control mode of the inverter. **Notes:** - This parameter is displayed if [G6.5.12-Ramp selector] = AEMO and [SG10.1.19-Visma enable] = Enabled. - Only users with a role higher than Admin can modify this parameter.	Yes
G6.5.24-AEMO max losses = 2	1 to 10	Allows to configure the maximum number of communications losses to stop the inverter according to AEMO standard configuration. **Notes:** -This parameter is displayed if [G6.5.12-Ramp selector = AEMO]. - Only users with a role equal to or higher than Power Electronics staff can modify this parameter.	Yes
G6.5.24a-AEMO alt max losses = 2	1 to 10	Allows to configure the maximum number of communications losses to stop the inverter according to AEMO alternative configuration. **Notes:** - This parameter is displayed if [G6.5.12-Ramp selector] = AEMO and [SG10.1.19-Visma enable] = Enabled. - Only users with a role higher than Admin can modify this parameter.	Yes
G6.5.25-AEMO max time = 60 min	1 to 600 min	Allows to configure the maximum time between consecutive communication losses according to AEMO standard configuration. **Notes:** -This parameter is displayed if [G6.5.12-Ramp selector = AEMO]. - Only users with a role equal to or higher than Power Electronics staff can modify this parameter.	Yes
G6.5.25a-AEMO alt max time = 60 min	1 to 600 min	Allows to configure the maximum time between consecutive communication losses according to AEMO alternative configuration. **Notes:** - This parameter is displayed if [G6.5.12-Ramp selector] = AEMO and [SG10.1.19-Visma enable] = Enabled. - Only users with a role higher than Admin can modify this parameter.	Yes

[8] The maximum value of the range is 5000 ms or the value of [G6.5.23-AEMO timeout 1st loss] if it is less than 5000 ms.
[9] The minimum value of the range is 1 s or the value of [G6.5.22-AEMO detect loss], rounded up to the second highest integer, if this is greater than 1 s.
[10] The minimum value of the range is 1.000 s or the value of [G6.5.22-AEMO detect loss], rounded up to the highest integer, if this is greater than 1.000 s.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G6.5.26-AEMO comm state = Ok	Ok Heartbeat lost Communication fault	Shows the status of AEMO communications. OPTION: Ok — DESCRIPTION: The heartbeat is received correctly. OPTION: Heartbeat lost — DESCRIPTION: The heartbeat is lost. OPTION: Communication fault — DESCRIPTION: Communications failure. **Note:** This parameter is displayed if [G6.5.12-Ramp selector = AEMO] and is read-only (RO).	Yes
G6.5.27-AEMO losses counter = 0	0 to 65535	Allows to count the number of AEMO communications losses. **Note:** This parameter is displayed if [G6.5.12-Ramp selector = AEMO] and is read-only (RO).	Yes
G6.5.28-AEMO config selector = Standard	Standard Alternative	Allows to select AEMO configuration as Standard or Alternative. OPTION: Standard — DESCRIPTION: AEMO standard mode for VISMA. OPTION: Alternative — DESCRIPTION: AEMO alternative mode for VISMA. **Note:** it is shown if [G8.0.2-Control type] = Visma. **Note:** this parameter is displayed if [G6.5.12-Ramp selector] = AEMO and [SG10.1.19-Visma enable] = Enabled.	Yes

EN

6.7. Group G8: Control

This group allows the configuration of the control of the inverter, the network and the batteries (in case of PCSM/K).

Note: Only users with a role equal to or higher than Admin (qualified customer) can modify these parameters.

6.7.1. Subgroup G8.0: Control mode

This subgroup allows to configure the inverter control types and sub-types, both in current source (CSI) and VISMA (VSI) modes.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.0.1-Control type = CSI	CSI VSI	Allows to select the control type of the inverter. OPT. 0 — MODE: CSI — DESCRIPTION: Current control type. OPT. 1 — MODE: VSI — DESCRIPTION: Voltage control type. **Note**: this option will be displayed if [SG1.10.1-Black start enable = Enabled] and [SG10.1.19-Visma enable = Enabled].	No[11]
G8.0.2-Control subtype = CSI	CSI VISMA FAC	Allows to select the control subtype of the inverter. OPT. 0 — MODE: CSI — DESCRIPTION: Current source control subtype. OPT. 1 — MODE: VISMA — DESCRIPTION: VISMA control subtype. **Note**: this option will be displayed if [SG10.1.19-Visma enable = Enabled]. OPT. 2 — MODE: FAC — DESCRIPTION: AC source control subtype. **Note**: this option will be displayed if [SG1.10.1-Black start enable = Enabled] or [SG10.1.19-Visma enable = Disabled].	No

[11] If [SG10.1.20-Enable seamless] = Enabled, then “set to run” will be YES.

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G8.0.3-CSI P control mode = MPPt</td><td>MPPt
Id
P
Vdc
Pac
Multi Pac</td><td>Allows setting the active power control mode for CSI current source mode. The control modes are as shown in the table below:
<table>
<tr><th>OPT.</th><th>MODE</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>MPPt</td><td>The inverter is always looking for the maximum power point. This option only applies to HEM/K inverters.</td></tr>
<tr><td>1</td><td>Id</td><td>The inverter maintains the active current output set in [G8.1.1 Id reference] constant. This option only applies to HEM/K inverters.</td></tr>
<tr><td>2</td><td>P</td><td>The inverter maintains the active power output set in [G8.1.1 P reference] constant. This option only applies to HEM/K inverters.</td></tr>
<tr><td>3</td><td>Vdc</td><td>The inverter maintains the DC voltage set in [G8.1.1 Vdc reference] constant. This option only applies to HEM/K inverters.</td></tr>
<tr><td>4</td><td>Pac</td><td>The inverter maintains the active power reference output set in [G8.1.1.1.x-Pac reference x] constant. This option only applies to PCSM/K and MPCSM/K inverters. In this case, this will be the defect value of the parameter.</td></tr>
<tr><td>5</td><td>Multi Pac</td><td>The inverter maintains the active power reference output set for each busbar according to the values set in [G8.1.1.1.x-Pac reference x] constant. This option only applies to PCSM/K and MPCSM/K inverters.</td></tr>
</table></td><td>No</td></tr>
<tr><td>G8.0.4-CSI Q control mode = Q</td><td>Cos Phi
Q
I
P Cos Phi</td><td>Allows setting the reactive power control mode for CSI current source mode. The control modes are as shown in the table below:
<table>
<tr><th>OPT.</th><th>MODE</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Cos Phi</td><td>The inverter maintains the cos(phi) output set in [G8.1.1.2-Cos phi reference] constant.</td></tr>
<tr><td>1</td><td>Q</td><td>The inverter maintains the reactive power output set in [G8.1.1.2-Q reference] constant.</td></tr>
<tr><td>2</td><td>I</td><td>The inverter maintains the reactive current output set in [G8.1.1.2-Iq reference] constant.</td></tr>
<tr><td>3</td><td>P Cos Phi</td><td>This control mode enables reactive power to be supplied automatically according to the characteristic curve cos φ = f (P). This characteristic curve is uniquely defined by the linear interpolation of three points: A, B and C, which are the six configurable points in [G8.1.1.4-P1 Cos phi], [G8.1.1.5-P2 Cos phi], [G8.1.1.6-P3 Cos phi] and [G8.1.1.7-Cos phi 1], [G8.1.1.8-Cos phi 2], [G8.1.1.9-Cos phi 3].
Note: the representation of the curve cos φ = f (P) is shown at the end of the following subgroup.</td></tr>
</table></td><td>No</td></tr>
<tr><td>G8.0.5-P VSI control = Pac VISMA</td><td>Pac VISMA
Freq VISMA</td><td>Allows to configure active control modes for VSI control type with VISMA control subtype. These controls are P (active power) and f (frequency).
<table>
<tr><th>OPT.</th><th>MODE</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Pac VISMA</td><td>Pac reference.</td></tr>
<tr><td>1</td><td>Freq VISMA</td><td>Freq reference.
This parameter is an increase in frequency with respect to [SG10.1.7-Nominal Frequency].</td></tr>
</table>
Note: this parameter will be displayed if [SG10.1.19-Visma enable = Enabled].</td><td>Yes</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G8.0.6-Q VSI control = Vac</td><td>VAC FAC
Vd FAC
Vac
Q
Iq</td><td>Allows to configure reactive control modes for VSI control type.
Note: this parameter is shown when VISMA control is enabled ([SG10.1.19-Visma enable] or [G8.0.2-Control subtype] = FAC.
<table>
<tr><th>OPT.</th><th>MODE</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>VAC FAC</td><td>Voltage reference for VSI control type with FAC control subtype. The inverter will regulate the closed-loop voltage to this reference.
No current limit or load sharing applies. It is designed to work in the laboratory or in controlled situations.
Note: this option will be displayed if [G8.0.2-Control subtype = FAC].</td></tr>
<tr><td>1</td><td>Vd FAC</td><td>Modulation index for VSI control type with FAC control subtype. AC voltage will depend on FDC (open loop control).
Note: this option will be displayed if [G8.0.2-Control subtype = FAC].</td></tr>
<tr><td>2</td><td>Vac</td><td>Voltage reference for VSI control type with VISMA control subtype.
The reference is in % with respect to the nominal.
Note: this option will be displayed if [SG10.1.19-Visma enable = Enabled].</td></tr>
<tr><td>3</td><td>Q</td><td>Reactive power reference for VSI control type with VISMA control subtype.
The reference is in % with respect to the nominal.
Note: this option will be displayed if [SG10.1.19-Visma enable = Enabled].</td></tr>
<tr><td>4</td><td>Iq</td><td>Reactive current reference for VSI control type with VISMA control subtype.
The reference is in % with respect to the nominal.
Note: this option will be displayed if [SG10.1.19-Visma enable = Enabled].</td></tr>
</table></td><td>Yes</td></tr>
<tr><td>G8.0.7-Control Theta = PLL</td><td>PLL
FAC
SWING EQ</td><td>Allows to configure of the FAC subtype angle control modes.
<table>
<tr><th>OPT.</th><th>MODE</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>PLL</td><td>A PLL control is used for the generation of the voltage angle.
Note: this option will be displayed if [G8.0.2-Control subtype = CSI].</td></tr>
<tr><td>1</td><td>FAC</td><td>A random angle with known frequency is set for the generation of the voltage angle.
Note: this option will be displayed if [G8.0.2-Control subtype = FAC].</td></tr>
<tr><td>2</td><td>SWING EQ</td><td>The oscillation equation is used for the generation of voltage angle.
Note: this option will be displayed if [G8.0.2-Control subtype = VISMA].</td></tr>
</table></td><td>No</td></tr>
</table>

EN

6.7.2. Subgroup G8.1: References

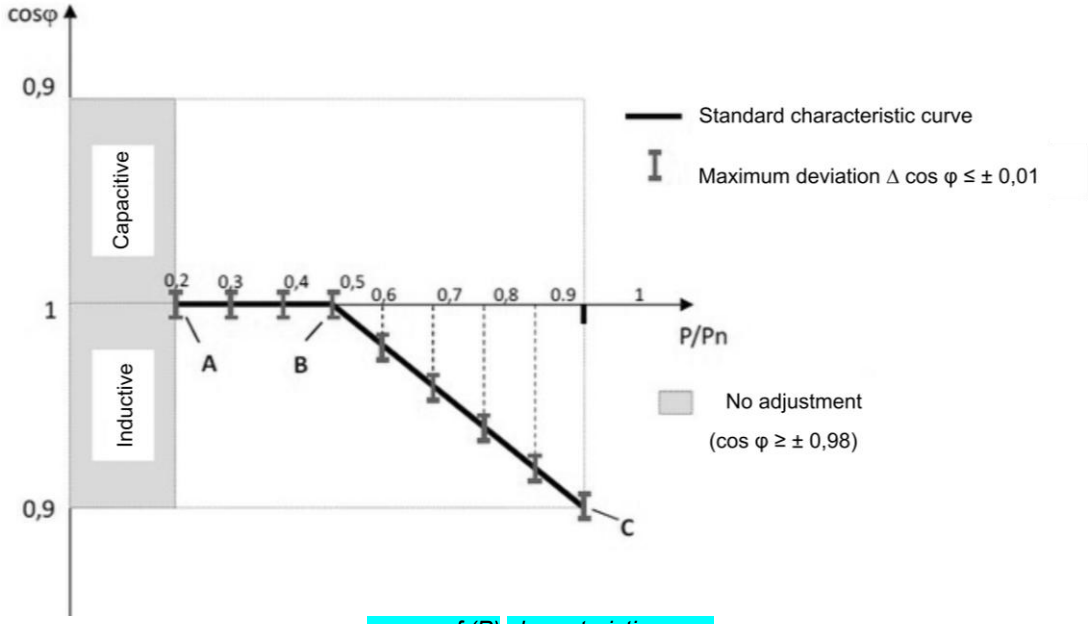
This subgroup allows to configure the references of the inverter for the CSI and VSI control modes.

Subgroup G8.1.1-CSI Control

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G8.1.1.1-P reference = 0.00 %</td><td>-100.00 to 100.00 %</td><td rowspan="4">The reference for the active power depends on which control mode has been chosen in [G8.0.3-CSI P control mode].
Thus, parameter G8.1.1.1 will vary as shown in the following table:
<table><tr><th>G8.0.3</th><th>G8.1.1.1</th><th>DESCRIPTION</th></tr><tr><td>MPPt</td><td>Not shown</td><td>-</td></tr><tr><td>I</td><td>Id reference</td><td>Sets an active current reference in percentage.</td></tr><tr><td>P</td><td>P reference</td><td>Sets an active power reference in percentage.</td></tr><tr><td>Vdc</td><td>Vdc reference</td><td>Sets a DC voltage reference in volts.</td></tr><tr><td>Pac</td><td>Pac reference</td><td>Sets an active power reference in percentage.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G8.1.1.1-Pac reference = 0.00 %</td><td>-100.00 to 100.00 %</td><td>Yes</td></tr>
<tr><td>G8.1.1.1-Id reference = 100.00 %</td><td>-100.00 to 100.00 %</td><td>Yes</td></tr>
<tr><td>G8.1.1.1-Vdc reference = 1100 V</td><td>0 to 1500 V</td><td>Yes</td></tr>
<tr><td>G8.1.1.1.x-Pac reference x = 0.00 %</td><td>-100.00 to 100.00 %</td><td>Parameter for MPCS, where x is the module number and varies between 1 and 4, in multi Pac control mode.
If G8.1.1.1 = P reference, a set of parameters to set the P reference for each module will be shown (P1 reference, P2 reference, …).</td><td>Yes</td></tr>
<tr><td>G8.1.1.2-Cos phi reference= 1.000</td><td>0.001 to 1.000</td><td rowspan="3">The reference for the reactive power depends on which control mode has been chosen in [G8.0.4-CSI Q control mode].
Thus, parameter G8.1.1.2 will vary as shown in the following table:
<table><tr><th>G8.0.4</th><th>G8.1.1.2</th><th>DESCRIPTION</th></tr><tr><td>Cos phi</td><td>Cos phi reference</td><td>Allows setting a cos(phi) absolute reference.</td></tr><tr><td>Q</td><td>Q reference</td><td>Allows setting a reactive power reference in percentage.</td></tr><tr><td>I</td><td>Iq reference</td><td>Allows setting a reactive current reference in percentage.</td></tr><tr><td>P Cos Phi</td><td>-</td><td>-</td></tr></table></td><td>Yes</td></tr>
<tr><td>G8.1.1.2-Iq reference = 0.00 %</td><td>-100.00 to 100.00 %</td><td>Yes</td></tr>
<tr><td>G8.1.1.2-Q reference = 0.00 %</td><td>-100.00 to 100.00 %</td><td>Yes</td></tr>
<tr><td>G8.1.1.3-Excitation = Capacitive</td><td>Capacitive
Inductive</td><td>Allows to define the type of excitation used in the algorithm.</td><td>Yes</td></tr>
<tr><td>G8.1.1.4-P1 Cos phi = 20.00 %</td><td>0.00 to 100.00 %</td><td rowspan="4">Configure the X axis of the curve cos φ = f (P).
Note: These parameters are displayed if [G8.0.4-CSI Q control mode = P Cos Phi].</td><td>Yes</td></tr>
<tr><td>G8.1.1.5-P2 Cos phi = 50.00 %</td><td>[G8.1.1.4-P1 Cos phi] to 100.00 %</td><td>Yes</td></tr>
<tr><td>G8.1.1.6-P3 Cos phi = 90.00 %</td><td>[G8.1.1.5-P2 Cos phi] to 100.00 %</td><td>Yes</td></tr>
<tr><td>G8.1.1.6a-P4 Cos phi = 95.00 %</td><td>G8.1.1.6-P3 Cos phi to 100.00 %</td><td>Yes</td></tr>
<tr><td>G8.1.1.7-Cos phi 1 = 1.000</td><td rowspan="4">0.000 to 1.000</td><td rowspan="4">Configure the Y axis of the curve cos φ = f (P).
Note: These parameters are displayed if [G8.0.4-CSI Q control mode = P Cos Phi].</td><td>Yes</td></tr>
<tr><td>G8.1.1.8-Cos phi 2 = 1.000</td><td>Yes</td></tr>
<tr><td>G8.1.1.9-Cos phi 3 = 0.900</td><td>Yes</td></tr>
<tr><td>G8.1.1.9a-Cos phi 4 = 0.900</td><td>Yes</td></tr>
</table>

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.1.1.10-V activation = 105.00 %	70.00 to 110.00 %	Allows to define the minimum voltage to activate the algorithm. **Note:** This parameter is displayed if [G8.0.4-CSI Q control mode = P Cos Phi].	Yes
G8.1.1.11-V deactivation = 98.00 %	0.00 to 100.00 %	Allows to define the minimum voltage to deactivate the algorithm. **Note:** This parameter is displayed if [G8.0.4-CSI Q control mode = P Cos Phi].	Yes

The graphic below shows the characteristic curve cos φ = f (P).



cos φ = f (P) characteristic curve

Subgroup G8.1.2-VSI control

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.1.2.1-P reference = 0.00 %	-100.00 to 100.00 %	Allows to configure the P reference (active power) for VSI control type with VISMA control subtype.	Yes
G8.1.2.2-Freq increment = 0.000 Hz	-32.768 to 32.767 Hz	Allows to configure the increment of the frequency reference from nominal for VSI control type with VISMA control subtype.	Yes
G8.1.2.3-Freq sequence = Positive Sequence	Positive Sequence Negative Sequence	Allows to configure the grid sequence generated for the VSI control type with VISMA control subtype. OPT. / MODE / DESCRIPTION 0 / Positive Sequence / Direct sequence voltage is generated (RST). 1 / Negative Sequence / Reverse sequence voltage is generated (RTS).	No
G8.1.2.4-Q reference = 0.00 %	-100.00 to 100.00 %	Allows to configure the Q reference (reactive power) for VSI control type with VISMA control subtype. **Note**: this parameter is displayed if [G8.0.6-Q VSI control] ≠ Iq.	Yes
G8.1.2.4-Iq Reference = 0.00 %	-100.00 to 100.00 %	Allows setting the reactive current reference Iq for VSI voltage source mode and VISMA subtype. **Note**: this parameter is displayed if [G8.0.6-Q VSI control] = Iq.	Yes
G8.1.2.5-Vac Reference = 100.00 %	85.00 to 115.00 %	Allows to configure the Vac reference (AC voltage) for VSI control type with VISMA control subtype.	Yes
G8.1.2.6-Vd Reference = 0.00 %	0.00 to 120.00 %	Allows to configure the Vd reference in open loop (direct axis voltage) for VSI control type. **Notes**: – This parameter will be displayed if [G8.0.6-Q VSI control = Vd FAC]. – Only users with a role higher than Admin can modify this parameter.	Yes

6.7.3. Subgroup G8.2: Advanced config

G8.2.1- Control Loops

G8.2.1.1- Sync-freq meas

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.1.1-Kp pll = 10.0 %	0.0 to 100.0 %	Sets the PLL proportional constant.	Yes
G8.2.1.1.2-Ki pll = 10.0 %	0.0 to 100.0 %	Sets the PLL integral constant.	Yes
G8.2.1.1.3-Kp freq = 10.0 %	0.0 to 100.0 %	Sets the value of Kp for frequency control.	Yes
G8.2.1.1.4-Ki freq = 10.0 %	0.0 to 100.0 %	Sets the value of Ki for frequency control.	Yes

G8.2.1.2- Swing equation

This subgroup allows configuring the advanced control of the VISMA machine.

Note: these parameters will be displayed if [SG10.1.19-Visma enable = Enabled].

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G8.2.1.2.1-H = 0.1 s</td><td>0.1 to 10.0 s</td><td rowspan="2">Allows to configure the inertia constant of VISMA machine.
Note:
-These parameters are shown if [G1.3-User] ≥ Qualified customer.
-The parameter with value with a single decimal place (default value = 0.1 s) is hidden from Power Electronics remote management tool from version 4.2.1, but its Modbus address (88257) can be accessed via SCADA.</td><td>Yes</td></tr>
<tr><td>G8.2.1.2.1-H = 0.10 s</td><td>0.01 to 10.00 s</td><td>Yes</td></tr>
<tr><td>G8.2.1.2.2-Kd = 30</td><td>1 to 1000</td><td>Allows to configure the damping gain of VISMA machine.</td><td>Yes</td></tr>
<tr><td>G8.2.1.2.3-w0 = Meas</td><td>Fix
Meas
Meas decoupled
Internal meas</td><td>Allows to configure the damping type of VISMA machine.
<table><tr><th>OPT.</th><th>TYPE</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>Fix</td><td>Allows to configure a fixed frequency for the damping feedback of the oscillation equation.
The reference applied is the one set at [G8.1.2.2-Freq increment].</td></tr><tr><td>1</td><td>Meas</td><td>Allows to configure a measured frequency for the damping feedback of the oscillation equation.</td></tr><tr><td>2</td><td>Meas decoupled</td><td>Allows to configure a measured frequency for the damping feedback of the oscillation equation with an algorithm that decouples the action of the inverter itself.</td></tr><tr><td>3</td><td>Internal meas</td><td>The reference Wref will be obtained by feedback of the rotor speed calculated and passed through a low pass filter. The cutoff frequency of this filter will be configurable by Power Electronics staff.
Note: when this option is selected, parameter G8.2.1.2.3a will be enabled.</td></tr></table>
Note: the read and write privileges changed to role Power Electronics staff in version 4.10.0. From version 5.3.0, this parameter is not available.</td><td>Yes</td></tr>
<tr><td>G8.2.1.2.3a-Low pass filter = 0.25 Hz</td><td>0.01 to 300.00 Hz</td><td>Allows to set the cut-off frequency of the rotor speed filtering. This filtered rotor speed will be used as input of the speed reference in the Damping power computation.
Note: this parameter is displayed when [G8.2.1.2.2.3-w0] = Internal meas.</td><td>Yes</td></tr>
</table>

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.2.4-P feedback configP = GRID	GRID Direct Sequence	Allows to configure the type of active power feedback of the swing equation. OPT. / TYPE / DESCRIPTION 0 / GRID / Uses the measured voltage. 1 / Direct Sequence / Uses the direct sequence of the measure voltage. **Note**: This parameter is displayed if [SG10.1.4-Inverter type = PCS/MPCS]. From version 5.3.0, this parameter is not available.	Yes

G8.2.1.3- Vac loops

This subgroup allows to configure the AC voltage control.

Note: parameters G8.2.1.3.3 to G8.2.1.3.8 will be displayed if [SG10.1.19-Visma enable = Enabled].

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SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.3.1-Kp Vac = 10.00 %	0.00 to 100.00 %	Allows to configure the proportional gain of the AC voltage controller.	Yes
G8.2.1.3.2-Ki Vac = 10.00 %	0.00 to 100.00 %	Allows to configure the integral gain of the AC voltage controller.	Yes
G8.2.1.3.2b-Kerr Vac = 0 %	0.00 to 100.00 %	Parameterization for dynamic PI. Improved transient regulation.	Yes
G8.2.1.3.3-Kp Iq = 10.00 %	0.00 to 100.00 %	Allows to configure the proportional gain of the reactive current controller.	Yes
G8.2.1.3.4-Ki Iq = 10.00 %	0.00 to 100.00 %	Allows to configure the integral gain of the reactive current controller.	Yes
G8.2.1.3.4b-Kerr Iq = 0 %	0.00 to 100.00 %	Parameterization for dynamic PI. Improved transient regulation.	Yes
G8.2.1.3.5-Kp freq Vac loop = 10.00 %	0.00 to 100.00 %	Allows to configure the proportional gain of the frequency current controller.	Yes
G8.2.1.3.6-Ki freq Vac loop = 10.00 %	0.00 to 100.00 %	Allows to configure the integral gain of the frequency current controller.	Yes
G8.2.1.3.7-Vac loop fb config = Direct Sequence	GRID Direct Sequence	Allows to configure the voltage AC feedback type. OPT. / TYPE / DESCRIPTION 0 / GRID / Uses the measured voltage. 1 / Direct Sequence / Uses the direct sequence of the measure voltage.	Yes
G8.2.1.3.8-Iq config = Slow	Slow Fast	Allows to configure the reactive current feedback type. OPT. / TYPE / DESCRIPTION 0 / Slow / Allows to configure the Iq current control in the DSP. 1 / Fast / Allows to configure the Iq current control in the FPGA. **Note:** This parameter only applies if the inverter has one module.	Yes

G8.2.1.4- Vdc loops

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.4.1-Kp Vdc = 5.0 %	0.0 to 100.0 %	Sets the DC voltage loop proportional constant.	Yes
G8.2.1.4.2-Ki Vdc = 10.0 %	0.0 to 100.0 %	Sets the DC voltage loop integral constant.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.4.3-Kp Vdc night = 5.0 %	0.0 to 100.0 %	Sets the Kp value for Vdc during night. If this parameter is set to 0, the PID cannot react to the night mode Q demand. This control keeps the DC bus voltage stable in case of reactive variations that may be required from the inverter.	Yes
G8.2.1.4.4-Ki Vdc night = 10.0 %	0.0 to 100.0 %	Sets the Ki value for Vdc during night. If this parameter is set to 0, the PID cannot react to the night mode Q demand. This control keeps the DC bus voltage stable in case of reactive variations that may be required from the inverter.	Yes
G8.2.1.4.5-Kp Mod = 10.0 %	0.0 to 100.0 %	Proportional constant PI control gains for the inverter current.	Yes
G8.2.1.4.6-Ki Mod = 10.0 %	0.0 to 100.0 %	Integral constant PI control gains for the inverter current.	Yes
G8.2.1.4.7-Kerror Vdc MPPt = 50.00 %	0.00 to 200.00 %	Gain which is applied for the increase in the regulation speed of the field voltage bus controller in case of loss of irradiance.	Yes
G8.2.1.4.8-Sat Kp Vdc MPPt = 50.00 %	0.00 to 200.00 %	Maximum limit to which the Kp value is allowed to increase.	Yes
G8.2.1.4.9-Sat Ki Vdc MPPt = 50.00 %	0.00 to 200.00 %	Maximum limit to which the Ki value is allowed to increase.	Yes

Note: To avoid an oscillation in the DC bus from appearing, parameters [G8.2.1.4.1-Kp Vdc] and [G8.2.1.4.2-Ki Vdc] must be configured according to the values set in the operation and start-up gradients [G4.2.1-Start].

G8.2.1.5- Current loops

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.5.1-Kp I = 10.0 %	0.0 to 100.0 %	Sets the current loop proportional constant.	Yes
G8.2.1.5.2-Ki I = 10.0 %	0.0 to 100.0 %	Sets the current loop integral constant.	Yes
G8.2.1.5.3-Kp I inverse = 10.0 %	0.0 to 100.0 %	Sets the proportional constant PI control gains for the inverter current.	Yes
G8.2.1.5.4-KI I inverse = 10.0 %	0.0 to 100.0 %	Sets the integral constant PI control gains for the inverter current.	Yes
G8.2.1.5.5-Inverse sequence = Yes	No Yes	Enables or disables current injection with inverse sequence.	Yes
G8.2.1.5.6-Kres inverse VRT = 1.0 %	0.0 to 100.0 %	Resonant filter gain adjustment for VRT events. **Note:** This parameter only applies if [G4.3.1.1-Enable] ≠ Disabled.	Yes
G8.2.1.5.7-Kp I inverse VRT = 10.0 %	0.0 to 100.0 %	Sets the proportional gain of the reverse current loop control PI for VRT events.	Yes
G8.2.1.5.8-Ki I inverse VRT = 10.0 %	0.0 to 100.0 %	Sets the integral gain of the reverse current loop control PI for VRT events.	Yes

G8.2.1.6- Power sharing

This subgroup allows to configure the control of active and reactive power distribution between inverters in the same plant by Droop control.

The equation applied is:

$$X'=X+kX*(Yref-Ymeas)/Yavailable$$

X = V or f

Y = P or Q

Ymeas = value of the power delivered at each moment.

Yavailable = value so that the droop is proportional to the available power.

Note: these parameters will be displayed if [SG10.1.19-Visma enable = Enabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.6.1-kV(Q) = 5.00 %	-100.00 to 100.00 %	Allows to configure the VQ droop gain.	Yes
G8.2.1.6.2-kV(P) = 0.00 %	-100.00 to 100.00 %	Allows to configure the VP droop gain.	Yes
G8.2.1.6.3-kf(P) = 2.00 %	-100.00 to 100.00 %	Allows to configure the fP droop gain.	Yes
G8.2.1.6.4-kf(Q) = 0.00 %	-100.00 to 100.00 %	Allows to configure the fQ droop gain.	Yes
G8.2.1.6.5-P = 0.00 %	-100.00 to 100.00 %	Allows the balancing to be centered at the expected P.	Yes
G8.2.1.6.6-Q = 0.00 %	-100.00 to 100.00 %	Allows the balancing to be centered at the expected Q.	Yes
G8.2.1.6.7-Sharing mode = Absolut	Absolut Relative	Allows to configure the power sharing of the inverter. OPT. / TYPE / DESCRIPTION 0 / Absolut / Allows to configure the load sharing respect to the nominal power of the inverter. 1 / Relative / Allows to configure the load sharing respect to the relative power of the inverter. **Note:** The number of stopped modules is taken into account for load sharing.	Yes

G8.2.1.7-Speed limits

This subgroup allows configuring the controllers for frequency or speed limiting.

Note: these parameters will be displayed if [SG10.1.19-Visma enable = Enabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.7.1-Kp speed limit = 10.00 %	0.00 to 100.00 %	Allows setting the proportional gain of the frequency or speed limiting controller.	Yes
G8.2.1.7.2-Ki speed limit = 10.00 %	0.00 to 100.00 %	Allows setting the integral gain of the frequency or speed limiting controller.	Yes

G8.2.1.8-Vac limits

This subgroup allows configuring the controllers for AC voltage limiting.

Note: these parameters will be displayed if [SG10.1.19-Visma enable = Enabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.8.1-Kp Vac limit = 10.00 %	0.00 to 100.00 %	Allows setting the proportional gain of the AC voltage limiting controller.	Yes
G8.2.1.8.2-Ki Vac limit = 10.00 %	0.00 to 100.00 %	Allows setting the integral gain of the AC voltage limiting controller.	Yes

G8.2.1.9-Module Balancing

This subgroup allows configuring a regulation that balances Id and Iq between modules.

Note: these parameters will be displayed if [SG10.1.19-Visma enable = Enabled].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.9.1-Enable = Yes	No Yes	Allows to enable the balancing of currents between modules. **Note:** This parameter only applies if [SG4.3-Open loop = Yes].	Yes
G8.2.1.9.2-Kp dVdq = 10.00 %	0.00 to 100.00 %	Proportional configuration of the balancing loop.	Yes
G8.2.1.9.3-Ki dVdq = 10.00 %	0.00 to 100.00 %	Integral configuration of the balancing loop.	Yes

G8.2.1.10: Dynamic Limit

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.10.1-Kp Id = 10.00%	0.00 to 100.00%	Value of the Kp of the PI of the fast limit of Id.	Yes
G8.2.1.10.2-Ki Id = 10.00%	0.00 to 100.00%	Value of the Ki of the PI of the fast limit of Id.	Yes
G8.2.1.10.3-Kerr = 0.00%	0.00 to 100.00%	Parameterization for dynamic PI. Improved transient regulation.	Yes
G8.2.1.10.4-K Pac = 10.00%	0.00 to 100.00%	Gain value to act on Pac and limit current.	Yes
G8.2.1.10.5-K Vac = 10.00%	0.00 to 100.00%	Gain value to act on Vac and limit current.	Yes
G8.2.1.10.6-Mode = Speed	Speed Angle	Mode of action of the Id limit. It acts on the speed or on the angle.	Yes

G8.2.1.11: Overcurrent limit

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.11.0-Enable = Yes	No Yes	Allows to enable or disable the overcurrent control in VISMA. By default, it is enabled. It would only be disabled when the FPGA voltage measurement does not work and the unit must be re-started, losing all the dynamic performance of the unit.	Yes
G8.2.1.11.1-Kp I Limit = 100.00%	0.00 to 100.00%	Parameterization of the Kp of the PI for overcurrent control.	Yes
G8.2.1.11.2-Ki I limit = 100.00%	0.00 to 100.00%	Parameterization of the PI Ki for overcurrent control.	Yes
G8.2.1.11.3-Time limit = 10 ms	0 to 1000 ms	Time at which the overcurrent control is triggered.	Yes
G8.2.1.11.4-Threshold = 1750 A	100 to 2200 A	Threshold at which the overcurrent control is activated. Its value may be automatically adjusted according to the following conditions: • If the parameter has NOT been manually modified, its value is automatically adjusted if [SG10.1.19-Visma enable] = Enabled. In this case, the value is calculated based on [SG10.1.9b-VISMA Max Iac], as follows: [SG10.1.9b-VISMA Max Iac] / [G8.2.1.11.4-Threshold] ≤ 836 A / 1750 A Between 836 A and 920 A / Linear interpolation between 1750 A and 1536 A > 920 A / 1536 A • In the following cases, the parameter retains the previously configured value: ○ The above conditions are not met. ○ The parameter has been manually modified.	Yes

G8.2.1.12: Homopolar

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.1.12.1-Kp	0.00 to 100.00%	Parameterization of the Kp of the PR for the homopolar control of the fundamental harmonic.	Yes
G8.2.1.12.2-Kres 1h	0.00 to 100.00%	Parameterization of the Kp of the PR for the homopolar control of the fundamental harmonic.	Yes
G8.2.1.12.3-Kres 3h	0.00 to 100.00%	Parameterization of the Kp of the PR for the homopolar control of the third harmonic.	Yes

G8.2.2: Alt Control Loops

Note: these parameters will be displayed if [G1.3-User] ≥ Qualified customer. They are enabled when [G8.2.3-Enable Alt. Config] = Yes

G8.2.2.2: Alt Swing equation

This subgroup allows configuring the advanced control of the VISMA machine in the alternative configuration.

Note: these parameters will be displayed if [SG10.1.19-Visma enable] = Enabled.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.2.2.1-Alternative H = 0.10 s	0.01 to 10.00 s	Allows to configure the inertia constant of VISMA machine in the alternative configuration.	Yes
G8.2.2.2.2-Alternative Kd = 30	1 to 1000	Allows to configure the damping gain of VISMA machine in the alternative configuration.	Yes

G8.2.2.3: Alt Vac loops

This subgroup allows to configure the AC voltage control in the alternative configuration.

Note: these parameters will be displayed if [G1.3-User] ≥ Power Electronics staff.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.2.3.1-Alternative Kp Vac = 10.00 %	0.00 to 100.00%	Allows to configure the proportional gain of the AC voltage controller of the VISMA machine in the alternative configuration.	Yes
G8.2.2.3.2-Alternative Ki Vac = 10.00 %	0.00 to 100.00%	Allows to configure the integral gain of the AC voltage controller of the VISMA machine in the alternative configuration.	Yes
G8.2.2.3.3-Alternative Kp Iq = 10.00 %	0.00 to 100.00%	Allows to configure the proportional gain of the reactive current controller of the VISMA machine in the alternative configuration. **Note:** this parameter is only displayed if [SG10.1.19-Visma enable] = Enabled.	Yes
G8.2.2.3.4-Alternative Ki Iq = 10.00 %	0.00 to 100.00%	Allows to configure the integral gain of the reactive current controller of the VISMA machine in the alternative configuration. **Note:** this parameter is only displayed if [SG10.1.19-Visma enable] = Enabled.	Yes

Others:

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.2.3-Enable Alt. Config. = No	No Yes	Allows to activate the alternative configuration. **Note:** this parameter is shown if [G1.3-User] ≥ Qualified customer.	Yes

6.7.4. Subgroup G8.3: Grid

This group permits the configuration of the parameters necessary to define the inverter connection point.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.3.1-SCR = 20.000 %	Disabled = 20.001 0.000 to 20.000 %	Defines the short-circuit radius.	Yes
G8.3.2-X/R = 10.000 %	0.000 to 10.000 %	Defines the X/R ratio of the short-circuit impedance. **Note:** This parameter is only displayed if [G8.3.1 = 0.000 to 20.000].	Yes
G8.3.3-To POI XI = 0.000 %	0.000 to 60.000 %	These are line parameters. They are defined by Power Electronics for each plant and are related to the distance between the POI and the plant. **Note:** Do not modify this parameter without authorization from Power Electronics.	Yes
G8.3.4-To POI R = 0.000 %	0.000 to 60.000 %		Yes

6.7.5. Subgroup G8.4: Batteries

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff] and [SG10.1.4-Inverter type = PCS/MPCS].

G8.4.1- Model

The static battery model consists of a single resistor in series with a voltage source. This model is used to estimate the Open Circuit Voltage (OCV), in other words, the voltage at which the battery returns zero current. The aim is to improve the logic behavior of the inverter in a simple way.

The result of this model is applied for start conditions [G2.9: Busbar Start], stop conditions [G2.10: Busbar Stop], and active balancing [G4.11: Active balancing]. The problem it tries to solve is the rebound effect that can cause a multitude of starts and stops due to the conditions applied with the direct voltage of the batteries or the balancing error due to setpoint differences between both busbars.

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.4.1.1-R1 = Disabled	Disabled = 0.000 0.001 to 60.000 %	Sets the internal resistance of the batteries connected to the first bus of MPCS / MPCSK inverter. For HEM / HEMK inverters set this parameter to zero. **Note:** This parameter is only displayed if [SG10.1.14 Number of busbar ≥ 1].	Yes
G8.4.1.2-R2 = Disabled	Disabled = 0.000 0.001 to 60.000 %	Sets the internal resistance of the batteries connected to the second bus of MPCS / MPCSK inverter. For HEM / HEMK inverters set this parameter to zero. **Note:** This parameter is only displayed if [SG10.1.14 Number of busbar ≥ 2].	Yes
G8.4.1.3-R3 = Disabled	Disabled = 0.000 0.001 to 60.000 %	Sets the internal resistance of the batteries connected to the third bus of MPCS / MPCSK inverter. For HEM / HEMK inverters set this parameter to zero. **Note:** This parameter is only displayed if [SG10.1.14-Number of busbar ≥ 3].	Yes
G8.4.1.4-R4 = Disabled	Disabled = 0.000 0.001 to 60.000 %	Sets the internal resistance of the batteries connected to the fourth bus of MPCS / MPCSK inverter. For HEM / HEMK inverters set this parameter to zero. **Note:** This parameter is only displayed if [SG10.1.14-Number of busbar ≥ 4].	Yes

G8.4.2-Busbar Power

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.4.2.1-Busbar 1 Power = 200.0 %	10.0 to 500.0 %	Defines the percentage of power in busbar DC 1.	Yes
G8.4.2.2-Busbar 2 Power = 200.0 %	10.0 to 500.0 %	Defines the percentage of power in busbar DC 2.	Yes
G8.4.2.3-Busbar 3 Power = 200.0 %	10.0 to 500.0 %	Defines the percentage of power in busbar DC 3.	Yes
G8.4.2.4-Busbar 4 Power = 200.0 %	10.0 to 500.0 %	Defines the percentage of power in busbar DC 4.	Yes
G8.4.2.5-Busbar 1 Ratio C = 0.25	0.04 to 5.00	Defines the DC ratio C of the batteries connected to busbar 1.	Yes
G8.4.2.6-Busbar 2 Ratio C = 0.25	0.04 to 5.00	Defines the DC ratio C of the batteries connected to busbar 2	Yes
G8.4.2.7-Busbar 3 Ratio C = 0.25	0.04 to 5.00	Defines the DC ratio C of the batteries connected to busbar 3.	Yes
G8.4.2.8-Busbar 4 Ratio C = 0.25	0.04 to 5.00	Defines the DC ratio C of the batteries connected to busbar 4.	Yes

6.7.6. Subgroup G8.5: Seamless transition

This subgroup allows to configure the delay applied when switching between CSI and VSI control modes, ensuring a seamless transition.

Note: this subgroup is displayed if [G1.3-User] ≥ Power Electronics staff, [SG10.1.19-Visma enable] = Enabled, and [SG10.1.20-Enable seamless] = Enabled.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G8.5.1-Active control type = CSI	CSI VSI	Shows the control type configured in [G8.0.1-Control type]. When a switch between CSI and VSI is commanded, this parameter retains the previous value until the configured delay expires. **Note**: this parameter is read-only (RO).	Yes
G8.5.2-Delay type = Automatic	Manual Automatic	Allows to choose whether the delay applied during the transition between control modes is managed manually using [G8.5.5-Remaining delay] or automatically using [G8.5.4-Automatic delay].	Yes
G8.5.3-Delay = 2.0 s	0.1 to 6000.0 s	Allows to set the delay time that must elapse before a control mode change is applied when [G8.5.2-Delay type] = Manual. The inverter will wait for this time before activating the new CSI/VSI mode. **Note**: this parameter is shown if [G8.5.2-Delay type] = Automatic.	Yes
G8.5.4-Automatic delay = 1.0 s	0.1 to 6000.0 s	Shows the delay time that the system will automatically apply during the transition between control modes when [G8.5.2-Delay type] = Automatic. The inverter will wait for this time before activating the new CSI/VSI mode. **Note**: this parameter is read-only (RO).	Yes
G8.5.5-Remaining delay = 1.0 s	0.1 to 6000.0 s	Shows the remaining time of the delay currently in progress during a transition between control modes. When a change from CSI to VSI or vice versa is commanded, the inverter starts a countdown based on [G8.5.3-Delay] or [G8.5.4-Automatic delay], and this parameter indicates how much time remains before the new control mode takes effect. **Note**: this parameter is read-only (RO).	Yes

6.8. Group G9: IMD

This group allows the configuration of the insulation monitoring device (IMD) information. This meter is able to measure the earth resistance of the inverter. If this is too low, the inverter is configured internally to go to a non-risk situation until the technical staff can perform a shunt to earth. There are two types of insulation testers: two-channel (one channel to measure AC and one channel to measure DC) and five-channel (one channel to measure AC and 4 channels to measure DC).

Notes:

- Normally, for PCSM/K inverter, the five-channel option is used. For HEM/K inverter, the two-channel option is used.
- This group is displayed if [G1.3-User ≥ Power Electronics staff].

6.8.1. Subgroup G9.1: General

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G9.1.1-Enable = Yes	No Yes	Enables / disable the IMI device. Note: only users with role Developer can modify this parameter.	Yes
G9.1.2-Autodiagnosis Period = 1440 min	Disabled = 5 6 to 1440 min	Configures the IMI selfdiagnosis period. This time configured, the IMI will start its selfdiagnosis to check if it still measures properly, measuring an intern and known resistance. **Note:** This parameter is only displayed if [G9.1.1-Enable = Yes].	Yes
G9.1.3-Autodiagnosis Forze = No	No Yes	Allows to force a selfdiagnosis at any time. **Note:** This parameter is only displayed if [G9.1.1-Enable = Yes].	Yes

6.8.2. Subgroup G9.2: Channels

G9.2.1-Channel AC

Note: Parameters within this subgroup are only displayed if [G9.1.1-Enable = Yes] and [G9.2.1.1-Enable = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G9.2.1.1-Enable = Yes	No Yes	Enables / disables the AC channel of the IMI to start considering what the IMI measures in this channel. **Note:** only users with role Developer can modify this parameter.	Yes
G9.2.1.2-Fault Threshold = 5.0 kOhms	0.5 to 1500.0 kOhms	Configures the fault resistance threshold for AC channel. If the IMI measures a resistance under this threshold, then it triggers the respective fault.	Yes
G9.2.1.3-Fault Recover Thrhld = 115 %	101 to 130 %	Defines the hysteresis to clear the isolation fault for AC channel.	Yes
G9.2.1.4-Warning Threshold = 50.0 kOhms	0.5 to 2000.0 kOhms	Defines the warning resistance threshold for AC channel. If the IMI measures a resistance under this threshold, then it triggers the respective warning.	Yes
G9.2.1.5-No Measure Timeout = 3000 min	1 to 6000 min	Defines the maximum time that an available channel can be without measuring before having a failure. If this time is expired, then the fault “F229:Timeout IMI Channel” will be triggered.	Yes

G9.2.2-Channel DC 1

Note: Parameters within this subgroup are only displayed if [G9.1.1-Enable = Yes] and [G9.2.2.1-Enable = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G9.2.2.1-Enable = Yes	No Yes	Enables / disables the DC 1 channel of the IMI to start considering what the IMI measures in this channel. **Note:** Only users with role Developer can modify this parameter.	Yes
G9.2.2.2-Fault Threshold = 5.0 kOhms	0.5 to 1500.0 kOhms	Configures the fault resistance threshold for DC 1 channel. If the IMI measures a resistance under this threshold, then it is triggered the respective fault.	Yes
G9.2.2.3-Fault Recover Thrhld = 115 %	101 to 130 %	Defines the hysteresis to clear the isolation fault for DC 1 channel.	Yes
G9.2.2.4-Warning Threshold = 50.0 kOhms	0.5 to 2000.0 kOhms	Defines the warning resistance threshold for DC 1 channel. If the IMI measures a resistance under this threshold, then it triggers the respective warning.	Yes
G9.2.2.5-No Measure Timeout = 3000 min	1 to 6000 min	Defines the maximum time that an available channel can be without measuring before having a failure. If this time is expired, then the fault "F229:Timeout IMI Channel" will be triggered.	Yes

G9.2.3- Channel DC 2

Note: Parameters within this subgroup are only displayed if [G9.1.1-Enable = Yes] and [G9.2.3.1-Enable = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G9.2.3.1-Enable = Yes	No Yes	Enables / disables the DC 2 channel of the IMI to start considering what the IMI measures in this channel. **Note:** only users with role Developer can modify this parameter.	Yes
G9.2.3.2-Fault Threshold = 5.0 kOhms	0.5 to 1500.0 kOhms	Configures the fault resistance threshold for DC2 channel. If the IMI measures a resistance under this threshold, then it triggers the respective fault.	Yes
G9.2.3.3-Fault Recover Thrhld = 115 %	101 to 130 %	Defines the hysteresis to clear the isolation fault for DC2 channel.	Yes
G9.2.3.4-Warning Threshold = 50.0 kOhms	0.5 to 2000.0 kOhms	Defines the warning resistance threshold for DC2 channel. If the IMI measures a resistance under this threshold, then it triggers the respective warning.	Yes
G9.2.3.5-No Measure Timeout = 3000 min	1 to 6000 min	Defines the maximum time that an available channel can be without measuring before having a failure. If this time is expired, then the fault "F229:Timeout IMI Channel" will be triggered.	Yes

G9.2.4- Channel DC 3

Note: Parameters within this subgroup are only displayed if [G9.1.1-Enable = Yes] and [G9.2.4.1-Enable = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G9.2.4.1-Enable = Yes	No Yes	Enables / disables the DC 3 channel of the IMI to start considering what the IMI measures in this channel. **Note:** only users with role Developer can modify this parameter.	Yes
G9.2.4.2-Fault Threshold = 5.0 kOhms	0.5 to 1500.0 kOhms	Configures the fault resistance threshold for DC3 channel. If the IMI measures a resistance under this threshold, then it triggers the respective fault.	Yes
G9.2.4.3-Fault Recover Thrhld = 115 %	101 to 130 %	Defines the hysteresis to clear the isolation fault for DC3 channel.	Yes
G9.2.4.4-Warning Threshold = 50.0 kOhms	0.5 to 2000.0 kOhms	Defines the warning resistance threshold for DC3 channel. If the IMI measures a resistance under this threshold, then it triggers the respective warning.	Yes
G9.2.4.5-No Measure Timeout = 3000 min	1 to 6000 min	Defines the maximum time that an available channel can be without measuring before having a failure. If this time is expired, then the fault "F229:Timeout IMI Channel" will be triggered.	Yes

EN

G9.2.5- Channel DC 4

Note: Parameters within this subgroup are only displayed if [G9.1.1-Enable = Yes] and [G9.2.5.1-Enable = Yes].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G9.2.5.1-Enable = Yes	No Yes	Enables / disables the DC 4 channel of the IMI to start considering what the IMI measures in this channel. **Note:** only users with role Developer can modify this parameter.	Yes
G9.2.5.2-Fault Threshold = 5.0 kOhms	0.5 to 1500.0 kOhms	Configures the fault resistance threshold for DC 4 channel. If the IMI measures a resistance under this threshold, then it triggers the respective fault.	Yes
G9.2.5.3-Fault Recover Thrhld = 115 %	101 to 130 %	Defines the hysteresis to clear the isolation fault for DC 4 channel.	Yes
G9.2.5.4-Warning Threshold = 50.0 kOhms	0.5 to 2000.0 kOhms	Defines the warning resistance threshold for DC 4 channel. If the IMI measures a resistance under this threshold, then it triggers the respective warning.	Yes
G9.2.5.5-No Measure Timeout = 3000 min	1 to 6000 min	Defines the maximum time that an available channel can be without measuring before having a failure. If this time is expired, then the fault "F229:Timeout IMI Channel" will be triggered.	Yes

6.8.3. Subgroup G9.3: Maneuver

Note: Parameters within this subgroup are only displayed if [G9.1.1-Enable = Yes].

G9.3.0-Manual Maneuver

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G9.3.0.1-Enable = Disabled	Disabled Enabled	Enable / disable the manual maneuver.	Yes
G9.3.0.2-Measure Channel = AC	AC DC 1 DC 2 DC 3 DC 4 No measuring	Sets the channel to be measured.	Yes
G9.3.0.3-GFDI Output = Open	Open Closed Shooted Undetermined	Sets the GFDI status. STATUS / DESCRIPTION Open – The GFDI is open. Closed – The GFDI is closed. Shooted – The GFDI output indicating overcurrent trip is activated. Undetermined – GFDI signals both open and closed, or neither **Nota**: este parámetro se muestra si [SG10.6.1-DU Type] = GFDI + IMD.	Yes
G9.3.1-Timeout Measure = 40 min	20 to 80 min	Configures a time within a new measure of the IMI is expected. If this time has passed, then a new fault is triggered to inform that the IMI cannot finish its measuring algorithm.	Yes
G9.3.2-Timeout GFDI Feedback = 5.0 s	0.1 to 10.0 s	Configures the time within the GFDI digital input feedback is expected. If this time has passed, then a new fault is triggered to inform that the action of the GFDI and its feedback is not consistent. **Note**: this parameter is shown if [SG10.6.1-DU Type] = GFDI + IMD.	Yes
G9.3.3-Minimum Channel Voltag = 100 V	25 to 1000 V	Configures the minimum voltage of the bus that needs the IMI to start measuring the isolation resistance.	Yes
G9.3.4-GFDI Forze Pending = No	No Yes	Allows to reset the daily GFDI maneuver pending flag. **Note:** This parameter is displayed if [SG10.6.1-DU Type] = GFDI + IMD.	Yes
G9.3.5-GFDI Init Time = 10 s	1 to 20 s	Allows to configure how long the system can wait for the system to boot after a power-up. During this time, the system ignores the status of the digital outputs of the GFDI device. **Note:** This parameter is displayed if [SG10.6.1-DU Type] = GFDI + IMD.	Yes

6.8.4. Subgroup G9.4: Advanced

Note: Parameters within this subgroup are only displayed if [G9.1.1-Enable] = Yes and if [G1.3-User] ≥ Power Electronics staff.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G9.4.1-Rp Error Offset = 1000 Ohms	500 to 2000 Ohms	Allows to set the offset defined by Rp1-Rp2.	Yes
G9.4.2-Rp Error Variation = 10 %	1 to 100 %	Allows to set the variation defined by Rp1-Rp2.	Yes
G9.4.3-Estabilization Delta = 35	5 to 500	Sets the number of raw allowed in 10 seconds.	Yes
G9.4.4-Estabiliz Linear Error = 6	1 to 100	Sets the linear error allowed in RAW.	Yes
G9.4.5-Estabiliz Timeout = 600 s	60 to 1800 s	Sets the pulse stabilization timeout.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G9.4.6-Tau Number = 5	1 to 10	Sets the number of Tau to give the pulse as stabilized.	Yes
G9.4.7-Tau Variation = 10 %	1 to 100 %	Sets the Tau variation calculated to give Tau as correct.	Yes
G9.4.8-Reboot IMD = No	No Yes	Sends reboot to IMI.	Yes
G9.4.9-Rp Negative Min = - 1500 Ohms	-10000 to -500 Ohms	Allows to set the selection of the minimum negative resistance (Rp) to be expected. This means that with a true 0 ohms, the IMD circuit can read an Rp<0, which will eventually saturate to 0, due to measurement inaccuracies and calibrations. However, if a very low Rp<0 is obtained, it is probably due to an abrupt bus change or a negatively synchronized MMPT. This protection allows to eliminate sporadic false negatives to 0 in the measurement.	Yes
G9.4.10-Stable Needed T = 0 ms	0 to 10000 ms	Defines the time to elapse before indicating that a stabilized pulse is stable.	Yes
G9.4.11-Filter N samples = 10	1 to 20	Allows configuring the number of samples of the MAV (Moving Average Filter) applied to the IMD insulation resistance measurement. **Note**: MAV filtering does not alter the IMD's internal measurement algorithm or its timing, applying only to the resistance value used for displaying and evaluating warnings and faults.	Yes
G9.4.12-Filter Tmax = 10 min	1 to 3600 min	Allows setting the maximum sampling time T of the MAV filter applied to the IMD insulation resistance measurement.	Yes

EN

6.9. Group G10: DU

This group allows to configure which channels of the DC cabinet are physically connected and which are going to be used to calculate the currents of the inverter. The DC cabinet has up to 40 channels for current measurement.

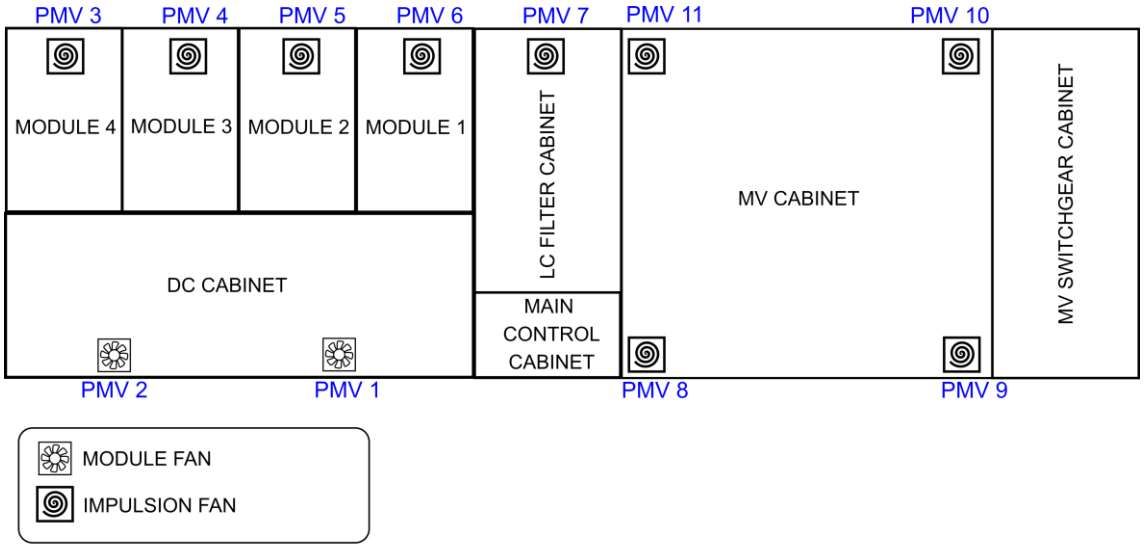
Note: This group is displayed if [G1.3-User ≥ Power Electronics staff] and [SG10.6.1-DU Type ≠ Disabled].

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G10.1-Channels 1 to 10 = XXXXXXXXXX</td><td rowspan="4">0000000000 to XXXXXXXXXX</td><td rowspan="4">Allows to configure the channels of the DU. Each bit (from right to left) represents a channel, as described in the following table as an example for the group of the first 10 channels.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Channel 1</td><td>5</td><td>Channel 6</td></tr>
<tr><td>1</td><td>Channel 2</td><td>6</td><td>Channel 7</td></tr>
<tr><td>2</td><td>Channel 3</td><td>7</td><td>Channel 8</td></tr>
<tr><td>3</td><td>Channel 4</td><td>8</td><td>Channel 9</td></tr>
<tr><td>4</td><td>Channel 5</td><td>9</td><td>Channel 10</td></tr>
</table></td><td>Yes</td></tr>
<tr><td>G10.2-Channels 11 to 20 = XXXXXXXXXX</td><td>Yes</td></tr>
<tr><td>G10.3-Channels 21 to 30 = XXXXXXXXXX</td><td>Yes</td></tr>
<tr><td>G10.4-Channels 31 to 40 = XXXXXXXXXX</td><td>Yes</td></tr>
</table>

6.10.Group G11: Fans

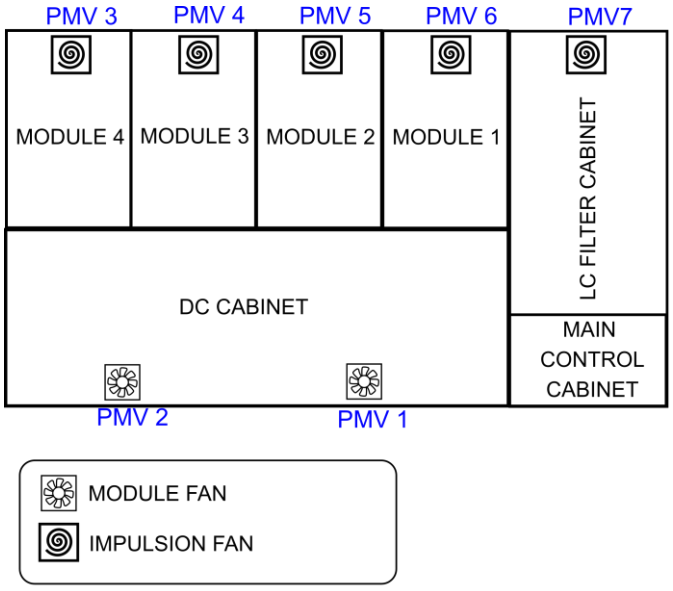
Inverters can support the control of up to eleven fans that communicate with the central control via RS485. The distribution and location of the fans depend on the product as shown in the following pictures.

In case of **HEM / PCSM** the distribution is as follows:



HEM / PCSM inverters' fan distribution

While for **HEMK / PCSK** the distribution is shown below:



HEMK / PCSK inverters' fan distribution

Each fan has an associated software identifier (ID) for Modbus access. The following table lists each fan in the different cabinets with its corresponding identification in the software.

CABINET	ELECTRIC DIAGRAM NOMENCLATURE	ZONE IDENTIFICATION AND SOFTWARE PARAMETERS	MODBUS IDENTIFICATION
DC cabinet	PMV 1	Fan DU 1 (DU1)	ID 10
	PMV 2	Fan DU 2 (DU2)	ID 11
Modules cabinet	PMV 3	Fan module 4 (M4)	ID 12
	PMV 4	Fan module 3 (M3)	ID 13
	PMV 5	Fan module 2 (M2)	ID 14
	PMV 6	Fan module 1 (M1)	ID 15
LC filter cabinet	PMV 7	Fan LC filter (LC)	ID 16
MV cabinet*	PMV 8	Fan MT1 (MT1)	ID 17
	PMV 9	Fan MT2 (MT2)	ID 18
	PMV 10	Fan MT3 (MT3)	ID 19
	PMV 11	Fan MT4 (MT4)	ID 20

*Only HEM / PCSM inverters.

There are three fan manufacturers: **EBM, TESOER and DELTA**. Each fan has an associated reference as shown in the table below:

REFERENCE	MANUFACTURER	POWER SUPPLY
L7812A	EBM	Three phase
L7793B	EBM	Single phase
L8821A	TESOER	Three phase
L8822	TESOER	Single phase
L9210	DELTA	Three phase
L9169	DELTA	Single phase
L9587	DELTA	Three phase

Regardless of the fan reference number, fans from different fan manufacturers must not be mixed within the same zone. It is possible to install fans of different references in the same product as long as the condition of not mixing different fans in the same zone is respected.

The following table shows the fan installation possibilities for the different zones.

REFERENCE	MANUFACTURER	ZONES			
		MT	MODULES	LC FILTER	DC CABINET
L7812A	EBM				
L7793B	EBM				
L8821A	TESOER				
L8822	TESOER				
L9210	DELTA				
L9169	DELTA				
L9587	DELTA				

If fans from different manufacturers are mixed in the same area, one of the following faults will be triggered: F230, F231 or F232 (see fan related faults).

It should be noted that each time a fan is changed, the default values must be checked and readjusted to the values given in the tables below. There are two exceptions: DELTA L9169 and L9210 fans, which must be set to the values given below.

Subgroup G11.3: DU Configuration

Parameters	By default	DELTA L9169
[G11.3.2-Setpoint]	80%	94%

Subgroup G11.4: MT Configuration

Parameters	By default	DELTA L9210
[G11.4.2-Setpoint]	80%	92%
[G11.4.7-Minimum speed]*	50%	76%
[G11.4.8-Maximum speed]*	80%	92%
[G11.4.9-Minimum setpoint]	50%	76%

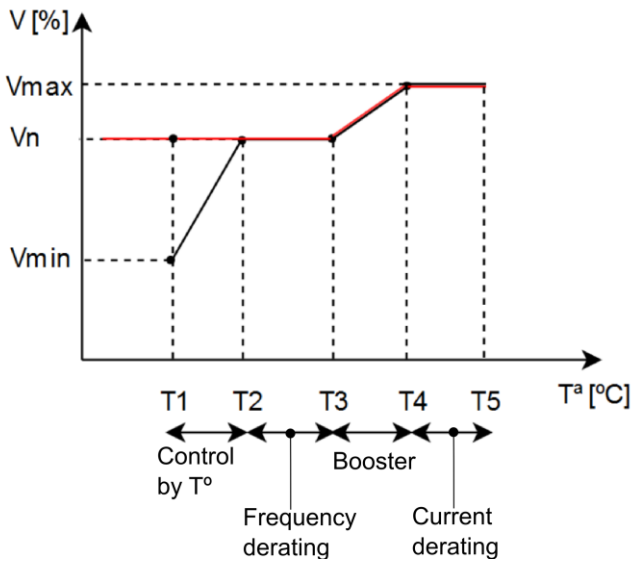
* Only visible in "Temperature" control mode. Parameters not available from version 2.1.0.

NOTICE

The products leave the factory with the ventilation configured. If a fan is to be replaced, it may be necessary to reconfigure it. To do so, please refer to the corresponding documentation.

In parallel, as for the ventilation system, the software includes the booster algorithm. This algorithm increases the fan speed when certain temperature conditions are fulfilled and after frequency derating is completed. Booster is only available to modules cabinet. Parameters to configure the booster algorithm are located in the [G11.5: Config Modules] subgroup and it is only shown if [G11.5.1-Mode] = Run.

The next picture shows the operation of the booster algorithm:



Booster algorithm

On the other hand, the logic implemented for the communication lost protection between the main control board and the fans changes in this product version. When communication between these devices is lost, a warning or a fault will be triggered, depending on the status of the inverter:

- If the inverter is stopped (status of the DSP is stopped) or working (status of the DSP is on), then a warning will be triggered (W161: Fan impulsion MT, W162: Fan DU zone, W163: Fan zone filter LC or W164: Fan zone modules).
- In other situations, the respective fault will be triggered (F204: Fan impulsion MT, F205: Fan DU zone, F206: Fan zone filter LC or F207: Fan zone modules).

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

EN

6.10.1. Subgroup G11.1: Detection

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G11.1.1a-Zone Search = None</td><td>None
DU zone
MT zone
Modules zone</td><td>Allows to choose which fan zone needs to be searched.
Note: MT zone option is only available for HEM / PCSM inverter. This parameter is only displayed if [G11.1.4-Fan to detect= EBM-Papst] and [G11.1.3-Detection Mode = Automatic + zone].</td><td>Yes</td></tr>
<tr><td>G11.1.1ab-Fan to search = Fan DU 1</td><td>Fan DU 1
Fan DU 2
Fan module 1
Fan module 2
Fan module 3
Fan module 4
Fan LC filter
Fan MT 1
Fan MT 2
Fan MT 3
Fan MT 4</td><td>Allows to choose which fan needs to be searched.
Note: Fan MT 1, 2, 3 and 4 options are only available for HEM / PCSM inverter. This parameter is only displayed if [G11.1.4-Fan to detect = Tesoer] and [G11.1.3-Detection Mode = Automatic + individual].</td><td>Yes</td></tr>
<tr><td>G11.1.1b-Start Search = No</td><td>No
Yes</td><td>Starts the automatic search algorithm.</td><td>Yes</td></tr>
<tr><td>G11.1.2-Search State = Disabled</td><td>Disabled
Not detected
Detected
Searching
Automatically
Searching Manually</td><td>Indicates the status of the fan search.</td><td>Yes</td></tr>
<tr><td>G11.1.3-Detection Mode = Automatic + individual</td><td>Automatic + zone
Automatic + individual
Manual</td><td>Allows to choose if the process to search the fans is manual or automatic, but since the version 4.0.0, this parameter cannot be modified, it is fixed and the process to search will be always “Automatic + individual”.
Note: This parameter is only displayed if [G11.1.4-Fan to detect = EBM-Papst].</td><td>Yes</td></tr>
<tr><td>G11.1.4-Fan to detect = EBM-Papst</td><td>EBM-Papst
Tesoer
Delta</td><td>Allows to know which fans provider is going to be used.</td><td>Yes</td></tr>
<tr><td>G11.1.5-Search Request = Unknown</td><td>0 to 21</td><td>Sets the search request. These are states of the internal state machine of the fans.
<table>
<tr><th>OPT.</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Unknown</td><td>Unknown status in the search process.</td></tr>
<tr><td>1</td><td>Update Master config</td><td>Tries to synchronize the RS485 communication properties between the master (microcontroller of the main control board) and the slaves (the fans).</td></tr>
<tr><td>2</td><td>Set password 1</td><td rowspan="3">It is specific to EBM fans. Sets an internal password to allow modification of certain password-protected parameters.</td></tr>
<tr><td>3</td><td>Set password 2</td></tr>
<tr><td>4</td><td>Set password 3</td></tr>
<tr><td>5</td><td>Set default config</td><td>It is specific to EBM fans. Allows to default to the fan's internal parameters.</td></tr>
<tr><td>6</td><td>Reset to apply config</td><td>It is specific to EBM fans. Allows to reset the ventilator in order to save the new configuration made.</td></tr>
<tr><td>7</td><td>Broadcast id 1</td><td>Search fan by broadcast.</td></tr>
<tr><td>8</td><td>Reset</td><td>Command to reset fan.</td></tr>
</table></td><td>Yes</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td></td><td></td><td>
<table>
<tr><th>OPT.</th><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>9</td><td>Search by S/N</td><td>Search fan by serial number.</td></tr>
<tr><td>10</td><td>Reset by S/N</td><td>Command to reset the fan by its serial number.</td></tr>
<tr><td>11</td><td>Check ID</td><td>Check ID fan.</td></tr>
<tr><td>12</td><td>Check default ID</td><td>Check ID of the fan which is communicating.</td></tr>
<tr><td>13</td><td>Set default configuration</td><td>Allows to default to the fans internal parameters.</td></tr>
<tr><td>14</td><td>Iterative id search</td><td>Iterative search for the fan by its ID.</td></tr>
<tr><td>15</td><td>Set new ID</td><td>Assign new ID to the fan.</td></tr>
<tr><td>16</td><td>Check new ID</td><td>Check that the ID has been correctly assigned to the fan.</td></tr>
<tr><td>17</td><td>Check Delta default ID</td><td>It is specific to Delta fans. Checks if it communicates with the default ID of these fans.</td></tr>
<tr><td>18</td><td>Set new Delta baudrate</td><td>It is specific to Delta fans. Sets correctly the baud rate of these fans (one of the RS485 properties). If the baud rate is not 19200, the microcontroller forces the fan to have this baud rate to allow communication.</td></tr>
<tr><td>19</td><td>Iterative id search</td><td>Iterative search for the fan by its ID.</td></tr>
<tr><td>20</td><td>Set new ID</td><td>Assign new ID to the fan.</td></tr>
<tr><td>21</td><td>Check new Delta ID</td><td>Check that the ID has been correctly assigned to the fan.</td></tr>
</table>
</td><td></td></tr>
<tr><td>G11.1.6-Fan 1 SN = 0000000000</td><td>0000000000 to 995900ZZZZ</td><td>Sets the serial number of the fan 1.
Note: This parameter is only displayed if [G11.11- Avaliables Fan Nº ≥ 1] and ([G11.1.4-Fan to detect = EBM-Papst] or [G11.1.3-Detection Mode = Manual]).</td><td>Yes</td></tr>
<tr><td>G11.1.7-SN 1 Found = Not detected</td><td>Not detected
Searching
Detected</td><td>Indicates if the fan 1 has been detected.
Note: This parameter is only displayed if [G11.11- Avaliables Fan Nº ≥ 1]</td><td>Yes</td></tr>
<tr><td>G11.1.8-Fan 2 SN = 0000000000</td><td>0000000000 to 995900ZZZZ</td><td>Sets the serial number size of fan 2.
Note: This parameter is only displayed if [G11.11- Avaliables Fan Nº ≥ 2] and ([G11.1.4-Fan to detect = EBM-Papst] or [G11.1.3-Detection Mode = Manual]).</td><td>Yes</td></tr>
<tr><td>G11.1.9-SN 2 Found = Not detected</td><td>Not detected
Searching
Detected</td><td>Indicates if the fan 2 has been detected.
Note: This parameter is only displayed if ([G11.1.1a-Zone Search = None] or [G11.1.1a-Zone Search = DU zone] or [G11.1.1a-Zone Search = MT zone] or [G11.1.1a-Zone Search = Modules zone]) and ([G11.1.3-Detection Mode = Automatic + zone] or [G11.1.4-Fan to detect = EBM-Papst]).</td><td>Yes</td></tr>
<tr><td>G11.1.10-Fan 3 SN = 0000000000</td><td>0000000000 to 995900ZZZZ</td><td>Sets the serial number size of fan 3.
Note: This parameter is only displayed if ([G11.1.4-Fan to detect = EBM-Papst] or [G11.1.3-Detection Mode = Manual]) and [G11.11- Avaliables Fan Nº ≥ 3].</td><td>Yes</td></tr>
<tr><td>G11.1.11-SN 3 Found = Not detected</td><td>Not detected
Searching
Detected</td><td>Indicates if the fan 3 has been detected.
Note: This parameter is only displayed if ([G11.1.1a-Zone Search = None] or [G11.1.1a-Zone Search = MT zone] or [G11.1.1a-Zone Search = Modules zone]) and ([G11.1.3-Detection Mode = Automatic + zone] or [G11.1.4-Fan to detect = EBM-Papst]).</td><td>Yes</td></tr>
<tr><td>G11.1.12-Fan 4 SN = 0000000000</td><td>0000000000 to 995900ZZZZ</td><td>Sets the serial number size of fan 4.
Note: This parameter is only displayed if ([G11.1.4-Fan to detect = EBM-Papst] or [G11.1.3-Detection Mode = Manual]) and [G11.11- Avaliables Fan Nº ≥ 4].</td><td>Yes</td></tr>
</table>

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.1.13-SN 4 Found = Not detected	Not detected Searching Detected	Indicates if the fan 4 has been detected. **Note:** This parameter is only displayed if ([G11.1.1a-Zone Search = None] or [G11.1.1a-Zone Search = MT zone] or [G11.1.1a-Zone Search = Modules zone]) and ([G11.1.3-Detection Mode = Automatic + zone] or [G11.1.4-Fan to detect = EBM-Papst]).	Yes
G11.1.14-Fan 5 SN = 0000000000	0000000000 to 995900ZZZZ	Sets the serial number size of fan 5. **Note:** This parameter is only displayed if [G11.1.3-Detection Mode = Manual] and [G11.11- Avaliables Fan N° ≥ 5].	Yes

6.10.2. Subgroup G11.2: Model assign

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.1-DU Fan 1 Model = EBM-Papst	EBM-Papst Tesoer Delta	They allow configuring the fan models of the different inverter zones.	Yes
G11.2.2-DU Fan 2 Model = EBM-Papst			
G11.2.3-Module Fan 1 Model = EBM-Papst			
G11.2.4-Module Fan 2 Model = EBM-Papst			
G11.2.5-Module Fan 3 Model = EBM-Papst			
G11.2.6-Module Fan 4 Model = EBM-Papst			
G11.2.7-Filter LC Fan Model = EBM-Papst			
G11.2.8-MT Fan 1 Model = EBM-Papst			
G11.2.9-MT Fan 2 Model = EBM-Papst			
G11.2.10-MT Fan 3 Model = EBM-Papst			
G11.2.11-MT Fan 4 Model = EBM-Papst			

G11.2.1-Fan 1

This group of parameters is not available from version 5.0.0.

Note: These parameters are only displayed if [G11.9- Avaliables Fan Nº ≥ 1].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.1.1-Asigned Zone = DU zone	DU zone Global zone	Sets the assigned zone.	Yes
G11.2.1.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK.	No
G11.2.1.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: • Emb past. • Tesoer. • Delta.	Yes

G11.2.2-Fan 2

This group of parameters is not available from version 5.0.0.

Note: These parameters are only displayed if [G11.9- Avaliables Fan Nº ≥ 2].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.2.1-Asigned Zone = DU zone	DU zone Global zone	Sets the assigned zone.	Yes
G11.2.2.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK	No
G11.2.2.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: • Emb past • Tesoer • Delta	Yes

G11.2.3-Fan 3

This group of parameters is not available from version 5.0.0.

Note: These parameters are only displayed if [G11.9- Avaliables Fan Nº ≥ 3].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.3.1-Asigned Zone = Modules zone	Modules zone Global zone	Sets the assigned zone	Yes
G11.2.3.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK	No
G11.2.3.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: • Emb past. • Tesoer. • Delta.	Yes

EN

G11.2.4-Fan 4

This group of parameters is not available from version 5.0.0.

Note: These parameters are only displayed if [G11.9- Avaliables Fan Nº ≥ 4].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.4.1-Asigned Zone = Modules zone	Modules zone Global zone	Sets the assigned zone.	Yes
G11.2.4.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK	No
G11.2.4.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: • Emb past. • Tesoer. • Delta.	Yes

G11.2.5-Fan 5

This group of parameters is not available from version 5.0.0.

Note: these parameters are only displayed if [G11.9- Avaliables Fan Nº ≥ 5].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.5.1-Asigned Zone = Modules zone	Modules zone Global zone	Sets the assigned zone	Yes
G11.2.5.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK	No
G11.2.5.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: • Emb past • Tesoer • Delta	Yes

G11.2.6-Fan 6

This group of parameters is not available from version 5.0.0.

Note: these parameters are only displayed if [G11.9- Avaliables Fan Nº ≥ 6].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.6.1-Asigned Zone = Modules zone	Modules zone Global zone	Sets the assigned zone	Yes
G11.2.6.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK	No
G11.2.6.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: • Emb past. • Tesoer. • Delta.	Yes

G11.2.7-Fan 7

This group of parameters is not available from version 5.0.0.

Note: these parameters are only displayed if [G11.9- Avaliables Fan Nº ≥ 7].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.7.1-Asigned Zone = LC filter zone	LC filter zone Global zone	Sets the assigned zone	Yes
G11.2.7.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK	No
G11.2.7.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: • Emb past. • Tesoer. • Delta.	Yes

EN

G11.2.8-Fan 8

This group of parameters is not available from version 5.0.0.

Note: These parameters are only displayed if [G11.9- Avaliables Fan Nº ≥ 8].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.8.1-Asigned Zone = MT zone	MT zone Global zone	Sets the assigned zone.	Yes
G11.2.8.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK.	No
G11.2.8.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: • Emb past. • Tesoer. • Delta.	Yes

G11.2.9-Fan 9

This group of parameters is not available from version 5.0.0.

Note: these parameters are only displayed if [G11.9- Avaliables Fan Nº ≥ 9].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.9.1-Asigned Zone = MT zone	MT zone Global zone	Sets the assigned zone.	Yes
G11.2.9.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK.	No
G11.2.9.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: • Emb past. • Tesoer. • Delta.	Yes

G11.2.10-Fan 10

This group of parameters is not available from version 5.0.0.

Note: these parameters are only displayed if [G11.9- Avaliables Fan Nº ≥ 10].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.10.1-Asigned Zone = MT zone	MT zone Global zone	Sets the assigned zone.	Yes
G11.2.10.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK.	No
G11.2.10.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: • Emb past. • Tesoer. • Delta.	Yes

G11.2.11-Fan 11

This group of parameters is not available from version 5.0.0.

Note: these parameters are only displayed if [G11.9- Avaliables Fan Nº ≥ 11].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.2.11.1-Asigned Zone = MT zone	MT zone Global zone	Sets the assigned zone.	Yes
G11.2.11.2-Test = No	No Yes	Allows to launch a test to the fan. The fan is commanded to rotate to check which fan it is communicating with and if its status is OK.	No
G11.2.11.3-Type = Emb-Papst	Emb-Papst Tesoer Delta	Defines the type of fan used: • Emb past. • Tesoer. • Delta.	Yes

6.10.3. Subgroup G11.3: DU config

This subgroup is used to configure the DU zone ventilation system, i.e. all properties that define the behavior of the fans that are installed in that zone.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.3.1-Mode = Run	Disabled Run Fixed	Configures the fans mode to start working. **STATUS** / **DESCRIPTION** Disable: Disable the zone ventilation. **Note**: this option will only be available if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges]. Run: The zone fans will rotate when the inverter is ON. Fixed: Fans rotate at all times, independent of the inverter status.	Yes
G11.3.2-Setpoint = 80 %	80 to 100 %	Configures the speed of the fans. **Note**: if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 100 %.	Yes
G11.3.3-Delay off after run = 1200 s	1200 to 3600 s	Configures the time after which the fans should stop when they work with Run mode. **Note**: if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 3600 s.	Yes

6.10.4. Subgroup G11.4: MT config

This subgroup is used to configure the MT zone ventilation system, i.e. all properties that define the behavior of the fans that are installed in that zone.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.4.1-Mode = Run	Disabled Run Fixed Temperature	Configures the fans mode to start working. **STATUS** / **DESCRIPTION** Disable: Disable the zone ventilation. **Note**: this option will only be available if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges]. Run: The zone fans will rotate when the inverter is ON. Fixed: Fans rotate at all times, independent of the inverter status. Temperature: The fans rotate when the temperature conditions configured in the parameters are met, following a straight line between the minimum and maximum speed and temperature values. **Note**: this option is disabled since version 3.3.0.	Yes
G11.4.2-Setpoint = 80 %	80 to 100 %	Configures the speed of the fans. **Note**: if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 100 %.	Yes
G11.4.3-Delay off after run = 2400 s	2400 to 3600 s	Configures the time after which the fans should stop when they work with Run mode. **Note**: if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 3600 s.	Yes

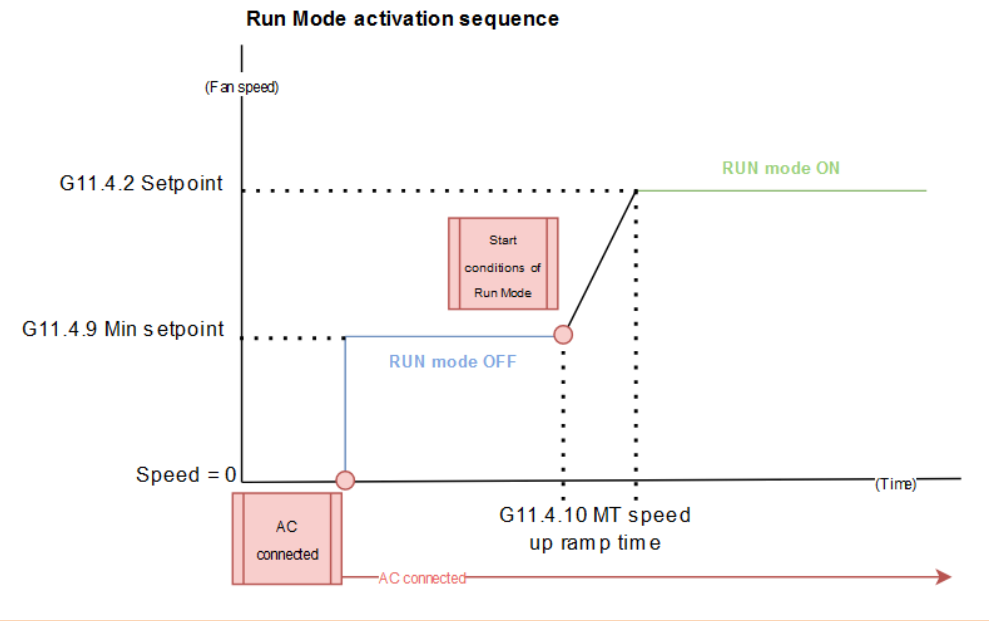
EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.4.5-Tª Activation = 100 ºC	80 to 140 ºC	Defines the maximum temperature when they work with the temperature mode. **Note**: This parameter is only displayed if [G11.4.1-Mode] = Temperature[12].	Yes
G11.4.6-Tª Deactivation = 60 ºC	20 to 100 ºC	Defines the minimum temperature when they work with the temperature mode. **Note**: This parameter is only displayed if [G11.4.1-Mode] = Temperature[12].	Yes
G11.4.7-Minimum speed = 50	50 to [G11.4.8-Maximum speed]	Defines the minimum speed at which the fan starts working when the mode is configured as temperature mode. **Note**: This parameter is only displayed if [G11.4.1-Mode] = Temperature[12].	Yes
G11.4.8-Maximum speed = 80	[G11.4.7-Minimum speed] to 100	Defines the maximum speed when the mode is configured as temperature mode. **Note**: This parameter is only displayed if [G11.4.1-Mode] = Temperature[12].	Yes
G11.4.9-Minimum setpoint = 50%	50 % to [G11.4.2-Setpoint]	Defines the minimum speed at which the MV cabinet fans would spin when the system is connected to AC. The next condition is added to the already developed MV switchgear ventilation algorithm: – If [G11.4.1-Mode = Run], the MV transformer is connected and the inverter is not running (this means that the IGBTs are off and the time configured in [G11.4.3-Delay off after run] is expired), the fans of this area should remain spinning at the speed configured in G11.4.9 parameter. **Note**: if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 100 %.	Yes
G11.4.10-Speed up ramp time = 120 s	0 to 120 s	Defines the ramp time to increase the fan speed from the minimum to the configured setpoint when IGBTs are on. When the conditions to stop the "Run" and "Temperature" mode in G11.4.1 are fulfilled, the fans will change its setpoint to the one configured in G11.4.9 parameter and they will remain spinning at this setpoint until the conditions to trigger the "Run" or "Temperature[12]" mode are active again. This parameter configures the ramp to switch from the G11.4.9 setpoint to the new setpoint of the "Run/Temperature[12]". See the graphs for this parameter below this table. **Note:** This parameter is displayed if [G11.4.1-Mode = Run].	Yes
G11.4.11-Speed down ramp time = 1 s	0 to 3600 s	Defines the ramp time to decrease the fan speed from the configured setpoint to the minimum speed when IGBTs are off. When the conditions to stop the "Run" and "Temperature[12]" mode from G11.4.1 are fulfilled, the fans will change its setpoint to the one configured in G11.4.9 parameter. This parameter configures the ramp to switch from the previous setpoint to the new [G11.4.9 setpoint]. See the graphs for this parameter below this table. **Notes:** - This parameter is displayed if [G11.4.1-Mode = Run]. - If the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 3600 s.	Yes

[12] This option is disabled since version 3.3.0.

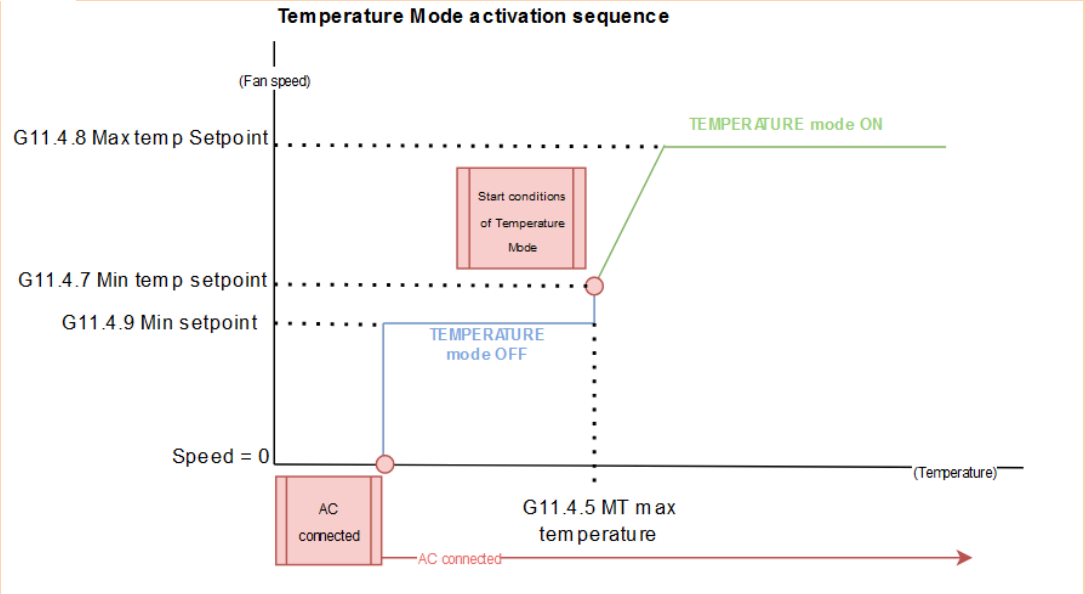
Next pictures show the behavior of the fans from MV cabinet fans when the RUN mode is configured in relation to the parameter [G11.4.10-Speed up ramp time].

– If parameter [G11.4.1-Mode] is configured as “Run” the behavior is as follows:



Activation sequence of the MV cabinet fans in Run mode

– On the other hand, if [G11.4.1-Mode] is configured as “Temperature[13]” the behavior is as follows:

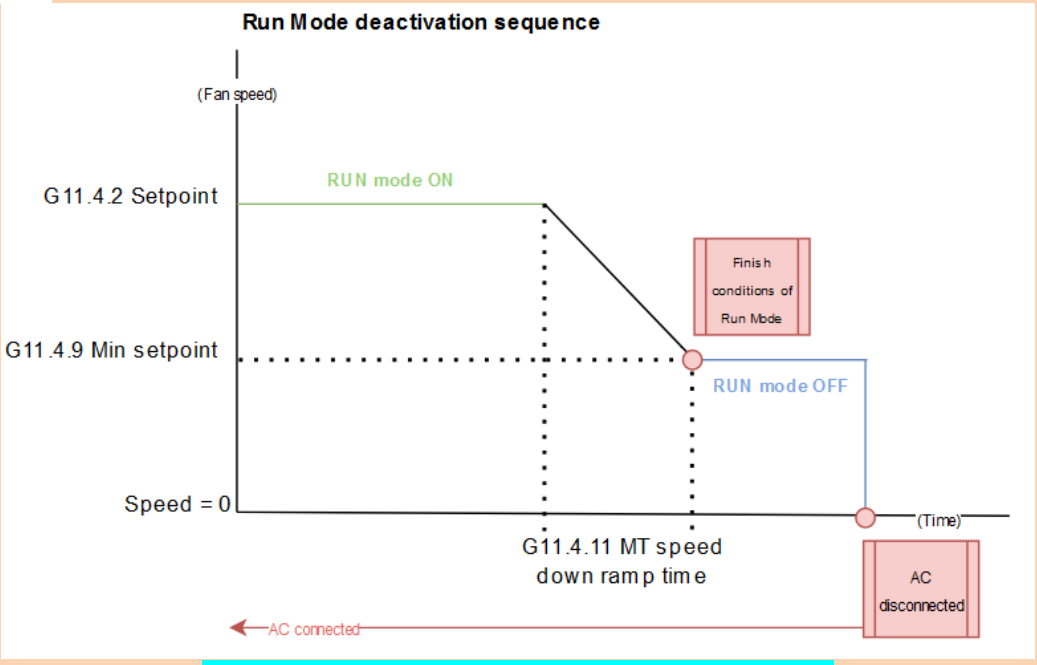


Activation sequence of the MV cabinet fans in Temperature mode

[13] This option is disabled since version 3.3.0.

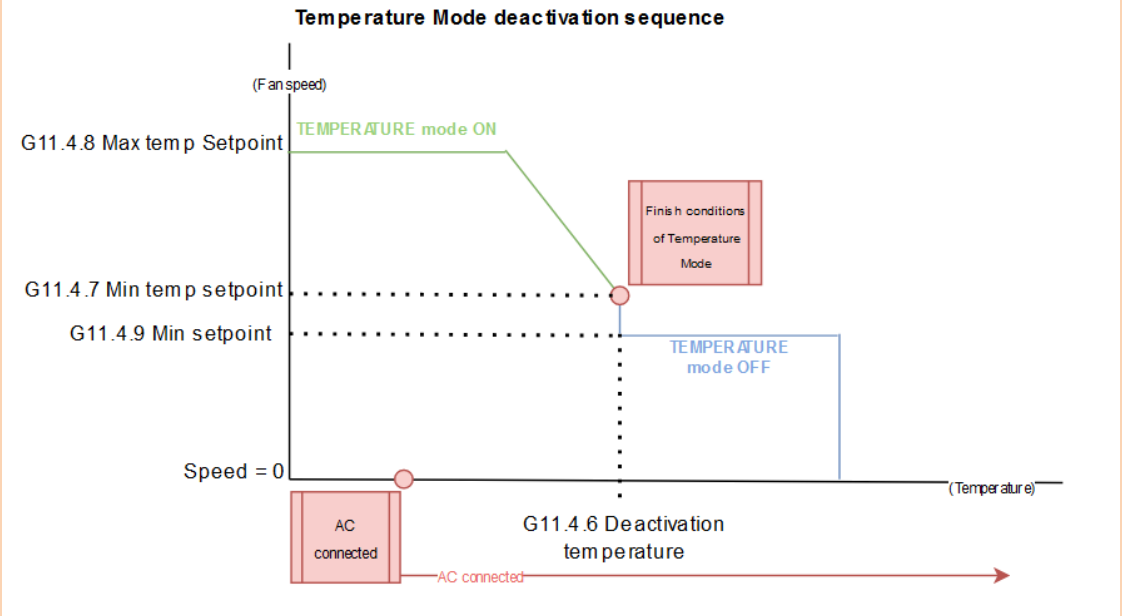
Similarly, the behaviour of the fans from MV cabinet in relation to parameter [G11.4.11-Speed down ramp time] is shown below:

– If parameter [G11.4.1-Mode] is configured as “Run” the behavior is as follows:



Deactivation sequence of the MV cabinet fans in Run mode

– On the other hand, if [G11.4.1-Mode] is configured as “Temperature[14]” the behavior is as follows:



Deactivation sequence of the MV cabinet fans in Temperature mode

[14] This option is disabled since version 3.3.0.

6.10.5. Subgroup G11.5: Config Modules

This subgroup is used to configure the modules zone ventilation system, i.e. all properties that define the behavior of the fans that are installed in that zone.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.5.0-Control = General	General Individual	Allows to configure if the fans are controlled individually or not. **Note:** This parameter is only displayed if [G11.5.1-Mode ≠ Temperature]. **STATUS** / **DESCRIPTION** General: All four fans of the four modules are controlled at the same time. Individual: All 4 fans are controlled individually; different conditions can be configured. **Note**: this option will only be available if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges].	Yes
G11.5.1-Mode = Temperature	Disabled Run Fixed Temperature	Configures the mode. **STATUS** / **DESCRIPTION** Disable: Disable the zone ventilation. **Note**: this option will only be available if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges]. Run: The zone fans will rotate when the inverter is ON. Fixed: Fans rotate at all times, independent of the inverter status. Temperature: The fans rotate when the temperature conditions are given, which are configured with the following parameters, following a straight line between the minimum and maximum values of speed and temperature.	Yes
G11.5.2-Setpoint = 80 %	80 to 100 %	Configures the speed of the fans. **Note:** This parameter is only displayed if [G11.5.1-Modo] ≠ Temperature. **Note**: if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 100 %.	Yes
G11.5.2a-Setpoint module 1 = 80 %	0 to 100 %	Configures the speed of the fans. **Note**: This parameter is only displayed if [G11.5.0-Control = Individual].	Yes
G11.5.2b-Setpoint module 2 = 80 %	0 to 100 %	Configures the speed of the fans. **Note**: This parameter is only displayed if [G11.5.0-Control = Individual].	Yes
G11.5.2c-Setpoint module 3 = 80 %	0 to 100 %	Configures the speed of the fans. **Note**: This parameter is only displayed if [G11.5.0-Control = Individual].	Yes
G11.5.2d-Setpoint module 4 = 80 %	0 to 100 %	Configures the speed of the fans. **Note**: This parameter is only displayed if [G11.5.0-Control = Individual].	Yes
G11.5.3-Delay off after run = 1200 s	1200 to 3600 s	Configures the time after which the fans should stop when they work with Run mode. **Note**: if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 3600 s.	Yes
G11.5.5-Tª minimum = 70 ºC	20 ºC to [G11.5.6-Tª maximum]	Defines the minimum temperature when they work with the temperature mode. **Note**: This parameter is only displayed if [G11.5.1-Mode = Temperature].	Yes
G11.5.6-Tª maximum = 96 ºC	[G11.5.5-Tª minimum] to 140 ºC	Defines the maximum temperature when they work with the temperature mode. **Note**: This parameter is only displayed if [G11.5.1-Mode = Temperature].	Yes
G11.5.7-Minimum speed = 50 %	50 % to [G11.5.8-Maximum speed]	Defines the minimum speed at which the fan start working when the mode is configured as temperature mode. **Note**: This parameter is only displayed if [G11.5.1-Mode = Temperature].	Yes

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SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.5.8-Maximum speed = 100 %	[G11.5.7-Minimum speed] to 100 %	Defines the maximum speed when the mode is configured as temperature mode. **Note**: This parameter is only displayed if [G11.5.1-Mode = Temperature].	Yes
G11.5.9-Run Booster Speed = 100 %	0 to 100 %	Configure the speed of the fans at which they will rotate when the booster algorithm is active, being the control mode of these fans, Run mode configured in [G11.5.1-Mode]. **Note**: This parameter is only displayed if [G11.5.1-Mode = Run].	Yes

6.10.6. Subgroup G11.6: Config LC filter

This subgroup is used to configure the LC filter zone ventilation system, i.e. all properties that define the behavior of the fans that are installed in that zone.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.6.1-Mode = Run	Disabled Run Fixed	Configures the fans mode to start working. **STATUS** / **DESCRIPTION** Disable: Disable the zone ventilation. **Note**: this option will only be available if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges]. Run: The zone fans will rotate when the inverter is ON. Fixed: Fans rotate at all times, independent of the inverter status.	Yes
G11.6.2-Setpoint = 80 %	80 to 100 %	Configures the speed of the fans. **Note**: if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 100 %.	Yes
G11.6.3-Delay off after run = 1200 s	1200 to 3600 s	Configures the time after which the fans should stop when they work with Run mode. **Note**: if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 3600 s.	Yes

6.10.7. Subgroup G11.7: Global config

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G11.7.1-Mode = Run</td><td>Disabled
Run
Fixed
Temperature</td><td>Configures the fans mode to start working.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disable</td><td>Disable the zone ventilation.
Note: this option will only be available if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges].</td></tr><tr><td>Run</td><td>The zone fans will rotate when the inverter is ON.</td></tr><tr><td>Fixed</td><td>Fans rotate at all times, independent of the inverter status.</td></tr><tr><td>Temperature</td><td>The fans rotate when the temperature conditions configured in the parameters are met, following a straight line between the minimum and maximum speed and temperature values.
Note: this option is disabled since version 3.3.0.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G11.7.2-Setpoint = 80 %</td><td>80 to 100 %</td><td>Configures the speed of the fans.
Note: if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 100 %.</td><td>Yes</td></tr>
<tr><td>G11.7.3-Delay off after run = 2400 s</td><td>2400 to 3600 s</td><td>Configures the time after which the fans should stop when they work with Run mode.
Note: if the correct password has been entered in [SG4.35-Pwd Fans Extend ranges], the range of this parameter will be 0 to 3600 s.</td><td>Yes</td></tr>
<tr><td>G11.7.6-Tª Deactivation = 60 ºC</td><td>20 to 100 ºC</td><td>Defines the minimum temperature when they work with the temperature mode.
Note: This parameter is only displayed if [G11.7.1-Mode] = Temperature[15].</td><td>Yes</td></tr>
<tr><td>G11.7.5-Tª Activation = 100 ºC</td><td>80 to 140 ºC</td><td>Defines the maximum temperature when they work with the temperature mode.
Note: This parameter is only displayed if [G11.7.1-Mode] = Temperature[15].</td><td>Yes</td></tr>
<tr><td>G11.7.7-Minimum speed = 50</td><td>50 to [G11.7.8-Maximum speed]</td><td>Defines the minimum speed at which the fan starts working when the mode is configured as temperature mode.
Note: This parameter is only displayed if [G11.7.1-Mode] = Temperature[15].</td><td>Yes</td></tr>
<tr><td>G11.7.8-Maximum speed = 80</td><td>[G11.7.7-Minimum speed] to 100 %</td><td>Defines the maximum speed when the mode is configured as temperature mode.
Note: This parameter is only displayed if [G11.7.1-Mode] = Temperature[15].</td><td>Yes</td></tr>
<tr><td>G11.7.9-Ramp-up time = 15 s</td><td>11 to 65535 s</td><td rowspan="2">Defines the time to reach a higher or lower setpoint for Delta and EBM-Papst fans. When the speed setpoint of the fan changes, the algorithm waits for the ramp-up/down time from these parameters before monitoring the actual speed and comparing it to the setpoint.
After these times, it will be checked if the actual speed is beyond the limits and if it is, the fault “F228: Fan setpoint” will be triggered. Each instance of this fault notifies the fan number that has failed.
• Ramp-up is applied when the actual speed setpoint is lower than the needed speed setpoint.
• Ramp-down is applied when the actual speed setpoint is higher than the needed speed setpoint.</td><td>Yes</td></tr>
<tr><td>G11.7.10-Ramp-down time = 80 s</td><td>61 to 65535 s</td><td>Yes</td></tr>
</table>

[15] This option is disabled since version 3.3.0.

6.10.8. Subgroup G11.8: Additional Reg Read

This group allows to consult the fan Modbus addresses that are not currently displayed in Power Electronics remote management tool in groups [G11: Fans] and / or [SV11: Fans].

Note: parameters G11.8.1 to G11.8.14 can only be seen and modified by users with role Developer.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.8.1-Enable = No	No Yes	Allows to enable or disable the fan custom address functionality.	Yes
G11.8.2-Type = Input Register	Input Register Holding Register	Allows to configure the type of register. OPTION: Input Register – DESCRIPTION: Type of register to be consulted is read. OPTION: Holding register – DESCRIPTION: Type of register to be consulted is read / write. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes].	Yes
G11.8.3-Address = 0x01 Hex	0x00 to 0xFFFF Hex	Allows to configure the address of the fan Modbus map that needs to be asked for. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes].	Yes
G11.8.4-Register Fan 1 = 0	Not available = 6666 0 to 65535	Sets the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes] and [G11.11-Avaliables Fan Nº ≥ 1].	Yes
G11.8.5-Register Fan 2 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes] and [G11.11-Avaliables Fan Nº ≥ 2].	Yes
G11.8.6-Register Fan 3 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes] and [G11.11-Avaliables Fan Nº ≥ 3].	Yes
G11.8.7-Register Fan 4 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes] and [G11.11-Avaliables Fan Nº ≥ 4].	Yes
G11.8.8-Register Fan 5 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes] and [G11.11-Avaliables Fan Nº ≥ 5].	Yes
G11.8.9-Register Fan 6 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes] and [G11.11-Avaliables Fan Nº ≥ 6].	Yes
G11.8.10-Register Fan 7 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes] and [G11.11-Avaliables Fan Nº ≥ 7].	Yes
G11.8.11-Register Fan 8 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes] and [G11.11-Avaliables Fan Nº ≥ 8].	Yes
G11.8.12-Register Fan 9 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes] and [G11.11-Avaliables Fan Nº ≥ 9].	Yes
G11.8.13-Register Fan 10 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes] and [G11.11-Avaliables Fan Nº ≥ 10].	Yes
G11.8.14-Register Fan 11 = 0	Not available = 6666 0 to 65535	Shows the result of the configured register. **Note:** This parameter is only displayed if [G11.8.1-Enable = Yes] and [G11.11-Avaliables Fan Nº ≥ 11].	Yes

6.10.9. Subgroup G11.8: Fan test

Note: This subgroup is displayed if [G1.3-User] ≥ Power Electronics staff.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.9.1-DU Fan 1 Test = No	No Yes	Allow a test to be performed on the required fan. The corresponding fan is commanded to rotate to check which fan it is communicating with and if its status is OK.	No
G11.9.2-DU Fan 2 Test = No			
G11.9.3-Module Fan 1 Test = No			
G11.9.4-Module Fan 2 Test = No			
G11.9.5-Module Fan 3 Test = No			
G11.9.6-Module Fan 4 Test = No			
G11.9.7-LC Filter Fan Test = No			
G11.9.8-MT Fan 1 Test = No			
G11.9.9-MT Fan 2 Test = No			
G11.9.10-MT Fan 3 Test = No			
G11.9.11-MT Fan 4 Test = No			

EN

6.10.10. Subgroup G11.10: Lost frames

This group allows configuring the limit of frames lost by each fan and the monitoring time of these frames, which establish the activation conditions of the W196 warning.

Note: This subgroup is displayed if [G1.3-User] ≥ Power Electronics staff.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.10.1-Lost Frames Counter = 5	1 to 65535	Allows to set the limit of incorrect frames in a period defined by parameter [G11.10.2-Monitor time].	No
G11.10.2-Monitoring time = 60 s	1 to 65535	Allows to set the monitoring time for incorrect frames.	No
G11.10.3-Warning196 lowerer = No	No Yes	Allows to remove W196 from all fans.	No

Others

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G11.11-Avaliables Fan Nº = 11	7 to 11	Shows the number of the impulsion fans of the inverter.	No
G11.12-Fault Cond. LV Time = 10 s	10 to 600 s	Indicates the time in which, in the event of a voltage drop below the threshold defined in [G11.13- Fault Cond. LV Voltage], the inverter fails.	No
G11.13-Fault Cond. LV Voltage = 70 %	70 to 100 %	Indicates the percentage of the operating voltage above which undervoltage is considered, and the time to failure defined in [G11.12 Fault Cond. LV Time] begins to apply.	No
G11.14-Setpoint deviation err = 30 %	0 to 50 %	Allows to configure the admissible error between the fan setpoint and the actual speed.	No
G11.15-Setpoint error delay = 60 s	1 to 120 s	Once the conditions of triggering the fault “F228: Fan setpoint” are met, it is necessary to wait the time configured in this parameter before trigger the F228.	No

6.11.Group G12: SD card

All the inverters are equipped with an SD card that is physically connected to the main control board. The main function of the SD card is to store information for further post-processing and to analyse the behaviour of the inverter.

In this group, some general properties of the SD can be configured (such as its enablement), as well as some important data on the card status. In addition, it is possible to configure the custom log. It consists in a log that stores the data of a series of previously configured parameters, as well as everything related to the Blackbox (logs supplied by the module FPGAs). The following table shows the content of each log.

Note: This subgroup is displayed if [G1.3-User ≥ Power Electronics staff].

LOG NUMBER	DESCRIPTION
Log 1	This log saves information about grid current, grid voltage, reactive and active power, limits, and frequency. It is stored every 0,5 seconds. Information is obtained from the following parameters: • SV3.2.4-Grid current I1 LV • SV3.2.5-Grid current I2 LV • SV3.2.6-Grid current I3 LV • SV3.2.1-Grid voltage RS LV • SV3.2.2-Grid voltage ST LV • SV3.2.3-Grid voltage TR LV • G4.10.3-Vbus reference • G4.10.2-Q reference • SV3.2.7-P LV • SV3.2.8-Q LV • SV3.2.12-Grid freq • G3.1.1-P limit Discharge • G3.1.2-Q limit
Log 2	This log saves information about grid current. Variables from this log will be saved per module. It is stored every 0,5 seconds. Information is obtained from the following parameters: • SV12.1.5-Debug • SV12.1.6-Inputs/Outputs • SV12.1.2.4-DC Current • SV12.1.2.1-Current R • SV12.1.2.2-Current S • SV12.1.2.3-Current T
Log 3	This log saves information about the temperature measurement of the main inductance sensors, ambient temperature of each module PCB, internal temperature of the PCB, air intake temperature of the air entering the inverter, humidity of the PCB of each module, humidity measured on the main control board and temperature of the medium voltage transformer. It is stored every 30 seconds. Information is obtained from the following parameters: • SV1.6.11-Ind PT100 (1) • SV1.6.12-Ind PT100 (2) • SV1.6.13-Ind PT100 (3) • SV1.6.14-Ind PT100 (4) • SV12.1.3.1-Ambient Temp • SV12.2.3.1-Ambient Temp • SV12.3.3.1-Ambient Temp • SV12.4.3.1-Ambient Temp • SV12.5.3.1-Ambient Temp • SV12.6.3.1-Ambient Temp • SV12.7.3.1-Ambient Temp • SV12.8.3.1-Ambient Temp • SV1.6.1-Internal T° • SV1.6.4-Admission T° • SV1.6.5-Admission T° Av • SV12.1.7-Humidity • SV12.2.7-Humidity • SV12.3.7-Humidity • SV12.4.7-Humidity • SV12.5.7-Humidity • SV12.6.7-Humidity • SV12.7.7-Humidity • SV12.8.7-Humidity • SV1.6.8-MV PT100 (1) • SV1.6.9-MV PT100 (2) • SV1.6.10-MV PT100 (3)

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LOG NUMBER	DESCRIPTION
Log 4	This log saves information about the DC voltage of the busbars, the input current to the DC cabinet, the reference voltage and current, cos phi, and the reference reactive power. It is stored every 0,5 seconds. Information is obtained from the following parameters: • SV2.13-DC volt last start 1 • SV2.14-DC volt last start 2 • SV2.15-DC volt last start 3 • SV2.16-DC volt last start 4 • SV2.5-I DC 1 • SV2.6-I DC 2 • SV2.7-I DC 3 • SV2.8-I DC 4 • SV4.1.5-Current reference • G8.1.1.1-Pac reference • G8.1.1.1-Id reference • G8.1.1.1-Vdc reference • G8.1.1.2-Cos phi reference • G8.1.4-Iq reference • G8.1.4-Q reference
Log 5	This log saves information about the temperature of each IGBT in each module. It is stored every 10 seconds. Information is obtained from the following parameters: • SV12.1.3.3-Temperature R1 • SV12.1.3.4-Temperature R2 • SV12.1.3.5-Temperature R3 • SV12.1.3.6-Temperature S1 • SV12.1.3.7-Temperature S2 • SV12.1.3.8-Temperature S3 • SV12.1.3.9-Temperature T1 • SV12.1.3.10-Temperature T2 • SV12.1.3.11-Temperature T3
Events Log	This log saves information about the current faults and warnings, DSP status, status of each module, IMI measurements, P and Q control mode. Information is obtained from the following parameters: • SV1.1.1-Current fault • SV1.1.3-Current warning • SV1.1.4-Current status • SV12.1.1-Status • SV12.2.1-Status • SV12.3.1-Status • SV12.4.1-Status • SV12.5.1-Status • SV12.6.1-Status • SV12.7.1-Status • SV12.8.1-Status • SV9.2.1.3-Resistor • SV9.2.2.3-Resistor • SV9.2.3.3-Resistor • SV9.2.4.3-Resistor • SV9.2.5.3-Resistor • SV1.7.1-P control • SV1.7.2-Q control
Custom Log	This log saves information about Modbus address. It is stored every 1 second. Information is obtained from the following parameters: • G12.1.2-Modbus address 1 • G12.1.3-Modbus address 2 • G12.1.4-Modbus address 3 • G12.1.5-Modbus address 4 • G12.1.6-Modbus address 5 • G12.1.7-Modbus address 6 • G12.1.8-Modbus address 7 • G12.1.9-Modbus address 8 • G12.1.10-Modbus address 9 • G12.1.11-Modbus address 10 • G12.1.12-Modbus address 11 • G12.1.13-Modbus address 12 • G12.1.14-Modbus address 13 • G12.1.15-Modbus address 14 • G12.1.16-Modbus address 15

6.11.1. Subgroup G12.0: General

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G12.0.1-SD card status = OK</td><td>OK
Error
Not formatted
Not mounted
Not initalized
Formatted SD
SD Full</td><td>Displays the status of the SD card.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>OK</td><td>SD card is ready</td></tr><tr><td>Error</td><td>SD card has an error</td></tr><tr><td>Not formatted</td><td>SD card is not formatted</td></tr><tr><td>Not mounted</td><td>SD card is not mounted</td></tr><tr><td>Not initalized</td><td>SD card has not been initialized</td></tr><tr><td>Formatted SD</td><td>SD card has been formatted</td></tr><tr><td>SD Full</td><td>The SD card is full</td></tr></table></td><td>Yes</td></tr>
<tr><td>G12.0.2-SD operation = Init SD card</td><td>Init SD card
Format SD card
Update status
Disable SD
Force Delete Old Files</td><td>Sets the task to execute with the SD card.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Init SD card</td><td>Initialises the SD card.</td></tr><tr><td>Format SD card</td><td>Formats the SD card partition.</td></tr><tr><td>Update status</td><td>Updates the status of the card; if it is mounted and accessible, OK will appear.</td></tr><tr><td>Disable SD</td><td>Disables the SD card.</td></tr><tr><td>Force Delete Old Files</td><td>Allows forced deletion of files retained prior to the period configured in [G12.0.7- Days to keep files].</td></tr></table></td><td>Yes</td></tr>
<tr><td>G12.0.3-Apply changes = No</td><td>No
Yes</td><td>Allows applying or discarding the modifications made to the SD card.</td><td>Yes</td></tr>
<tr><td>G12.0.4-Logger enable = Yes</td><td>No
Yes</td><td>Besides having the possibility to store a customized set of parameters, the SD card will always keep a record of the following information:
Full list of faults.
Full list of warnings.
Input event in OVRT mode.
Insulation resistance:
For grounded pole: Value before starting.
For isolated poles: Registered value in case of fault; registered value in case of an emergency stop (fire mode or bus short-circuit). Maximum and minimum daily value.</td><td>Yes</td></tr>
<tr><td>G12.0.5-SD test stress = No</td><td>No
Yes</td><td>This parameter is used for development tests.</td><td>Yes</td></tr>
<tr><td>G12.0.6-Used Capacity = 0.00 %</td><td>0.00 to 100.0%
SD Problem = 100.01;</td><td>Displays the % used capacity of the SD card that is installed in the main control board.</td><td>Yes</td></tr>
<tr><td>G12.0.7-Days to keep files = 30 days</td><td>15 to 90 days</td><td>Allows configuration of the file retention period on the SD card and performs daily automatic deletion of files that exceed this period.</td><td>Yes</td></tr>
<tr><td>G12.0.8-Supervisor time = 15000 ms</td><td>500 to 15000 ms</td><td>Indicates the maximum time allowed for the SD card to sequence correctly. If this time is exceeded, the warning “W180:SD corrupted” is generated.</td><td>Yes</td></tr>
<tr><td>G12.0.9-Supervisor max time = 0 ms</td><td>0 to 65535 ms</td><td>Displays the maximum time used by the SD to perform its sequence.</td><td>Yes</td></tr>
</table>

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SCREEN	RANGE	DESCRIPTION	SET ON RUN
G12.0.10-Task of Maximeter = RTX Idle	Unknow RTX Idle RTX Timers NetCore NetEth0 Watchdog Menu Debug Debug Task Serve Master Serial 1 Master Serial 2 Master Serial 3 GPIO Tools Periodic EEPROM Network Autotest Autotest Polling Fault Manager Fault Obs Sub DSP Fast DSP Slow FPGA Comms DSP SCI SD Manager SD Log Msg SD Delete MVI Temperatures I2C Isolation Maneuver Event Manager Caiso SD Supervisor SD event Enable SD Card Enable Broker AntiPID Blackbox SD Log Sys Recovery manager TCP logger	Corresponds to the task associated with the maximum access or usage time of the SD file system.	Yes

6.11.2. Subgroup G12.1: Parameters logger

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G12.1.1-Enable = Yes	No Yes	Allows to enable or disable the inverter custom log. The custom log is saved on the SD card. The content of this log is the one configured in this group. Up to 15 variables information can be saved, configured as follows. Each of the parameters listed below indicates the Modbus address of the parameter. There are some defined by default and others that are free.	Yes
G12.1.2-Modbus address 1 = 1000	Off = 0 1 to 9999	Show the Modbus addresses 1 to 20.	Yes
G12.1.3-Modbus address 2 = 1001			
G12.1.4-Modbus address 3 = 1002			
G12.1.5-Modbus address 4 = 1003			
G12.1.6-Modbus address 5 = 6505			
G12.1.7-Modbus address 6 = 6540			
G12.1.8-Modbus address 7 = 6575			
G12.1.9-Modbus address 8 = 6610			
G12.1.10-Modbus address 9 = 2345			
G12.1.11-Modbus address 10 = 2346			
G12.1.12-Modbus address 11 = 2347			
G12.1.13-Modbus address 12 = 2348			
G12.1.14-Modbus address 13 = 2351			
G12.1.15-Modbus address 14 = 2352			
G12.1.16-Modbus address 15 = 1336			
G12.1.17-Modbus address 16 = 1337			
G12.1.18-Modbus address 17 = 1338			
G12.1.19-Modbus address 18 = 1339			
G12.1.20-Modbus address 19 = 1340			
G12.1.21-Modbus address 20 = Off			

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6.11.3. Subgroup G12.2: Blackbox

This functionality consists of a logger that feeds the module and central FPGA (one blackbox per module and one central blackbox per inverter), which contains information about its electrical measures and other interesting data that can be used to analyze its behavior. This group allows to configure and check this functionality and its status.

There are two options to download the blackbox from the module FPGAs and the FPGA central: manual download or automatic download.

The blackbox automatic download allows that when a type of event occurs in the inverter (a fault, module status, etc), the module blackbox or all the blackboxes are automatically downloaded, and the result log is saved in the SD card.

In the G12.2.1 subgroup, the automatic download of the blackbox can be configured. It is important to underscore two issues about this:

- When the automatic download is enabled and active, then CAISO functionality is deactivated and vice versa. If CAISO is activated, then the automatic download cannot be enabled.
- When the automatic download needs to be activated, it is necessary to set the type of fault that will trigger a new download. This means, which type of module fault (if any) and which type of system fault (if any). Can be set both (FPGA fault and system fault) or only one of them (FPGA fault or system fault) or any of them, which will mean that the automatic download is disabled.

In the G12.2 subgroup, the user can access to command a manual download of the module and central FPGA blackbox and can check the status of the download process.

Note: The automatic Blackbox download must not be set as “Lockable” through parameter ([G12.2.1.4-Flt Mod Action Lvl] ≠ Lockable).

G12.2.1: Autodownload

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G12.2.1.1-Flt Sys Trigger En = Yes</td><td>No
Yes</td><td>Enables the automatic download functionality of the blackbox when a system fault is triggered.
If enabled, when the type of fault configured in the following parameters occurs in the inverter, then all the black boxes will be automatically downloaded, and the information will be saved in the SD memory.</td><td>Yes</td></tr>
<tr><td>G12.2.1.2-Level Fault Sys Typ = Special</td><td>Special
DU+Abrupt stop
DU+Controlled stop
Abrupt stop
Controlled stop</td><td>In case automatic download is enabled, this parameter allows to select with which minimum fault type, this functionality is launched.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Special</td><td>Open the circuit breaker and isolate the inverter (opens DC and AL low / medium voltage). To manual reset, Power Electronics password is required.</td></tr><tr><td>DU+Abrupt stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>DU+Controlled stop</td><td>Open all DC switch disconnectors.</td></tr><tr><td>Abrupt stop</td><td>The inverter is stopped as soon as possible by not switching the IGBTs</td></tr><tr><td>Controlled stop</td><td>The inverter is stopped following a configurable stop ramp.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G12.2.1.3-Flt Mod Trigger En = Yes</td><td>No
Yes</td><td>This parameter and the next one allow configuring the automatic downloading of the blackbox when the event is a module fault. Consequently, only the module where the event occurred will be downloaded.
This parameter enables / disables this functionality.
If enabled, when the fault type configured in the following parameter occurs in the inverter, then the module blackbox of module that has failed will be automatically downloaded and the information will be saved in the SD.</td><td>Yes</td></tr>
<tr><td>G12.2.1.4-Flt Mod Action Lvl = Normal</td><td>Normal
Lockable
Isolated
Device</td><td>This parameter allows configuring the type of module fault that will cause the module to be unloaded. The configuration of each module fault can be seen in G5.8.</td><td>Yes</td></tr>
<tr><td>G12.2.2-Blackbox download = No</td><td>0 to 10</td><td>Blackbox functionality consists of sending part of the FPGA information to the microprocessor in order to restore it in case of fault.
To consult this information, the user must download the csv file that is stored in the SD.
Configuring the following options defines the data download in the Blackbox:<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>No</td><td>No download.</td></tr><tr><td>All</td><td>Total download.</td></tr><tr><td>Module 1
…
Module 10</td><td>Select module to download.</td></tr></table>Note: These options are displayed if [SG10.1.12-Number of modules < ‘n’] being ‘n’ the number of the module.</td><td>Yes</td></tr>
<tr><td>G12.2.3-Download Status = Download Idle</td><td>Download Idle
Download Ongoing</td><td>Indicates the status of the download.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Download Idle</td><td>The download is idle.</td></tr><tr><td>Download Ongoing</td><td>The download is in process.</td></tr></table></td><td>Yes</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G12.2.4-Last/Actual Download = All Mod by Manual Download</td><td>0 to 20</td><td>Shows which is the last blackbox that has been downloaded, or which is the blackbox that is downloading, in case the system is in process of downloading.
<table>
<tr><th>OPT.</th><th>STATUS</th><th>OPT.</th><th>STATUS</th></tr>
<tr><td>1</td><td>All Mod by Manual Download</td><td>11</td><td>Module 1 by Auto. Download</td></tr>
<tr><td>2</td><td>Module 1 by Manual Download</td><td>12</td><td>Module 2 by Auto. Download</td></tr>
<tr><td>3</td><td>Module 2 by Manual Download</td><td>13</td><td>Module 3 by Auto. Download</td></tr>
<tr><td>4</td><td>Module 3 by Manual Download</td><td>14</td><td>Module 4 by Auto. Download</td></tr>
<tr><td>5</td><td>Module 4 by Manual Download</td><td>15</td><td>Module 5 by Auto. Download</td></tr>
<tr><td>6</td><td>Module 5 by Manual Download</td><td>16</td><td>Module 6 by Auto. Download</td></tr>
<tr><td>7</td><td>Module 6 by Manual Download</td><td>17</td><td>Module 7 by Auto. Download</td></tr>
<tr><td>8</td><td>Module 7 by Manual Download</td><td>18</td><td>Module 8 by Auto. Download</td></tr>
<tr><td>9</td><td>Module 8 by Manual Download</td><td>19</td><td>Central by Manual Download</td></tr>
<tr><td>10</td><td>All Mod by Automatic Download</td><td>20</td><td>Central by Auto. Download</td></tr>
</table></td><td>Yes</td></tr>
<tr><td>G12.2.5-Download Progress M1 = 0 %</td><td rowspan="8">0 to 100 %;
Error Timeout = 101
SD Not Available = 102</td><td rowspan="8">Show the download progress status for each of the modules (from 1 (M1) to 8 (M8) of the inverter.</td><td rowspan="8">Yes</td></tr>
<tr><td>G12.2.6-Download Progress M2 = 0 %</td></tr>
<tr><td>G12.2.7-Download Progress M3 = 0 %</td></tr>
<tr><td>G12.2.8-Download Progress M4 = 0 %</td></tr>
<tr><td>G12.2.9-Download Progress M5 = 0 %</td></tr>
<tr><td>G12.2.10-Download Progress M6 = 0 %</td></tr>
<tr><td>G12.2.11-Download Progress M7 = 0 %</td></tr>
<tr><td>G12.2.12-Download Progress M8 = 0 %</td></tr>
<tr><td>G12.2.13-Download Progress C = 0 %</td><td>0 to 100 %;
Error Timeout = 101
SD Not Available = 102</td><td>Shows the status of the download progress of the central FPGA.</td><td>Yes</td></tr>
</table>

6.11.4. Subgroup G12.3: Vcc protection

This subgroup implements a new protection that disables the SD card when the voltage conditions are not sufficient to ensure that the SD card can function properly.

Note: These parameters are only displayed if [G1.3-User] ≥ Power Electronics staff.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G12.3.1-VCC prot thresh = 22.00 V	21.00 to 23.00 V	Configures the minimum voltage below which plus a hysteresis [G12.3.2-VCC prot hyste], the SD card would be disable.	Yes
G12.3.2-VCC prot hyste = 0.50 V	0.10 to 0.50 V	Configures the hysteresis. If the power supply voltage of the main control board is below the sum of this hysteresis and the voltage configured in the previous parameter, the SD card will be disabled.	Yes

6.12.Group G13: Normative

This group allows the configuration of variables related to functionalities implemented by regulation. It allows to enable and/or disable the fast logger and as well as to adjust this functionality.

6.12.1. Subgroup G13.1: Fast Logger

This subgroup allows the configuration of variables related to the CAISO (California Independent System Operator) and IEEE 2800 standards. The purpose of these standards is to obtain information on what happened during a period prior to a certain defined event and what happened during a period after this event. This information must be stored on the SD card (information for the last 30 days), so that it can be accessed for root cause analysis.

The events that are considered to trigger this functionality are:

- Power stages faults
- Frequency variations that are over a certain configurable value (FRT).
- LVRT/HVRT events that lead to a reactive power injection to grid.
- Transitory grid overvoltage that leads to a cessation of reactive power injection
- Faults that cause the stop of the inverter

The data that should be saved per event is:

- Inverter exit voltage per phase
- Inverter exit current per phase
- Grid frequency
- Inverter fault code
- Inverter warning code
- DC voltage
- DC current
- Active power
- Reactive power

The synchronization between the inverters that belong to the same plant would is by means of a GPS (hardware).

Notes:

- Parameters within this subgroup are displayed if [G13.1.1-Enable Fast Logger = Enabled].
- This group is displayed if [G1.3-User ≥ Qualified customer]. Only users with a role equal or higher than Admin can modify these parameters except G13.1.13 and G13.1.16 which require a role higher than Admin.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G13.1.1-Enable Fast Logger = Enabled	Disabled Enabled	Allows the Fast Logger functionality to be enabled (if it is not enabled, the rest of the parameters of this group are hidden), but it doesn't activate it. **Note:** This parameter is displayed if [G12.2.1.1-Flt Sys Trigger En = No] or [G12.2.1.3-Flt Mod Trigger = No].	Yes

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G13.1.2-Fast Logger selector = Disabled</td><td>Disabled
CAISO
IEEE</td><td>Allows the Fast Logger standard to be selected. When an event occurs, a new log is created, and it is saved in the SD card.
Note: this parameter is shown if [G12.2.1.1-Flt Sys Trigger En] = Yes and [G12.2.1.3-Flt Mod Trigger En] = Yes.</td><td>Yes</td></tr>
<tr><td>G13.1.3-Pre-shoot samples = 150 ms</td><td>0 to 5000 ms</td><td>Allows the pre-event log duration to be selected, i.e. the recording captured before the event that triggers the Fast Logger; this data is stored on the SD card for subsequent analysis. Both its default value and its maximum depend on the standard selected in [G13.1.2-Fast Logger selector]. Together with [G13.1.4-Post-shoot samples] it shares a total capture window, so its maximum is not fixed but rather the space left free by G13.1.4 within that window (maximum = total window − G13.1.4).
<table>
<tr><th>Standard (G13.1.2)</th><th>Default value</th><th>Max. with G13.1.4 at its default value</th><th>Total window</th></tr>
<tr><td>CAISO</td><td>150 ms</td><td>500 ms</td><td>1500 ms</td></tr>
<tr><td>IEEE</td><td>1000 ms</td><td>1000 ms</td><td>5000 ms</td></tr>
</table></td><td>Yes</td></tr>
<tr><td>G13.1.4-Post-shoot samples = 1000 ms</td><td>0 to 5000 ms</td><td>Allows the post-event log duration to be selected, i.e. the recording captured after the event that triggers the Fast Logger; this data is stored on the SD card for subsequent analysis. Both its default value and its maximum depend on the standard selected in [G13.1.2-Fast Logger selector]. Together with [G13.1.3-Pre-shoot samples] it shares a total capture window, so its maximum is not fixed but rather the space left free by G13.1.3 within that window (maximum = total window − G13.1.3).
<table>
<tr><th>Standard (G13.1.2)</th><th>Default value</th><th>Max. with G13.1.3 at its default value</th><th>Total window</th></tr>
<tr><td>CAISO</td><td>1000 ms</td><td>1350 ms</td><td>1500 ms</td></tr>
<tr><td>IEEE</td><td>4000 ms</td><td>4000 ms</td><td>5000 ms</td></tr>
</table></td><td>Yes</td></tr>
<tr><td>G13.1.5-Shoot over frequency = 0.50 Hz</td><td>Off = 0.00;
0.01 to 5.00 Hz</td><td>Configures the frequency level that will shoot an over frequency event.</td><td>Yes</td></tr>
<tr><td>G13.1.6-Shoot sub frequency = -0.50 Hz</td><td>Off = 0.00;
0.01 to 5.00 Hz</td><td>Configures the frequency level that will shoot a sub frequency event.</td><td>Yes</td></tr>
<tr><td>G13.1.7-Shoot LVRT = Enabled</td><td>Disabled
Enabled</td><td>Enables or disables the Fast Logger functionality when a LVRT event takes place.</td><td>Yes</td></tr>
<tr><td>G13.1.8-Shoot OVRT = Enabled</td><td>Disabled
Enabled</td><td>Enables or disables the Fast Logger functionality when an OVRT event takes place.</td><td>Yes</td></tr>
<tr><td>G13.1.9-Shoot fault = Enabled</td><td>Disabled
Enabled</td><td>Enables or disables the Fast Logger functionality when a system fault takes place.</td><td>Yes</td></tr>
<tr><td>G13.1.10-Event force = No</td><td>No
Yes</td><td>Forces an event logger to verify the right behavior of the algorithm, for testing purpose.</td><td>Yes</td></tr>
<tr><td>G13.1.11-Sample time = 1 ms</td><td>1 to 10 ms</td><td>Configures the sample time of the log.
Note: this parameter is not available from version 5.21.0 onwards.</td><td>Yes</td></tr>
<tr><td>G13.1.12-Logger mode = Normative</td><td>Normative
Debug</td><td>Logger mode (normative or Debug) In this moment only is available normative mode.</td><td>Yes</td></tr>
<tr><td>G13.1.13-Counter events/day = 0</td><td>0 to 65535</td><td>Indicates the number of events that have occurred during the day. Each new day returns to 0 this counter.</td><td>Yes</td></tr>
<tr><td>G13.1.14-Sync force = No</td><td>No
Yes</td><td>Forces the synchronization with the GPS when it is installed.</td><td>Yes</td></tr>
<tr><td>G13.1.15-Clock source = GPS</td><td>GPS
Internal</td><td>Chooses the clock source from GPS or internal (RTC) to store the time in the log.</td><td>Yes</td></tr>
<tr><td>G13.1.16-GPS status = GPS OK</td><td>GPS OK
PPS ERROR
COMMS ERROR
GPS NOT AVAILABLE</td><td>Indicates the status of the synchronization process with the GPS.
Note: This parameter is displayed if [G13.1.15-Clock source = GPS].</td><td>Yes</td></tr>
<tr><td>G13.1.17-Time domain = RTC</td><td>RTC
PTP</td><td>Allows the timestamp domain to be configured as RTC (s/ms) or PTP (ns).</td><td>Yes</td></tr>
</table>

6.13.Group G14: Custom modbus

This group allows accessing 120 variables consecutively (even if they are not at consecutive Modbus addresses). This is useful for SCADA systems that need to read or write certain variables in particular, consecutively.

Modbus address configuration is only allowed if it has been enabled internally; a password must first be set:

- In parameter SG1.1.12, by setting the password 1710, the group is enabled to configure the Modbus addresses. When the password is removed, then the G14 parameters show the actual values of the parameters previously configured. Those can be read if they are read or write and those can be written if they are read / write. Those that are read/write, can be modified from this group and from the corresponding group where they are placed.
- The access and modification of the parameters of this group depends on the access role configured in parameter [SG14.2.1-Custom Modbus Level].

Note: Only users with a role equal or higher than Admin (qualified customer) can modify these parameters.

EN

6.13.1. Subgroup G14.1: Custom modbus 1

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G14.1.1-Custom 1 = 0	0 to 65535	Defines addresses 1 to 60 of the customized Modbus map. **Note:** Only users with a role equal or higher than Admin (qualified customer) can modify these parameters.	Yes
G14.1.2-Custom 2 = 0			
G14.1.3-Custom 3 = 0			
G14.1.4-Custom 4 = 0			
G14.1.5-Custom 5 = 0			
G14.1.6-Custom 6 = 0			
G14.1.7-Custom 7 = 0			
G14.1.8-Custom 8 = 0			
G14.1.9-Custom 9 = 0			
G14.1.10-Custom 10 = 0			
...			
G14.1.55-Custom 55 = 0			
G14.1.56-Custom 56 = 0			
G14.1.57-Custom 57 = 0			
G14.1.58-Custom 58 = 0			
G14.1.59-Custom 59 = 0			
G14.1.60-Custom 60 = 0			

6.13.2. Subgroup G14.2: Custom modbus 2

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G14.2.1-Custom 61 = 0	0 to 65535	Defines addresses 61 to 120 of the customized Modbus map. **Note:** Only users with a role equal or higher than Admin (qualified customer) can modify these parameters.	Yes
G14.2.2-Custom 62 = 0			
G14.2.3-Custom 63 = 0			
G14.2.4-Custom 64 = 0			
G14.2.5-Custom 65 = 0			
G14.2.6-Custom 66 = 0			
G14.2.7-Custom 67 = 0			
G14.2.8-Custom 68 = 0			
G14.2.9-Custom 69 = 0			
G14.2.10-Custom 70 = 0			
…			
G14.2.55-Custom 115 = 0			
G14.2.56-Custom 116 = 0			
G14.2.57-Custom 117 = 0			
G14.2.58-Custom 118 = 0			
G14.2.59-Custom 119 = 0			
G14.2.60-Custom 120 = 0			

6.14. Group G15: Silent mode

This group allows the silent mode to be configured when the inverter is running, whereby the noise level is reduced by reducing the ventilation depending on the temperature.

Note: this group is displayed if [G1.3-User] ≥ Qualified customer. Parameters from [G15.1.2-Silent mode level] to [G15.3.17-50ºC derating] will only be displayed if the algorithm is enabled ([G15.1.1-Enable silent mode] = Enabled).

6.14.1. Subgroup G15.1: Configuration

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G15.1.1-Enable silent mode = Disabled</td><td>Disabled
Enabled</td><td>Allows to enable the silent mode.
Note: this parameter is displayed if [G1.3-User] ≥ Power Electronics staff.</td><td>Yes</td></tr>
<tr><td>G15.1.2-Silent mode level = Disabled</td><td>Disabled
Level 1
Level 2
Level 3
Level 4
Level 5</td><td>When the silent mode is enabled, it allows to activate and select the noise reduction level.
For each level, specific fan speed settings are applied for each zone and a reduction in the power delivered by the inverter (derating) is applied, which activates the warning [W56: Silent mode derating].
These settings are fixed at levels 1, 2, and 3, while at levels 4 and 5, they can be configured in the parameter groups [G15.2: Level 4 configuration] and [G15.3: Level 5 configuration].
<table>
<tr><th rowspan="2">Fan zone</th><th colspan="3">Speed adjustment</th></tr>
<tr><th>LEVEL 1</th><th>LEVEL 2</th><th>LEVEL 3</th></tr>
<tr><td>DC cabinet (DU)</td><td>80 %</td><td>70 %</td><td>60 %</td></tr>
<tr><td>Modules: [G11.5.1-Mode] = Run</td><td>80 %</td><td>70 %</td><td>60 %</td></tr>
<tr><td>Modules: [G11.5.1-Mode] = Temperature</td><td>80 % * (max. speed)</td><td>70 % * (max. speed)</td><td>60 % * (max. speed)</td></tr>
<tr><td>LC filter</td><td>80 %</td><td>80 %</td><td>If [SV1.6.4-Admission Tº] < 25 ºC: 60 % **</td></tr>
<tr><td>MT</td><td>80 %</td><td>70 %</td><td>60 %</td></tr>
</table>
* The configured Speed-Temperature curve is applied.
** If [SV1.6.4-Admission Tº] ≥ 25 ºC, it is configured by [G15.1.6-Behaviour].
<table>
<tr><th rowspan="2">Value in °C of [SV1.6.4-Admission Tº]</th><th colspan="3">Derating adjustment</th></tr>
<tr><th>LEVEL 1</th><th>LEVEL 2</th><th>LEVEL 3</th></tr>
<tr><td>10</td><td>100 %</td><td>100 %</td><td>100 %</td></tr>
<tr><td>20</td><td>100 %</td><td>100 %</td><td>95 %</td></tr>
<tr><td>25</td><td>100 %</td><td>100 %</td><td>90 %</td></tr>
<tr><td>30</td><td>100 %</td><td>100 %</td><td>90 %</td></tr>
<tr><td>40</td><td>100 %</td><td>90 %</td><td>80 %</td></tr>
<tr><td>45</td><td>92 %</td><td>85 %</td><td>78 %</td></tr>
<tr><td>50</td><td>84 %</td><td>79 %</td><td>75 %</td></tr>
</table>
For intermediate values of the temperatures indicated in the table above, the derating values will be applied by linear interpolation.
The operation of the 5 levels of silent mode will apply up to 50°C. When the ambient temperature exceeds this value, fault [F245: Over temp silent mode] will be activated, deactivating when it is below 51°C minus the hysteresis defined in [G15.1.7-Threshold].
Note: noise reduction levels cannot be activated if any ventilation zone is set to [G11.x.1-Mode] = Fixed.</td><td>Yes</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>G15.1.3-Temp offset level 1 = 0 ºC</td><td rowspan="3">0 to 10 ºC</td><td rowspan="3">They allow setting a temperature offset for 1, 2 and 3 noise reduction levels, shifting the derating curve (defined in [G15.1.2-Silent mode level]) to the left (more conservative side) by the degrees that are set, in case the components get too hot and field adjustments are necessary.
Note: they are shown if [G3.1-User] ≥ Power Electronics staff.</td><td rowspan="3">Yes</td></tr>
<tr><td>G15.1.4-Temp offset level 2 = 0 ºC</td></tr>
<tr><td>G15.1.5-Temp offset level 3 = 0 ºC</td></tr>
<tr><td>G15.1.6-Behaviour = Increase speed</td><td>Stop
Increase speed</td><td>Allows to configure the ventilation behavior of the LC filter zone of level 3 when the maximum allowed temperature (25 ºC) is exceeded. It allows to choose between two options:<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Stop</td><td>Stop the inverter and thus ensure compliance with noise regulations. Trigger fault [F244: T LC Silent mode].</td></tr><tr><td>Increase speed</td><td>Keep the inverter running at the cost of increasing the speed setpoint to 80 % by means of a fixed hysteresis (set in [G15.1.7-Threshold]) around this temperature so that the speed changes of this fan are not reduced.</td></tr></table></td><td>Yes</td></tr>
<tr><td>G15.1.7-Threshold = 5 ºC</td><td>2 to 5 ºC</td><td>Allows to set the temperature hysteresis for deactivation of fault [F245: Over temp silent mode], as well as fault [F244: T LC Silent Mode] and the speed limit of level 3 in the LC filter zone.
Note: it is shown if [G3.1-User] ≥ Power Electronics staff.</td><td>Yes</td></tr>
<tr><td>G15.1.8-Enable config = Disabled</td><td>Disabled
Enabled</td><td>Enables or disables the configuration of levels 4 and 5 to allow or prevent modification of their default values in the parameters of groups [G15.2: Level 4 Configuration] and [G15.3: Level 5 Configuration].
Note: it is displayed if [G1.3-User] ≥ Emp. Power Electronics. It can only be edited if [G1.3-User] ≥ Developer.</td><td>No</td></tr>
</table>

6.14.1. Subgroup G15.2: Level 4 configuration

This subgroup allows Level 4 silent mode to be configured.

Note: these parameters can only be modified if they have been enabled internally in [G15.1.8-Enable config] = Enabled.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G15.2.1-Load default values = No	No Yes	Allows the parameter values between [G15.2.2-10ºC fan speed] and [G15.2.17-50ºC derating] to be restored to their default settings.	No
G15.2.2-10ºC fan speed = 60 %	60 % to 100 %	These parameters enable the ventilation speed of the modules, DC cabinet, and MV cabinet zones to be configured, as well as the derating for each temperature point ([SV1.6.4-Admission Tº]) in Level 4 silent mode. When the ambient temperature is between two defined values, the associated speed and derating values will be determined by interpolation. The speed of the LC filter fans will follow the configured ventilation speed, except when [SV1.6.4-Admission Tº] is higher than 20 °C and the configured speed is lower than 80 %. In this case, the fan speed will be progressively increased by interpolation so that it reaches 80 % when the temperature reaches 25 °C, remaining at that value for higher temperatures. To ensure a safe configuration, each fan speed has an associated derating value. Therefore, applying a fan speed setpoint will limit the maximum derating, while applying a derating setpoint will limit the minimum fan speed.	No
G15.2.3-10ºC derating = 100 %	50 % to 100 %		
G15.2.4-20ºC fan speed = 60 %	60 % to 100 %		
G15.2.5-20ºC derating = 95 %	50 % to 100 %		
G15.2.6-25ºC fan speed = 60 %	60 % to 100 %		
G15.2.7-25ºC derating = 90 %	50 % to 100 %		
G15.2.8-30ºC fan speed = 70 %	60 % to 100 %		
G15.2.9-30ºC derating = 100 %	50 % to 100 %		
G15.2.10-35ºC fan speed = 70 %	60 % to 100 %		
G15.2.11-35ºC derating = 95 %	50 % to 100 %		
G15.2.12-40ºC fan speed = 80 %	60 % to 100 %		
G15.2.13-40ºC derating = 100 %	50 % to 100 %		
G15.2.14-45ºC fan speed = 100 %	60 % to 100 %		
G15.2.15-45ºC derating = 97 %	50 % to 100 %		
G15.2.16-50ºC fan speed = 100 %	60 % to 100 %		
G15.2.17-50ºC derating = 93 %	50 % to 100 %		

EN

6.14.1. Subgroup G15.3: Level 5 configuration

This subgroup allows Level 5 silent mode to be configured.

Note: these parameters can only be modified if they have been enabled internally in [G15.1.8-Enable config] = Enabled.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
G15.3.1-Load default values = No	No Yes	Allows the parameter values between [G15.3.2-10ºC fan speed] and [G15.3.17-50ºC derating] to be restored to their default settings.	No
G15.3.2-10ºC fan speed = 70 %	60 % to 100 %	These parameters enable the ventilation speed of the modules, DC cabinet, and MV cabinet zones to be configured, as well as the derating for each temperature point ([SV1.6.4-Admission Tº]) in Level 5 silent mode. When the ambient temperature is between two defined values, the associated speed and derating values will be determined by interpolation. The speed of the LC filter fans will follow the configured ventilation speed, except when [SV1.6.4-Admission Tº] is higher than 20 °C and the configured speed is lower than 80 %. In this case, the fan speed will be progressively increased by interpolation so that it reaches 80 % when the temperature reaches 25 °C, remaining at that value for higher temperatures. To ensure a safe configuration, each fan speed has an associated derating value. Therefore, applying a fan speed setpoint will limit the maximum derating, while applying a derating setpoint will limit the minimum fan speed.	No
G15.3.3-10ºC derating = 100 %	50 % to 100 %		
G15.3.4-20ºC fan speed = 70 %	60 % to 100 %		
G15.3.5-20ºC derating = 100 %	50 % to 100 %		
G15.3.6-25ºC fan speed = 70 %	60 % to 100 %		
G15.3.7-25ºC derating = 100 %	50 % to 100 %		
G15.3.8-30ºC fan speed = 70 %	60 % to 100 %		
G15.3.9-30ºC derating = 100 %	50 % to 100 %		
G15.3.10-35ºC fan speed = 70 %	60 % to 100 %		
G15.3.11-35ºC derating = 95 %	50 % to 100 %		
G15.3.12-40ºC fan speed = 80 %	60 % to 100 %		
G15.3.13-40ºC derating = 100 %	50 % to 100 %		
G15.3.14-45ºC fan speed = 100 %	60 % to 100 %		
G15.3.15-45ºC derating = 97 %	50 % to 100 %		
G15.3.16-50ºC fan speed = 100 %	60 % to 100 %		
G15.3.17-50ºC derating = 93 %	50 % to 100 %		

7. SERVICE GROUP PARAMETERS

7

7.1. Group SG1: Service

7.1.1. Subgroup SG1.1: uP service

This group includes variables mainly used by the uP to configure certain functionalities that depend primarily on this device.

Subgroup SG1.1.1: Watchdog

This group allows configuring the behavior of the watchdog of the microcontroller (uC).

Note: parameters [SG1.1.1.1-Watchdog time] to [SG1.1.1.5-Watchdog period] can only be edited with the Developer role.

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.1.1.1-Watchdog time = 20000 ms	500 to 60000 ms	Allows setting the watchdog monitoring time.	Yes
SG1.1.1.2-Watchdog FTP time = 30000 ms	500 to 60000 ms	Allows setting the FTP time of the watchdog.	Yes
SG1.1.1.3-Active watchdog = Regular	Regular FTP	Allows to configure the type of active watchdog.	Yes
SG1.1.1.4-WD Max retries = 0	0 to 65535	Allows to confirm the number of consecutive retries before restarting in those where the result is wrong.	Yes
SG1.1.1.5-Watchdog period = 0 ms	0 to 65535 ms	Allows to configure the watchdog monitoring period.	Yes
SG1.1.1.6-Retries IRQ blocked = 1	1 to 10	Allows to configure the number of consecutive lockouts per IRQ to generate the F47.	Yes
SG1.1.1.7-Time IRQ blocked = 5 ms	1 to 10 ms	Allows to configure the Operating System blocking time by IRQ to detect F47.	Yes
SG1.1.1.8-Recovery network = No	No Yes	Allows to reset the Network task once it is in fault, getting the communications to work again and thus avoiding the inverter to stop due to F47.	Yes

Others:

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.1.1-New modbus map = No	No Yes	Set a new Modbus map.	Yes
SG1.1.2-MAC protection key = 0	0 to 65535	Set the MAC protection key.	Yes

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG1.1.3-Init factory defaults = 0</td><td>0 to 65535</td><td>Set all parameters to factory values.
Init Factory: This should be avoided unless exceptional circumstances occur.
This option deletes all data, including calibration values of currents and voltages.
It sets all parameters to their default value, except the MAC address.
If the new SW version changes the measurement system, it will be necessary to restore the calibrations and calibrate afterwards.
If it does not change, avoid this step if the inverter is not to be calibrated.</td><td>Yes</td></tr>
<tr><td>SG1.1.4-Init service group = No</td><td>No
Yes</td><td>Initialize service groups.</td><td>Yes</td></tr>
<tr><td>SG1.1.5-NTP IP address = 98.175.203.200</td><td>1.0.0.1 to 255.255.255.254</td><td>NTP, Network Time Protocol is a protocol for clock synchronization between different inverters. This parameter allows to define the IP address of the NTP server that will be used for time synchronization.
IP address format is: A.B.C.D.
To adjust this address, introduce a value in each of the four parameters that compose it.</td><td>Yes</td></tr>
<tr><td>SG1.1.5b-NTP time request = 5 min</td><td>1 to 30 min</td><td>Allows to configure a delay for a periodic synchronization through GPS.</td><td>Yes</td></tr>
<tr><td>SG1.1.6-System reboot = 0</td><td>0 to 65535</td><td>Reboot the system once a password is entered. Similar to turning off and on the control.</td><td>Yes</td></tr>
<tr><td>SG1.1.10-TCP logger = No</td><td>No
Yes</td><td>This parameter is reserved for future use.</td><td>Yes</td></tr>
<tr><td>SG1.1.11-Selector config = Key</td><td>Key
Remote
Key + Remote</td><td>Allows configuring when the bus charge should start (connection to Disconnection Unit), depending on the selector status.
There are two inputs that can be used as a selector: The physical key (in the AC cabinet door) or a virtual selector that is configured in SG1.1.11 according to the following options.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Key</td><td>It will work according to the position of the physical key. The selector key must be placed and in ON position. The bus will charge as soon as charge conditions are detected.</td></tr>
<tr><td>Remote</td><td>Regardless the position of the key, the bus charging will start according to the setting of [SG1.4.2-Selector status EEPR], which must be set to “Yes”.</td></tr>
<tr><td>Key + remote</td><td>It is a combination of the previous options.
The selector key must be placed and in ON position. Besides, [SG1.4.2-Selector status EEPR] must be “Yes”. Then, the bus charge will start.</td></tr>
</table></td><td>Yes</td></tr>
<tr><td>SG1.1.12-Custom modbus key = 0</td><td>0 to 65535</td><td>Password to edit the customized Modbus map.</td><td>Yes</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG1.1.15-Keep alive status = Normal</td><td>Disabled
Normal
Special</td><td>V1500 inverters have a redundant control system; two control boards are installed and, if one fails, the controller will enable the other.
The control board sends a “keep alive” status to the controller to show it is operating correctly. If the controller stops receiving the “keep alive” status, it will cut power supply and turn on the redundant control board.
This parameter defines the time lapse that the controller can wait before enabling the redundant device:
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Keep alive status is not sent. This option should only be used in inverters without control redundancy.</td></tr><tr><td>Normal</td><td>The main control board sends the “keep alive” status every 15 ms.</td></tr><tr><td>Special</td><td>A longer time lapse is defined (adjusted internally). This option is useful when a longer task is going to take place (e.g. remote upgrade), the control board lets the controller know it must maintain the power supply on during this time.</td></tr></table>Note: this parameter is not available from version 2.2.0.</td><td>Yes</td></tr>
<tr><td>SG1.1.17-Humidity/Temp sensor = Yes</td><td>No
Yes</td><td>This parameter allows enabling / disabling humidity and temperature protections for the electronic board and the active heating.
If SG1.1.17 = Yes, protections will be enabled.
To disable protections, set SG1.1.17 = No.
Note: this parameter is not available from version 2.2.0.</td><td>Yes</td></tr>
<tr><td>SG1.1.19-Watchdog[1] time = 10000 ms</td><td>500 to 60000 ms</td><td>Sets the watchdog time.
Note: this parameter has been moved to [SG1.1.1.1-Watchdog time] from version 4.1.1.</td><td>Yes</td></tr>
<tr><td>SG1.1.20-Watchdog FTP time = 30000 ms</td><td>500 to 60000 ms</td><td>Sets the watchdog FTP time.
Note: this parameter has been moved to [SG1.1.1.2-Watchdog FTP time] from version 4.1.1.</td><td>Yes</td></tr>
<tr><td>SG1.1.21-Active watchdog = Regular</td><td>Regular
FTP</td><td>Sets the watchdog active type.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Regular</td><td>Normal watchdog operation. Applies the timeout from parameter [SG1.1.19-Watchdog time].</td></tr><tr><td>FTP</td><td>Special watchdog operation. Applies the timeout from parameter [SG1.1.20-Watchdog FTP time]. Different time allowed before watchdog is skipped when downloading files from SD via FTP.</td></tr></table>Note: this parameter has been moved to [SG1.1.1.3-Active watchdog] from version 4.1.1.</td><td>Yes</td></tr>
<tr><td>SG1.1.22-Watchdog max retries = 0</td><td>0 to 65535</td><td>Number of consecutives retries before restarting in those where the result is wrong.
Note: this parameter has been moved to [SG1.1.1.4-WD Max retries] from version 4.1.1.</td><td>Yes</td></tr>
<tr><td>SG1.1.23-Watchdog period = 0 ms</td><td>0 to 65535 ms</td><td>Sets the watchdog period in ms.
Note: this parameter has been moved to [SG1.1.1.5-Watchdog period] from version 4.1.1.</td><td>Yes</td></tr>
<tr><td>SG1.1.24-DSP Comms Threshold = 50 %</td><td>11 to 100 %</td><td>Indicates the percentage of bad frames between uP and DSP from which the fault F48: Internal comms will be triggered.</td><td>Yes</td></tr>
</table>

[1] A watchdog is a peripheral that periodically checks that the state of the microcontroller is correct. If not, it restarts it.

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.1.25-Filter Reset No Flt = Yes	No Yes	If this parameter is set to "Yes", in case of receiving a fault reset request without being in fault status, it is ignored without any action being taken. **Note:** This parameter can be edited with the Developer role.	Yes
SG1.1.26-I DU Factor = 100 %	0 to 100 %	Allows to configure the channel factor of the DC cabinet that is applied to the difference between the average of all channels and the real current value. This is used to calculate the measure and displayed it properly in the "SV10: DU" group.	Yes
SG1.1.27-DU BP overcurrent = 100 A	0 to 200 A; Disabled = 201	Allows to configure the limit of the total DU current (DC cabinet) measured in the channels when the system is not switching. This parameter is only available when the system is [SG10.1.4-Inverter type = Basic BUSPLUS] and [SG10.1.4-Inverter type= Advanced BUSPLUS]. If the DU current is over the limit here configured, then fault "F240: Du BP overcurrent" will trigger.	Yes
SG1.1.28-Threshold Wrn SensHT = 5	1 to 30	Allows to configure a threshold for the consecutive errors of the temperature and humidity sensor which are displayed in [SV6.2.9.1-Sens T/H Errors I2C].	Yes
SG1.1.29-Serial Number pass = 0	0 to 65535	Sets the password that must be entered in [G1.9-Serial number] in order to change the serial number of the inverter.	Yes

7.1.2. Subgroup SG1.2: DSP service

Subgroup SG1.2a: VRT Fault Avoid

Group to auto reset modules and avoid fault in a voltage drop event.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.2.a2-Sudden V drop = 0.079	0.000 to 1.000	Sets the sudden voltage drop with respect to the voltage measured by the central board when a voltage dip occurs. **Note:** –This parameter is only displayed if [SG1.2.a1-VRT Mode] = Mode 2. –This parameter is not available from version 4.2.0.	Yes
SG1.2.a3-Sudden f drop = 0.100	0.000 to 1.000	Sets the sudden frequency drop, with respect to the frequency measurement of the central board when a voltage dip occurs. **Note:** –This parameter is only displayed if [SG1.2.a1-VRT Mode] = Mode 2. –This parameter is not available from version 4.2.0.	Yes
SG1.2.a4-Sudden n drop = 2	1 to 10	Automatically resets inverter module(s) (FPGA) when there are 3 or more ([SG1.2.a 4-Sudden n drop]) modules in overcurrent warning ([W65:AC overcurrent]).	Yes
SG1.2.a5-Recovery = 100 %	0 to 150 %	Factor that will be applied to the reference after the module reset.	Yes
SG1.2.a6-Re-entry block Time = 100 ms	1 to 100 ms	Allows to configure the time at the output of an LVRT or HVRT during which the previous reference to the event starts to recover with the configured ramp but blocking the detection of a new VRT. It allows avoiding false VRTs when retrieving references with very aggressive gradients in weak grids.	Yes

Subgroup SG1.2.b: FFAC

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.2.b.1-IGBTs blocking time = 15	0 to 1000	This parameter is used to define a delay, in milliseconds, between restarts in case of a voltage event. The number of restarts is defined in [SG1.3.2-VRT Retries].	Yes
SG1.2.b.2-Vac feed = DSP/FPGA	No DSP/FPGA FPGA FPGA 2	Allows defining who determines the module output voltage. **Note:** The default value is DSP/FPGA and must not be changed unless explicitly stated by R&D department.	Yes
SG1.2.b.3-VAC Module Gain = 0	-1000 to 1000	Sets the Vac module gain. **Notes**: –This parameter is displayed if [SG1.2.b.2 = yes]. –This parameter is not available from version 2.2.0.	Yes
SG1.2.b.4-Vac Angle Offset = 0	-100 to 100	Allows to set a delay to the voltage offset angle. **Notes**: –This parameter is displayed if [SG1.2.b.2 = yes]. –This parameter is not available from version 2.2.0.	Yes
SG1.2.b.5-FF reference = FPGA	DSP FPGA	Sets the FF references. **Note:** this parameter is not available from version 2.2.0.	Yes
SG1.2.b.6-Current decoupling = No	No Yes	Sets if there are current decoupling. **Note:** this parameter is not available from version 2.2.0.	Yes

Subgroup SG1.2c: P Balancing

Group to set parameters of power balance algorithm. This algorithm allows to avoid oscillations in the busbar when the grid operates in inverse sequence.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.2.c.1-Balancing Mode = Mode 1	Disabled Mode 1 Mode 2	This parameter allows to stablish the working mode of the power balance algorithm. **Option** / **Description** Disabled: Balancing mode is disabled. Mode 1: Balancing mode is activated when **voltage increases** due to inverse sequence exceeds a value of increase calculated internally. Mode 2: Balancing mode is activated when **voltage value** due to inverse sequence exceeds a threshold stablished internally.	Yes
SG1.2.c.2-Balancing Time = 0.10 s	0.10 to 1.00 s	Time (seconds) during which the power balance will be activated. **Note**: This parameter will remain hidden when SG1.2.c1 is off (0).	Yes
SG1.2.c.3-Balancing Gain = 1.00	0.01 to 1.00	Constant to control the balancing of P. It allows to set higher or lower balancing (100% would correspond to the maximum possible balancing). **Note**: This parameter will remain hidden when SG1.2.c1 is off (0).	Yes
SG1.2.c.4-Balancing Threshold = 0.05	0.01 to 1.00	Threshold for activating the balance of P. This same threshold can be configured for both mode 1 and mode 2. **Note**: This parameter will remain hidden when SG1.2.c1 is off (0).	Yes

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Subgroup SG1.2.d: Saturation limit

Group to configure the saturation limit by modulation index. Allows to configure the threshold from which the inverter will limit the references to avoid excessive overmodulation, adapting this to the hardware limits of the inverter, in such a way that the inverter is protected from situations with a relation Vdc < Vac*√2.

The limit is activated at the highest threshold and the value of P or Q of the inverter is reduced according to the configured priority until the modulation index is lower than the middle threshold. The algorithm shall stop limiting when the modulation index falls below the lowest threshold for the configured time in steady state.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.2.d.1-Enable = Enabled	Disabled Enabled	Allows enabling or disabling the algorithm that limits the saturation by the modulation index.	Yes
SG1.2.d.2-Low threshold = 1.02	1.00 to 1.05	Allows to configure the threshold above which the limit is disabled.	Yes
SG1.2.d.3-Middle threshold = 1.05	SG1.2.d.2-Low threshold to 1.10	Allows to configure the working threshold. It is the modulation index in which the inverter will work in steady state.	Yes
SG1.2.d.4-High threshold = 1.10	SG1.2.d.3-Middle threshold to 1.20	Allows to configure the activation threshold of the limit.	Yes
SG1.2.d.5-Output delay = 2 s	1 to 10 s	Allows to configure the time that's the modulation index must be lower than the threshold configured in SG1.2.d.2-Low threshold.	Yes
SG1.2.d.6-Input gradient = 10.0 %/s	0.1 to 3000.0 %/s	Allows to configure the limit of the operating gradient.	Yes
SG1.2.d.7-Recovery gradient = 0.1 %/s	0.1 to 3000.0 %/s	Allows to configure the recovery gradient of the limit.	Yes
SG1.2.d.8-Priority = P	P Q	Allows to set the algorithm priority to P or Q.	Yes

Others:

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.2.1-Bus discharge = Yes	No Yes	Enable or disable the manual bus discharge. **Note**: this parameter is not available from version 2.2.0.	Yes
SG1.2.3-Open night dc cont = No	No Yes	Open DC contactor at night and close back in the morning. **Note**: this parameter is not available from version 2.2.0.	Yes
SG1.2.4-Nº max faults = 10	0 to 20	Sets the number of attempts to restart the module. Then the DSP launches W116 for 30 min, then restarts auto reset sequences indefinitely. Set the number of attempts to restart the module after any fault. If the module fails to restart SG1.2.4 times, the DSP will not attempt to restart it until the next day. This parameter also sets the number of retries for the modules to activate the AC contactors before the fault F88 is triggered.	Yes
SG1.2.4b-Reset time = 30 min	No Yes	Sets the reset retries counter.	Yes

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG1.2.5-DU open Vmax = 10 %</td><td>0 to 60 %</td><td>Set the maximum voltage to open the Disconnecting Unit.
Notes:
–Do not modify this parameter without authorization from the R&D department.
–This parameter is not available from version 2.2.0.</td><td>Yes</td></tr>
<tr><td>SG1.2.6-Eficiency = 98.0 %</td><td>90.0 to 100.0 %</td><td>Defines the accuracy of the DC limit per module.</td><td>Yes</td></tr>
<tr><td>SG1.2.8-Mod. Start fault = 6</td><td>1 to 7</td><td>Number of modules that must fail at start before triggering a fault and stopping the inverter.
Notes:
–Default and maximum values of this parameter are set in Modbus addresses 43386 and 43387, respectively. By default, its default value is 6 and maximum value 7.
–This parameter is not available from version 2.2.0.</td><td>Yes</td></tr>
<tr><td>SG1.2.9-Disab IMI Diagnos = Yes</td><td>No
Yes</td><td>Disables the processes associated with IMI during diagnosis of the inverter/PCS.
Note: only users with role Developer can modify this parameter.</td><td>Yes</td></tr>
<tr><td>SG1.2.10-AC current reg = Autodetection</td><td>Autodetection
Direct
Indirect</td><td>Allows to configure the current values which will be registered for CAISO.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Autodetection</td><td>The software registers the current values directly or indirectly depending on the status of the AC current transformers of the inverter.
Note: this option is only available if [SG10.1.19-Visma enable] = Disabled.</td></tr>
<tr><td>Direct</td><td>The current values registered are those measured from the AC current transformers.</td></tr>
<tr><td>Indirect</td><td>The current values registered are estimated from other internal current measurements of the inverter.
Note: this option is only available if [SG10.1.19-Visma enable] = Disabled.</td></tr>
</table></td><td>Yes</td></tr>
<tr><td>SG1.2.11-CTs Autodetection = 111</td><td>000 to XXX</td><td>Allows to configure how to perform the current autodetection of the CTs through a bits register.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Calculates the error of the measurement of the CTs with respect to the calculated current for the F173/W50.</td></tr>
<tr><td>1</td><td>Calculates the error of the measurement of the CTs with respect to calculated current for the control (expected current).</td></tr>
<tr><td>2</td><td>Calculates the offset of the measurement of the CTs when the inverter is stopped.</td></tr>
</table></td><td>Yes</td></tr>
<tr><td>SG1.2.21-Filter Vdc = 25.0 Hz</td><td>1.0 to 150.0 Hz</td><td>Shows the DC measure, a filter is used. This parameter sets the cut frequency of the filter.</td><td>Yes</td></tr>
<tr><td>SG1.2.22-MV board enable = Yes</td><td>No
Yes</td><td>Enables triggering “MV comms” when there is no medium voltage board.
Notes:
–This parameter is used for test benches without MV board, and we want to avoid being bothered by the fault. Do not modify this parameter without express approval from R&D department.
–This parameter is not available from version 2.2.0.</td><td>Yes</td></tr>
</table>

EN

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.2.24-En. Undue LVRT = Yes	No Yes	Enables triggering “Unable Reconnect” when the algorithm for fast reconnections is running. **Note:** This parameter is used in case the algorithm does not work correctly in the field. **Please, contact R&D department before using this parameter. Do not modify it without express approval.**	Yes
SG1.2.25-Num max cons LVRT = 5	0 to 10	Defines the maximum number of times that the system can avoid a fault status (code 17), while the LVRT event is taking place during the time configured in [SG1.2.26-max time cons LVRT] and before failing because of an improper LVRT event.	Yes
SG1.2.26-max time cons LVRT = 2 s	0 to 100 s	Defines the time frame reserved for consecutive fault status (code 17). If the system goes to a fault status, the number of times configured in [SG1.2.25-Num max cons LVRT] during the time configured in this parameter, the system will fail due to an improper LVRT event.	Yes
SG1.2.27-Opening Delay Filt C = 7 s	Disabled = 0; 1 to 10 s	Allows to configure the opening delay of the LC filter contactors in case of the inverter fault or after a stop.	Yes
SG1.2.28-K1 Pmin = 133	100 to 1000	Defines the K gain of the [G2.2.2-Min P] when there are a stopped module and a closed DC switch disconnector.	Yes
SG1.2.29-K2 Pmin = 200	100 to 1000	Defines the K gain of the [G2.2.2-Min P] when there are two stopped modules and a closed DC switch disconnector.	Yes
SG1.2.30-K3 Pmin = 400	100 to 1000	Defines the K gain of the [G2.2.2-Min P] when there are three stopped modules and a closed DC switch disconnector.	Yes
SG1.2.31-F165 Reset count. = No	No Yes	Allows to reset the counter of the double-signal modulation protection .	Yes

7.1.3. Subgroup SG1.3: FPGA service

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG1.3.1-FPGA Faults En. = XXXXXXXXXXX</td><td>00000000000 to XXXXXXXXXXX</td><td>Indicates which faults are enabled on the FPGA. Allows individual faults to be disabled.
<table>
<tr><th>BIT</th><th>FAULT</th></tr>
<tr><td>0</td><td>F67:Overcurrent DC</td></tr>
<tr><td>1</td><td>F71 to F83: Desaturation</td></tr>
<tr><td>2</td><td>F98:Idc derivation</td></tr>
<tr><td>3</td><td>F99: Mod. Iac derivation</td></tr>
<tr><td>4</td><td>F108: Mod. High Vdc</td></tr>
<tr><td>5</td><td>F109: Mod. Low Vdc</td></tr>
<tr><td>6</td><td>F114: Measure Ir</td></tr>
<tr><td>7</td><td>F115: Measure Is</td></tr>
<tr><td>8</td><td>F116: Measure It</td></tr>
<tr><td>9</td><td>F117: Measure Idc</td></tr>
<tr><td>10</td><td>F119 to F124: Desaturation</td></tr>
</table></td><td>No</td></tr>
</table>

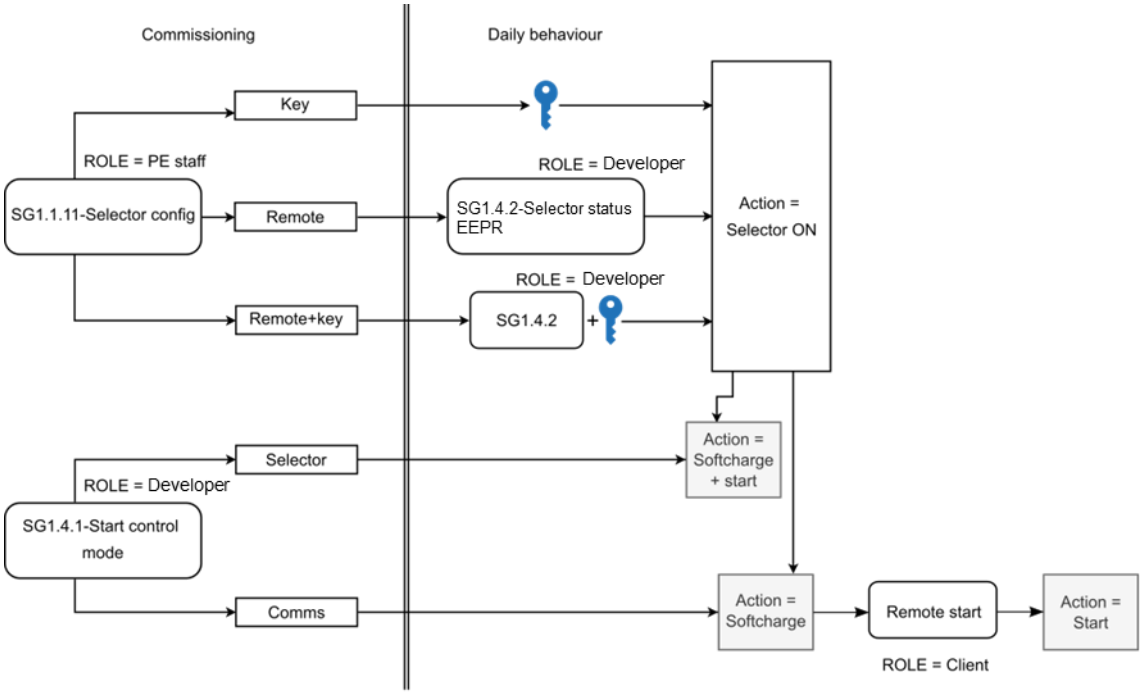
<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG1.3.2-VRT Retries = 5</td><td>0 to 10</td><td>In case of a voltage event, overcurrent will be detected. The module will attempt to restart as many times as the value set in this parameter and with the delay between retries defined in [SG1.2.15-IGBTs blocking time].
If the maximum number of retries has been reached, and the module has not been able to restart, fault [F65: Overcurrent AC] will be triggered.
Note: This parameter should not be modified, please contact Power Electronics before making any changes.</td><td>Yes</td></tr>
<tr><td>SG1.3.3-Stop measures = No</td><td>No
Yes</td><td>When the module is stopped, no current should be measured and group SV12 should display:
SV12.1.2.1-Current R = 0.0 A
SV12.1.2.2-Current S = 0.0 A
SV12.1.2.3-Current T = 0.0 A
If any sensor measures a value other than zero, then there is a problem in the current measurement. This parameter helps to identify where the problem lies.
If set to NO (default value), the inverter will force to zero the values of SV12.1.2.1, SV12.1.2.2 and SV12.1.2.3 so they always display 0.0 A, even if there is a problem in any of the current measurement elements (sensor, adapter board or even the central board).</td><td>Yes</td></tr>
<tr><td>SG1.3.4-Modulation type = Single</td><td>Single
Double</td><td>Setting for the type of modulation.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Single</td><td>Dynamic changes between single and double modulation Double modulation keeps two DC-Link semi buses the same, so IGBTs support the same voltage. But single modulation has better results in THDi emission. This option checks the voltage of the DC-Link so that when a Vsemibus is higher than 655 V when Q=0 then the double modulation will be followed.</td></tr><tr><td>Double</td><td>Keeps better balance between two DC-Link semi-buses.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.3.5-Vector Balance = 0XXX</td><td>0000 to XXXX</td><td>Sets the group enable some method to vector balance.<table><tr><th>BIT</th><th>STATUS</th></tr><tr><td>0</td><td>SV balance</td></tr><tr><td>1</td><td>SV balance central</td></tr><tr><td>2</td><td>SV unbalanced</td></tr><tr><td>3</td><td>FFVdc/2</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.3.6-K Vector Balance = 150</td><td>0 to 1000</td><td>Allows selection of vector balanced gain with single signal modulation.</td><td>Yes</td></tr>
<tr><td>SG1.3.6a-K Vector Balance VSI = 50</td><td>0 to 1000</td><td>Allows selection of vector balanced gain for VSI control type.</td><td>Yes</td></tr>
<tr><td>SG1.3.7-K FF Semibus = 0</td><td>0 to 1000</td><td>Allows selection of the gain of the single signal modulated semi bus feed-forward.</td><td>Yes</td></tr>
<tr><td>SG1.3.9-Mod Blackbox Fsample = 10000 Hz</td><td>0 to 10000 Hz</td><td>Sampling frequency of blackbox module.</td><td>Yes</td></tr>
<tr><td>SG1.3.10-En reduction alg = Disabled</td><td>Disabled
Enabled</td><td>Allows to enable or disable the reference reduction algorithm with which it is possible to change the Idc from the power module received by the DSP. This algorithm is not configurable, only can be enable or disable.
Note: this parameter is not available from version 4.2.0.</td><td>Yes</td></tr>
</table>

EN

7.1.4. Subgroup SG1.4: Start control

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG1.4.1-Start control mode = Selector</td><td>Selector
Comms</td><td>Selects the mode for start control (selector or communications) and will execute the soft charge.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Selector</td><td>The inverter will start automatically once start conditions are fulfilled.
This is the automatic start mode (only by switching SG1.4.2 to Yes, if the start conditions are met (G2.X) the inverter would reach RUN or ON status.</td></tr><tr><td>Comms</td><td>The inverter will start if start conditions are met and once it has received the START command either from the display or from communications.
This is the manual mode, once the selector (SG1.4.2=Yes) is activated, the inverter performs only the soft charge. Then, in order to start injecting power, the StartLocalControl command (register XXXX) or the green button on the display must be applied.</td></tr></table>Note: this parameter can only be edited with the Developer role.</td><td>Yes</td></tr>
<tr><td>SG1.4.2-Selector status EEPR = Yes</td><td>No
Yes</td><td>Defines whether the selector value should be stored in the EEPROM or not.
Note:
–This parameter will only be displayed if SG1.1.11 is set to ‘Remote’ or ‘Key + remote’.
–From version 5.0.0, the default value of this parameter is always Yes.
–This parameter can only be edited with the Developer role.</td><td>Yes</td></tr>
</table>

The following diagram shows the types of inverter startup indicating how the start-up operation should be performed on the HEM inverter.



Start-up operation in HEM inverters

7.1.5. Subgroup SG1.5: Modules configuration

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.5.1-Module 1 state = Enabled	Disabled Enabled Lockable Reserved	Displays the status of module 1. **Note:** This parameter is displayed if [SV1.3.1.4-Number of modules ≥ 1].	Yes
SG1.5.2-Module 2 state = Enabled	Disabled Enabled Lockable Reserved	Displays the status of module 2. **Note:** This parameter is displayed if [SV1.3.1.4-Number of modules ≥ 2].	Yes
SG1.5.3-Module 3 state = Enabled	Disabled Enabled Lockable Reserved	Displays the status of module 3. **Note:** This parameter is displayed if [SV1.3.1.4-Number of modules ≥ 3].	Yes
SG1.5.4-Module 4 state = Enabled	Disabled Enabled Lockable Reserved	Displays the status of module 4. **Note:** This parameter is displayed if [SV1.3.1.4-Number of modules ≥ 4].	Yes

EN

7.1.6. Subgroup SG1.6: Fault GFDI Reset

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.6.1-Attempt 1 Date = 01/01/2000 00:00	01/01/2000 00:00 to 31/12/2127 23:59	F41 can be reset manually. This parameter shows the date of the first attempt.	Yes
SG1.6.2-Attempt 2 Date = 01/01/2000 00:00	01/01/2000 00:00 to 31/12/2127 23:59	F41 can be reset manually. This parameter shows the date of the second attempt.	Yes
SG1.6.3-Reset Counter = No	No Yes	Reset the number of remaining GFDI reset retries.	Yes
SG1.6.4-Remaining Retries = 2	0 to 2	Displays the number of retries remaining to do a manual reset of the F41 before it locks into that fault.	Yes
SG1.6.5-Disable Measure = No	No Yes	Allows to perform a GFDI fault reset without the need for insulation measurements.	Yes

7.1.7. Subgroup SG1.7: Warning LED

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG1.7.1-Warning LED enable = Turn off Warning LED</td><td>Turn off Warning LED
Turn on Warning LED</td><td>Configures the warning LED to turn on or turn off when an external warning is triggered with the exception of warnings “W158: General Isolation”, “W159: Intern Isolation” and “W160: Extern Isolation”.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Turn off Warning LED</td><td>If this option is configured, then all external warnings should not turn or flash the LED. There are three exceptions: warnings “W158: General Isolation”, “W159: Intern Isolation” and “W160: Extern Isolation”. These warnings must always turn on or flash the warning LED when they are triggered.</td></tr>
<tr><td>Turn on Warning LED</td><td>If this option is configured, then all external warnings should turn on or flash the LED.</td></tr>
</table></td><td>No</td></tr>
</table>

7.1.8. Subgroup SG1.8: MV cell service

These parameters are only displayed if [SG10.1.3-Product type = HEMK] and [SG10.7.3-MV Skid = Yes, Motorized].

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG1.8.1-MV cell fault time = 10 s</td><td>1 to 30 s</td><td>Allows setting a timeout in MV switchgear automatic operation feedback failure. The failure in the operation process will trigger the fault once the timeout has elapsed with no cell operation.</td><td>Yes</td></tr>
<tr><td>SG1.8.2-MV cell switch status = Idle</td><td>Idle
Open
Closed</td><td>Enables the closing and/or opening maneuver of the MV switchgear, provided that the disconnector spring is charged.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Idle</td><td>No action is commanded to the MV switchgear, it is idle.</td></tr><tr><td>Open</td><td>Commands the action of opening the MV switchgear disconnector.</td></tr><tr><td>Closed</td><td>Commands the action of closing the MV switchgear disconnector.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.8.3- MV cell spring status = Unknown</td><td>Discharged
Charged
Unknown</td><td>Shows the MV switchgear spring status.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Discharged</td><td>MV switchgear spring is discharged.</td></tr><tr><td>Charged</td><td>MV switchgear spring is charged.</td></tr><tr><td>Unknown</td><td>Unknown status.</td></tr></table></td><td>No</td></tr>
<tr><td>SG1.8.4- MV gen switch status = Idle</td><td>Idle
Open
Closed</td><td>Shows the MV switchgear general switch status.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Idle</td><td>No action is commanded to the MV switchgear, it is idle.</td></tr><tr><td>Open</td><td>Commands the action of opening the MV switchgear disconnector.</td></tr><tr><td>Closed</td><td>Commands the action of closing the MV switchgear disconnector.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.8.5-MV earth switch state = Unknown</td><td>Open
Closed
Unknown</td><td>Shows the MV switchgear earth general switch status.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Open</td><td>Commands the action of opening the MV switchgear disconnector.</td></tr><tr><td>Closed</td><td>Commands the action of closing the MV switchgear disconnector.</td></tr><tr><td>Unknown</td><td>Unknown status.</td></tr></table></td><td>Yes</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG1.8.6-MV cell premag status = Premag off</td><td>Premag off
Premag check cell status
Premag check frequency MV
Premag check mod Vac MV
Premag close C filter
Premag start modules
Premag black start
Premag check Vac to close
Premag close MV
Premag done
Premag fault</td><td>Shows the current status of the premagnetization process.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Premag off</td><td>Premagnetization is not in progress.</td></tr><tr><td>Premag check cell status</td><td>Premagnetization is starting. The status of MV switchgear is checked.</td></tr><tr><td>Premag check frequency MV</td><td>Premagnetization is starting. The frequency of medium voltage is checked.</td></tr><tr><td>Premag check mod Vac MV</td><td>Premagnetization is starting. The voltage module at medium voltage is checked.</td></tr><tr><td>Premag close C filter</td><td>Premagnetization is starting. The filter is closed.</td></tr><tr><td>Premag start modules</td><td>Premagnetization is starting. The modules required for premagnetization are started up.</td></tr><tr><td>Premag black start</td><td>Premagnetization is starting. The necessary voltage is generated on the ramp</td></tr><tr><td>Premag check Vac to close</td><td>Premagnetization is starting. Synchronisation is performed in phase.</td></tr><tr><td>Premag close MV</td><td>Premagnetization is starting. MV switchgear is closed.</td></tr><tr><td>Premag done</td><td>Premagnetization is starting. Premagnetization is complete.</td></tr><tr><td>Premag fault</td><td>Premagnetization is starting. Premagnetization has failed.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG1.8.7-En. Shutdown premag = No</td><td>No
Yes</td><td>Allows the inverter to be connected to AC without premagnetising.
Note: This manoeuvre must be authorised by Power Electronics.</td><td>Yes</td></tr>
<tr><td>SG1.8.8-MV Cell op delay time = 2400 s</td><td>60 to 7200 s</td><td>Corresponds to the time that the system waits for the MV switchgear to open, once the conditions for opening are met.</td><td>Yes</td></tr>
<tr><td>SG1.8.9-Forze cond CL1 = No</td><td>No
Yes</td><td>Allows to force start conditions when the system is the inverter HEMK with motorized MV Skid that has voltage measure. These inverters need the CL1 contactor to be closed in order to carry out the premagnetization of the MV switchgear and be able to start up.
If this parameter is set to “Yes”, then the system will ignore the premagnetization event and will start when the low voltage conditions are met as long as CL1 contactor is not closed. In order to close the MV cell, at least when CL1 contactor must be closed.</td><td>Yes</td></tr>
</table>

7.1.9. Subgroup SG1.9: Faults initialization

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.9.1-General Init = No	No Yes	Corresponds to the default initialization of the configuration of all faults and warnings. Also set the initialization of parameters included in subgroups “G5.7.1-Faults”, “G5.7.2- Warnings”, “G5.8-Faults module config” and “G5.9-Faults busbar config”.	Yes
SG1.9.2-Faults Init = No	No Yes	Corresponds to the default initialization of the configuration of all faults included in subgroup “G5.7.1-Faults”.	Yes
SG1.9.3-Warnings Init = No	No Yes	Corresponds to the default initialization of the configuration of all warnings included in subgroup “G5.7.2- Warnings”.	Yes
SG1.9.4-Init Module fault = No	No Yes	Corresponds to default all module faults configuration included in “G5.8-Faults module”.	Yes
SG1.9.5-Init busbar fault = No	No Yes	Corresponds to default all busbar faults configuration included in “G5.9-Faults busbar config”.	Yes

EN

7.1.10. Subgroup SG1.10: VSI Services

This subgroup allows the configuration of advanced options for VSI control type.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.10.1-Black start enable = Disabled	Disabled Enabled	Enables or disables the option of start in Blackstart for VSI control type.	Yes
SG1.10.2-Force F13 enable = Disabled	Disabled Enabled	Allows to enable the F13 fault when an only module fails and stop the inverter.	Yes

7.1.11. Subgroup SG1.11: Autodiagnosis

This subgroup allows the AC/DC self-diagnostics configuration.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.11.1-Enable DC Diag = After fault/pwrup	Disabled Always enabled After fault/pwrup	Enables or disables the DC autodiagnosis. **OPTION** / **DESCRIPTION** Disabled: Autodiagnosis disabled. **Note:** DC Autodiagnosis will be re-enabled automatically 24 hours after disabled. Always enabled: The autodiagnosis is executed always the inverter starts. After fault/pwrup: The autodiagnosis is executed only after a failure or power up.	Yes
SG1.11.2-Enable AC Diag = Always enabled	Disabled Always enabled	Enables or disables the AC autodiagnosis. **OPTION** / **DESCRIPTION** Disabled: Autodiagnosis disabled. **Note:** AC Autodiagnosis will be re-enabled automatically 24 hours after disabled. Always enabled: The autodiagnosis is executed always the inverter starts.	Yes
SG1.11.3-N mod simult charge = 2	1 to SG10.1.12-Number of modules	Allows to configure the number of modules that will perform soft charging simultaneously.	Yes
SG1.11.4-Autodiag DC delay = 1440 min	1 to 1440 min	Allows to configure the DC autodiagnosis delay, i.e., the time to reactivate the DC autodiagnostic after disabling it. **Note**: Only users with role Developer can see and modify this parameter.	Yes
SG1.11.5-Autodiag AC delay = 1440 min	Disabled = 0; 1 to 1440 min	Allows to configure the AC self-test delay, i.e. the time to reactivate the AC self-test after it has been deactivated. **Note**: only users with Developer role can view and modify this parameter.	Yes

7.1.12. Subgroup SG1.13: PTP-DSP

This subgroup allows the adaptive exponential moving average (EMA) filter to be configured, which smooths the time synchronization error (clock offset) between the central control board's microcontroller (µP) and the DSP. The filter classifies each sample according to its synchronization error and applies a different smoothing factor (alpha) depending on the band in which that error falls.

Notes:*

- Synchronization between the uP and the DSP is part of the time synchronization chain PTP master clock–PSAMS–uP–DSP. For information about the PSAMS time source and PTP status, refer to section I.4.5 of Annex I: PSAMS.
- These parameters are only displayed if [G1.3-User] = Developer.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG1.13.1-EMA Alpha Outlier = 0.03	0.00 to 1.00	Configures the EMA filter alpha value applied when the synchronization error exceeds the outlier threshold ([SG1.13.6-Error Outlier Thr]). The lower it is, the more an isolated outlier is ignored, preventing a one-off spike from affecting synchronization.	Yes
SG1.13.2-EMA Alpha High = 0.25	0.00 to 1.00	Configures the EMA filter alpha value applied when the synchronization error is in the high error band [SG1.13.7-EMA High Thr], or when an outlier persists for 3 or more consecutive samples and is interpreted as a real change. The higher it is, the faster the tracking of time variations.	Yes
SG1.13.3-EMA Alpha Normal = 0.15	0.00 to 1.00	Configures the EMA filter alpha value applied when the synchronization error is in the normal error band [SG1.13.8-EMA Normal Thr]. It balances response speed and noise suppression.	Yes
SG1.13.4-EMA Alpha Low = 0.05	0.00 to 1.00	Configures the EMA filter alpha value applied when the synchronization error is in the low error band [SG1.13.9-EMA Low Thr]. It increases smoothing as the system approaches synchronization.	Yes
SG1.13.5-EMA Alpha Very Low = 0.03	0.00 to 1.00	Configures the EMA filter alpha value applied when the synchronization error falls below the low threshold [SG1.13.9-EMA Low Thr]. It maximizes stability and minimizes error fluctuations once synchronization is established.	Yes
SG1.13.6-Error Outlier Thr = 30 µs	0 to 100 µs	Configures the synchronization error threshold above which a sample is considered an outlier and the corresponding alpha value [SG1.13.1-EMA Alpha Outlier] is applied.	Yes
SG1.13.7-EMA High Thr = 10 µs	0 to 100 µs	Configures the synchronization error threshold above which the high alpha value [SG1.13.2-EMA Alpha High] is applied.	Yes
SG1.13.8-EMA Normal Thr = 5 µs	0 to 100 µs	Configures the synchronization error threshold above which the normal alpha value [SG1.13.3-EMA Alpha Normal] is applied.	Yes
SG1.13.9-EMA Low Thr = 2 µs	0 to 100 µs	Configures the synchronization error threshold above which the low alpha value [SG1.13.4-EMA Alpha Low] is applied. Below this threshold, the very low alpha value [SG1.13.5-EMA Alpha Very Low] is used.	Yes
SG1.13.10-EMA Reset Thr = 100 µs	0 to 1000 µs	Configures the synchronization error threshold above which the EMA filter is reset, enabling synchronization to be quickly recovered after a large time deviation.	Yes
SG1.13.11-RTT Tolerance = 100 µs	10 to 200 µs	Configures the accepted margin for the round-trip time (RTT) of the timestamp exchange. The sample is only accepted if its RTT falls within this margin.	Yes
SG1.13.12-Precision Threshold = 15 µs	2 to 20 µs	Configures the minimum accuracy required for synchronization. Once synchronization is established, the sample is only accepted if the time difference between µP and DSP is lower than this value.	Yes
SG1.13.13-Max Rej Samples = 8	1 to 50	Configures the maximum number of consecutive samples that can be rejected before synchronization is reset.	Yes

EN

7.2. Group SG2: Debug

This group includes debug variables used internally to debug functionalities.

7.2.1. Subgroup SG2.1: uP debug

SG2.1.1: uP Debug vars

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG2.1.1.1-Micro debug 1 = 0	0 to 65535	Microcontroller debug variables for testing and programming.	Yes
SG2.1.1.2-Micro debug 2 = 0	0 to 65535		Yes
SG2.1.1.3-Micro debug 3 = 0	0 to 65535		Yes
SG2.1.1.4-Micro debug 4 = 0	0 to 65535		Yes
SG2.1.1.5-Micro debug 5 = 0	0 to 65535		Yes
SG2.1.1.6-Micro debug 6 = 0	0 to 65535		Yes
SG2.1.1.7-Micro debug 7 = 0	0 to 65535		Yes
SG2.1.1.8-Micro debug 8 = 0	0 to 65535		Yes
SG2.1.1.9-Micro Visual 1 = 0	0 to 65535	Microcontroller visual variables for testing and programming.	Yes
SG2.1.1.10-Micro Visual 2 = 0	0 to 65535		Yes
SG2.1.1.11-Micro Visual 3 = 0	0 to 65535		Yes
SG2.1.1.12-Micro Visual 4 = 0	0 to 65535		Yes
SG2.1.1.13-Micro Visual 5 = 0	0 to 65535		Yes
SG2.1.1.14-Micro Visual 6 = 0	0 to 65535		Yes
SG2.1.1.15-Micro Visual 7 = 0	0 to 65535		Yes
SG2.1.1.16-Micro Visual 8 = 0	0 to 65535		Yes

SG2.1.2: uP Debug vars

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG2.1.2.1-Init micro mode 1 = No data</td><td>No data
Watchdog
Power on
Sw reset
Hardfault</td><td rowspan="3">Sets the initial mode of the microcontroller, according to the following table:<table><tr><th>OPTION</th><th>NORMATIVE</th></tr><tr><td>No data</td><td>There is no information.</td></tr><tr><td>Watchdog</td><td>Start by watchdog.</td></tr><tr><td>Power on</td><td>Start by power on.</td></tr><tr><td>Sw reset</td><td>Start by software reset.</td></tr><tr><td>Hardfault</td><td>Start by hardfault.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG2.1.2.2-Init micro mode 2 = No data</td><td>No data
Watchdog
Power on
Sw reset
Hardfault</td><td>Yes</td></tr>
<tr><td>SG2.1.2.3-Init micro mode 3 = No data</td><td>No data
Watchdog
Power on
Sw reset
Hardfault</td><td>Yes</td></tr>
<tr><td>SG2.1.2.4-Hardfault LR = 0x00</td><td>0x00 to 0xFFFF0000</td><td>Sets LR (Link Register) register allocation.</td><td>Yes</td></tr>
<tr><td>SG2.1.2.5-Hardfault PC = 0x00</td><td>0x00 to 0xFFFF0000</td><td>Sets PC (Program Counter) register allocation.</td><td>Yes</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG2.1.2.6-Last Running Task = RTX Idle</td><td>Unknow
RTX Idle
RTX Timers
NetCore
NetEth0
Watchdog
Menu Debug
Debug Task Serve
Master Serial 1
Master Serial 2
Master Serial 3
GPIO Tools
Periodic
EEPROM
Network
Autotest
Autotest Polling
Fault Manager
Fault Obs Sub
DSP Fast
DSP Slow
FPGA Comms
DSP SCI
SD Manager
SD Log Msg
SD Delete
MVI Temperatures
I2C
Isolation Maneuver
Event Manager
Caiso
SD Supervisor
SD event Enable
SD Card Enable
Broker
AntiPID
Blackbox
SD Log Sys
Recovery manager
TCP logger</td><td>This parameter is for internal use by the developer. What each of the options indicates is where the uP has been before jumping the uP watchdog.</td><td>Yes</td></tr>
<tr><td>SG2.1.2.7-Last task ThrID = 0x00</td><td>0x00 to 0xFFFF0000</td><td>Sets the last running task Thread ID before the reset.</td><td>Yes</td></tr>
<tr><td>SG2.1.2.8-Rt Watchdog Enable = 0000000000000000</td><td>0000000000000000 to XXXXXXXXXXXXXXXX</td><td>Sets the Watchdog Enable Mask before Reset.<table><tr><th>BIT</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>Modules Init</td></tr><tr><td>1</td><td>Network</td></tr><tr><td>2</td><td>Periodic</td></tr><tr><td>3</td><td>EEPROM</td></tr><tr><td>4</td><td>FPGA Comms</td></tr><tr><td>5</td><td>DSP Slow</td></tr><tr><td>6</td><td>DSP Fast</td></tr><tr><td>7</td><td>Diagnostic</td></tr><tr><td>8</td><td>Faults Sub Obs</td></tr></table></td><td>Yes</td></tr>
</table>

EN

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG2.1.2.9-Rt Watchdog Update = 0000000000000000</td><td>0000000000000000 to XXXXXXXXXXXXXXXX</td><td>Sets the tasks of the microcontroller that have updated their watchdog.
<table><tr><th>BIT</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>Modules Init</td></tr><tr><td>1</td><td>Network</td></tr><tr><td>2</td><td>Periodic</td></tr><tr><td>3</td><td>EEPROM</td></tr><tr><td>4</td><td>FPGA Comms</td></tr><tr><td>5</td><td>DSP Slow</td></tr><tr><td>6</td><td>DSP Fast</td></tr><tr><td>7</td><td>Diagnostic</td></tr><tr><td>8</td><td>Faults Sub Obs</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG2.1.2.10-Reset Modbus Addr = 0</td><td>0 to 65535</td><td>Sets the address from the last Modbus access before reset.</td><td>Yes</td></tr>
<tr><td>SG2.1.2.11-Reset Modbus Value = 0</td><td>0 to 65535</td><td>Sets the value from the last Modbus access before reset.</td><td>Yes</td></tr>
<tr><td>SG2.1.2.12-Reset Modbus IP = 0.0.0.0</td><td>1.0.0.0 to 255.255.255.255</td><td>Displays the last IP that accessed the inverter before a system reset was triggered.</td><td>Yes</td></tr>
</table>

SG2.1.3: uP Tasks

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG2.1.3.1-Selected Task = Unknow	Unknow RTX Idle RTX Timers NetCore NetEth0 Watchdog Menu Debug Debug Task Serve Master Serial 1 Master Serial 2 Master Serial 3 GPIO Tools Periodic EEPROM Network Autotest Autotest Polling Fault Manager Fault Obs Sub DSP Fast DSP Slow FPGA Comms DSP SCI SD Manager SD Log Msg SD Delete MVI Temperatures I2C Isolation Maneuver Event Manager Caiso SD Supervisor SD event Enable SD Card Enable Broker AntiPID Blackbox SD Log Sys Recovery manager TCP logger PTP I28K logger Log queue SD	This parameter is for internal use by the developer. What each of the options indicates is where the uP has been before jumping the uP watchdog.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG2.1.3.2-CPU Usage = 0.00 %	0.00 to 100.00 %	Sets selected task CPU Use.	Yes
SG2.1.3.3-Max CPU Usage = 0.00 %	0.00 to 100.00 %	Sets selected task CPU use maximum.	Yes
SG2.1.3.4-Reset Max CPU Usage = No	No Yes	Sets selected task CPU reset maximum.	Yes

SG2.1.4:uP Log SYS

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG2.1.4.1-Debug logging level = INFO log level</td><td>DEBUG log level
INFO log level
WARNING log level
ERROR log level
CRITICAL log level</td><td>Allows to select the quantity of depuration frames stored on the SD card.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>DEBUG log level</td><td>Maximum quantity of frames stored.</td></tr><tr><td>INFO log level</td><td>Frames from the CRITICAL, ERROR, WARNING and this log level are stored.</td></tr><tr><td>WARNING log level</td><td>Frames from the CRITICAL, ERROR and this log level are stored.</td></tr><tr><td>ERROR log level</td><td>Frames from the CRITICAL and this log level are stored.</td></tr><tr><td>CRITICAL log level</td><td>Minimum quantity of frames stored.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG2.1.4.2-Changes in faultsss log = Enabled</td><td>Disabled
Enabled</td><td>Enables or disables the register in SD changes in MCF variables. Some of the information stored in the SD card, in particular, in the so-called “SYS log” is related with the modifications of certain defined parameters configuration by an external Modbus /TCP device. The log shows which parameter has been modified, the previous and current value, besides the IP who has made this change. With this parameter, it is possible to enable/disable this functionality. This means, enables/disables these events to be stored in the SD card.</td><td>Yes</td></tr>
<tr><td>SG2.1.4.3-Lim frames/s Modbus = 20</td><td>1 to 100</td><td>Allows to configure the maximum number of traces per second that can be stored in the “SYS log” in order to avoid wrong behavior of the SD card, regarding the modifications of certain defined parameters configuration functionality. When this limit is reached, the message “Max Modbus Events/s reached" is shown in the SYS log in order to notify the user that no more information would be included during that second. After this, the system would take up the saving data algorithm.</td><td>Yes</td></tr>
<tr><td>SG2.1.4.4-Maximeter frames/s = 0</td><td>0 to 65535</td><td>Shows the number of traces per second that has been reached. This data is updated every second. This parameter is used for debugging purposes to monitor the Modbus variations with which the DUT (Device Under Test) is working.</td><td>Yes</td></tr>
<tr><td>SG2.1.4.5-Code logging enable = 0000000000000000</td><td>0000000000000000 to XXXXXXXXXXXXXXXX</td><td>Allows enabling or disabling the generation of logs for internal software modules. The available options are:<table><tr><th>BIT</th><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>0</td><td>FILE OPS</td><td>SD card write module log.</td></tr><tr><td>1</td><td>PTP DSP SD LOG</td><td>SD card log of the PTP synchronization between the µP and the DSP via SCI.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG2.1.4.6-Enanle TCP log = No</td><td>No
Yes</td><td>Allows enabling or disabling the real-time transmission of system logs over TCP.</td><td>Yes</td></tr>
<tr><td>SG2.1.5-Fault code invalid = 0</td><td>0 to 65535</td><td>Shows fault code invalid.</td><td>Yes</td></tr>
</table>

EN

7.2.2. Subgroup SG2.2: DSP debug

In this version of software, the visualization of internal DSP references and limits that can be monitored using the visualization debug parameters from this subgroup are also added. It is highly recommended to include these parameters in the custom log from group G12.1: Parameters logger, when the plant contains inverters that limit active power with no reason.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG2.2.1-Config DSP 1 = 0	0 to 65535	DSP debug variables for testing and programming.	Yes
SG2.2.2-Config DSP 2 = 0	0 to 65535		Yes
SG2.2.3-Config DSP 3 = 0	0 to 65535		Yes
SG2.2.4-Config DSP 4 = 0	0 to 65535		Yes
SG2.2.5-Config DSP 5 = 0	0 to 65535		Yes
SG2.2.6-Config DSP 6 = 0	0 to 65535		Yes
SG2.2.7-Config DSP 7 = 0	0 to 65535		Yes
SG2.2.8-Config DSP 8 = 0	0 to 65535		Yes
SG2.2.9-Config % DSP 1 = 0.00 %	-25000 to 25000		Yes
SG2.2.10-Config % DSP 2 = 0.00 %	-25000 to 25000		Yes
SG2.2.11-Config % DSP 3 = 0.00 %	-25000 to 25000		Yes
SG2.2.12-Config % DSP 4 = 0.00 %	-25000 to 25000		Yes
SG2.2.13-Visual DSP 1 = 0	0 to 65535		Yes
SG2.2.14-Visual DSP 2 = 0	0 to 65535		Yes
SG2.2.15-Visual DSP 3 = 0	0 to 65535		Yes
SG2.2.16-Visual DSP 4 = 0	0 to 65535		Yes
SG2.2.17-Visual DSP 5 = 0	0 to 65535		Yes
SG2.2.18-Visual DSP 6 = 0	0 to 65535		Yes
SG2.2.19-Visual DSP 7 = 0	0 to 65535		Yes
SG2.2.20-Visual DSP 8 = 0	0 to 65535		Yes
SG2.2.21-Visual % DSP 1 = 0.00 %	-25000 to 25000		Yes
SG2.2.22-Visual % DSP 2 = 0.00 %	-25000 to 25000		Yes
SG2.2.23-Visual % DSP 3 = 0.00 %	-25000 to 25000		Yes
SG2.2.24-Visual % DSP 4 = 0.00 %	-25000 to 25000		Yes

7.2.3. Subgroup SG2.3: FPGA debug

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG2.3.1-Config FPGA 1 = 0	0 to 65535	FPGA debug variables for testing and programming.	Yes
SG2.3.2-Config FPGA 2 = 0	0 to 65535		Yes
SG2.3.3-Config FPGA 3 = 0	0 to 65535		Yes
SG2.3.4-Config FPGA 4 = 0	0 to 65535		Yes
SG2.3.5-Config FPGA 5 = 0	0 to 65535		Yes
SG2.3.6-Config FPGA 6 = 0	0 to 65535		Yes
SG2.3.7-Config FPGA 7 = 0	0 to 65535		Yes
SG2.3.8-Config FPGA 8 = 0	0 to 65535		Yes
SG2.3.9-Visual FPGA 1 = 0	0 to 65535	Displays FPGA debug variables.	Yes
SG2.3.10-Visual FPGA 2 = 0	0 to 65535		Yes
SG2.3.11-Visual FPGA 3 = 0	0 to 65535		Yes
SG2.3.12-Visual FPGA 4 = 0	0 to 65535		Yes
SG2.3.13-Visual FPGA 5 = 0	0 to 65535		Yes
SG2.3.14-Visual FPGA 6 = 0	0 to 65535		Yes
SG2.3.15-Visual FPGA 7 = 0	0 to 65535		Yes
SG2.3.16-Visual FPGA 8 = 0	0 to 65535		Yes

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7.2.4. Subgroup SG2.4: DU debug

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG2.4.1-Config DU 1 = 0	0 to 65535	DU debug variables for testing and programming.	Yes
SG2.4.2-Config DU 2 = 0	0 to 65535		Yes
SG2.4.3-Config DU 3 = 0	0 to 65535		Yes
SG2.4.4-Config DU 4 = 0	0 to 65535		Yes
SG2.4.5-Visual DU 1 = 0	0 to 65535	Displays DU debug variables.	Yes
SG2.4.6-Visual DU 2 = 0	0 to 65535		Yes
SG2.4.7-Visual DU 3 = 0	0 to 65535		Yes
SG2.4.8-Visual DU 4 = 0	0 to 65535		Yes

7.3. Group SG3: Remote log

This group configures the remote log (Grafana) and enables this functionality. The remote log allows to monitor variables in real time and to be able to graph them to analyze the behavior of the inverter over time. It is an alternative to the SD.

7.3.1. Subgroup SG3.1: Conexion

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG3.1.1-Enable log = Disabled	Disabled Enabled	Enable log.	Yes
SG3.1.2-TimeOut send log = 1 s	1 to 65535 s	Displays the timeout send log.	Yes
SG3.1.3-Server access mode = IP	IP Name	Displays the server access mode.	Yes
SG3.1.4-Port server = 8080	0 to 65535	Port server.	Yes
SG3.1.5-Server name =	-	Sets the server's name.	Yes
SG3.1.6-Subdomain name =	-	Sets the subdomain name.	Yes
SG3.1.7-IP log server = 18.101.61.62	0.0.0.0 to 255.255.255.255	Shows IP log server.	Yes
SG3.1.8-Log websocket status = Not connected	Not connected Connecting Sending header Waiting for response Processing connection Connected Timeout	Shows the log web socket status.	Yes
SG3.1.9-Log websocket timeout = 5 s	1 to 65535 s	Shows the log web socket timeout.	No
SG3.1.10-Local port = 1011	0 to 65535	Shows the local port.	Yes

7.3.2. Subgroup SG3.2: Configurable variables

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG3.2.1a-ID_1 local PCB = 0	0 to 65535	Configure which extra addresses need to be displayed in Grafana, in addition to the ones you already have fixed.	Yes
SG3.2.1a-ID_2 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_3 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_4 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_5 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_6 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_7 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_8 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_9 local PCB = 0	0 to 65535		Yes
SG3.2.1a-ID_10 local PCB = 0	0 to 65535		Yes

EN

7.4. Group SG4: Test mode

This group includes parameters to configure certain conditions when not working with the real inverter, in order to simulate some states and/or measurements.

7.4.1. Subgroup SG4.40: Troubleshooting mode

This subgroup allows to configure the troubleshooting mode.

Note: it is only displayed if [G1.3-User mode ≥ developer].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG4.40.1-State = Disabled	Disabled = 0 1 to 65535	Enables access to the troubleshooting mode.	Yes
SG4.40.2-Receiv heartbeat = No	No Yes	Displays whether a heartbeat is being received.	Yes
SG4.40.3-Heartbeat reg = 0	0 to 65535	Displays the number of heartbeats received.	Yes
SG4.40.4-Heartbeat loss time = 10 s	1 to 30 s	Set the time to detect that the heartbeat has been lost.	Yes
SG4.40.5-End time = 5 min	1 to 30 min	Allows to set the time to end the troubleshooting mode since the last heartbeat received.	Yes

7.4.2. Subgroup SG4.45: Busbar setpoint fakes

This subgroup, available only in DEMO compilation mode, allows to simulate the power measurements used by the system by replacing the measurements sent by the DSP with user-configurable values.

Note: it is only displayed if [G1.3-User mode] ≥ Developer.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG4.45.1-Enable simulation = 0	0 to 1	Allows to enable or disable the simulation of busbar power measurements. OPT. 0: Simulation is disabled. The system uses the measurements sent by the DSP. OPT. 1: Simulation is enabled. The system uses the simulated values configured in parameters from [SG4.45.2-Busbar 1 P sim] to [SG4.45.6-Q sim].	Yes
SG4.45.2-Busbar 1 P sim = 0.00 %	-100.00 to 100.00 %	Allows to configure the simulated active power value of Busbar 1. When simulation is enabled, this value is displayed in [SV1.11.1-Busbar1 P ref].	Yes
SG4.45.3-Busbar 2 P sim = 0.00 %	-100.00 to 100.00 %	Allows to configure the simulated active power value of Busbar 2. When simulation is enabled, this value is displayed in [SV1.11.2-Busbar2 P ref].	Yes
SG4.45.4-Busbar 3 P sim = 0.00 %	-100.00 to 100.00 %	Allows to configure the simulated active power value of Busbar 3. When simulation is enabled, this value is displayed in [SV1.11.3-Busbar3 P ref].	Yes
SG4.45.5-Busbar 4 P sim = 0.00 %	-100.00 to 100.00 %	Allows to configure the simulated active power value of Busbar 4. When simulation is enabled, this value is displayed in [SV1.11.4-Busbar4 P ref].	Yes
SG4.45.6-Q sim = 0.00 %	-100.00 to 100.00 %	Allows to configure the simulated reactive power value. When simulation is enabled, this value is displayed in [SV1.11.5-Busbars Q ref].	Yes
SG4.45.7-Sharing P sim = 0.00 %	-100.00 to 100.00 %	Allows to configure the simulated value for the active power sharing setpoint.	Yes
SG4.45.8-Sharing Q sim = 0.00 %	-100.00 to 100.00 %	Allows to configure the simulated value for the reactive power sharing setpoint.	Yes

7.4.3. Subgroup SG4.46: Idc Compensation

This subgroup allows configuration of DC compensation for the signal generated in VISMA V-F mode.

Note: this group is only displayed if [G1.3-User mode] ≥ Developer.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG4.46.1-Enable Idc Offset = No	No Yes	Allows to enable DC current compensation in AC current measurement.	Yes
SG4.46.2-Calculate Idc Offset = No	No Yes	Allows to enable DC current compensation calculation. When set to NO and [SG4.46.1-Enable Idc offset] = Yes, the values configured in [SG4.46.6-Conf. Idc Offset R], [SG4.46.7-Conf. Idc Offset S] and [SG4.46.8-Conf. Idc Offset T] are applied.	Yes
SG4.46.3-Vis. Idc Offset R = 0	-2048 to 2048	Displays the calculated offset for the R phase. **Note:** this parameter is read-only (RO).	Yes
SG4.46.4-Vis. Idc Offset S = 0	-2048 to 2048	Displays the calculated offset for the S phase **Note:** this parameter is read-only (RO).	Yes
SG4.46.5-Vis. Idc Offset T = 0	-2048 to 2048	Displays the calculated offset for the T phase **Note:** this parameter is read-only (RO).	Yes
SG4.46.6-Conf. Idc Offset R = 0	-2048 to 2048	Allows configuration of the calculated offset for the R phase	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG4.46.7-Conf. Idc Offset S = 0	-2048 to 2048	Allows configuration of the calculated offset for the S phase	Yes
SG4.46.8-Conf. Idc Offset T = 0	-2048 to 2048	Allows configuration of the calculated offset for the T phase	Yes

Others:

- **Note:** These parameters (except for [SG4.20.1-Test frequency centr] and [SG4.20.2-Test frequency delta]) can only be modified with the role Developer. In addition, parameters from SG4.28 to SG4.44 can only be seen with the role Developer.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG4.1-Force start conditions = 000	000 to XXX	Allows to force the start conditions. BIT 0: Force start conditions uP. BIT 1: Force start conditions DSP. BIT 2: Force start conditions emergency stop.	Yes
SG4.2-Start delays = Yes	No Yes	When set to Yes, allows entering the start delay according to the normative.	Yes
SG4.3-Open loop = No	No Yes	Configure open loop control. **Note:** This parameter is shown if SG4.10 and SG4.11 are configured.	Yes
SG4.4-Open loop stop = Ramp	Ramp Instant	Configurate open loop stop. **Note:** This parameter is shown if SG4.3 = Yes.	Yes
SG4.5-Short circuit = No	No Yes	Enables short circuit mode.	Yes
SG4.6-AC source = No	No Yes	Enables or disables AC source mode. **Note:** This parameter is shown if SG4.13, SG4.14 and SG4.15 are configured.	Yes
SG4.7-Cold commissioning = No	No Yes	Enables cold commissioning. If this mode is selected, modules that will work as AC source must be configured in SG4.17.	Yes
SG4.8-Resistors mode = No	No Yes	Enables resistors mode.	Yes
SG4.9-Ref Vd = 1.000	0.000 to 1.000	Sets the active power reference for open loop control. **Note:** This parameter is shown if SG4.3 = Yes.	Yes
SG4.10-Ref Vq = 0.000	-1.000 to 1.000	Sets the reactive power reference for open loop control. **Note:** This parameter is shown if SG4.3 = Yes.	Yes
SG4.11-Ref Vd Inv = 0.000	-1.000 to 1.000	Sets the active power reference for open loop control in inverse sequence. **Note:** This parameter is shown if SG4.3 = Yes.	Yes
SG4.12-Ref Vq Inv = 0.000	-1.000 to 1.000	Sets the reactive power reference for open loop control in inverse sequence. **Note:** This parameter is shown if SG4.3 = Yes.	Yes
SG4.13-Ref Vac = 690 V	50 to 900 V	Sets voltage reference. **Note:** This parameter is shown if SG4.6 = Yes.	Yes
SG4.14-Grid = Yes Hz	Yes = 39.9; 40.0 to 70.0 Hz	If this parameter is set to Yes, the inverter uses the grid normally (this should always be the setting on field). Otherwise, a grid of 40 to 70 Hz can be simulated, which is useful for testing (e.g. set as AC source, cold commissioning…). Ensure to set SG4.14 back to 'Yes' once tests have been finished.	Yes
SG4.15-Sequence = Positive	Positive Negative	If [SG4.14-Grid] is other than Yes, use this parameter to configure the grid sequence as positive or negative.	Yes

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<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG4.16-FF Vdc = No</td><td>No
Yes</td><td>Sets the Vdc feedforward.
Note: This parameter is shown if [SG4.6-AC source = Yes] and [SG4.3-Open loop = Yes].</td><td>Yes</td></tr>
<tr><td>SG4.17-AC source mod select = 0000000000</td><td>0000000000 to XXXXXXXXXX</td><td>Sets which modules function as AC source.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Enable module 1</td><td>8</td><td>Not available</td></tr>
<tr><td>1</td><td>Lock module 1</td><td>9</td><td>Not available</td></tr>
<tr><td>2</td><td>Enable module 2</td><td>10</td><td>Not available</td></tr>
<tr><td>3</td><td>Enable module 2</td><td>11</td><td>Not available</td></tr>
<tr><td>4</td><td>Enable module 3</td><td>12</td><td>Not available</td></tr>
<tr><td>5</td><td>Lock module 3</td><td>13</td><td>Not available</td></tr>
<tr><td>6</td><td>Enable module 4</td><td>14</td><td>Not available</td></tr>
<tr><td>7</td><td>Lock module 4</td><td>15</td><td>Not available</td></tr>
</table>
Note: These options are shown if [SV1.3.1.4-Number of modules < n] being ‘n’ the module number.</td><td>Yes</td></tr>
<tr><td>SG4.18-Grid protections = No</td><td>No
Yes</td><td>Shows if grid protections are enables.</td><td>Yes</td></tr>
<tr><td>SG4.19-DC source = No</td><td>No
Yes</td><td>The inverter acts as a DC source regulating the DC BUS voltage.</td><td>Yes</td></tr>
<tr><td>SG4.20.1-Test frequency centr = Disabled</td><td>Disabled
50 Hz
60 Hz</td><td>Allows to disable DC control.</td><td>Yes</td></tr>
<tr><td>SG4.20.2-Test frequency delta = 0.000</td><td>-30.000 to 30.000</td><td>Shows the test frequency delta.
Note: This parameter is shown if [SG4.20.1-Test frequency centr ≥ 50 Hz].</td><td>Yes</td></tr>
<tr><td>SG4.21-IMI Comms = Enabled</td><td>Disabled
Enabled</td><td>Enable/disabled the IMI Comms.</td><td>Yes</td></tr>
<tr><td>SG4.22-Disable DC BK = No</td><td>No
Yes</td><td>Shows if the DC BK is enabled.</td><td>Yes</td></tr>
<tr><td>SG4.23-W90 disable = No</td><td>No
Yes</td><td>Disable W90 when IGBTs temperature is over Drive Select max temperature allowed.</td><td>Yes</td></tr>
<tr><td>SG4.24-Modulation = Field</td><td>Field
Test</td><td>Allows to change configure the modulation type on power stages. Field applies last implement modulation on SW. Tester applies ANPC-A modulation in order to avoid sudden explosion (limits bus voltage to 1000 V).</td><td>No</td></tr>
<tr><td>SG4.25-Disable NTC/PT faults = No</td><td>No
Yes</td><td>Allows to enable/disable NTC/PT100 not connected faults. The uP will send the bit to the DSP and disable uP faults related to this functionality.</td><td>No</td></tr>
<tr><td>SG4.26-Force module discharge = No</td><td>No
Yes</td><td>Allows DSP to activate active or deactivate the heating mode even when no conditions are met.
Note: this parameter is intended for testing and diagnostic purposes and should not be used during normal operation.</td><td>Yes</td></tr>
<tr><td>SG4.27-Force active heating = No</td><td>No
Yes</td><td>This allows to force active heating mode by acting on the µP ↔ DSP communication.
<table>
<tr><th>BIT</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Force µP conditions: forces the bit that the µP sends to the DSP to enable active heating.</td></tr>
<tr><td>1</td><td>Force DSP conditions: forces the feedback bit that the DSP sends to the uP, so that the uP interprets the mode as active/inactive and executes the associated actions.</td></tr>
</table>
Note: this parameter is intended for testing and diagnostic purposes and should not be used during normal operation.</td><td>Yes</td></tr>
<tr><td>SG4.28-DU Error type = Fault</td><td>Fault
Warning</td><td>Sets the action when the fans fault in the DC cabinet zone.
<table>
<tr><th>OPTION</th><th>DESCRIPTION</th></tr>
<tr><td>Fault</td><td>“F205: Fan DU zone” will be displayed.</td></tr>
<tr><td>Warning</td><td>“W162: Fan DU zone” will be displayed.</td></tr>
</table></td><td>Yes</td></tr>
</table>

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG4.29-MT Error type = Fault	Fault Warning	Sets the action when the fans fault in the MV cabinet zone. OPTION: Fault — DESCRIPTION: "F204: Fan impulsion MT" will be displayed. OPTION: Warning — DESCRIPTION: "W161: Fan impulsion MT" will be displayed. **Note:** Only HEM / PCSM inverter [SG10.1.3-Product type = HEM].	Yes
SG4.30-Module Error type = Fault	Fault Warning	Sets the action when the fans fault in the modules zone. OPTION: Fault — DESCRIPTION: "F207: Fan zone modules" will be displayed. OPTION: Warning — DESCRIPTION: "W164: Fan zone modules" will be displayed.	Yes
SG4.31-LC filter error type = Fault	Fault Warning	Sets the action when the fans fault in the LC filter zone. OPTION: Fault — DESCRIPTION: "F206: Fan zone filter LC" will be displayed. OPTION: Warning — DESCRIPTION: "W163: Fan zone filter LC" will be displayed.	Yes
SG4.32-Global error type = Fault	Fault Warning	Configures whether the global ventilation fault is a fault or a warning.	Yes
SG4.33-Antipid delay value = 1440 min	0 to 65535 min	Allows to configure the delay time between each retry. The Anti-PID algorithm includes some protections against wrong behavior due to external causes. Thus, in case it has been tried to start the Anti-PID by three times, but it is not possible to finish the activation sequence, the system does not try it anymore until the next day.	Yes
SG4.34-Fans 485 Enable = Yes	No Yes	Enables or disables the functionality of the fans connected with RS485. **Note:** this parameter, in versions prior to 3.0.0, used to be [G11.11-Fans 485 Enable].	Yes
SG4.35-Pwd Fans Extend ranges = Disabled	Disabled = 0; 1 to 65535	Allows to enter a password to extend ventilation ranges (for test purposes only). If the password introduced is correct, the parameter will show Enabled.	Yes
SG4.36-Enable F164 = Enabled	Disabled Enabled	Enables or disables [F164: Excessive Maneuvers] (for test purposes only).	Yes
SG4.37 Password for Inhibited mixed fans fault after fault reset = No	No = 0; 1 to 65535	Allows to enter a password to inhibit reported faults due to mixed fans in the inverter. If the password introduced is correct, the parameter will show Yes. **Note:** consult Power Electronics for password.	Yes
SG4.38-Enable F241 = Enabled	Disabled Enabled	Allows to inhibit the reporting of [F241:Delta T max DU]. **Note:** from version 4.1.0, F241 is deprecated and it has been replaced by W241.	Yes
SG4.39-Autoverification mode = Disabled	Disabled = 0 1 to 65535	Allows entering a password to access the self-test mode. If the password entered is correct, the parameter will show Enabled.	Yes
SG4.41-Manual frames set = 0	0 to 65535	It allows to set the counter of correct and lost frames and to verify the reset to 0 after an overflow (for tests only).	Yes
SG4.42-Enable fault 202 = Enabled	Disabled Enabled	Allows to enable or disable [F202: Taux overtemperature] (for testing only).	Yes
SG4.43-Close DU = Disabled	Disabled Enabled	Allows to simulate that the DU has been closed to report [F45: General Isolation].	Yes
SG4.44-SD Delete Retry time = 60 min	1 to 60 min	Allows configuring the retry wait time for the SD card file deletion task when the SD card is not available.	Yes
SG4.47-Disch With Closed cell = No	No Yes	Allows configuring alternative conditions for the transition to Discharge. If you select "YES", the circuit will close with the switchgear closed if there are no low or medium voltage readings.	Yes

EN

7.5. Group SG5: Filter maintenance

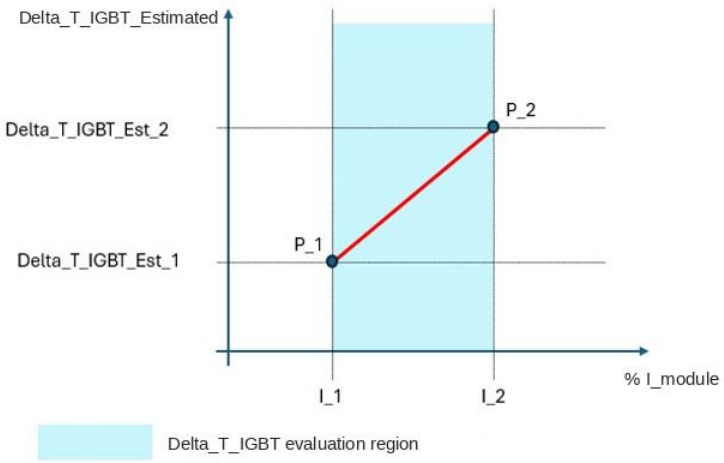
7.5.1. Subgroup SG5.1: Modules filter mainten

This subgroup allows detecting the need to carry out maintenance tasks on the air filters or on the ventilation system of the power stages by detecting abnormal temperature increases in the IGBTs.

For each of the active modules, the temperature increases between the hottest IGBT of the module and the ambient temperature measured at the inverter air admission [SV1.6.4-Admission temperature] (Delta_T_IGBT_Real) is calculated. This calculation is performed with a periodicity of one minute and is only performed if the following conditions are met simultaneously:

- Depending on the type of inverter:
  - In PCSM and PCSK: inverter discharging ([SV12.x.2.8-P] > 0 kW).
  - In HEM and HEMK: inverter running (ON).
- Module current [SV12.x.2] higher than a percentage (configurable in [SG5.1.2-I1]) of the nominal current set in the drive select.

This calculated temperature increase (Delta_T_IGBT_real) is compared with an estimated temperature increase value (Delta_T_IGBT_Estimated), calculated by interpolation in a function defined by two configurable points: P_1 ([SG5.1.2-I1], [SG5.1.3-Delta temperature 1]) and P_2 ([SG5.1.4-I2], [SG5.1.5-Delta temperature 2]). The function would have the following form:



Delta_T_IGBT_Estimated as a function of module current

If Delta_T_IGBT_actual ≥ Delta_T_IGBT_Estimated, the counter [SG5.1.x-Mod.X issues counter] will be incremented by one unit, representing the number of minutes in which the module is in an abnormal state.

When the counter is greater than the threshold configurable in [SG5.1.6-Activation threshold], the warning [W194: Module filter] will be generated, alerting the user of the need to perform maintenance on that module. The [W194: Module Filter] message will remain activated as long as the counter is above the threshold.

The counter will continue to increment at each iteration where Delta_T_IGBT_actual ≥ Delta_T_IGBT_Estimated until it reaches a saturation level beyond which it can no longer increment.

The saturation level shall be equal to twice the threshold set for triggering the warning [W194: Module filter].

Likewise, the counter will be decremented by one unit in any of the following cases until it reaches zero:

- If Delta_T_IGBT_Real < Delta_T_IGBT_Estimated
- If the module current or positive power percentage conditions are no longer met.

When the counter falls below the threshold, the warning [W194: Module filter] will be deactivated.

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG5.1.1-Enable = Disabled</td><td>Disabled
Safe enabled
Customizable</td><td>This allows to enable the filter maintenance warning for the [W194: Module filter] modules and select the configuration mode for the IGBT thermal monitoring algorithm.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Warning disabled.</td></tr><tr><td>Safe enabled</td><td>Enables the warning and allows to set only [SG5.1.3-Delta temperature 1] between 10 ºC and 80 °C. The parameter [SG5.1.5-Delta temperature 2] automatically adopts the same value as [SG5.1.3-Delta temperature 1] and remains hidden.</td></tr><tr><td>Customizable</td><td>Enables the warning and allows to adjust parameters from [SG5.1.2-I1] to [SG5.1.5-Delta temperature 2].
Note: this option is only visible to users with the Developer role.</td></tr></table>Note: even if the warning is enabled, it is only triggered when the operating conditions and thresholds configured in the other parameters of the group are met.</td><td>Yes</td></tr>
<tr><td>SG5.1.2-I1 = 80 %</td><td>0 to [SG5.1.4-I2]</td><td>Allows to set the percentage of module current with respect to the nominal current configured in the drive select corresponding to the first point of the temperature rise estimation curve.
Note: this is shown if [SG5.1.1-Enable] = Customizable and can only be modified by users with the Developer role.</td><td>Yes</td></tr>
<tr><td>SG5.1.3-Delta temperature 1 = 50 ºC</td><td>0 to 300 ºC</td><td>Allows to set the value of the temperature rise estimation of the hottest IGBT of the module corresponding to the current [SG5.1.2-I1].
Notes:
- If [SG5.1.1-Enable] = Safe enabled, the adjustment range is limited to between 10 ºC and 80 °C.
- If [SG5.1.1-Enable] = Customizable, it can only be modified by users with the Developer role.</td><td>Yes</td></tr>
<tr><td>SG5.1.4-I2 = 100 %</td><td>[SG5.1.2-I1] to 100 %</td><td>Allows to set the percentage of module current with respect to the nominal current configured in the drive select corresponding to the second point of the temperature rise estimation curve.
Note: this is shown if [SG5.1.1-Enable] = Customizable and can only be modified by users with the Developer role.</td><td>Yes</td></tr>
<tr><td>SG5.1.5-Delta temperature 2 = 100 ºC</td><td>[SG5.1.3-Delta temperature 1] to 300 ºC</td><td>Allows to set the value of the temperature rise estimation of the hottest IGBT of the module corresponding to the current [SG5.1.4-I2].
Notes:
- If [SG5.1.1-Enable] = Safe enabled, its value is set to the value configured in [SG5.1.3-Delta temperature 1] and remains hidden.
- If [SG5.1.1-Enable] = Customizable, it can only be modified by users with the Developer role.</td><td>Yes</td></tr>
<tr><td>SG5.1.6-Activation threshold = 10 min</td><td>1 to 60 min</td><td>Allows to set the maintenance warning activation threshold [W194: Module filter], if it is enabled.</td><td>Yes</td></tr>
<tr><td>SG5.1.7-Mod1 issues counter = 0 min</td><td rowspan="4">0 to 65535 min</td><td rowspan="4">Allows to display the value of the time counter with abnormal thermal behavior corresponding to module “n”.

Notes:
- These parameters are shown if [SG10.1.12-Number of modules] ≥ n, where “n” is the module number.
- They are read-only (RO)..</td><td rowspan="4">Yes</td></tr>
<tr><td>SG5.1.8-Mod2 issues counter = 0 min</td></tr>
<tr><td>SG5.1.9-Mod3 issues counter = 0 min</td></tr>
<tr><td>SG5.1.10-Mod4 issues counter = 0 min</td></tr>
</table>

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7.5.2. Subgroup SG5.2: Filter fault

This subgroup allows to configure the time-based criteria for triggering the [F246: Clean filters] fault.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG5.2.1-Filter Fault = 3 días	1 to 30 días	Allows to configure the maximum number of consecutive days that the [W194: Module Filter] warning can remain active before triggering [F246: Clean filters] fault. **Note**: if [W194: Module Filter] warning is not reported during a day of operation, the counter is reset.	Yes
SG5.2.2-Day First Warning = 01/01/2000 00:00	01/01/2000 00:00 to 31/12/2127 23:59	Shows the date and time of the first warning [W194: Module Filter] corresponding to the current sequence of consecutive days with the warning active. Format: DD/MM/YYYY HH:MM:SS. **Notes**: - This parameter is read-only (RO). - It is only shown if its value is not 0. - If the warning is not reported on a given operating day, the date/time is reset (a new sequence begins when it is reported again). - See section "SV7.1.2: Faults History" for more information on the internal interpretation of the date and time.	Yes

7.6. Group SG6: Calibrations

This group allows to adjust the measurements of different points of the inverter to avoid measurement and calculation errors. It is necessary to calibrate them. From this group it is possible to launch the calibrations of the DU, average voltage, power and VSI control type.

7.6.1. Subgroup SG6.1: Calibration DU

The calibration of the current and voltage measurements of the acquisition DU is carried out in the following way:

1) **Preconditions**:
   - **Zero current in all channels:** battery strings must be disconnected or idle, with no current flowing.
   - **DU communication OK:** the inverter must be operational with no active communication faults with the DU.
   - Set up a user account with the role "Power Electronics staff" or higher.
2) **Check the initial status.** Check the [SG6.1.2-Status] parameter: if it shows "Required" or "In progress", wait until the ongoing calibration is completed before starting a new one; if it shows "Not calibrated" or "Completed", the system is ready for calibration.
3) **Start the calibration.** Write "Yes" in parameter [SG6.1.1-Calibrate DU]. Immediately after accepting the command, this parameter automatically returns to "No".
4) **Monitor the status progression in [SG6.1.2-Status].** The normal sequence is "Sin calibrar" → "Requested " → "On going" → "Completed". The process is fully automatic and lasts a few seconds.

5) **Verify the result.** Once the calibration is completed, with the strings still without current, verify that all active measurement channels display 0.0 A or a value very close to zero. Optionally, check parameters from [SG6.1.3-Offset Channel 1] to [SG6.1.42-Offset Channel 40] to review the measured deviation for each channel individually:
   - Typical values are below ±1 A.
   - Unusually high deviations in a specific channel may indicate an issue with that sensor.

- **Note:** calibration does not have a fixed mandatory periodicity. It is recommended to perform it:
  - After the initial commissioning of the inverter.
  - If required, after restarting the inverter.
  - After a communication fault, once communication has been restored.
  - After replacing the DU or any associated measurement board.
  - If residual current is observed in the channels with the strings disconnected.

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SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG6.1.1-Calibrate DU = No	No Yes	Request a calibration of all enabled DU Channels by taking its offset in 0.	Yes
SG6.1.2-Status = Sin Calibrar	Sin Calibrar Requested On going Completed	Shows the calibration status. **OPTION** / **DESCRIPTION** Sin Calibrar: The DC cabinet has no active calibration. Readings do not apply offset correction. Requested: The request has been accepted. The DC cabinet is preparing the calibration process. On going: Calibration in progress. The DC cabinet is measuring the deviation of each channel. Completed: Calibration completed successfully. The channels apply the correction automatically.	Yes
SG6.1.3-Offset Channel 1 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 1. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.1-Channels 1 to 10 = 000000000X].	Yes
SG6.1.4-Offset Channel 2 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 2. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.1-Channels 1 to 10 = 00000000X0].	Yes
SG6.1.5-Offset Channel 3 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 3. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.1-Channels 1 to 10 = 0000000X00].	Yes
SG6.1.6-Offset Channel 4 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 4. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.1-Channels 1 to 10 = 000000X000].	Yes
SG6.1.7-Offset Channel 5 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 5. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.1-Channels 1 to 10 = 00000X0000].	Yes
SG6.1.8-Offset Channel 6 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 6. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.1-Channels 1 to 10 = 0000X00000].	Yes
SG6.1.9-Offset Channel 7 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 7. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.1-Channels 1 to 10 = 000X000000].	Yes
SG6.1.10-Offset Channel 8 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 8. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.1-Channels 1 to 10 = 00X0000000].	Yes
SG6.1.11-Offset Channel 9 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 9. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.1-Channels 1 to 10 = 0X00000000].	Yes
SG6.1.12-Offset Channel 10 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 10. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.1-Channels 1 to 10 NAND X000000000].	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG6.1.13-Offset Channel 11 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 11. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.2-Channels 11 to 20 = 000000000X].	Yes
SG6.1.14-Offset Channel 12 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 12. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.2-Channels 11 to 20 = 00000000X0].	Yes
SG6.1.15-Offset Channel 13 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 13. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.2-Channels 11 to 20 = 0000000X00].	Yes
SG6.1.16-Offset Channel 14 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 14. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.2-Channels 11 to 20 = 000000X000].	Yes
SG6.1.17-Offset Channel 15 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 15. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.2-Channels 11 to 20 = 00000X0000].	Yes
SG6.1.18-Offset Channel 16 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 16. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.2-Channels 11 to 20 = 0000X00000].	Yes
SG6.1.19-Offset Channel 17 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 17. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.2-Channels 11 to 20 = 000X000000].	Yes
SG6.1.20-Offset Channel 18 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 18. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.2-Channels 11 to 20 = 00X0000000].	Yes
SG6.1.21-Offset Channel 19 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 19. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.2-Channels 11 to 20 = 0X00000000].	Yes
SG6.1.22-Offset Channel 20 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 20. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.2-Channels 11 to 20 = X000000000].	Yes
SG6.1.23-Offset Channel 21 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 21. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.3-Channels 21 to 30 = 000000000X].	Yes
SG6.1.24-Offset Channel 22 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 22. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.3-Channels 21 to 30 = 00000000X0].	Yes
SG6.1.25-Offset Channel 23 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 23. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.3-Channels 21 to 30 = 0000000X00].	Yes
SG6.1.26-Offset Channel 24 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 24. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.3-Channels 21 to 30 = 000000X000].	Yes
SG6.1.27-Offset Channel 25 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 25. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.3-Channels 21 to 30 = 00000X0000].	Yes
SG6.1.28-Offset Channel 26 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 26. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.3-Channels 21 to 30 = 0000X00000].	Yes
SG6.1.29-Offset Channel 27 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 27. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.3-Channels 21 to 30 = 000X000000].	Yes
SG6.1.30-Offset Channel 28 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 28. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.3-Channels 21 to 30 = 00X0000000].	Yes
SG6.1.31-Offset Channel 29 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 29. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.3-Channels 21 to 30 = 0X00000000].	Yes
SG6.1.32-Offset Channel 30 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 30. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.3-Channels 21 to 30 = X000000000].	Yes
SG6.1.33-Offset Channel 31 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 31. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.4-Channels 31 to 40 = 000000000X].	Yes
SG6.1.34-Offset Channel 32 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 32. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.4-Channels 31 to 40 = 00000000X0].	Yes
SG6.1.35-Offset Channel 33 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 33. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.4-Channels 31 to 40 = 0000000X00].	Yes
SG6.1.36-Offset Channel 34 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 34. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.4-Channels 31 to 40 = 000000X000].	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG6.1.37-Offset Channel 35 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 35. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.4-Channels 31 to 40 = 00000X0000].	Yes
SG6.1.38-Offset Channel 36 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 36. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.4-Channels 31 to 40 = 0000X00000].	Yes
SG6.1.39-Offset Channel 37 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 37. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.4-Channels 31 to 40 = 000X000000]	Yes
SG6.1.40-Offset Channel 38 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 38. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.4-Channels 31 to 40 = 00X0000000].	Yes
SG6.1.41-Offset Channel 39 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 39. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.4-Channels 31 to 40 NAND 0X00000000].	Yes
SG6.1.42-Offset Channel 40 = 0.0 A	-3276.8 to 3276.7 A	Current offset adjustment on channel 40. **Note:** This parameter is displayed if [SG6.1.2-Status = Completed] and [G10.4-Channels 31 to 40 NAND X000000000].	Yes

EN

7.6.2. Subgroup SG6.2: MV calibration

This subgroup allows the calibration of the average voltage measurement in the inverter that requires it. This measurement is critical for switching processes with premagnetization. The inverter carries out automatic calibrations autonomously when the medium voltage cubicle is closed.

Note: These parameters are only displayed if [SG10.1.3-Product type = HEM) and [SG10.7.3-MV Skid = Yes, Motorized] or [SG10.7.4-Volt Meas MV Skid = No].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG6.2.1-Status = Not calibrated	Not calibrated Positive seq calibrated Negative seq calibrated Calibration complete	Medium volage calibration is essential for the premagnetization process. The inverter must be correctly calibrated in, at least, the voltage sequence in which transformer premagnetization will be carried out. This parameter shows the calibration status of the MV voltages (for HEM inverters). **OPTION** / **DESCRIPTION** Not calibrated / Inverter has not been calibrated. Positive seq calibrated / Inverter is calibrated in positive sequence. Negative seq calibrated / Inverter is calibrated in negative sequence. Calibration complete / Inverter is calibrated in both sequences.	Yes
SG6.2.2-Update = No	No Yes Updating	Allows to force a calibration process manually. **Note:** In "Updating" status it is not possible to force a manual calibration process until the update is finished.	Yes
SG6.2.3-Reset = No	No Yes	Allows resetting the medium voltages calibration.	Yes
SG6.2.4-Gain MVr pos = 2047	1 to 4095	Allows to visualize the gain of the R-phase in positive sequence.	Yes
SG6.2.5-Gain MVs pos = 2047	1 to 4095	Allows to visualize the gain of the S-phase in positive sequence.	Yes
SG6.2.6-Gain Mvt pos = 2047	1 to 4095	Allows to visualize the gain of the T-phase in positive sequence.	Yes
SG6.2.7-Gain MVr neg = 2047	1 to 4095	Allows to visualize the gain of the R-phase in negative sequence.	Yes
SG6.2.8-Gain MVs neg = 2047	1 to 4095	Allows to visualize the gain of the S-phase in negative sequence.	Yes
SG6.2.9-Gain Mvt neg = 2047	1 to 4095	Allows to visualize the gain of the T-phase in negative sequence.	Yes
SG6.2.10-Seq pos angle = 3109	1 to 4095	Allows to visualize the gain of the phase in positive sequence.	Yes
SG6.2.11-Seq neg angle = 3109	1 to 4095	Allows to visualize the gain of the phase in negative sequence.	Yes
SG6.2.12-Estado interno = 0x00	0x00 to 0xFFFF	Shows the calibrations that have been performed: 0x0000 = Not calibrated. 0x0010 = Positive sequence calibrated. 0x0020 = Negative sequence calibrated. 0x0030 = Calibration complete. 0x0001 = Wired calibrated. Indicates whether the calibration has automatically reversed the measurement phases due to wiring problems.	Yes

7.6.3. Subgroup SG6.3: Power Calibrations

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG6.3.1-Current Angle = 0	-1000 to 1000	Allows to set a delay to the current angle, due to measure acquisition time. The calibration of this angle is done in terms of counts, 360º is equivalent to 1024 beads. E.g.: SG6.5.8 = 7 →angle of 2,46º. **Note: The use of this parameter is reserved for testing.** Do not modify it without express authorization from Power Electronics.	Yes
SG6.3.2-Thermal Drift = 102%	0 to 150 %	Corresponds to the thermal calibration of the sensors. **Note:** This parameter must not be changed unless explicitly stated by R&D department.	Yes
SG6.3.3-PQ Gain = 100.0 %	90.0 to 110.0 %	Allows to configure the gain value of P and Q which modifies the actual power of the inverter. It mainly affects the current and power loops as well as the maximum power delivered by the inverter. Incorrect settings can be dangerous.	Yes
SG6.3.4-Q C Gain = 100.0 %	90.0 to 110.0 %	Allows to configure the gain value of Q to compensate the filter C of the inverter. Incorrect settings may result in unwanted reactive compensation.	Yes
SG6.3.5-PQ visualization gain = 100.00 %	50.00 to 150.00 %	Allows to configure the gain value of P and Q which can be displayed without modifying the operation of the inverter.	Yes
SG6.3.6-Voltage angle = 0	-1000 to 1000	Allows to configure a delay in the voltage angle due to the measurement acquisition time.	Yes
SG6.3.7-Central I gain = 100.0 %	90.0 to 110.0 %	Allows to configure the gain value to be applied to the current measured at the output of the inverter and read at the main control board.	Yes
SG6.3.8-Central I angle = 0	-1000 to 1000	Allows to configure a delay in the current angle measured at the output of the inverter and read at the main control board due to the measurement acquisition time.	Yes

7.6.4. Subgroup SG6.4: VSI Calibrations

This subgroup allows the calibration of different measurements for VSI control type.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG6.4.1-Cal Vac Modulo = 1.000	0.000 to 1.500	Allows to configure the gain to calibrate the direct sequence voltage generation.	Yes
SG6.4.2-Cal Vac Theta = 0	-1000 to 1000	Allows to configure the phase to calibrate the direct sequence voltage generation.	Yes

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7.6.5. Subgroup SG6.5: CTs Phase Calibration

This group allows adjusting the sequence of the current measurement phases of the CTs (which will be used in VISMA control).

Note: parameters SG6.5.5 to SG6.5.11 are read-only parameters (RO).

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG6.5.1-Ir Connector = Channel A	Channel A Channel B Channel C	The current transformer control board has three connectors (J5, J6, and J7). Each connector, by default, is assigned to the current of one phase (R, S or T): **SG6.5.1** = Channel A → connector J5 → current Ir (connection of phase R). **SG6.5.2** = Channel B → connector J6 → current Is (connection of phase S) **SG6.5.3** = Channel C → connector J7 → current It (connection of phase T) It is possible to change the connection of each connector to the busbar current via software. If a wiring error is detected (crossed phases or phases not wired correctly), the default value of the corresponding parameter must be changed.	Yes
SG6.5.2-Is Connector = Channel B			Yes
SG6.5.3-It Connector = Channel C			Yes
SG6.5.4-Invert Ix Sign = 000	000 a XXX	Inverts the sign of the current. 3 bits are defined to handle the sign of the three currents so that each bit (from right to left) represents one current. According to the bit value, the sign of the current will be inverted or not (0= invert the sign; X=do not invert). Bit 0 → Inverts the sign of Ir. Bit 1 → Inverts the sign of Is. Bit 2 → Inverts the sign of It.	Yes
SG6.5.5-Measured Vr Phase = 0.0°	-180.0 to 180.0°	Shows the time phase of the voltage Vr.	Yes
SG6.5.6-Measured Vs Phase= 0.0°	-180.0 to 180.0°	Shows the time phase of the voltage Vs.	Yes
SG6.5.7-Measured Vt Phase = 0.0°	-180.0 to 180.0°	Shows the time phase of the voltage Vt.	Yes
SG6.5.8-Measured Ir Phase= 0.0°	-180.0 to 180.0°	Displays the time phase of the Ir current sensed by the low voltage current transformers.	Yes
SG6.5.9-Measured Is Phase= 0.0°	-180.0 to 180.0°	Displays the time phase of the Is current sensed by the low voltage current transformers.	Yes
SG6.5.10-Measured It Phase= 0.0°	-180.0 to 180.0°	Displays the time phase of the It current sensed by the low voltage current transformers.	Yes
SG6.5.11-RMS Curr. Homopolar = 0.00%	0.00 to 655.35%	RMS value in % with respect to the nominal value of the equipment of the zero sequence current calculated in the current transformers.	Yes

7.7. Group SG7: Diagnosis

This group allows to directly test certain functionalities of the uP to verify the correct functioning of its parts.

7.7.1. Subgroup SG7.0: Password

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.0.1-Password 1 = 0	0 to 65535	Sets the password 1.	Yes
SG7.0.2-Password 2 = 0	0 to 65535	Sets the password 2.	Yes

7.7.2. Subgroup SG7.1: EEPROM

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.1.0-Eeprom number = 0	0 to 1	Variables for EEPROM internal diagnostics, for debugging and verification. This parameter is used to configure which EEPROM is to be accessed for testing.	Yes
SG7.1.1-Eeprom offset = 0	0 to 4096	Variables for EEPROM internal diagnostics, for debugging and verification. This parameter is used to configure which EEPROM address is to be accessed for testing.	Yes
SG7.1.2-Eeprom value = 0	0 to 65535	Variables for EEPROM internal diagnostics, for debugging and verification. This parameter is used to write a value in range to the eeprom variable configured in the previous parameter.	Yes
SG7.1.3-Eeprom out = 0	0 to 65535	Returns the value stored in the EEPROM address configured in SG7.1.1.	Yes
SG7.1.4-Eeprom operations = Read	Read Write	Operations to make in EEPROM.	Yes

7.7.3. Subgroup SG7.2: TCP Client

SG7.2.1- Create Task

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.2.1.0-Create TCP Task = No	No Yes	Create TCP tasks.	Yes
SG7.2.1.2-Ip = 192.168.1.143	1.0.0.0 to 255.255.255.255	TCP Client diagnostic group variable. This variable configures the IP of the server.	Yes
SG7.2.1.3-Puerto = 502	0 to 65535	TCP Client diagnostic group variable. This variable configures the server port.	Yes
SG7.2.1.4-Periodo = 1000 ms	0 to 65535 ms	Configures the read/write period.	Yes
SG7.2.1.5-Timeout = 500 ms	0 to 65535 ms	Configures the timeout after request.	Yes
SG7.2.1.5-Retries = 1	0 to 5	Configures the number of attempts per request.	Yes
SG7.2.1.8-Slave ID = 1	0 to 255	Configures the ID identifier of the slave.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.2.1.9-Creation Result = Correct	Correct Error: Full Error: Parameters	Sets the result of creation task.	Yes
SG7.2.1.10-Task ID Creada = 0	0 to 65535	Configures the task creation ID.	Yes

SG7.2.2- Watch Task

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.2.2.1-Task ID = 0	0 to 65535	Selected ID to display task.	Yes
SG7.2.2.2-Satus = 0	0 to 65535	Indicates the status of the task.	Yes
SG7.2.2.3-Value = 0	0 to 65535	Write/read value of the task.	Yes
SG7.2.2.5-Eliminate Task = No	No Yes	Delete task.	Yes

7.7.4. Subgroup SG7.3: RS485

SG7.3.1- Config

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.3.1.1-Enable = No	No Yes	Enables the RS485 device.	Yes
SG7.3.1.2-Slave id 1 = 0	0 to 255	Configures ID RS485 of the slave 1.	Yes
SG7.3.1.3-Slave id 2 = 0	0 to 255	Configures ID RS485 of the slave 2.	Yes
SG7.3.1.4-Slave id 3 = 0	0 to 255	Configures ID RS485 of the slave 3.	Yes
SG7.3.1.5-Slave id 4 = 0	0 to 255	Configures ID RS485 of the slave 4.	Yes
SG7.3.1.6-Slave id 5 = 0	0 to 255	Configures ID RS485 of the slave 5.	Yes
SG7.3.1.7-Slave id 6 = 0	0 to 255	Configures ID RS485 of the slave 6.	Yes

SG7.3.2- Read/Write

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG7.3.2.0-Comms status 1 = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 1.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The Modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG7.3.2.1-Comms status 2 = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 2.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The Modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG7.3.2.2-Comms status 3 = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 3.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The Modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG7.3.2.3-Comms status 4 = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 4.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The Modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG7.3.2.4-Comms status 5 = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 5.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The Modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG7.3.2.5-Comms status 6 = Disabled</td><td>Disable
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows communication status of device 6.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Communications disabled.</td></tr><tr><td>Comms OK</td><td>After performing the test, communications are ready.</td></tr><tr><td>Timeout</td><td>Timeout occurred after performing the Communications test.</td></tr><tr><td>Error CRC</td><td>After performing the test, the communication does not work correctly. CRC failed.</td></tr><tr><td>Invalid modbus code</td><td>The Modbus code used is not supported.</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG7.3.2.6-Read device 1 = 0</td><td>0 to 65535</td><td>Indicates the value read by RS485 by slave 1.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.8-Read device 3 = 0</td><td>0 to 65535</td><td>Indicates the value read by RS485 by slave 3.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.10-Read device 5 = 0</td><td>0 to 65535</td><td>Indicates the value read by RS485 by slave 5.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.11-Read device 6 = 0</td><td>0 to 65535</td><td>Indicates the value read by RS485 by slave 6.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.12-Write device 1 = 0</td><td>0 to 65535</td><td>Indicates the value written by RS485 by slave 1.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.13-Write device 2 = 0</td><td>0 to 65535</td><td>Indicates the value written by RS485 by slave 2.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.14-Write device 3 = 0</td><td>0 to 65535</td><td>Indicates the value written by RS485 by slave 3.</td><td>Yes</td></tr>
<tr><td>SG7.3.2.17-Write device 6 = 0</td><td>0 to 65535</td><td>Indicates the value written by RS485 by slave 6.</td><td>Yes</td></tr>
</table>

7.7.5. Subgroup SG7.4: Network

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.4.1-Create Log Msg = 0	0 to 65535	Sets the parameter to create the log message.	Yes

7.7.6. Subgroup SG7.5: SD

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG7.5.1-Enable SD ops = No</td><td>No
Yes</td><td>Enable and mount SD card in target.</td><td>Yes</td></tr>
<tr><td>SG7.5.2-Write log version = No</td><td>No
Yes</td><td>Sets the parameter to write log version and IP in file.</td><td>Yes</td></tr>
<tr><td>SG7.5.4-Write log warnings = No</td><td>No
Yes</td><td>Sets the parameter to write log warning in SD file.</td><td>Yes</td></tr>
<tr><td>SG7.5.5-SD status operations = Disabled</td><td>Disabled
Mountting sd
SD correct
Write file ok
Write file error
SD not inserted
SD error
SD formatting
SD format ok
SD format error</td><td>Shows the result of different operations in SD card.
<table>
<tr><th>STATUS</th><th>DESCRIPTION</th></tr>
<tr><td>Disabled</td><td>SD disabled.</td></tr>
<tr><td>Mountting SD</td><td>The SD is being mounted (previous step to save logs on the SD).</td></tr>
<tr><td>SD correct</td><td>SD is detected correctly and does not have any faults.</td></tr>
<tr><td>Write file ok</td><td>The SD has been written successfully.</td></tr>
<tr><td>Write file error</td><td>There has been an error writing to the SD.</td></tr>
<tr><td>SD not inserted</td><td>No presence of SD has been detected.</td></tr>
<tr><td>SD error</td><td>The SD has some kind of error.</td></tr>
<tr><td>SD formatting</td><td>The SD is being formatted.</td></tr>
<tr><td>SD format ok</td><td>The SD has been formatted correctly.</td></tr>
<tr><td>SD format error</td><td>The SD has not been formatted correctly.</td></tr>
</table></td><td>Yes</td></tr>
</table>

7.7.7. Subgroup SG7.6: Dsp Comms

SG7.6.1- Statisc Comms

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.6.1.1-Spi Select = Spi Slow	Spi Slow Spi Fast Spi Slow to Spi Fast	Selects SPI to show statistics of communications.	Yes
SG7.6.1.3-erroneous frames = 0	0 to 65535	Indicates the number of incorrect frames in the communication between uP and DSP.	Yes
SG7.6.1.4-Succesfull Frames = 0	0 to 65535	Indicates the number of correct frames in the communication between uP and DSP.	Yes
SG7.6.1.5-Error Rate Frames = 0	0 to 65535	Error frame rate between successful frames during one second.	Yes
SG7.6.1.6-Frames per sec = 0	0 to 65535	Total number of frames during a time window of one second.	Yes

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SG7.6.2- Frame Integrity

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.6.2.1-Frame Select = Fast Parameters to DSP	Fast Parameters to DSP Fast Parameters to UP Slow Parameters to DSP Slow Parameters to UP	Sets the select type of frame. OPTION / DESCRIPTION Fast Parameters to DSP: Sending parameters using SPI fast communication channel from uP to DSP. Fast Parameters to UP: Sending parameters using SPI fast communication channel from DSP to uP. Slow Parameters to DSP: Sending parameters using SPI slow communication channel from uP to DSP. Slow Parameters to UP: Sending parameters using SPI slow communication channel from DSP to uP.	Yes
SG7.6.2.3-Amount of param. = 0	0 to 65535	Total configured parameters of the selected frame (SG7.6.2.1).	Yes
SG7.6.2.4-Diagnosis Value = 0	0 to 65535	Parameter sent and received with a known modification.	Yes
SG7.6.2.5-Diagnosis Value Ack = 0	0 to 65535	Configures the Tx frame sending period for the diagnostic mode of uP to FPGA, uP to central FPGA communications.	Yes

SG7.6.3-Mirror Test

This subgroup allows to check if the configuration carried out in the microcontroller (MCU) is received by the DSP properly, and vice versa. It also allows to check if the MCU receives the configuration made by the DSP device. This subgroup is only for testing purposes.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.6.3.1-Frame Select = Spi Slow	Spi Slow Spi Fast	Allows to select the frame (or channels) enabled to communicate between uP and DSP using the Spi protocol. The difference between slow and fast Spi is the speed at which they communicate between uP and DSP. “Spi Slow” communicates every 8 ms and “Spi Fast” every 4 ms.	Yes
SG7.6.3.2-Register ID = 0	0 to 65535	Sets the DSP Comms register ID to ask to DSP.	Yes
SG7.6.3.3-RAW Value = 0	0 to 65535	Displays the raw value requested from the DSP.	Yes
SG7.6.3.4-Fault selected = F1: Watchdog	Check complete list in tab Failures	Select the fault to check the configuration.	Yes
SG7.6.3.5-Fault Conf. RAW = 0	0 to 65535	Displays the raw fault setting value requested from the DSP.	Yes

7.7.8. Subgroup SG7.7: FPGA Comms

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.7.1-Enable = No	No Yes	Enables or disables the FPGA communications.	Yes
SG7.7.2-Period Tx = 100 ms	1 to 255 ms	Configures the Tx frame sending period for the diagnostic mode of uP to FPGA, uP to central FPGA communications.	Yes
SG7.7.3-Eco Variable Tx = 0	0 to 65535	Configures the variable to be sent from the uP to the central FPGA.	Yes

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG7.7.4-Eco Variable Rx = 0</td><td>0 to 65535</td><td>Returns the value of the variable sent from the uP to the FPGA. That is, what the central FPGA has received it sends to the uP.</td><td>Yes</td></tr>
<tr><td>SG7.7.6-Status = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Error Driver</td><td>Shows the current status of the communication between uP and central FPGA, of the central control.<table><tr><th>STATUS</th><th>DESCRIPTION</th></tr><tr><td>Disabled</td><td>Test has not been produced.</td></tr><tr><td>Comms OK</td><td>Frames between uP and central FPGA were sent and received successfully.</td></tr><tr><td>Timeout</td><td>There has been a timeout in the communication between uP and central FPGA, the frame could not be processed by any of the devices.</td></tr><tr><td>Error CRC</td><td>The communication between- uP and central FPGA does not work correctly. CRC failed.</td></tr><tr><td>Error driver</td><td>The communication between- uP and central FPGA does not work correctly. driver failed.</td></tr></table></td><td>Yes</td></tr>
</table>

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7.7.9. Subgroup SG7.8: Watchdog

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.8.1-Enable = No	No Yes	Sets if Watchdog is enabled or not.	Yes
SG7.8.2-Refresh = Yes	No Yes	Allows to force a uP reset by watchdog. Allows developer level debugging and verification of the behavior of the uP.	Yes
SG7.8.3-Update Mask Reset = 1	0 to 1	Sets the task that was running before the watchdog was triggered.	Yes
SG7.8.4-Enable Mask Reset = 1	0 to 1	Sets the task that was running before the watchdog was triggered.	Yes

7.7.10. Subgroup SG7.9: CPU Usage

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG7.9.1-Current CPU usage = 0.0 %	0.0 to 100.0 %	Indicates the current CPU usage of the inverter's uP.	Yes
SG7.9.2-Max CPU usage = 0.0 %	0.0 to 100.0 %	Indicates the maximum CPU usage of the inverter's uP.	Yes
SG7.9.3-Reset max CPU usage = No	No Yes	Parameter to reset maximum allowed CPU usage in the inverter uP (SG7.9.2).	Yes

7.8. Group SG8: Modbus Memory Access

This group allows to directly access the memory of the microcontroller and to check its correct operation.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG8.1-Password 1 = 0	0 to 65535	Password must be set to activate diagnostic mode.	Yes
SG8.2-Password 2 = 0	0 to 65535	Password must be set to activate diagnostic mode.	Yes
SG8.3-Var 1 Address High = 0x2400 Hex	0x00 to 0xFFFF Hex	Upper part of the memory address to be tested on.	Yes
SG8.4-Var 1 Address Low = 0x00 Hex	0x00 to 0xFFFF Hex	Lower part of the memory address to be tested on.	Yes
SG8.5-Var 1 Tipo = Uint8	Uint8 Uint16	Data type to be configured.	Yes
SG8.7-Var 2 Address High = 0x2400 Hex	0x00 to 0xFFFF Hex	Parameter to be able to set the high part of the memory address on which the test is to be performed.	Yes
SG8.8-Var 2 Address Low = 0x00 Hex	0x00 to 0xFFFF Hex	Parameter to be able to set the low part of the memory address on which the test is to be performed.	Yes
SG8.9-Var 2 Tipo = Uint8	Uint8 Uint16	Data type to be configured.	Yes
SG8.10-Var 2 Value = 0	0 to 65535	Value to be written to the memory address configured in the previous parameters (Var2 address high and Var2 address low).	Yes
SG8.11-Read memory address = No	No Yes	Action to be taken on the configured memory address.	Yes
SG8.12-Write memory address = No	No Yes	Action to be taken on the configured memory address.	Yes

7.9. Group SG9: Reset counters

There are counters that allow to monitor certain properties of the inverter, such as energy counters or connection / disconnection counters to the MV switchgear. From this group it is possible to reset these counters to 0 and start counting again.

7.9.1. Subgroup SG9.1: Rst energy counter

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG9.1.1-Rst energy counter = No	No Yes	Allows to reset manually the energy counter to 0.	Yes

7.9.2. Subgroup SG9.2: Rst run time counters

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG9.2.1-Rst run time tot coun = No	No Yes	Allows to reset the run time total counters to 0 by hand.	Yes
SG9.2.2-Rst run time par coun = No	No Yes	Allows to reset the run time partial counters to 0 by hand.	Yes

7.9.3. Subgroup SG9.3: Rst grid conn counters

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG9.3.1-Reset tot grid conns = No	No Yes	Allows to reset the total number of grid connections counter to 0 by hand.	Yes
SG9.3.2-Reset part grid conns = No	No Yes	allows to reset the partial number of grid connections counter to 0 by hand.	Yes

7.9.4. Subgroup SG9.4: Rst MV cell counters

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG9.4.1-Rst no premag closure = No	No Yes	Allows to reset the partial number of MV no premag closures counter to 0 by hand.	Yes
SG9.4.1a-Rst manual closure = No	No Yes	Allows the counter [SV5.5.1a-Manual cmd closures] to be reset. When set to Yes, the uP requests a reset from the DSP and keeps the request active until the DSP resets the counter. After the reset, the request is released. **Notes:** - This parameter is only shown to users with the Developer role. - It is only shown if [SG10.1.3-Product type] = HEM or [SG10.1.3-Product type] = HEMK and [SG10.7.3-MT Skid] = Yes. Motorized. - The counter is reset to 0 and the value is saved for persistence	Yes
SG9.4.1b-Rst uncommanded clos = No	No Yes	Allows the counter [SV5.5.1b-Uncommanded closures] to be reset. When set to Yes, the uP requests a reset from the DSP and keeps the request active until the DSP resets the counter. After the reset, the request is released. **Notes**: - This parameter is only shown to users with the Developer role. - It is only shown if [SG10.1.3-Product type = HEM] or [SG10.1.3-Product type] = HEMK and [SG10.7.3-MT Skid] = Yes. Motorized. - The counter is reset to 0 and the value is saved for persistence.	Yes
SG9.4.2-Rst premag closures = No	No Yes	Allows to reset the partial number of MV switchgear premag closures counter to 0 by hand.	Yes
SG9.4.3-Rst loaded openings = No	No Yes	Allows to reset the partial number of MV switchgear loaded openings counter to 0 by hand.	Yes
SG9.4.4-Rst not loaded open = No	No Yes	Allows to reset the partial number of MV switchgear not loaded openings counter to 0 by hand.	Yes

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7.10.Group SG10: Drive select

This group configures the hardware (from the nominal properties of the inverter, to the enabling of possible options installed, depending on the final application) of the inverter so that the software works according to the inverter characteristics configured.

7.10.1. Subgroup SG10.1: General

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG10.1.1-Password = 0</td><td>0 to 65535</td><td>Password to enable drive select configuration.</td><td>Yes</td></tr>
<tr><td>SG10.1.2-Force defaults = No</td><td>No
Yes</td><td>Sets all values to factory default values.</td><td>No</td></tr>
<tr><td>SG10.1.3-Product type = HEMK</td><td>HEMK
HEM</td><td>Defines the type of inverter HEMK / MPCSK or HEM / MPCSM.</td><td>No</td></tr>
<tr><td>SG10.1.4-Inverter type = Solar inverter</td><td>Solar inverter
Basic BUSPLUS
Advanced BUSPLUS
PCS/MPCS</td><td>Shows the application of the inverter.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Solar inverter</td><td>Shows when the inverter is a solar inverter (HEM or HEMK).</td></tr><tr><td>Basic BUSPLUS</td><td>Shows when the inverter is a Basic BUSPLUS, allows to connect up to two 1200 kW DC/DC or four 525 kW DC/DC.</td></tr><tr><td>Advanced BUSPLUS</td><td>Shows when the inverter is an Advanced BUSPLUS, allows to connect up to two 1200 kW DC/DC or four 525 kW DC/DC.</td></tr><tr><td>PCS/MPCS</td><td>Shows when the inverter is a battery inverter (PCSK/MPCKS).</td></tr></table>Concerning DC switch disconnectors management in the case the inverter is set as bus-plus model according to configuration (SG10.1.4 = Basic BUSPLUS or SG10.1.4 = Advanced BUSPLUS), the operation of all DC switch disconnectors from DC cabinet is required to be performed at the same time. So that:
All DC switch disconnectors will close at the same time (for example: during night to day transition by auto night).
It is forbidden to run the inverter with one open DC switch disconnector and the rest of them as closed. This would trip an “F167: Open DU” fault.
In the other handspring loading maneuver over DC switch disconnectors must be done in a staggered way.
Now a module’s fault that isolates its DC-side then will generate an inverter-level “DU open” type fault, disabling all modules, triggering F167, setting working modules in status “W118: Disable mod. SC” and faulty module in its own fault status.</td><td>No</td></tr>
</table>

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG10.1.5a-Bus Plus = No</td><td>No
1 x 525 KW
2 x 525 KW
3 x 525 KW
4 x 525 KW
5 x 525 KW
6 x 525 KW
7 x 525 KW
8 x 525 KW
1 x 1200 KW
2 x 1200 KW
3 x 1200 KW
4 x 1200 KW</td><td>Shows the DC/DC converter power connected to Bus Plus terminals.<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>No</td><td>No Bus Plus.</td></tr><tr><td>1 x 525 KW</td><td>1 DC/DC converter of 525 kW.</td></tr><tr><td>2 x 525 KW</td><td>2 DC/DC converter of 525 kW.</td></tr><tr><td>3 x 525 KW</td><td>3 DC/DC converter of 525 kW.</td></tr><tr><td>4 x 525 KW</td><td>4 DC/DC converter of 525 kW.</td></tr><tr><td>5 x 525 KW</td><td>5 DC/DC converter of 525 kW.</td></tr><tr><td>6 x 525 KW</td><td>6 DC/DC converter of 525 kW.</td></tr><tr><td>7 x 525 KW</td><td>7 DC/DC converter of 525 kW.</td></tr><tr><td>8 x 525 KW</td><td>8 DC/DC converter of 525 kW.</td></tr><tr><td>1 x 1200 KW</td><td>1 DC/DC converter of 1200 kW.</td></tr><tr><td>2 x 1200 KW</td><td>2 DC/DC converter of 1200 kW.</td></tr><tr><td>3 x 1200 KW</td><td>3 DC/DC converter of 1200 kW.</td></tr><tr><td>4 x 1200 KW</td><td>4 DC/DC converter of 1200 kW.</td></tr></table>Note: This parameter is only displayed if [SG10.1.4 Inverter type] is Bus Plus Basic or Bus Plus Advanced.</td><td>No</td></tr>
<tr><td>SG10.1.6-Nominal Vac = 660 V</td><td>208 to 800 V</td><td>Sets the nominal AC voltage of the inverter.</td><td>No</td></tr>
<tr><td>SG10.1.7-Nominal Frequency = 60.00 Hz</td><td>40.00 to 70.00 Hz</td><td>Sets the frequency at which the inverter will operate.</td><td>No</td></tr>
<tr><td>SG10.1.9-Iac max = 920 A</td><td>30 to 1000 A</td><td>Configures the maximum AC current at which the inverter will work.</td><td>No</td></tr>
<tr><td>SG10.1.9b-VISMA Max Iac = 828 A</td><td>30 to 1000 A</td><td>Allows configuring the maximum AC current per power stage at an ambient temperature of 40 °C in VISMA control mode.
Note: this parameter is displayed if [SG10.1.19-Visma enable] = Enabled.</td><td>No</td></tr>
<tr><td>SG10.1.10-Idc max = 1200 A</td><td>300 to 1250 A</td><td>Configures the maximum DC current at which the inverter will work.</td><td>No</td></tr>
<tr><td>SG10.1.11-Nominal S = 1052 kVA</td><td>200 to 1500 kVA</td><td>Allows to configure the nominal apparent power (S) of each power stage. The Nominal S at which the inverter will operate is equal to the value of this parameter multiplied by the number of enabled modules ([SG10.1.12-Number of modules]).</td><td>No</td></tr>
<tr><td>SG10.1.12-Number of modules = 4</td><td>1 to 8</td><td>Allow configuring the number of the modules that the inverter has.</td><td>No</td></tr>
<tr><td>SG10.1.13-Available modules = 000000XXXX</td><td>0000000000 to XXXXXXXXXX</td><td>Defines the number of the modules connected physically.<table><tr><th>BIT</th><th>STATUS</th><th>BIT</th><th>STATUS</th></tr><tr><td>0</td><td>Module 1</td><td>6</td><td>Not available</td></tr><tr><td>1</td><td>Module 2</td><td>7</td><td>Not available</td></tr><tr><td>2</td><td>Module 3</td><td>8</td><td>Not available</td></tr><tr><td>3</td><td>Module 4</td><td>9</td><td>Not available</td></tr><tr><td>4</td><td>Not available</td><td>--</td><td>-</td></tr></table>Note: These options are displayed if [SV1.3.1.4-Number of modules] < n, being ‘n’ the module number.</td><td>No</td></tr>
<tr><td>SG10.1.14-Number of busbar = 1</td><td>1 to 4</td><td>Configures the number of DC busbars that the inverter has.
Note: this parameter is displayed if [SV1.3.1.1-Inverter type] = PCS/MPCS.</td><td>No</td></tr>
<tr><td>SG10.1.15-HW version number = HW version 1.4</td><td>HW version 1.0
HW version 1.1
HW version 1.2
HW version 1.3
HW version 1.4</td><td>Defines the version number of the hardware (there are differences between the hardware version defined).</td><td>No</td></tr>
<tr><td>SG10.1.16-Number of DC/DC = 0</td><td>0 to 4</td><td>Defines the number of DC/DC connected to the inverter.</td><td>No</td></tr>
</table>

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<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG10.1.17-DC/DC comunication = 0000</td><td>0000 to XXXX</td><td>Indicates to which module the DC/DC is connected.<table><tr><th>BIT</th><th>STATUS</th><th>BIT</th><th>STATUS</th></tr><tr><td>0</td><td>Module 1</td><td>2</td><td>Module 3</td></tr><tr><td>1</td><td>Module 2</td><td>3</td><td>Module 4</td></tr></table></td><td>No</td></tr>
<tr><td>SG10.1.18-Altitude</td><td>0 to 5000</td><td>Sets the altitude in meters.</td><td>No</td></tr>
<tr><td>SG10.1.19-Visma enable = Disabled</td><td>Disabled
Enabled</td><td>Allows to enable or disable VISMA control subtype.
Note: this parameter is displayed if [SG10.1.4-Inverter type] = PCS/MPCS.</td><td>No</td></tr>
<tr><td>SG10.1.20-Enable seamless = Disabled</td><td>Disabled
Enabled</td><td>Enables the transition between control modes (CSI and VSI) in RUN.
Note: this parameter is displayed if [SG10.1.19-Visma enable] = Enabled. If [SG10.1.19-Visma enable] = Disabled, the value of this parameter will be forced to Disabled.</td><td>No</td></tr>
</table>

7.10.2. Subgroup SG10.2: Protections

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG10.2.1-Overcurrent threshol = 1800 A	0 to 2200 A	Sets the PDINT overcurrent limit for Vdc of the inverter.	No
SG10.2.2-Igbts disc temp thre = 104 ºC	20 to 150 ºC	Allows to set the temperature derating point of the IGBTs in discharge (injecting to mains).	No
SG10.2.3-Igbts charge temp th = 100 ºC	20 to 150 ºC	Allows to set the temperature derating point of the IGBTs under load (consuming from the mains).	No
SG10.2.4-Offset RST1 discharg = 9 ºC	0 to 20 ºC	Allows to set the temperature offset of the IGBTs R1 S1 and T1 with respect to R2 R3 S2 S3 T2 T3 due to the location of the temperature probe when the inverter is in discharge (injecting into the grid). Expressed in Celsius.	No
SG10.2.5-Offset RST3 charge = 8 ºC	0 to 20 ºC	Allows to set the temperature offset of the IGBTs R3, S3 and T3 with respect to R2, R1, S2, S1, T1 and T2 due to the location of the temperature probe when the inverter is on load (consuming from the mains).	No
SG10.2.6-Delta T charge = 4ºC	0 to 10 ºC	Allows to configure the temperature decrease to be applied to the ventilation curve during loading of the inverter.	Yes
SG10.2.7-Filt C Overcur Thres = Disabled	1000 to 2000 A; Disabled = 2001 A	Allows to configure the overcurrent protection threshold of the filter C. If this threshold is exceeded, F175 will be trigger.	No

7.10.3. Subgroup SG10.3: Limits

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG10.3.1-Inductive I limit = 920 A	0.0 to 6553.5 A	Displays the module inductive current limit calculated from parameter [SG10.3.7-Inductive cosphi]. **Note**: this parameter is read-only (RO).	No
SG10.3.2-Capacitive I limit = 920 A	0.0 to 6553.5 A	Displays the module capacitive current limit calculated from parameter [SG10.3.8-Capacitive cosphi]. **Note**: this parameter is read-only (RO).	No
SG10.3.3-Max P limit = 100.00 %	0.00 to 100.00 %	Parameters SG10.3.3, SG10.3.4 and SG10.3.5 set the maximum P, Q and S limits, respectively, with respect to the inverter's nominal value. They also limit the maximum value that parameters [G3.1.1-P limit discharge], [G3.1.1b-P limit charge] and [G3.1.4-S limit] can take. This means that if a user attempts to set in G3 a higher value than the one set in the drive select limit, it will be rejected and overridden by the value of the drive select.	No
SG10.3.4-Max Q limit = 100.00 %	0.00 to 100.00 %	Example: If [SG10.3.3-Max P limit] = 90 % and user tries to set [G3.1.1-P limit discharge] to 95 %, the resulting configuration will be [G3.1.1-P limit discharge] = 90 %. **Note**: P, Q and S may not have the expected value by default. Ensure they are correctly set, or output power may not be as expected. From software version 5.0.0 onwards, parameter [SG10.3.5-Max S Limit] is also used in the calculation of the intake temperature derating. Under AC overvoltage conditions, its configured value determines the apparent power limit used to calculate the module current limit: – When [SG10.3.5-Max S Limit] = 100%, an additional limitation may be applied under overvoltage conditions. – When [SG10.3.5-Max S Limit] ≥ 110%, the behavior is equivalent to that of previous versions.	No
SG10.3.5-Max S limit = 100.00 %	0.00 to 100.00 %		No
SG10.3.6-VRT reactive I = 920 A	0 to 1000 A	Allows setting the maximum reactive current per module (P-Q curve per module) that the system can provide during the LVRT / HVRT events.	No
SG10.3.7-Inductive cosphi= 0.50	0.00 a 1.00	Allows setting the inductive limit of the inverter for AC voltage 1 pu.	No
SG10.3.8-Capacitive cosphi= 0.50	0.00 a 1.00	Allows setting the capacitive limit of the inverter for AC voltage 1 pu.	No

7.10.4. Subgroup SG10.4: Transformer

This group allows configuring parameters related to the medium voltage transformer.

Note: These parameters are displayed if [SG10.1.3-Product type = HEMK] and [SG10.7.3-MV Skid = Yes, Motorized].

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG10.4.1-Cell type = V	V LV LVL	Defines the MV cell type topology. OPTION: V – DESCRIPTION: Protection cell (V). OPTION: LV – DESCRIPTION: Input + Protection cell (LV). OPTION: LVL – DESCRIPTION: Input + Protection + Output cell (LVL).	No
SG10.4.2-MV Voltage = 34500 V	6600 to 36000 V	Allows defining the medium voltage value.	No

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG10.4.3-Pn = 4206 kW	0 to 5000 kW	Allows setting the medium voltage transformer nominal power. **Note**: This parameter is displayed if [SG10.1.3-Product type = HEM].	No
SG10.4.4-Xcc = 10.00 %	0.00 to 20.00 %	Allows setting the medium voltage transformer short-circuit reactance, in percentage. **Note**: This parameter is displayed if [SG10.1.3-Product type = HEM].	No
SG10.4.5-Pfe = 7.00 kW	0.00 to 100.00 kW	Allows the adjustment of MV transformer core losses for subsequent calculations. **Note**: This parameter is displayed if [SG10.1.3-Product type = HEM].	No
SG10.4.6-Pcu = 23.0 kW	0.0 to 100.0 kW	Allows setting the medium voltage transformer copper losses for subsequent calculations. **Note**: This parameter is displayed if [SG10.1.3-Product type = HEM].	No
SG10.4.7-Test frequency = 60 Hz	40 to 70 Hz	Allows setting the medium voltage transformer manufactured frequency. **Note**: This parameter is displayed if [SG10.1.3-Product type = HEM].	No
SG10.4.8-Open current = 0.500 %	0.000 to 10.000 %	Allows setting the medium voltage transformer open current in percentage. **Note**: This parameter is displayed if [SG10.1.3-Product type = HEM].	No

7.10.5. Subgroup SG10.5: Filter

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG10.5.0-New HW Filter = No	No Yes	Allows to configure the new HW filter.	Yes
SG10.5.1-R (mOhm) = 5.00 mOhms	0.00 to 200.00 mOhms	Sets the value of the LCL filter main inductance resistance. **Note**: from version 4.11.0, this is a read-only parameter.	No
SG10.5.2-L (mH) = 0.073 mH	0.000 to 20.000 mH	Sets the value of the LCL filter main inductance. **Note**: from version 4.11.0, this is a read-only parameter and its default value depends on [SG10.5.0-New HW Filter]: • if SG10.5.0=No, the value default will be forced to 0.073 mH. • If SG10.5.0=Yes, the default value will be 0.130 mH.	No
SG10.5.3-C (uF) = 123.00 µF	0.00 to 200.00 µF	Sets the value of the capacitance of the LCL filter per module. **Note**: from version 4.11.0, this is a read-only parameter and its default value depends on [SG10.5.0-New HW Filter]: • if SG10.5.0=No, the value default will be forced to 123.00 uF. • If SG10.5.0=Yes, the default value will be 81.00 uF.	No
SG10.5.4-Lc (uH) = 10.00	0.00 to 200.00	Sets the value of the notch inductance of the LCL filter. **Note**: from version 4.11.0, this is a read-only parameter and its default value depends on [SG10.5.0-New HW Filter]: • if SG10.5.0=No, the value default will be forced to 10.0 uH. • If SG10.5.0=Yes, the default value will be 14.00 uH.	No
SG10.5.5-Benchs Number C = 4	1 to 4	Number of benches that can be connected to the product. The default value depends on the number of modules of the inverter configured in the parameter [SG10.1.12-Number of modules]. **SG10.1.12** / **Default value of SG10.5.5** 1 / 1 2 / 3 3 / 4 4 / 4 **Note**: this is a read-only parameter.	No

7.10.6. Subgroup SG10.6: DU

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG10.6.1-DU Type = GFDI + IMD	Disabled IMD GFDI + IMD	Sets DU Type. **OPTION** / **DESCRIPTION** Disabled: DU disabled. IMD: Isolated pole option. GFDI + IMD: Grounded poles.	No
SG10.6.2-DU Fuses = Positive Pole	Positive Pole Both Poles Custom	Sets DU Fuses. **OPTION** / **DESCRIPTION** Positive Pole: Grounded poles. Both Poles: Isolated poles option. Custom: Allows to be custom	No
SG10.6.3-DU Channel Numbers = 3	1 to4	Sets the number of DU channels to be connected and used.	No

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7.10.7. Subgroup SG10.7: Optional

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG10.7.1-Econmode = No	No Yes	Defines if the optional ECONMODE transformer is installed. Only compatible with HEM or MPCSM (no HEMK). The system has a transformer to feed auxiliar system using grid. It has an external power supply provided by the customer. **Note:** This parameter is displayed if [SG10.1.3-Product type ≠ HEMK].	No
SG10.7.2-Anti-PID = No	No Yes	Defines the ANTI-PID device is installed. Only compatible with DU Type (SG10.6.1) = grounded pole. **Note**: This parameter is displayed if [SG10.6.1-DU Type]] = IMD.	No
SG10.7.3-MV Skid = No	No Yes, Not Motorized Yes, Motorized	Parameter indicating that the HEMK inverter is connected to a Power Electronics MV Skid. **OPTION** / **DESCRIPTION** No: The inverter is not connected to MV Skid. Yes, Not Motorized: The inverter is connected to MV Skid. The protection cell of the MV Skid does not have the motorised circuit breaker, so it can be opened remotely but not closed remotely. Yes, Motorized: The inverter is connected to MV Skid. The protection cell of the MV Skid has a motorised circuit breaker, so it can be opened and closed remotely. **Note:** This parameter is displayed if [SG10.1.3-Product type = HEMK].	No
SG10.7.4-Volt Meas MV Skid = No	No Yes	Configures if MV Skid has MV voltage measure. If it is configured in "Yes", then the MV Skid transformer premagnetization functionality is available. **Note:** This parameter is displayed if [SG10.1.3-Product type = HEMK] and [SG10.7.3-MV Skid] = Yes, Motorized].	No
SG10.7.5-Skid MV Transf PT100 = No	No Yes	Defines if a HEMK inverter is connected to a MV Skid with MV transformer PT100 measure. **Notes:** - If [SG10.7.9-Cooling Type] = KNAN, this value is forced to Yes and cannot be modified. -This parameter is displayed if [SG10.7.3-MV Skid] ≠ No and [SG10.1.3-Product type] = HEMK.	No
SG10.7.6-UPS = No	No Yes	Defines if a HEMK inverter is connected to a MV Skid with a UPS device that feeds auxiliars of the inverter and is refrigerated with air conditioning. **Note:** This parameter is displayed if [SG10.7.3-MV Skid] ≠ No and [SG10.1.3-Product type] = HEMK.	No

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG10.7.7-MV Cell Manufacturer = OMZ	OMZ Siemens SFA	Allows to configure the MV cell manufacturer so that the SW can adapt its response to the Watchdog signal. **OPTION** / **DESCRIPTION** OMZ – Cell manufacturer MV OMZ. SIEMENS – Cell manufacturer MV Siemens. SFA – Cell manufacturer SFA **Note:** This parameter is displayed if [SG10.7.3-MV Skid ≠ No] and [SG10.1.3-Product type = HEMK].	No
SG10.7.8-Temp. board = Disabled	Disabled Enabled	Allows to enable or disable the specific temperature measurement boards for monitoring the temperature of BUSPLUS Advanced switch disconnectors. These temperatures are displayed in the [SV1.17: Bus Plus temp] group. **Note**: This parameter is shown and set to Enabled only when all of the following conditions are met simultaneously: - [SG10.1.4-Inverter type] = BUSPLUS Advanced. - [SG10.1.15-HW version number] = HW version 1.4. Otherwise, the parameter is not shown and is forced to Disabled.	No
SG10.7.9-Cooling Type = ONAN	ONAN KNAN	Allows the inverter's power derating thresholds to be configured automatically, depending on the option selected (sub-groups [G3.3.2: Derating Tª adm disc] and [G3.3.4: Derating T adm charge]) automatically, depending on the ambient and intake temperatures, measured at connector J35 on the central control board and displayed in [SV1.6.4-Admission Tº]. **OPTION** / **DESCRIPTION** ONAN – Not defined in this version. KNAN – It allows to configure the following parameters: • [G3.3.2.5-Adm. I2 discharge] = 86% • [G3.3.4.5-Admission I2 charge] = 86%. It also enables the [F187: SKID transformer overtemperature] fault report, configurable in [G5.4.8-Skid Transformer Admission]. **Note:** this parameter is shown if [SG10.7.3-MV Skid] ≠ No and [SG10.1.3-Inverter type] = HEMK.	No

7.11.Group SG14: Access Settings

This group allows the management of external users with access to the inverter.

7.11.1. Subgroup SG14.1: Ethernet

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG14.1.1-White List IP 1 = 192.168.1.143	1.0.0.0 to 255.255.255.255	Configures the access IP of the external inverter allowed to connect to the inverter.	Yes
SG14.1.2-User Level IP 1 = Customer	Customer Comissioning Qualified customer Power Electronics staff Developer Black List	Configures the access role for the IP configured in parameter [SG14.1.1-White List IP 1].	Yes
SG14.1.3-White List IP 2 = 192.168.1.143	1.0.0.0 to 255.255.255.255	Configures the access IP of the external inverter allowed to connect to the inverter.	Yes
SG14.1.4-User Level IP 2 = Customer	Customer Comissioning Qualified customer Power Electronics staff Developer Black List	Configures the access role for the IP configured in parameter [SG14.1.3-White List IP 2].	Yes
SG14.1.5-White List IP 3 = 192.168.1.143	1.0.0.0 to 255.255.255.255	Configures the access IP of the external inverter allowed to connect to the inverter.	Yes

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG14.1.6-User Level IP 3 = Customer	Customer Comissioning Qualified customer Power Electronics staff Developer Black List	Configures the access role for the IP configured in parameter [SG14.1.5-White List IP 3].	Yes
SG14.1.7-White List IP 4 = 192.168.1.143	1.0.0.0 to 255.255.255.255	Configures the access IP of the external inverter allowed to connect to the inverter.	Yes
SG14.1.8-User Level IP 4 = Customer	Customer Comissioning Qualified customer Power Electronics staff Developer Black List	Configures the access role for the IP configured in parameter [SG14.1.7-White List IP 4].	Yes
SG14.1.9-White List IP 5 = 192.168.1.143	1.0.0.0 to 255.255.255.255	Configures the access IP of the external inverter allowed to connect to the inverter.	Yes
SG14.1.10-User Level IP 5 = Customer	Customer Comissioning Qualified customer Power Electronics staff Developer Black List	Configures the access role for the IP configured in parameter [SG14.1.9-White List IP 5].	Yes
SG14.1.11-White List IP 6 = 192.168.1.143	1.0.0.0 to 255.255.255.255	Configures the access IP of the external inverter allowed to connect to the inverter.	Yes
SG14.1.12-User Level IP 6 = Customer	Customer Comissioning Qualified customer Power Electronics staff Developer Black List	Configures the access role for the IP configured in parameter [SG14.1.11-White List IP 6].	Yes

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7.11.2. Subgroup SG14.2: Custom Modbus

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG14.2.1-Custom Modbus Level = Power Electronics staff	Customer Comissioning Qualified customer Power Electronics staff Developer	Defines the role with which parameters configured in G14 can be viewed or modified via its mirror address. It does not modify them via their original Modbus address. **Note:** The Modbus mirror address is a resource used internally at programming level to obtain parameter information, according to the required specifications, without the need to modify the original Modbus address.	Yes

7.12. Group SG15: Communications

7.12.1. Subgroup SG15.1: PPC/SCADA

This group allows for proper debugging of communications faults at the Wellingthon North plant.

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG15.1.1-Offset time = 0 ms	0 to 65535 ms	Temporary compensation between communication losses.	Yes
SG15.1.2-Consecutive failures = 0	0 to 5	Amount of losses until failure.	Yes
SG15.1.3-AEMO real losses = 0	0 to 65535	AEMO actual loss counter. **Note**: this parameter is read-only (RO).	Yes

7.12.2. Subgroup SG15.2: Custom modbus master

This group of parameters allows configuring the central control board to operate as a configurable Modbus master, which can be assigned to different communication buses. It allows defining the slave, the type of operation, and the Modbus addresses involved, as well as performing periodic register readings and writing to holding registers with validation through a subsequent read.

<table>
<tr><th>SCREEN</th><th>RANGE</th><th>DESCRIPTION</th><th>SET ON RUN</th></tr>
<tr><td>SG15.2.0-Enable master modbus = Disabled</td><td>Disabled
Custom master modbus 1 enable
Custom master modbus 2 enable
Custom master modbus 3 enable</td><td>Allows enabling or disabling Modbus communications in master mode and selecting the custom Modbus master that will be active.</td><td>Yes</td></tr>
<tr><td>SG15.2.1-Slave id = 0</td><td>0 to 255</td><td>Allows configuring the Modbus address of the slave with which communication will be established.</td><td>Yes</td></tr>
<tr><td>SG15.2.2a-Comms status = Disabled</td><td>Disabled
Comms OK
Timeout
Error CRC
Invalid modbus code</td><td>Shows the status of the master-slave communication for the selected slave and master.</td><td>Yes</td></tr>
<tr><td>SG15.2.2b-Correct frame = 0</td><td>0 to 65535</td><td>Displays the number of correct Modbus frames exchanged by the custom master.</td><td>Yes</td></tr>
<tr><td>SG15.2.2c-Wrong frame = 0</td><td>0 to 65535</td><td>Displays the number of wrong Modbus frames exchanged by the custom master.</td><td>Yes</td></tr>
<tr><td>SG15.2.3-Function code = Read Holding Register</td><td>Read Holding Register
Read Input Register</td><td>Allows selecting the Modbus function code that the master will use to make read requests to the slave.
<table><tr><th>OPTION</th><th>DESCRIPTION</th></tr><tr><td>Read Holding Register</td><td>Select reading of holding registers</td></tr><tr><td>Read Input Register</td><td>Select reading of input registers</td></tr></table></td><td>Yes</td></tr>
<tr><td>SG15.2.4-Modbus address = 0</td><td>0 to 65535</td><td>Allows configuration of the Modbus address of the register to be consulted on the slave.</td><td>Yes</td></tr>
<tr><td>SG15.2.5-Number of registers = 1</td><td>1 to 20</td><td>Allows configuration of the number of consecutive Modbus registers to be read or written.</td><td>Yes</td></tr>
<tr><td>SG15.2.6-Register 1 = 0</td><td>0 to 65535</td><td rowspan="11">Show the values associated with the configured Modbus registers, up to a maximum of 20 registers.
Note: the number of available and visible registers depends on the value configured in the [SG15.2.5-Number of registers].</td><td>Yes</td></tr>
<tr><td>SG15.2.7-Register 2 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG15.2.8-Register 3 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG15.2.9-Register 4 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG15.2.10-Register 5 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG15.2.11-Register 6 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG15.2.12-Register 7 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG15.2.13-Register 8 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG15.2.14-Register 9 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
<tr><td>SG15.2.15-Register 10 = 0</td><td>0 to 65535</td><td>Yes</td></tr>
</table>

SCREEN	RANGE	DESCRIPTION	SET ON RUN
SG15.2.16-Register 11 = 0	0 to 65535		Yes
SG15.2.17-Register 12 = 0	0 to 65535		Yes
SG15.2.18-Register 13 = 0	0 to 65535		Yes
SG15.2.19-Register 14 = 0	0 to 65535		Yes
SG15.2.20-Register 15 = 0	0 to 65535		Yes
SG15.2.21-Register 16 = 0	0 to 65535		Yes
SG15.2.22-Register 17 = 0	0 to 65535		Yes
SG15.2.23-Register 18 = 0	0 to 65535		Yes
SG15.2.24-Register 19 = 0	0 to 65535		Yes
SG15.2.25-Register 20 = 0	0 to 65535		Yes

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8. PARAMETER INITIALIZATION

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It is possible to initialize parameters (restore them to their default value) one by one or by blocks. For this purpose, there are different initialization modes. Up next, the modes are listed and a summary of which parameters are initialized for each mode is shown:

- **Init G1.1:**
  - Corresponds to parameter [G1.1-Init parameters].
  - Initializes all the parameters that belong to the Gx configuration groups which must be reset to their default value, except for the configuration related to warnings and faults.
  - Recommended after upgrading to a new software version. It is also often used by the SAT to initialize algorithms when there are problems in the field.
- **Init SG1.1.4:**
  - Corresponds to parameter **[SG1.1.4-Init service group]**
  - Initializes parameters from the Service Group except for those marked as NO INIT and SG10 (different initialization).
  - Recommended after upgrading to a new software version.
- **Factory reset SG1.1.3:**
  - Corresponds to parameter **[SG1.1.3-Init factory defaults]**
  - Password protected.
  - It is usually for internal use, for the Validation area, and in Production.
  - It starts practically everything, including configuration and service groups.
- **Init SG10.1.2:**
  - Corresponds to parameter **[SG10.1.2-Force defaults]**.
  - Initializes the Service Group [SG10: Drive Select].
  - It is not recommended to perform this initialization, unless requested by Power Electronics. If this is the case, after initialization, the equipment configuration must be checked manually to confirm that everything is correct and matches the hardware.

Note: to see one by one the parameters that are initialized in each mode (column “Inicializado”), please refer to the information provided for each SW version. It is important to note that regardless of the mode, all those variables whose value is not stored in memory are returned to their default value after a power up (column “Variable no almacenada en memoria”).

ANNEX I. PSAMS

I.1. Introduction

This annex provides an operational overview of the PSAMS: what it is, the role it performs within the inverter, the protections it applies autonomously, and the actions that an authorized user can perform on it. It is intended for authorized technical personnel responsible for system administration and maintenance. It covers the following information about the PSAMS:

- The position of the PSAMS between the external network and the inverter internal network.
- Network segmentation, traffic control, and communication protection.
- Authentication and authorization of users accessing the configuration.
- The features that an authorized user can access on the PSAMS and the meaning of each read and configuration parameter.
- Normal behavior and possible errors that may occur.
- The operational responsibilities and precautions of the administrator and/or maintenance personnel.

The following are outside the scope of this annex:

- Operation of the inverter, BCI, SCADA, or PPC, except in relation to the PSAMS.
- Use of the interface, menus, and controls of the Power Electronics remote management tool.

  Note: to perform operations using the Power Electronics remote management tool, refer to the manual for that application. This annex only explains the purpose of each action on the PSAMS and the expected result.

I.2. PSAMS description

The PSAMS (*Power Station Advanced Monitoring System*) is a cybersecurity gateway integrated into the inverter. It acts as the single entry point between devices on the external network (such as the Power Electronics remote management tool, SCADA, or PPC) and the trusted internal network, where the inverter central control board and the BCI are located. In this way, it reduces the exposure of both devices to an untrusted network. The device divides the installation into **three network zones**:

DOMAIN	DEVICES (EXAMPLES)	DESCRIPTION
WAN	Power Electronics remote management tool, SCADA, and PPC	Untrusted external network containing devices external to Power Electronics.
Boundary	PSAMS	Applies access control, forwarding, and communication protection.
LAN	Inverter control board and BCI	Trusted internal network isolated by the PSAMS.

I.3. PSAMS features

This section presents the PSAMS main features within the inverter. The associated read and configuration operations are described in section I.4.

I.3.1. Temporary acquisition of inverter logs

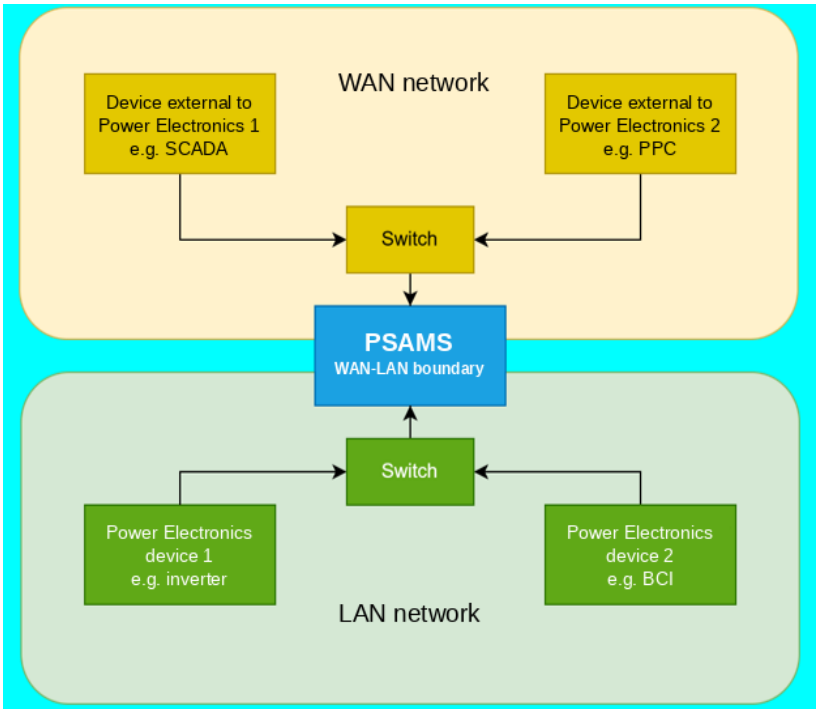
When an event occurs, the inverter sends the associated logs to the PSAMS, which stores them temporarily for at least 90 days.

I.3.2. Inverter access control

Under factory settings, the PSAMS blocks all communication between the WAN and the LAN. Communication is enabled only when a forwarding rule has been explicitly configured and authorized.

Modbus/TCP communication between the PSAMS and the inverter central control board or BCI is not encrypted because it takes place within the LAN, a trusted internal domain protected by the PSAMS.

Section I.4.4 describes how forwarding and masquerading rules between the WAN and LAN are read and configured.



WAN-LAN architecture diagram

I.3.3. Communication with the Power Electronics remote management tool for PSAMS configuration

I.3.3.1. Communication interface

Communication between the Power Electronics remote management tool and the PSAMS follows a client-server architecture, in which the remote management tool acts as the client and the PSAMS acts as the server. Requests sent from the client to the server are made through a REST API over HTTPS, which protects the channel with TLS 1.3 and ensures the confidentiality, integrity and authenticity of the data exchanged.

I.3.3.2. Identification, authentication, and capability control

The user accesses the PSAMS through a temporary personal certificate presented by the Power Electronics remote management tool. The PSAMS validates the certificate and authorizes reading or modifying its configuration according to the permissions included in the certificate. The authentication model, permissions, and rejection conditions are described in section I.4.2.

I.3.3.3. Secure software update

Through the Power Electronics remote management tool, an authorized user can update the software of the available PSAMS devices. Before the software package is installed, the PSAMS verifies its authenticity and integrity. Section I.4.6 describes the process and the monitoring parameters.

I.4. Reading and configuring PSAMS features

From the Power Electronics remote management tool, a user can read or modify the PSAMS configuration according to the permissions they have. For each feature, this section describes the meaning of its parameters, whether they are read-only or also configurable, their allowed values, and their dependencies.

Each feature is protected by a permission: the read permission allows reading its parameters, and the configuration permission allows reading and modifying them.

I.4.1. Functional blocks and permissions

The PSAMS parameters and operations are grouped into the functional blocks listed below, which are described in detail in the referenced sections. The table summarizes the purpose of each block and the available read and configuration operations.

FUNCTIONAL BLOCK	DESCRIPTION	READ OPERATIONS	CONFIGURATION OPERATIONS
Network (I.4.3)	Defines the addressing and status of the WAN and LAN interfaces.	Read interface addressing, link status, and DNS settings.	Modify addressing, DNS settings, and MAC address.
Firewall (I.4.4)	Controls communication between WAN and LAN devices through forwarding and masquerading rules.	Read the active firewall state and configured rules.	Manage forwarding and masquerading rules.
Date and time (I.4.5)	Determines the PSAMS time reference and therefore the timestamping of all logs.	Read system time, time source, time zone and offset, PTP status, NTP status, and adjust the RTC.	Configure the synchronization source, time zone, NTP server list, and adjust the RTC.
Remote software update (I.4.6)	Manages the complete software update lifecycle.	Read available PSAMS devices and, for each device, its descriptor, update status, versions, and version and partition information.	Upload the software package and start installation.
General information (I.4.7)	Displays general PSAMS information as label/value pairs.	Read general information displayed as label/value pairs.	–

I.4.2. Authentication and permissions

Access to any PSAMS feature is governed by the following rules:

- Access is provided **exclusively through mutual TLS 1.3 (mTLS)**: both the PSAMS and the client present their certificates. **User credentials, tokens, and application sessions are not used**.
- Permissions **are extracted from the presented certificate itself**; the PSAMS does not query them in a database.

The PSAMS access process is as follows:

1) The user signs in to the Power Electronics remote management tool.
2) The tool requests a temporary personal certificate from a Power Electronics intermediate server.
3) The intermediate server returns a certificate containing the permissions for the user.
4) The remote management tool presents the certificate to the PSAMS over a secure connection.
5) The PSAMS **validates the certificate** and restricts operations to the authorized permissions.

The PSAMS rejects access in the following cases:

SITUATION	PSAMS RESPONSE
No user certificate is presented.	Access is denied due to a missing user certificate.
The presented certificate is not a user certificate.	Access is denied because the certificate type is incorrect.
The certificate is valid but does not include the permission required for the feature.	Access is denied because the permissions are insufficient.

I.4.3. Network configuration

The PSAMS has two independent Ethernet interfaces: one connected to the **WAN** and the other to the **LAN**. This block allows the IP addressing of both interfaces to be read and configured.

I.4.3.1. Network parameters

The following parameters allow the PSAMS network interfaces to be read and configured.

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	ALLOWED VALUES / EXAMPLE
Interface	Number identifying the interface whose parameters are to be read or configured.	Read-only	There are two interfaces: - 0: eth0 (WAN) - 1: eth1 (LAN)
Interface name	Name used by the system to identify the interface.	Read-only	eth0 and eth1
Connected	Indicates whether the physical link is established. This is the first parameter to check if communication is lost, as it distinguishes a physical connection problem from a configuration problem.	Read-only	True / False
DHCP enabled	Allows a DHCP server to assign the interface network configuration automatically.	Configuration	True / False
IP address	IP address assigned to the interface.	Configuration	IPv4 address
Subnet mask	Mask of the subnet to which the interface belongs.	Configuration	IPv4 mask
Gateway IP	IP address of the gateway for that network.	Configuration	IPv4 address

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PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	ALLOWED VALUES / EXAMPLE
MAC address	MAC address of the PSAMS network interface being configured.	Configuration	48 bits, 12 hexadecimal digits arranged in 6 pairs separated by colons (:). Example: 00:1a:2b:3c:4d:5e
Primary DNS	Primary name server used by the interface. Required when NTP servers are configured by name rather than by IP address.	Configuration	IPv4 address. Example: 8.8.8.8
Secondary DNS	Alternative name server used when the primary DNS server does not respond.	Configuration	IPv4 address. Example: 8.8.4.4

I.4.3.2. Parameter dependencies

The configuration of the above parameters is subject to the following dependencies:

- **Manual configuration of the [IP address] parameter only applies when the [DHCP enabled] parameter = False**. If [DHCP enabled] = True, the network configuration is provided by the DHCP server ([DHCP enabled] parameter = True).
- The **[Gateway]** parameter does not need to be configured for interface eth1, as this interface belongs to the internal LAN, which is isolated by the PSAMS itself.
- The **[Primary DNS]** and **[Secondary DNS]** parameters are only used when a server is configured by its name (for example, an NTP server identified as ntp.company.com instead of by an IP address). If the servers are configured by IP addresses, these parameters are not used.

The PSAMS rejects the operation in the following cases:

- **Nonexistent interface**: the identifier specified in the [Interface] parameter does not match any available interface.
- **Incorrect format**: the values entered in the [IP address], [Subnet mask], [Gateway], [MAC address], [Primary DNS], or [Secondary DNS] parameters do not match the expected format. In this case, the entire modification is rejected, and the previous configuration is kept.

I.4.4. Firewall: forwarding and masquerading

The PSAMS manages a **single address translation table** organized into two chains:

- **Prerouting**: contains **forwarding** rules (destination address translation, DNAT), which explicitly authorize incoming traffic from the WAN to a LAN device.
- **Postrouting**: contains **masquerading** rules (source address translation, NAT), which replace the packet source address before transmission.

I.4.4.1. Reading the current state and configuration

The PSAMS provides two queries that allow the firewall state to be checked before modifying its configuration:

FEATURE	RETURNED INFORMATION	PURPOSE
Current firewall state	Rules **actually active** in the PSAMS, in text format.	Check that the configuration has been applied. This is the **diagnostic** view.
Firewall configuration	Structured list of defined forwarding and masquerading rules.	Review and audit defined entries. This is the **configuration** view.

Note: after creating or deleting a rule, it is recommended to check the **current firewall state** and not only the firewall configuration. If the two views do not match, there is a problem applying the rules.

I.4.4.2. Forwarding configuration

A forwarding entry authorizes communication between a device on the WAN and a device on the LAN. Each entry consists of the following fields:

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PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	ALLOWED VALUES	REQUIRED
Protocol	Transport protocol allowed for this entry.	Configuration	TCP or UDP	Yes
Source port (WAN)	Port on which the PSAMS listens on its WAN interface. This is the port to which the external device must connect.	Configuration	Integer	Yes
Source IP (WAN)	IP address of the WAN device to which the rule is restricted. **If omitted, the rule accepts any source**.	Configuration	IPv4 address	No
Destination IP (LAN)	IP address of the LAN device to which traffic is forwarded.	Configuration	IPv4 address	Yes
Destination port (LAN)	Port on the LAN device to which traffic is forwarded.	Configuration	Integer	Yes

I.4.4.3. Forwarding entry logic

- The fields form an **indivisible block**: they are repeated as many times as entries are created.
- While no entry exists, communication between the devices on the WAN and those on the LAN remains **disabled**. This is the case, for example, for communication between an external device and the inverter control board.
- When an entry is **deleted**, the corresponding communication is disabled again.
- **Leaving [Origin IP (WAN)] unspecified opens the port to any device on the WAN**. It must always be specified unless expressly justified.

The PSAMS rejects the operation in the following cases:

- **Invalid destination IP:** the address specified in [Destination IP (LAN)] has an invalid format.
- **Invalid origin IP:** the address specified in [Source IP (WAN)] has an invalid format.
- **Unsupported protocol:** the value specified in [Protocol] is neither TCP nor UDP.
- **Duplicate entry:** an entry with the same [Protocol], [Source port (WAN)], and [Source IP (WAN)] values already exists.
- **Nonexistent entry**: when deleting an entry, no entry matches the specified [Protocol], [Source port (WAN)], and [Source IP (WAN)] parameters. The destination fields are not used to find the entry to be deleted.

I.4.4.4. Masquerading configuration

Masquerading allows the PSAMS to replace the source address of the packets it forwards with its own address on the outgoing interface. As a result, the destination device receives the traffic as if it came from the PSAMS rather than from the device that originated it.

The replacement is applied once the packet's output interface has been determined and before the packet is transmitted. It only affects packets that meet the configured conditions.

The affected traffic is selected using one or both of the following criteria, which apply to both the LAN and the WAN. These criteria determine which packets masquerading is applied to, but only their source IP address is replaced.

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	ALLOWED VALUES
Masquerading Interface	Output interface on which masquerading is applied. The rule matches packets whose output interface corresponds to the configured value.	Configuration	Interface name, e.g. eth1
Target IP	Destination IPv4 address used to select the packets to which masquerading is applied. The rule matches packets whose destination address corresponds to the configured value, but does not modify that address.	Configuration	IPv4 address

Note: At least one of the two parameters must be configured. If neither is specified, the operation is rejected.

The PSAMS also rejects the operation in the following cases:

- **Invalid destination IP**: the address specified in the [Target IP] parameter does not have a valid format.
- **Duplicate rule**: a masquerading rule with the same values already exists.
- **Nonexistent rule**: when deleting a rule, no rule exactly matches the values specified in the [Masquerading interface] and [Target IP] parameters.

I.4.4.5. Forwarding configuration example between a SCADA and the central control board

PSAMS configuration

A typical application of the PSAMS is to allow a device on the WAN to access a device on the LAN by means of an explicit forwarding rule. For example, Modbus/TCP communication needs to be enabled between a SCADA system with IP address 192.168.25.12 and a control board with IP address 192.168.204.202.

The forwarding entry would consist of the following parameters:

PARAMETER	VALUE
Protocol	TCP
Source port (WAN)	502
Source IP (WAN)	192.168.25.12
Destination IP (LAN)	192.168.204.202
Destination port (LAN)	502

Once this entry has been created, the PSAMS allows the SCADA system on the WAN to establish Modbus/TCP connections to the control board on the LAN. As long as the entry remains configured, the corresponding traffic is authorized. If the entry is deleted, the PSAMS blocks communication between the two devices again.

This behavior maintains the separation between the WAN and the LAN, enabling only the communications explicitly defined by the administrator.

SCADA configuration

Once a forwarding entry has been configured, the external device must establish communication using **the IP address of the PSAMS WAN interface** and the source port configured in the rule. It must not use the IP address of the destination device on the LAN directly. The PSAMS receives the communication on the WAN interface and forwards it, according to the defined rule, to the IP address and port of the destination device on the LAN.

As long as the forwarding entry remains configured, authorized Modbus/TCP traffic can cross the WAN-LAN boundary. If the entry is deleted, access to the internal device is blocked again.

Note: the configuration of external applications is outside the scope of this annex. Refer to the documentation of each system.

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I.4.5. Date and time

This block determines where the PSAMS obtains its time reference. This reference timestamps event logs and therefore determines their traceability.

The system clock always runs in UTC. The time zone is only a display preference: it changes the local time displayed and the offset reported, but not the UTC time used to log events. It can therefore be adjusted at any time without affecting the traceability of the logs.

I.4.5.1. Reading the system time

These parameters show the current PSAMS time in different formats, as well as the source it comes from.

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	EXAMPLE (+INFO)
Local time	Current time expressed in the configured time zone, including its offset.	Read-only	2026-06-01T16:23:45+02:00
UTC time	Current PSAMS time in UTC.	Read-only	2026-06-01T14:23:45Z
UTC epoch	Seconds elapsed since January 1, 1970 at 00:00:00 UTC, excluding leap seconds.	Read-only	1748788425
Time source [*]	Source from which the PSAMS is obtaining the time at the moment. In automatic mode, the effective sub-source is shown in parentheses.	Read-only	PTP, NTP, RTC, auto(PTP), auto(NTP), auto(RTC)

[*]**Note:** [Time source] parameter indicates the source actually used by the PSAMS, which may differ from the configured source. The value *auto(RTC)* means that the device operates without external synchronization; time may drift and compromise log traceability. This parameter must be checked after any network change.

I.4.5.2. Selecting the time synchronization source

This parameter determines where the PSAMS obtains its time reference from.

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	ALLOWED VALUES
Time Synchronization Source	PSAMS time synchronization source.	Configuration	Single selection from: Automatic, PTP, NTP, or RTC (local hardware clock)

The behavior of each option available for this parameter is as follows:

- **Automatic**: the PSAMS uses PTP preferentially and NTP if PTP is unavailable. If it is synchronized via NTP and PTP becomes available again, it automatically switches to PTP. If neither is available, it free-runs using the internal clock.
- **PTP**: the clock is synchronized with a PTP master clock on the network.
- **NTP**: the clock is synchronized with the configured NTP servers.
- **RTC (internal clock)**: free-running operation using the local hardware clock, **without external synchronization**. This is the only option that allows the time to be set manually.

The PSAMS rejects the operation if a synchronization source other than the available options is specified.

I.4.5.3. Display time zone

These parameters define the time zone used to display the local time and its corresponding offset from UTC.

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	EXAMPLE (+INFO)
Time zone	Display time zone expressed as an IANA database name.	Configuration	Europe/Madrid
UTC offset	Current offset of the time-zone time from UTC.	Read-only	+02:00

Notes:

- **Daylight saving time is applied automatically** from the [Time zone] definition: it requires no user intervention and explains why the reported offset changes throughout the year.
- If a zone name that does not exist in the IANA database is specified in the [Time zone] parameter, the operation is rejected.
- **Changing the time zone does not alter the PSAMS clock**, only its display.

I.4.5.4. PTP status

These parameters allow the PTP synchronization status to be checked. The protocol is enabled through the synchronization source selection (section I.4.5.2).

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	READ EXAMPLE
Enabled	Indicates whether the PTP protocol is available for use.	Read-only	True / False
Status	Indicates whether PTP is synchronized with the master clock.	Read-only	Synchronized
Master clock ID	Identifier of the master clock with which it is synchronized.	Read-only	00:1B:21:AB:CD:EF
Offset from master	Time offset from the master clock.	Read-only	125
Current PTP timestamp	Date and time supplied by PTP.	Read-only	2024-01-15T10:30:45.123456Z

I.4.5.5. NTP configuration and status

These parameters allow the NTP servers to be configured and the synchronization status with them to be checked.

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	EXAMPLE (+INFO)
Enabled	Indicates whether NTP is available for use.	Read-only	True / False
Status	PSAMS synchronization status relative to the server.	Read-only	Synchronized
Servers [*]	List of servers from which the time is obtained by NTP. They are configured as a set: each modification replaces the complete list.	Configuration	"0.pool.ntp.org", "1.pool.ntp.org", "2.pool.ntp.org". Accepts an IP address or server name
Status message	Text details of the synchronization status.	Read-only	Server synchronized
Last sync	Date and time of the last effective synchronization.	Read-only	2026-06-01T14:20:15Z
Stratum	Hierarchical distance of the server from the reference time source. A lower value indicates a closer, more accurate reference.	Read-only	2

[*] **The server list is subject to the following conditions:**

- Each entry must be **a valid IP address or a valid server name**. A malformed value causes the complete operation to be rejected.
- **The list cannot be empty**: at least one server must be specified. An empty list causes the operation to be rejected.
- If servers are configured by **name**, the **DNS** servers of the corresponding interface must be configured (see section I.4.3.2).
- Configuring NTP servers **does not mean** that PSAMS is using them, as this depends on the selected time source and PTP availability. The **[Time source]** parameter must be checked (section I.4.5.1).

I.4.5.6. Real-time clock (RTC)

These parameters allow the PSAMS hardware clock to be read and adjusted. This clock keeps the time when the device is not powered.

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	EXAMPLE (+INFO)
Datetime	Hardware clock date and time. The value is kept in UTC, not in local time.	Configuration	2026-06-01T14:23:45Z
Epoch	Number of seconds elapsed since January 1, 1970 at 00:00:00 UTC.	Read-only	1748788425

Note: the **[Datetime] parameter can only be set when the [Time synchronization source] parameter is configured as RTC**. If the [Time synchronization source] parameter is configured as PTP or NTP, the operation is rejected, because the synchronization service would immediately correct the value. The adjustment sets the system clock and **stores it in the RTC**, so that it is retained after a power outage.

The PSAMS rejects the operation if the specified date and time do not have the expected format.

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I.4.6. Remote software update

This block allows reading the available PSAMS devices, selecting the device to be updated, and monitoring the update process.

Descriptor of the device to be updated

These parameters show the characteristics of the device to be updated and must be read before starting the upload. These parameters do not reflect the update status: their values do not change during the process.

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	EXAMPLE (+INFO)
ID	Number that identifies the PSAMS device.	Read-only	0
Name	Device name.	Read-only	Controller
Accepted file format	Extension and format required for the update package.	Read-only	.bin
Max file size bytes	Maximum package size allowed, in bytes. **It must be checked before starting the upload.**	Read-only	16.777.216
Connection available during start	Indicates whether the connection to the PSAMS is maintained while installation starts. **If the value is False, a temporary loss of connection must be expected.**	Read-only	True / False

The PSAMS rejects the operation if the value indicated in [Identifier] does not correspond to an available PSAMS device.

Process phases

The update process goes through the following phases, which are shown in the [State] parameter (see *Monitoring parameters*):

PHASE	MEANING	PROGRESS INFORMATION
Idle	No update is in progress.	No
Uploading	The package is being transferred to the device.	Yes (in bytes)
Validating	The package signature and integrity are being checked.	Optionally (as a %)
Ready	The package has been validated and is ready to be installed.	No
Installing	The package is being written.	Optionally (as a %)
Rebooting	The device is rebooting with the new version.	No

Note: the absence of progress information does not mean that the process has stopped; the phase may not be quantifiable.

Monitoring parameters

These parameters allow the update progress to be monitored.

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	EXAMPLE (+INFO)
State	Phase the update process is in. The possible phases are described in *Process phases*	Read-only	Uploading
Last upgrade successful	Indicates whether the last completed update finished successfully. **Its value does not change while an attempt is in progress.**	Read-only	True / False / No data
Message [*]	Details of the failure of the last completed update.	Read-only	Text / No data

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	EXAMPLE (+INFO)
Progress completed / total	Amount processed and total amount, in the unit indicated.	Read-only	45 of 100
Progress unit	Unit in which the progress is expressed.	Read-only	bytes or %

[*] **Note:** a failure message remains visible while the device is idle and also during subsequent attempts. It is only cleared when an update completes successfully. Therefore, seeing a failure message during an update in progress does not mean that the update has failed: it may correspond to the previous attempt.

Version and slot information

The PSAMS maintains two software slots (partitions) so that the previous version is retained after each update.

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	EXAMPLE (+INFO)
Current version	Software version currently running.	Read-only	1.4.0
Previous version	Last known stable version.	Read-only	1.3.0
Target version	Version prepared for installation. **No data is shown until a software package has been uploaded and validated successfully.**	Read-only	1.5.0
Current slot	Slot from which the software is running.	Read-only	slot-a
Next slot	Slot selected for the next boot. **No data if no switchover is pending.**	Read-only	slot-b
Slote state	Status of each slot.	Read-only	Booted / Active / Inactive
Slote version	Software version installed in each slot.	Read-only	1.4.0
Installed at	Date and time when the software was installed in the slot.	Read-only	2026-08-10T09:12:04Z
Healthy	Indicates whether the slot is considered suitable for booting.	Read-only	True / False
Attempts left	Number of remaining boot attempts before the slot is discarded.	Read-only	3

Uploading the software package

The upload is subject to the following conditions:

- The file must match the **format and maximum size** declared in the device descriptor.
- The upload **always starts from the beginning of the file**, so an interrupted transfer must be repeated in full.
- **Only one upload per device is allowed at a time**. Simultaneous requests for the same device are rejected **without affecting the upload in progress**.
- The upload is completed and **the package is validated before the operation is confirmed**. If the operation is confirmed successfully, the package is validated and ready for installation.

Note: the PSAMS does not allow an update to be requested if the user does not have the required permissions.

If an error occurs during the software package upload process, the following messages are displayed:

REJECTION MESSAGE	DESCRIPTION
Invalid image	The package is malformed, incomplete, or empty.
Device not found	The device identifier specified in [ID] does not exist.
Upgrade in progress	An upload or installation is already in progress for that device.
Image too large	The size exceeds the maximum declared by the device.
Unsupported media type	The upload was not sent as binary content.
Device incorrect	The software does not correspond to the device.
Verify error	The software does not correspond to the device.
Device remote upgrade error	The remote update or the update storage failed.
Communication error	Communication with the update target could not be established.

Starting installation

This operation starts the installation of the previously uploaded and validated package. The confirmation does not indicate that the installation has finished, only that the request has been accepted successfully. The restart and the possible loss of connection occur afterwards. Before starting the installation, the [Connection available during start] parameter must be read. Once the process has started, reading of the status must be resumed when the device becomes accessible again.

The PSAMS rejects the installation start in the following cases:

- The user does not have the required configuration permission.
- The specified device does not exist.
- No software package has been uploaded and validated.
- An update is already in progress.
- The device could not start the package installation.

I.4.7. General system information

This block shows general information about the PSAMS as label/value pairs. The available labels may vary depending on the device configuration.

PARAMETER	DESCRIPTION	READ-ONLY / CONFIGURATION	EXAMPLE (+INFO)
Total	Number of entries included in the query.	Read-only	8
Total available	Total number of configured entries.	Read-only	8
Label	Entry name.	Read-only	Software Version
Value	Current value associated with the entry label.	Read-only	1.2.3.4

The available labels are the following:

LABEL	DESCRIPTION
Communication	Indicates whether the PSAMS is connected to the Power Electronics remote management tool.
Software Version	Software version currently running on the device.
Build	Identifier that distinguishes a specific software build from other builds.
Platform	Hardware architecture of the PSAMS device running the software.
Date / Time	Current date and time on the PSAMS.
Hostname	Name used to identify the device on the network and in the operating system.

I.5. Operating scenarios

The following scenarios illustrate the use of the features described in section I.4 in typical operation and maintenance situations.

I.5.1. Initial commissioning with no forwarding rules configured

If no forwarding rule is configured, the PSAMS blocks communication between the WAN and LAN. During initial commissioning, **this absence of communication is the expected behavior and does not necessarily indicate a fault**.

Before enabling communication, perform the following checks:

1) Confirm that both interfaces are connected (section I.4.3.1).
2) Check [Time source] parameter and verify that the PSAMS time is correct (section I.4.5.1).
3) Create only the required forwarding entries and then verify the current firewall state (section I.4.4.1).

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I.5.2. Routine operation

During normal operation, periodically check that the active firewall rules match the configured rules and that all enabled communications remain necessary. Remove any rule that is no longer used.

I.5.3. Forwarding rule changes

Before adding or deleting a forwarding rule, identify the affected devices and communications. After applying the change, check the current firewall state and verify that only the intended communication remains enabled.

I.5.4. Software update

The user starts the process from the Power Electronics remote management tool and provides the signed software package. The PSAMS verifies the trust chain, the signature, and the package integrity. Installation continues only if all validations are completed successfully; otherwise, the package is rejected.

During the process, the [Current phase], [Completed/total progress], and [Progress unit] parameters can be read (section I.4.6). If [Connection available during start] indicates that the connection is not maintained, a temporary loss of connection should be expected.

I.6. Secure operation best practices

During PSAMS operation, the following practices are recommended:

- Maintain only the forwarding entries that are necessary and remove those that are no longer used.
- **Always specify the [Source IP (WAN)] address** in forwarding rules. Omitting it opens the port to any device on the WAN.
- Explicitly document and approve any use of unencrypted Modbus/TCP.
- Plan changes and keep evidence of the reasons that justify them.
- **Verify the active firewall state** after every rule change, not just the configuration.
- **Periodically check the [Time source] parameter**. A PSAMS operating without external synchronization may compromise log traceability.

I.7. Glossary

This glossary lists the terms and acronyms used in this annex.

TERM	DEFINITION
BCI	Battery system control component (Battery Control Interface).
CA	Certification authority that issues or signs digital certificates (Certificate Authority).
CRL	List of certificates revoked before their expiry date (Certificate Revocation List).
CSR	Certificate signing request sent by a device to a certification authority (Certificate Signing Request).
DHCP	Protocol for the automatic assignment of IP addresses (Dynamic Host Configuration Protocol).
DNAT	Translation of a packet's destination address; basis of forwarding. (Destination Network Address Translation).
DNS	Service that resolves domain names to IP addresses (Domain Name System).
Gateway	Component that interconnects networks and controls traffic between them.
IANA	Organization that maintains, among other resources, the time zone database (Internet Assigned Numbers Authority).
LAN	Local network that connects devices within the same physical or logical area (Local Area Network).
Masquerading	Replacement of a packet' source address with the address of the output interface.
Modbus/TCP	Application protocol used for PSAMS communications.
mTLS	TLS mode in which both endpoints present a certificate and authenticate each other (mutual TLS).
NAT	Network address translation (Network Address Translation).
NTP	Protocol used to synchronize time over a network (Network Time Protocol).
PPC	Automation platform that manages and optimizes plant operation (Power Plant Controller).
Prerouting / postrouting	Address translation table chains in which forwarding and masquerading rules are evaluated, respectively.
PTP	High-precision time synchronization protocol (Precision Time Protocol).
RTC	Device hardware clock that maintains the time when the main power supply is unavailable (Real-Time Clock).
SCADA	Platform used to remotely monitor, control, and automate industrial processes (Supervisory Control and Data Acquisition).
TLS 1.3	Protocol used to provide cryptographic protection for the communication channel (Transport Layer Security).
UTC	Global reference time scale (Coordinated Universal Time).
WAN	Network that connects devices that are not necessarily on the same local network or at the same physical location (Wide Area Network).